4WD Robot

keyestudio WiKi

Jun 03, 2026

KS0559 KEYESTUDIO 4WD ROBOT V2.0

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Note

Documents are available in multiple languages.

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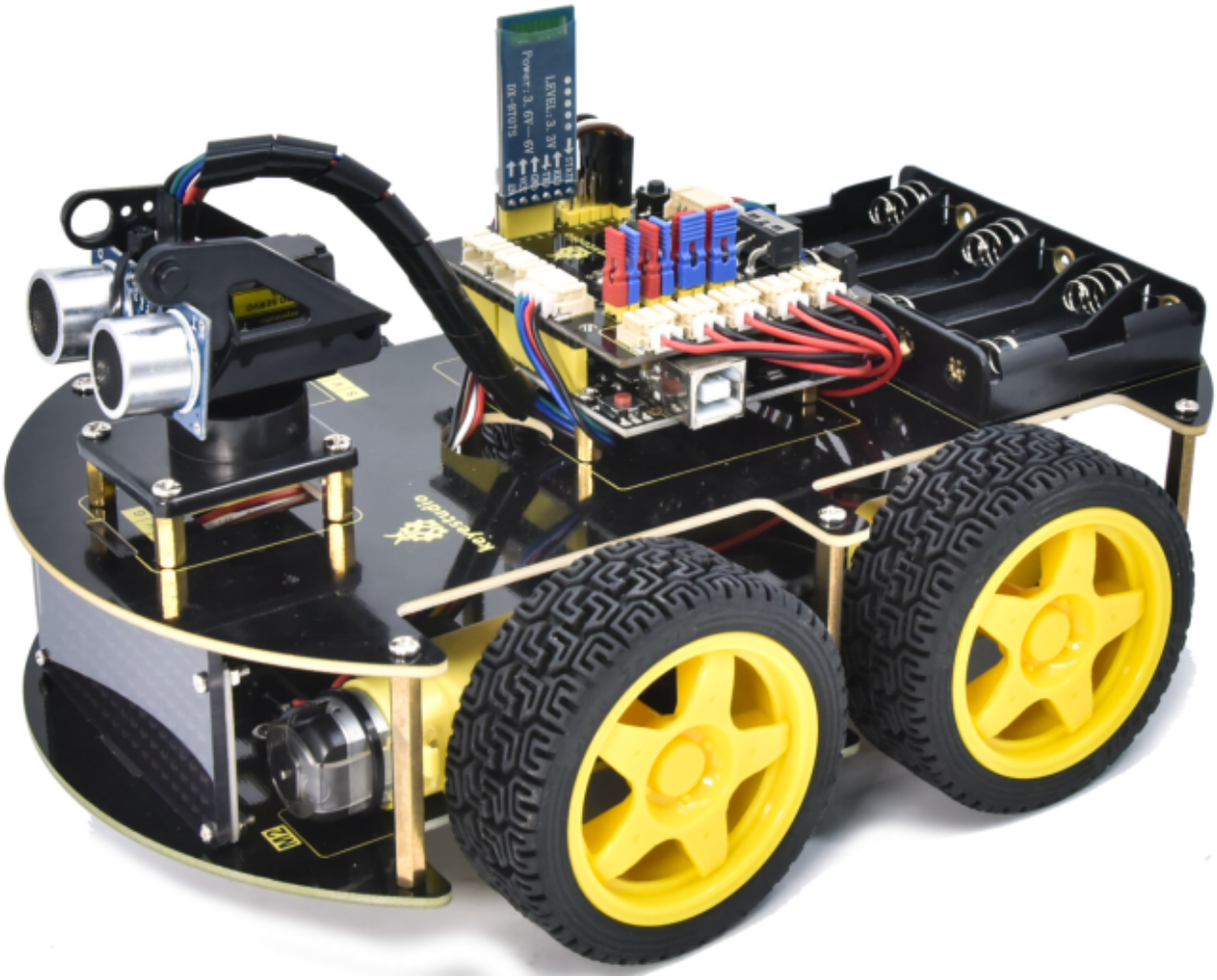


CHAPTER

ONE

1.1 INTRODUCTION

The newly upgraded 4WD Multi-purpose car is one of the most popular Arduino sets. It not only has a beautiful appearance but also powerful functions. In addition to the usual functions of line tracking, obstacle avoidance and remote control, it also uses facial expressions to make lovely expressions. Furthermore, you empower to expand your own projects, or add other sensors to complete your exploration.

Most of the components are connected via screws and copper pillars, thereby you can build your own robot through a few simple assembly steps.

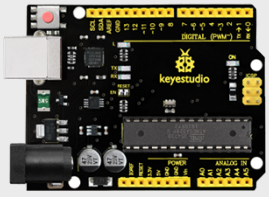
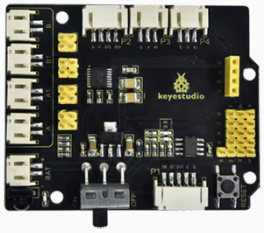
What's more, the basic knowledge such as line tracking, obstacle avoidance sensors, ultrasonic ranging, Bluetooth communication, motor drivers as well as IR remote controls are accessible for us to learn.

This product boasts 17 learning projects, from simple to complex, which will guide you to make an Arduino robot.

CHAPTER

TWO

1.2 KIT LIST

#	Name	QTY	Picture
1	Keyestudio V4.0 Board(UNO compatible)	1	
2	Keyestudio Motor Driver Expansion Board	1	
3	Bluetooth Module	1	
4	Red LED Module	1	
5	Ultrasonic Sensor	1	
6	Keyestudio Line Tracking Sensor	1	

continues on next page

Table 1 – continued from previous page

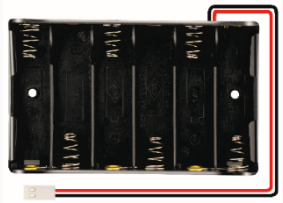
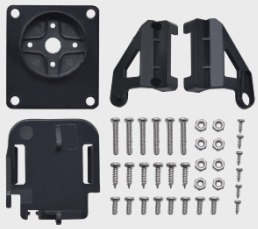
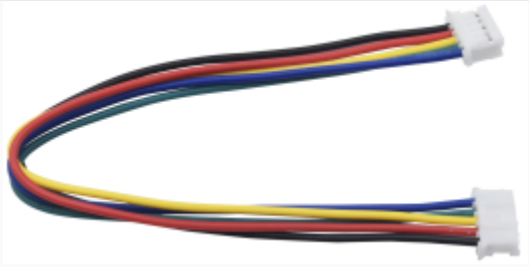
#	Name	QTY	Picture
7	Keyestudio 8*16 LED Board (4Pin Connection Wire)	1	
8	Servo	1	
9	IR Remote Control	1	
10	USB Cable	1	
11	18650 Battery Holder	1	
12	6 AA Battery Holder	1	
13	Cradle Head	1	
14	150MM JST-PH2.0MM-5P 24AWG Wire	1	

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Table 1 – continued from previous page

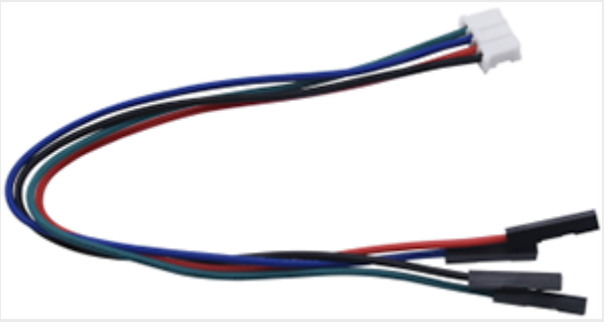
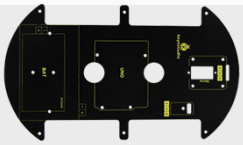
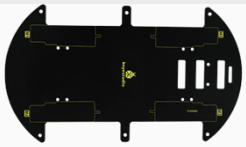
#	Name	QTY	Picture
15	150MM PH2.0mm-4P to 2.54 Dupont Wire	1	
16	Acrylic Board	1	
17	Keyestudio Top Board	1	
18	Keyestudio Base Board	1	
19	Fixed Parts	4	

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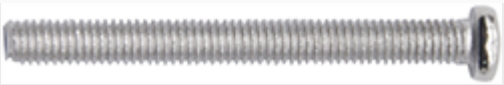
Table 1 – continued from previous page

#	Name	QTY	Picture
20	Wheels	4	
21	M3*10MM Dual-pass Copper Pillar	10	
22	M3*40MM Dual-pass Copper Pillar	4	
23	M3*30MM Round Head Screw	8	
24	M3*6MM Round Head Screw	40	
25	M3 Nickle-plated Nut	16	
26	M2*8MM Round Head Screw	6	

continues on next page

Table 1 – continued from previous page

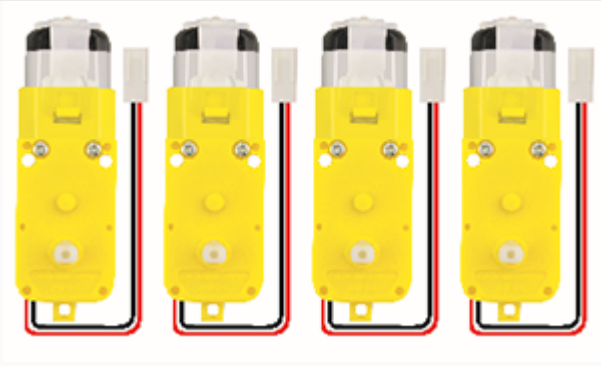
#	Name	QTY	Picture
27	M3*8MM Round Head Screw	4	
28	M2 Nickle-plated Nut	6	
29	M3*10MM Flat Head Screw	3	
30	Motor	4	
31	Screwdriver	1	
32	Black Tie	6	
33	Winding Pipe	1	
34	20CM 3pin F-F Dupont Cable	1	

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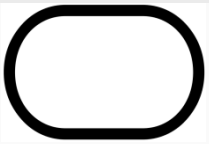
Table 1 – continued from previous page

#	Name	QTY	Picture
35	Decorative Card	1	
36	Blue Jumper Cap	4	
37	Red Jumper Cap	4	
38	Tracking Runway	1	

CHAPTER

THREE

1.3 FEATURES

- Multi-purpose function: Obstacle avoidance, following, IR remote control, Bluetooth control, confined with a circle, gravity control as well as facial emoticons display.
- Simple assembly: No soldering circuit required, simple assembly.
- Sturdy construction : 4WD construction, 4 premium motors and wheels.
- High extension: It can expand other sensors and modules via a motor driver expansion board.
- Multiple controls: IR remote control, App control(iOS and Android system)
- Basic programmingC language code of Arduino IDE.

CHAPTER

FOUR

1.4 SPECIFICATION

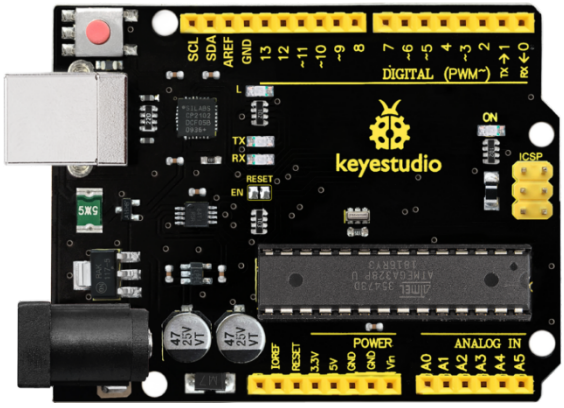
- Working voltage: 5V
- Input voltage: 6-9V
- Maximum output current: 2A
- Maximum power dissipation: 25W (T=75℃)
- Motor speed: 5V 200 rpm
- Motor drive mode: DRV8833 motor driver
- Ultrasonic induction angle: <15 degrees
- Ultrasonic detection distance: 2cm-400cm
- Infrared remote control distance: 10M (measured)
- Bluetooth remote control distance: 50M(measured)
- Bluetooth control: support Android and iOS system

CHAPTER

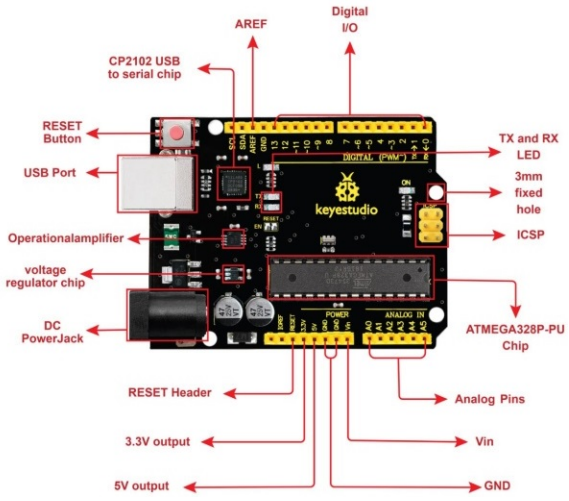
FIVE

1.5 DEVELOPMENT BOARD

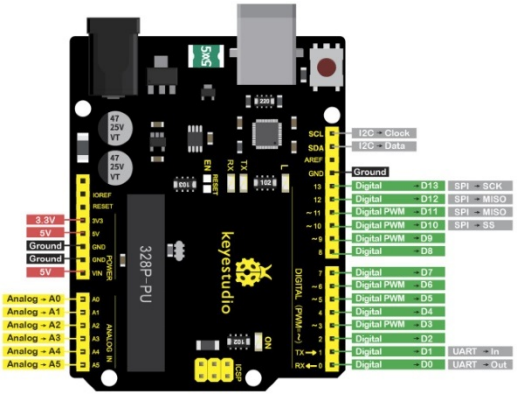
You need to know that keyestudio V4.0 development board is the core of this 4WD BT Multi-purpose Car V2.0.



Keyestudio V4.0 development board is an Arduino Uno -compatible board, which is based on ATmega328P MCU, and with a CP2102 chip as a UART-to-USB converter.



It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz quartz crystal, a USB connection, a power jack, 2 ICSP headers and a reset button.



It contains everything needed to support the microcontroller: simply connect it to a computer with a USB cable or power it via an external DC power jack (DC 7-12V) or via female headers Vin/GND (DC 7-12V) to get started.

Microcontroller	ATmega328P-PU
Operating Voltage	5V
Input Voltage (recommended)	DC 7-12V
Digital I/O Pins	14 (D0-D13) (of which 6 provide PWM output)
PWM Digital I/O Pins	6 (D3, D5, D6, D9, D10, D11)
Analog Input Pins	6 (A0-A5)
DC Current per I/O Pin	20 mA
DC Current for 3.3V Pin	50 mA
Flash Memory	32 KB (ATmega328P-PU) of which 0.5 KB used by bootloader
SRAM	2 KB (ATmega328P-PU)
EEPROM	1 KB (ATmega328P-PU)
Clock Speed	16 MHz
Onboard LED	D13

CHAPTER

SIX

1.6 MOTOR SHIELD

Description

The 8833 board adopts the 8833 motor driver chip driven by two-channel H bridges , and the maximum driving current of a single channel is up to 1.5A, and the terminal of PH2.0 is used.

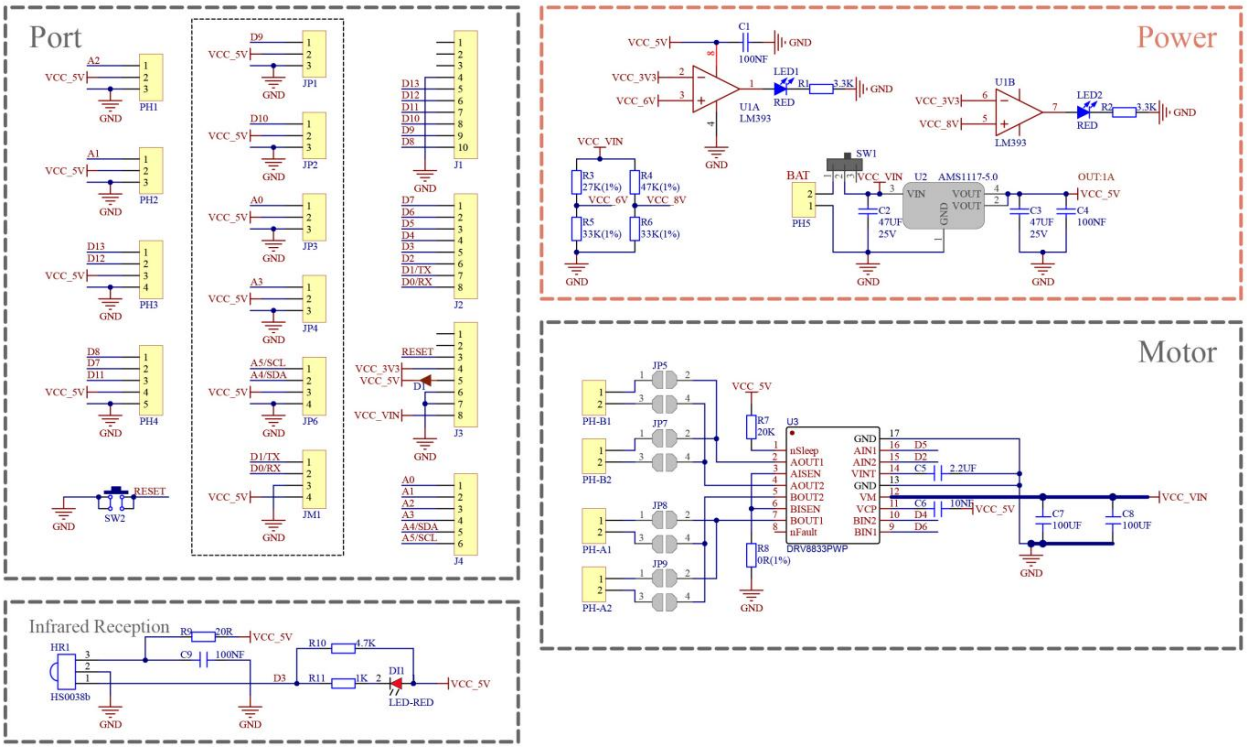
The IR receiver module is also integrated on the board. Furthermore, there are many PH2.0 expansion interfaces on the board such as ultrasonic interface, analog interface, three-channel tracking interface as well as pin interfaces with Bluetooth and motor driver.

Specification

Voltage	USB 5VDC 6-9V
Current	The maximum output is 3A, the actual output is about 1A (it's 2A when the motor is loaded)
Maximum power	27WThe actual value is about 9W
Working temperature	-10~50 Degree
Dimension	69x56x18mm
Weight	25.5g
Environmental protection attributes	ROHS

Function

Schematic Diagram


Port
Power
Motor
Infrared Reception
PH1
PH2
PH3
PH4
SW2
JP1
JP2
JP3
JP4
JP6
JM1
J1
J2
J3
J4
SW1
PH5
U2
AMS1117-5.0
U1A
U1B
LM393
LED1
LED2
U3
DRV8833PWP
PH-B1
PH-B2
PH-A1
PH-A2
JP5
JP7
JP8
JP9
HR1
HS0038b
LED-RED
VCC_5V
VCC_3V3
VCC_6V
VCC_8V
VCC_VIN
GND

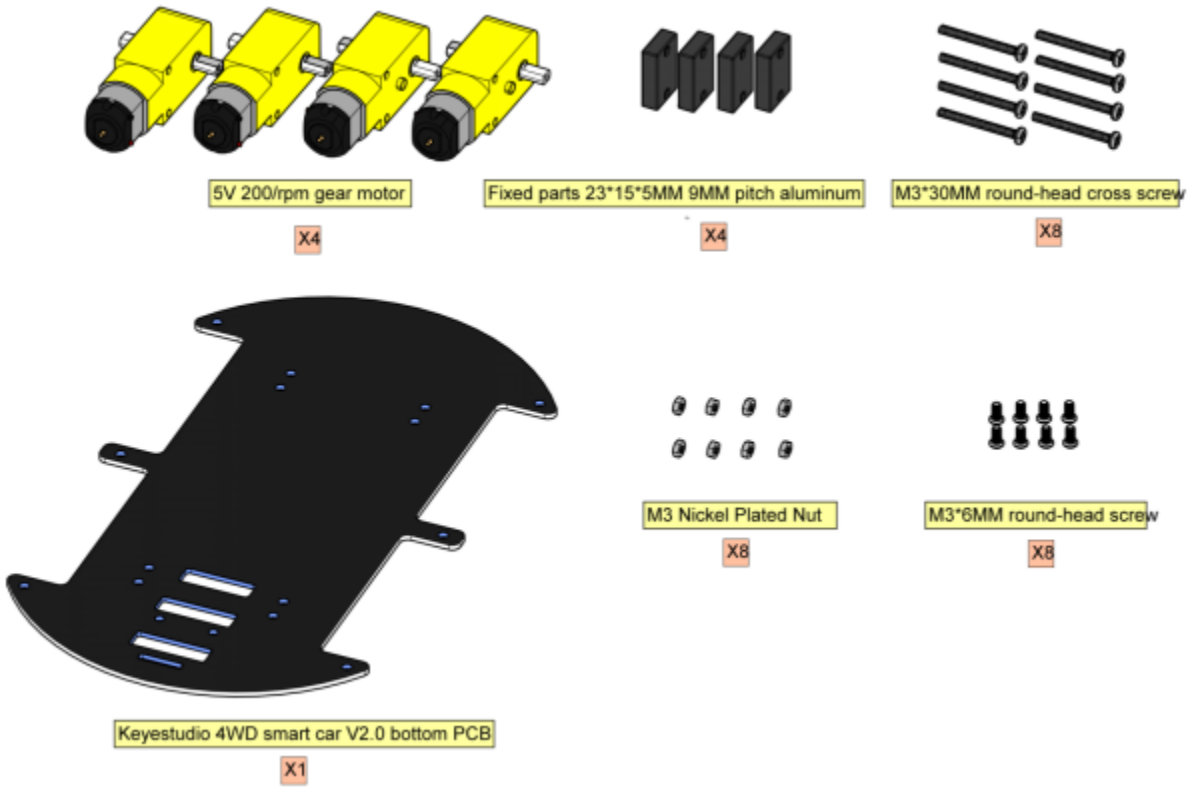
CHAPTER

SEVEN

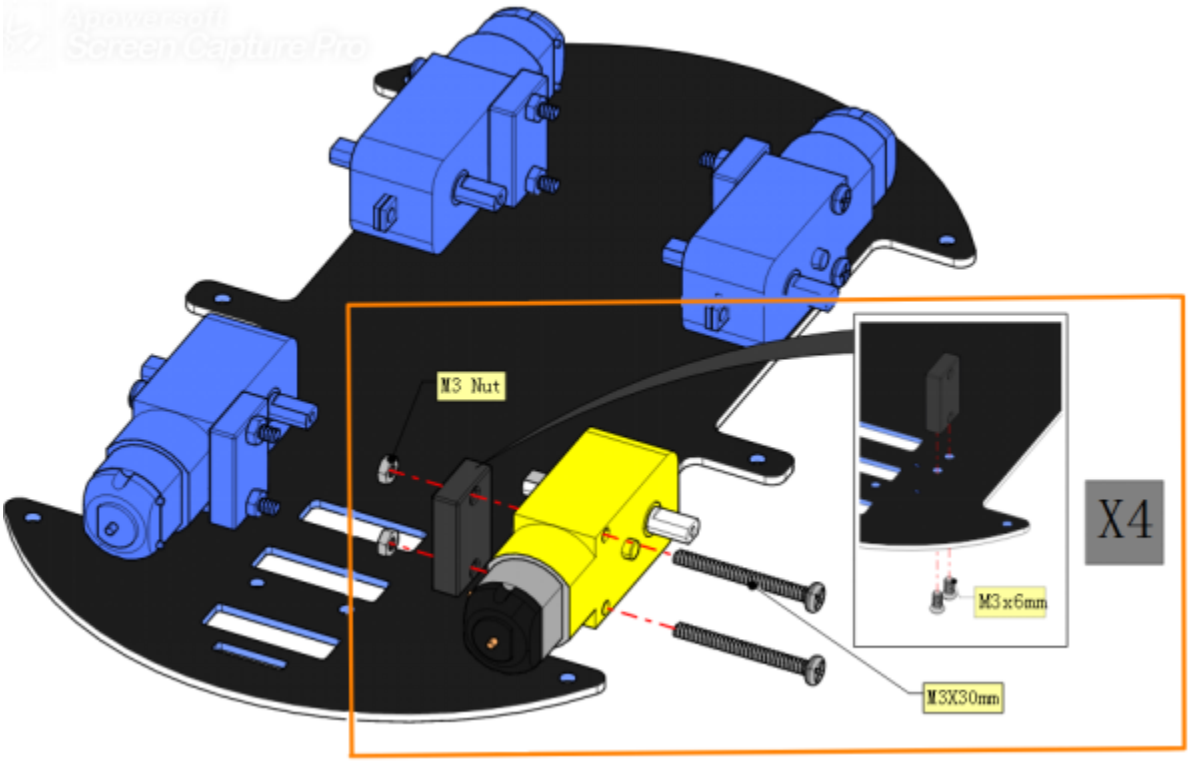
2. PRODUCT INSTALLATION

7.1 part 1

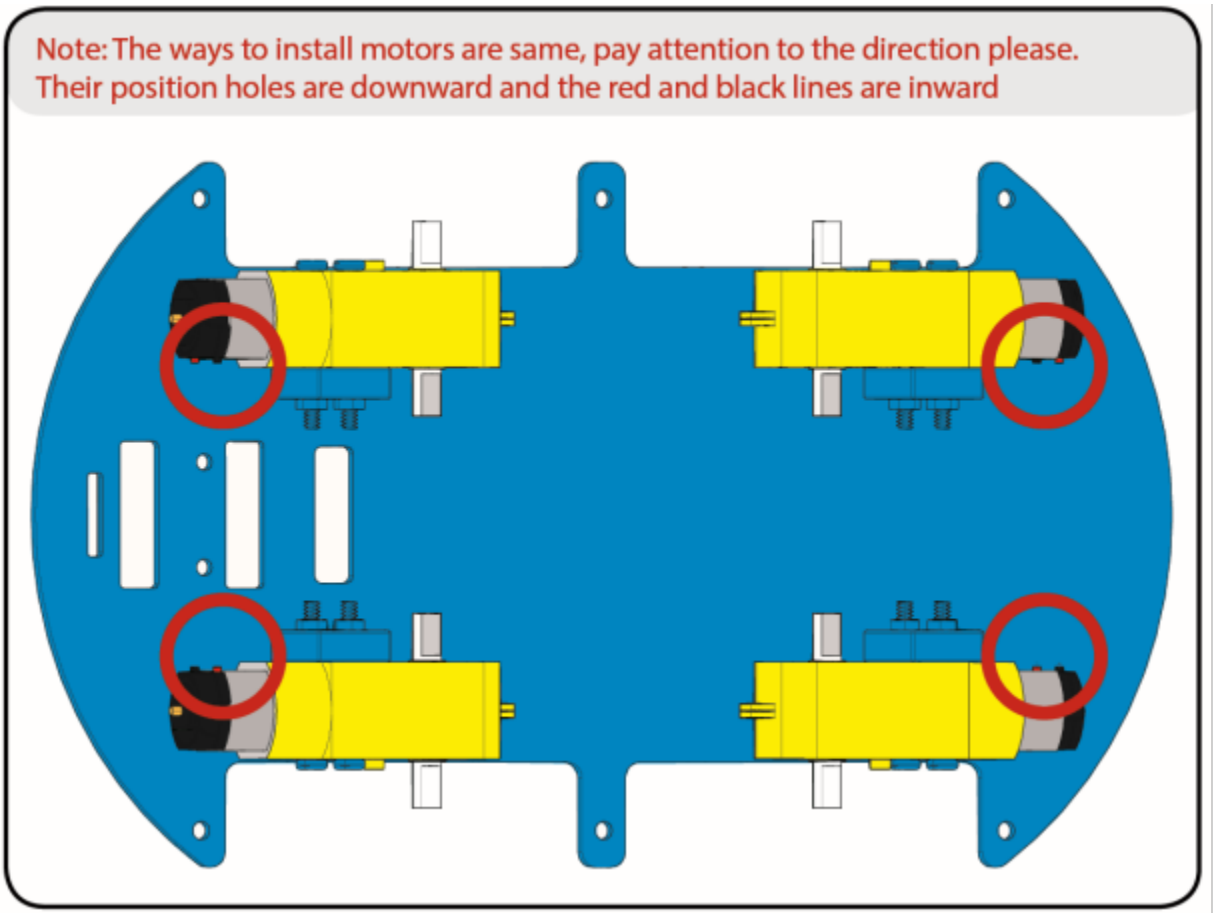
1.1. Components Needed.



1.2. Installation Diagram.

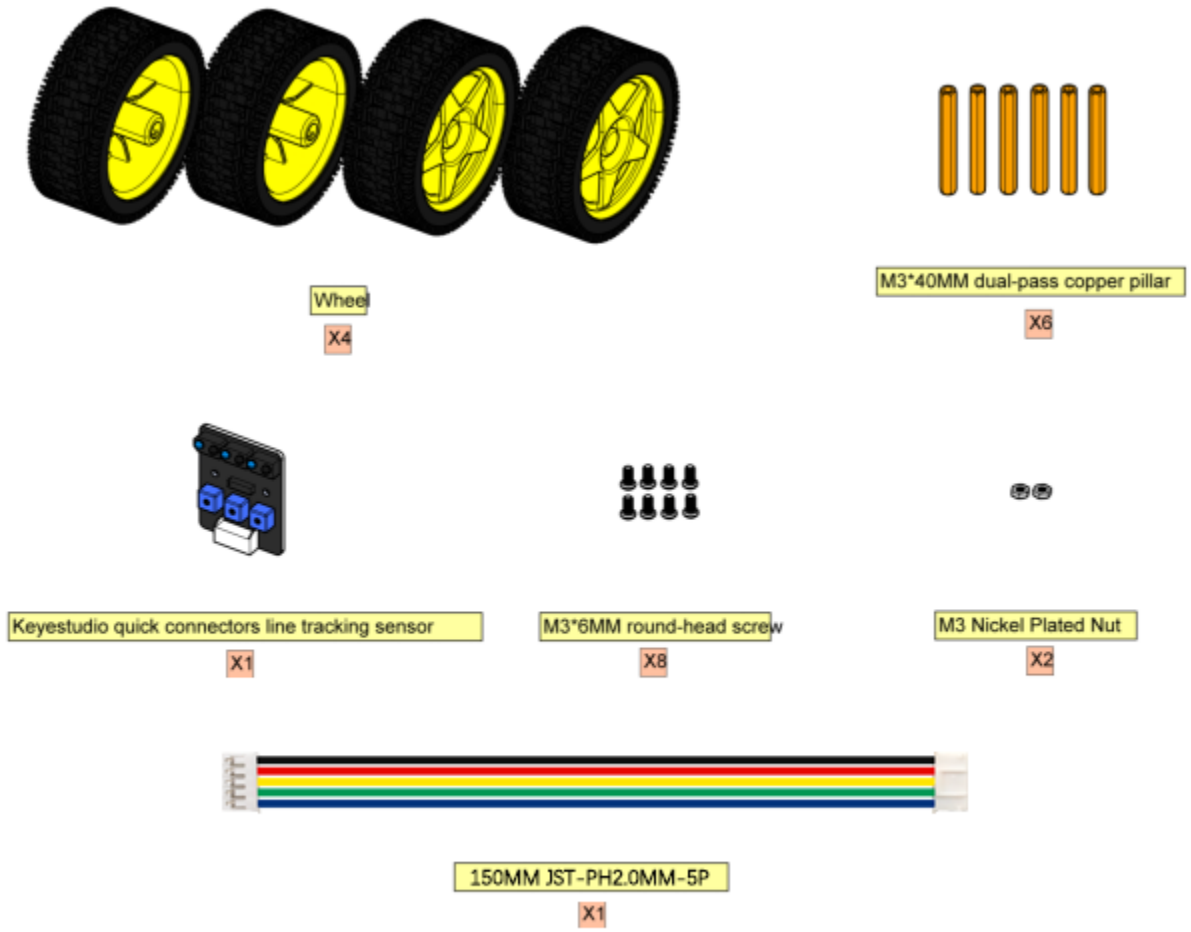


1.3. Prototype.

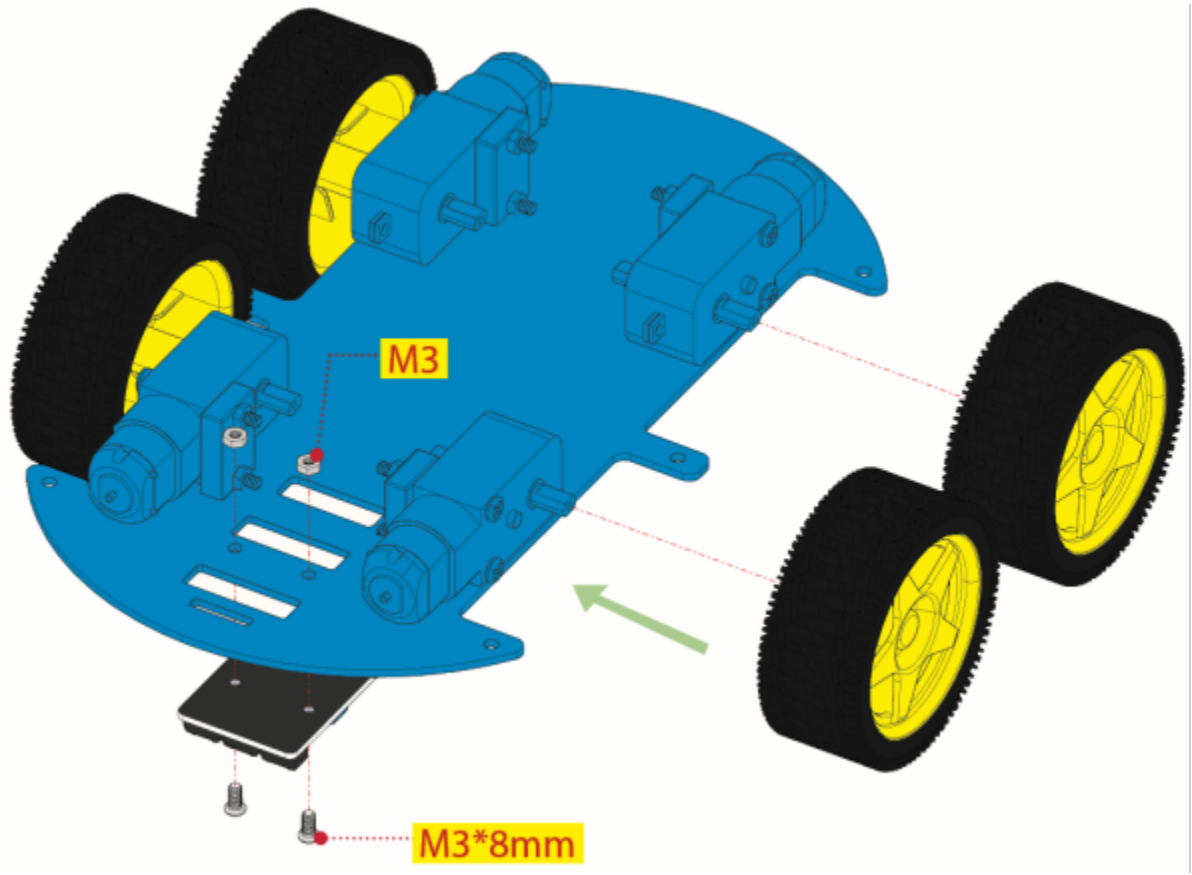


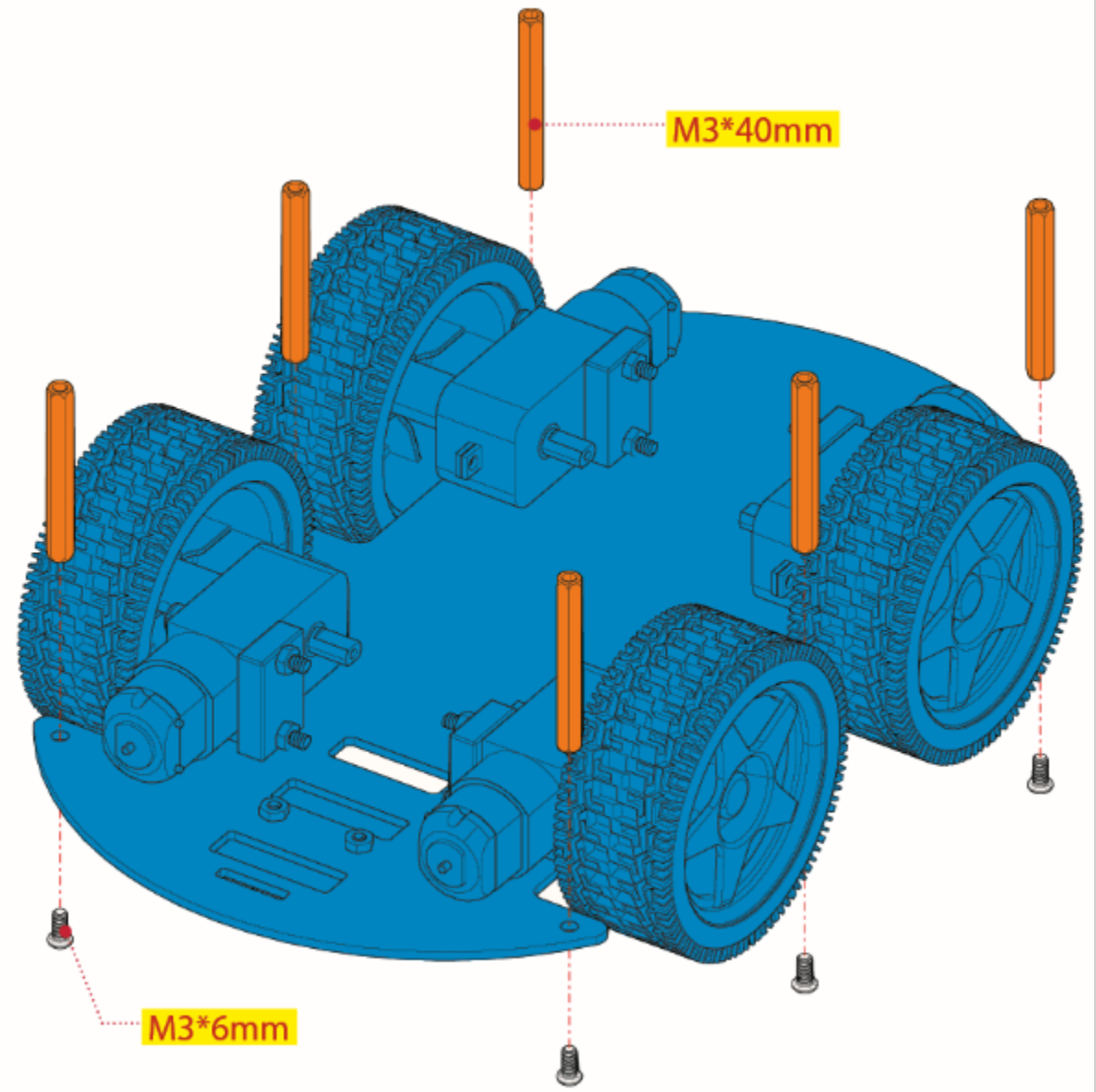
7.2 part 2

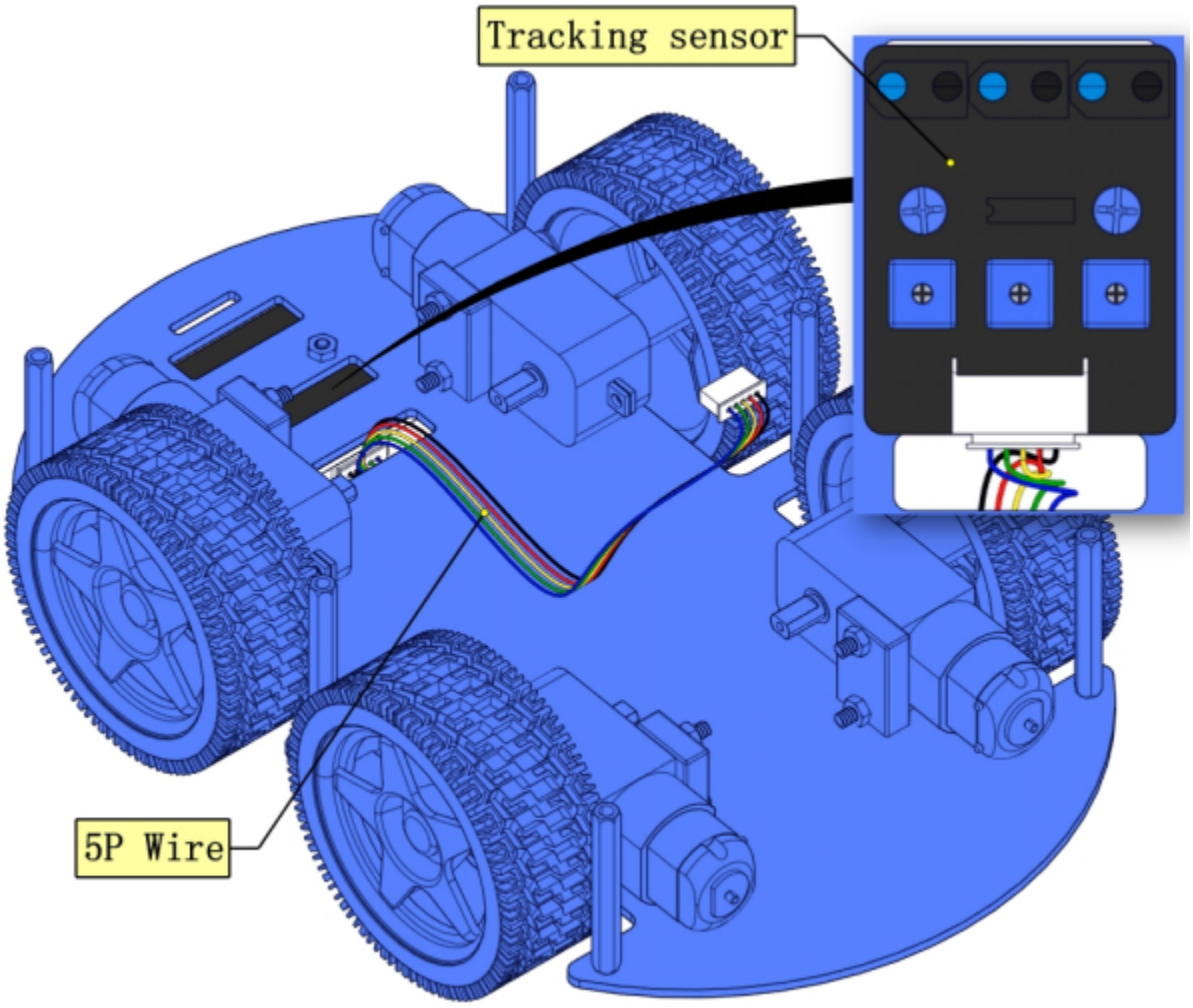
2.1. Components Needed.



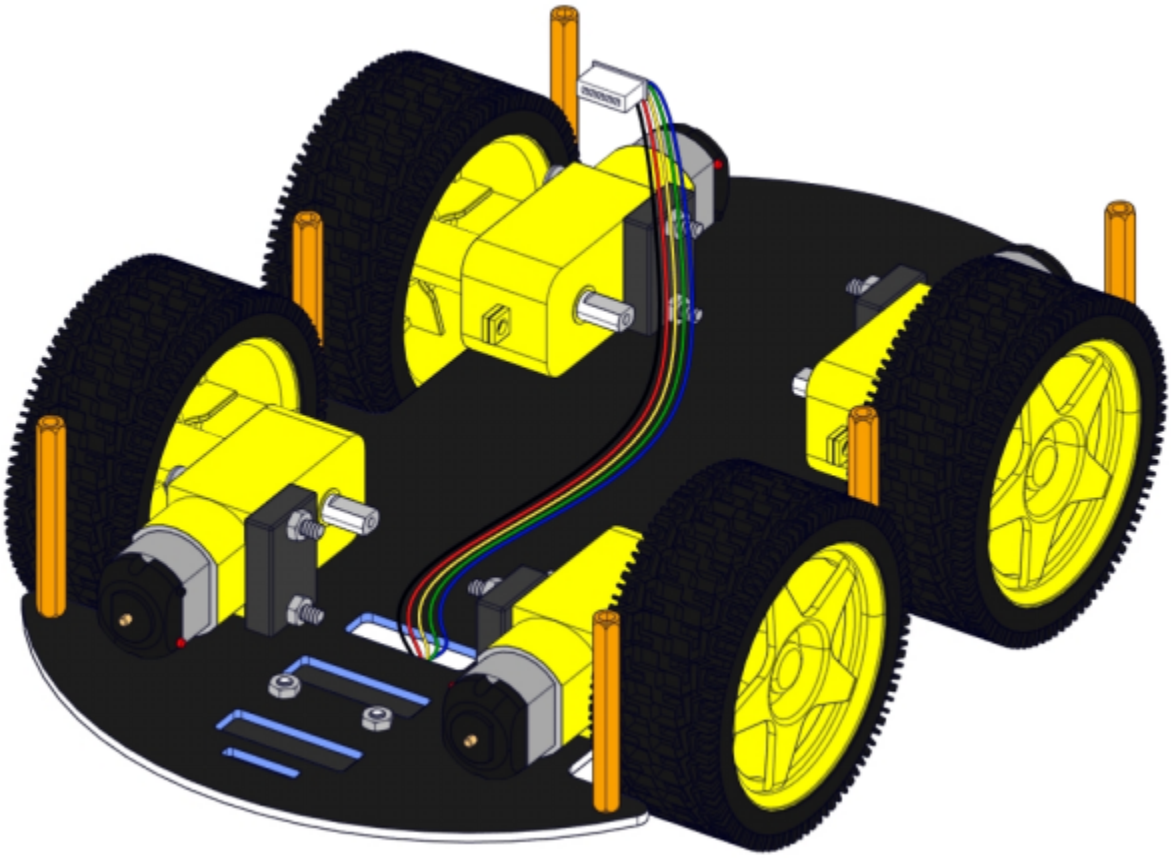
2.2. Installation Diagram.


M3
M3*8mm


M3*40mm
M3*6mm

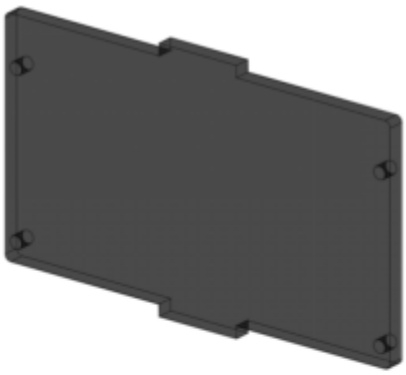


2.3. Prototype.



7.3 part 3

3.1. Components Needed.

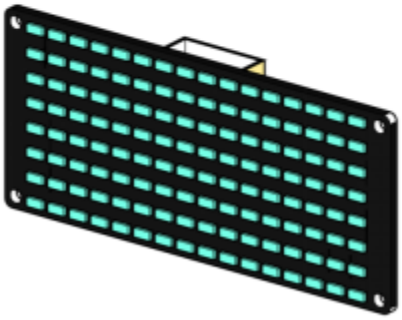


The front acrylic board of 4WD car LED screen

X1

M2 Nickel Plated Nut

X4



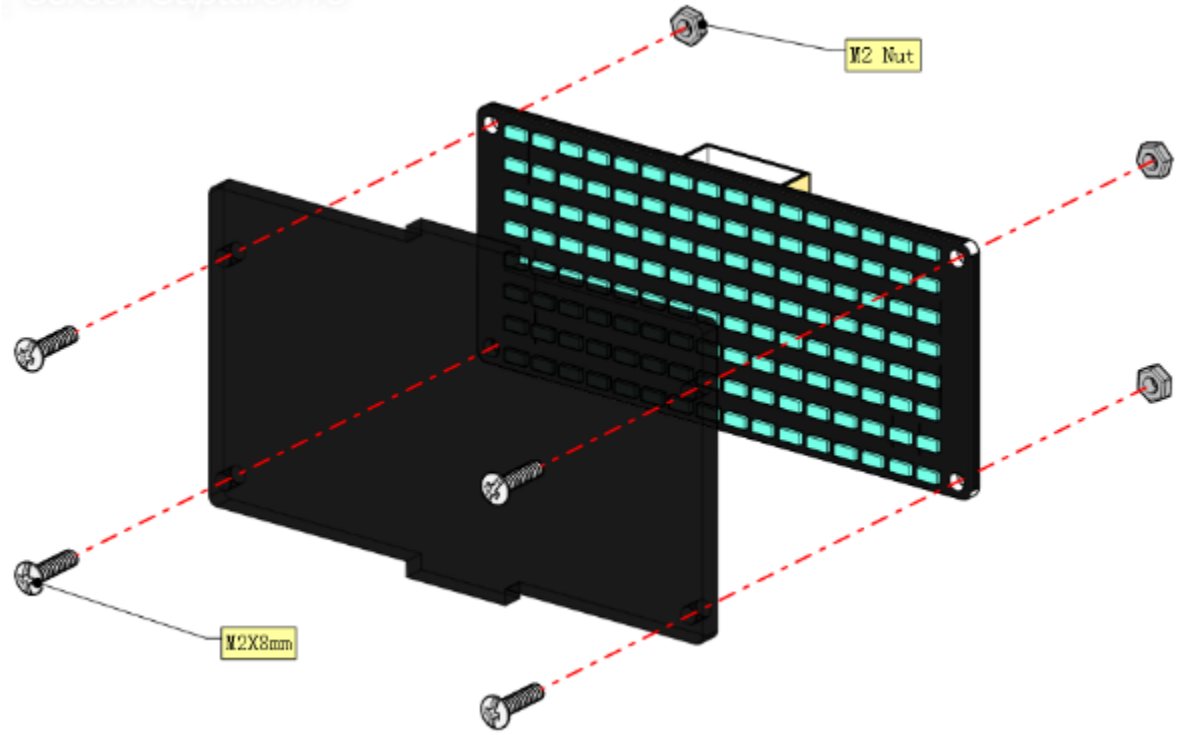
Keyestudio 8X16 LED light board

X1

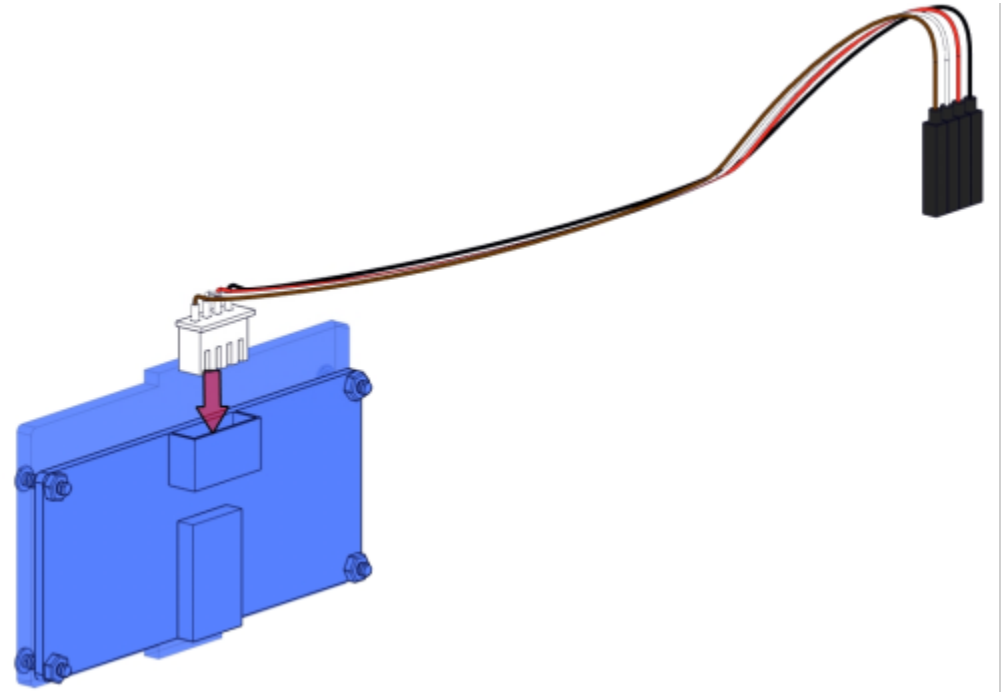
M2X8MM round-head screw

X4

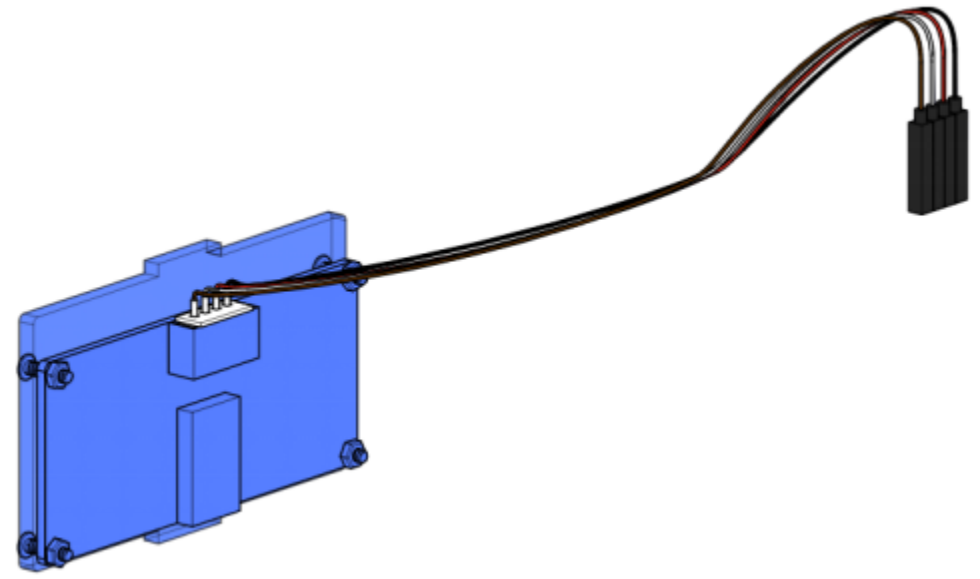
3.2. Installation Diagram.



3.3. Plug Wire.

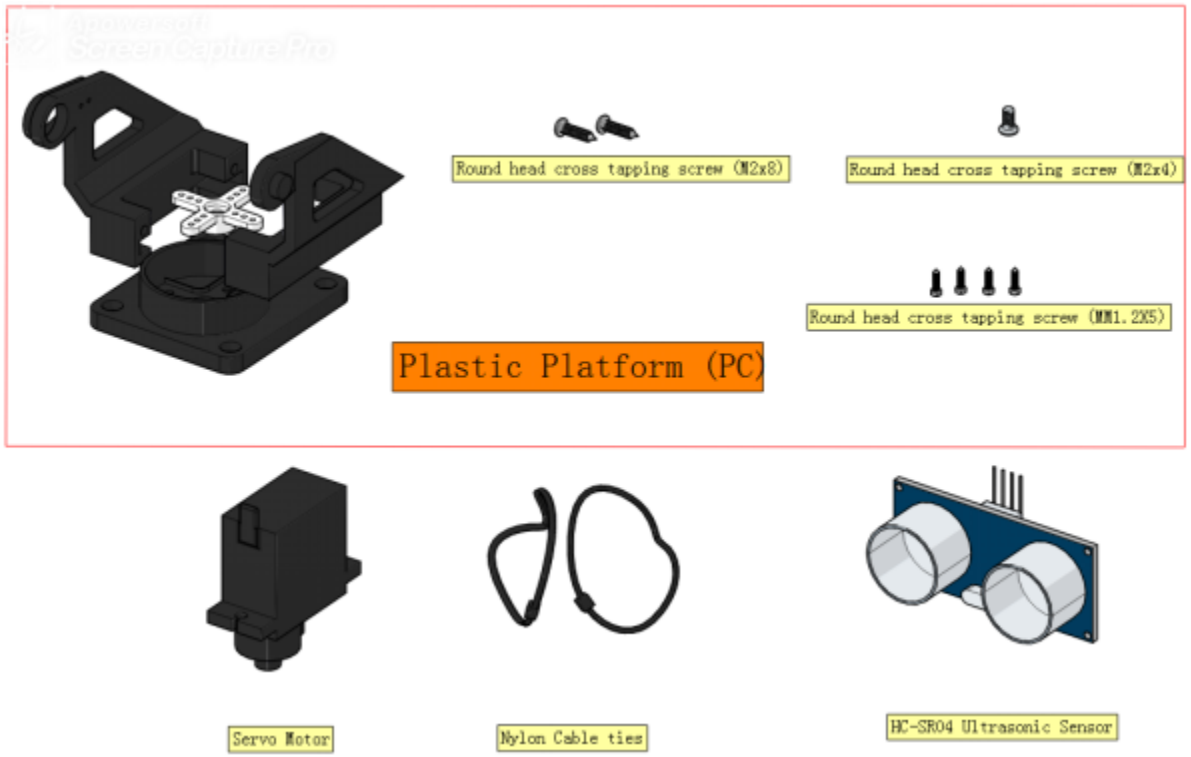


3.4. Prototype.



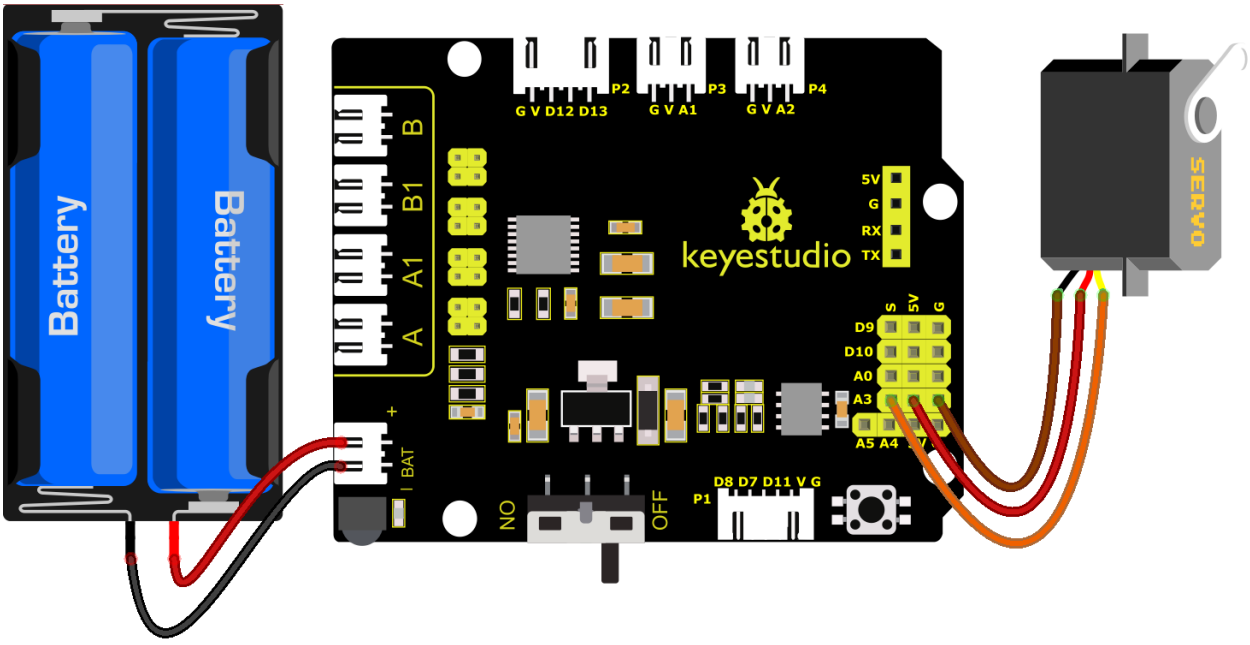
7.4 part 4

4.1. Components Needed.



4.2. Rotate the servo to 90 degrees.

Servo	8833 Motor Driver Expansion Board
Brown	G
Red	5V
Yellow	A3

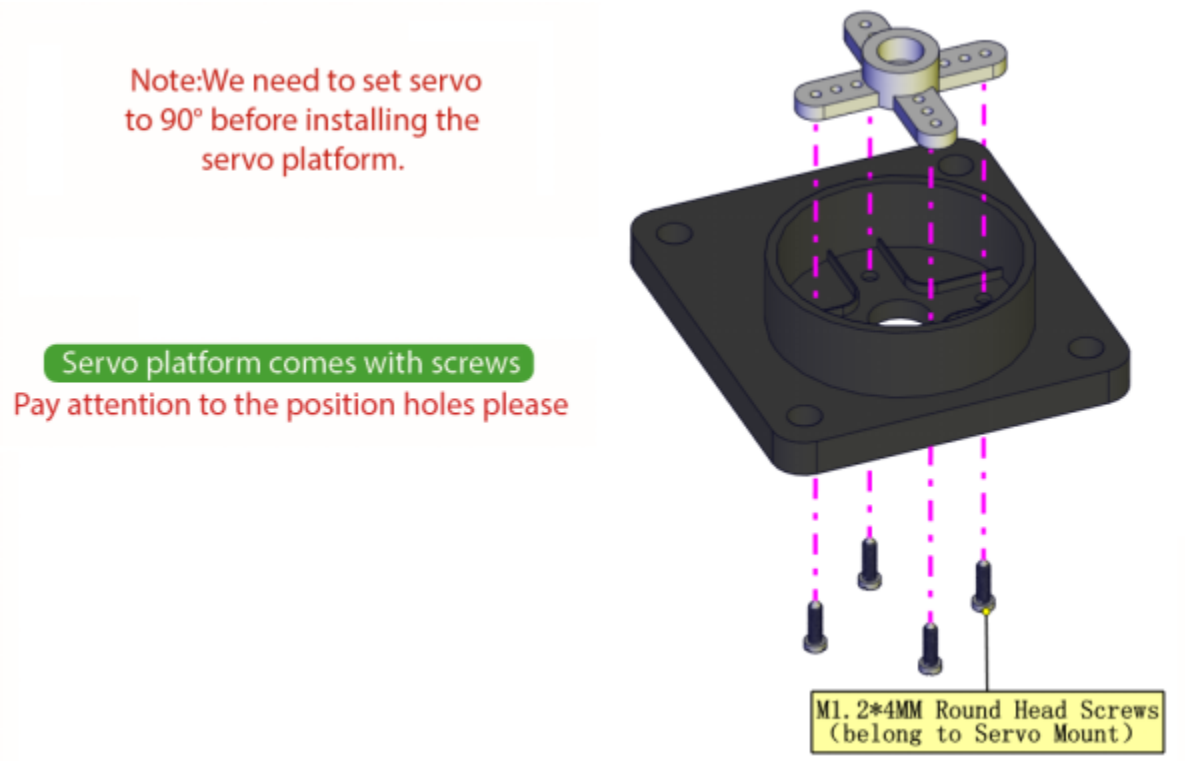


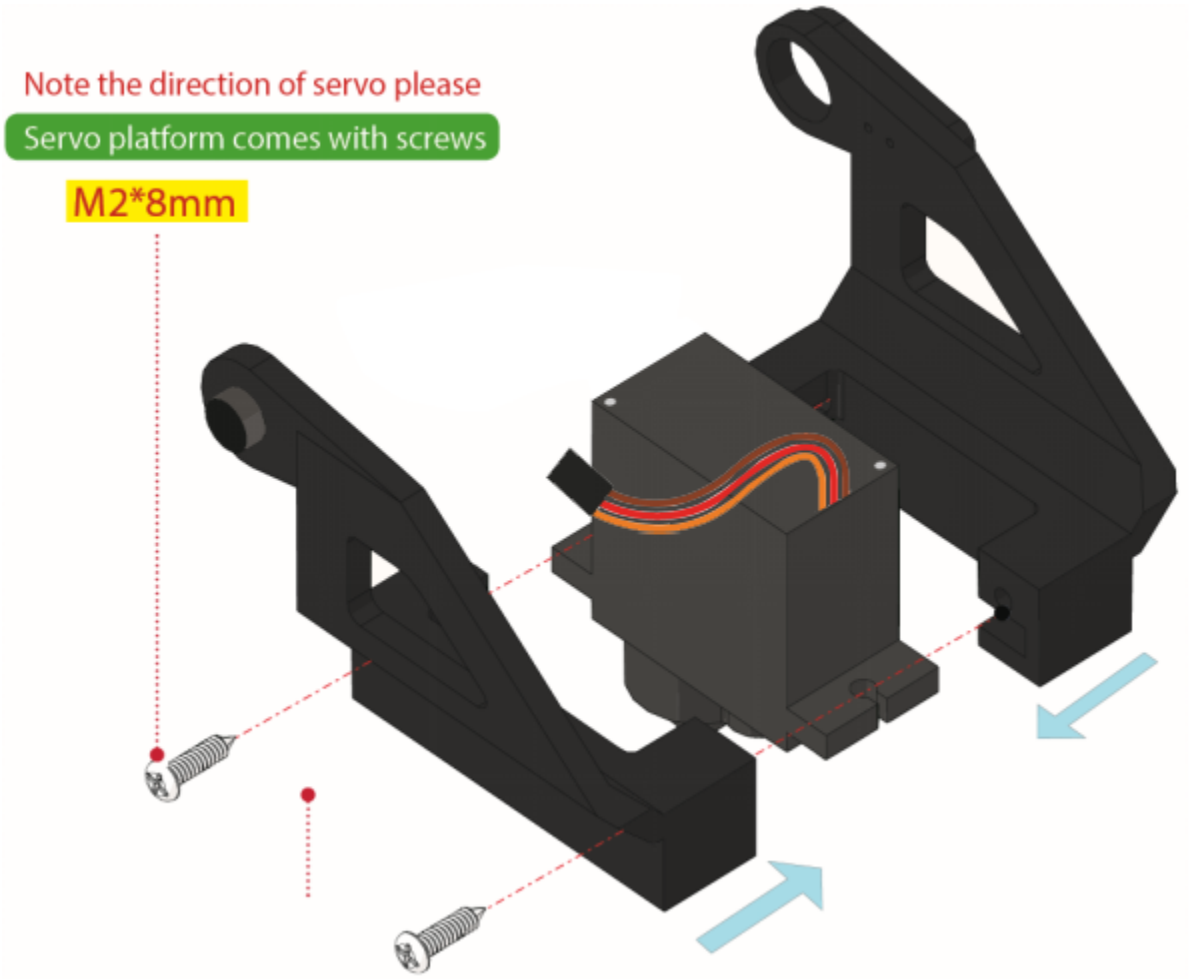
To adjust the code of the servo,please select it according to the course.

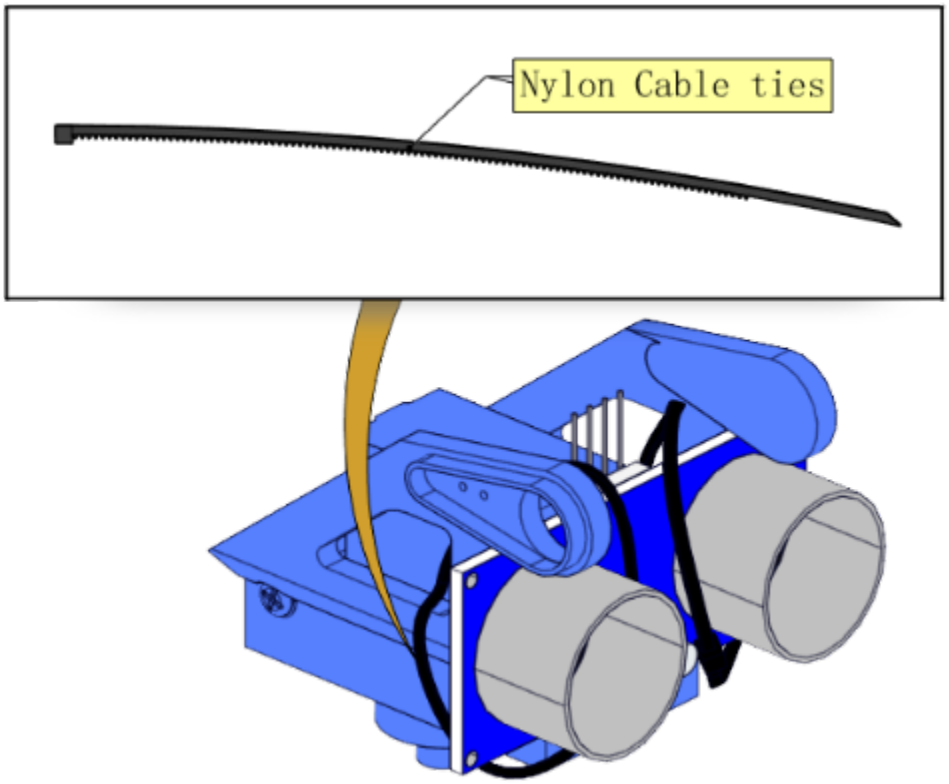
1.**Arduino:**Download the code file:Arduino

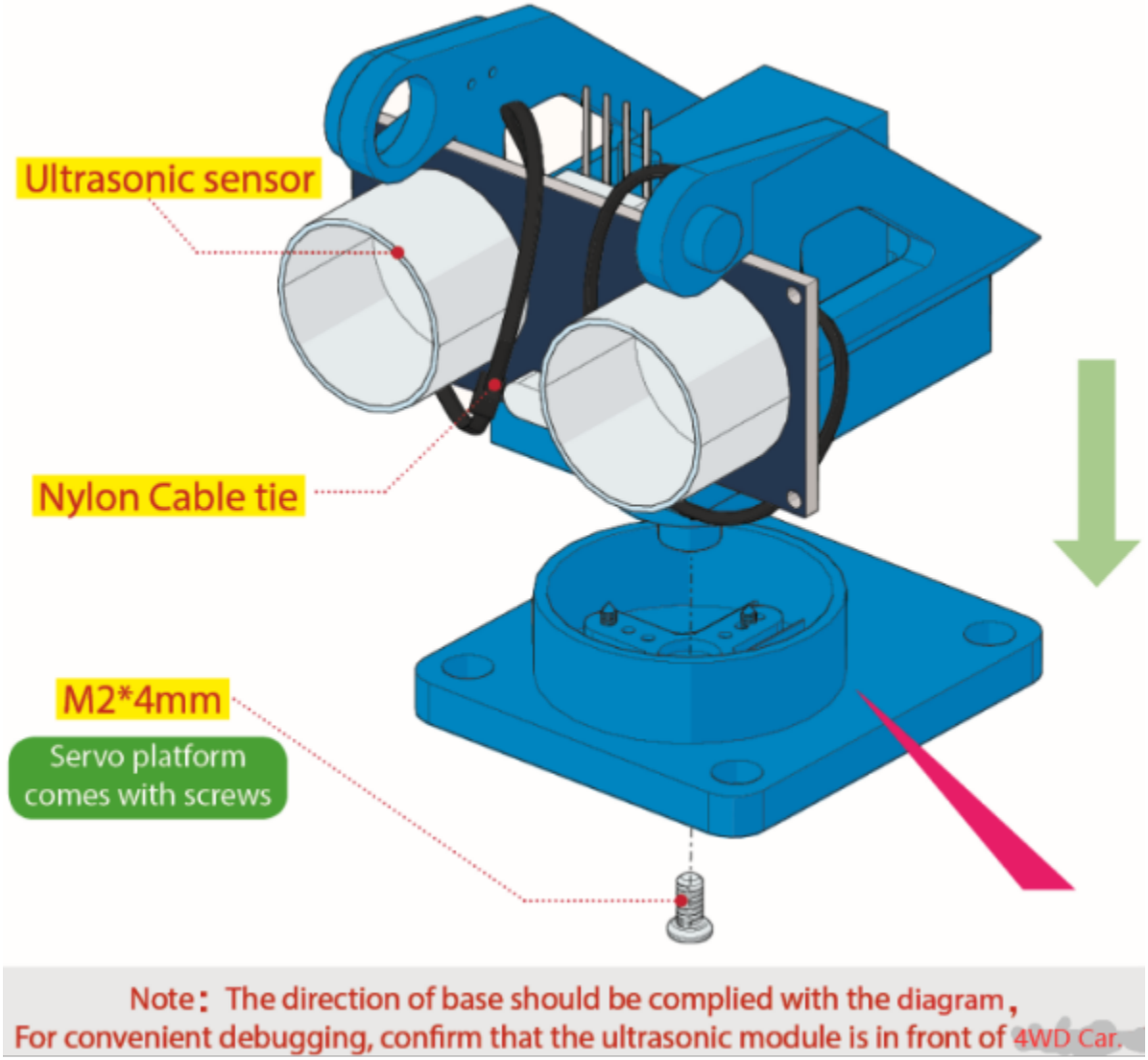
2.**Kidsblock:**Download the code file:Kidsblock

4.3. Installation Diagram.

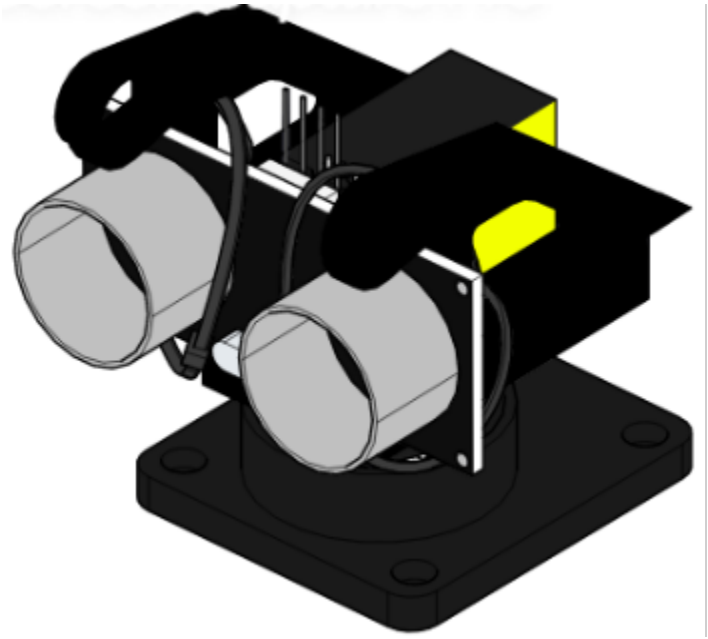



Note the direction of servo please
Servo platform comes with screws
M2*8mm


Nylon Cable ties

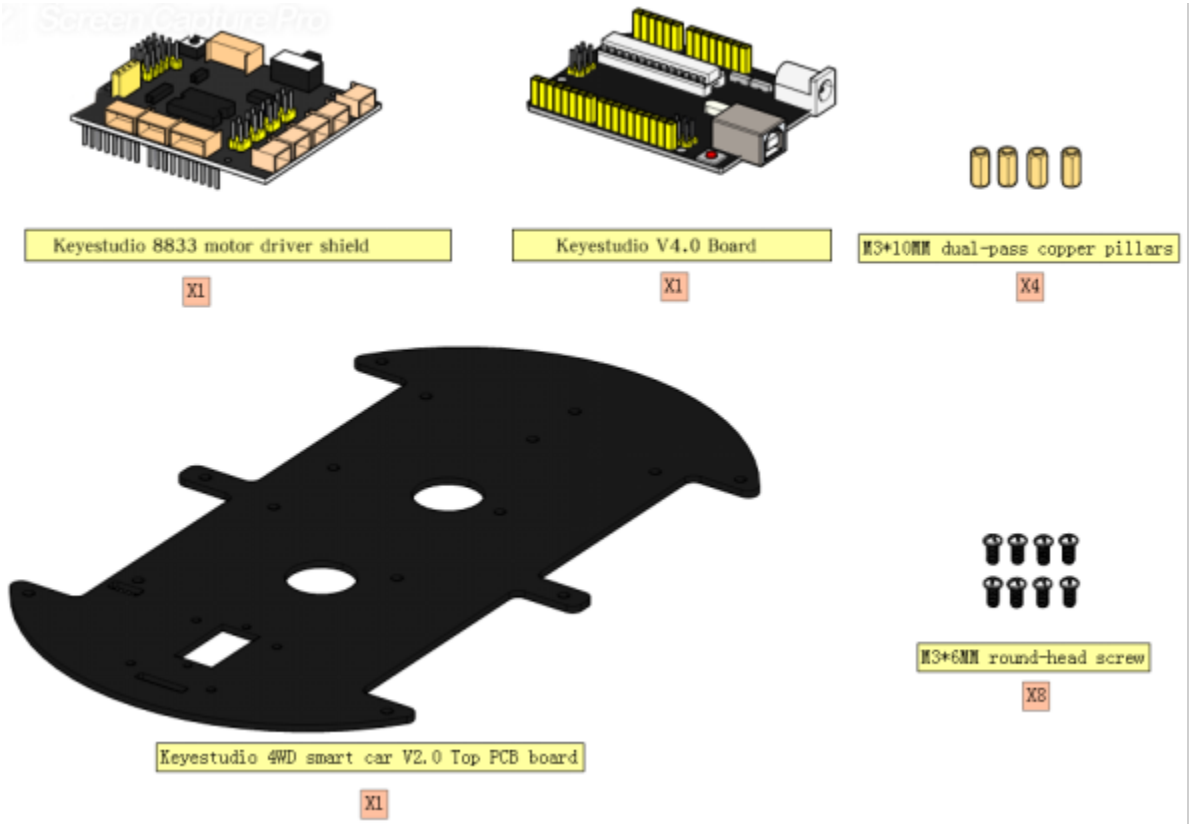


4.4. Prototype.

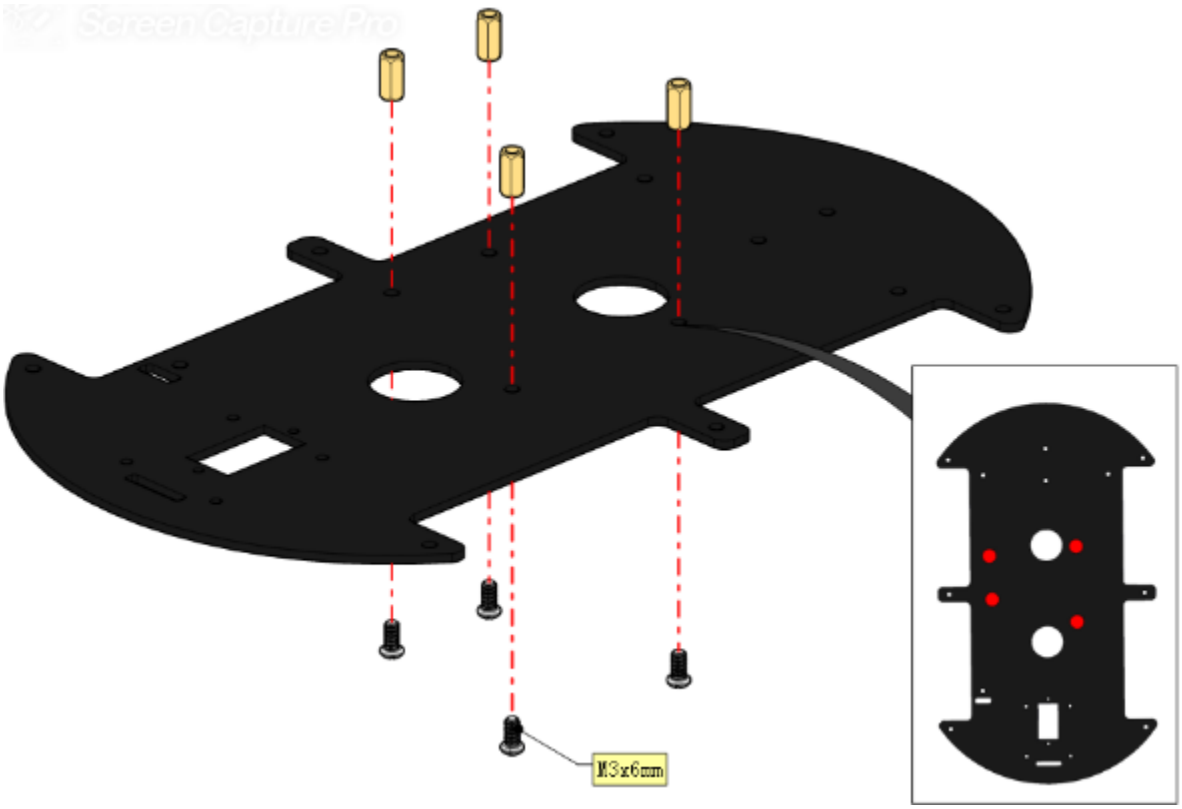


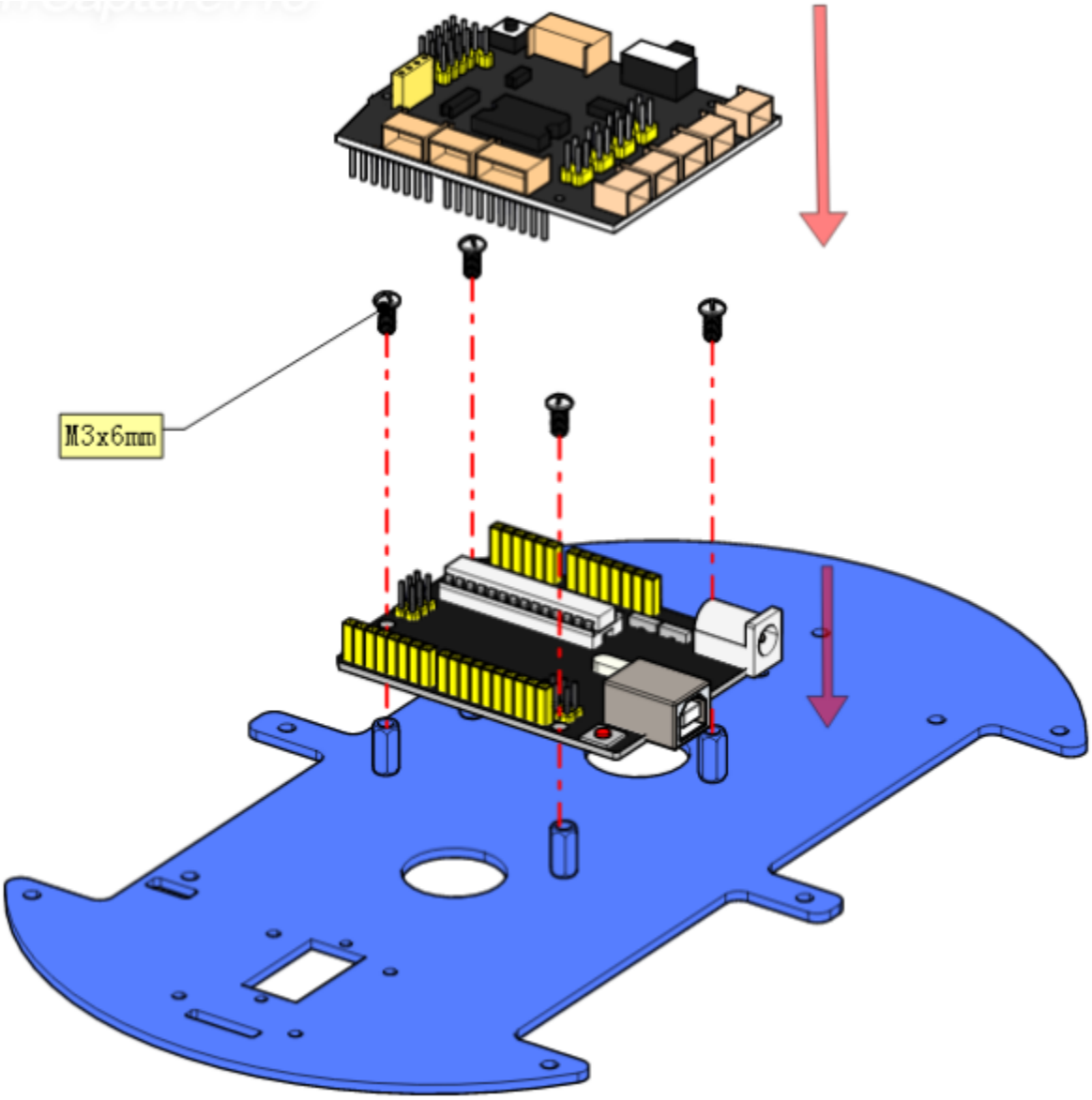
7.5 part 5

5.1. Components Needed.

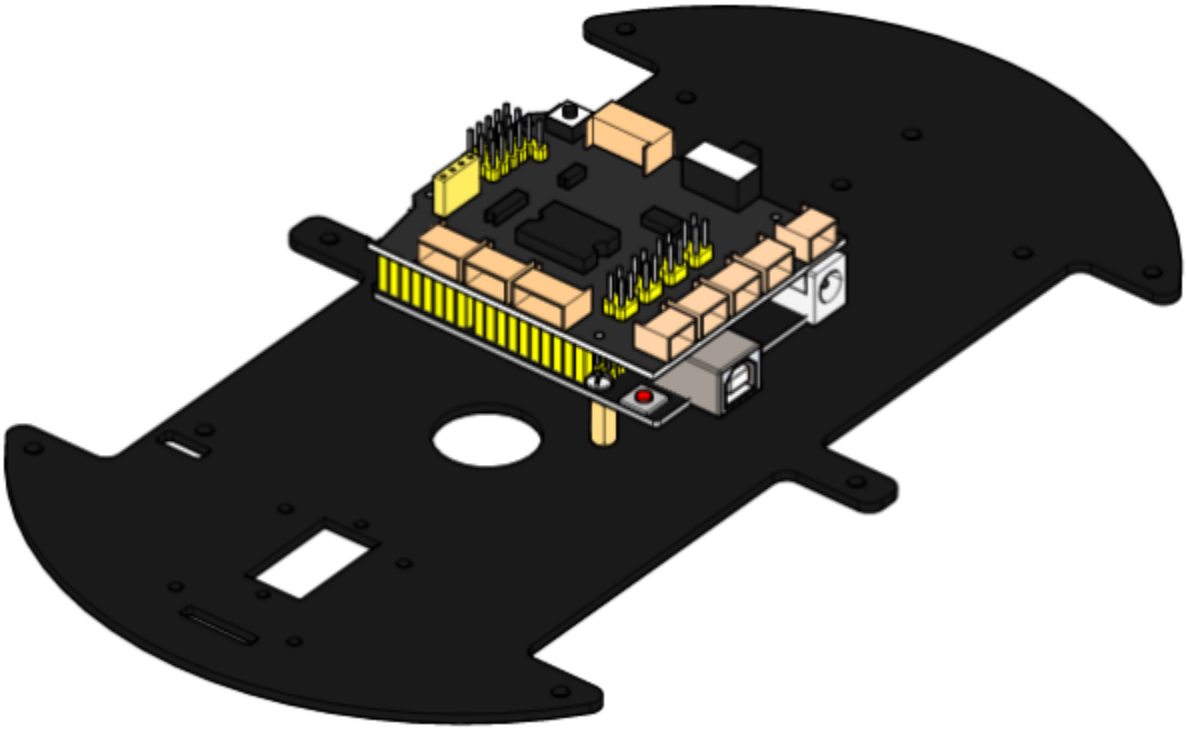


5.2. Installation Diagram.


M3x6mm



5.3. Prototype.



7.6 part 6

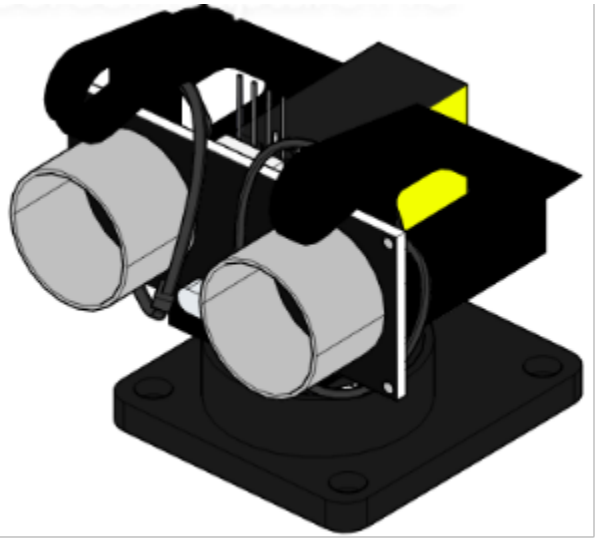
6.1. Components Needed.

M3*10MM dual-pass copper pillars

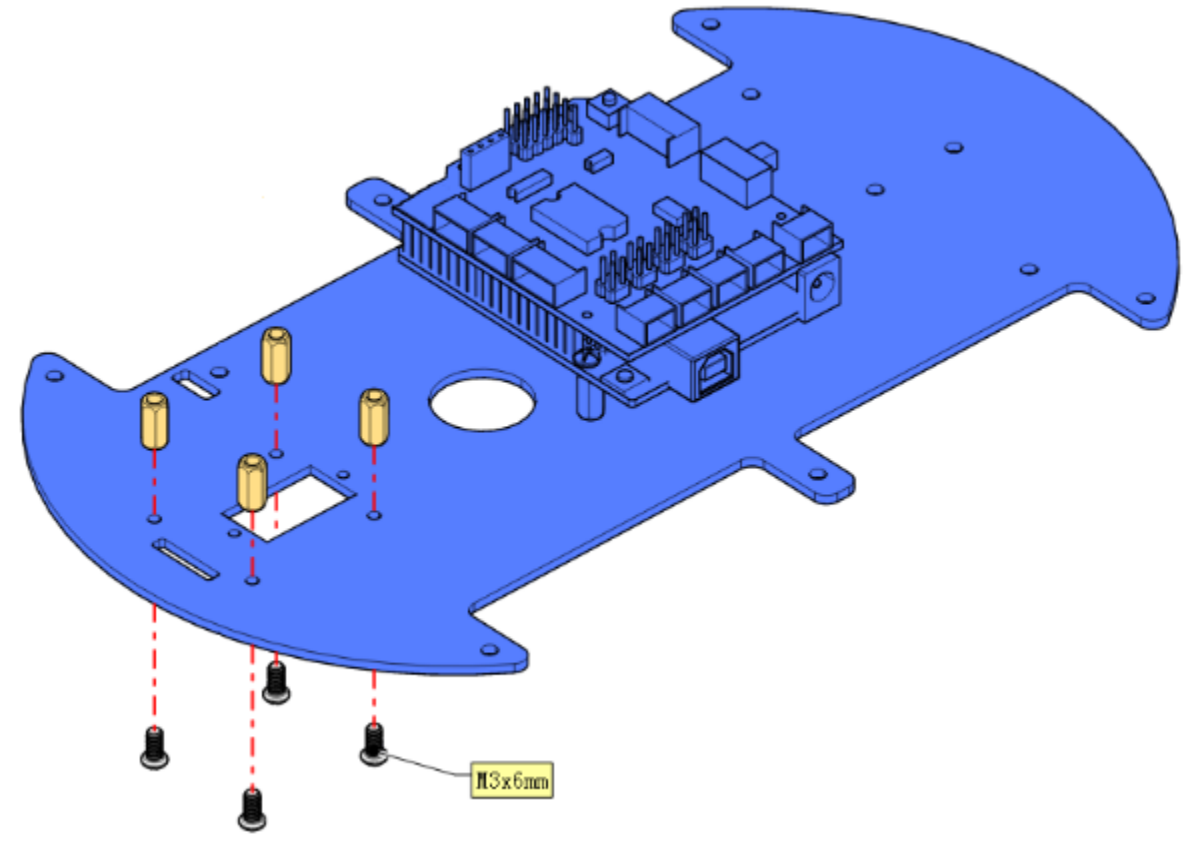
X4

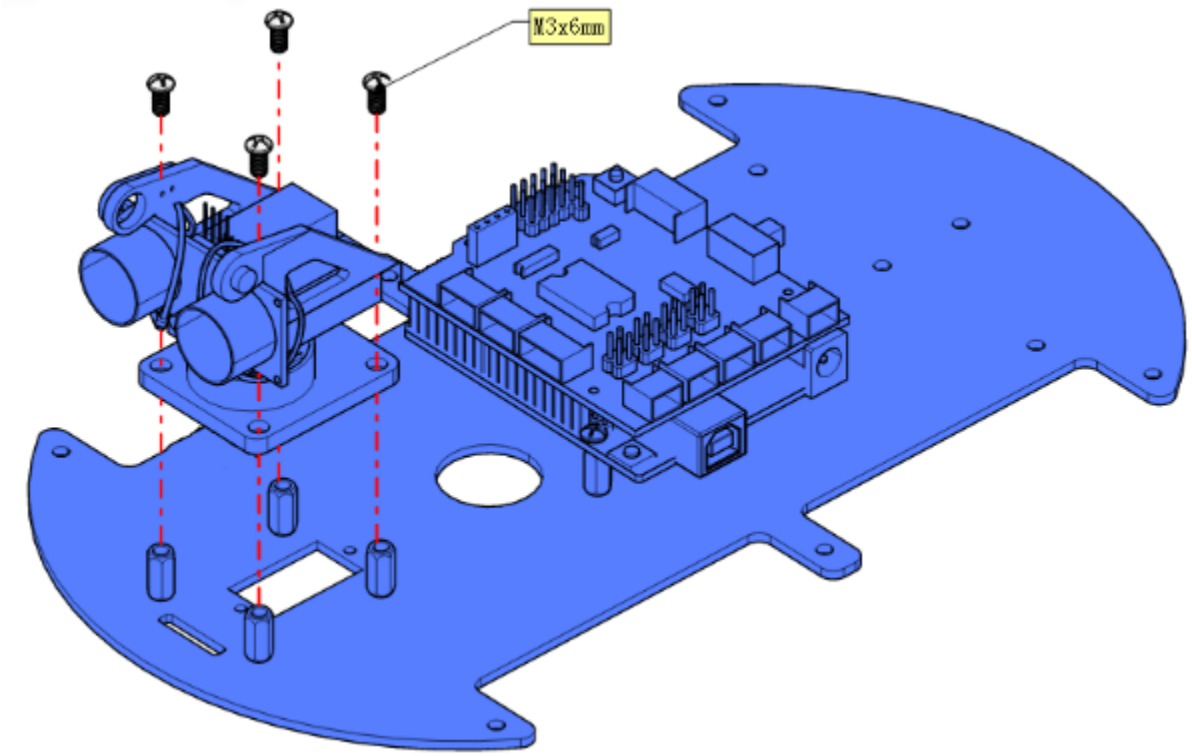
M3*6MM round-head screw

X8

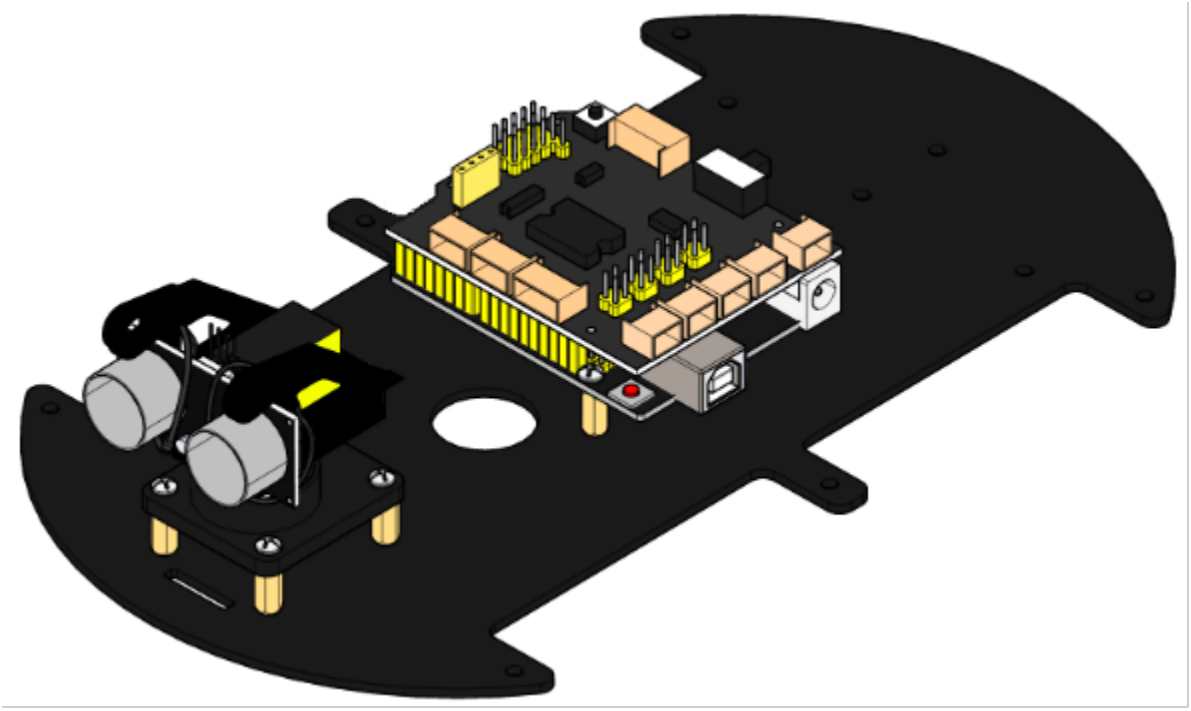


6.2. Installation Diagram.



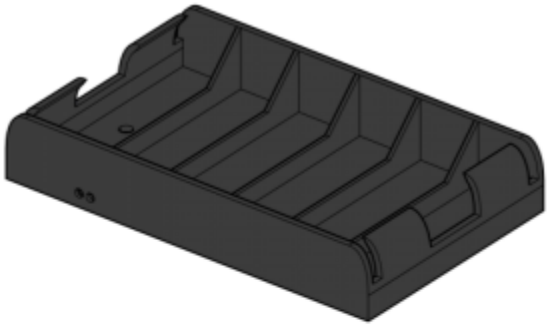


6.3. Prototype.



7.7 part 7

7.1. Components Needed.



battery pack

X1

M3*10MM flat screw

X2

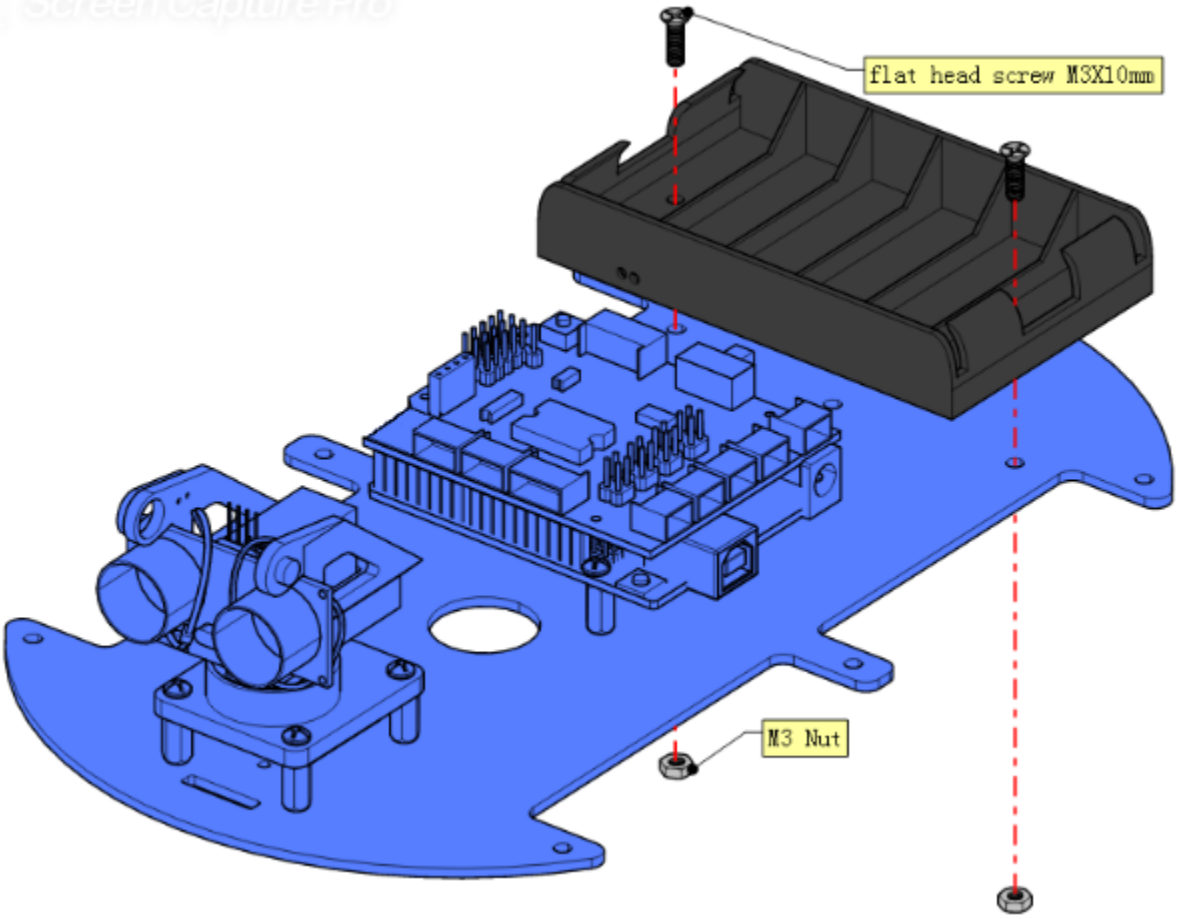
M3 Nickel Plated Nut

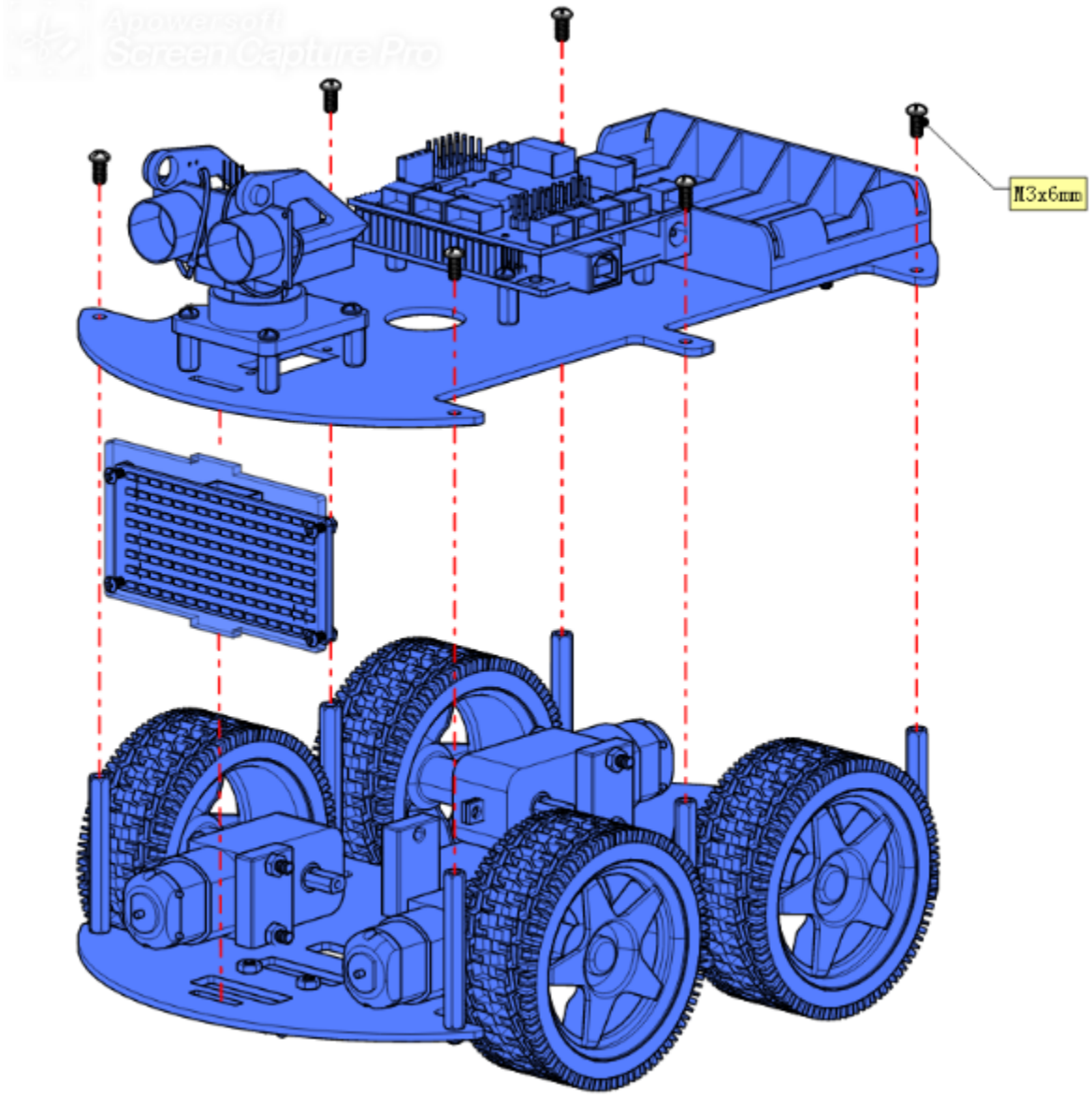
X2

M3*6MM round-head screw

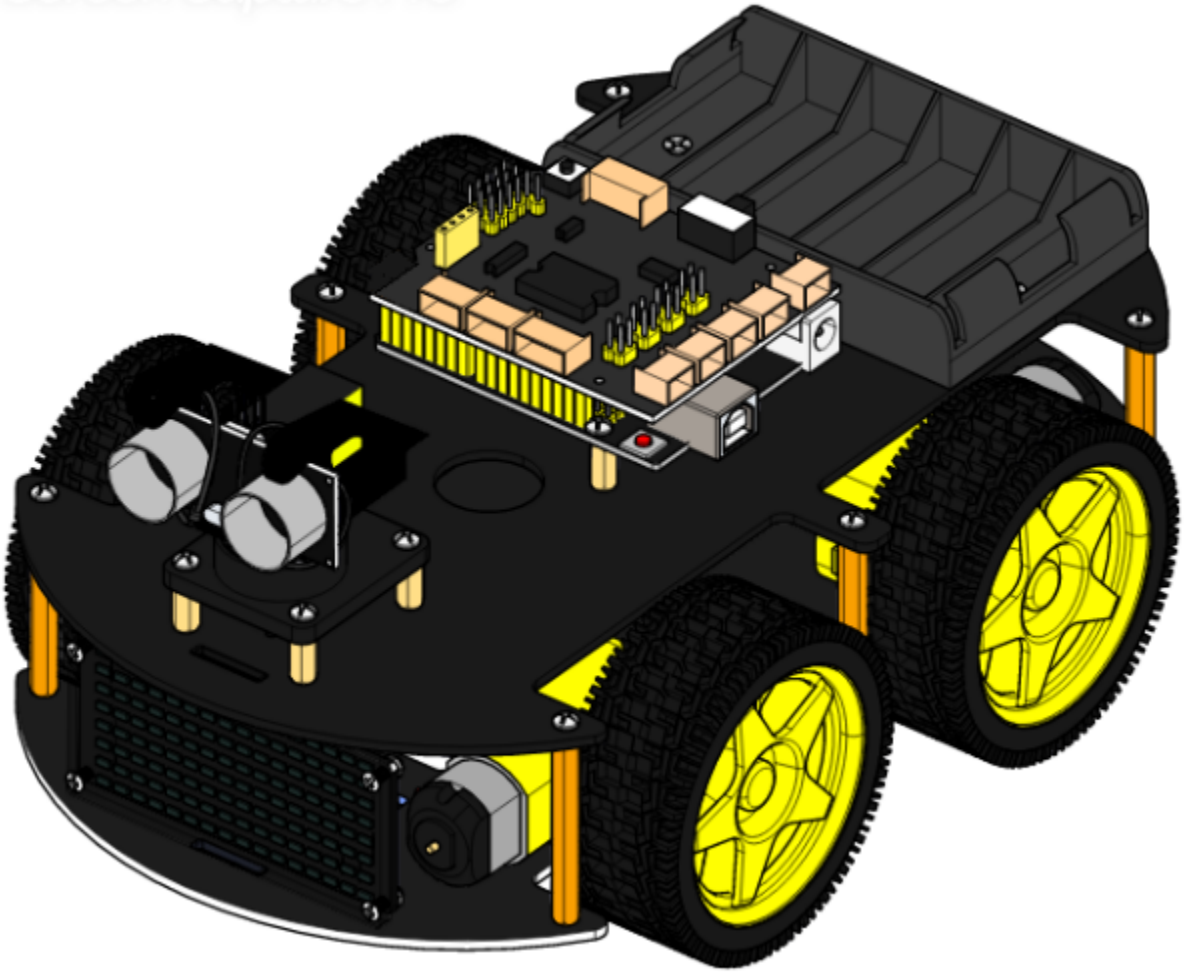
X6

7.2. Installation Diagram.


flat head screw M3X10mm
M3 Nut

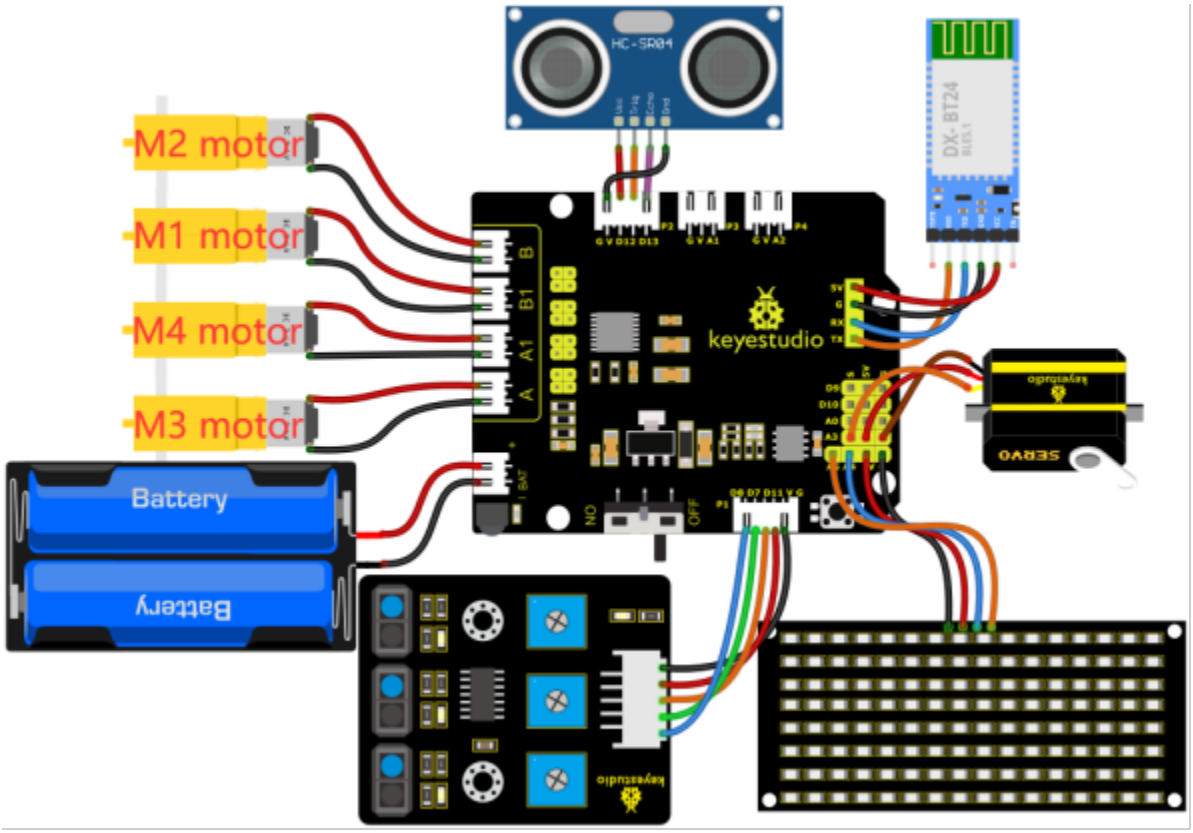


7.3. Prototype.

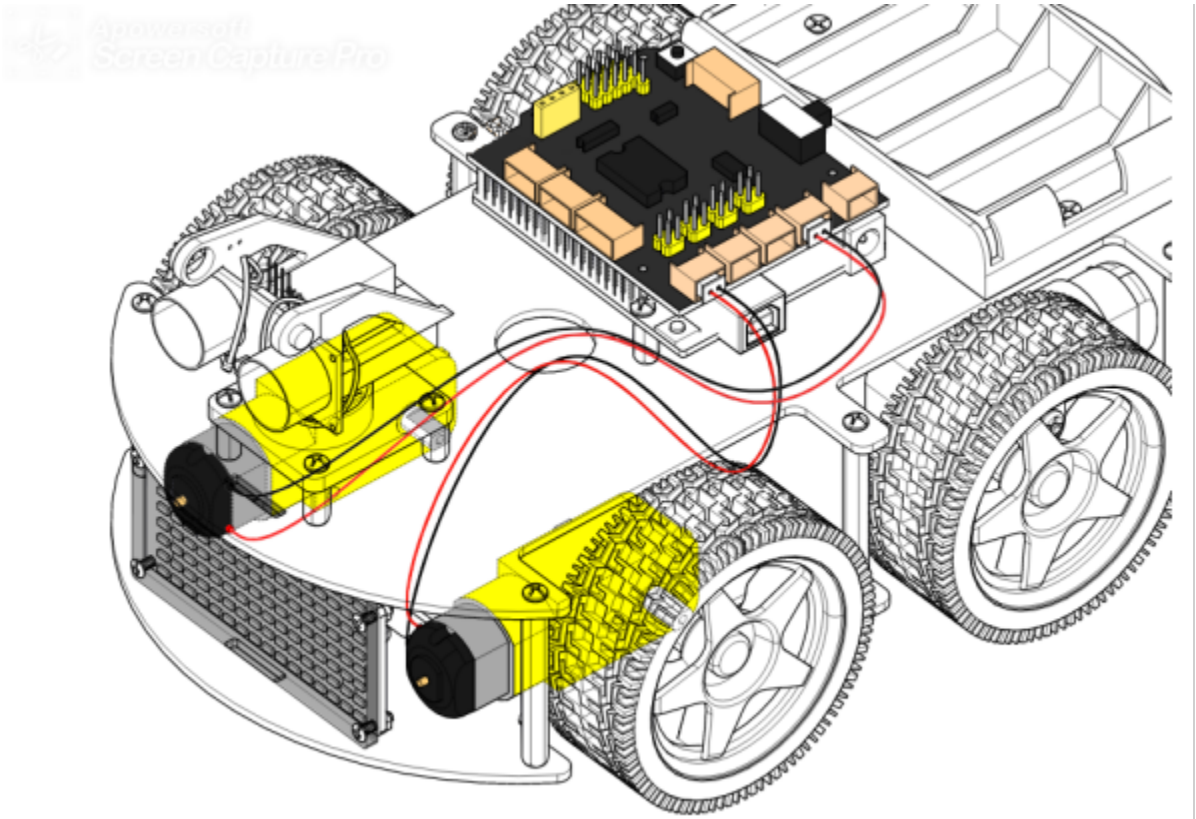


7.8 part 8

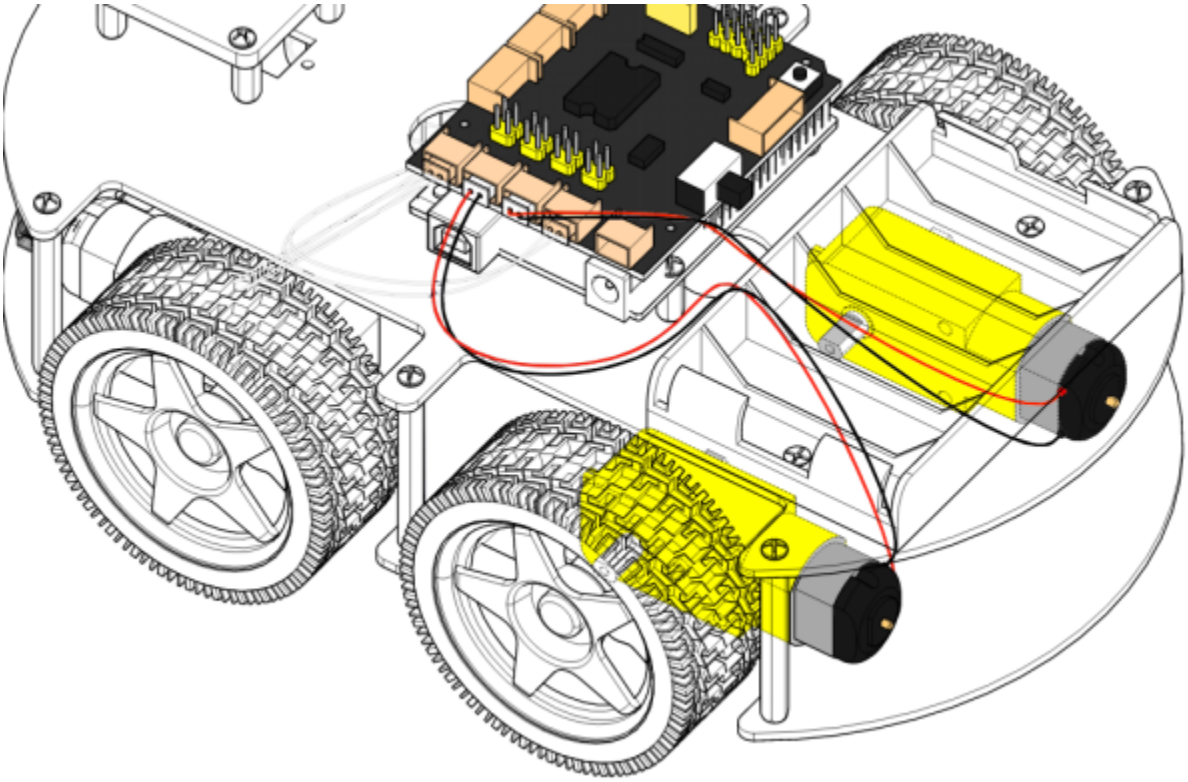
8.1. Wiring Diagram.



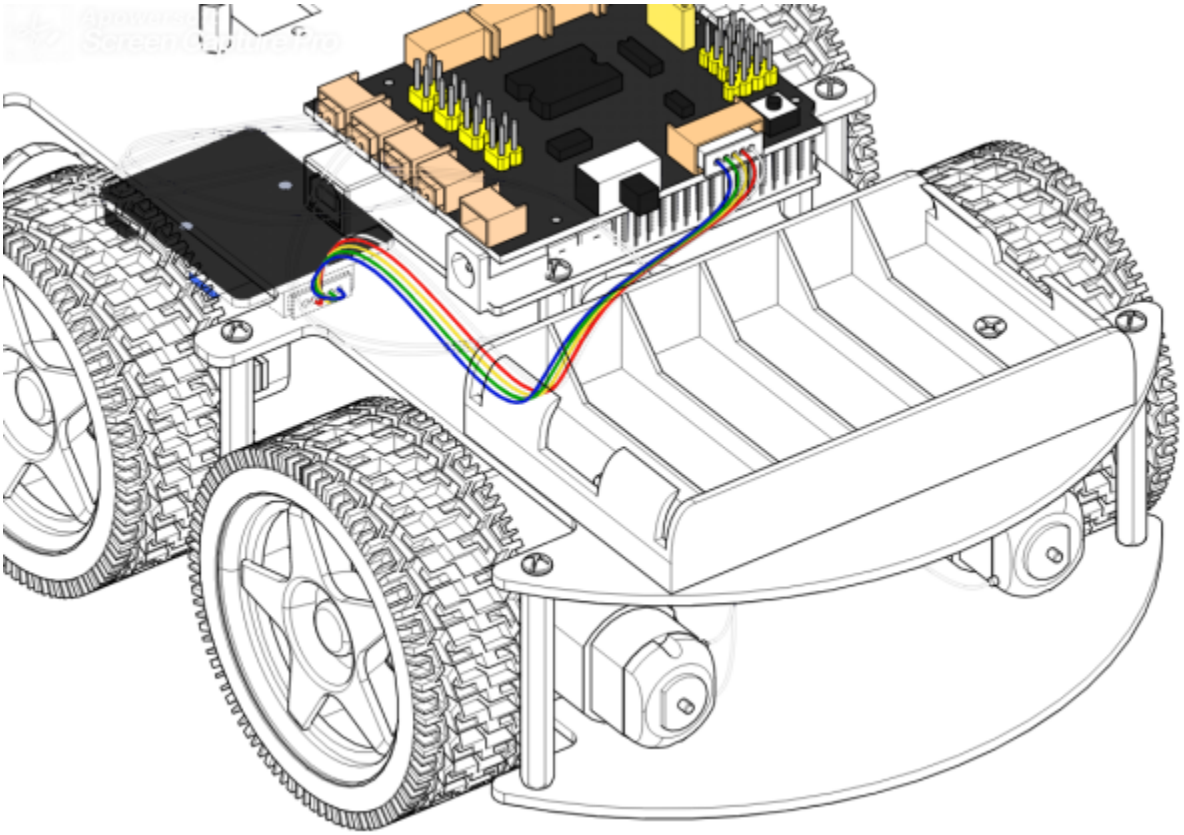
8.2. M2 Motor is connected to B interface,M3 Motor is connected to A interface.



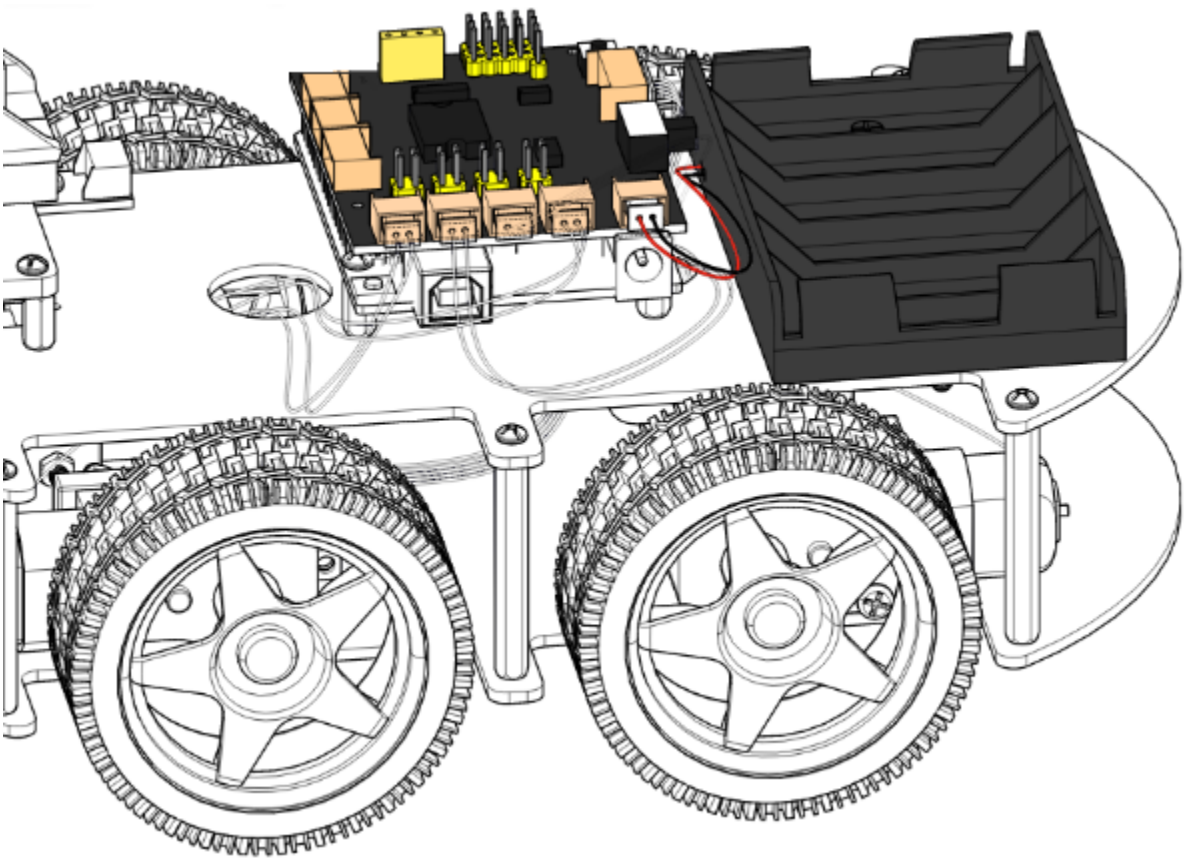
8.3. M1 Motor is connected to B1 interfaceM4 Motor is connected to A1 interface.



8.4. The wiring of the 3-channel line-tracking sensor.

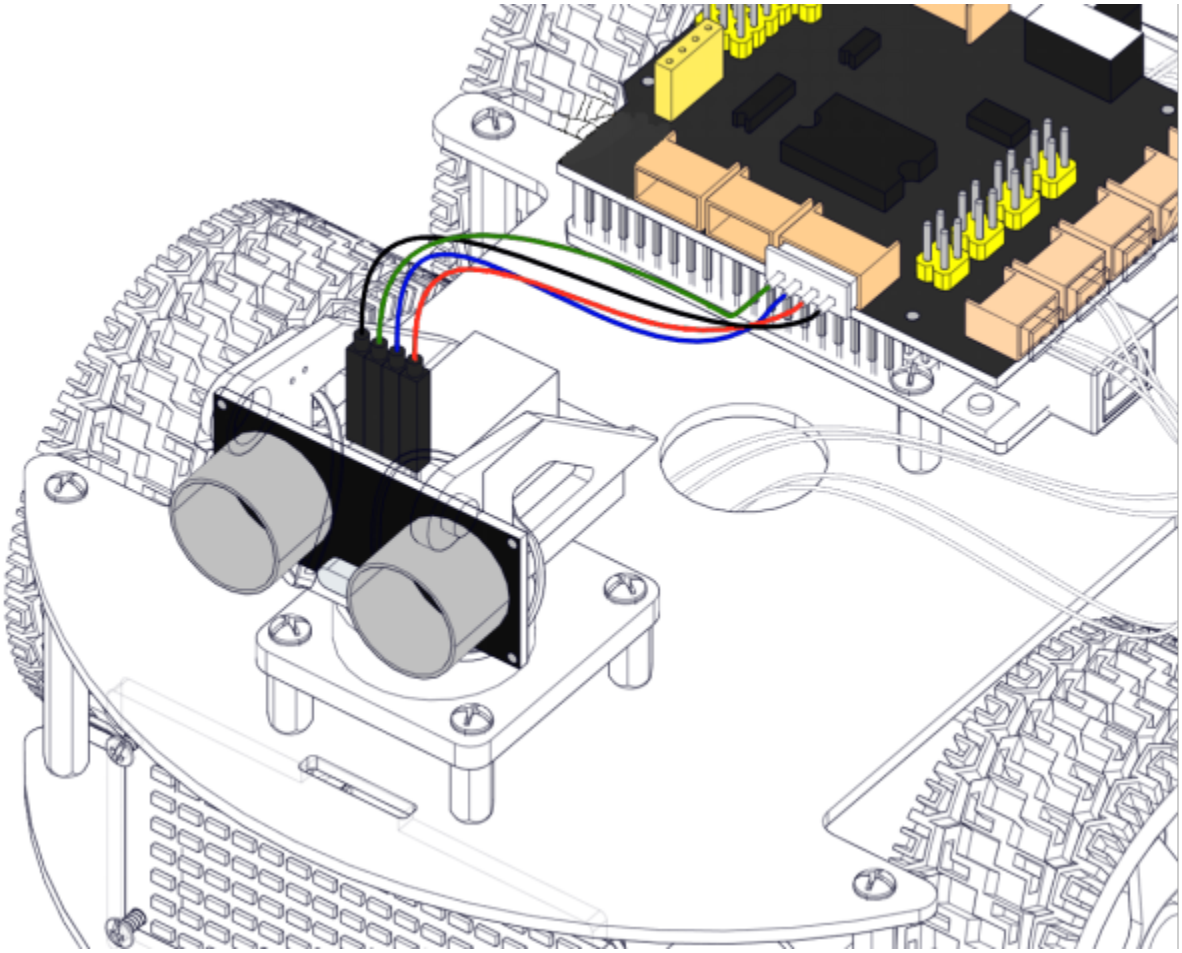


8.5. The power supply is connected to BAT.



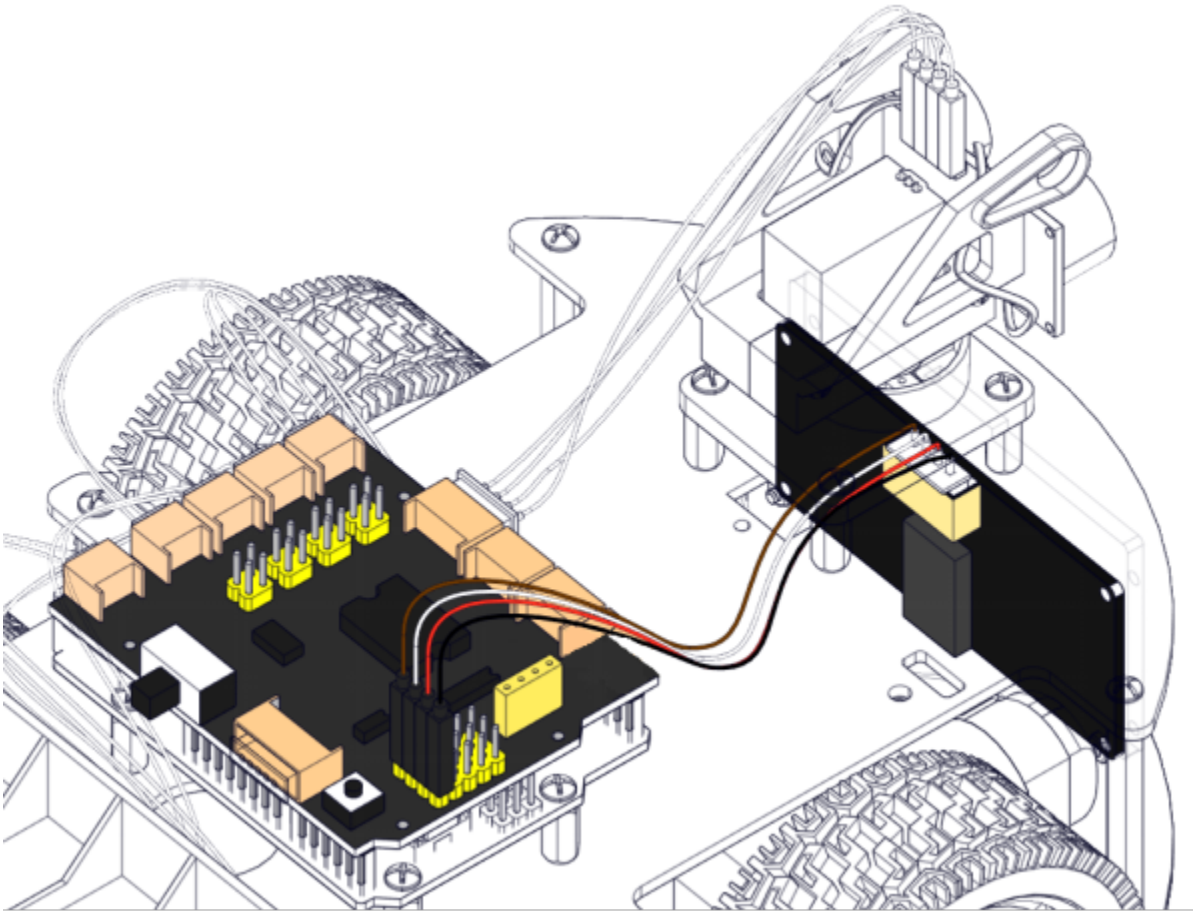
8.6. The wiring of the ultrasonic sensor.

Ultrasonic Sensor		8833 Motor Driver Expansion Board
Vcc		V
Trig		D12
Echo		D13
Gnd		G



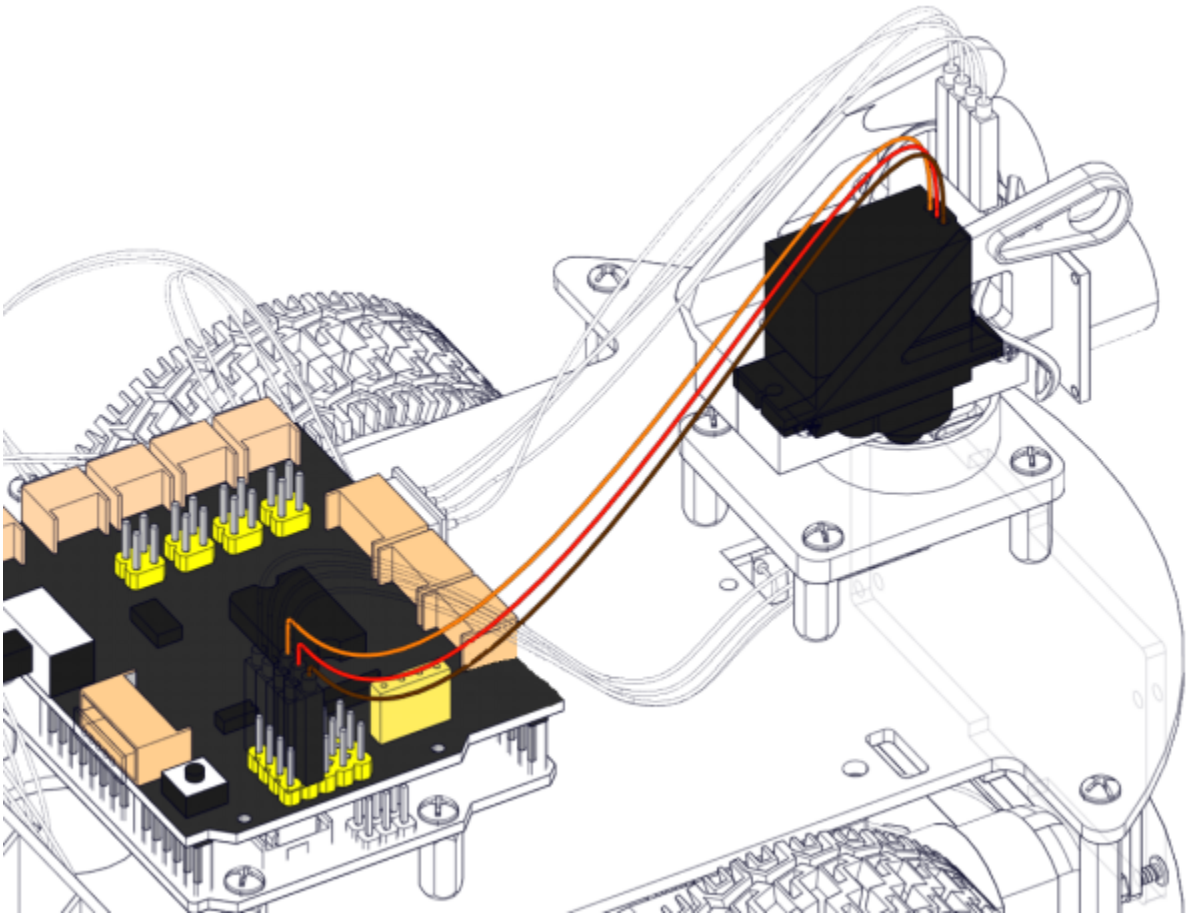
8.7. The wiring of the 8*16 LED Board.

8*16 LED Board		8833 Motor Driver Expansion Board
GND		G
VCC		5V
SDA		A4
SCL		A5

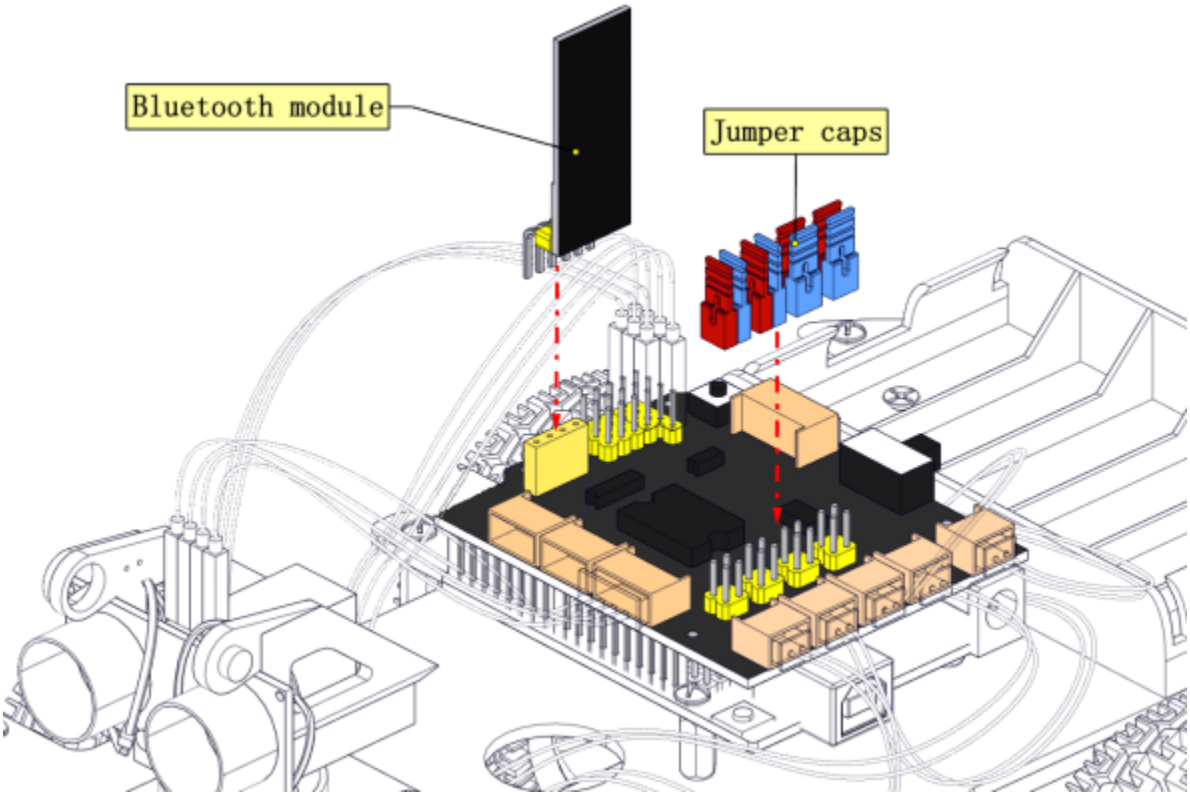


8.8. The wiring of the servo.

Servo	8833 Motor Driver Expansion Board
Brown	G
Red	5V
Yellow	A3

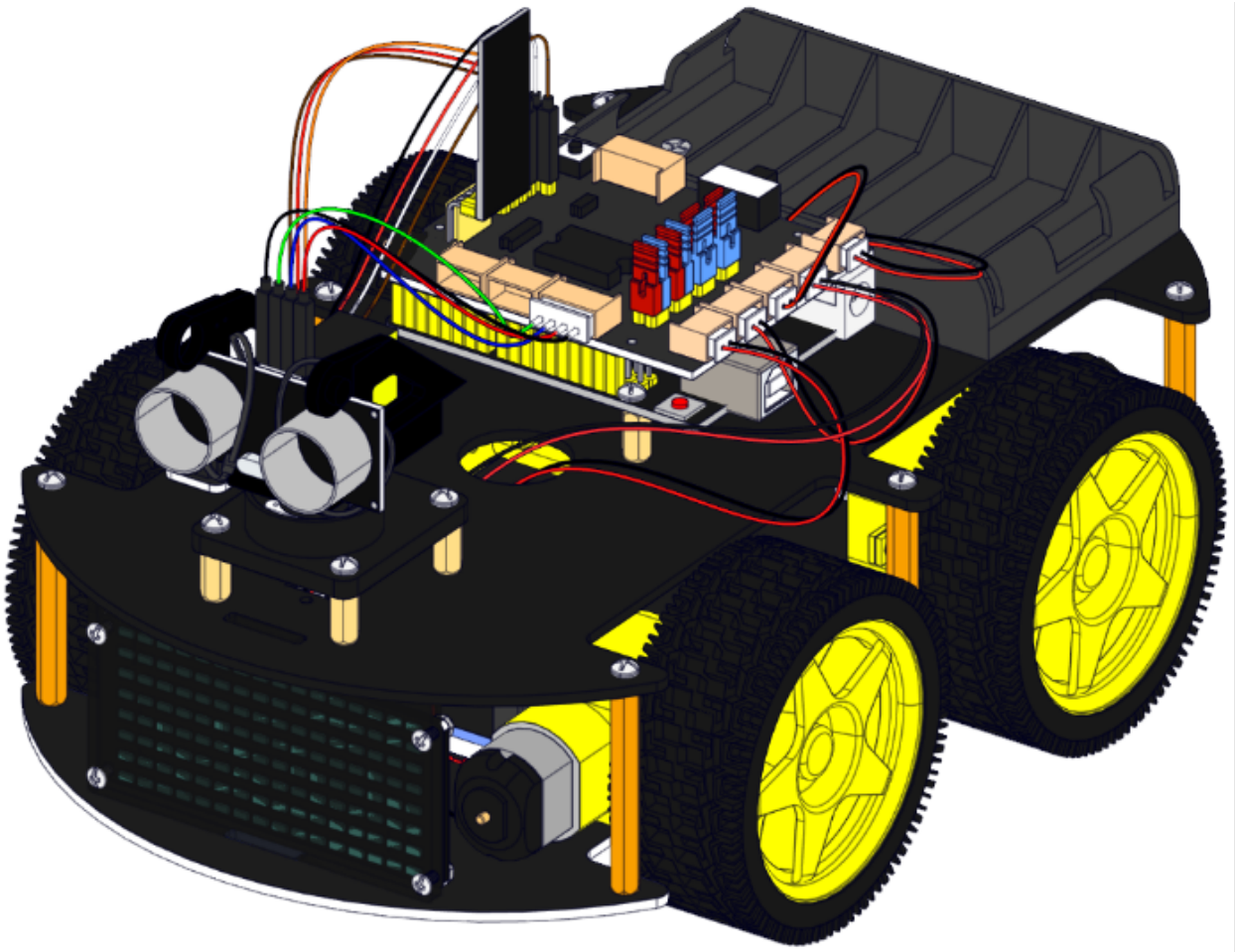


8.9. Plug in the Bluetooth module and jumper caps.



Note: Before uploading the test code, you need to remove the Bluetooth module, otherwise the code will fail to be uploaded.Connect the Bluetooth module after uploading the code successfully.

8.10. Complete Prototype.



CHAPTER

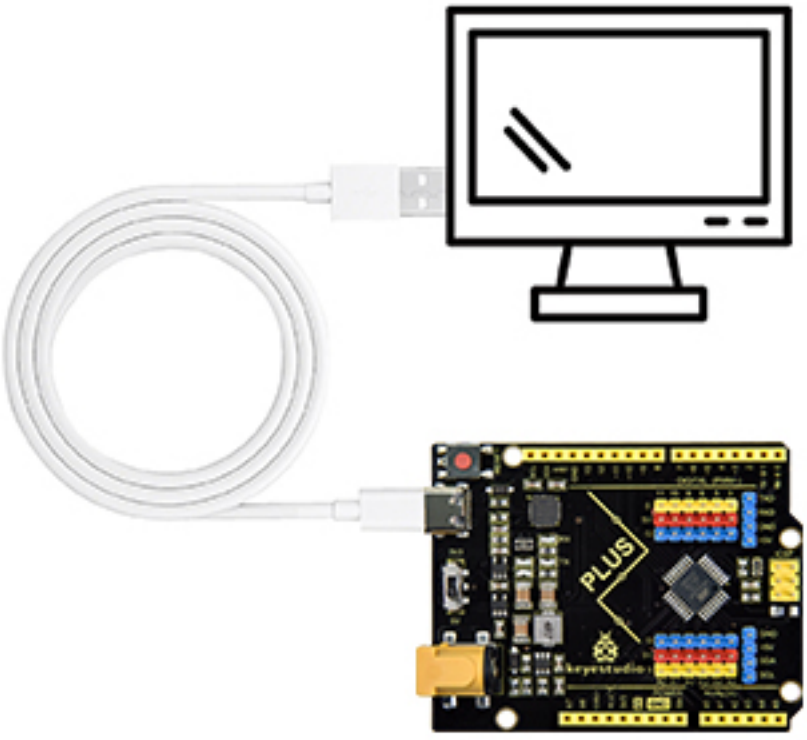
EIGHT

3. DRIVER INSTALLATION

8.1 3.1 Windows System

8.1.1 1 Checking the driver

1Connect the motherboard to the computer.



2Open Device ManagerOpen the device manager, if the prompt **"Silicon Labs CP210x USB to UART Bridge (COMx)"** appears to prove that the driver has been installed, please skip the **"Driver installation"** part.

8.1.2 2 Manual driver installation

1Driver download

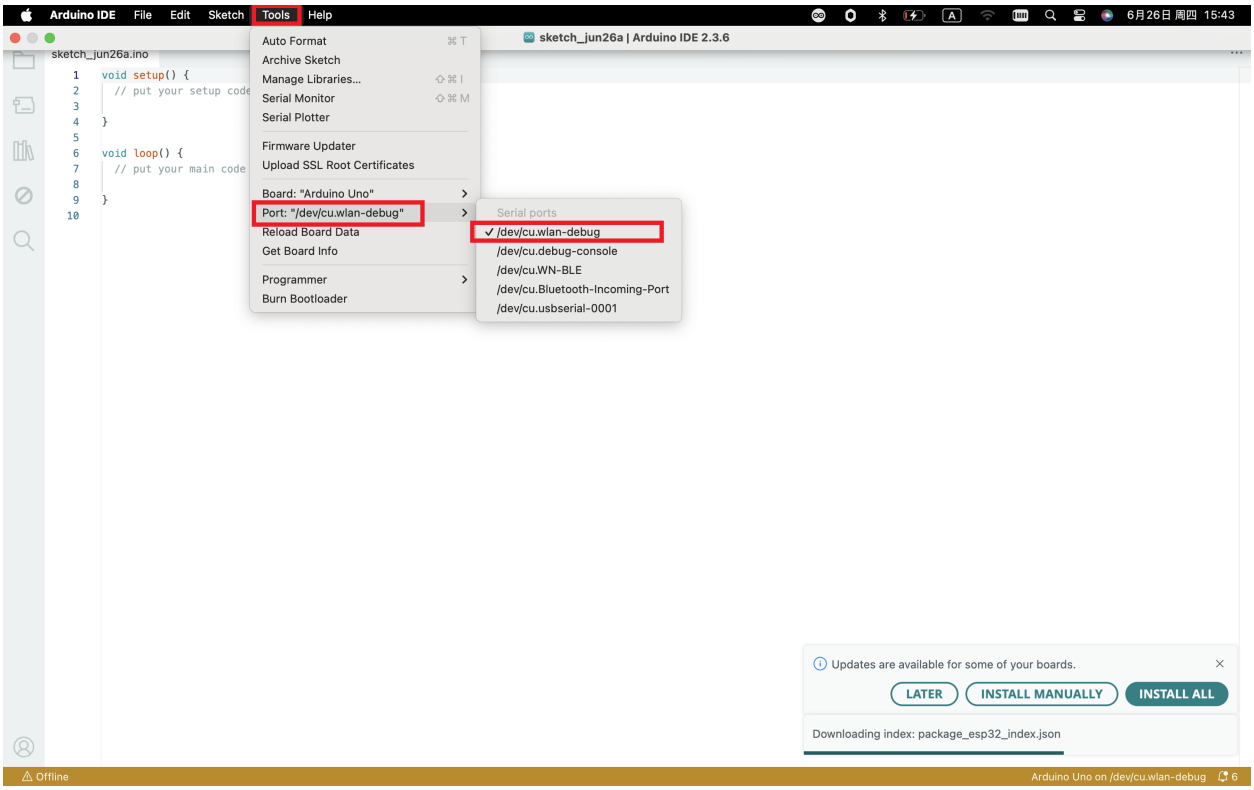
- Windowns System:Windowns System driver

2Connect the motherboard to the computer, open the device manager, if there is a yellow exclamation mark in front of the driver in the picture, it proves that the driver is not installed, please download the driver and install it manually.

8.2 3.2 MAC System

8.2.1 1 Checking the driver

Connect the development board to the computer, according to [Tools] —> [Port] to select the development board port (Note: If you can not confirm which port is the development board, please connect the motherboard to take pictures to record all the ports, and then unplug the development board to re-take pictures to record all the ports, and then compare to find the disappeared ports, and then unplug the motherboard after the disappeared ports is the port of the board, and then select the port on the line)If you can not recognize the port, please replace the computer USB port or around the phone cable to re-recognize the port, if it still does not work refer to the following steps to install the driver.

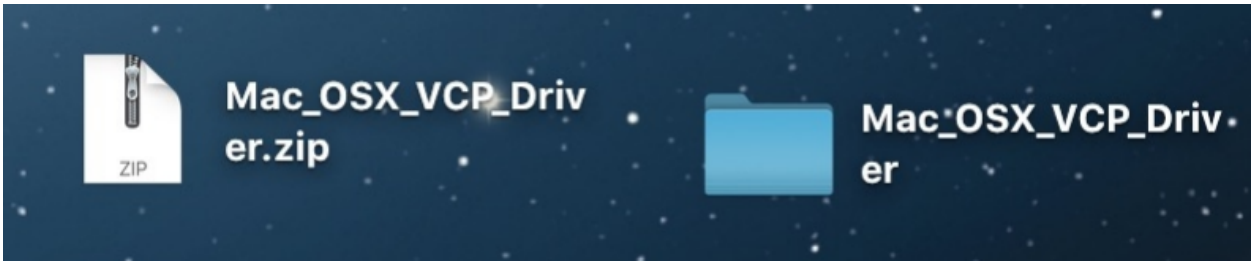


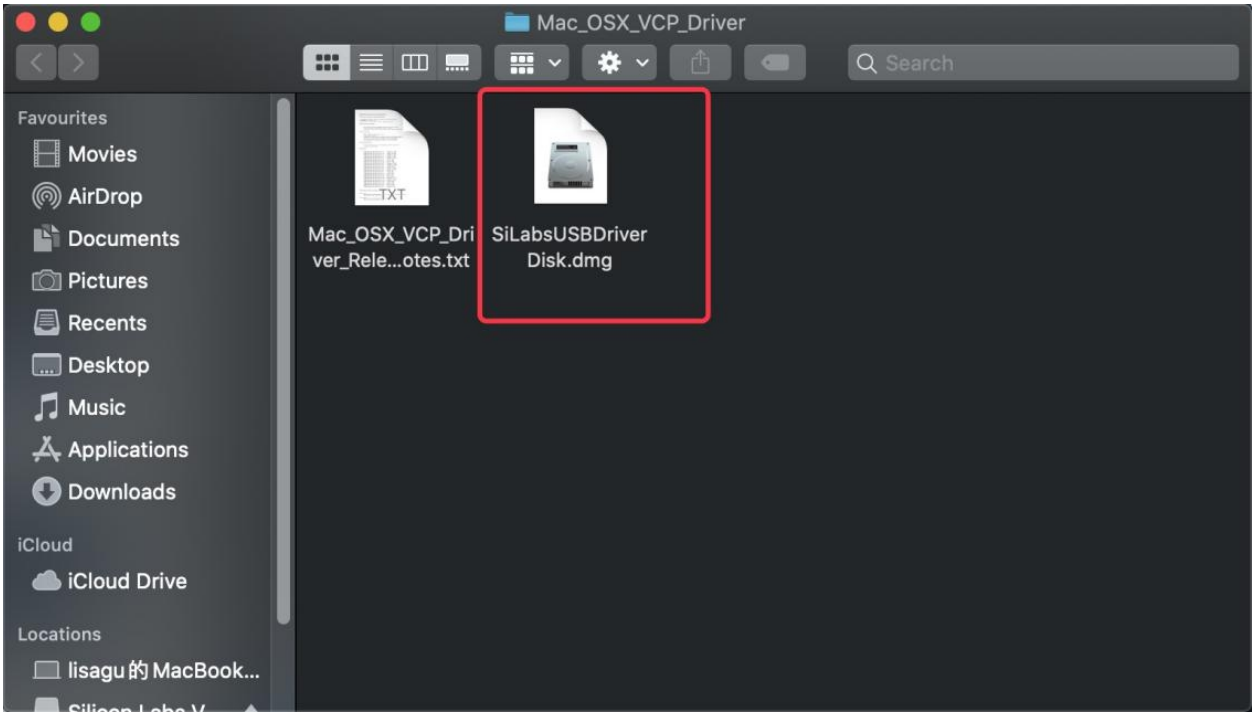
8.2.2 2 Manual driver installation

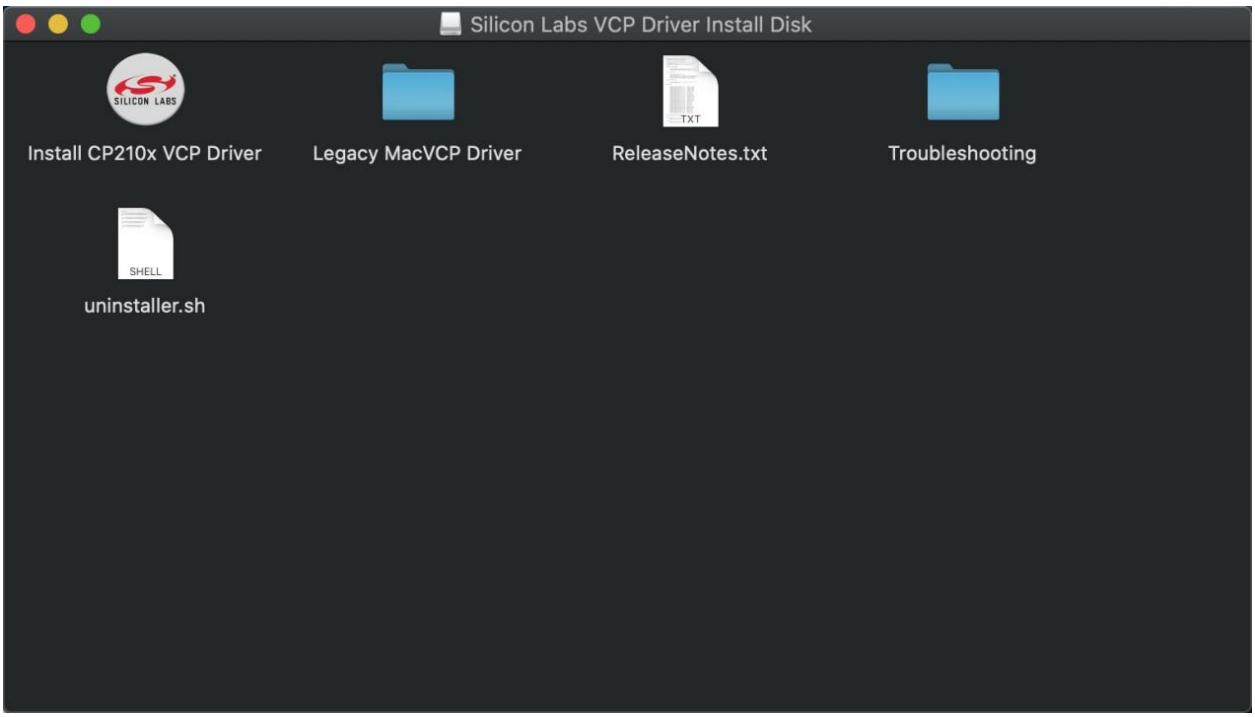
1Driver download

Mac System:Mac System driver

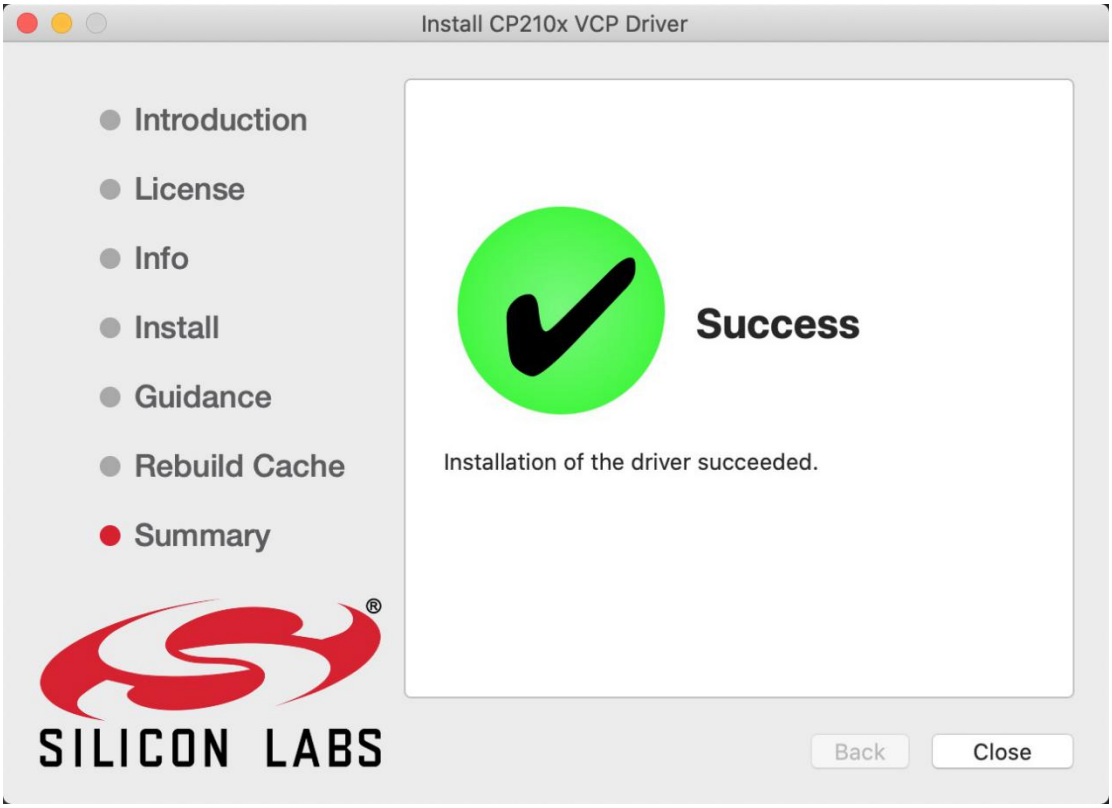
2double-click to decompress the downloaded driver zip package


Mac_OSX_VCP_Driver.zip
ZIP
Mac_OSX_VCP_Driver


Mac_OSX_VCP_Driver
Search
Favourites
Movies
AirDrop
Documents
Pictures
Recents
Desktop
Music
Applications
Downloads
iCloud
iCloud Drive
Locations
lisagu的MacBook...
TXT
Mac_OSX_VCP_Driver_Rele...otes.txt
SiLabsUSBDriverDisk.dmg

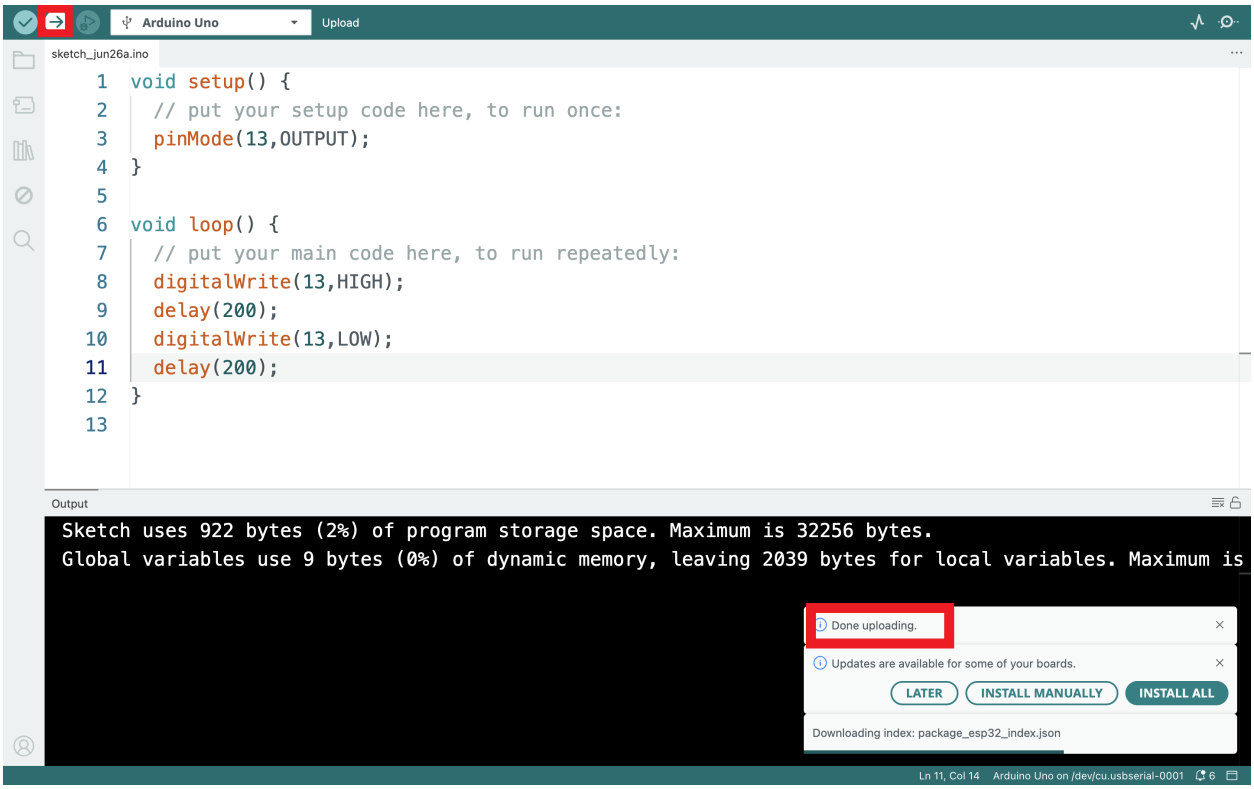

Silicon Labs VCP Driver Install Disk
SILICON LABS
Install CP210x VCP Driver
Legacy MacVCP Driver
TXT
ReleaseNotes.txt
Troubleshooting
SHELL
uninstaller.sh

3After that keep **"Next "** installing until the installation is complete

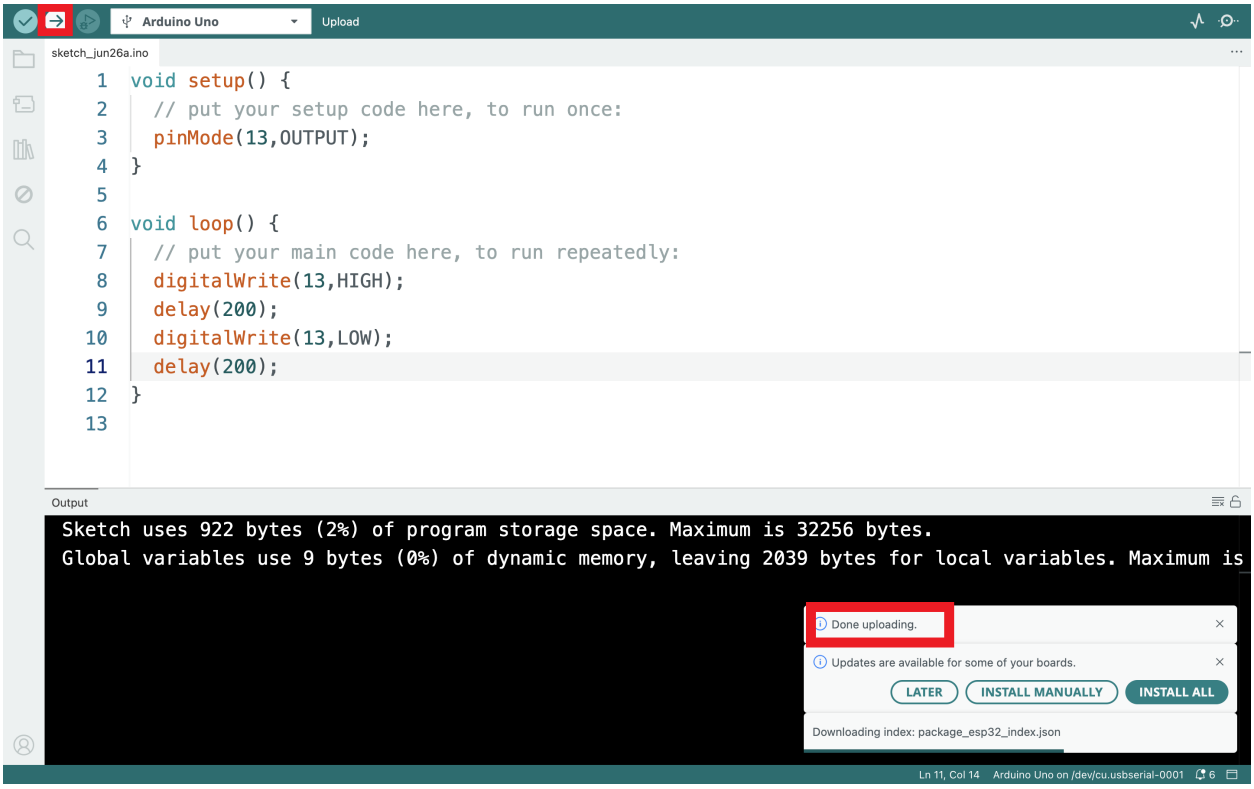


At this point, the port can be recognized by plugging in the board again.

4Then go to the Arduino IDE, click on "Tools", select the board Arduino Uno and the recognized development board port



16.Click to upload code and show “Done uploading”.



CHAPTER

NINE

4. ARDUINO TUTORIAL

9.1 Arduino - Read Me First

9.1.1 1. Data download

Arduino information contains library files project code and APP,please click to download for follow-up sttudy!!!

Data download:Arduino Data

9.1.2 2. Software Download

When we get control board, we need to download Arduino IDE and driver firstly.

You could download Arduino IDE from the official website:https://www.arduino.cc/en/software.

There are various versions for Arduino,just download a suitable version for your system,we will take WINDOWS system as an example to show you how to download and install.

You just need to click JUSTDOWNLOAD,then click the downloaded file to install it.

And when the ZIP file is downloaded,you can directly unzip and start it.

9.1.3 3. Set Arduino IDE

1Connecting the board to the computer.

9.1.4 4. Add Library

What are Libraries ?

Libraries are a collection of code that makes it easy for you to connect to a sensor,display, module, etc.

There are hundreds of additional libraries available on the Internet for download.

We will introduce the most simple way for you to add libraries .

9.2 Arduino - Projects

Note: In this course, the interface of each sensor / module marked with (G,-, GND) indicates the negative pole, G is connected to G, or GND of sensor shield or control board; "V" is positive pole and connected with V, VCC or 5V.

9.2.1 Project 1: LED Blink

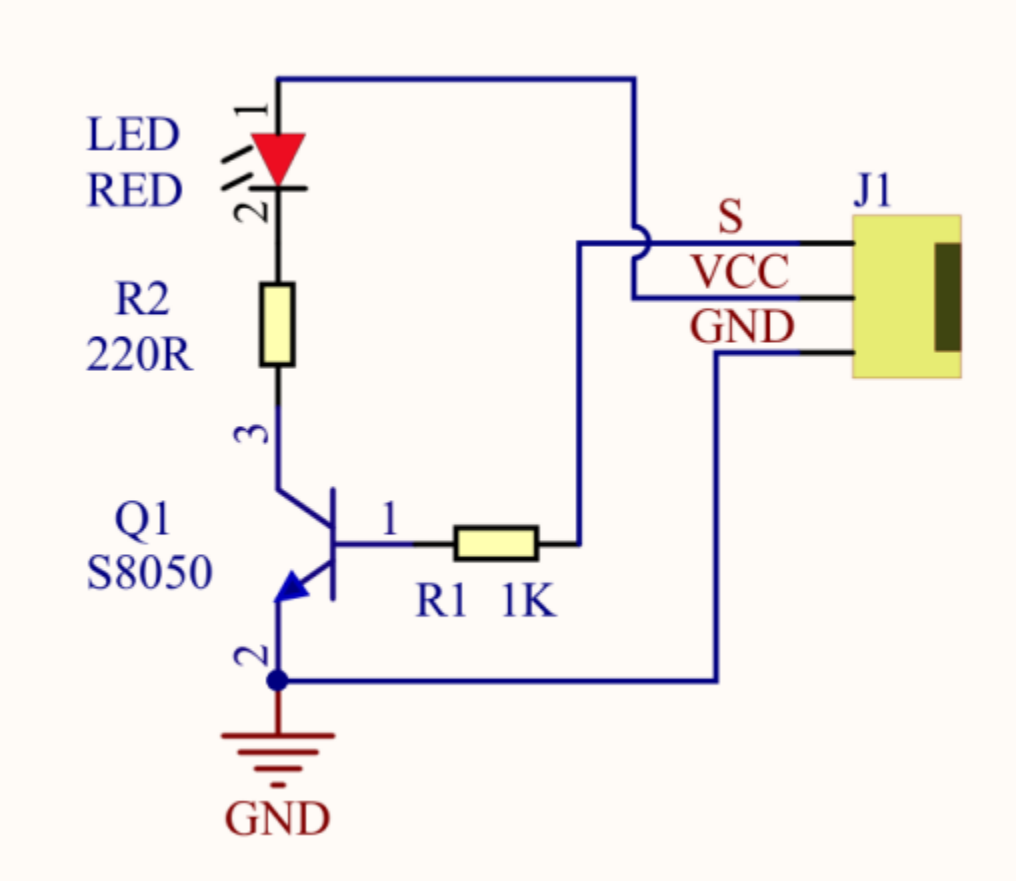
1.Description

For starters and enthusiasts, LED Blink is a fundamental program. LED, the abbreviation of light emitting diodes, consists of Ga, As, P, N chemical compounds and so on.

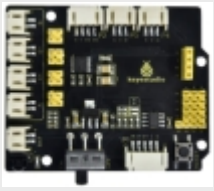
The LED can flash in diverse color by altering the delay time in the test code. When in control, power on GND and VCC, the LED will be on if S end is in high level, otherwise it will go off.

2.Specification

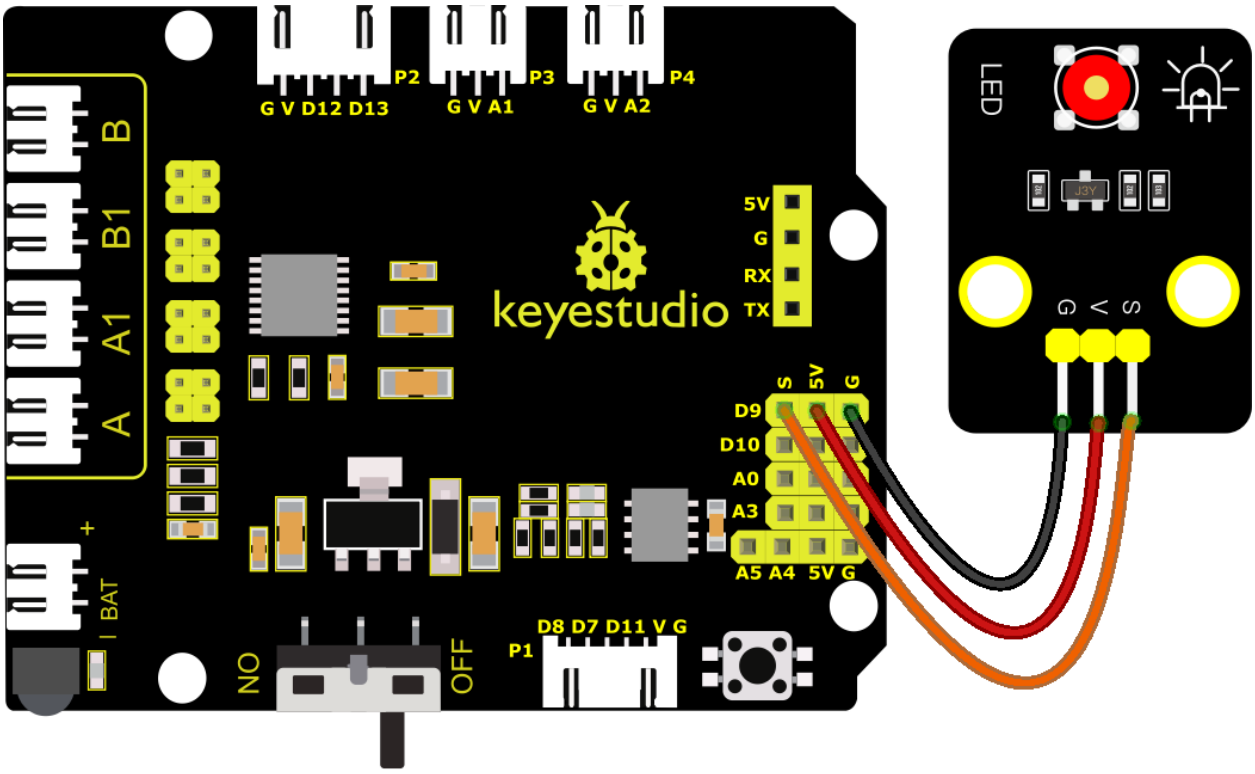
- Control interface: digital port
- Working voltage: DC 3.3-5V
- Pin spacing: 2.54mm
- LED display color: red



3.Components

Development Board *1	8833 Motor Driver *1	Red LED Module*1
		
3P Dupont Wire*1	USB Cable*1	

4.Wiring Diagram



As can be seen from the above figure, the Keyestudio 8833 motor Shield is stacked on the Keyestudio 4.0 development board.

The pin G, V and S of the LED module are connected to G, 5V and D9 of the expansion board respectively.

5.Test Code

```
//*****************************************************************************
/*
keyestudio 4wd BT Car
lesson 1.1
Blink
http://www.keyestudio.com
*/
void setup()
{
  pinMode(9, OUTPUT);// initialize digital pin 9 as an output.
}

void loop() // the loop function runs over and over again forever
{
  digitalWrite(9, HIGH); // turn the LED on (HIGH is the voltage level)
   delay(1000); // wait for a second
   digitalWrite(9, LOW); // turn the LED off by making the voltage LOW
   delay(1000); // wait for a second
}
//*****************************************************************************
```

6.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, and use a USB cable to connect the computer to power the board. After powering on, you will see the LED connected to the D9 will be on and off.

7.Code Explanation

pinMode(9OUTPUT) - This function can denote that the pin is INPUT or OUTPUT

digitalWrite(9HIGH) - When pin is OUTPUT, we can set it to HIGH(output 5V) or LOW(output 0V)

8.Extension Practice

We have succeeded in blinking LED. Next, let's observe what will happen to the LED if we modify the delay time.

```
//*****************************************************************************
/*
 keyestudio 4wd BT Car
 lesson 1.2
 delay
 http://www.keyestudio.com
*/
void setup()
{
  // initialize digital pin 11 as an output.
  pinMode(9, OUTPUT);
}
// the loop function runs over and over again forever
void loop()
{
  digitalWrite(9, HIGH); // turn the LED on (HIGH is the voltage level)
  delay(100); // wait for 0.1 second
  digitalWrite(9, LOW); // turn the LED off by making the voltage LOW
  delay(100); // wait for 0.1 second
}
//*****************************************************************
```

The test result shows that the LED flashes faster. Therefore, the delaying time enables to affect the flash frequency of the LED.

9.2.2 Project 2: Adjust LED Brightness

1.Description

In previous lesson, we control LED on and off and make it blink.

In this project, we will control LED's brightness through PWM simulating breathing effect.

PWM is a means of controlling the analog output via digital means. Digital control is used to generate square waves with different duty cycles (a signal that constantly switches between high and low levels) to control the analog output. In general, the input voltages of ports are 0V and 5V.

What if the 3V is required? Or a switch among 1V, 3V and 3.5V? We cannot change resistors constantly. For this reason, we resort to PWM.

For the Arduino digital port voltage output, there are only LOW and HIGH, which correspond to the voltage output of 0V and 5V. You can define LOW as 0 and HIGH as 1, and let the Arduino output five hundred 0 or 1 signals within 1s.

If all of the output five hundred are 1, that is 5V; if all of which are 0, that is 0V. If output 010101010101 in this way then the output port is 2.5V, which is like showing movie. The movie we watch are not completely continuous. It actually outputs 25 pictures per second. In this case, the human can't see it, neither does PWM. If we want different voltage, we need to control the ratio of 0 and 1. The more 0,1 signals output per unit time, the more accurate the control.

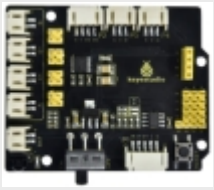
PWM is a technology that uses digital methods to obtain analog quantities. Digital control enables to form a square wave, the square wave signal only has on and off two states (high and low). A voltage ranging from 0 to 5V can be simulated by controlling the ratio of on to off duration. The time spent on (technically called high level) is called pulse width, so PWM is also called pulse width modulation.

The green vertical bars represent one period of the square wave. The value written in each analogWrite(value) corresponds to a percentage, which is also called Duty Cycle. This percentage refers to the ratio of time occupied by the high level in a cycle, that is, duty cycle = high level time/cycle time.

In the figure, from top to bottom, the duty cycle of the first square wave is 0%, and the corresponding value is 0, and the LED brightness is the lowest, that is, off state. The longer the high level lasts, the brighter it will be. Therefore, the value of the last duty cycle of 100% is 255, and the LED is the brightest. 50% is the brightest half, and 25% is darker.

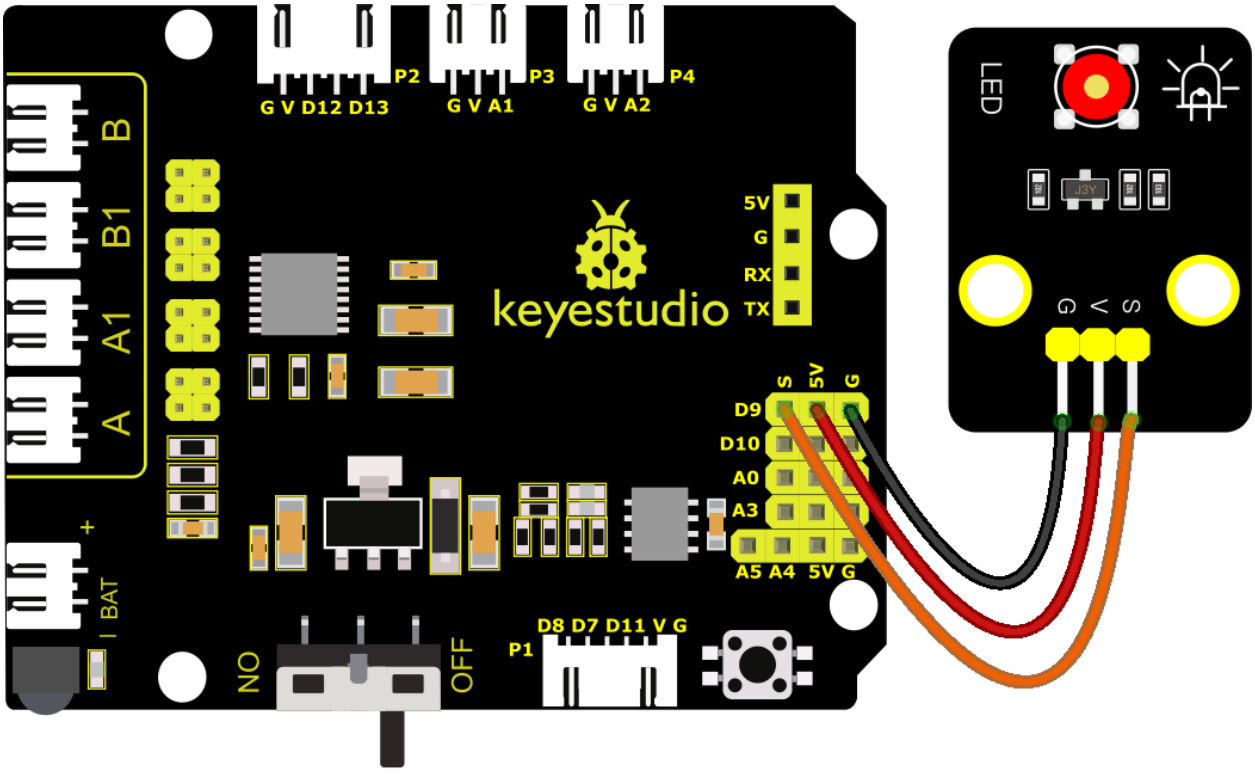
PWM is more used to adjust the brightness of LED lights or the rotation speed of the motors, and the wheels speed driven by the motors can be easily controlled. When playing with some Arduino robots, the benefits of PWM can be better reflected.

2.Components

Development Board *1	8833 Motor Driver *1	Red LED Module*1
		
3P Dupont Wire*1	USB Cable*1	

3.Wiring Diagram

Keep the wiring-up unchanged.



4.Test Code

```
//*****************************************************************************
/*
 keyestudio 4wd BT Car
 lesson 2.1
 pwm
 http://www.keyestudio.com
*/
int ledPin = 9; // Define the LED pin at D9
int value;

void setup () {
  pinMode (ledPin, OUTPUT); // initialize ledpin as an output.
}

void loop () {
  for (value = 0; value <255; value = value + 1)
  {
    analogWrite (ledPin, value); // LED lights gradually light up
    delay (5); // delay 5ms
  }
  for (value = 255; value> 0; value = value-1)
  {
    analogWrite (ledPin, value); // LED gradually goes out
    delay (5); // delay 5ms
  }
}
```

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(continued from previous page)

```
//*****************************************************************
```

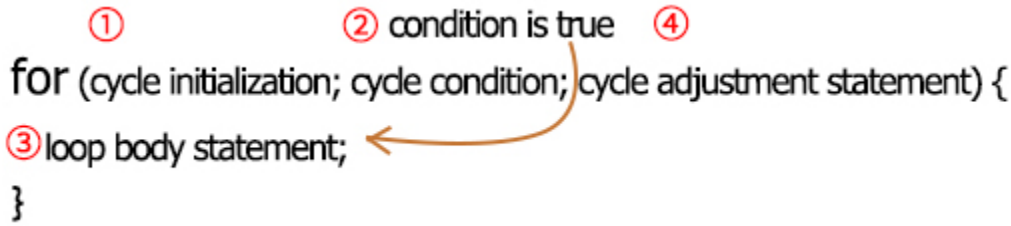
5.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, and use a USB cable to connect the computer to power the board. After powering on, you will see that the LED gradually changes from bright to dark, like human's breath, rather than turning on and off immediately.

6.Code Explanation

If we need to repeat a certain statement, we could use for statement.

For statement format is shown below:



FOR cyclic sequence:

Round 11 → 2 → 3 → 4

Round 22 → 3 → 4

...

Until number 2 is not established, "for"loop is over,

After knowing this order, go back to code:

for (int value = 0; value < 255; value=value+1)

for (int value = 255; value >0; value=value-1)

The two"for"statements make value increase from 0 to 255, then reduce from 255 to 0, then increase to 255...infinitely loop.

There is a new function in the following —— analogWrite().

We know that digital port only has two state of 0 and 1. So how to send an analog value to a digital value? Here, this function is needed. Let's observe the Arduino board and find 6 pins marked"~"which can output PWM signals.

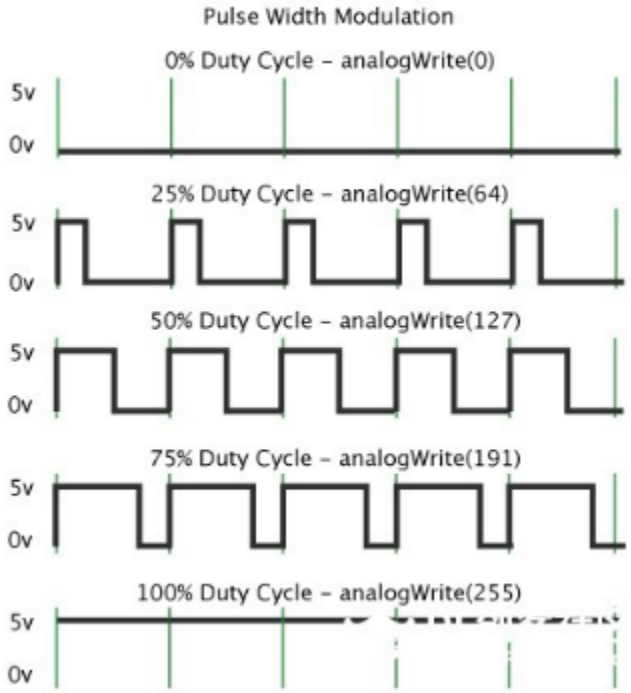
Function format as follows:

analogWrite(pin,value)

analogWrite() is used to write an analog value from 0~255 for PWM port, so the value is in the range of 0~255. Attention that you can only write the digital pins with PWM function, such as pin 3, 5, 6, 9, 10, 11.

PWM is a technology to obtain analog quantity through digital method. Digital control forms a square wave, and the square wave signal only has two states of turning on and off (that is, high or low levels). By controlling the ratio of the duration of turning on and off, a voltage varying from 0 to 5V can be simulated. The time turning on(academically referred to as high level) is called pulse width, so PWM is also called pulse width modulation.

Through the following five square waves, let's learn more about the PWM.



In the above figure, the green line represents a period, and the value of analogWrite() corresponds to a percentage which is called Duty Cycle as well. Duty cycle implies the ratio of time occupied by the high level in the cycle. From top to bottom, the duty cycle of first square wave is 0% and its corresponding value is 0.

The LED brightness is lowest, that is, light off. The more time the high level lasts, the brighter the LED. Therefore, the last duty cycle is 100%, which corresponds to 255, and LED is the brightest. And 50% is the brightest half, 25% means darker.

PWM is more used for adjusting the LED's brightness or the rotation speed of motors.

It plays a vital role in controlling smart robot cars. I believe that you cannot wait to learn the next project.

7.Extension Practice

Let's modify the value of delay and remain the pin unchanged, then observe how the LED changes.

```
//*************************************************************
/*
 keyestudio 4wd BT Car
 lesson 2.2
 pwm
 http://www.keyestudio.com
*/
int ledPin = 9; // Define the LED pin at D9
void setup () {
   pinMode(ledPin, OUTPUT); // initialize ledpin as an output.
}

void loop () {
   for (int value = 0; value <255; value = value + 1) {
     analogWrite (ledPin, value); // LED lights gradually light up
     delay (30); // delay 30MS
   }
```

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```
  for (int value = 255; value> 0; value = value-1) {
    analogWrite (ledPin, value); // LED gradually goes out
    delay (30); // delay 30MS
  }
}
//**********************************************************
```

Upload the code to the development board, then the LED will blink more slowly.

9.2.3 Project 3: Line Tracking Sensor

1.Description

The tracking sensor is actually an infrared sensor. The component used here is the TCRT5000 infrared tube. Its working principle is to use different reflectivity of infrared light to colors, then convert the strength of the reflected signal into a current signal.

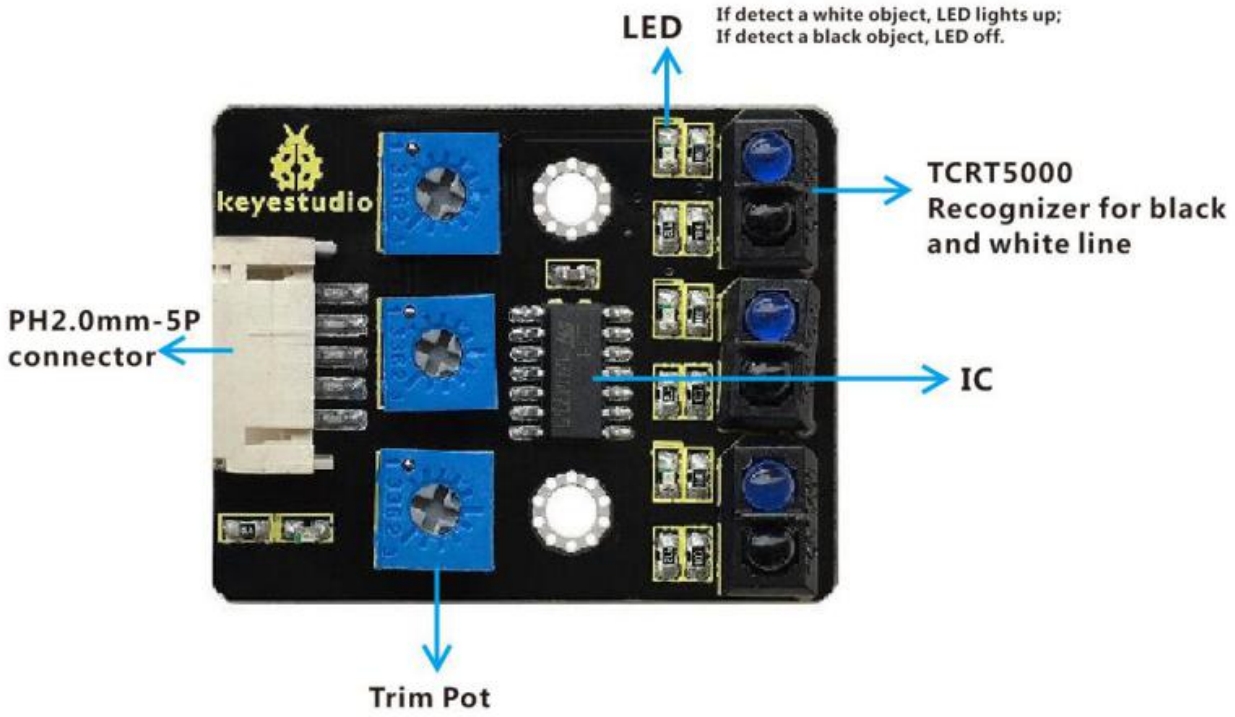
During the process of detection, black is active at HIGH level while white is active at LOW level. The detection height is 0-3 cm.

Keyestudio 3-channel line tracking module has integrated 3 sets of TCRT5000 infrared tubes on a board, which is more convenient for wiring and control.

By rotating the adjustable potentiometer on the sensor, it can adjust the detection sensitivity of the sensor.

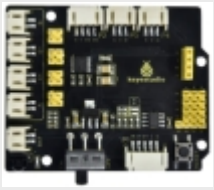
2.Specification

- Operating Voltage: 3.3-5V (DC)
- Interface: 5PIN
- Output Signal: Digital signal
- Detection Height: 0-3 cm

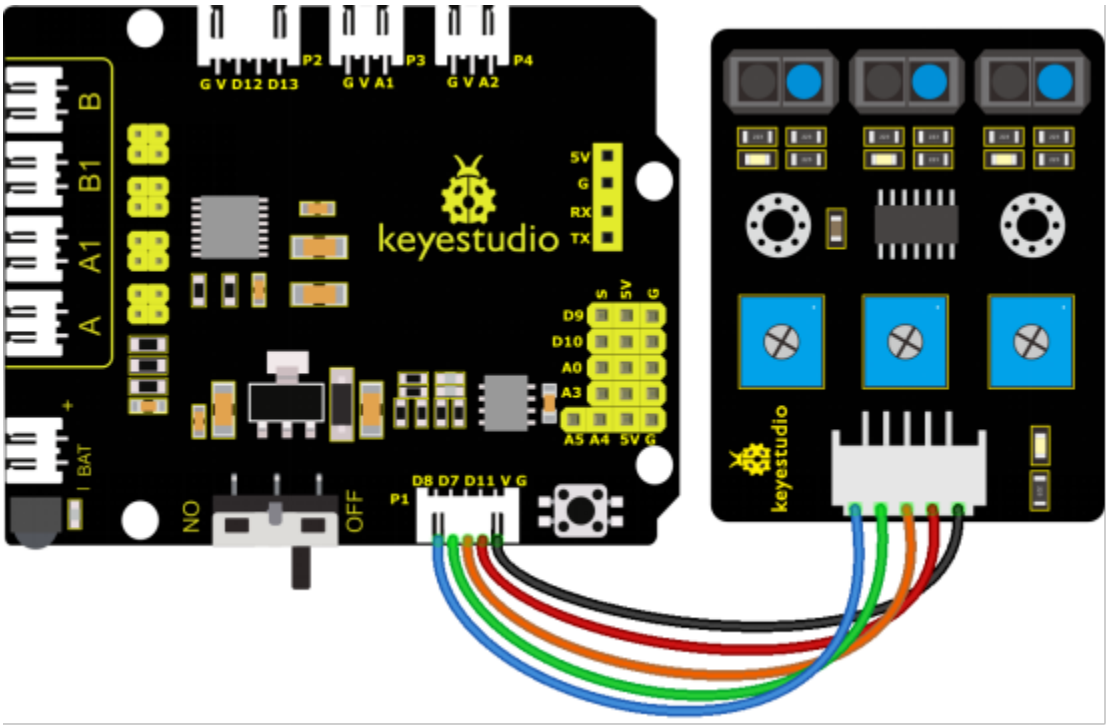


Note:Before testing, rotate the potentiometer on the sensor to adjust the detection sensitivity. The sensitivity is best when adjusting the LED to a threshold between ON and OFF.

3.Components

Development Board *1	8833 Motor Driver *1	Red LED Module*1	Line Tracking Sensor*1
			
5P Dupont Wire*1	USB Cable*1	3P Dupont Wire*1	

4.Wiring Diagram



G, V, S1, S2 and S3 of the line tracking sensor are connected to GGND), VVCC), D11, D7 and D8 of the sensor expansion board.

5.Test Code

```
//*****************************************************************************
/*
keyestudio 4wd BT Car
lesson 3.1
 Line Track sensor
 http://www.keyestudio.com
*/
int L_pin = 11;  //pins of  left line tracking sensor
int M_pin = 7;  //pins of  middle line tracking sensor
int R_pin = 8;  //pins of  right  line tracking sensor
int val_L,val_R,val_M;// define the variable value of three sensors

void setup()
{
  Serial.begin(9600); // initialize serial communication at 9600 bits per second
  pinMode(L_pin,INPUT); // make the L_pin as an input
  pinMode(M_pin,INPUT); // make the M_pin as an input
  pinMode(R_pin,INPUT); // make the R_pin as an input
}

void loop()
{
  val_L = digitalRead(L_pin);//read the L_pin:
```

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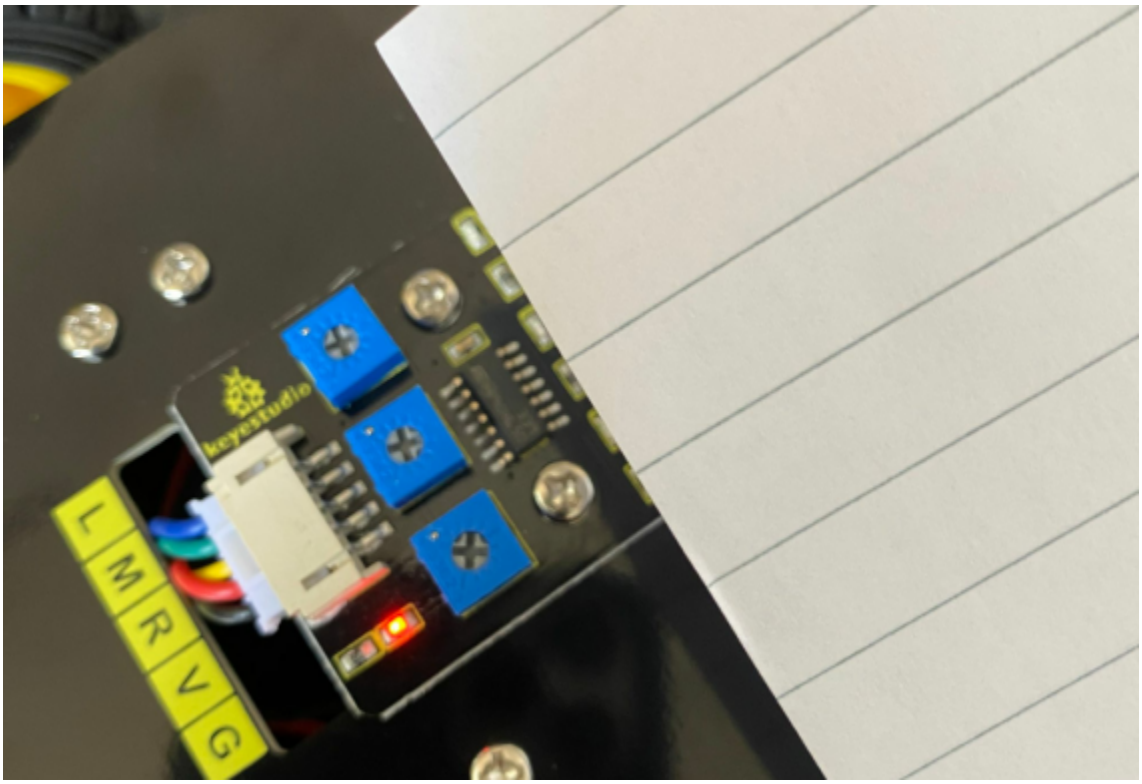
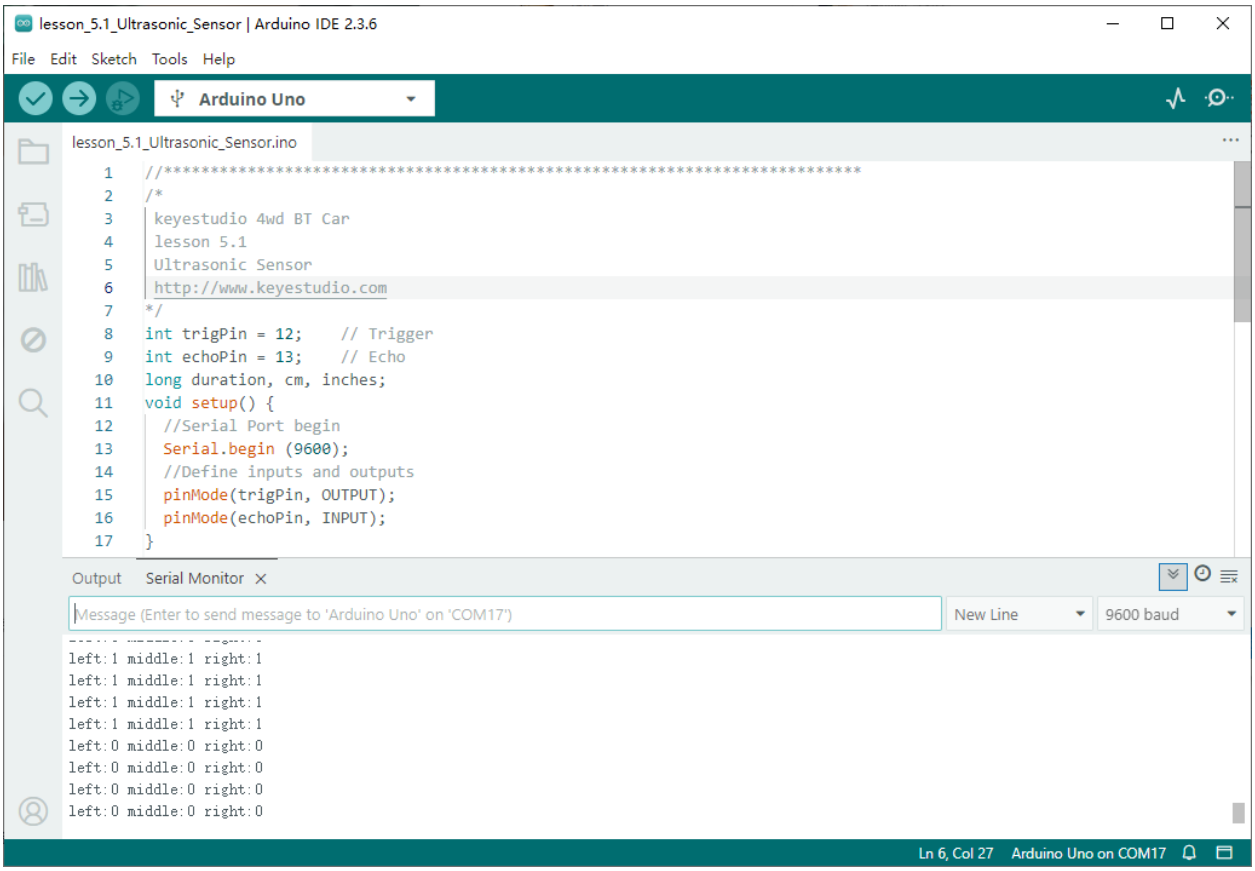
```
  val_R = digitalRead(R_pin);//read the R_pin:
  val_M = digitalRead(M_pin);//read the M_pin:
  Serial.print("left:");
  Serial.print(val_L);
  Serial.print(" middle:");
  Serial.print(val_M);
  Serial.print(" right:");
  Serial.println(val_R);
  delay(500);// delay in between reads for stability
}
//**************************************************************************
```

6.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, and use a USB cable to connect the computer to power the board.

After powering on, open the serial monitor and you will view status of three line tracking sensors. When no signals are received, the value is 1. If we cover the sensor with a white paper, the value will be 0.





7.Code Explanation

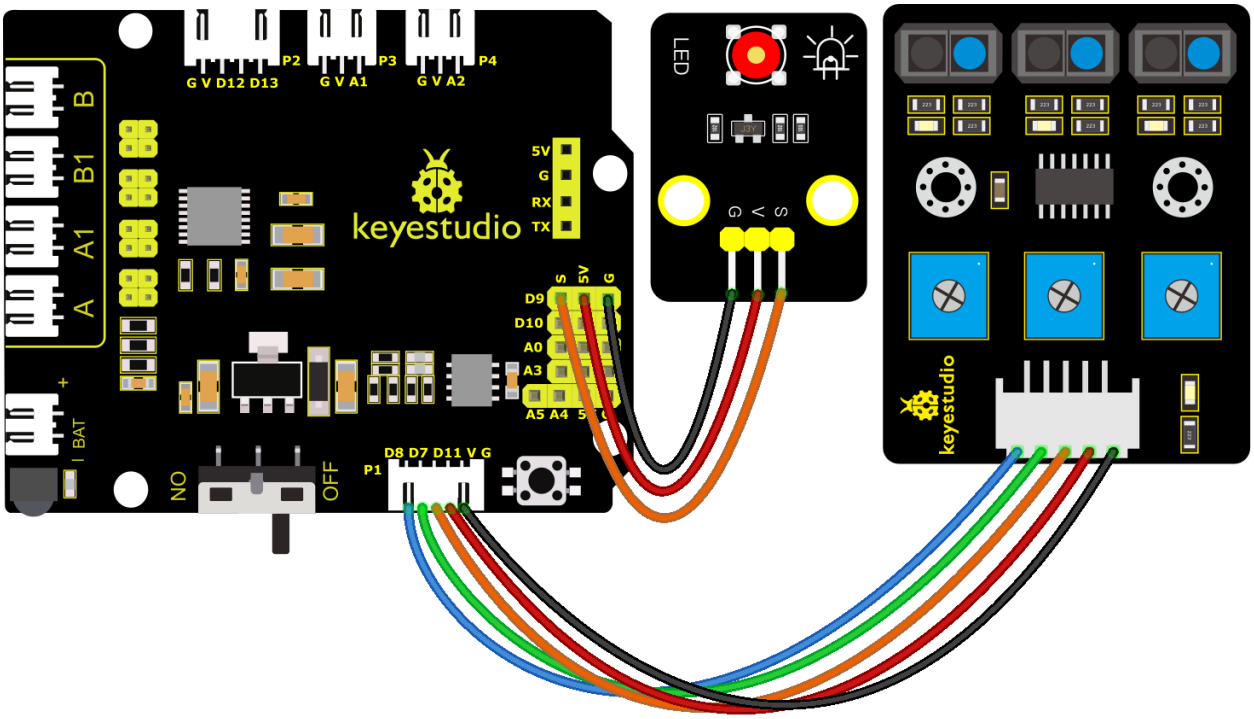
Serial.begin(9600) - Initialize serial port, set baud rate to 9600

pinMode -Define the pin as input or output mode

digitalRead -Read the state of pin, which are generally HIGH and LOW level

8.Extension Practice

After knowing its working principle, you can connect an LED to D9 so as to control LED by it.




```
/*
keyestudio 4wd BT Car
lesson 3.2
 Line Track Sensor LED
 http://www.keyestudio.com
*/
int L_pin = 11;  //pins of  left line tracking sensor
int M_pin = 7;  //pins of  middle line tracking sensor
int R_pin = 8;  //pins of  right line tracking sensor
int val_L,val_R,val_M;// define the variables of three sensors

void setup()
{
  Serial.begin(9600); // initialize serial communication at 9600 bits per second
  pinMode(L_pin,INPUT); // make the L_pin as an input
  pinMode(M_pin,INPUT); // make the M_pin as an input
  pinMode(R_pin,INPUT); // make the R_pin as an input
  pinMode(9, OUTPUT);
}

void loop()
{
  val_L = digitalRead(L_pin);//read the L_pin:
  val_R = digitalRead(R_pin);//read the R_pin:
  val_M = digitalRead(M_pin);//read the M_pin:
  Serial.print("left:");
  Serial.print(val_L);
  Serial.print(" middle:");
  Serial.print(val_M);
  Serial.print(" right:");
```

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```
  Serial.println(val_R);
  delay(500);// delay in between reads for stability
  if ((val_L == LOW) || (val_M == LOW) || (val_R == LOW))//if left line tracking sensor␣
↪detects signals
  {
    Serial.println("HIGH");
    digitalWrite(9, HIGH);//LED is off
  }
  else//if left line tracking sensor doesn't detect signals
  {
    Serial.println("LOW");
    digitalWrite(9, LOW);//LED is off
  }
 }
//*****************************************************************************
```

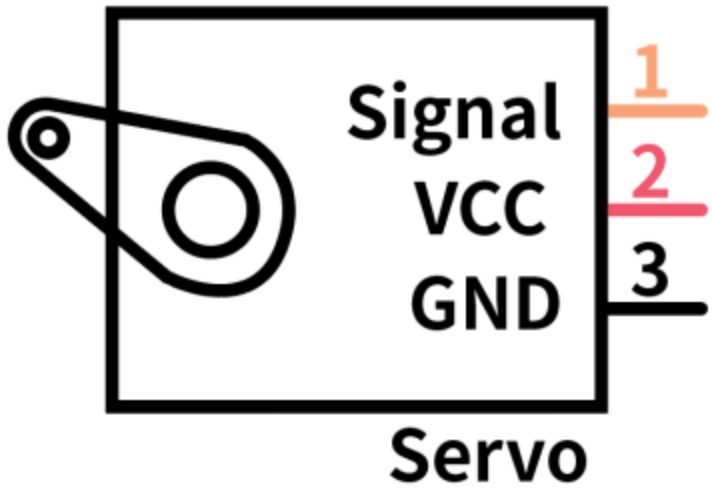
After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, and use a USB cable to connect the computer to power the board.

After powering on, make a paper close to the sensor, then we can find the LED light up when covering the line tracking sensor.

9.2.4 Project 4 Servo Control

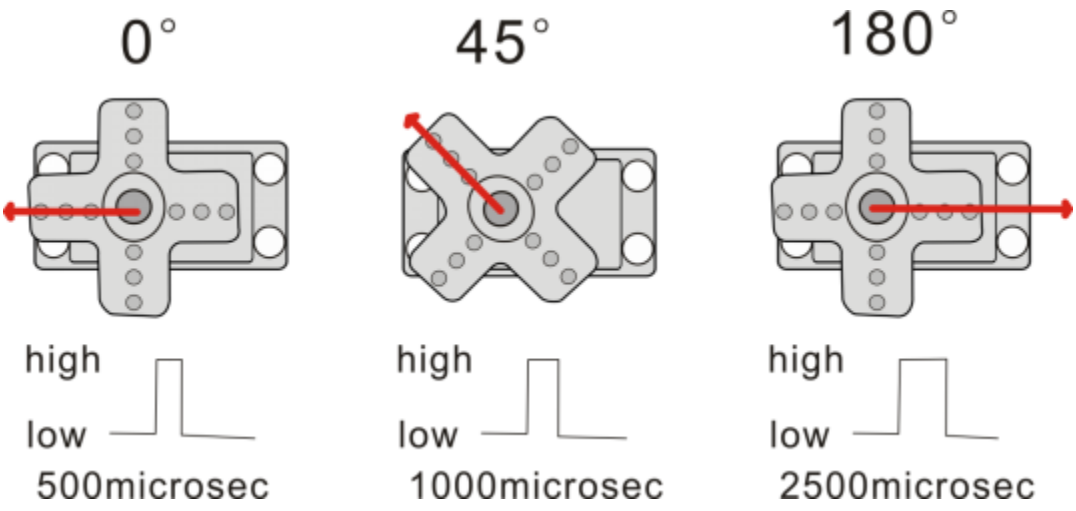
1.Description

Servo motor is a position control rotary actuator. It mainly consists of a housing, a circuit board, a core-less motor, a gear and a position sensor. Its working principle is that the servo receives the signal sent by MCUs or receivers and produces a reference signal with a period of 20ms and width of 1.5ms, then compares the acquired DC bias voltage to the voltage of the potentiometer and obtain the voltage difference output.



In general, servo has three lines in brown, red and orange. The brown wire is grounded, the red one is a positive pole line and the orange one is a signal line.

The rotation angle of servo motor is controlled by regulating the duty cycle of PWM (Pulse-Width Modulation) signal. The standard cycle of PWM signal is 20ms (50Hz). Theoretically, the width is distributed between 1ms-2ms, but in fact, it's between 0.5ms-2.5ms. The width corresponds the rotation angle from 0° to 180°. But note that for different brand motors, the same signal may have different rotation angles.



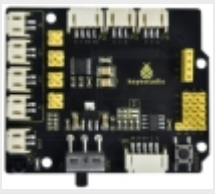
The corresponding servo angles are shown below:

High level time	Servo angle
0.5ms	0 degree
1ms	45 degree
1.5ms	90 degree
2ms	135 degree
2.5ms	180 degree

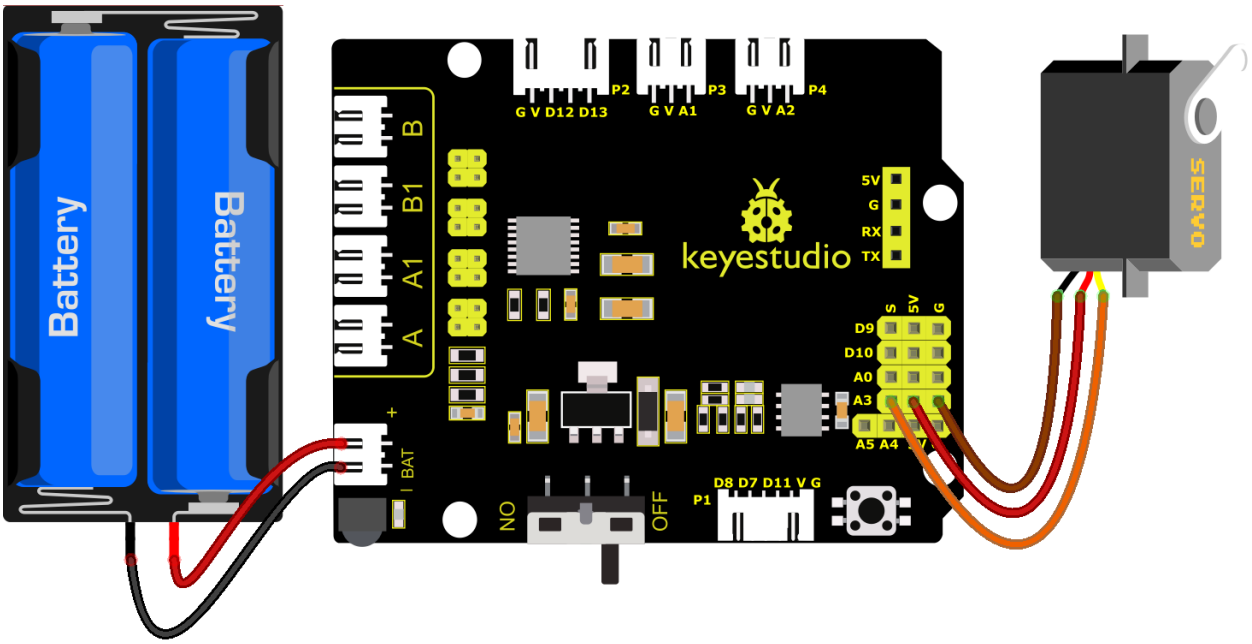
2.Specification

- Working voltage: DC 4.8V ~ 6V
- Operating angle range: about 180 ° (at 500 → 2500 sec)
- Pulse width range: 500 → 2500 sec
- No-load speed: 0.12 ± 0.01 sec / 60 (DC 4.8V) 0.1 ± 0.01 sec / 60(DC 6V)
- No-load current: 200 ± 20mA (DC 4.8V) 220 ± 20mA (DC 6V)
- Stopping torque: 1.3 ± 0.01kg · cm (DC 4.8V) 1.5 ± 0.1kg · cm (DC6V)
- Stop current: 850mA (DC 4.8V) 1000mA (DC 6V)
- Standby current: 3 ± 1mA (DC 4.8V) 4 ± 1mA (DC 6V)

3.Components

Development Board *1	8833 Motor Driver *1	Servo*1
		
18650 Battery Holder*1	USB Cable*1	18650 Battery*2self-provided

4.Wiring Diagram



Wiring note: The servo is connected to GGNDVVCCand A3, the brown line of servo is linked with Gnd(G), the red one is connected to 5v(V) and the orange one is attached to A3.

The servo is obliged to connect to the external power due to its high demand for driving servo current. Generally, the current of development board is not big enough. If without connecting the external power, the development board could be burnt.

5.Test Code

```
//*****************************************************************************
/*
keyestudio 4wd BT Car
lesson 4.1
Servo
http://www.keyestudio.com
*/
#define servoPin A3  //servo Pin
int pos; //the angle variable of servo
int pulsewidth; //pulse width variable of servo

void setup() {
  pinMode(servoPin, OUTPUT);  //set the pins of servo to output
  procedure(0); //set the angle of servo to 0 degree
}

void loop() {
  for (pos = 0; pos <= 180; pos += 1) { // goes from 0 degrees to 180 degrees
    // in steps of 1 degree
    procedure(pos);              // tell servo to go to position in variable 'pos'
    delay(15);                   //control the rotation speed of servo
  }
```

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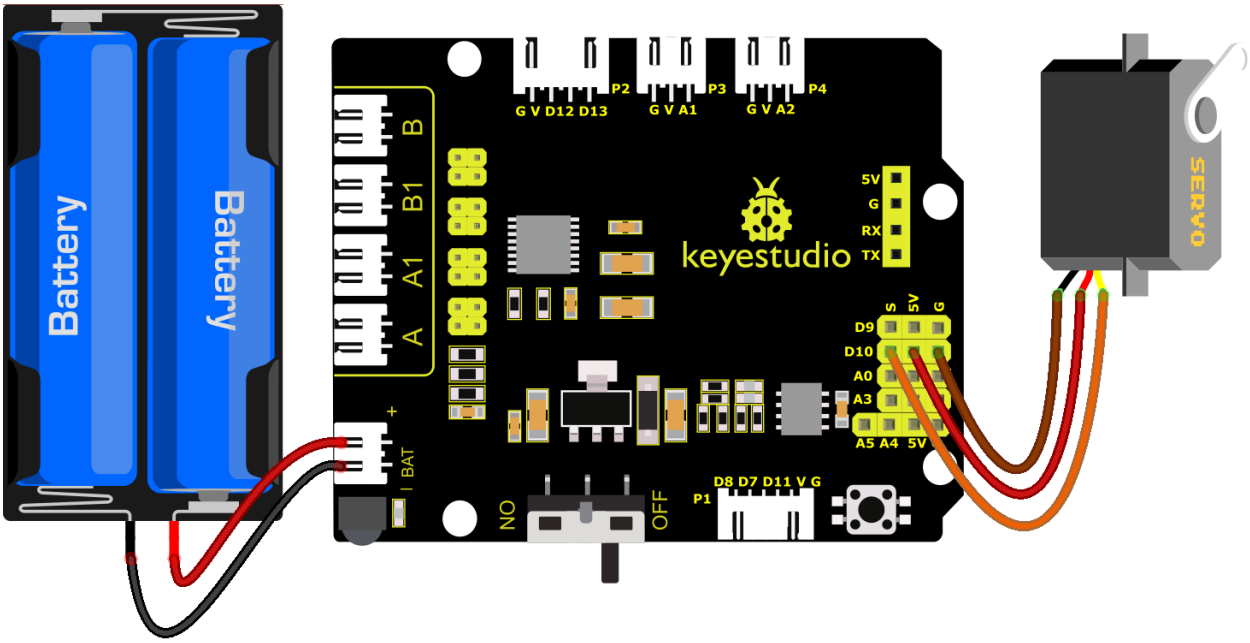
```
  for (pos = 180; pos >= 0; pos -= 1) { // goes from 180 degrees to 0 degrees
    procedure(pos);              // tell servo to go to position in variable 'pos'
    delay(15);
  }
}

//function to control servo
void procedure(int myangle) {
  pulsewidth = myangle * 11 + 500;  //calculate the value of pulse width
  digitalWrite(servoPin,HIGH);
  delayMicroseconds(pulsewidth);   //The duration of high level is pulse width
  digitalWrite(servoPin,LOW);
  delay((20 - pulsewidth / 1000));  //the cycle is 20ms, the low level last for the rest␣
↪of time
}
//*********************************************************************************
```

6.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, and power on the external power. After powering on, turn the dip switch to the "ON" end, then servo will swing in the range of 0° to 180°.

7.Extension Practice

What's more, we empower to control the servo via library file. Please refer to the link:https://www.arduino.cc/en/Reference/Servo.




```
//*********************************************************************************
/*
 keyestudio 4wd BT Car
 lesson 4.2
```

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```
 Servo
 http://www.keyestudio.com
*/
#include <Servo.h>
Servo myservo;  // create servo object to control a servo
// twelve servo objects can be created on most boards
int pos = 0;    // variable to store the servo position

void setup() {
  myservo.attach(A3);  // attaches the servo on pin A3 to the servo object
}
void loop() {
  for (pos = 0; pos <= 180; pos += 1) { // goes from 0 degrees to 180 degrees
    // in steps of 1 degree
    myservo.write(pos);              // tell servo to go to position in variable 'pos'
    delay(15);                       // waits 15ms for the servo to reach the position
  }
  for (pos = 180; pos >= 0; pos -= 1) { // goes from 180 degrees to 0 degrees
    myservo.write(pos);              // tell servo to go to position in variable 'pos'
    delay(15);                       // waits 15ms for the servo to reach the position
  }
}
//**************************************************************************
```

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, and power on the external power. After powering on, turn the dip switch to the "ON" end, then servo will swing in the range of 0° to 180° too. We usually control it by library file.

8.Code Explanation

Arduino comes with **#include <Servo.h>** (servo function and statement

The following are some common statements of the servo function:

1). **attachinterface**——Set interface of the servo

2). **writeangle**——Used to set rotation angle of servo, and the set angle range is from 0° to 180°

3). **read**——used to read angle of servo, namely, reading the command value of"write()"

4). **attached**——Judge if the parameter of servo is sent to its interface

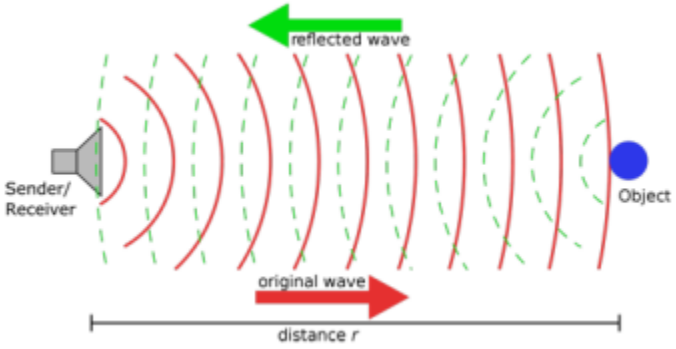
Note: The above written format is"servo variable name, specific statement", for instance: myservo.attach(9).

9.2.5 Project 5 Ultrasonic Sensor

1.Description

The HC-SR04 ultrasonic sensor uses sonar to determine distance to an object like what bats do. It offers excellent non-contact range detection with high accuracy and stable readings in an easy-to-use package. It comes complete with

an ultrasonic transmitter and receiver modules.

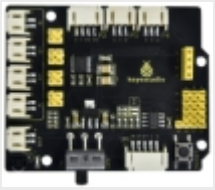


The HC-SR04 or the ultrasonic sensor is being used in a wide range of electronics projects for creating obstacle detection and distance measuring application as well as various other applications. Here we have brought the simple method to measure the distance with arduino and an ultrasonic sensor and how to use the ultrasonic sensor with Arduino.

2.Specification

- Working Voltage :+5V DC
- Quiescent Current : <2mA
- Working Current: 15mA
- Effectual Angle: <15°
- Distance Range : 2cm – 300 cm
- Precision : 0.3 cm
- Measuring Angle: 30 degree
- Trigger Input Pulse width: 10uS

3.Components

Development Board *1	8833 Motor Driver *1	Red LED Module*1	Ultrasonic Sensor*1
			
4P Dupont Wire*1	USB Cable*1	3P Dupont Wire*1	

4.Working Principle

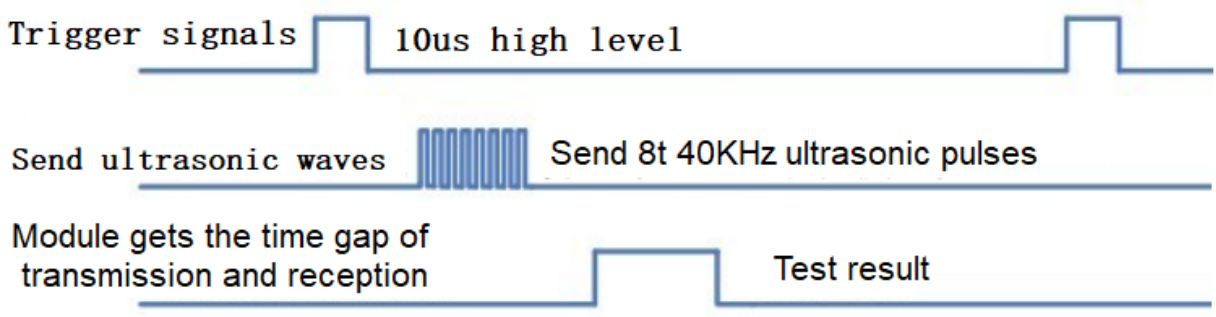
As the above picture shown, it is like two eyes. One is transmitting end, the other is receiving end.

The ultrasonic module will emit the ultrasonic waves after triggering a signal. When the ultrasonic waves encounter the object and are reflected back, the module outputs an echo signal, so it can determine the distance of the object from the time difference between the trigger signal and the echo signal.

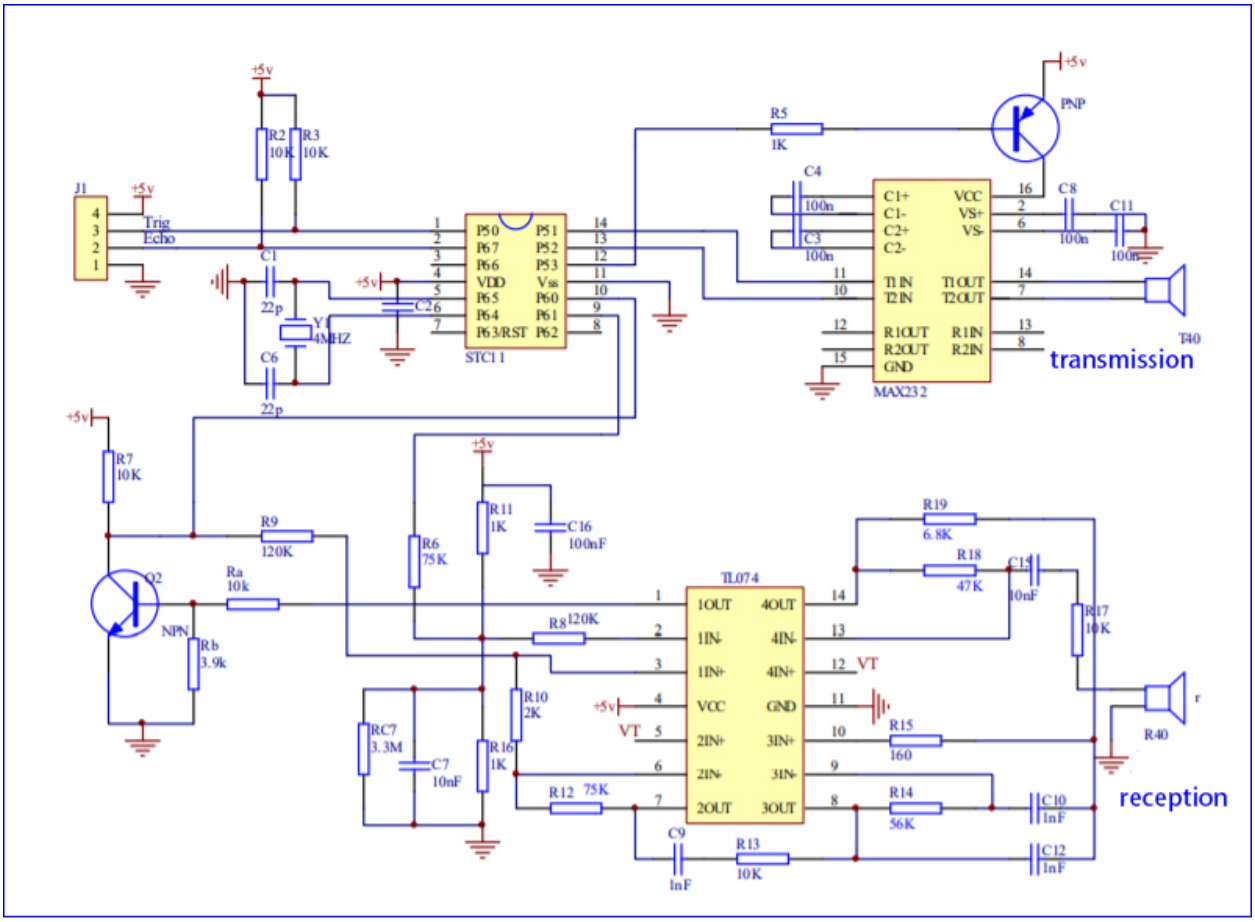
The t is the time that emitting signal meets obstacle and returns. And the propagation speed of sound in the air is about 343m/s, and distance = speed * time. However, the ultrasonic wave emits and comes back, which is 2 times of distance. Therefore, it needs to be divided by 2, the distance measured by ultrasonic wave = (speed * time)/2.

Use method and chart of ultrasonic module:

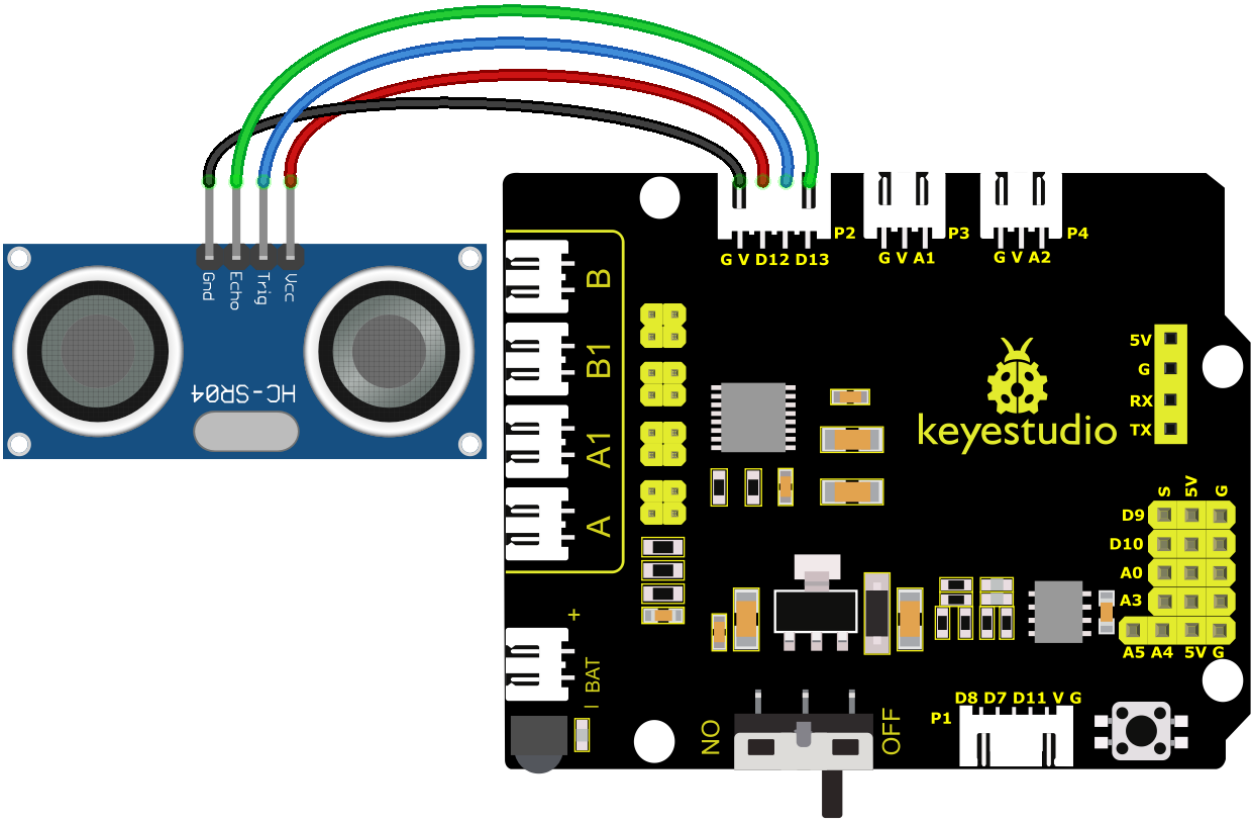
1. Use the GPIO pin to give a high level signal of at least 10s to the Trig pin of SR04, which can trigger it to detect distance.
2. After triggering, the module will automatically send eight 40KHz ultrasonic pulses and detect whether there is a signal return. This step will be completed automatically by the module.
3. If the signal returns, the Echo pin will output a high level, and the duration of the high level is the time from the transmission of the ultrasonic wave to the return.



Circuit diagram of ultrasonic sensor:


+5v
R2 10K
R3 10K
J1
Trig
Echo
C1
22p
Y1 4MHZ
C6
22p
C2
STC11
P50
P67
P66
VDD
P65
P64
P63/RST
P51
P52
P53
Vss
P60
P61
P62
R5 1K
PNP
C4 100n
C3 100n
C1+
C1-
C2+
C2-
VCC
VS+
VS-
T1 IN
T2 IN
T1 OUT
T2 OUT
R1OUT
R2OUT
GND
R1IN
R2IN
MAX232
C8 100n
C11 100n
T40
transmission
R7 10K
R9 120K
Q2
NPN
Ra 10k
Rb 3.9k
R6 75K
R11 1K
C16 100nF
R8 120K
R10 2K
RC7 3.3M
C7 10nF
R16 1K
R12 75K
C9 1nF
R13 10K
TL074
1OUT
1IN-
1IN+
VCC
2IN+
2IN-
2OUT
4OUT
4IN-
4IN+
GND
3IN+
3IN-
3OUT
VT
R19 6.8K
R18 47K
C15 10nF
R17 10K
R15 160
R14 56K
C10 1nF
C12 1nF
R40
reception

5.Wiring Diagram



VCC, Trig, Echo and Gnd of the ultrasonic sensor are connected to 5V(V), D12, D13 and Gnd(G)

6.Test Code

```
//*****************************************************************************
/*
 keyestudio 4wd BT Car
 lesson 5.1
 Ultrasonic Sensor
 http://www.keyestudio.com
*/
int trigPin = 12;    // Trigger
int echoPin = 13;    // Echo
long duration, cm, inches;
void setup() {
  //Serial Port begin
  Serial.begin (9600);
  //Define inputs and outputs
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
}

void loop() {
  // The sensor is triggered by a HIGH pulse of 10 or more microseconds.
  // Give a short LOW pulse beforehand to ensure a clean HIGH pulse:
```

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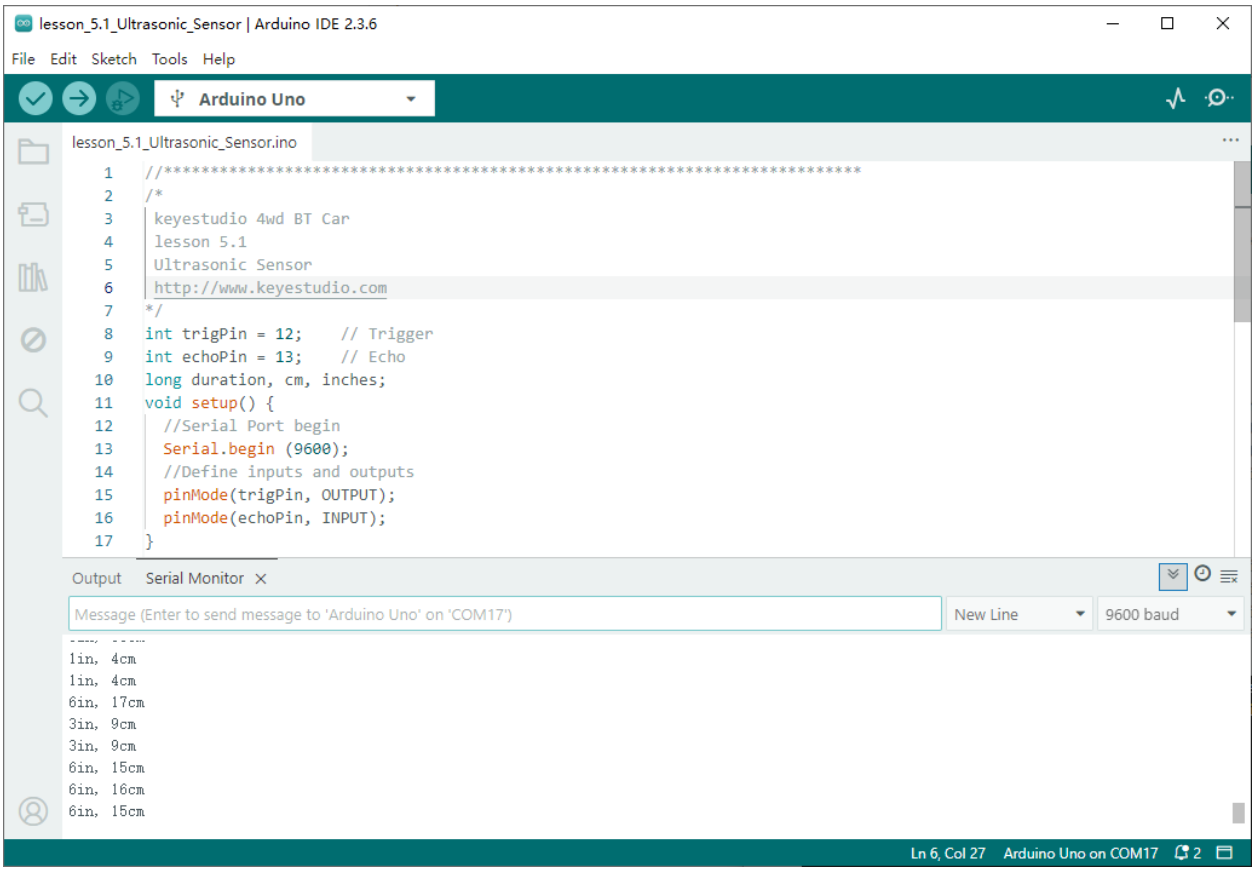
```
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
   // Read the signal from the sensor: a HIGH pulse whose
  // duration is the time (in microseconds) from the sending
  // of the ping to the reception of its echo off of an object.
  duration = pulseIn(echoPin, HIGH);
   // Convert the time into a distance
  cm = (duration/2) / 29.1;     // Divide by 29.1 or multiply by 0.0343
  inches = (duration/2) / 74;   // Divide by 74 or multiply by 0.0135
  Serial.print(inches);
  Serial.print("in, ");
  Serial.print(cm);
  Serial.print("cm");
  Serial.println();
  delay(250);
}
//**************************************************************************
```

7.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, then connect the computer via a USB cable to power the board. After powering on, open the serial monitor and set baud rate to 9600.

The detected distance will be displayed, and the unit is cm and inch. Hinder the ultrasonic sensor by hand, the displayed distance value gets smaller.



8.Code Explanation

int trigPin- this pin is defined to transmit ultrasonic waves,generally output.

int echoPin - this is defined as the pin of reception, generally input.

cm = (duration/2) / 29.1-

inches = (duration/2) / 74-

We can calculate the distance by using the following formula:

distance = (traveltime/2) x speed of sound

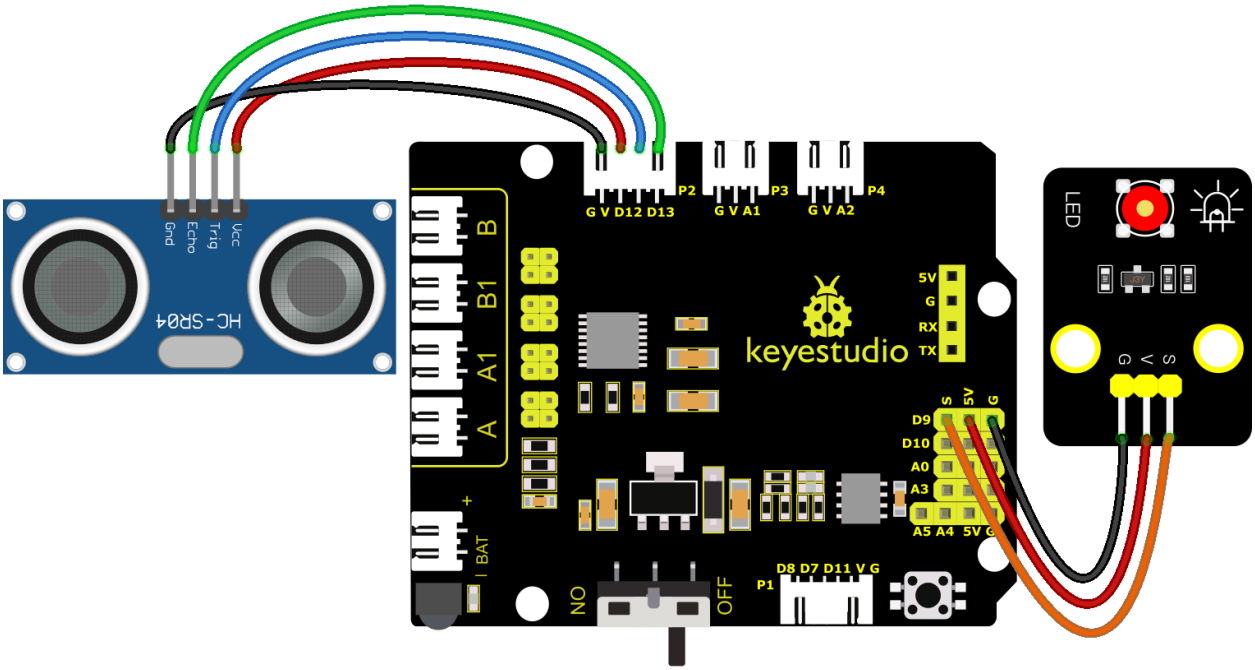
The speed of sound is: 343m/s = 0.0343 cm/us = 1/29.1 cm/us

Or in inches: 13503.9in/s = 0.0135in/us = 1/74in/us

We need to divide the traveltime by 2 for we have to take into account that the wave was sent, hit the object, and then returned back to the sensor.

9.Extension Practice

We have just measured the distance displayed by the ultrasonic. How about controlling the LED with the measured distance? Let's try it and connect an LED light module to the D9 pin.




```
//*****************************************************************
/*
 keyestudio 4wd BT Car
 lesson 5.2
 Ultrasonic LED
 http://www.keyestudio.com
*/
int trigPin = 12;    // Trigger
int echoPin = 13;    // Echo
long duration, cm, inches;

void setup() {
  Serial.begin (9600);  //Serial Port begin
  pinMode(trigPin, OUTPUT);  //Define inputs and outputs
  pinMode(echoPin, INPUT);
}

void loop()
{
  // The sensor is triggered by a HIGH pulse of 10 or more microseconds.
  // Give a short LOW pulse beforehand to ensure a clean HIGH pulse:
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  // Read the signal from the sensor: a HIGH pulse whose
  // duration is the time (in microseconds) from the sending
  // of the ping to the reception of its echo off of an object.
  duration = pulseIn(echoPin, HIGH);
  // Convert the time into a distance
```

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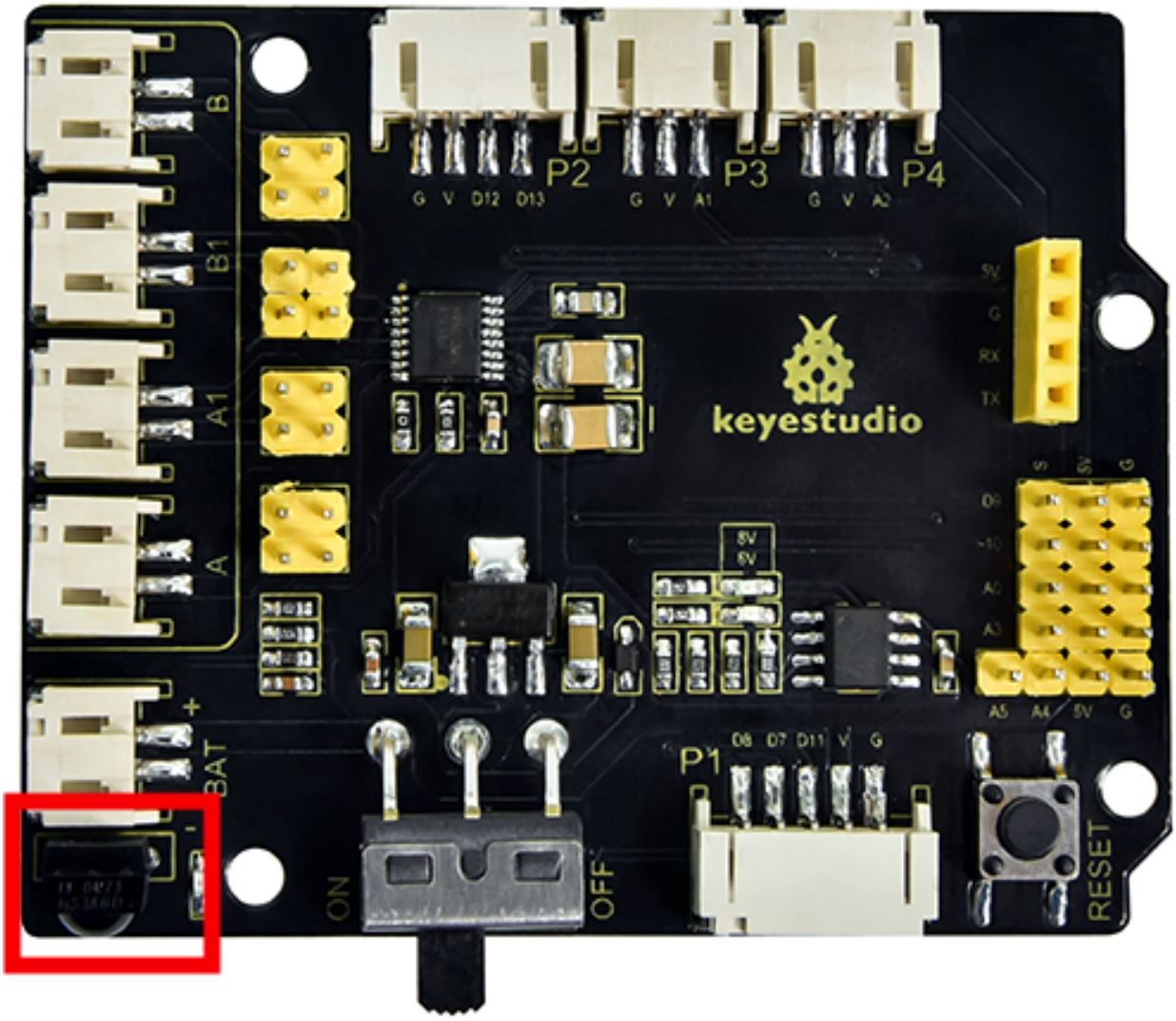
```
  cm = (duration/2) / 29.1;     // Divide by 29.1 or multiply by 0.0343
  inches = (duration/2) / 74;   // Divide by 74 or multiply by 0.0135
  Serial.print(inches);
  Serial.print("in, ");
  Serial.print(cm);
  Serial.print("cm");
  Serial.println();
  delay(250);
  if (cm>=2 && cm<=10)
  {
    Serial.println("HIGH");
    digitalWrite(9, HIGH);
  }
  else
  {
    Serial.println("LOW");
    digitalWrite(9, LOW);
  }
}
//*****************************************************************
```

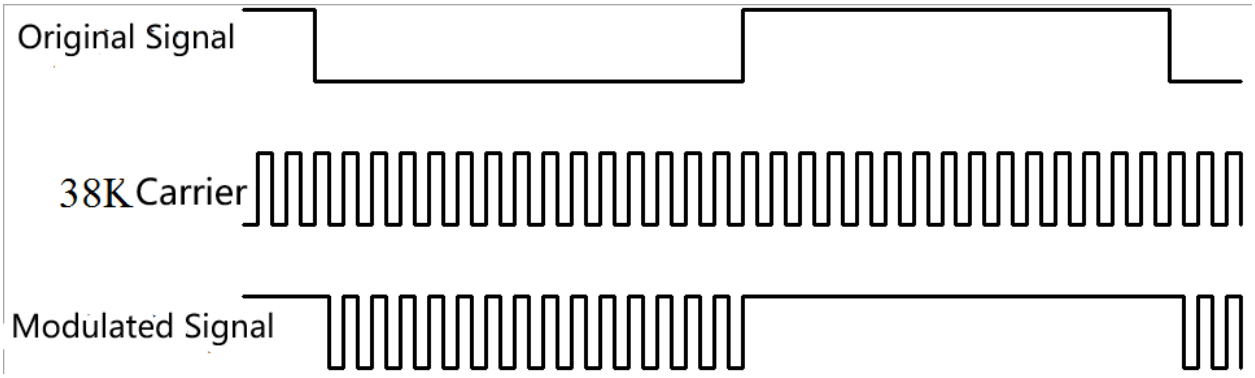
After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, then connect the computer via a USB cable to power the board. After powering on, block the ultrasonic sensor by hand(the distance is between 2-10cm), then check if the LED is on.

9.2.6 Project 6 IR Reception



1.Description

There is no doubt that infrared remote control is ubiquitous in daily life. It is used to control various household appliances, such as TVs, stereos, video recorders and satellite signal receivers. Infrared remote control is composed of infrared transmitting and infrared receiving systems, that is, an infrared remote control and infrared receiving module and a single-chip microcomputer capable of decoding.



The 38K infrared carrier signal emitted by remote controller is encoded by the encoding chip in the remote controller. It is composed of a section of pilot code, user code, user inverse code, data code, and data inverse code. The time interval of the pulse is used to distinguish whether it is 0 or 1 signal and the encoding is made up of these 0, 1 signals.

The user code of the same remote control is constant while the data code can distinguish the key.

When the remote control button is pressed, the remote control sends out an infrared carrier signal. When the IR receiver receives the signal, the program will decode the carrier signal and determines which key is pressed. The MCU decodes the received 01 signal, thereby judging what key is pressed by the remote control.

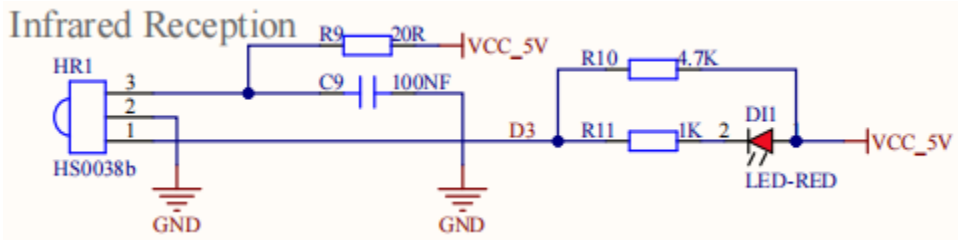
Infrared receiver we use is an infrared receiver module. Mainly composed of an infrared receiver head, which is a device that integrates reception, amplification, and demodulation. Its internal IC has completed demodulation, and can achieve from infrared reception to output and be compatible with TTL signals.

Additionally, it is suitable for infrared remote control and infrared data transmission. The infrared receiving module made by the receiver has only three pins, signal line, VCC and GND. It is very convenient to communicate with Arduino and other microcontrollers.

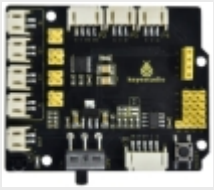
2.Specification

- Operating Voltage: 3.3-5VDC)
- Output Signal: Digital signal
- Receiving Angle: 90 degrees
- Frequency: 38khz
- Receiving Distance: 10m

The picture shows the real product and circuit diagram of the infrared receiver.



3.Components

Development Board *1	8833 Motor Driver *1	Red LED Module*1
		
3P Dupont Wire*1	USB Cable*1	

Since the 8833 board integrates with the IR receiver, it doesn't need wiring up. Pins of IR receiver module are G(GND, VVCCand D3.

4.Test Code

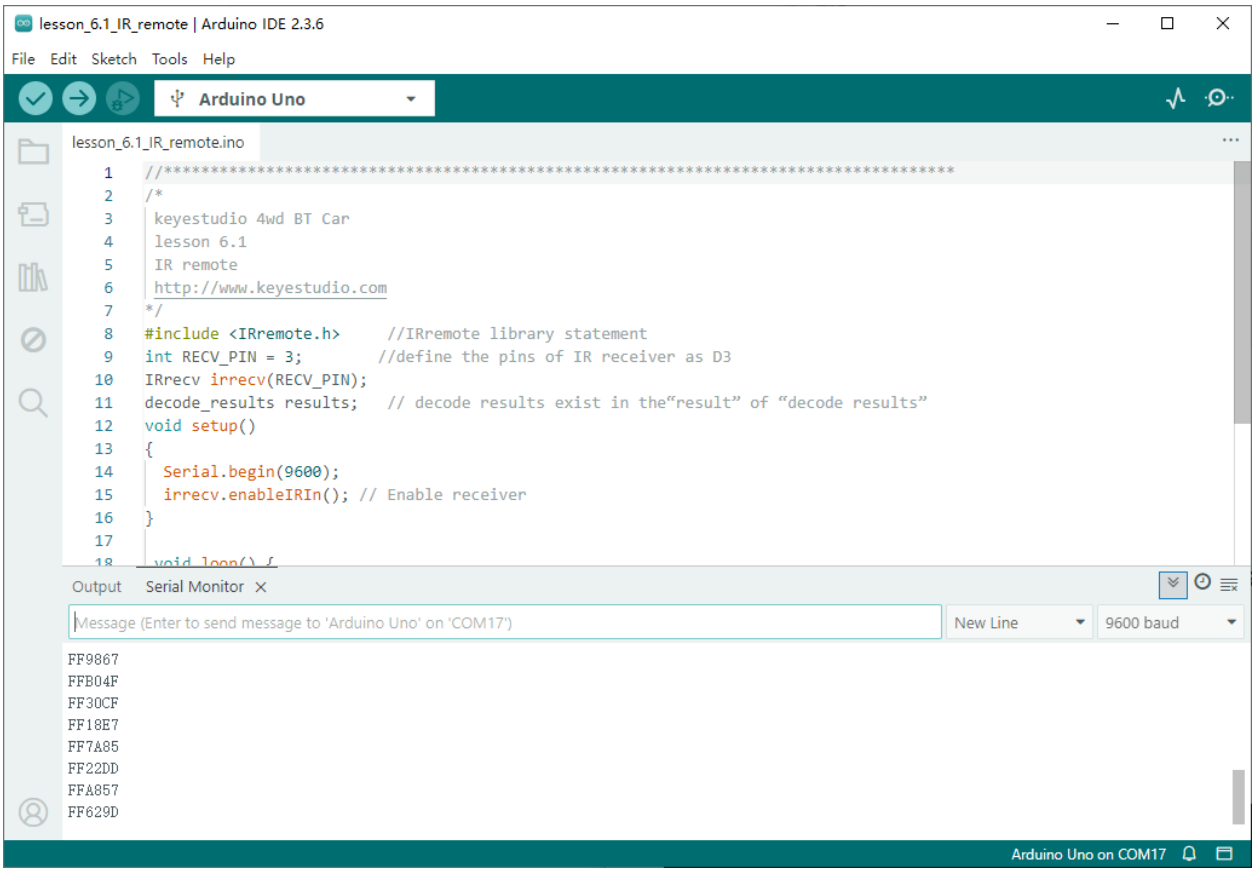
```
//*****************************************************************************************
/*
 keyestudio 4wd BT Car
 lesson 6.1
 IR remote
 http://www.keyestudio.com
*/
#include <IRremote.h>      //IRremote library statement
int RECV_PIN = 3;         //define the pins of IR receiver as D3
IRrecv irrecv(RECV_PIN);
decode_results results;   // decode results exist in the“result” of “decode results”
void setup()
{
  Serial.begin(9600);
  irrecv.enableIRIn(); // Enable receiver
}

 void loop() {
  if (irrecv.decode(&results))//decode successfully, receive a set of infrared signals
   {
     Serial.println(results.value, HEX);//Wrap word in 16 HEX to output and receive code
     irrecv.resume(); // Receive the next value
   }
   delay(100);
 }
//*****************************************************************************************
```

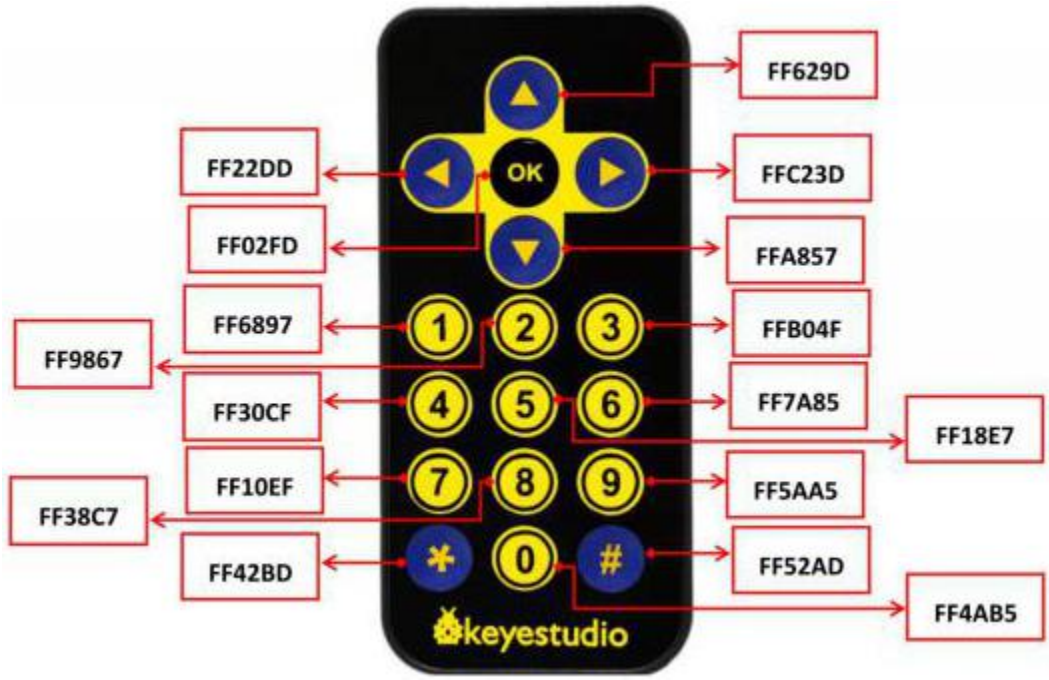
5.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, then connect the computer via a USB cable to power the board. After powering on, open the serial monitor and set baud rate to 9600.

Take out the remote control, and send signal to the infrared receiver sensor. You can see the key value of the corresponding key, if the key time is too long, FFFFFFFF is prone to garbled characters.



The keys value of Keyestudio remote control are shown below.



6.Code Explanation

irrecv.enableIRIn(): After enabling IR decoding, the IR signals will be received,

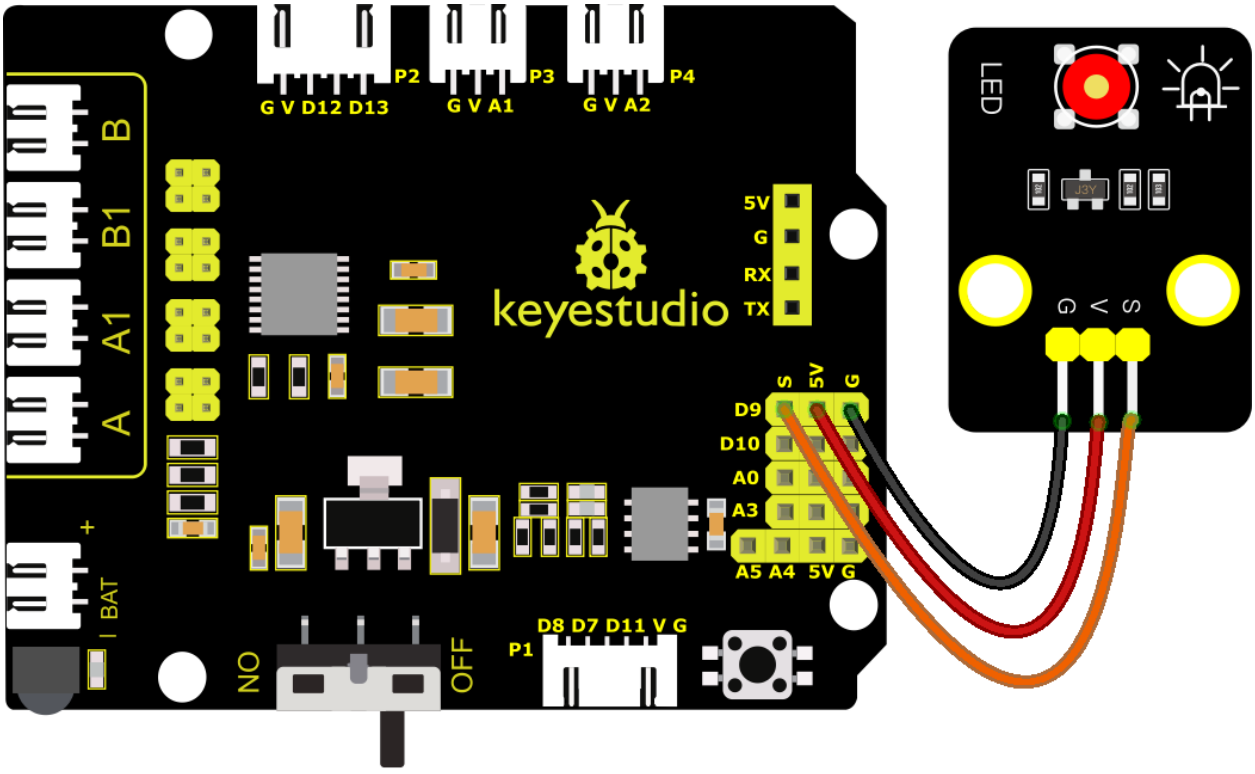
decode(): The function“decode()”will check continuously to make sure if decoding successfully.

irrecv.decode(&results): after decoding successfully, this function will come back to “true”, and keep result in “results”. After decoding the IR signals, run the resume()function and continue to receive the next signal.

7.Extension Practice

We have decoded the key value of the IR remote control. How about controlling LED by the measured value? We could design an experiment.

Attach an LED to D9, then press the keys of remote control to make LED light on and off.




```
//*****************************************************************************************************
/*
keyestudio 4wd BT Car
lesson 6.2
IR remote LED
http://www.keyestudio.com
*/
#include <IRremote.h>
int RECV_PIN = 3;//define the pin of IR receiver as D3
int LED_PIN = 9;//define the pin of LED as pin 9
int a=0;
IRrecv irrecv(RECV_PIN);
decode_results results;

void setup()
{Serial.begin(9600);
  irrecv.enableIRIn(); //Initialize the IR receiver
  pinMode(LED_PIN,OUTPUT);//set pin 9 of LED to OUTPUT
}

void loop() {
  if (irrecv.decode(&results))
  {
    if(results.value==0xFF02FD && (a==0)) //according to the above key value,␣
↪press“OK”on remote control , LED will be controlled
    {
      Serial.println("HIGH");
      digitalWrite(LED_PIN,HIGH);//LED will be on
```

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```
      a=1;
    }
    else if(results.value==0xFF02FD && (a==1)) //press again
    {
      Serial.println("LOW");
      digitalWrite(LED_PIN,LOW);//LED will go off
      a=0;
    }
    irrecv.resume(); // receive the next value
  }
}
//*********************************************************************************************
```

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, then connect the computer via a USB cable to power the board. After powering on, press the "**OK**" key on remote control can make the LED on and off.

9.2.7 Project 7 Bluetooth Remote Control

1.Description

There is a DX-BT24 5.1 Bluetooth module in this kit. This bluetooth module comes with 256Kb space and complies with V5.1BLE bluetooth specification, which supports AT commands. Users can change parameters such as the baud rate and device name of the serial port as required.

Furthermore, it supports UART interface and bluetooth serial port transparent transmission, which also contains the advantages of low cost, small size, low power consumption and high sensitivity for sending and receiving. Notably, it solely needs a few peripheral components to realize its powerful functions.

2.Specification

- Bluetooth protocol: Bluetooth Specification V5.1 BLE
- Working distance: In an open environment, it can achieve 40m ultra-long distance communication
- Operating frequency: 2.4GHz ISM band
- Communication interface: UART
- Bluetooth certification: Accord with FCC CE ROHS REACH certification standard
- Serial port parameters: 9600, 8 data bits, 1 stop bit, invalid bit, no flow control
- Power: 5V DC
- Operating temperature: –10℃ to +65℃

3.Application

The DX-BT24 module also supports the BT5.1 BLE protocol, which can be directly connected to iOS devices with BLE Bluetooth function, and supports resident running of background programs. It is mainly used in the field of short-distance data wireless transmission. It enables to avoid cumbersome cable connections and can directly replace serial cables.

Successful application areas of BT24 modules:

※ Bluetooth wireless data transmission;

※ Mobile phone, computer peripheral equipment;

※ Handheld POS equipment;

※ Wireless data transmission of medical equipment;

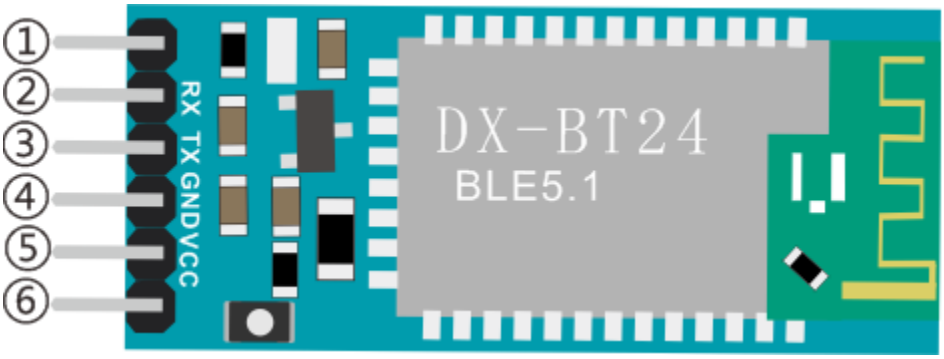
※ Smart home control;

※ Bluetooth printer;

※ Bluetooth remote control toys;

※ Shared bicycles;

4.Ports



STATEStatus pin

RXReceiving pin

TXsending pin

GNDGND

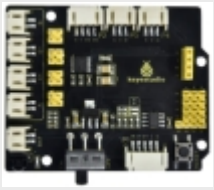
VCCPower

EN Enable pin

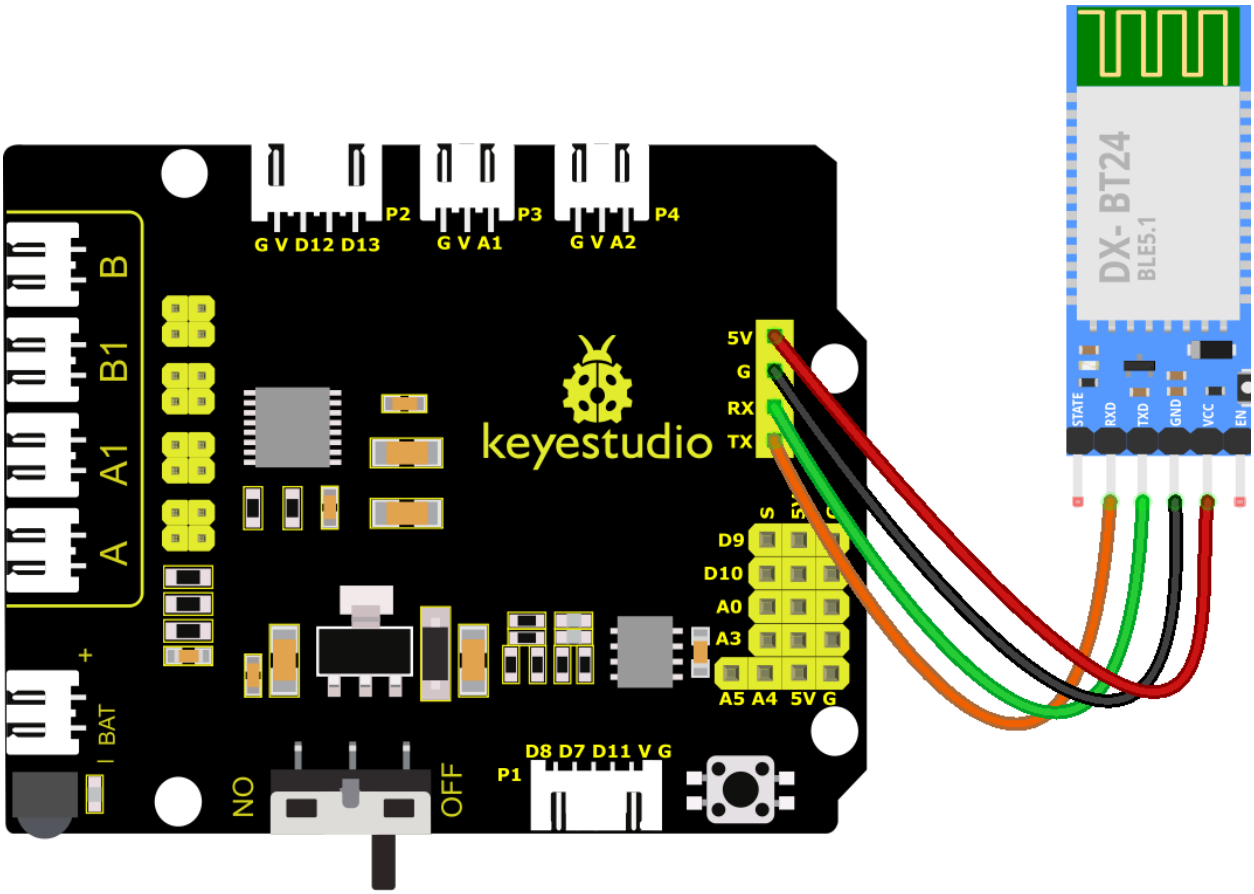
Connect the BT module to the development board.

Uno	BT24
TX	RX
RX	TX
VCC	5V
GND	GND

5.Components

Development Board *1	8833 Motor Driver *1	Red LED Module*1
		
3P Dupont Wire*1	USB Cable*1	DX-BT24 Bluetooth Module*1

6.Wiring Diagram



RXD, TXD, GND and VCC of the BT module are connected to TX, RX, G and 5V.

STATE and BRK of the BT module don't need connection.

Note:the direction of the BT module when inserting it onto the 8833 board. And don't insert it before uploading the code.

7.Test Code

Note: Before uploading the test code, you need to remove the Bluetooth module, otherwise the code will fail to be uploaded.Connect the Bluetooth module after uploading the code successfully.

```
//*****************************************************************************
/*
keyestudio 4wd BT Car
lesson 7.1
Bluetooth
http://www.keyestudio.com
*/
char ble_val; //character variable, used to store the value received by Bluetooth


void setup() {
  Serial.begin(9600);
```

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```
}
void loop() {
  if(Serial.available() > 0)  //make sure if there is data in serial buffer
  {
    ble_val = Serial.read();  //Read data from serial buffer
    Serial.println(ble_val);  //Print
  }
}
//**********************************************************************
```

8.Test Result

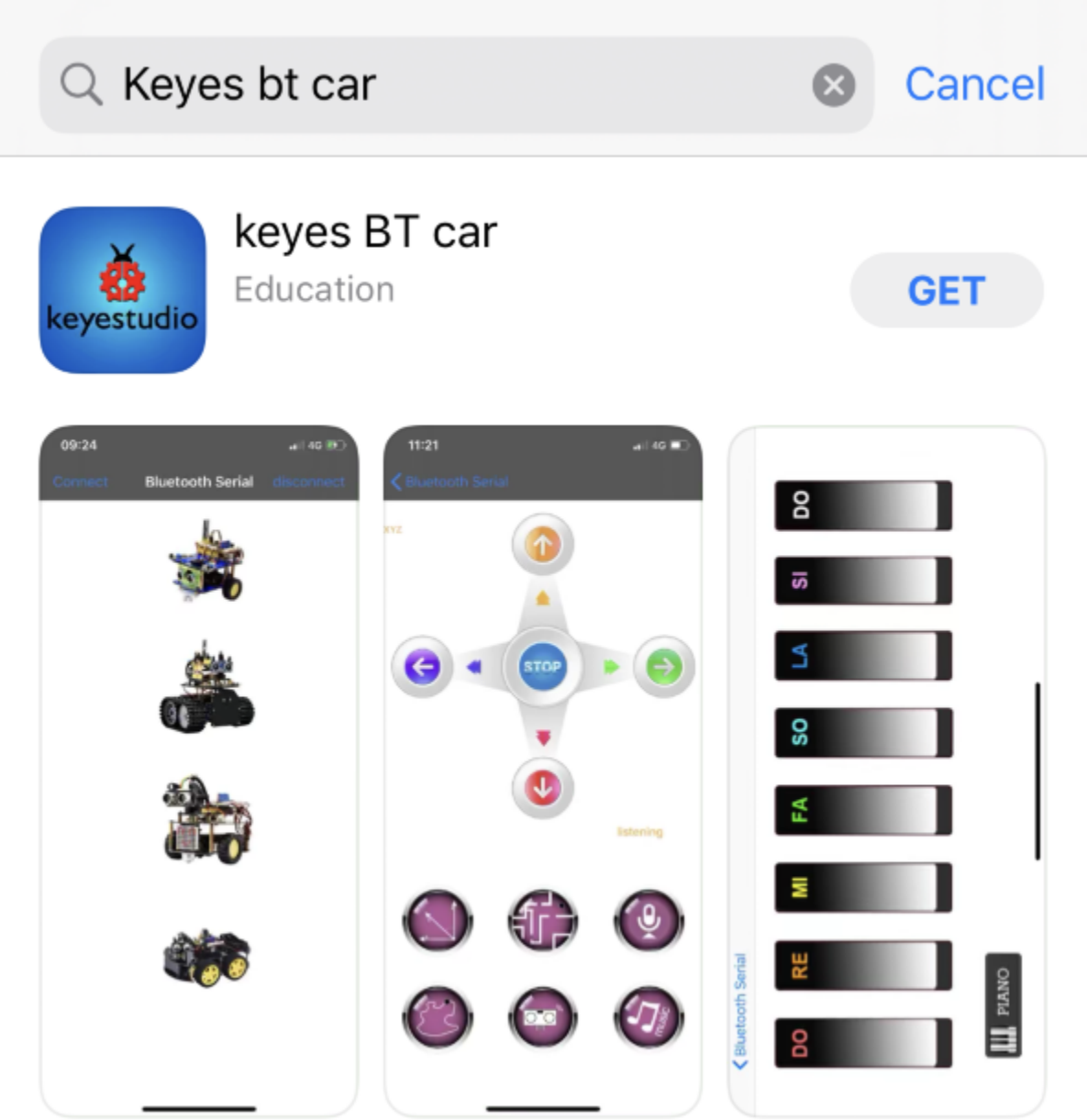
After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, then connect the computer via a USB cable to power the board. After powering on, insert the BT module and the LED will flash, then we need to download the BT app.

9.Download Bluetooth APP

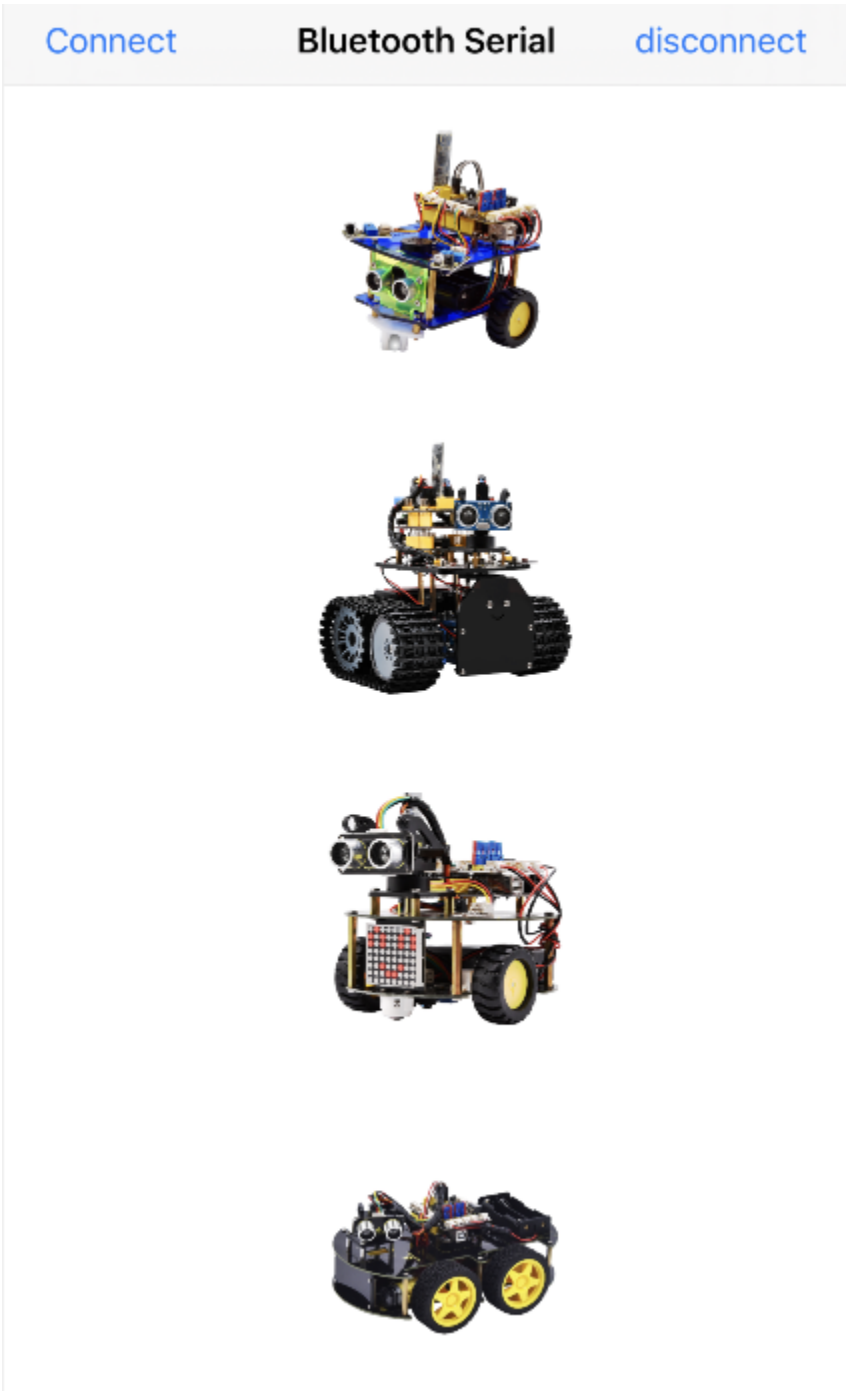
Apple system

(1). Open the App Store on the iPhone.

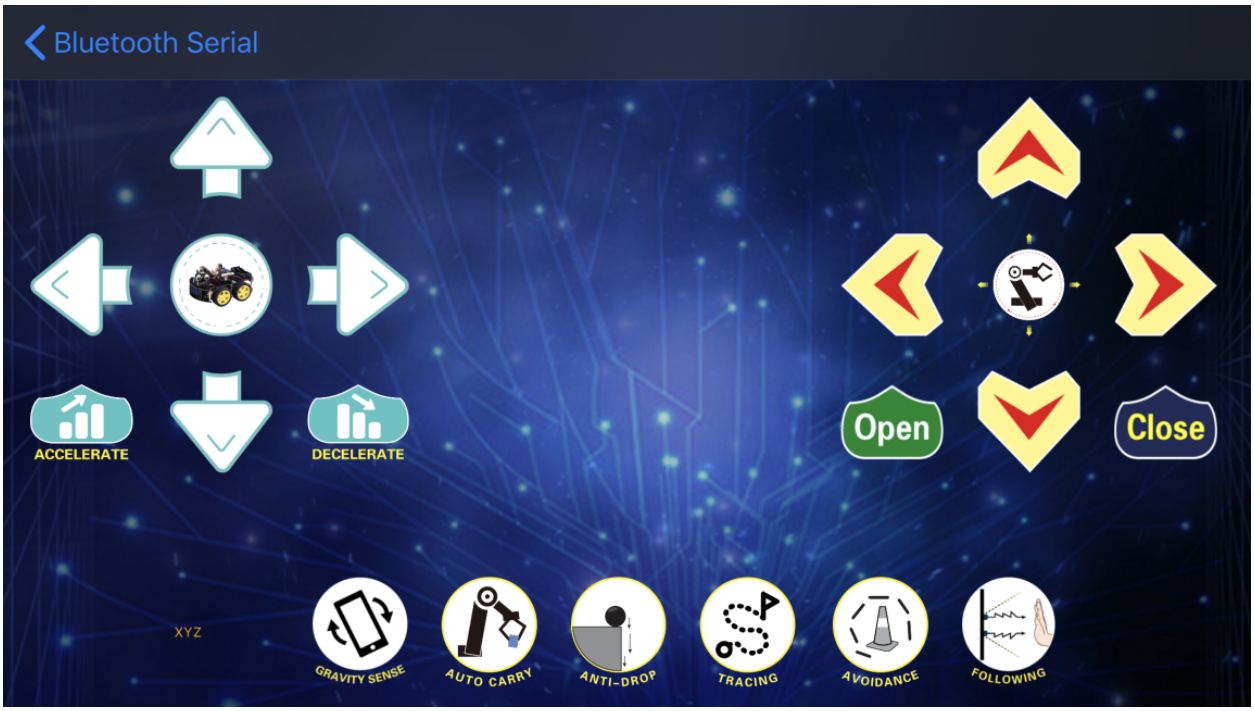
(2). Search keyes BT car and download the APP to your phone.



(3). After installation, enter its interface.

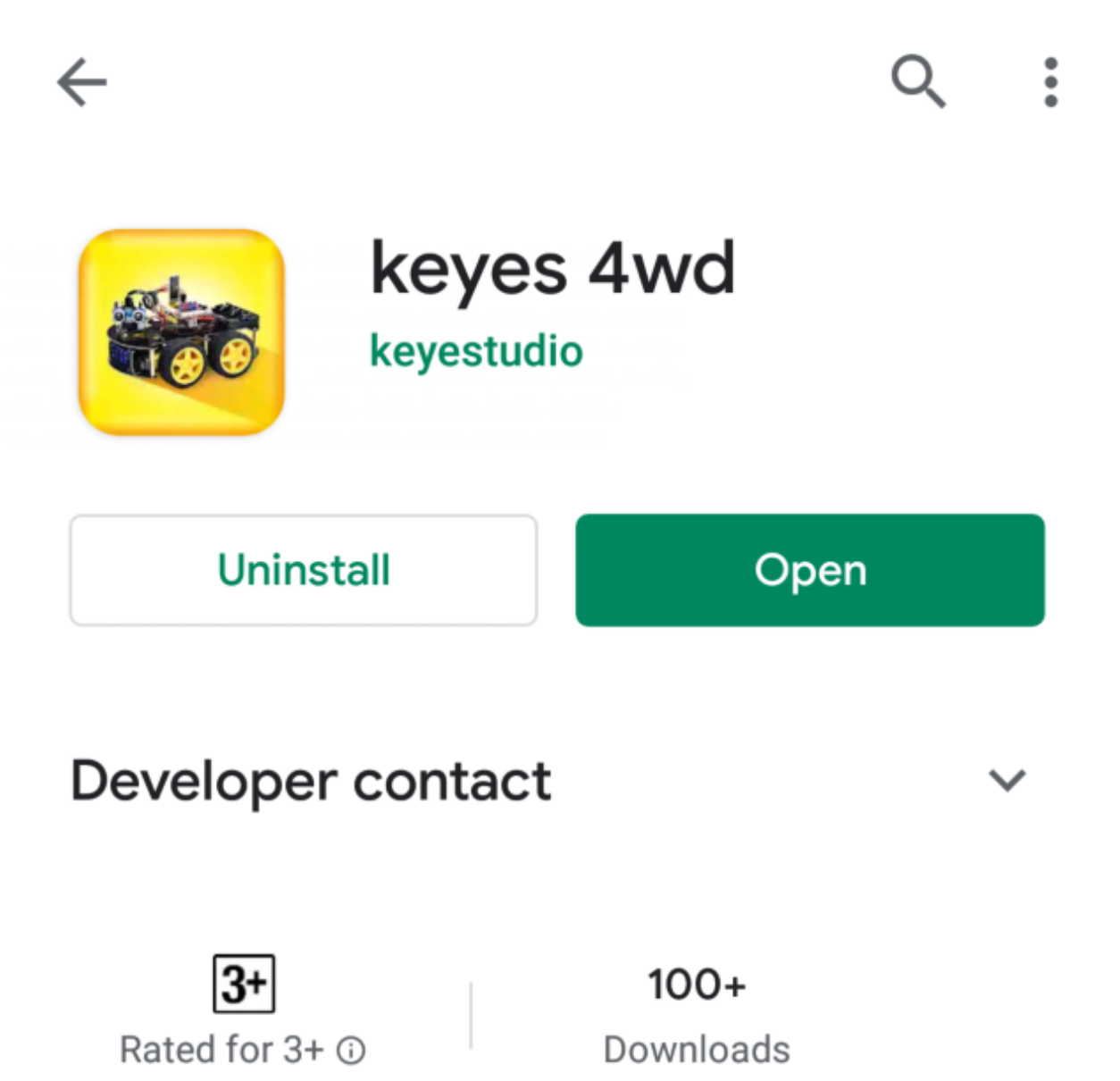


(4). Click "**Connect**" button in the upper left corner to automatically search for Bluetooth. When **BT24** is found, click "**Connect**" to connect Bluetooth, and then click to enter the control interface of 4WD smart car.



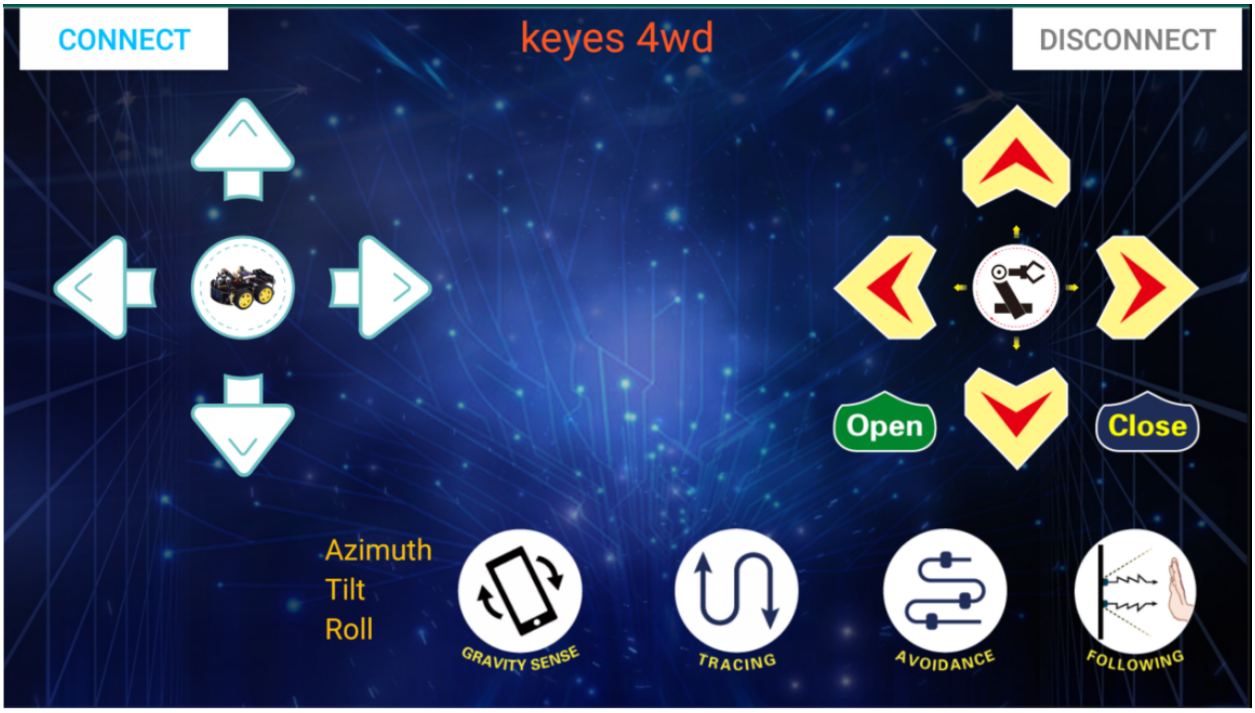
Android System

(1). Enter google play store to search for“keyes 4wd”.

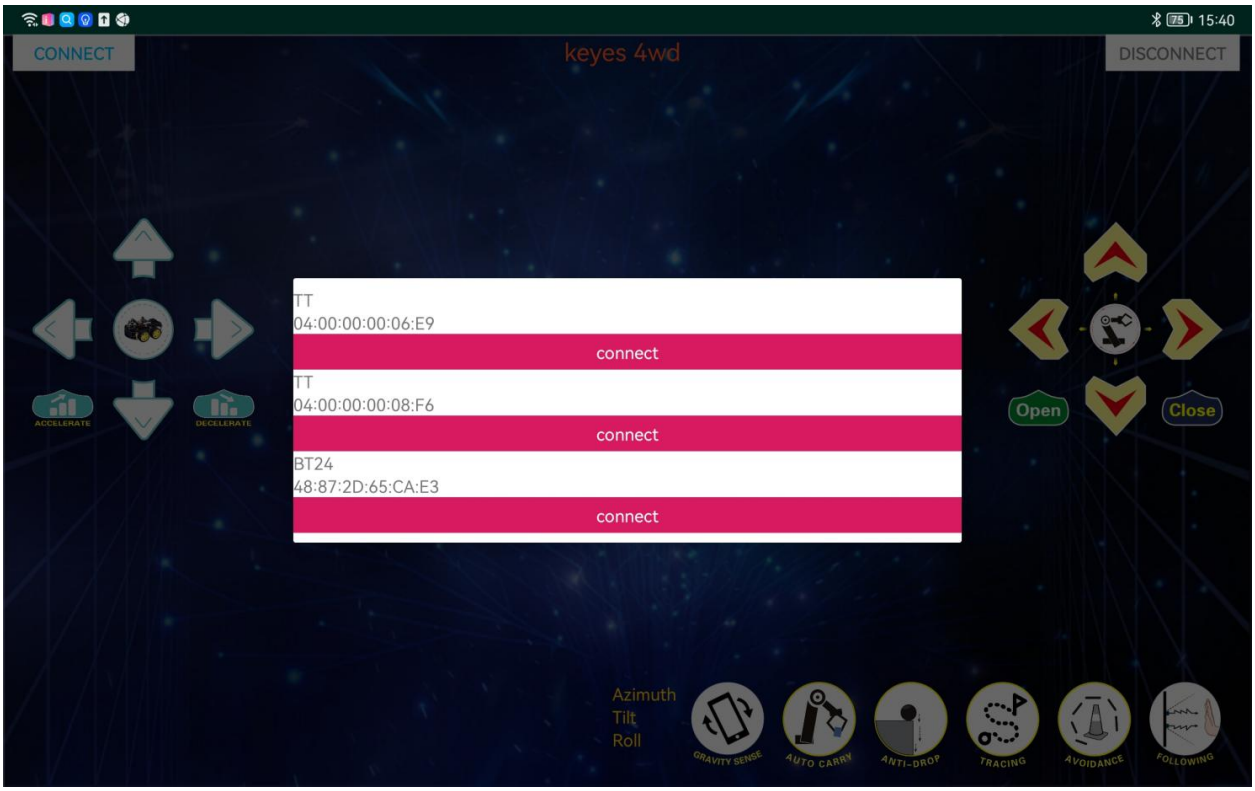


(2). The app icon is shown below after installation.

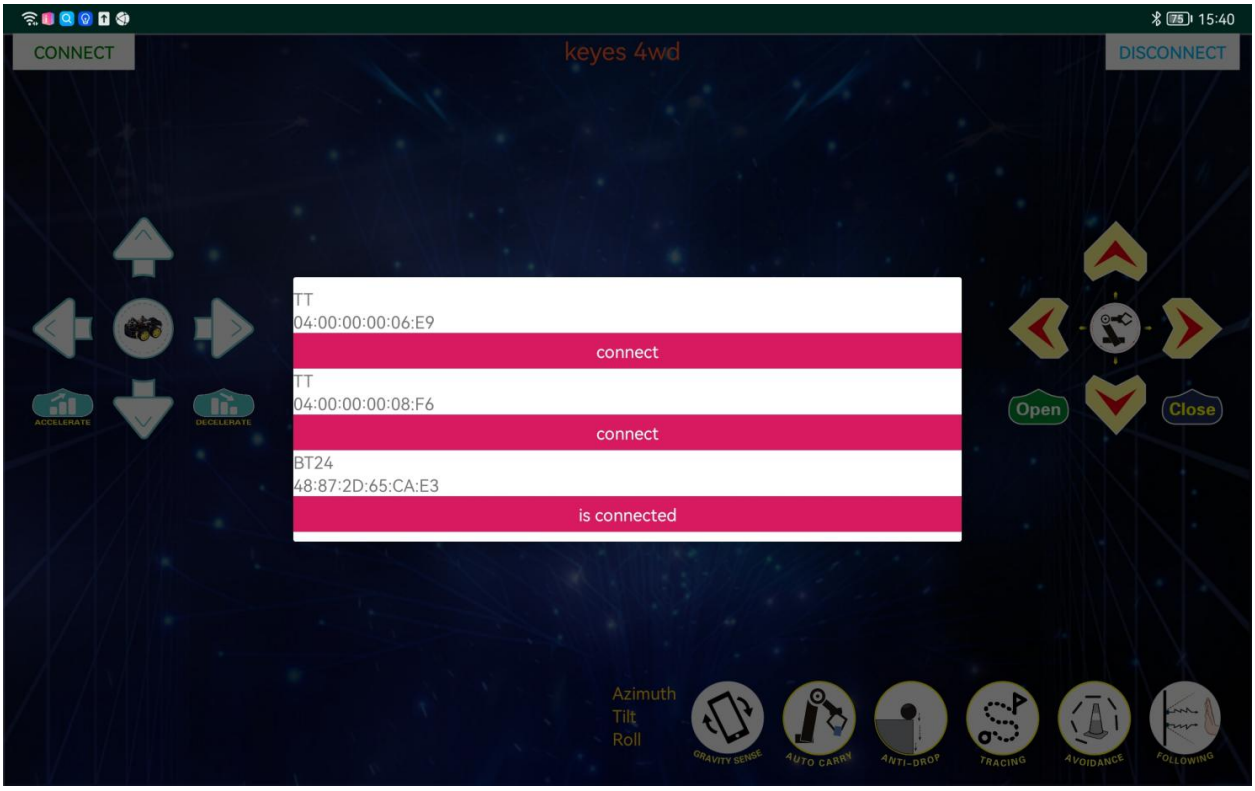
(3). Click app to enter the following page.



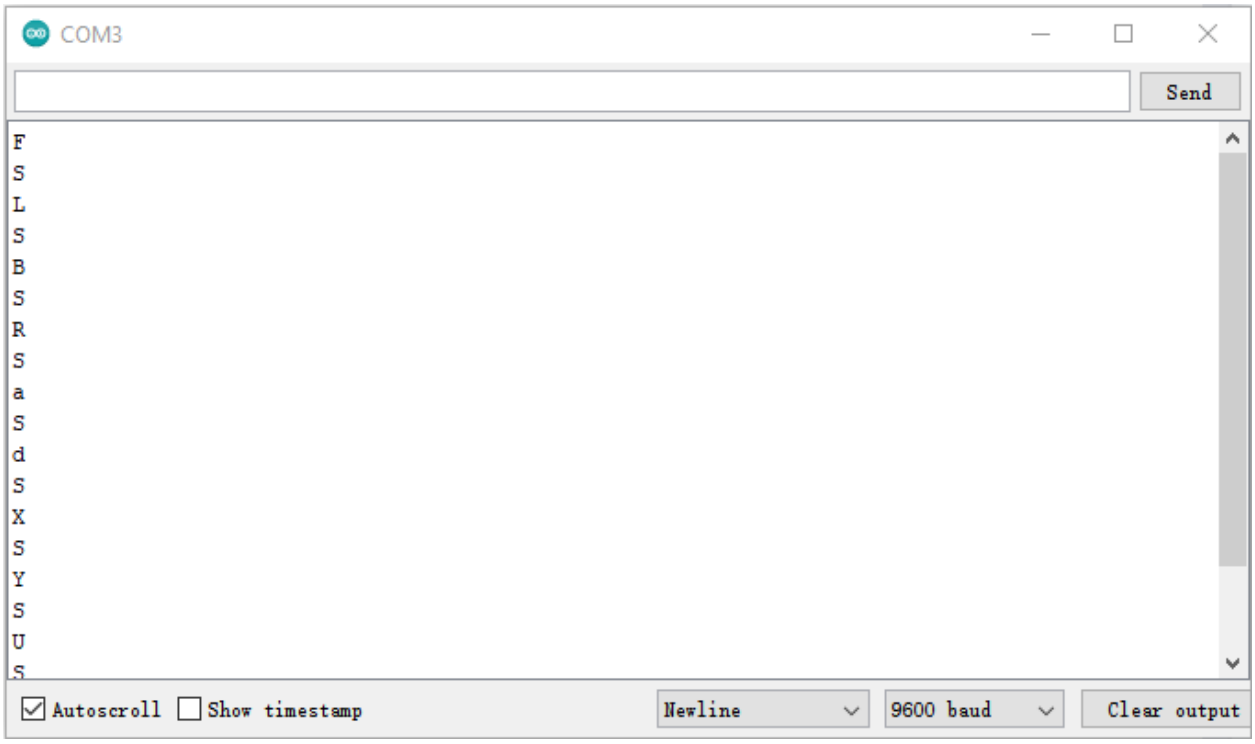
(4). After connecting Bluetooth, plug in power and LED indicator of Bluetooth module will flicker. Tap"**Connect**"to search the Bluetooth.

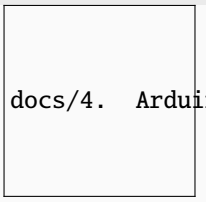


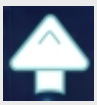
(5). When **BT24** is found, click "Connect" to connect Bluetooth. When "**Connect**" turns into "**is Connected**", it indicates that the Bluetooth connection is successful. As shown in the picture below, the Bluetooth LED becomes will stay on.



(6). After connecting Bluetooth module, open serial monitor to set baud rate to 9600. Pressing the button of the Bluetooth APP, and the corresponding characters will be displayed, as shown below:



Key	Function
	Pair DX-BT24 5.1 Bluetooth module
	Disconnect Bluetooth

	Control character	Control character
	Press: F Release: S	Press the button, the car goes front; release to stop
	Press: L Release: S	Press the button, the car turns left; release to stop
	Press: R Release: S	Press the button, the car turns right; release to stop
	Press: B Release: S	Press the button, the car goes back; release to stop
	Press: “a” Release: “S”	Click to speed up(maximum:255)
	Press: “d” Release: “S”	Click to slow down(minimum:0)
	Click to start the gravity sensing function of the mobile phone: click again to exit the gravity sensing control	
	Click to send“X”, click again to send“S”	Start line tracking function; click again to exit
	Click to send“Y”, click again to send“S”	Start ultrasonic avoiding function; click again to exit
	Click to send“U”, click again to send“S”	Start ultrasonic follow function; click again to exit
	Click to send“G”, click again to send“S”	Start restricting function; click again to exit

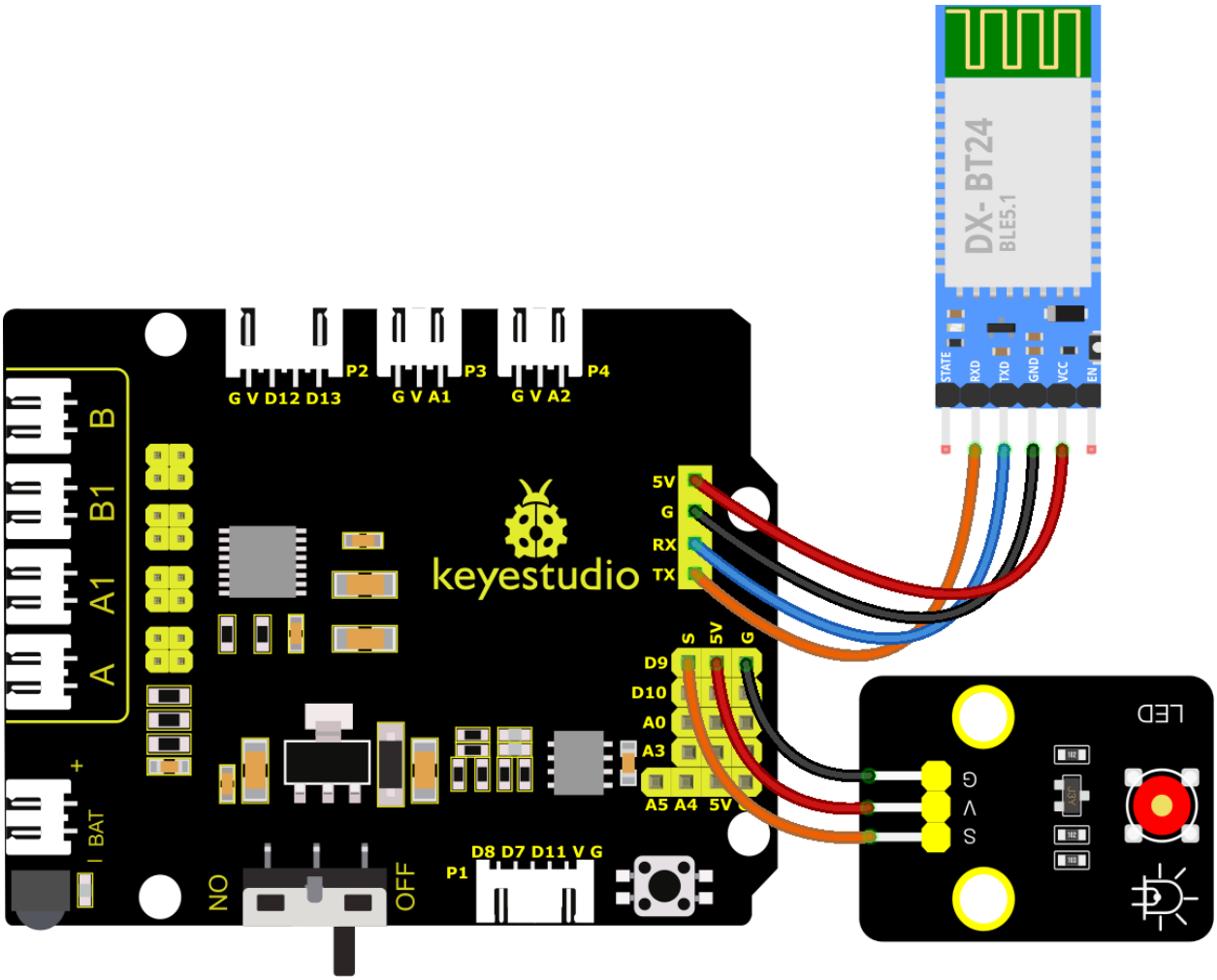
10.Code Explanation

Serial.available() : Return the number of characters currently remaining in the serial port buffer. Generally, this function is used to judge whether there is data in the buffer of the serial port. When Serial.available()>0, it means that the serial port has received data and can be read;

Serial.read() : Refers to taking out and reading a Byte of data from the serial port buffer. For example, if a device sends data to Arduino through the serial port, we can use Serial.read() to read the sent data.

11.Extension Practice

Here we look to use the command sent by the mobile phone to turn on or off an LED light. Looking at the wiring diagram, an LED is connected to the D9 pin.




```
//*****************************************************************************
/*
 keyestudio smart turtle robot
 lesson 7.2
 Bluetooth LED
 http://www.keyestudio.com
*/
int ledpin=9;
```

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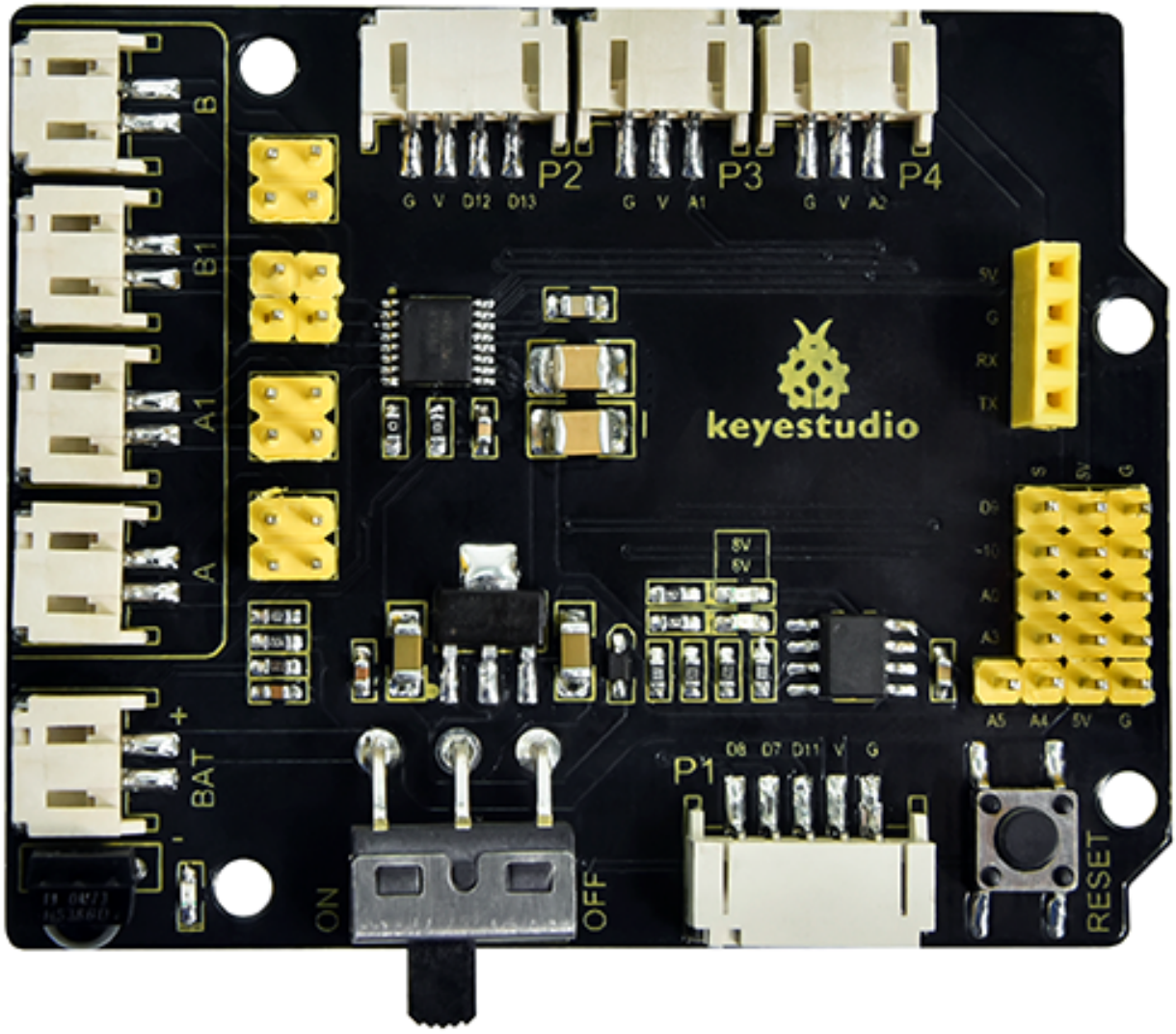
```
char ble_val;// An integer variable used to store the value received by Bluetooth

void setup()
{
  Serial.begin(9600);
  pinMode(ledpin,OUTPUT);
}

void loop()
{
  if (Serial.available() > 0) //Check whether there is data in the serial port cache
  {
    ble_val = Serial.read();  //Read data from the serial port cache
    Serial.print("DATA RECEIVED:");
    Serial.println(ble_val);
    if (ble_val == 'F') {
      digitalWrite(ledpin, HIGH);
      Serial.println("led on");
    }
    if (ble_val == 'B') {
      digitalWrite(ledpin, LOW);
      Serial.println("led off");
    }
   }
}
//*****************************************************************************
```

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, then connect the computer via a USB cable to power the board. After powering on, click and to control the LED turn on and turn off.

9.2.8 Project 8 Motor Driving and Speed Control



1.Description

There are many ways to drive motors. Our car uses the most commonly used DRV8833 motor driver chip, which provides a two-channel bridge electric drive solution for toys, printers and other integrated motor applications.

When we stack the Shield on the 4.0 development board and power on the BAT, then set the DIP switch to the ON end, the external power supply will power on the two boards at the same time. To facilitate wiring connections, the Shield comes with an anti-reverse port (PH2.0-2P-3P-4P-5P). You can connect the motors, power supply, and sensor modules directly to the Shield.

The Bluetooth interface of the Shield is fully compatible with the DX-BT24 5.1 Bluetooth module. When connecting the Bluetooth module, you solely need to plug it into the corresponding interface. At the same time, 2.54 row pins are used to draw out some unused digital and analog ports on the Shield, making it accessible for you to add other sensors and carry out extension experiments.

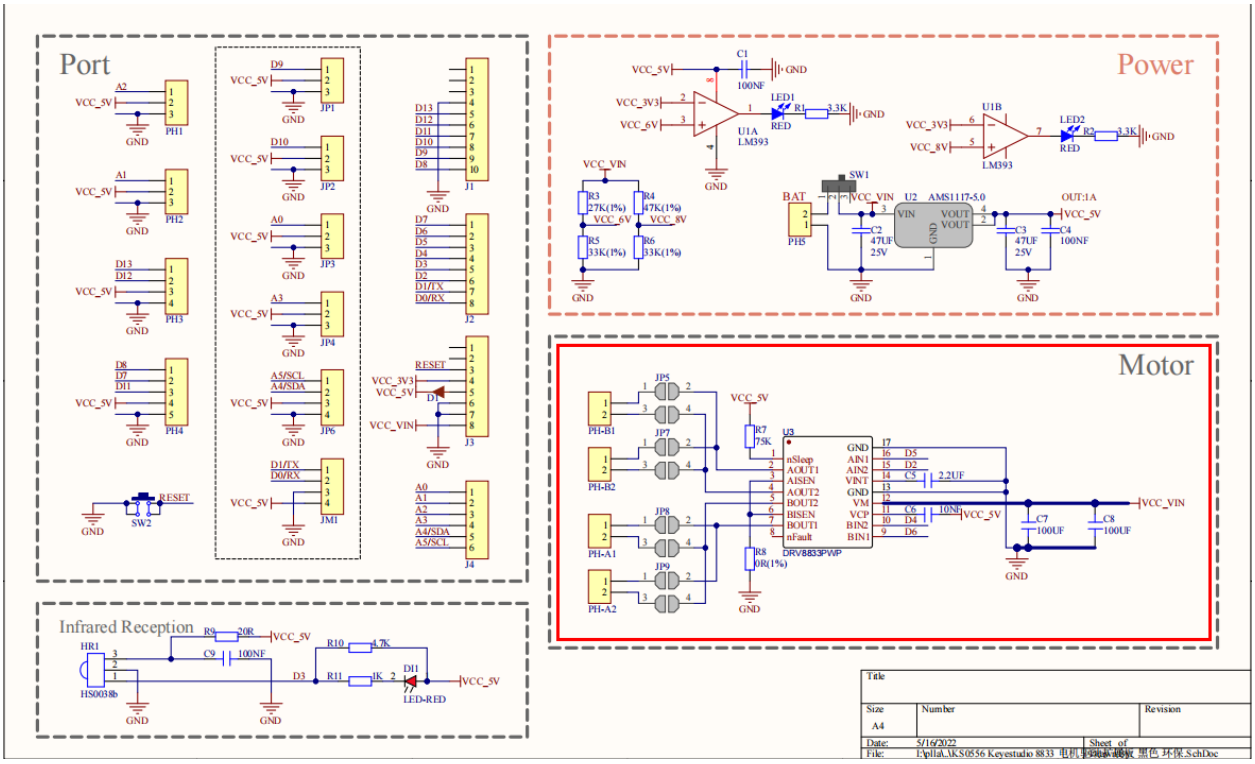
The expansion board can be connected to four DC motors. When the jumper cap is connected by default, the motors of ports A and A1 and B and B1 are connected in parallel and have the same motion law. 8 jumper caps can be used to control the rotation direction of the 4 motor interfaces.

For example, when the 2 jumper caps in front of B1 of the M1 motor change from transverse connection to longitudinal connection, the rotation direction of M1 motor will be opposite to the original rotation direction.

2.Specification

- Input voltage for logicDC 5V
- Input voltage for drivingDC 6-9 V
- Working current for logic<36mA
- Working current for driving<2A
- Maximum power dissipation25WT=75℃
- Input level for control signal high level is 2.3V<Vin<5V low level is -0.3V<Vin<1.5V
- Working temperature-25130℃

Keyestudio 8833 motor driver expansion board

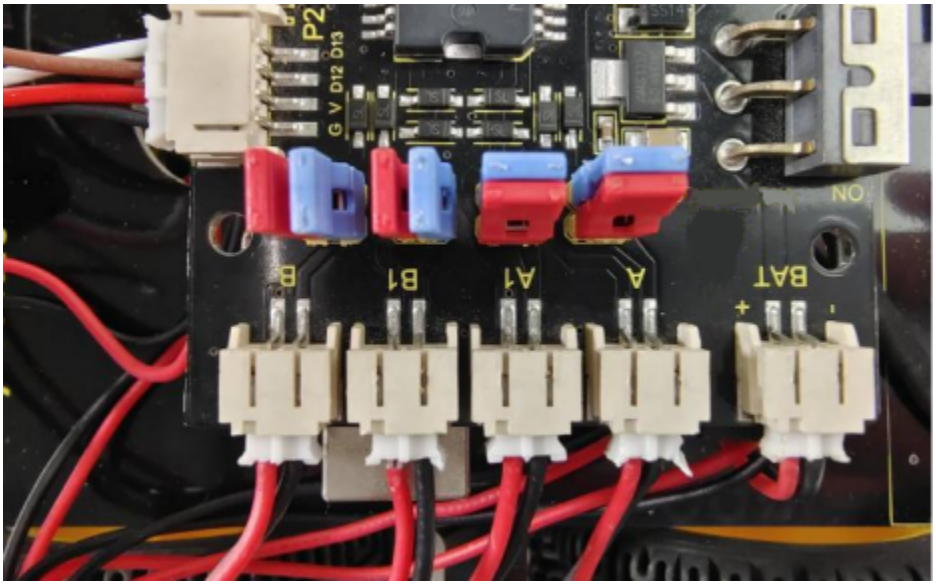


3.Working Principle

We use the same side parallel connection mode for the four motors, which can be regarded as two groups of motors. As shown in the wiring diagram, B and B1 are a group, and A and A1 are a group.

The motors in the same group should rotate in the same direction. If they are different, please adjust the corresponding jumper caps next to the terminal to change the direction.

As shown below, if the directions of A and A1 are different, adjust the direction of jumper caps until the motor movement direction of the same group is consistent.

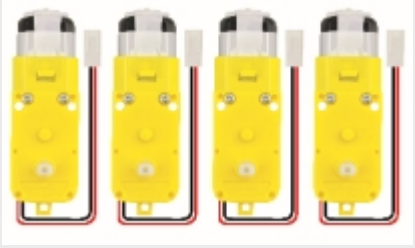


From the above diagram, it is known that the direction pin of A motor is D4, the speed pin is D6; D2 is the direction pin of B motor; and D6 is speed pin.

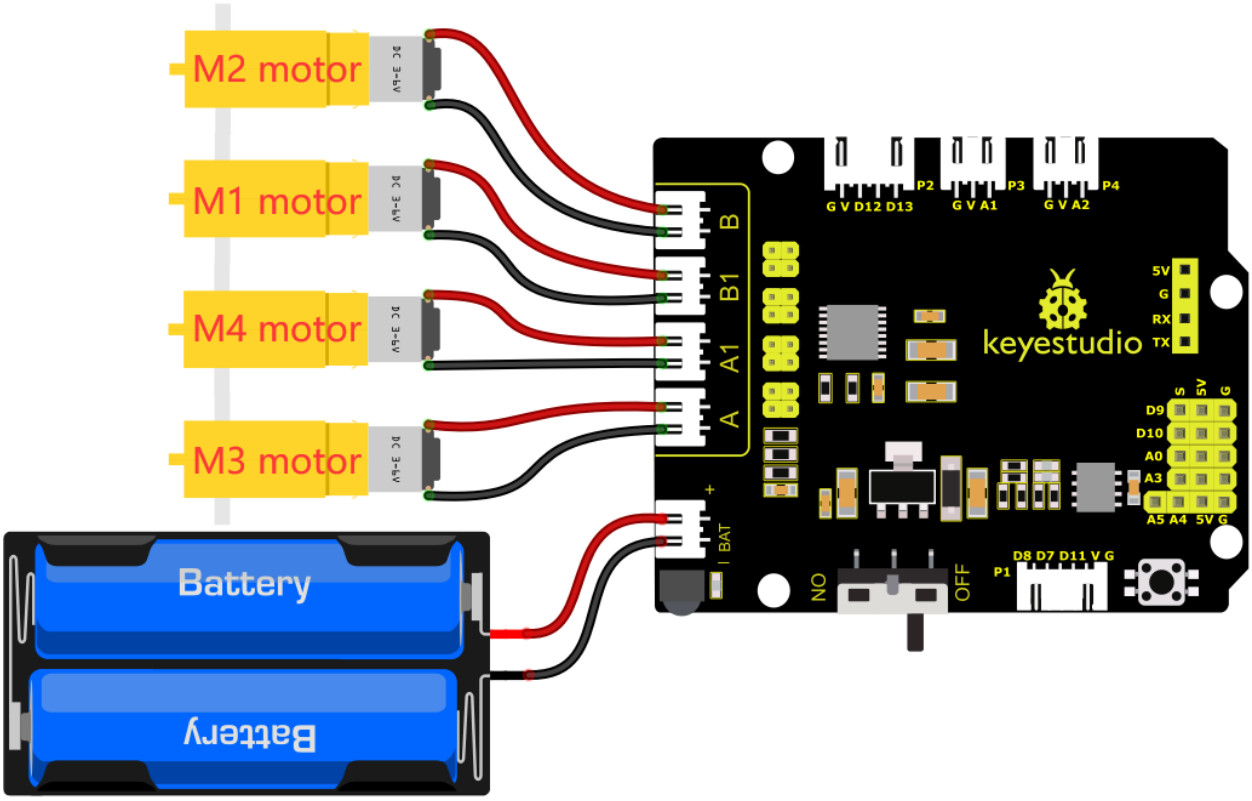
PWM drives the robot car. The PWM value is in the range of 0-255. When we set the direction to HIGH, the smaller the PWM number, the faster the rotation of the motor.

	D2	D5PWM	B Motor left	D4	D6PWM	A Motorright
Go forward	HIGH	255-200	Rotate clockwise	HIGH	255-200	Rotate clockwise
Go back	LOW	200	Rotate anticlockwise	LOW	200	Rotate anticlockwise
Turn left	HIGH	255-200	Rotate clockwise	LOW	200	Rotate anticlockwise
Turn right	LOW	200	Rotate anticlockwise	HIGH	255-200	Rotate clockwise

4.Components

Development Board *1	8833 Motor Driver *1	USB Cable*1
18650 Battery Holder*1	Motor*4	18650 Battery *2self-provided
		

5.Wiring Diagram



Connect the power supply to the BAT port.

6.Test Code

```
//*****************************************************************************
/*
 keyestudio 4wd BT Car
 lesson 8.1
 Motor driver shield
 http://www.keyestudio.com
*/
#define ML_Ctrl 2     //define the direction control pins of group B motor
#define ML_PWM 5   //define the PWM control pins of group B motor
#define MR_Ctrl 4    //define the direction control pins of group A motor
#define MR_PWM 6   //define the PWM control pins of group A motor

void setup()
{
  pinMode(ML_Ctrl, OUTPUT);//set direction control pins of group B motor to output
  pinMode(ML_PWM, OUTPUT);//set PWM control pins of group B motor to output
  pinMode(MR_Ctrl, OUTPUT);//set direction control pins of group A motor to output
  pinMode(MR_PWM, OUTPUT);//set PWM control pins of group A motor to output
}
void loop()
{
  //front
```

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```
  digitalWrite(ML_Ctrl,HIGH);//set the direction control pins of group B motor to HIGH
  analogWrite(ML_PWM,55);//set the PWM control speed of group B motor to 55
  digitalWrite(MR_Ctrl,HIGH);//set the direction control pins of group A motor to HIGH
  analogWrite(MR_PWM,55);// set the PWM control speed of group A motor to 55
  delay(2000);//delay in 2000ms
  //back
  digitalWrite(ML_Ctrl,LOW);//set the direction control pins of group B motor to LOW␣
↪level
  analogWrite(ML_PWM,200);// set the PWM control speed of group B motor to 200
  digitalWrite(MR_Ctrl,LOW);//set the direction control pins of group A motor to LOW␣
↪level
  analogWrite(MR_PWM,200);//set the PWM control speed of group A motor to 200
  delay(2000);//delay in 2000ms
  //left
  digitalWrite(ML_Ctrl,LOW);//set the direction control pins of group B motor to LOW␣
↪level
  analogWrite(ML_PWM,200);//set the PWM control speed of group B motor to 200
  digitalWrite(MR_Ctrl,HIGH);//set the direction control pins of group A motor to HIGH␣
↪level
  analogWrite(MR_PWM,55);//set the PWM control speed of group A motor to 200
  delay(2000);//delay in 2000ms
  //right
  digitalWrite(ML_Ctrl,HIGH);//set the direction control pins of group B motor to HIGH␣
↪level
  analogWrite(ML_PWM,55);//set the PWM control speed of group B motor to 55
  digitalWrite(MR_Ctrl,LOW);// set the direction control pins of group A motor to LOW␣
↪level
  analogWrite(MR_PWM,200);//set the PWM control speed of group A motor to 200
  delay(2000);//delay in 2000ms
  //stop
  digitalWrite(ML_Ctrl, LOW);// set the direction control pins of group B motor to LOW␣
↪level
  analogWrite(ML_PWM,0);//set the PWM control speed of group B motor to 0
  digitalWrite(MR_Ctrl, LOW);// set the direction control pins of group A motor to LOW␣
↪level
  analogWrite(MR_PWM,0);//set the PWM control speed of group A motor to 0
  delay(2000);// delay in 2000ms
}
//*****************************************************************************
```

7.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, then power on the external power and turn the DIP switch to ON, the car will go forward for 2s, back for 2s, turn left for 2s and right for 2s and stop for 2s

8.Code Explanation

digitalWrite(ML_Ctrl,LOW): The rotation direction of motor is decided by the high/low level and and the pins that decide rotation direction are digital pins.

analogWrite(ML_PWM,200): The speed of motor is regulated by PWM, and the pins that decide the speed of motor must be PWM pins.

9.Code Expansion

Adjust the speed that PWM controls the motor, hook up in the same way.

```
//*********************************************************************
/*
 keyestudio 4wd BT Car
 lesson 8.2
 Motor driver
 http://www.keyestudio.com
*/
#define ML_Ctrl 2     //define the direction control pins of group B motor
#define ML_PWM 5   //define the PWM control pins of group B motor
#define MR_Ctrl 4    //define the direction control pins of group A motor
#define MR_PWM 6   //define the PWM control pins of group A motor

void setup()
{
  pinMode(ML_Ctrl, OUTPUT);//set direction control pins of group B motor to output
  pinMode(ML_PWM, OUTPUT);//set PWM control pins of group B motor to output
  pinMode(MR_Ctrl, OUTPUT);//set direction control pins of group A motor to output
  pinMode(MR_PWM, OUTPUT);//set PWM control pins of group A motor to output
}
void loop()
{
  //front
  digitalWrite(ML_Ctrl,HIGH);//set the direction control pins of group B motor to HIGH
  analogWrite(ML_PWM,105);//set the PWM control speed of group B motor to 55
  digitalWrite(MR_Ctrl,HIGH);//set the direction control pins of group A motor to HIGH
  analogWrite(MR_PWM,105);// set the PWM control speed of group A motor to 55
  delay(2000);//delay in 2000ms
  //back
  digitalWrite(ML_Ctrl,LOW);//set the direction control pins of group B motor to LOW␣
↪level
  analogWrite(ML_PWM,150);// set the PWM control speed of group B motor to 200
  digitalWrite(MR_Ctrl,LOW);//set the direction control pins of group A motor to LOW␣
↪level
  analogWrite(MR_PWM,150);//set the PWM control speed of group A motor to 200
  delay(2000);//delay in 2000ms
  //left
  digitalWrite(ML_Ctrl,LOW);//set the direction control pins of group B motor to LOW␣
↪level
  analogWrite(ML_PWM,150);//set the PWM control speed of group B motor to 200
  digitalWrite(MR_Ctrl,HIGH);//set the direction control pins of group A motor to HIGH␣
↪level
  analogWrite(MR_PWM,105);//set the PWM control speed of group A motor to 200
  delay(2000);//delay in 2000ms
  //right
  digitalWrite(ML_Ctrl,HIGH);//set the direction control pins of group B motor to HIGH␣
↪level
  analogWrite(ML_PWM,105);//set the PWM control speed of group B motor to 55
  digitalWrite(MR_Ctrl,LOW);// set the direction control pins of group A motor to LOW␣
↪level
  analogWrite(MR_PWM,150);//set the PWM control speed of group A motor to 200
```

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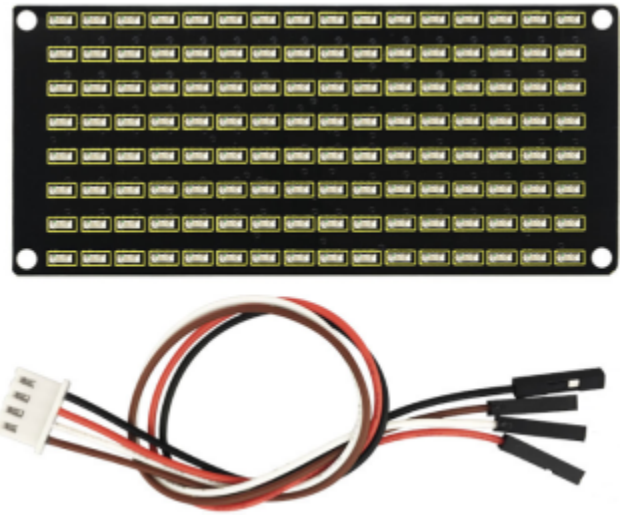
```
  delay(2000);//delay in 2000ms
  //stop
  digitalWrite(ML_Ctrl, LOW);// set the direction control pins of group B motor to LOW␣
↪level
  analogWrite(ML_PWM,0);//set the PWM control speed of group B motor to 0
  digitalWrite(MR_Ctrl, LOW);// set the direction control pins of group A motor to LOW␣
↪level
  analogWrite(MR_PWM,0);//set the PWM control speed of group A motor to 0
  delay(2000);// delay in 2000ms
}
//**********************************************************************
```

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, then power on the external power and turn the DIP switch to ON, then you find the speed of the motor is much slower.

Note:Low battery will lead to slow motor speed.

9.2.9 Project 9 Facial Expression LED Board



1.Description

How fun it is if a expression board is added to the robot. And the Keyestudio 8*16 LED board can do the trick. With the help of it, you could design facial expressions, images, patterns and other displays by yourselves.

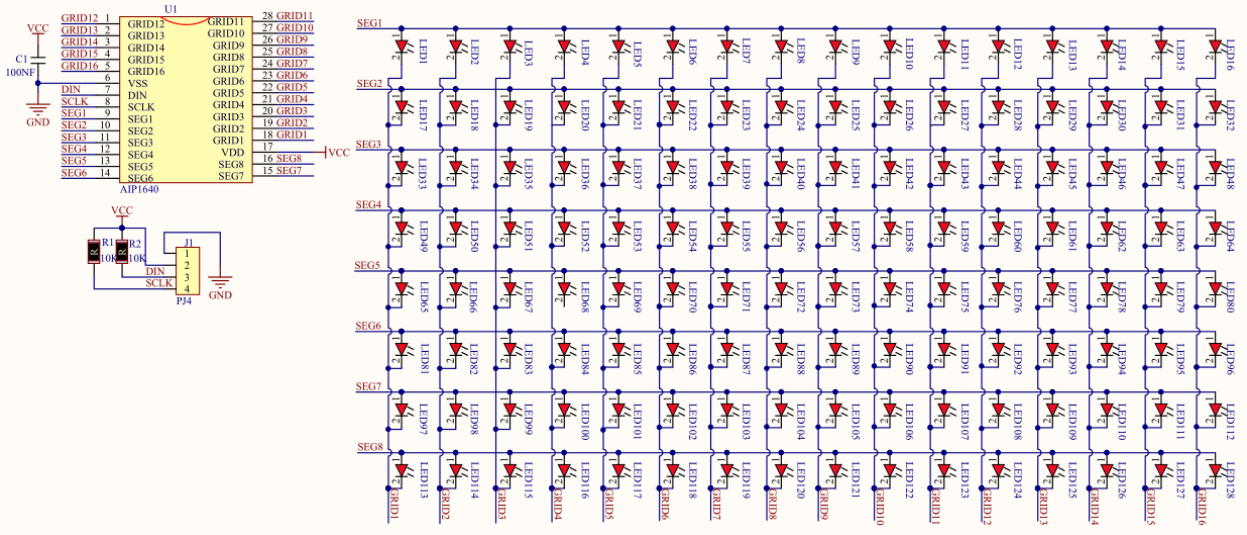
The 8*16 LED board comes with 128 LEDs. The data of the microprocessor(Arduino) communicates with the AiP1640 through a two-wire bus interface. Therefore, it can control the on and off of 128 LEDs on the module, so as to make the dot matrix on the module to display the pattern you need. A HX-2.54 4Pin cable is provided for your convenience of wiring.

2.Specification

- Working voltage: DC 3.3-5V
- Power loss: 400mW
- Oscillation frequency: 450KHz
- Drive current: 200mA

- Working temperature: -40~80℃
- Communication mode: I2C

3.Circuit Diagram



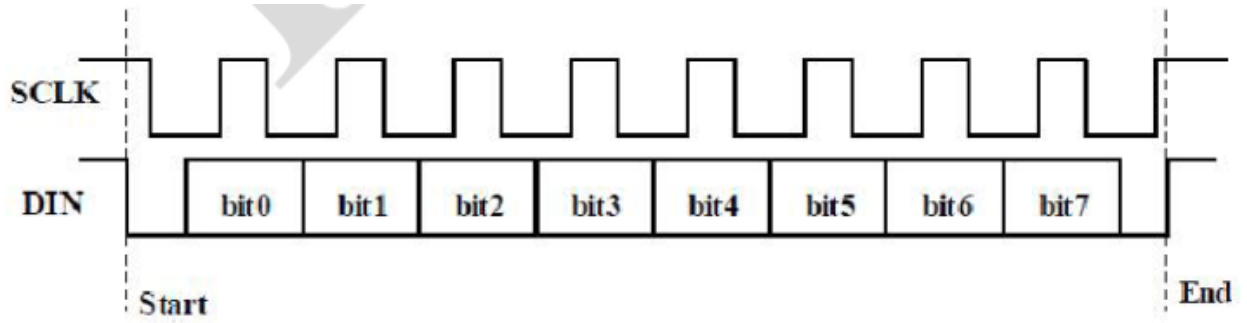
4.Working Principle

How to control each LED of the 8*16 dot matrix? It is known that each byte has 8 bits and each bit is 0 or 1. when it is 0, LED is off while when it is 1 LED is on. One byte can control one column of the LED,and naturally 16 bytes can control 16 columns of LEDs, that's the 8*16 dot matrix.

5.Pins description and communication protocol

The data of the microprocessor (Arduino) communicates with the AiP1640 through a two-wire bus cable.

The communication protocol diagram is as follows (SCLK) is SCL, (DIN) is SDA.



The starting condition for data input: SCL is high level and SDA changes from high to low.

For data command setting, there are methods as shown in the figure below.

In our sample program, select the way to **add 1 to the address automatically**, the binary value is 0100 0000 and the corresponding hexadecimal value is 0x40.

B7	B6	B5	B4	B3	B2	B1	B0	Description
0	1	Irrelevant choice, fill in 0			0	Irrelevant choice, fill in 0		add 1 to the address automatically
0	1				1			Fixed address
0	1			0				Universal mode
0	1			1				

For address command setting, the address can be selected as shown below.

The first 00H is selected in our sample program, and the binary number 1100 0000 corresponds to the hexadecimal 0xc0.

B7	B6	B5	B4	B3	B2	B1	B0	Display address
1	1	Irrelevant choice, fill in 0		0	0	0	0	00H
1	1			0	0	0	1	01H
1	1			0	0	1	0	02H
1	1			0	0	1	1	03H
1	1			0	1	0	0	04H
1	1			0	1	0	1	05H
1	1			0	1	1	0	06H
1	1			0	1	1	1	07H
1	1			1	0	0	0	08H
1	1			1	0	0	1	09H
1	1			1	0	1	0	0AH
1	1			1	0	1	1	0BH
1	1			1	1	0	0	0CH
1	1			1	1	0	1	0DH
1	1			1	1	1	0	0EH
1	1			1	1	1	1	0FH

The requirement for data input is that when SCL is at high level when inputting data, the signal on SDA must remain unchanged. Only when the clock signal on SCL is at low level, can the signal on SDA be changed. The input of data is the low bit first, and the high bit later.

The condition for the end of data transmission is that when SCL is at low level, SDA at low level and SCL at high level, the level of SDA becomes high.

Display control, set different pulse width, pulse width can be selected as shown in the figure below.

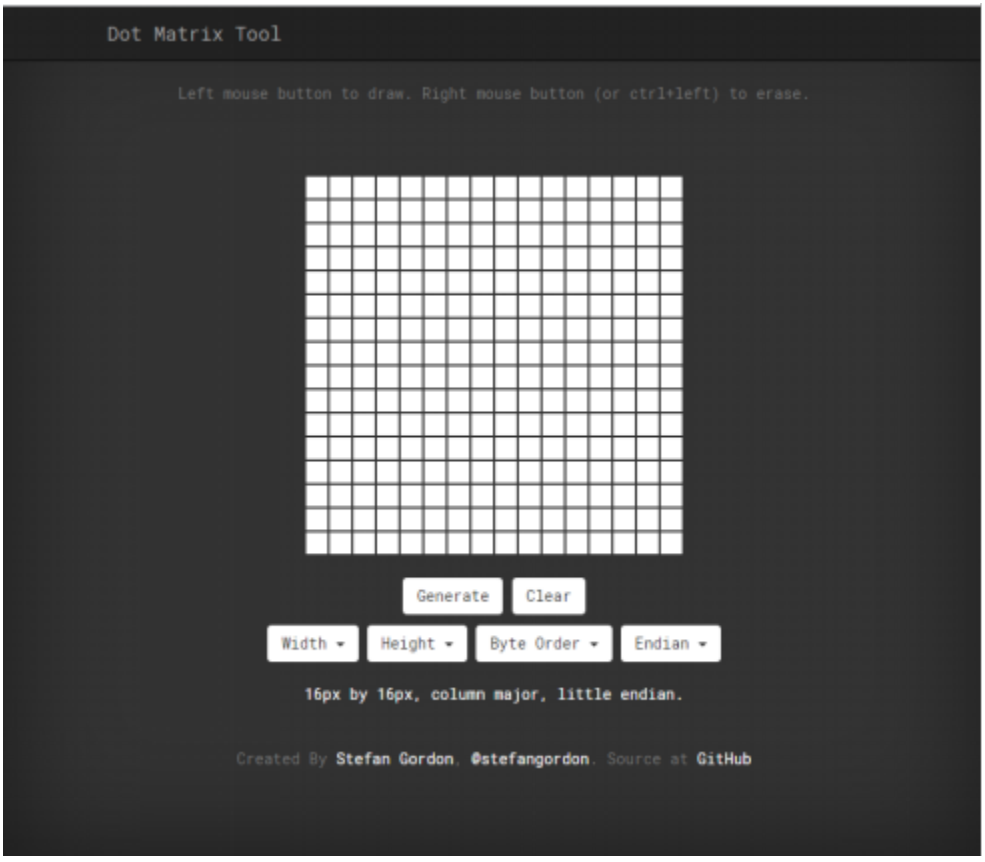
In the example, the pulse width is 4/16, and the hexadecimal corresponding to 1000 1010 is 0x8A.

B7	B6	B5	B4	B3	B2	B1	B0	Function	Description
1	0	Irrelevant choice, fill in 0		1	0	0	0	Clear quantity setting (Brightness setting)	Set pulse width to 1/16
1	0			1	0	0	1		Set pulse width to 2/16
1	0			1	0	1	0		Set pulse width to 4/16
1	0			1	0	1	1		Set pulse width to 10/16
1	0			1	1	0	0		Set pulse width to 11/16
1	0			1	1	0	1		Set pulse width to 12/16
1	0			1	1	1	0		Set pulse width to 13/16
1	0			1	1	1	1		Set pulse width to 14/16
1	0			0	X	X	X	Display switch setting	On
1	0			1	X	X	X		off

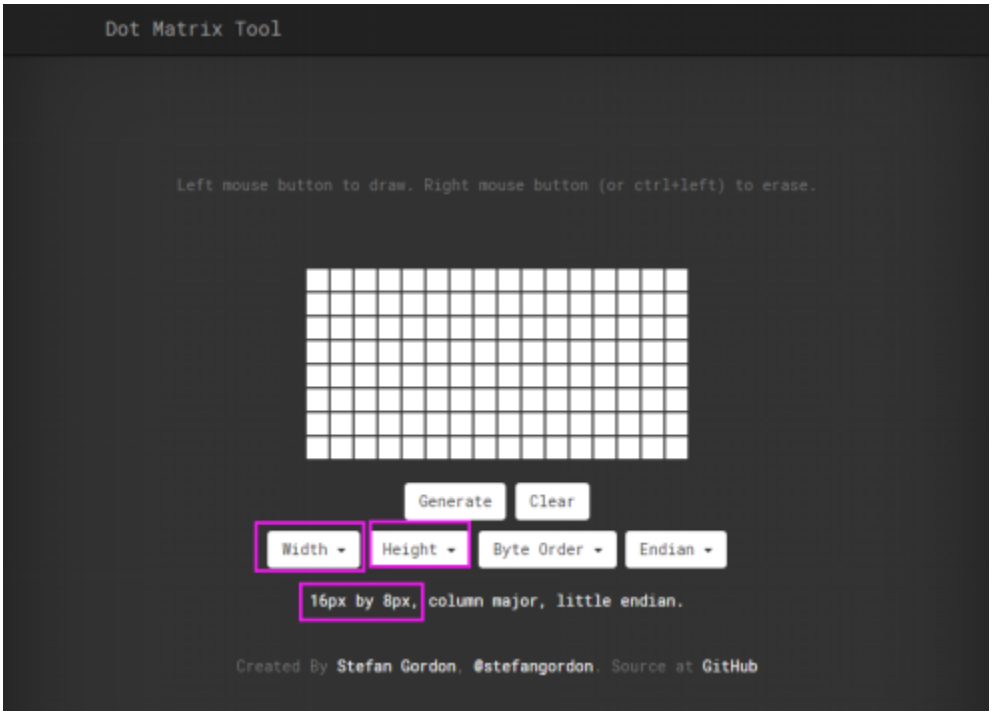
Instructions for the use of modulus tool

The dot matrix tool uses the online version, and the link is :http://dotmatrixtool.com/#

Enter the link and the page appears as shown below

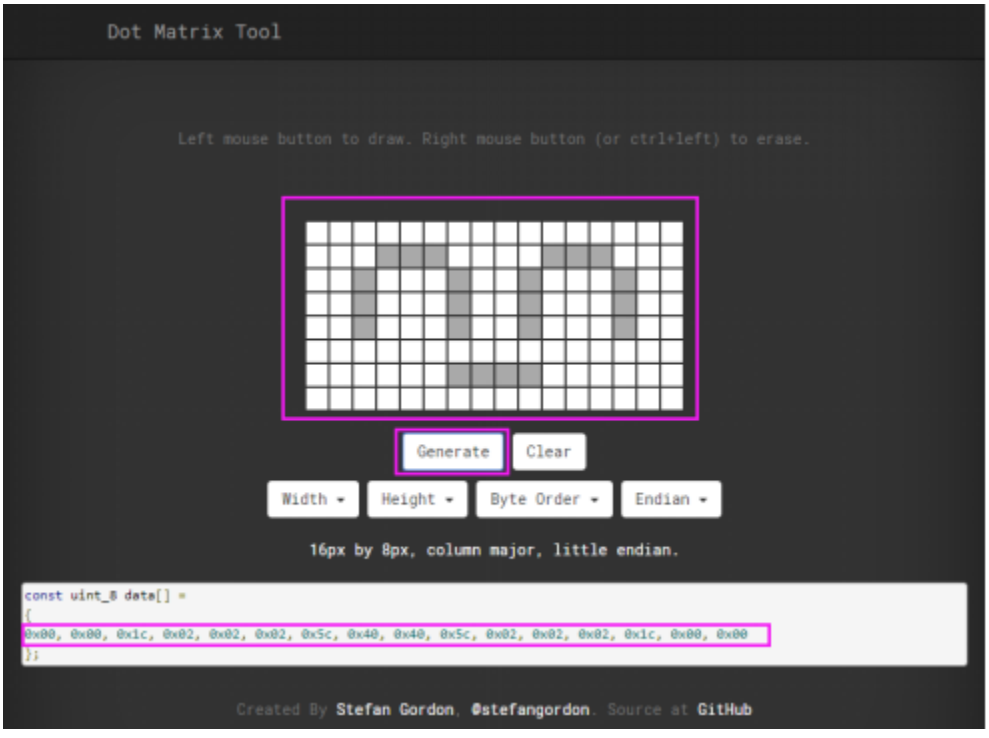


The dot matrix is 8*16, so adjust the height to 8 and width to 16, as shown in the figure below.



Generate hexadecimal data from the pattern

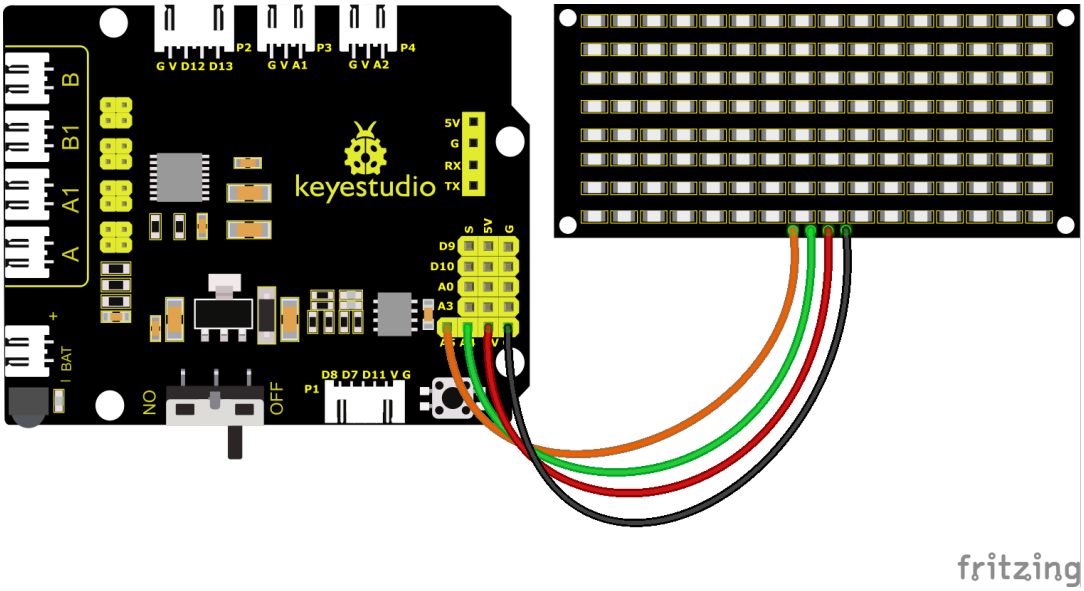
As shown in the figure below, press the left mouse button to select, right click to cancel; draw the pattern you want, click Generate, and the hexadecimal data we need will be generated.



6.Components

Development Board *1	8833 Motor Driver *1	USB Cable*1
USB Cable*1	HX-2.54 4P Dupont Wire 200mm *1	

7.Wiring Diagram



The GND, VCC, SDA, and SCL of the 8x16 LED light board are respectively connected to the keyestudio sensor expansion board-(GND), + (VCC), A4, A5 for two-wire serial communication.

(Note: Though it is connected with the IIC pin of Arduino, this module is not for IIC communication. And the IO port here is to simulate I2C communication and can be connected with any two pins).

8.Test Code

The code will show the smile face.

```
//**************************************************************************
/*
 keyestudio 4wd BT Car
  lesson 9.1
```

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```
  Matrix face
  http://www.keyestudio.com
*/
//Data from the smile pattern obtained from the touch tool
unsigned char smile[] = {0x00, 0x00, 0x1c, 0x02, 0x02, 0x02, 0x5c, 0x40, 0x40, 0x5c,␣
↪0x02, 0x02, 0x02, 0x1c, 0x00, 0x00};
#define SCL_Pin  A5  //Set the clock pin to A5
#define SDA_Pin  A4  //Set data pin to A4
void setup() {
  //Set pin to output
  pinMode(SCL_Pin, OUTPUT);
  pinMode(SDA_Pin, OUTPUT);
  //clear
  //matrix_display(clear);
}
void loop() {
  matrix_display(smile);  //display smiling expression pattern
}
//this function is used for dot matrix display
void matrix_display(unsigned char matrix_value[])
{
  IIC_start();  //the function that calls the data transfer start condition
  IIC_send(0xc0);  //select address

  for (int i = 0; i < 16; i++) //the pattern data is 16 bytes
  {
    IIC_send(matrix_value[i]); //Transmit the data of the pattern
  }
  IIC_end();   //End pattern data transmission
  IIC_start();
  IIC_send(0x8A);  //Display control, select 4/16 pulse width
  IIC_end();
}
//Conditions under which data transmission begins
void IIC_start()
{
  digitalWrite(SDA_Pin, HIGH);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, LOW);
}
//Indicates the end of data transmission
void IIC_end()
{
  digitalWrite(SCL_Pin, LOW);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, HIGH);
```

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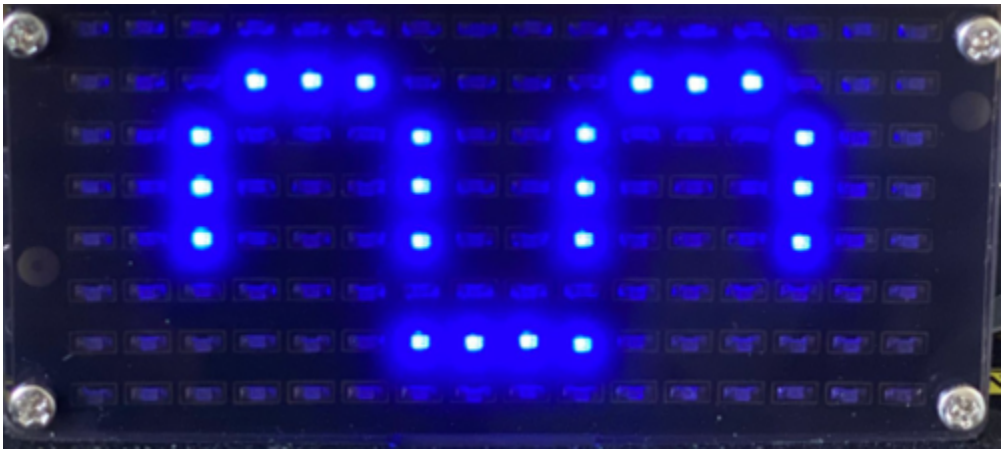
```
  delayMicroseconds(3);
}
//transmit data
void IIC_send(unsigned char send_data)
{
  for (byte mask = 0x01; mask != 0; mask <<= 1) //Each byte has 8 bits and is checked␣
↪bit by bit starting at the lowest level
  {
    if (send_data & mask) { //Sets the high and low levels of SDA_Pin depending on␣
↪whether each bit of the byte is a 1 or a 0
      digitalWrite(SDA_Pin, HIGH);
    } else {
      digitalWrite(SDA_Pin, LOW);
    }
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, HIGH); //Pull the clock pin SCL_Pin high to stop data␣
↪transmission
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, LOW); //pull the clock pin SCL_Pin low to change the SIGNAL of␣
↪SDA
  }
}
//*****************************************************************************
```

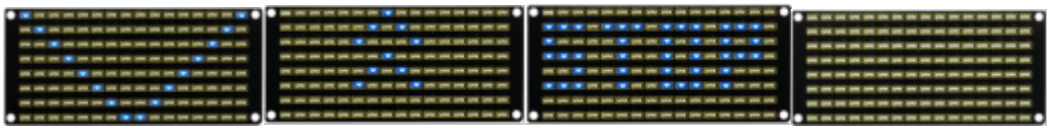
9.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, then turn the DIP switch to ON, a smile-shaped pattern will be displayed on the LED board.



10.Code Explanation

We use the modulus tool we just learned, http://dotmatrixtool.com/#, to make the dot matrix display the start pattern, going forward, and stop and then clear the pattern. The time interval is 2000 ms.



Code obtained from the module tool

Code for the pattern start:

0x01,0x02,0x04,0x08,0x10,0x20,0x40,0x80,0x80,0x40,0x20,0x10,0x08,0x04,0x02,0x01

Code for the pattern going forward:

0x00,0x00,0x00,0x00,0x00,0x24,0x12,0x09,0x12,0x24,0x00,0x00,0x00,0x00,0x00,0x00

Code for the pattern stepping back:

0x00,0x00,0x00,0x00,0x00,0x24,0x48,0x90,0x48,0x24,0x00,0x00,0x00,0x00,0x00,0x00

Code for the pattern turning left

0x00,0x00,0x00,0x00,0x00,0x00,0x44,0x28,0x10,0x44,0x28,0x10,0x44,0x28,0x10,0x00

Code for the pattern turning right

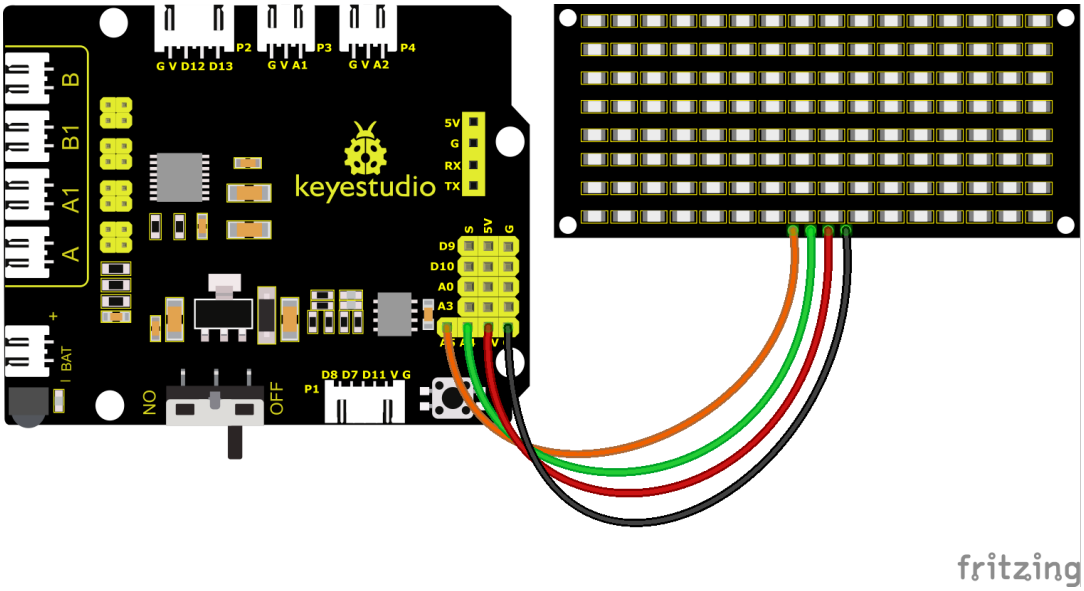
0x00,0x10,0x28,0x44,0x10,0x28,0x44,0x10,0x28,0x44,0x00,0x00,0x00,0x00,0x00,0x00

Code for the pattern stop

0x2E,0x2A,0x3A,0x00,0x02,0x3E,0x02,0x00,0x3E,0x22,0x3E,0x00,0x3E,0x0A,0x0E,0x00

Code to clear screen

0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00




```
//*****************************************************************************
/*
 keyestudio 4wd BT Car
  lesson 9.2
  Matrix face
  http://www.keyestudio.com
*/
//Data from the smile pattern obtained from the touch tool
unsigned char start01[] = {0x01, 0x02, 0x04, 0x08, 0x10, 0x20, 0x40, 0x80, 0x80, 0x40,␣
↪0x20, 0x10, 0x08, 0x04, 0x02, 0x01};
unsigned char front[] = {0x00, 0x00, 0x00, 0x00, 0x00, 0x24, 0x12, 0x09, 0x12, 0x24,␣
```

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```
↪0x00, 0x00, 0x00, 0x00, 0x00, 0x00};
unsigned char back[] = {0x00, 0x00, 0x00, 0x00, 0x00, 0x24, 0x48, 0x90, 0x48, 0x24, 0x00,
↪ 0x00, 0x00, 0x00, 0x00, 0x00};
unsigned char left[] = {0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x44, 0x28, 0x10, 0x44, 0x28,
↪ 0x10, 0x44, 0x28, 0x10, 0x00};
unsigned char right[] = {0x00, 0x10, 0x28, 0x44, 0x10, 0x28, 0x44, 0x10, 0x28, 0x44,␣
↪0x00, 0x00, 0x00, 0x00, 0x00, 0x00};
unsigned char STOP01[] = {0x2E, 0x2A, 0x3A, 0x00, 0x02, 0x3E, 0x02, 0x00, 0x3E, 0x22,␣
↪0x3E, 0x00, 0x3E, 0x0A, 0x0E, 0x00};
unsigned char clear[] = {0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,␣
↪0x00, 0x00, 0x00, 0x00, 0x00, 0x00};
#define SCL_Pin  A5  //Set the clock pin to A5
#define SDA_Pin  A4  //Set data pin to A4
void setup() {
  //Set pin to output
  pinMode(SCL_Pin, OUTPUT);
  pinMode(SDA_Pin, OUTPUT);
  //clear
  //matrix_display(clear);
}
void loop() {
    matrix_display(start01);  //Show start pattern
    delay(2000);
    matrix_display(front);    //Show forward pattern
    delay(2000);
    matrix_display(STOP01);   //Show stop pattern
    delay(2000);
    matrix_display(clear);    //Clear Screen
    delay(2000);
}
//this function is used for dot matrix display
void matrix_display(unsigned char matrix_value[])
{
  IIC_start();  //the function that calls the data transfer start condition
  IIC_send(0xc0);  //select address

  for (int i = 0; i < 16; i++) //the pattern data is 16 bytes
  {
    IIC_send(matrix_value[i]); //Transmit the data of the pattern
  }
  IIC_end();   //End pattern data transmission
  IIC_start();
  IIC_send(0x8A);  //Display control, select 4/16 pulse width
  IIC_end();
}
//Conditions under which data transmission begins
void IIC_start()
{
  digitalWrite(SDA_Pin, HIGH);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, LOW);
```

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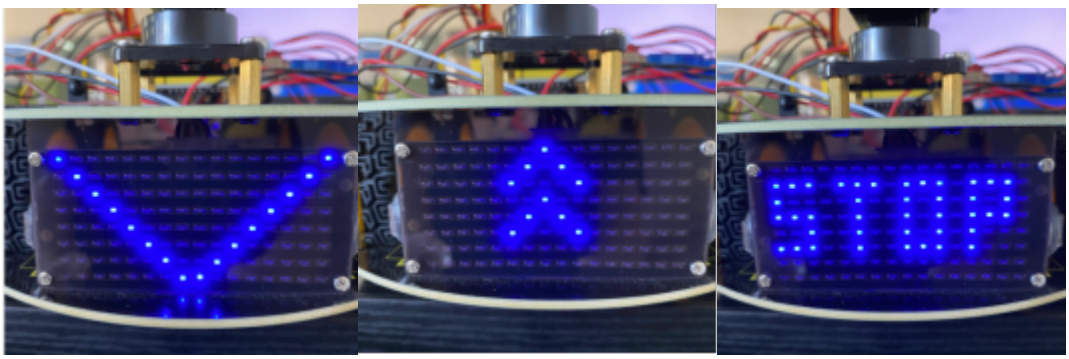
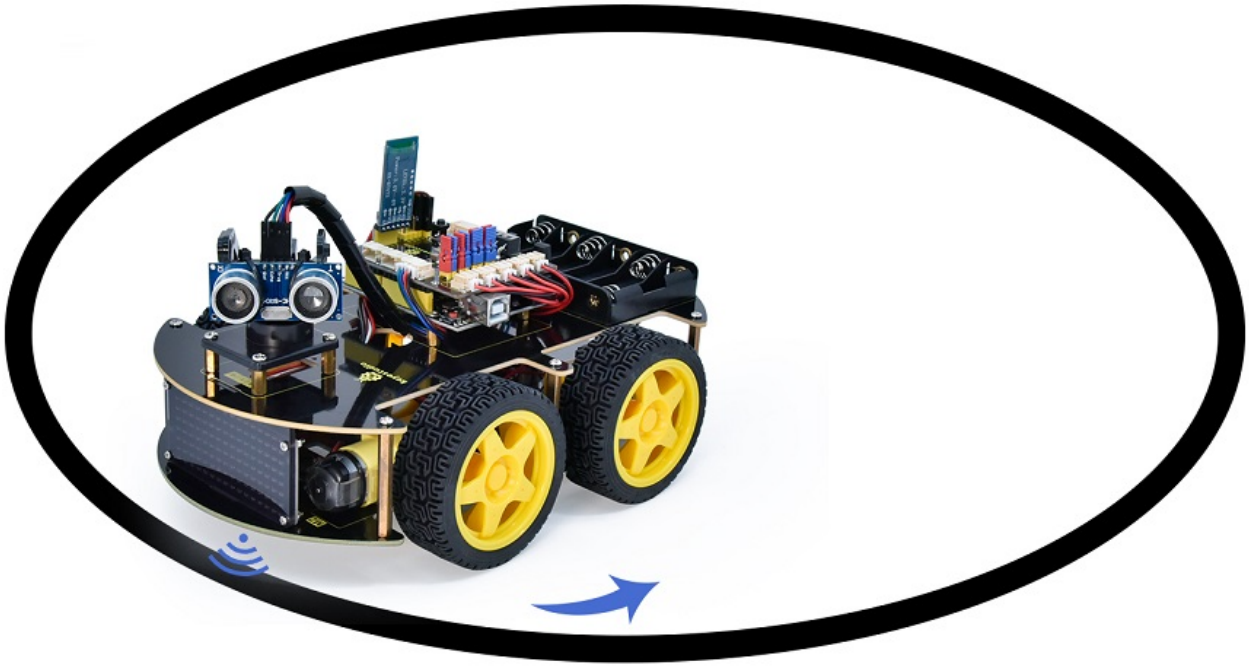
```
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, LOW);
}
//Indicates the end of data transmission
void IIC_end()
{
  digitalWrite(SCL_Pin, LOW);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, HIGH);
  delayMicroseconds(3);
}
//transmit data
void IIC_send(unsigned char send_data)
{
  for (byte mask = 0x01; mask != 0; mask <<= 1) //Each byte has 8 bits and is checked␣
↪bit by bit starting at the lowest level
  {
    if (send_data & mask) { //Sets the high and low levels of SDA_Pin depending on␣
↪whether each bit of the byte is a 1 or a 0
      digitalWrite(SDA_Pin, HIGH);
    } else {
      digitalWrite(SDA_Pin, LOW);
    }
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, HIGH); //Pull the clock pin SCL_Pin high to stop data␣
↪transmission
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, LOW); //pull the clock pin SCL_Pin low to change the SIGNAL of␣
↪SDA
  }
}
//*****************************************************************************
```

After uploading test code, the facial expression board shows these patterns orderly and repeats this sequence.



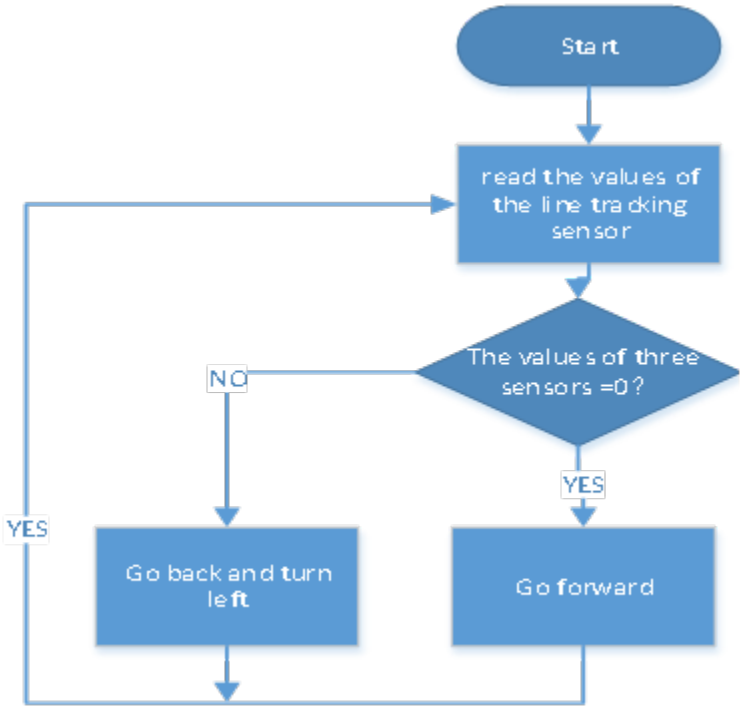
9.2.10 Project 10 Restricting Smart Car



1.Description

In this project, we look to combine the knowledge of a line tracking sensor and motor driver modules to make a restricting smart car. In the experiment, we aim to use the line tracking sensor to detect whether there is a black line around the smart car, and then control the rotation of the two motors according to the detection results in a way that lock the smart car in a circle drawn in black line.

2.Flow Chart

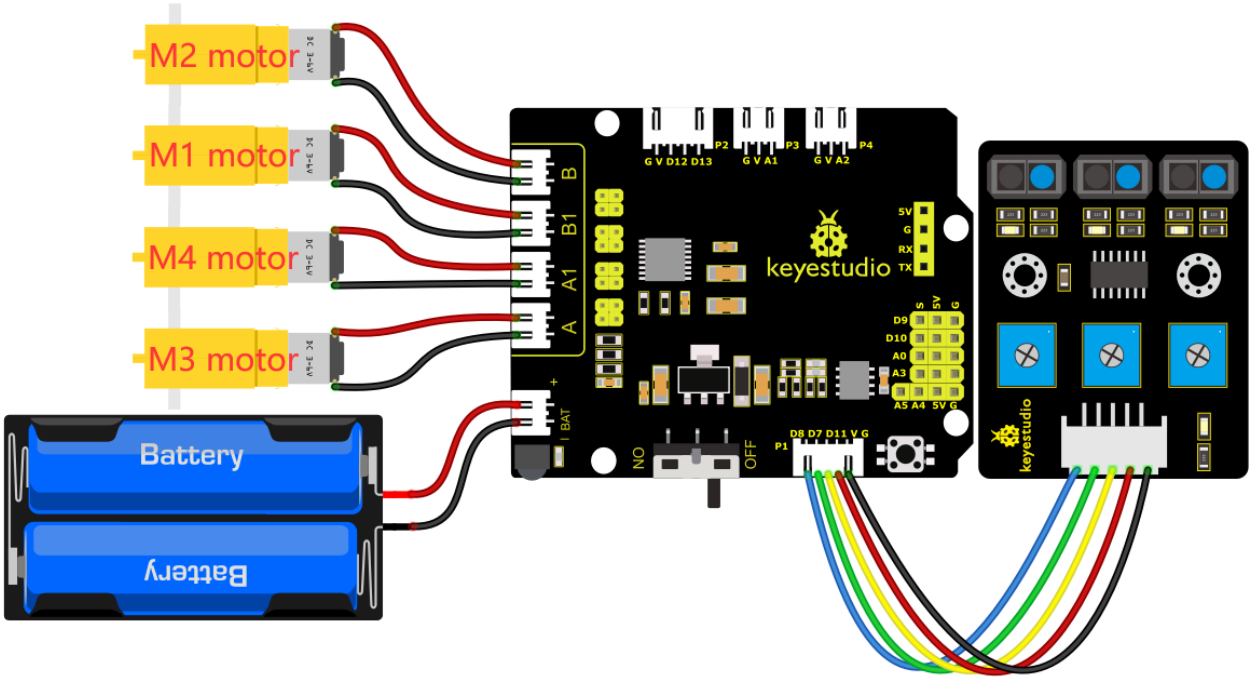


The specific logic of the restricting 4WD smart car is shown in the table.

Detection	Line-tracking sensor in the middle	Black line detected: in high level
		White line detected: in low level
	Line-tracking sensor on the left	Black line detected: in high level
		White line detected: in low level
	Line-tracking sensor on the right	Black line detected: in high level
		White line detected: in low level

Condition	Movement
All the three line-tracking sensors don' t detect black lines	Move forward
Any of the three line-tracking sensors detect black lines	Step back Then rotate left

3.Wiring Diagram



G, V, S1, S2 and S3 of the line tracking sensor are connected to GGND), VVCC), D11, D7 and D8 of the sensor expansion board.

The power is connected to the BAT port

4.Test Code

```
//*****************************************************************************
/*
 keyestudio 4wd BT Car
 lesson 10
 Restricting Smart Car
 http://www.keyestudio.com
*/
//Data from the smile pattern obtained from the touch tool
unsigned char start01[] = {0x01, 0x02, 0x04, 0x08, 0x10, 0x20, 0x40, 0x80, 0x80, 0x40,␣
↪0x20, 0x10, 0x08, 0x04, 0x02, 0x01};
#define SDA_Pin  A4  //Set data pin to A4
#define SCL_Pin  A5  //Set the clock pin to A5

int left_ctrl = 2;//define the direction control pins of group B motor
int left_pwm = 5;//define the PWM control pins of group B motor
int right_ctrl = 4;//define the direction control pins of group A motor
int right_pwm = 6;//define the PWM control pins of group A motor
int sensor_L = 11;//define the pin of left line tracking sensor
int sensor_M = 7;//define the pin of middle line tracking sensor
int sensor_R = 8;//define the pin of right line tracking sensor
int L_val,M_val,R_val;//define these variables

void setup() {
```

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```
  Serial.begin(9600);//start serial monitor and set baud rate to 9600
  pinMode(left_ctrl,OUTPUT);//set direction control pins of group B motor to OUTPUT
  pinMode(left_pwm,OUTPUT);//set PWM control pins of group B motor to OUTPUT
  pinMode(right_ctrl,OUTPUT);//set direction control pins of group A motor to OUTPUT
  pinMode(right_pwm,OUTPUT);//set PWM control pins of group A motor to OUTPUT
  pinMode(sensor_L,INPUT);//set the pins of left line tracking sensor to INPUT
  pinMode(sensor_M,INPUT);//set the pins of middle line tracking sensor to INPUT
  pinMode(sensor_R,INPUT);//set the pins of right line tracking sensor to INPUT
 //Set pin to output
  pinMode(SCL_Pin, OUTPUT);
  pinMode(SDA_Pin, OUTPUT);
  matrix_display(start01);//Show start pattern
}

void loop()
{
  tracking(); //run main program
}

void tracking()
{
  L_val = digitalRead(sensor_L);//read the value of left line tracking sensor
  M_val = digitalRead(sensor_M);//read the value of middle line tracking sensor
  R_val = digitalRead(sensor_R);//read the value of right line tracking sensor
  if ( L_val == 0 && M_val == 0 && R_val == 0 ) { //when no black lines are␣
↪detectedturtle car forward
    Car_front();
  }
  else { //Otherwise, if any of the patrol sensors detect a black line, back up and turn␣
↪left
    Car_back();
    delay(500);
    Car_left();
    delay(500);
  }
}

void Car_front()
{
  digitalWrite(left_ctrl, HIGH);
  analogWrite(left_pwm, 180);
  digitalWrite(right_ctrl, HIGH);
  analogWrite(right_pwm, 180);
}
void Car_back()
{
  digitalWrite(left_ctrl, LOW);
  analogWrite(left_pwm, 80);
  digitalWrite(right_ctrl, LOW);
  analogWrite(right_pwm, 80);
}
void Car_left()
```

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```
{
  digitalWrite(left_ctrl, LOW);
  analogWrite(left_pwm, 100);
  digitalWrite(right_ctrl, HIGH);
  analogWrite(right_pwm, 150);
}

void Car_Stop()
{
  digitalWrite(left_ctrl, LOW);
  analogWrite(left_pwm, 0);
  digitalWrite(right_ctrl, LOW);
  analogWrite(right_pwm, 0);
}

//this function is used for dot matrix display
void matrix_display(unsigned char matrix_value[])
{
  IIC_start();  //the function that calls the data transfer start condition
  IIC_send(0xc0);  //select address

  for (int i = 0; i < 16; i++) //the pattern data is 16 bytes
  {
    IIC_send(matrix_value[i]); //Transmit the data of the pattern
  }
  IIC_end();   //End pattern data transmission
  IIC_start();
  IIC_send(0x8A);  //Display control, select 4/16 pulse width
  IIC_end();
}
//Conditions under which data transmission begins
void IIC_start()
{
  digitalWrite(SDA_Pin, HIGH);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, LOW);
}
//Indicates the end of data transmission
void IIC_end()
{
  digitalWrite(SCL_Pin, LOW);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, HIGH);
  delayMicroseconds(3);
}
//transmit data
```

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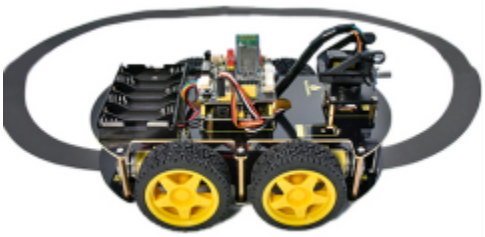
```
void IIC_send(unsigned char send_data)
{
  for (byte mask = 0x01; mask != 0; mask <<= 1) //Each byte has 8 bits and is checked␣
↪bit by bit starting at the lowest level
  {
    if (send_data & mask) { //Sets the high and low levels of SDA_Pin depending on␣
↪whether each bit of the byte is a 1 or a 0
      digitalWrite(SDA_Pin, HIGH);
    } else {
      digitalWrite(SDA_Pin, LOW);
    }
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, HIGH); //Pull the clock pin SCL_Pin high to stop data␣
↪transmission
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, LOW); //pull the clock pin SCL_Pin low to change the SIGNAL of␣
↪SDA
  }
}
//*****************************************************************************
```

5.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, power on the external power then turn the DIP switch to ON. Put the smart car in the black circle, then it will move solely in the circle.

9.2.11 Project 11 Line Tracking Smart Car

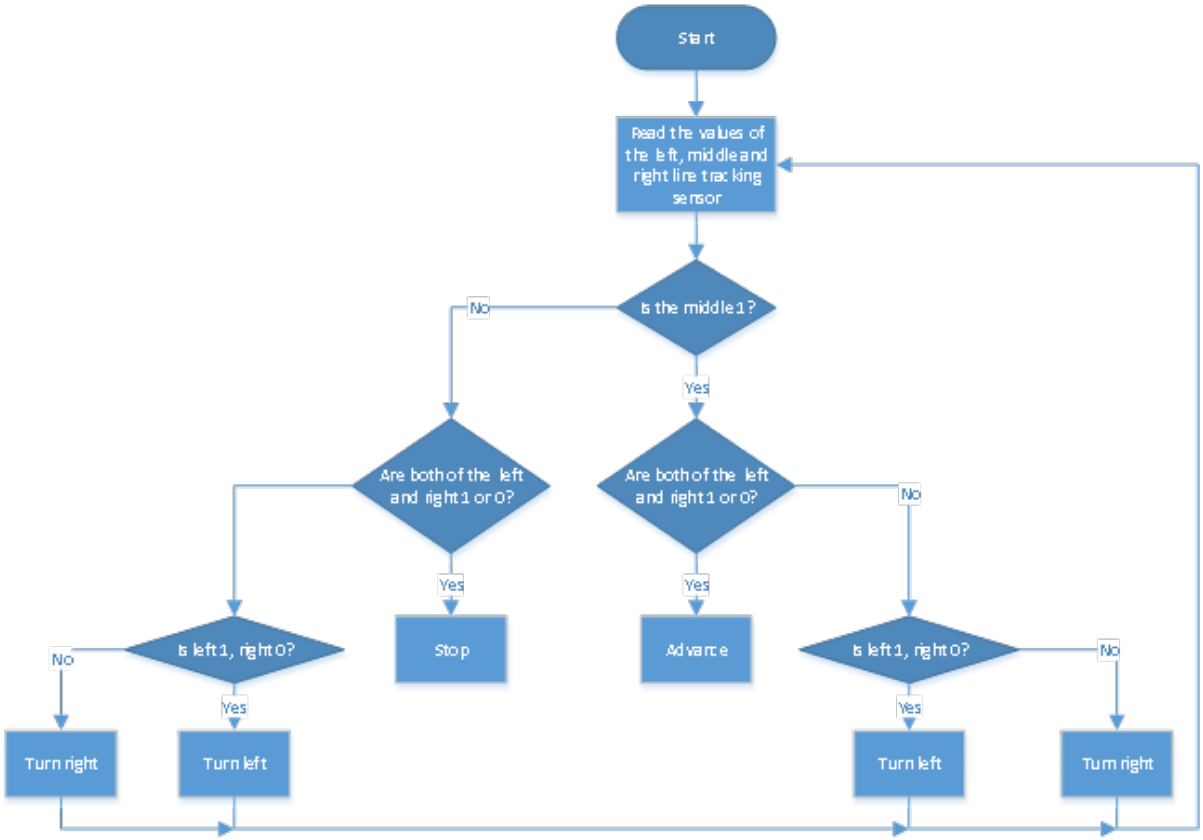


1.Description

Based on the working principle of the line tracking sensor, we empower to make a line tracking smart car.

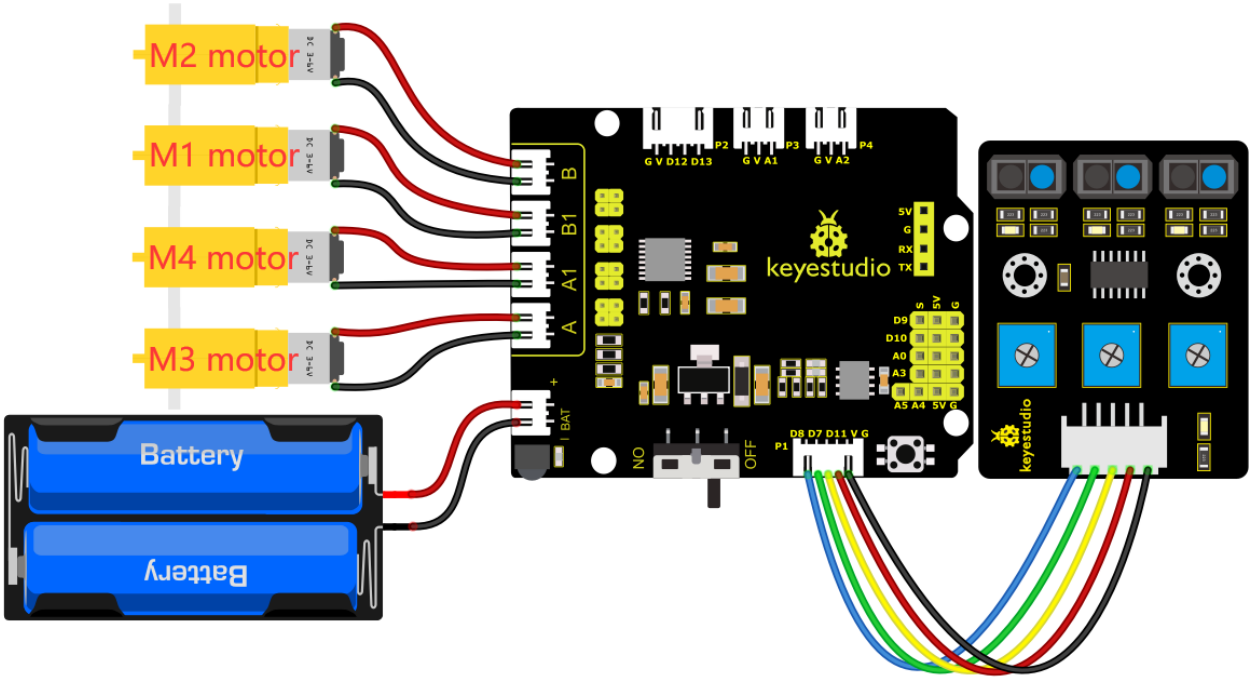
In this project, we detect whether there is a black line at the bottom of the smart car through a line tracking sensor, and then control the rotation of the two groups of motors according to the detection results in a way that controls the smart car to walk along the black line.

2.Flow Chart


Start
Read the values of the left, middle and right line tracking sensor
Is the middle 1?
No
Yes
Are both of the left and right 1 or 0?
Are both of the left and right 1 or 0?
Yes
No
Yes
Yes
Is left 1, right 0?
Stop
Advance
Is left 1, right 0?
No
No
Yes
Yes
Turn right
Turn left
Turn left
Turn right

Detection	Middle tracking sensor	detects black line: HIGH
		detects white line: LOW
	Left tracking sensor	detects black line: HIGH
		detects white line: LOW
	Right tracking sensor	detects black line: HIGH
		detects white line: LOW
Condition 1	Condition 2 Detect the left and the right tracking sensor	Status
Middle tracking sensor detects black line	left tracking sensor detects black line; right sensor detects white line	Turn left
	left tracking sensor detects white line; right sensor detects black line	Turn right
	left and right tracking sensor detect black line	Advance
	left and right tracking sensor detect white line	Advance
Middle tracking sensor detects white line	left tracking sensor detects black line; right sensor detects white line	Turn left
	left tracking sensor detects white line; right sensor detects black line	Turn right
	left and right tracking sensor detect black line	Stop
	left and right tracking sensor detect white line	Stop

3.Wiring Diagram



G, V, S1, S2 and S3 of the line tracking sensor are connected to GGND), VVCC), D11, D7 and D8 of the sensor expansion board.

The power is connected to the BAT port

4.Test Code

```
//*****************************************************************************
/*
 keyestudio 4wd BT Car
 lesson 11
 Tracking Car
 http://www.keyestudio.com
*/
//Data from the smile pattern obtained from the touch tool
unsigned char start01[] = {0x01, 0x02, 0x04, 0x08, 0x10, 0x20, 0x40, 0x80, 0x80, 0x40,␣
↪0x20, 0x10, 0x08, 0x04, 0x02, 0x01};
#define SDA_Pin  A4  //Set data pin to A4
#define SCL_Pin  A5  //Set the clock pin to A5

int left_ctrl = 2;//define the direction control pins of group B motor
int left_pwm = 5;//define the PWM control pins of group B motor
int right_ctrl = 4;//define the direction control pins of group A motor
int right_pwm = 6;//define the PWM control pins of group A motor
int sensor_L = 11;//define the pin of left line tracking sensor
int sensor_M = 7;//define the pin of middle line tracking sensor
int sensor_R = 8;//define the pin of right line tracking sensor
int L_val,M_val,R_val;//define these variables

void setup() {
```

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```
  Serial.begin(9600);//start serial monitor and set baud rate to 9600
  pinMode(left_ctrl,OUTPUT);//set direction control pins of group B motor to OUTPUT
  pinMode(left_pwm,OUTPUT);//set PWM control pins of group B motor to OUTPUT
  pinMode(right_ctrl,OUTPUT);//set direction control pins of group A motor to OUTPUT
  pinMode(right_pwm,OUTPUT);//set PWM control pins of group A motor to OUTPUT
  pinMode(sensor_L,INPUT);//set the pins of left line tracking sensor to INPUT
  pinMode(sensor_M,INPUT);//set the pins of middle line tracking sensor to INPUT
  pinMode(sensor_R,INPUT);//set the pins of right line tracking sensor to INPUT
  //Set pin to output
  pinMode(SCL_Pin, OUTPUT);
  pinMode(SDA_Pin, OUTPUT);
  matrix_display(start01);//Show start pattern
}

void loop()
{
  tracking(); //run main program
}

void tracking()
{
  L_val = digitalRead(sensor_L);//read the value of left line tracking sensor
  M_val = digitalRead(sensor_M);//read the value of middle line tracking sensor
  R_val = digitalRead(sensor_R);//read the value of right line tracking sensor

  if(M_val == 1){//if the state of middle one is 1, which means detecting black line

     if (L_val == 1 && R_val == 0) { //If a black line is detected on the left, but not␣
↪on the right, turn left
        left();
    }
     else if (L_val == 0 && R_val == 1) { //Otherwise, if a black line is detected on␣
↪the right and not on the left, turn right
      right();
    }
     else { //Otherwise,forward
      front();
    }
  }
  else { //No black lines detected in the middle
    if (L_val == 1 && R_val == 0) { //If a black line is detected on the left, but not␣
↪on the right, turn left
      left();
    }
    else if (L_val == 0 && R_val == 1) { //Otherwise, if a black line is detected on the␣
↪right and not on the left, turn right
      right();
    }
    else { //Otherwise,stop
      Stop();
    }
  }
```

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```
}
void front()//define the status of going forward
{
  digitalWrite(left_ctrl,HIGH);
  analogWrite(left_pwm,155);
  digitalWrite(right_ctrl,HIGH);
  analogWrite(right_pwm,155);
}
void back()//define the state of going back
{
  digitalWrite(left_ctrl,LOW);
  analogWrite(left_pwm,100);
  digitalWrite(right_ctrl,LOW);
  analogWrite(right_pwm,100);
}
void left()//define the left-turning state
{
  digitalWrite(left_ctrl, LOW);
  analogWrite(left_pwm, 100);
  digitalWrite(right_ctrl, HIGH);
  analogWrite(right_pwm, 155);
}
void right()//define the right-turning state
{
  digitalWrite(left_ctrl, HIGH);
  analogWrite(left_pwm, 155);
  digitalWrite(right_ctrl, LOW);
  analogWrite(right_pwm, 100);
}
void Stop()//define the state of stop
{
  digitalWrite(left_ctrl, LOW);
  analogWrite(left_pwm,0);
  digitalWrite(right_ctrl, LOW);
  analogWrite(right_pwm,0);
}

//this function is used for dot matrix display
void matrix_display(unsigned char matrix_value[])
{
  IIC_start();  //the function that calls the data transfer start condition
  IIC_send(0xc0);  //select address

  for (int i = 0; i < 16; i++) //the pattern data is 16 bytes
  {
    IIC_send(matrix_value[i]); //Transmit the data of the pattern
  }
  IIC_end();   //End pattern data transmission
  IIC_start();
  IIC_send(0x8A);  //Display control, select 4/16 pulse width
  IIC_end();
}
```

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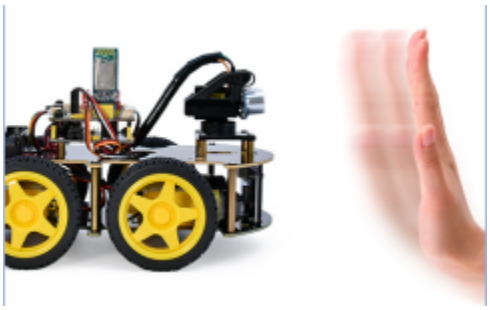
```
//Conditions under which data transmission begins
void IIC_start()
{
  digitalWrite(SDA_Pin, HIGH);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, LOW);
}
//Indicates the end of data transmission
void IIC_end()
{
  digitalWrite(SCL_Pin, LOW);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, HIGH);
  delayMicroseconds(3);
}
//transmit data
void IIC_send(unsigned char send_data)
{
  for (byte mask = 0x01; mask != 0; mask <<= 1) //Each byte has 8 bits and is checked␣
↪bit by bit starting at the lowest level
  {
    if (send_data & mask) { //Sets the high and low levels of SDA_Pin depending on␣
↪whether each bit of the byte is a 1 or a 0
      digitalWrite(SDA_Pin, HIGH);
    } else {
      digitalWrite(SDA_Pin, LOW);
    }
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, HIGH); //Pull the clock pin SCL_Pin high to stop data␣
↪transmission
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, LOW); //pull the clock pin SCL_Pin low to change the SIGNAL of␣
↪SDA
  }
}
//*****************************************************************************
```

5.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, power on the external power then turn the DIP switch to ON. Then the smart car will walk along the lines.

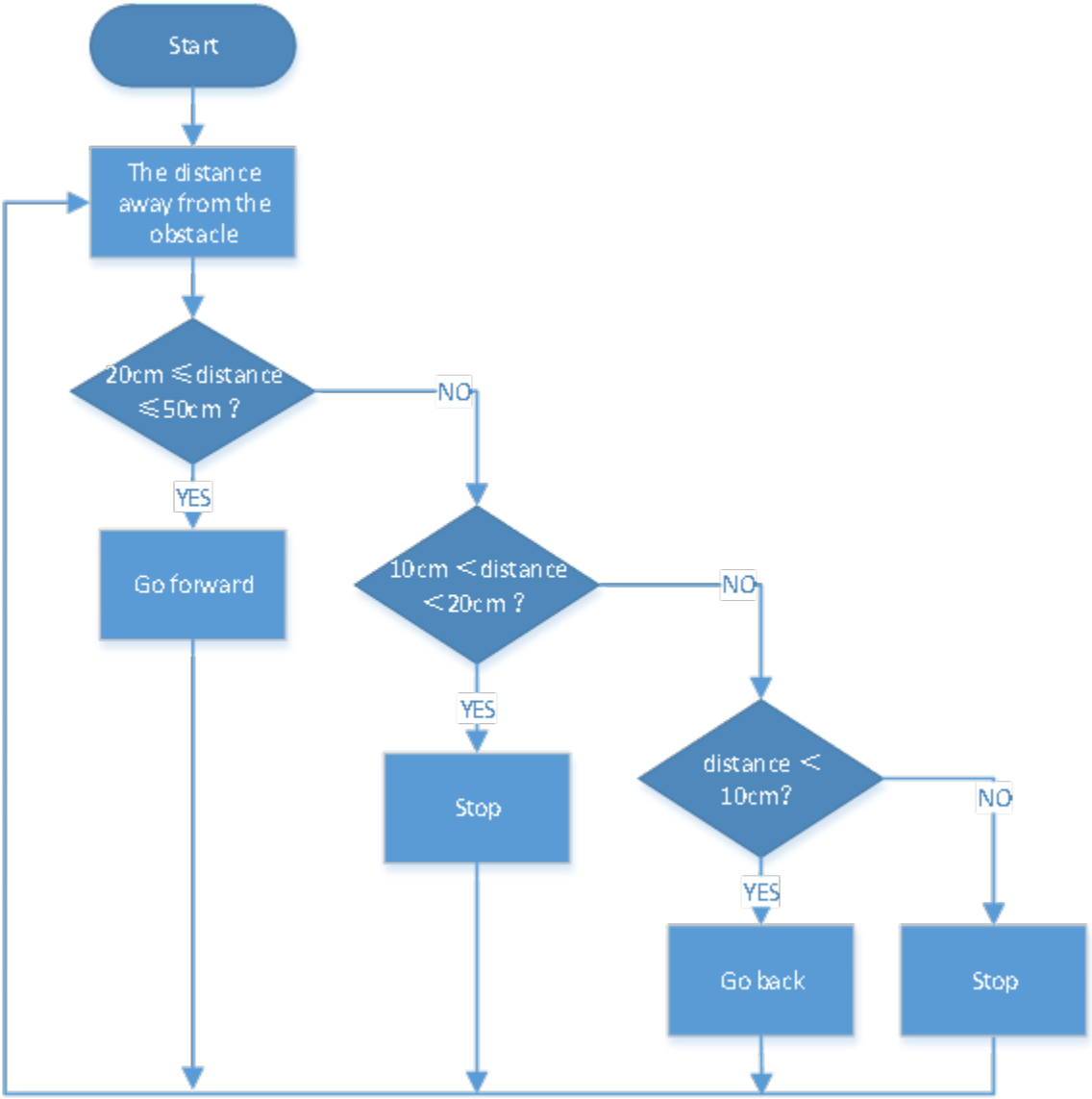
9.2.12 Project 12 Ultrasonic Following Smart Car



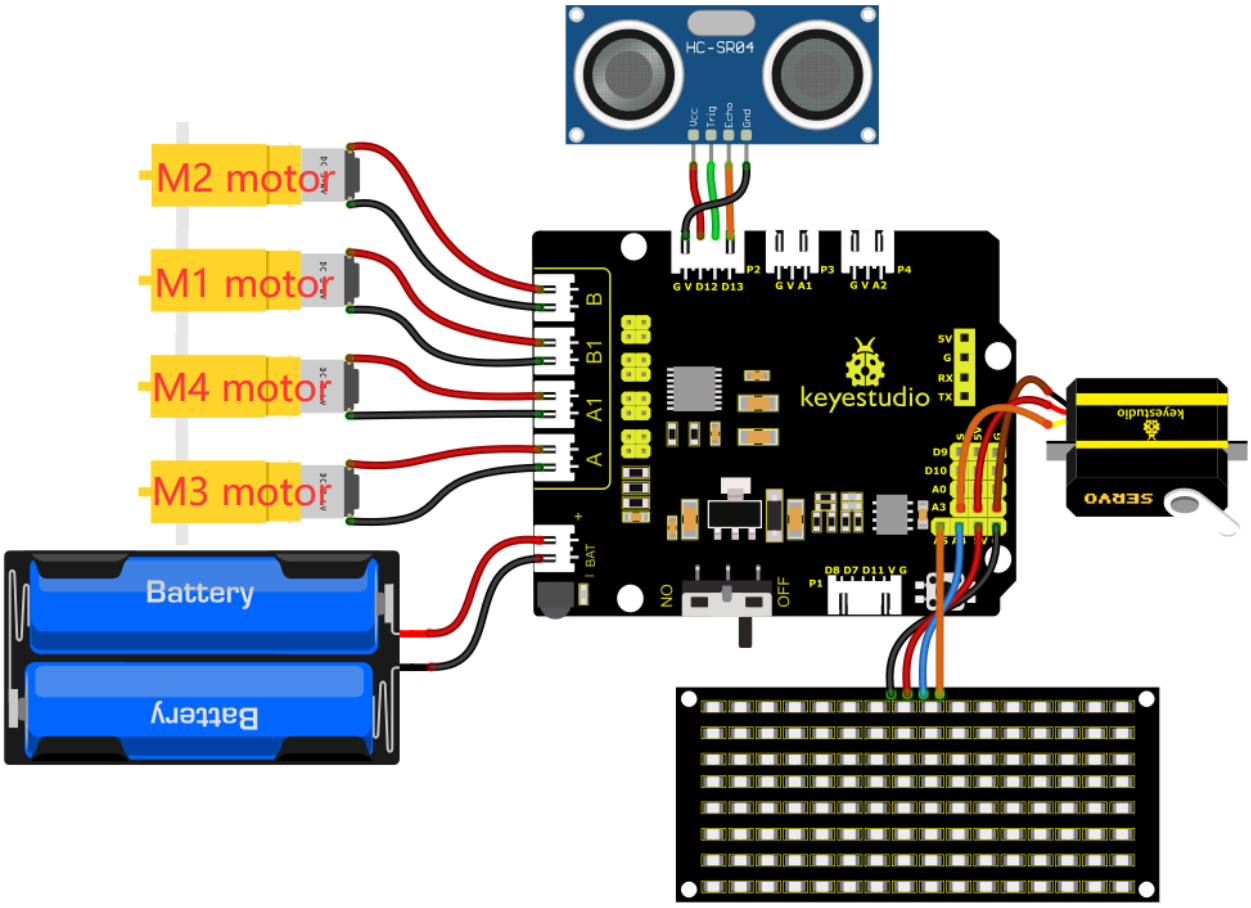
1.Description

In this project, we will look to detect the distance between the 4WD smart car and the obstacles ahead through an ultrasonic sensor to drive two motors in a way that make the car move and make the 8*8 LED board show a smile facial pattern.

2.Flow Chart


Start
The distance away from the obstacle
20cm ≤distance ≤50cm ?
NO
YES
Go forward
10cm <distance <20cm ?
NO
YES
Stop
distance < 10cm?
NO
YES
Go back
Stop

3.Wiring Diagram



Wiring up

1). GND, VCC, SDA and SCL of the 8*8 LED board are connected to GGND), VVCC), A4 and A5 of the expansion board.

2). VCC, Trig, Echo and Gnd of the ultrasonic sensor are connected to 5V(V), D12(S), D13(S) and Gnd(G)

3). The servo is connected to G, V and A3. The brown wire is interfaced with Gnd(G), the red wire is interfaced with 5V(V) and the orange wire is interfaced with A3.

4). The power is connected to the BAT port

4.Test Code

```
//*****************************************************************************
/*
 keyestudio 4wd BT Car
 lesson 12
 Flowing Car
 http://www.keyestudio.com
*/
#define SCL_Pin  A5  //Set the clock pin to A5
#define SDA_Pin  A4  //Set data pin to A4
```

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```
//Array, used to store the data of pattern, can be calculated by yourself or obtained␣
↪from the modulus tool
unsigned char smile[] = {0x00, 0x00, 0x1c, 0x02, 0x02, 0x02, 0x5c, 0x40, 0x40, 0x5c,␣
↪0x02, 0x02, 0x02, 0x1c, 0x00, 0x00};

const int servopin = A3;//Set the pin of the steering gear

#include "SR04.h" //define the function library of ultrasonic sensor
#define TRIG_PIN 12// set the signal of ultrasonic sensor to D12
#define ECHO_PIN 13// set the signal of ultrasonic sensor to D13
SR04 sr04 = SR04(ECHO_PIN,TRIG_PIN);
long distance;

int left_ctrl = 2;//define the direction control pins of group B motor
int left_pwm = 5;//define the PWM control pins of group B motor
int right_ctrl = 4;//define the direction control pins of group A motor
int right_pwm = 6;//define the PWM control pins of group A motor

void setup() {
  pinMode(left_ctrl,OUTPUT);//set direction control pins of group B motor to OUTPUT
  pinMode(left_pwm,OUTPUT);//set PWM control pins of group B motor to OUTPUT
  pinMode(right_ctrl,OUTPUT);//set direction control pins of group A motor to OUTPUT
  pinMode(right_pwm,OUTPUT);//set PWM control pins of group A motor to OUTPUT
  pinMode(TRIG_PIN, OUTPUT); //Set the trig pin to output
  pinMode(ECHO_PIN, INPUT); //Set the echo pin to input
  pinMode(SCL_Pin,OUTPUT);//Set the clock pin to output
  pinMode(SDA_Pin,OUTPUT);//Set the data pin to output
  servopulse(servopin,90);//Set the initial steering gear angle to 90°
  delay(500); //waits 500ms
  matrix_display(smile);  //display smiling expression pattern
}

void loop() {
  distance = sr04.Distance();//the distance detected by ultrasonic sensor
   if(distance <= 10)//if distance is less than 10
  {
    back();//go back
  }
  else if((distance > 10)&&(distance< 20 ))//if 10<distance<20
  {
    Stop();//stop
  }
  else if((distance >= 20)&&(distance <= 50))//if 20distance50
{
    front();//follow
  }
  else//otherwise
  {
    Stop();//stop
  }
}
```

(continues on next page)

(continued from previous page)

```
void front()//define the status of going front
{
  digitalWrite(left_ctrl,HIGH);
  analogWrite(left_pwm,100);
  digitalWrite(right_ctrl,HIGH);
  analogWrite(right_pwm,100);
}
void back()//define the status of going back
{
  digitalWrite(left_ctrl,LOW);
  analogWrite(left_pwm,150);
  digitalWrite(right_ctrl,LOW);
  analogWrite(right_pwm,150);
}
void left()//define the status of turning left
{
  digitalWrite(left_ctrl, LOW);
  analogWrite(left_pwm, 100);
  digitalWrite(right_ctrl, HIGH);
  analogWrite(right_pwm, 155);
}
void right()//define the status of right turning
{
  digitalWrite(left_ctrl, HIGH);
  analogWrite(left_pwm, 155);
  digitalWrite(right_ctrl, LOW);
  analogWrite(right_pwm, 100);
}
void Stop()//define the state of stop
{
  digitalWrite(left_ctrl, LOW);
  analogWrite(left_pwm,0);
  digitalWrite(right_ctrl, LOW);
  analogWrite(right_pwm,0);
}

void servopulse(int servopin,int myangle)//Steering gear running angle
{
  for(int i=0; i<30; i++)
  {
    int pulsewidth = (myangle*11)+500;
    digitalWrite(servopin,HIGH);
    delayMicroseconds(pulsewidth);
    digitalWrite(servopin,LOW);
    delay(20-pulsewidth/1000);
  }
}

//this function is used for dot matrix display
void matrix_display(unsigned char matrix_value[])
{
  IIC_start();  //the function that calls the data transfer start condition
```

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```
  IIC_send(0xc0);  //select address

  for (int i = 0; i < 16; i++) //the pattern data is 16 bytes
  {
    IIC_send(matrix_value[i]); //Transmit the data of the pattern
  }
  IIC_end();   //End pattern data transmission
  IIC_start();
  IIC_send(0x8A);  //Display control, select 4/16 pulse width
  IIC_end();
}
//Conditions under which data transmission begins
void IIC_start()
{
  digitalWrite(SDA_Pin, HIGH);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, LOW);
}
//Indicates the end of data transmission
void IIC_end()
{
  digitalWrite(SCL_Pin, LOW);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, HIGH);
  delayMicroseconds(3);
}
//transmit data
void IIC_send(unsigned char send_data)
{
  for (byte mask = 0x01; mask != 0; mask <<= 1) //Each byte has 8 bits and is checked␣
↪bit by bit starting at the lowest level
  {
    if (send_data & mask) { //Sets the high and low levels of SDA_Pin depending on␣
↪whether each bit of the byte is a 1 or a 0
      digitalWrite(SDA_Pin, HIGH);
    } else {
      digitalWrite(SDA_Pin, LOW);
    }
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, HIGH); //Pull the clock pin SCL_Pin high to stop data␣
↪transmission
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, LOW); //pull the clock pin SCL_Pin low to change the SIGNAL of␣
↪SDA
  }
}
```

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```
//*****************************************************************************
```

5.Test Result

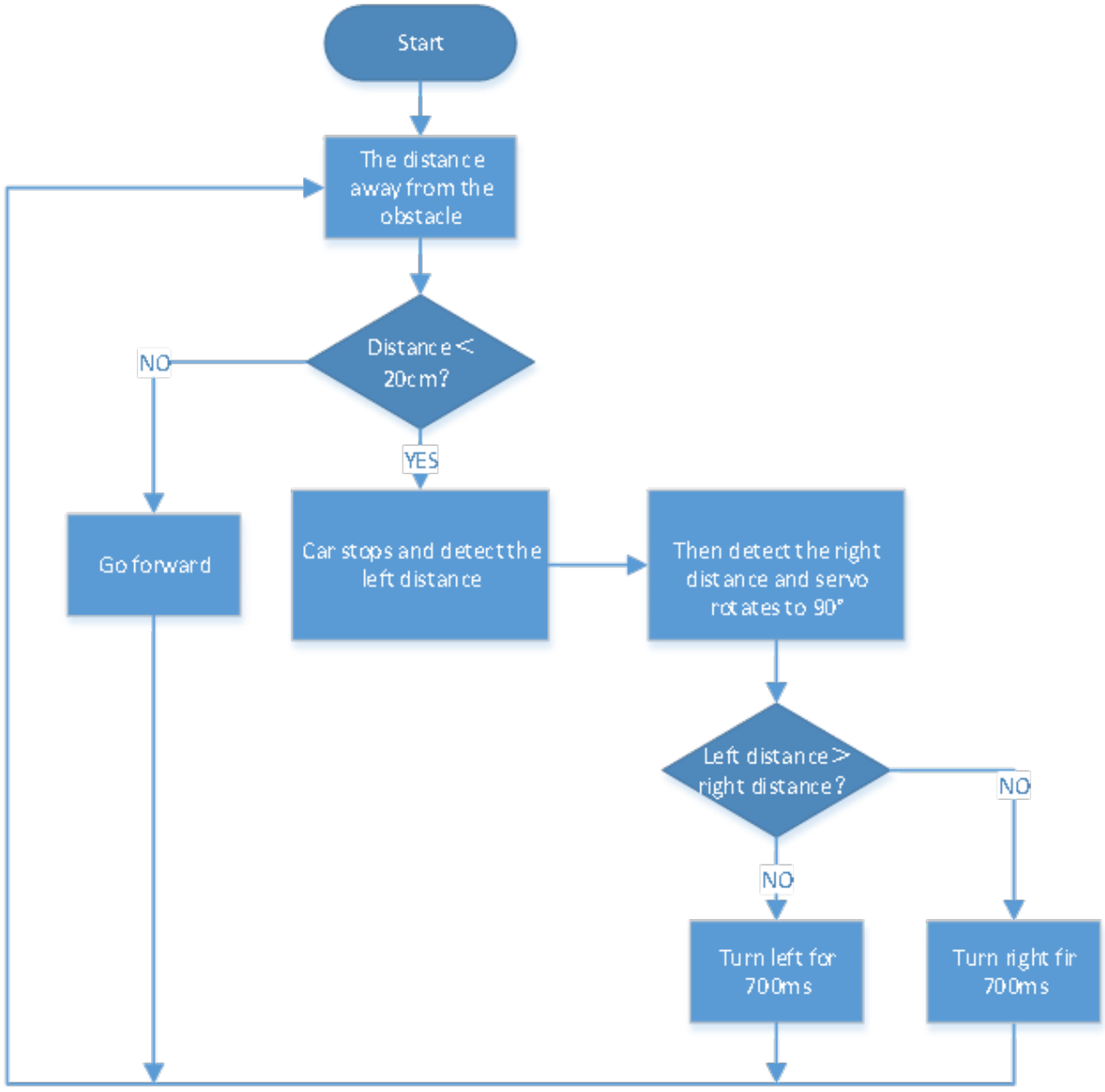
After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, power on the external power then turn the DIP switch to ON. Set the servo to 90°, the smart car will move with the obstacles and the 8X16 LED board will show “smile”.

9.2.13 Project 13 Ultrasonic Obstacle Avoidance Smart Car

1.Description

In this project, we aim to make an ultrasonic obstacle avoidance smart car. We will use the ultrasonic to detect the distance from the obstacle, which can be used to control the servo to rotate so as to make the car move. Meanwhile, the 8X16 LED board will display the corresponding status pattern.

2.Flow Chart

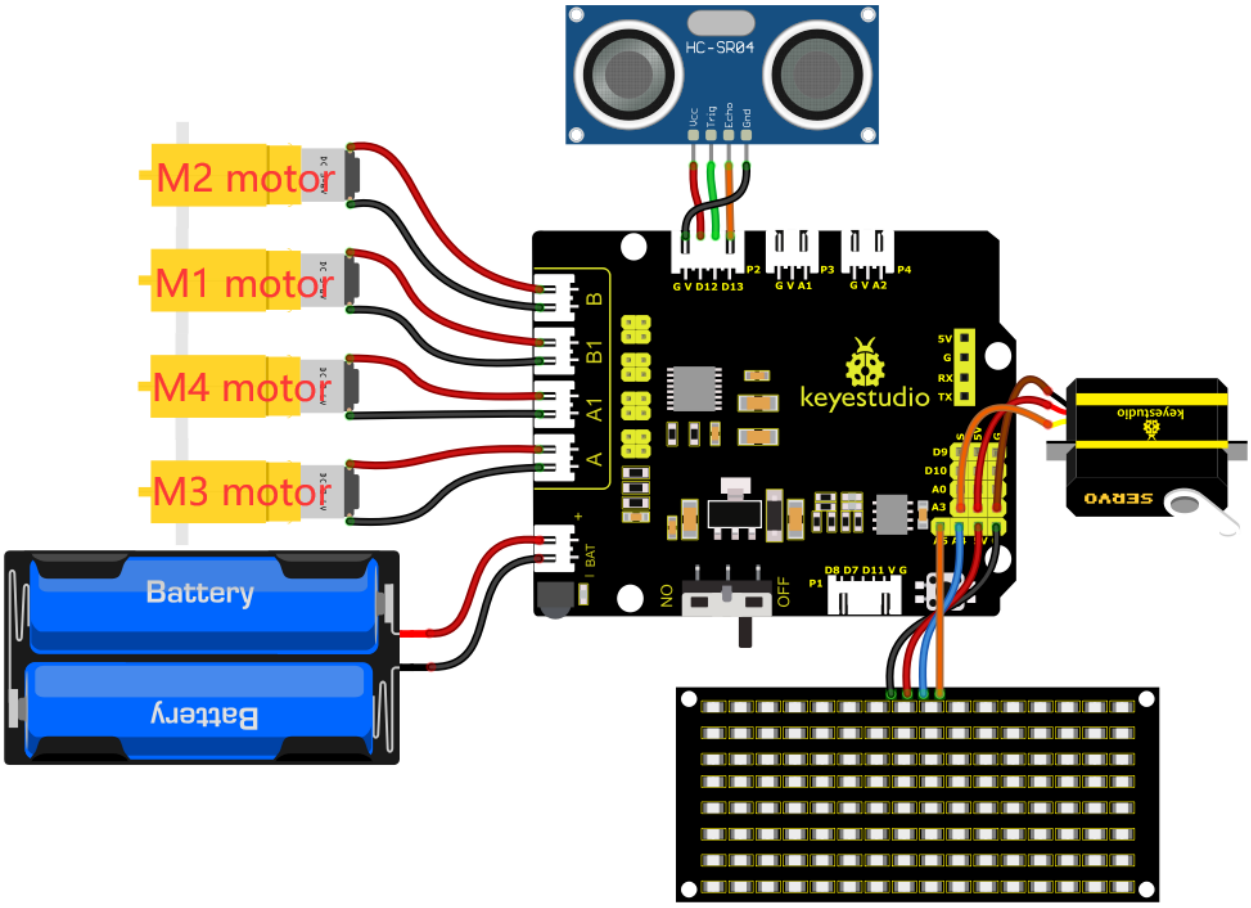


The specific logic of ultrasonic obstacle avoidance smart car is shown below:

<table>
<tr><td rowspan="3">Detection</td><td colspan="3">measured distance of front obstacle
set servo to 90°</td><td>distance (unit: cm)</td></tr>
<tr><td colspan="3">measured distance of left obstacle (set servo to 160°)</td><td>a1 (unit: cm)</td></tr>
<tr><td colspan="3">measured distance of right obstacle
(set servo to 20°)</td><td>a2 (unit: cm)</td></tr>
<tr><td>Setting</td><td colspan="4">set the initial angle of servo to 90°</td></tr>
<tr><td>Condition1</td><td colspan="4">Status</td></tr>
<tr><td rowspan="8">a < 20</td><td colspan="4">Stop for 1000ms; set the angle of servo to 160°, read a1, delay in 500ms; set the angle of servo to 20°, read a2, delay in 500ms</td></tr>
<tr><td colspan="2">Condition 2</td><td colspan="2">Status</td></tr>
<tr><td colspan="2" rowspan="3">a1 < 50
or
a2 < 50</td><td colspan="2">Compare a1 and a2</td></tr>
<tr><td>Condition 3</td><td>Status</td></tr>
<tr><td>a1 > a2</td><td>Set the angle of servo to 90°, rotate to left for 700ms and go front</td></tr>
<tr><td></td><td>a1 < a2</td><td colspan="2">Set the angle of servo to 90°, rotate to right for 700ms, go front</td></tr>
<tr><td>Condition 2</td><td colspan="3">Status</td></tr>
<tr><td rowspan="2">a1 ≥ 50
and
a2 ≥ 50</td><td rowspan="2">Random</td><td colspan="2">Set the angle of servo to 90°, rotate to left for 700ms and go front</td></tr>
<tr><td colspan="2">Set the angle of servo to 90°, rotate to right for 700ms, go front</td></tr>
</table>

Condition1	Status
a ≥ 20	Go forward

3.Wiring Diagram



1). GND, VCC, SDA and SCL of the 8*8 LED board module are connected to GGND), VVCC), A4 and A5 of the expansion board.

2). VCC, Trig, Echo and Gnd of the ultrasonic sensor are connected to 5V(V), D12(S), D13(S) and Gnd(G)

3). The servo is connected to G, V and A3. The brown wire is interfaced with Gnd(G), the red wire is interfaced with 5V(V) and the orange wire is interfaced with A3.

4). The power is connected to the BAT port

4.Test Code

```
//*****************************************************************************
/*
 keyestudio 4wd BT Car
 lesson 13
 Avoiding Car
 http://www.keyestudio.com
*/
#define SCL_Pin  A5  //Set the clock pin to A5
#define SDA_Pin  A4  //Set data pin to A4
//Array, used to store the data of pattern, can be calculated by yourself or obtained␣
↪from the modulus tool
unsigned char front[] = {0x00,0x00,0x00,0x00,0x00,0x24,0x12,0x09,0x12,0x24,0x00,0x00,
```

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```
↪0x00,0x00,0x00,0x00};
unsigned char left[] = {0x00,0x00,0x00,0x00,0x00,0x00,0x44,0x28,0x10,0x44,0x28,0x10,0x44,
↪0x28,0x10,0x00};
unsigned char right[] = {0x00,0x10,0x28,0x44,0x10,0x28,0x44,0x10,0x28,0x44,0x00,0x00,
↪0x00,0x00,0x00,0x00};
unsigned char STOP01[] = {0x2E,0x2A,0x3A,0x00,0x02,0x3E,0x02,0x00,0x3E,0x22,0x3E,0x00,
↪0x3E,0x0A,0x0E,0x00};
unsigned char clear[] = {0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
↪0x00,0x00,0x00,0x00};

int left_ctrl = 2;//define the direction control pins of group B motor
int left_pwm = 5;//define the PWM control pins of group B motor
int right_ctrl = 4;//define the direction control pins of group A motor
int right_pwm = 6;//define the PWM control pins of group A motor

#include "SR04.h"//define the library of ultrasonic sensor
#define TRIG_PIN 12// set the signal output of ultrasonic sensor to D12
#define ECHO_PIN 13//set the signal input of ultrasonic sensor to D13
SR04 sr04 = SR04(ECHO_PIN,TRIG_PIN);
long distance,a1,a2;//define three distance
const int servopin = A3;//set the pin of servo to A3

void setup() {
  pinMode(left_ctrl,OUTPUT);//set direction control pins of group B motor to OUTPUT
  pinMode(left_pwm,OUTPUT);//set PWM control pins of group B motor to OUTPUT
  pinMode(right_ctrl,OUTPUT);//set direction control pins of group A motor to OUTPUT
  pinMode(right_pwm,OUTPUT);//set PWM control pins of group A motor to OUTPUT
  pinMode(TRIG_PIN, OUTPUT); //Set the trig pin to output
  pinMode(ECHO_PIN, INPUT); //Set the echo pin to input
  servopulse(servopin,90);//the angle of servo is 90 degree
  delay(300);
  pinMode(SCL_Pin,OUTPUT);// Set the clock pin to output
  pinMode(SDA_Pin,OUTPUT);//Set the data pin to output
  matrix_display(clear);
}

void loop()
 {
  avoid();//run the main program
}

void avoid()
{
  distance=sr04.Distance(); //obtain the value detected by ultrasonic sensor

  if((distance < 20)&&(distance != 0))//if the distance is greater than 0 and less than␣
↪10

  {
    car_Stop();//stop
    matrix_display(clear);
    matrix_display(STOP01);//show stop pattern
```

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```
    delay(1000);
    servopulse(servopin,160);//servo rotates to 160°
    delay(500);
    a1=sr04.Distance();//measure the distance
    delay(100);
    servopulse(servopin,20);//rotate to 20 degree
    delay(500);
    a2=sr04.Distance();//measure the distance
    delay(100);
    servopulse(servopin,90);  //Return to the 90 degree position
    delay(500);
    if(a1 > a2)//compare the distance, if left distance is more than right distance
    {
      car_left();//turn left
      matrix_display(clear);
      matrix_display(left);    //display left-turning pattern
      servopulse(servopin,90);//servo rotates to 90 degree
      delay(700); //turn left 700ms
      matrix_display(clear);
      matrix_display(front);  //show forward pattern
    }
    else//if the right distance is greater than the left
    {
      car_right();//turn right
      matrix_display(clear);
      matrix_display(right);  //display right-turning pattern
      servopulse(servopin,90);//servo rotates to 90 degree
      delay(700);
      matrix_display(clear);
      matrix_display(front);  //show forward pattern
    }
  }
  else//otherwise
  {
    car_front();//go forward
    matrix_display(clear);
    matrix_display(front);  // show forward pattern
  }
}

void car_front()//car goes forward
{
  digitalWrite(left_ctrl,HIGH);
  analogWrite(left_pwm,155);
  digitalWrite(right_ctrl,HIGH);
  analogWrite(right_pwm,155);
}
void car_back()//go back
{
  digitalWrite(left_ctrl,LOW);
  analogWrite(left_pwm,100);
  digitalWrite(right_ctrl,LOW);
```

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(continued from previous page)

```
  analogWrite(right_pwm,100);
}
void car_left()//car turns left
{
  digitalWrite(left_ctrl, LOW);
  analogWrite(left_pwm, 100);
  digitalWrite(right_ctrl, HIGH);
  analogWrite(right_pwm, 155);
}
void car_right()//car turns right
{
  digitalWrite(left_ctrl, HIGH);
  analogWrite(left_pwm, 155);
  digitalWrite(right_ctrl, LOW);
  analogWrite(right_pwm, 100);
}
void car_Stop()//stop
{
  digitalWrite(left_ctrl,LOW);
  analogWrite(left_pwm,0);
  digitalWrite(right_ctrl,LOW);
  analogWrite(right_pwm,0);
}

void servopulse(int servopin,int myangle)//the running angle of servo
{
  for(int i=0; i<20; i++)
  {
    int pulsewidth = (myangle*11)+500;
    digitalWrite(servopin,HIGH);
    delayMicroseconds(pulsewidth);
    digitalWrite(servopin,LOW);
    delay(20-pulsewidth/1000);
  }
}

//this function is used for dot matrix display
void matrix_display(unsigned char matrix_value[])
{
  IIC_start();  //the function that calls the data transfer start condition
  IIC_send(0xc0);  //select address

  for (int i = 0; i < 16; i++) //the pattern data is 16 bytes
  {
    IIC_send(matrix_value[i]); //Transmit the data of the pattern
  }
  IIC_end();   //End pattern data transmission
  IIC_start();
  IIC_send(0x8A);  //Display control, select 4/16 pulse width
  IIC_end();
}
//Conditions under which data transmission begins
```

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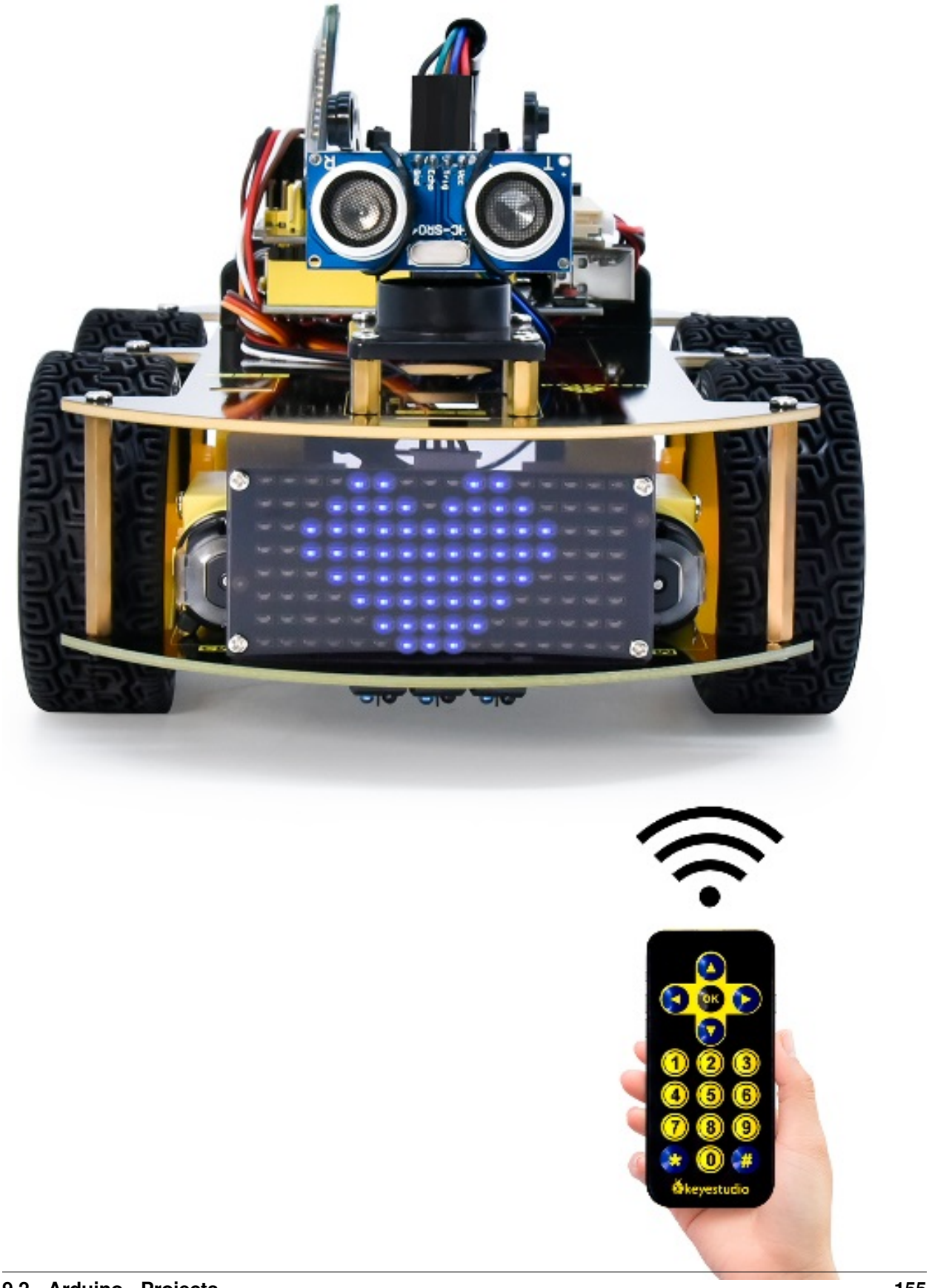
```
void IIC_start()
{
  digitalWrite(SDA_Pin, HIGH);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, LOW);
}
//Indicates the end of data transmission
void IIC_end()
{
  digitalWrite(SCL_Pin, LOW);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, HIGH);
  delayMicroseconds(3);
}
//transmit data
void IIC_send(unsigned char send_data)
{
  for (byte mask = 0x01; mask != 0; mask <<= 1) //Each byte has 8 bits and is checked␣
↪bit by bit starting at the lowest level
  {
    if (send_data & mask) { //Sets the high and low levels of SDA_Pin depending on␣
↪whether each bit of the byte is a 1 or a 0
      digitalWrite(SDA_Pin, HIGH);
    } else {
      digitalWrite(SDA_Pin, LOW);
    }
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, HIGH); //Pull the clock pin SCL_Pin high to stop data␣
↪transmission
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, LOW); //pull the clock pin SCL_Pin low to change the SIGNAL of␣
↪SDA
  }
}
//*****************************************************************************
```

5.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, power on the external power then turn the DIP switch to ON.

The smart car moves forward and automatically avoids obstacles. When there is no road ahead, the servo will drive the ultrasonic sensor to scan the left, middle and right distances, and the car will turn to the open side. Meanwhile, the 8X16 LED board will display the corresponding status pattern.

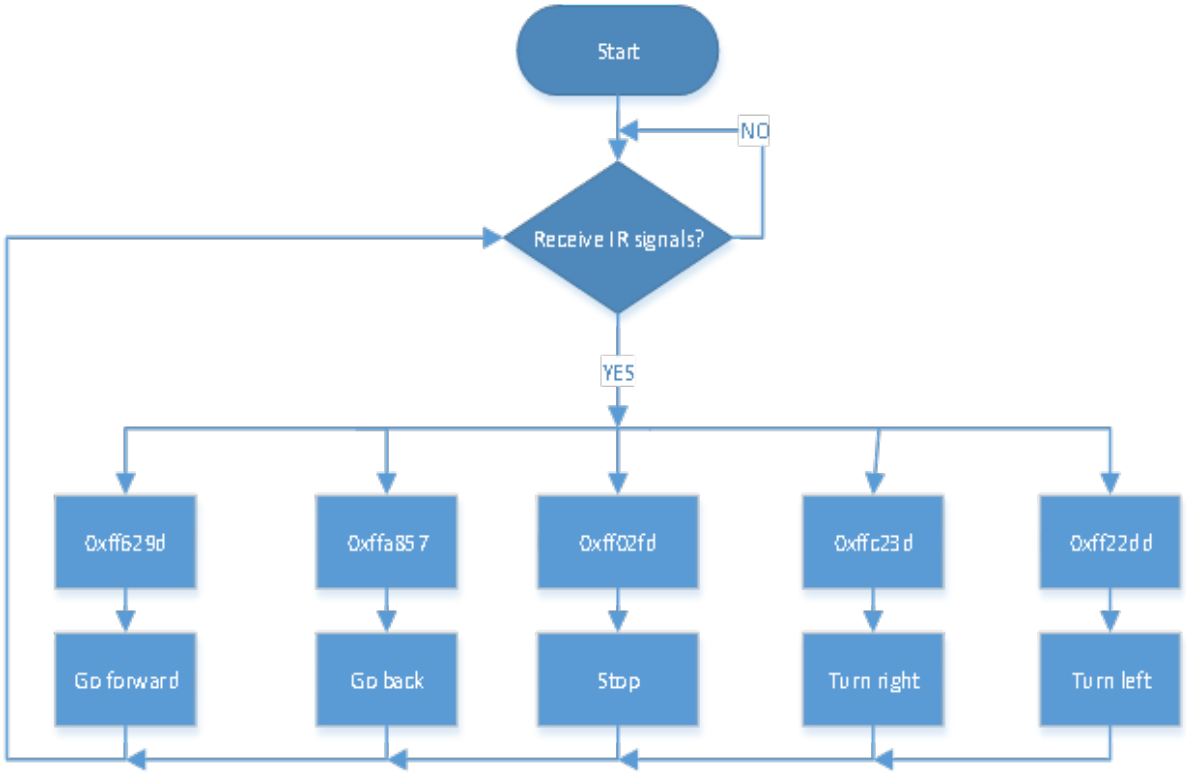
9.2.14 Project 14 IR Remote Control Smart Car



1.Description

In this project, we will make an IR remote control smart car and press the button on the IR remote control to drive the car to move.

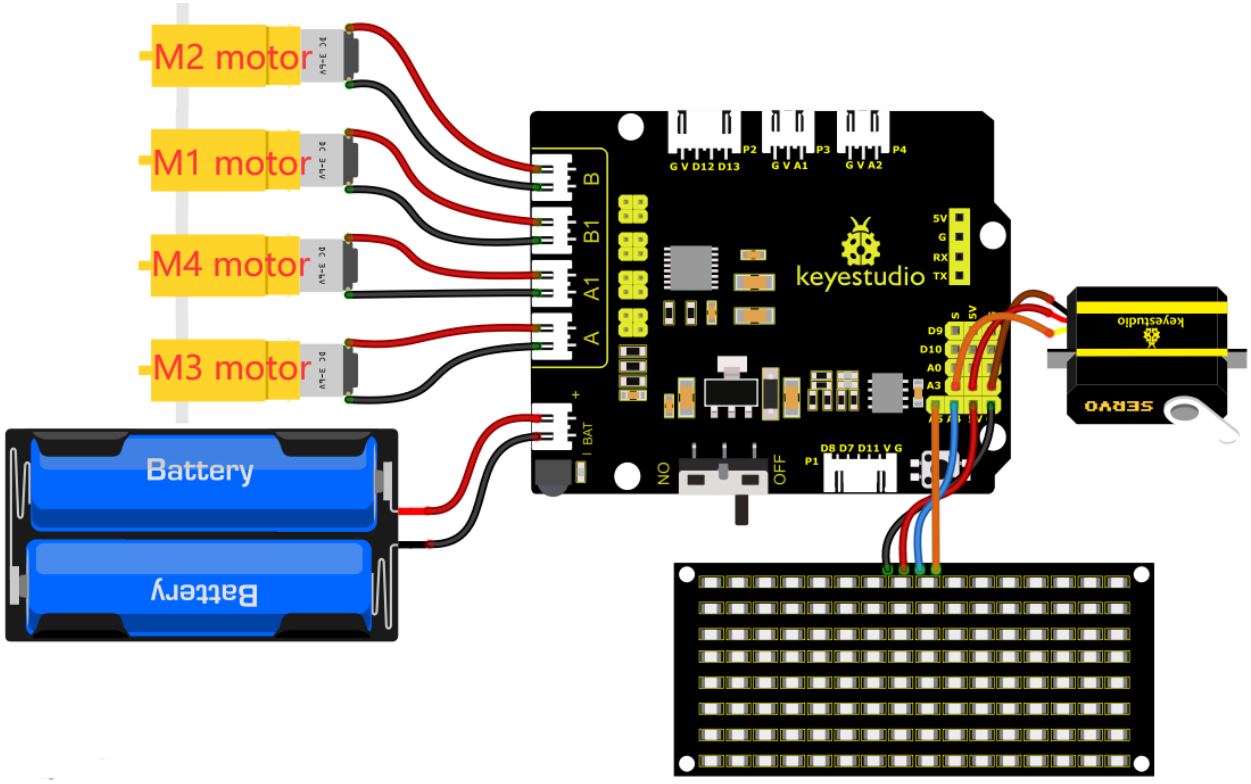
2.Flow Chart



The specific logic of IR remote control smart car is shown below:

Initial setup		LED board displays smile face
Remote control	Key value	Key state
	FF629D	Go front8*8 LED board shows front icon
	FFA857	Back8*8 LED board shows back icon
	FF22DD	Rotate to left8*8 LED board shows leftward icon
	FFC23D	Rotate to right8*8 LED board shows rightward icon
	FF02FD	Stop8*8 LED board shows“STOP”

3.Wiring Diagram



1). GND, VCC, SDA and SCL of the 8*8 LED board module are connected to GGND), VVCC), A4 and A5 of the expansion board.

2). As the IR receiver is integrated on the 8833 motor Shield, there is no need for additional wiring. The pins of the IR receiver on the 8833 board are G (GND), V (VCC) and D3 respectively.

3). The servo is connected to G, V and A3. The brown wire is interfaced with Gnd(G), the red wire is interfaced with 5V(V) and the orange wire is interfaced with A3.

4). The power is connected to the BAT port

4.Test Code

```
//*****************************************************************************
/*
keyestudio 4wd BT Car
lesson 14
IR remote Control Car
http://www.keyestudio.com
*/
#define SCL_Pin  A5  //Set the clock pin to A5
#define SDA_Pin  A4  //Set data pin to A4
//Array, used to store the data of pattern, can be calculated by yourself or obtained␣
↪from the modulus tool
unsigned char start01[] = {0x01,0x02,0x04,0x08,0x10,0x20,0x40,0x80,0x80,0x40,0x20,0x10,
↪0x08,0x04,0x02,0x01};
unsigned char front[] = {0x00,0x00,0x00,0x00,0x00,0x24,0x12,0x09,0x12,0x24,0x00,0x00,
↪0x00,0x00,0x00,0x00};
unsigned char back[] = {0x00,0x00,0x00,0x00,0x00,0x24,0x48,0x90,0x48,0x24,0x00,0x00,0x00,
↪0x00,0x00,0x00};
unsigned char left[] = {0x00,0x00,0x00,0x00,0x00,0x00,0x44,0x28,0x10,0x44,0x28,0x10,0x44,
↪0x28,0x10,0x00};
unsigned char right[] = {0x00,0x10,0x28,0x44,0x10,0x28,0x44,0x10,0x28,0x44,0x00,0x00,
↪0x00,0x00,0x00,0x00};
unsigned char STOP01[] = {0x2E,0x2A,0x3A,0x00,0x02,0x3E,0x02,0x00,0x3E,0x22,0x3E,0x00,
↪0x3E,0x0A,0x0E,0x00};
unsigned char clear[] = {0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
↪0x00,0x00,0x00,0x00};

#include <Arduino.h>
#include <IRremote.h>//function library of IR remote control
int RECV_PIN = 3;//set the pin of IR receiver to D3
IRrecv irrecv(RECV_PIN);
long irr_val;
decode_results results;

int left_ctrl = 2;//define the direction control pins of group B motor
int left_pwm = 5;//define the PWM control pins of group B motor
int right_ctrl = 4;//define the direction control pins of group A motor
int right_pwm = 6;//define the PWM control pins of group A motor

#include <Servo.h>
Servo servo_A3;//set the pin of servo to A3

unsigned char data_line = 0;
unsigned char delay_count = 0;
```

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```
void setup() {
  Serial.begin(9600);//
  // In case the interrupt driver crashes on setup, give a clue
  // to the user what's going on.
  Serial.println("Enabling IRin");
  irrecv.enableIRIn(); // Start the receiver
  Serial.println("Enabled IRin");
  pinMode(left_ctrl,OUTPUT);//set direction control pins of group B motor to OUTPUT
  pinMode(left_pwm,OUTPUT);//set PWM control pins of group B motor to OUTPUT
  pinMode(right_ctrl,OUTPUT);//set direction control pins of group A motor to OUTPUT
  pinMode(right_pwm,OUTPUT);//set PWM control pins of group A motor to OUTPUT
  servo_A3.attach(A3);
  servo_A3.write(90);//the angle of servo is 90 degree
  delay(300);
  pinMode(SCL_Pin,OUTPUT);// Set the clock pin to output
  pinMode(SDA_Pin,OUTPUT);//Set the data pin to output
  matrix_display(clear);
  matrix_display(start01); //display start01 expression pattern
}

void loop()
 {
  if (irrecv.decode(&results))
 {
    irr_val = results.value;
    Serial.println(irr_val, HEX);//serial prints the read IR remote signals
    switch(irr_val)
    {
      case 0xFF629D : car_front(); //Receive 0xFF629D,the car goes forward
      matrix_display(clear);
      matrix_display(front);
      break;

      case 0xFFA857 : car_back(); //Receive 0xFFA857,the car goes back
      matrix_display(clear);
      matrix_display(back);
      break;

      case 0xFF22DD : car_left(); //Receive 0xFF22DD,the car left rotates
      matrix_display(clear);
      matrix_display(left);
      break;

      case 0xFFC23D : car_right();//Receive 0xFFC23D,the car right rotates
      matrix_display(clear);
      matrix_display(right);
      break;

      case 0xFF02FD : car_Stop();//Receive 0xFF02FD,the car stops
      matrix_display(clear);
      matrix_display(STOP01);
      break;
```

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```
    }
        irrecv.resume(); // Receive the next value
  }
}

void car_front()//define the state of going front
{
  digitalWrite(left_ctrl,HIGH);
  analogWrite(left_pwm,105);
  digitalWrite(right_ctrl,HIGH);
  analogWrite(right_pwm,105);
}
void car_back()//define the status of going back
{
  digitalWrite(left_ctrl,LOW);
  analogWrite(left_pwm,150);
  digitalWrite(right_ctrl,LOW);
  analogWrite(right_pwm,150);
}
void car_left()//set the status of left turning
{
  digitalWrite(left_ctrl, LOW);
  analogWrite(left_pwm, 100);
  digitalWrite(right_ctrl, HIGH);
  analogWrite(right_pwm, 155);
}
void car_right()//set the status of right turning
{
  digitalWrite(left_ctrl, HIGH);
  analogWrite(left_pwm, 155);
  digitalWrite(right_ctrl, LOW);
  analogWrite(right_pwm, 100);
}
void car_Stop()//define the state of stop
{
  digitalWrite(left_ctrl,LOW);
  analogWrite(left_pwm,0);
  digitalWrite(right_ctrl,LOW);
  analogWrite(right_pwm,0);
}

//this function is used for dot matrix display
void matrix_display(unsigned char matrix_value[])
{
  IIC_start();  //the function that calls the data transfer start condition
  IIC_send(0xc0);  //select address

  for (int i = 0; i < 16; i++) //the pattern data is 16 bytes
  {
    IIC_send(matrix_value[i]); //Transmit the data of the pattern
  }
  IIC_end();   //End pattern data transmission
```

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```
  IIC_start();
  IIC_send(0x8A);  //Display control, select 4/16 pulse width
  IIC_end();
}
//Conditions under which data transmission begins
void IIC_start()
{
  digitalWrite(SDA_Pin, HIGH);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, LOW);
}
//Indicates the end of data transmission
void IIC_end()
{
  digitalWrite(SCL_Pin, LOW);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, HIGH);
  delayMicroseconds(3);
}
//transmit data
void IIC_send(unsigned char send_data)
{
  for (byte mask = 0x01; mask != 0; mask <<= 1) //Each byte has 8 bits and is checked␣
↪bit by bit starting at the lowest level
  {
    if (send_data & mask) { //Sets the high and low levels of SDA_Pin depending on␣
↪whether each bit of the byte is a 1 or a 0
      digitalWrite(SDA_Pin, HIGH);
    } else {
      digitalWrite(SDA_Pin, LOW);
    }
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, HIGH); //Pull the clock pin SCL_Pin high to stop data␣
↪transmission
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, LOW); //pull the clock pin SCL_Pin low to change the SIGNAL of␣
↪SDA
  }
}
//*****************************************************************************
```

5.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, power on the external power then turn the DIP switch to ON. Then we enable to use the IR remote control drive the car to move to and the 8X16 LED board will display the corresponding status pattern.

9.2.15 Project 15 Bluetooth Control Smart Car

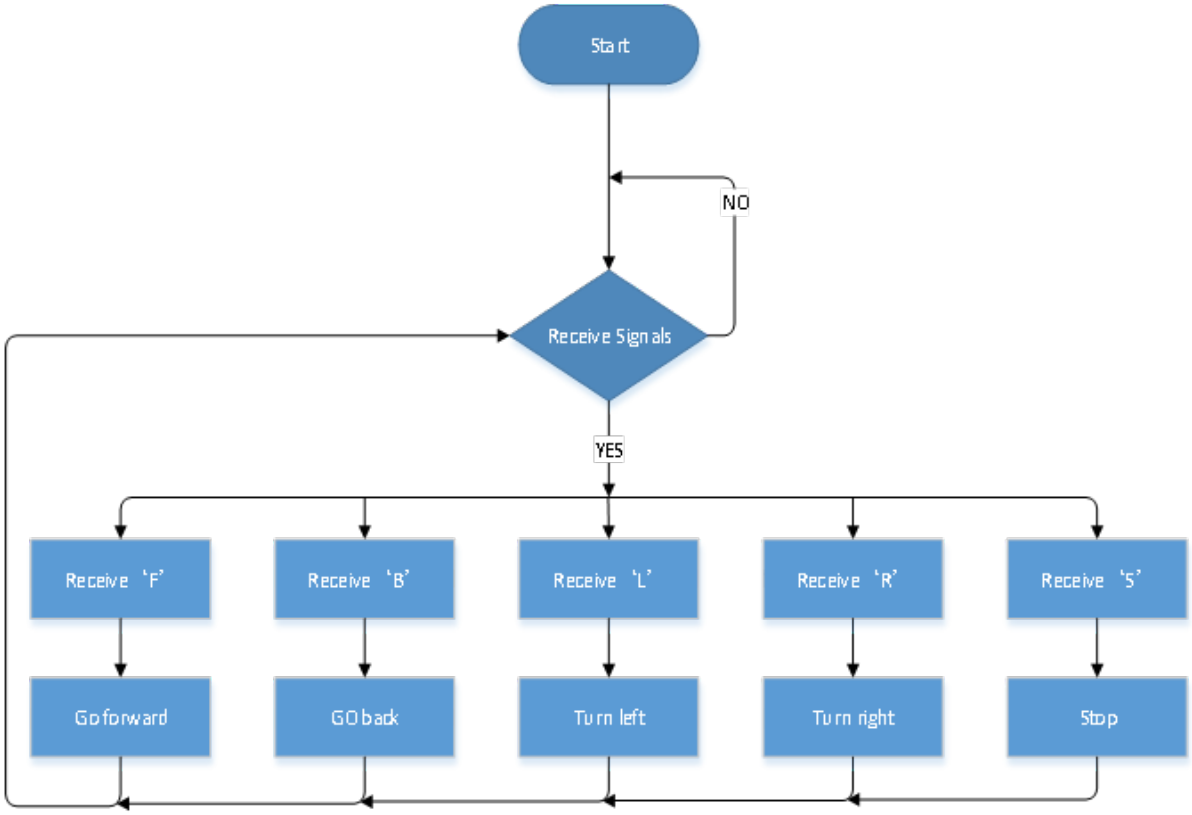
1.Description

We've learned the basic knowledge of Bluetooth. And in this lesson, we will make a Bluetooth control smart car. In this project, we aim to regard the mobile phone as the transmitter (host), and the smart car connected to the BT24 Bluetooth module (slave) as the receiver and use the mobile APP to control the smart car via the Bluetooth.

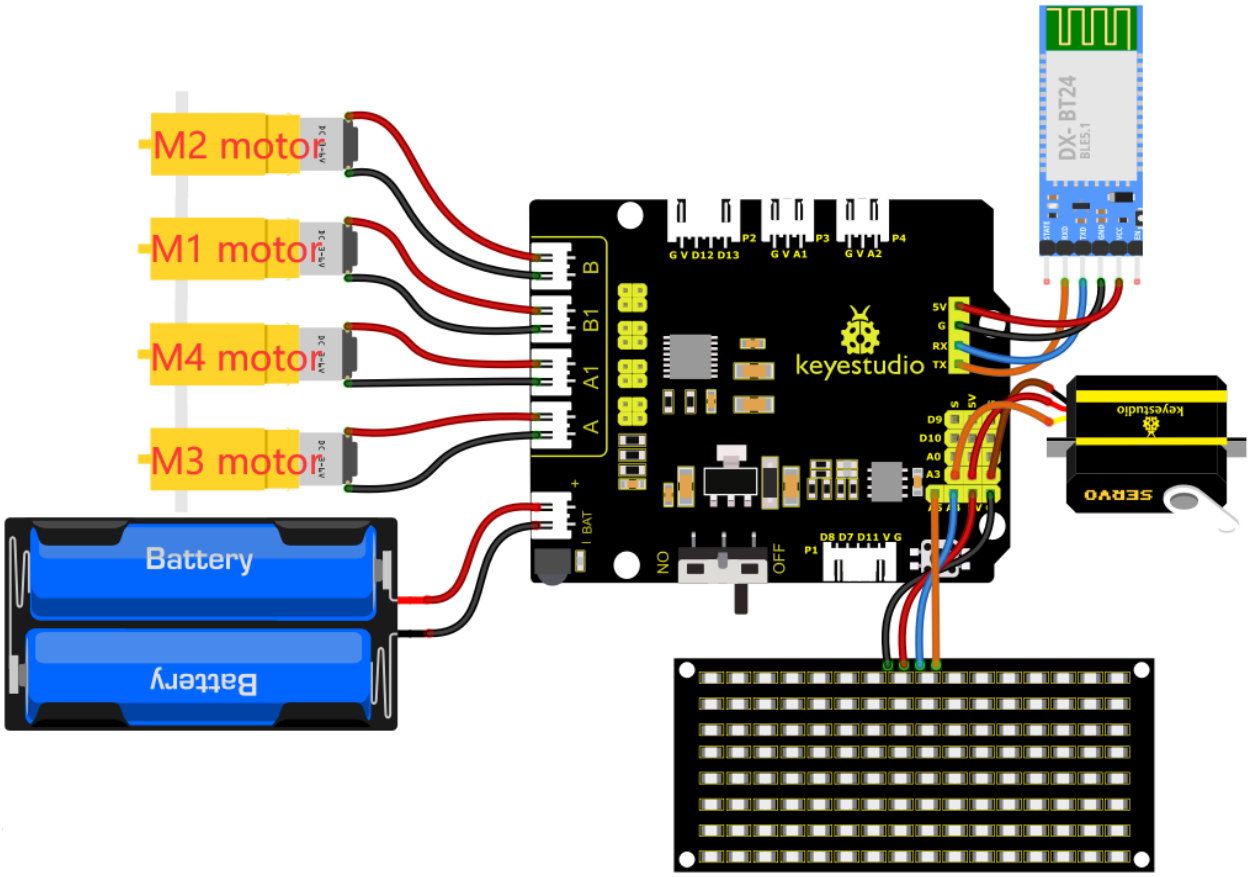
2.APP Control Button

	Control character	Control character
	Press: F Release: S	Press the button, the car goes front; release to stop
	Press: L Release: S	Press the button, the car turns left; release to stop
	Press: R Release: S	Press the button, the car turns right; release to stop
	Press: B Release: S	Press the button, the car goes back; release to stop
	Press: “a” Release: “S”	Click to speed up(maximum:255)
	Press: “d” Release: “S”	Click to slow down(minimum:0)
	Click to start the gravity sensing function of the mobile phone: click again to exit the gravity sensing control	
	Click to send“X”, click again to send“S”	Start line tracking function; click again to exit
	Click to send“Y”, click again to send“S”	Start ultrasonic avoiding function; click again to exit
	Click to send“U”, click again to send“S”	Start ultrasonic follow function; click again to exit
	Click to send“G”, click again to send“S”	Start restricting function; click again to exit

3.Flow Chart


Start
NO
Receive Signals
YES
Receive 'F'
Receive 'B'
Receive 'L'
Receive 'R'
Receive 'S'
Go forward
GO back
Turn left
Turn right
Stop

4.Wiring Diagram



1). GND, VCC, SDA and SCL of the 8*8 LED board are connected to GGND), VVCC), A4 and A5 of the expansion board.

2). The RXD, TXD, GND and VCC of the Bluetooth module are respectively connected to TX, RX, G and 5V on the 8833 motor Shield, while the STATE and BRK pins of the Bluetooth module do not need to be connected.

3). The servo is connected to G, V and A3. The brown wire is interfaced with Gnd(G), the red wire is interfaced with 5V(V) and the orange wire is interfaced with A3.

4). The power is connected to the BAT port

5.Test Code

Note: Before uploading the test code, you need to remove the Bluetooth module, otherwise the code will fail to be uploaded.Connect the Bluetooth module after uploading the code successfully.

```
//*****************************************************************************
/*
keyestudio 4wd BT Car
lesson 15
Bluetooth Control Car
http://www.keyestudio.com
*/
#define SCL_Pin  A5  //Set the clock pin to A5
#define SDA_Pin  A4  //Set data pin to A4
```

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```
//Array, used to store the data of pattern, can be calculated by yourself or obtained␣
↪from the modulus tool
unsigned char start01[] = {0x01,0x02,0x04,0x08,0x10,0x20,0x40,0x80,0x80,0x40,0x20,0x10,
↪0x08,0x04,0x02,0x01};
unsigned char front[] = {0x00,0x00,0x00,0x00,0x00,0x24,0x12,0x09,0x12,0x24,0x00,0x00,
↪0x00,0x00,0x00,0x00};
unsigned char back[] = {0x00,0x00,0x00,0x00,0x00,0x24,0x48,0x90,0x48,0x24,0x00,0x00,0x00,
↪0x00,0x00,0x00};
unsigned char left[] = {0x00,0x00,0x00,0x00,0x00,0x00,0x44,0x28,0x10,0x44,0x28,0x10,0x44,
↪0x28,0x10,0x00};
unsigned char right[] = {0x00,0x10,0x28,0x44,0x10,0x28,0x44,0x10,0x28,0x44,0x00,0x00,
↪0x00,0x00,0x00,0x00};
unsigned char STOP01[] = {0x2E,0x2A,0x3A,0x00,0x02,0x3E,0x02,0x00,0x3E,0x22,0x3E,0x00,
↪0x3E,0x0A,0x0E,0x00};
unsigned char clear[] = {0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
↪0x00,0x00,0x00};

int left_ctrl = 2;//define the direction control pins of group B motor
int left_pwm = 5;//define the PWM control pins of group B motor
int right_ctrl = 4;//define the direction control pins of group A motor
int right_pwm = 6;//define the PWM control pins of group A motor

const int servopin = A3;//set the pin of servo to A3

char BLE_val;

void setup() {
  Serial.begin(9600);//
  pinMode(left_ctrl,OUTPUT);//set direction control pins of group B motor to OUTPUT
  pinMode(left_pwm,OUTPUT);//set PWM control pins of group B motor to OUTPUT
  pinMode(right_ctrl,OUTPUT);//set direction control pins of group A motor to OUTPUT
  pinMode(right_pwm,OUTPUT);//set PWM control pins of group A motor to OUTPUT
  servopulse(servopin,90);//the angle of servo is 90 degree
  delay(300);
  pinMode(SCL_Pin,OUTPUT);// Set the clock pin to output
  pinMode(SDA_Pin,OUTPUT);//Set the data pin to output
  matrix_display(clear);
  matrix_display(start01); //display start01 expression pattern
}

void loop() {
   if(Serial.available()>0) {
    BLE_val = Serial.read();
    Serial.println(BLE_val);
  }
    switch(BLE_val)
    {
      case 'F' : car_front(); //Receive 'F',the car goes forward
      matrix_display(clear);
      matrix_display(front);
      break;
```

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```
      case 'B' : car_back(); //Receive 'B',the car goes back
      matrix_display(clear);
      matrix_display(back);
      break;

      case 'L' : car_left(); //Receive 'L',the car left rotates
      matrix_display(clear);
      matrix_display(left);
      break;

      case 'R' : car_right();//Receive 'R',the car right rotates
      matrix_display(clear);
      matrix_display(right);
      break;

      case 'S' : car_Stop();//Receive 'S',the car stops
      matrix_display(clear);
      matrix_display(STOP01);
      break;
}
}

void car_front()//define the state of going front
{
  digitalWrite(left_ctrl,HIGH);
  analogWrite(left_pwm,155);
  digitalWrite(right_ctrl,HIGH);
  analogWrite(right_pwm,155);
}
void car_back()//define the status of going back
{
  digitalWrite(left_ctrl,LOW);
  analogWrite(left_pwm,100);
  digitalWrite(right_ctrl,LOW);
  analogWrite(right_pwm,100);
}
void car_left()//set the status of left turning
{
  digitalWrite(left_ctrl, LOW);
  analogWrite(left_pwm, 100);
  digitalWrite(right_ctrl, HIGH);
  analogWrite(right_pwm, 155);
}
void car_right()//set the status of right turning
{
  digitalWrite(left_ctrl, HIGH);
  analogWrite(left_pwm, 155);
  digitalWrite(right_ctrl, LOW);
  analogWrite(right_pwm, 100);
}
void car_Stop()//define the state of stop
{
```

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```
  digitalWrite(left_ctrl,LOW);
  analogWrite(left_pwm,0);
  digitalWrite(right_ctrl,LOW);
  analogWrite(right_pwm,0);
}

void servopulse(int servopin,int myangle)//Steering gear running angle
{
  for(int i=0; i<30; i++)
  {
    int pulsewidth = (myangle*11)+500;
    digitalWrite(servopin,HIGH);
    delayMicroseconds(pulsewidth);
    digitalWrite(servopin,LOW);
    delay(20-pulsewidth/1000);
  }
}

//this function is used for dot matrix display
void matrix_display(unsigned char matrix_value[])
{
  IIC_start();  //the function that calls the data transfer start condition
  IIC_send(0xc0);  //select address

  for (int i = 0; i < 16; i++) //the pattern data is 16 bytes
  {
    IIC_send(matrix_value[i]); //Transmit the data of the pattern
  }
  IIC_end();   //End pattern data transmission
  IIC_start();
  IIC_send(0x8A);  //Display control, select 4/16 pulse width
  IIC_end();
}
//Conditions under which data transmission begins
void IIC_start()
{
  digitalWrite(SDA_Pin, HIGH);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, LOW);
}
//Indicates the end of data transmission
void IIC_end()
{
  digitalWrite(SCL_Pin, LOW);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, HIGH);
```

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```
  delayMicroseconds(3);
}
//transmit data
void IIC_send(unsigned char send_data)
{
  for (byte mask = 0x01; mask != 0; mask <<= 1) //Each byte has 8 bits and is checked␣
↪bit by bit starting at the lowest level
  {
    if (send_data & mask) { //Sets the high and low levels of SDA_Pin depending on␣
↪whether each bit of the byte is a 1 or a 0
      digitalWrite(SDA_Pin, HIGH);
    } else {
      digitalWrite(SDA_Pin, LOW);
    }
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, HIGH); //Pull the clock pin SCL_Pin high to stop data␣
↪transmission
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, LOW); //pull the clock pin SCL_Pin low to change the SIGNAL of␣
↪SDA
  }
}
//*****************************************************************************
```

6.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, power on the external power then turn the DIP switch to ON.

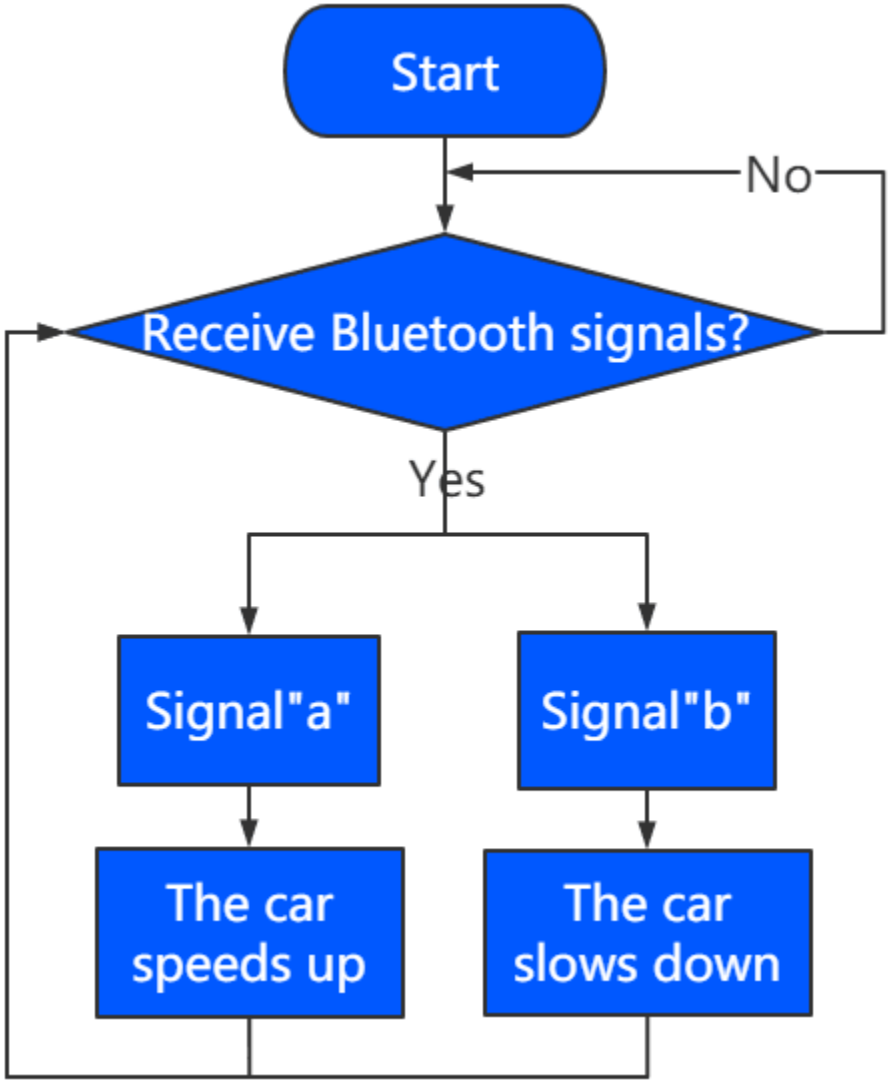
Inset the BT module and open your cellphone to connect the Bluetooth to control the smart car. The can will move forward, backward, turn left and right and stop. Also the 8*8 LED board will show the corresponding patterns.

9.2.16 Project 16 Bluetooth Speed Control Smart Car

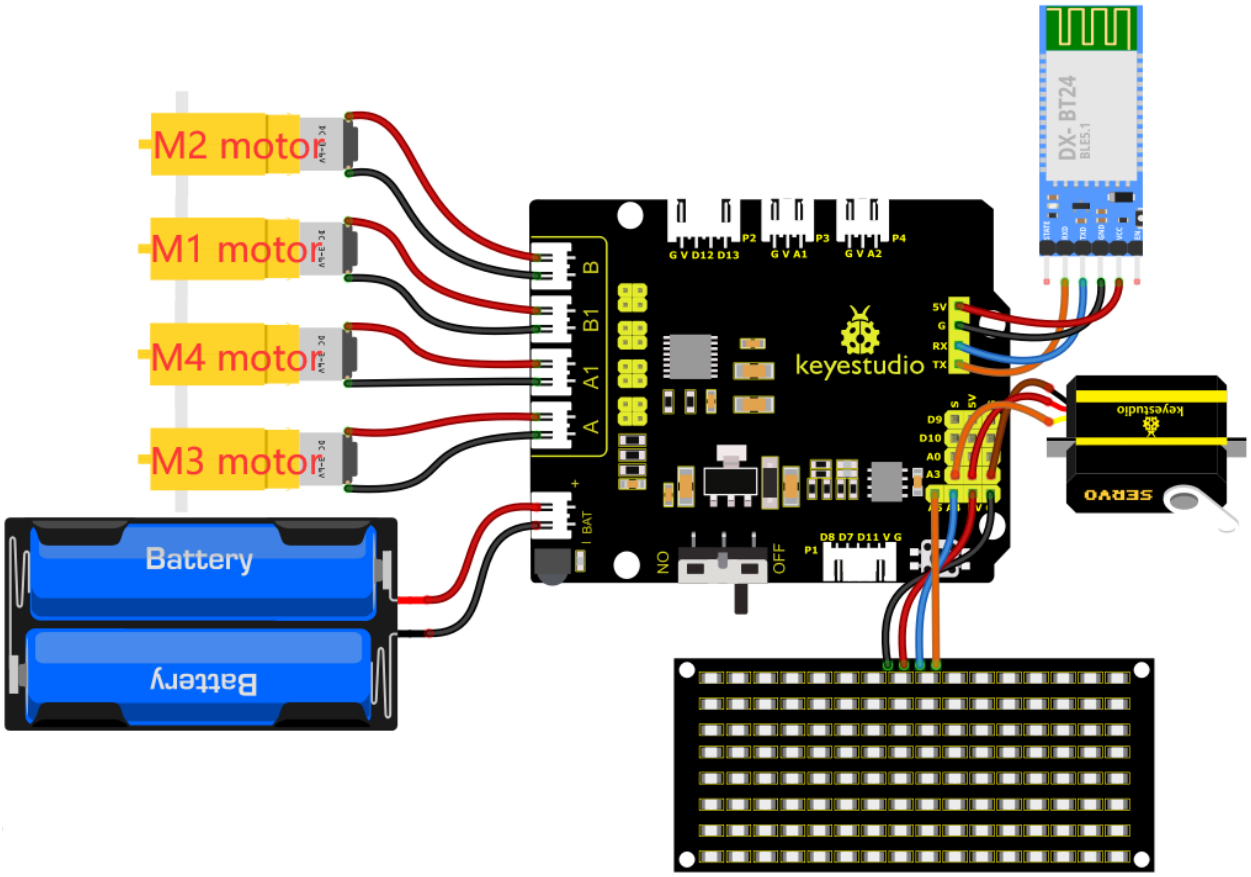
1.Description

In this project, we will use a Bluetooth to adjust the speed of the smart car. We empower to define a variable speeds and change it to change the speed of the smart car.

2.Flow Chart


Start
No
Receive Bluetooth signals?
Yes
Signal"a"
Signal"b"
The car speeds up
The car slows down

3.Wiring Diagram



1). GND, VCC, SDA and SCL of the 8*8 LED board are connected to GGND), VVCC), A4 and A5 of the expansion board.

2). The RXD, TXD, GND and VCC of the Bluetooth module are respectively connected to TX, RX, G and 5V on the 8833 motor Shield, while the STATE and BRK pins of the Bluetooth module do not need to be connected.

3). The servo is connected to G, V and A3. The brown wire is interfaced with Gnd(G), the red wire is interfaced with 5V(V) and the orange wire is interfaced with A3.

4). The power is connected to the BAT port

4.Test Code

Note: Before uploading the test code, you need to remove the Bluetooth module, otherwise the code will fail to be uploaded.Connect the Bluetooth module after uploading the code successfully.

```
//*****************************************************************************
/*
keyestudio 4wd BT Car
lesson 16
Bluetooth Speed Control Car
http://www.keyestudio.com
*/
#define SCL_Pin  A5  //Set the clock pin to A5
#define SDA_Pin  A4  //Set data pin to A4
```

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```
//Array, used to store the data of pattern, can be calculated by yourself or obtained␣
↪from the modulus tool
unsigned char start01[] = {0x01,0x02,0x04,0x08,0x10,0x20,0x40,0x80,0x80,0x40,0x20,0x10,
↪0x08,0x04,0x02,0x01};
unsigned char front[] = {0x00,0x00,0x00,0x00,0x00,0x24,0x12,0x09,0x12,0x24,0x00,0x00,
↪0x00,0x00,0x00,0x00};
unsigned char back[] = {0x00,0x00,0x00,0x00,0x00,0x24,0x48,0x90,0x48,0x24,0x00,0x00,0x00,
↪0x00,0x00,0x00};
unsigned char left[] = {0x00,0x00,0x00,0x00,0x00,0x00,0x44,0x28,0x10,0x44,0x28,0x10,0x44,
↪0x28,0x10,0x00};
unsigned char right[] = {0x00,0x10,0x28,0x44,0x10,0x28,0x44,0x10,0x28,0x44,0x00,0x00,
↪0x00,0x00,0x00,0x00};
unsigned char STOP01[] = {0x2E,0x2A,0x3A,0x00,0x02,0x3E,0x02,0x00,0x3E,0x22,0x3E,0x00,
↪0x3E,0x0A,0x0E,0x00};
unsigned char clear[] = {0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
↪0x00,0x00,0x00,0x00};
unsigned char speed_a[] =
{0x00,0x40,0x20,0x10,0x08,0x04,0x02,0xff,0x02,0x04,0x08,0x10,0x20,0x40,0x00,0x00};
unsigned char speed_d[] =
{0x00,0x02,0x04,0x08,0x10,0x20,0x40,0xff,0x40,0x20,0x10,0x08,0x04,0x02,0x00,0x00};

int left_ctrl = 2;//define the direction control pins of group B motor
int left_pwm = 5;//define the PWM control pins of group B motor
int right_ctrl = 4;//define the direction control pins of group A motor
int right_pwm = 6;//define the PWM control pins of group A motor

int speeds = 150; //Set the initial speed to 150

const int servopin = A3;//set the pin of servo to A3

char BLE_val;

void setup() {
  Serial.begin(9600);//
  pinMode(left_ctrl,OUTPUT);//set direction control pins of group B motor to OUTPUT
  pinMode(left_pwm,OUTPUT);//set PWM control pins of group B motor to OUTPUT
  pinMode(right_ctrl,OUTPUT);//set direction control pins of group A motor to OUTPUT
  pinMode(right_pwm,OUTPUT);//set PWM control pins of group A motor to OUTPUT
  servopulse(servopin,90);//the angle of servo is 90 degree
  delay(300);
  pinMode(SCL_Pin,OUTPUT);// Set the clock pin to output
  pinMode(SDA_Pin,OUTPUT);//Set the data pin to output
  matrix_display(clear);
  matrix_display(start01); //display start01 expression pattern
}

void loop() {
   if(Serial.available()>0) {
    BLE_val = Serial.read();
    Serial.println(BLE_val);
  }
    switch(BLE_val)
```

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```
    {
      case 'F' : car_front();
      matrix_display(clear);
      matrix_display(front);
      break;

      case 'B' : car_back();
      matrix_display(clear);
      matrix_display(back);
      break;

      case 'L' : car_left();
      matrix_display(clear);
      matrix_display(left);
      break;

      case 'R' : car_right();
      matrix_display(clear);
      matrix_display(right);
      break;

      case 'S' : car_Stop();
      matrix_display(clear);
      matrix_display(STOP01);
      break;

      case 'a' : speeds_a();
      matrix_display(clear);
      matrix_display(speed_a);
      break;

      case 'd' : speeds_d();
      matrix_display(clear);
      matrix_display(speed_d);
      break;
    }
}

void car_front()//define the state of going front
{
  digitalWrite(left_ctrl,HIGH);
  analogWrite(left_pwm,(255-speeds));
  digitalWrite(right_ctrl,HIGH);
  analogWrite(right_pwm,(255-speeds));
}
void car_back()//define the status of going back
{
  digitalWrite(left_ctrl,LOW);
  analogWrite(left_pwm,speeds);
  digitalWrite(right_ctrl,LOW);
  analogWrite(right_pwm,speeds);
}
```

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```
void car_left()//set the status of left turning
{
  digitalWrite(left_ctrl, LOW);
  analogWrite(left_pwm, speeds);
  digitalWrite(right_ctrl, HIGH);
  analogWrite(right_pwm, (255-speeds));
}
void car_right()//set the status of right turning
{
  digitalWrite(left_ctrl, HIGH);
  analogWrite(left_pwm, (255-speeds));
  digitalWrite(right_ctrl, LOW);
  analogWrite(right_pwm, speeds);
}
void car_Stop()//define the state of stop
{
  digitalWrite(left_ctrl,LOW);
  analogWrite(left_pwm,0);
  digitalWrite(right_ctrl,LOW);
  analogWrite(right_pwm,0);
}

void speeds_a() { //rapidly growing function
  while (1) {
    Serial.println(speeds);  //display speed information
    if (speeds < 255) { //Up to 255
      matrix_display(clear);
      matrix_display(speed_a);
      speeds++;
      delay(10);  //adjust the speed of growth
    }
    BLE_val = Serial.read();
    if (BLE_val == 'S') //Receive 'S',the car stops accelerating
    break;
  }
}
void speeds_d() { //velocity reduction function
  while (1) {
    Serial.println(speeds);  //display speed information
    if (speeds > 0) { //down to 0
      matrix_display(clear);
      matrix_display(speed_d);
      speeds--;
      delay(10);    //adjust the speed of deceleration
    }
    BLE_val = Serial.read();
    if (BLE_val == 'S') //Receive 'S',the car stops deceleration
    break;
}
}

void servopulse(int servopin,int myangle)//Steering gear running angle
```

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```
{
  for(int i=0; i<30; i++)
  {
    int pulsewidth = (myangle*11)+500;
    digitalWrite(servopin,HIGH);
    delayMicroseconds(pulsewidth);
    digitalWrite(servopin,LOW);
    delay(20-pulsewidth/1000);
  }
}

//this function is used for dot matrix display
void matrix_display(unsigned char matrix_value[])
{
  IIC_start();  //the function that calls the data transfer start condition
  IIC_send(0xc0);  //select address

  for (int i = 0; i < 16; i++) //the pattern data is 16 bytes
  {
    IIC_send(matrix_value[i]); //Transmit the data of the pattern
  }
  IIC_end();   //End pattern data transmission
  IIC_start();
  IIC_send(0x8A);  //Display control, select 4/16 pulse width
  IIC_end();
}
//Conditions under which data transmission begins
void IIC_start()
{
  digitalWrite(SDA_Pin, HIGH);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, LOW);
}
//Indicates the end of data transmission
void IIC_end()
{
  digitalWrite(SCL_Pin, LOW);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, HIGH);
  delayMicroseconds(3);
}
//transmit data
void IIC_send(unsigned char send_data)
{
  for (byte mask = 0x01; mask != 0; mask <<= 1) //Each byte has 8 bits and is checked␣
↪bit by bit starting at the lowest level
```

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```
  {
    if (send_data & mask) { //Sets the high and low levels of SDA_Pin depending on␣
↪whether each bit of the byte is a 1 or a 0
      digitalWrite(SDA_Pin, HIGH);
    } else {
      digitalWrite(SDA_Pin, LOW);
    }
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, HIGH); //Pull the clock pin SCL_Pin high to stop data␣
↪transmission
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, LOW); //pull the clock pin SCL_Pin low to change the SIGNAL of␣
↪SDA
  }
}
//*****************************************************************************
```

5.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, power on the external power then turn the DIP switch to ON. Pairing the APP with Bluetooth, the smart car can be controlled to move by the APP.

Press

, the car will speed up, press

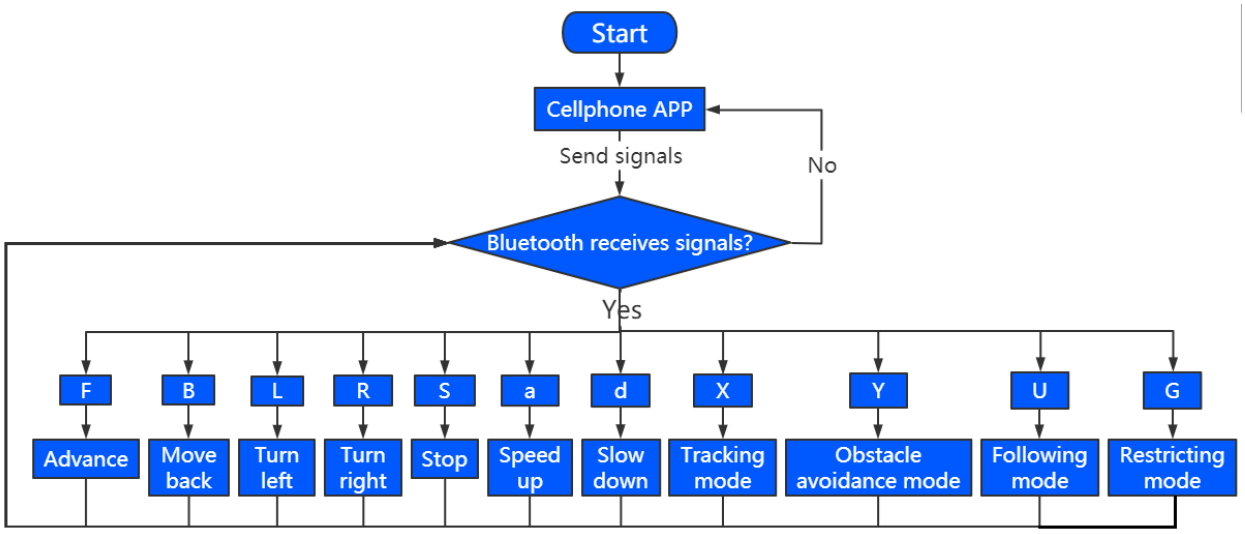
, the car will slow down, and the 8*16 LED board will display the corresponding status pattern of the smart car.

9.2.17 Project 17 Multi-purpose Bluetooth Smart Car

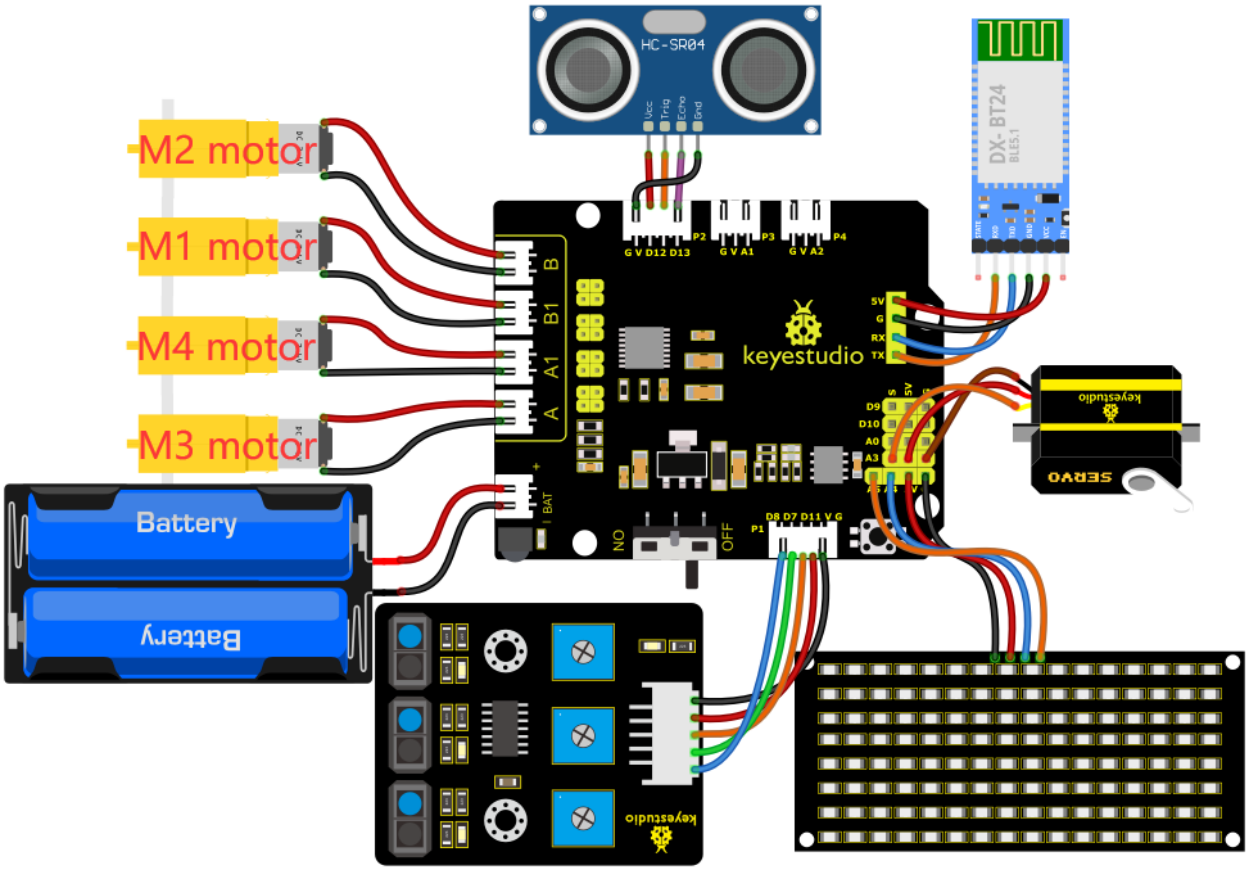
1.Description

In previous projects, the car only performs a single function. However, in this lesson, we will integrate all of its functions via a Bluetooth.

2.Flow Chart



3.Wiring Diagram



1). GND, VCC, SDA and SCL of the 8*8 LED board are connected to GGND), VVCC), A4 and A5 of the expansion board.

2). The RXD, TXD, GND and VCC of the Bluetooth module are respectively connected to TX, RX, G and 5V on the 8833 motor Shield, while the STATE and BRK pins of the Bluetooth module do not need to be connected.

3). The servo is connected to G, V and A3. The brown wire is interfaced with Gnd(G), the red wire is interfaced with 5V(V) and the orange wire is interfaced with A3.

4). G, V, S1, S2 and S3 of the line tracking sensor are connected to GGND), VVCC), D11, D7 and D8 of the sensor expansion board.

5). VCC, Trig, Echo and Gnd of the ultrasonic sensor are connected to 5V(V), D12(S), D13(S) and Gnd(G).

6). The power is connected to the BAT port

4.Test Code

Note: Before uploading the test code, you need to remove the Bluetooth module, otherwise the code will fail to be uploaded.Connect the Bluetooth module after uploading the code successfully.

```
//*****************************************************************************
/*
keyestudio 4wd BT Car
lesson 17
Bluetooth Multifunctional Car
http://www.keyestudio.com
```

(continues on next page)

(continued from previous page)

```
*/
#define SCL_Pin  A5  //Set the clock pin to A5
#define SDA_Pin  A4  //Set data pin to A4
//Array, used to store the data of pattern, can be calculated by yourself or obtained␣
↪from the modulus tool
unsigned char start01[] = {0x01,0x02,0x04,0x08,0x10,0x20,0x40,0x80,0x80,0x40,0x20,0x10,
↪0x08,0x04,0x02,0x01};
unsigned char front[] = {0x00,0x00,0x00,0x00,0x00,0x24,0x12,0x09,0x12,0x24,0x00,0x00,
↪0x00,0x00,0x00,0x00};
unsigned char back[] = {0x00,0x00,0x00,0x00,0x00,0x24,0x48,0x90,0x48,0x24,0x00,0x00,0x00,
↪0x00,0x00,0x00};
unsigned char left[] = {0x00,0x00,0x00,0x00,0x00,0x00,0x44,0x28,0x10,0x44,0x28,0x10,0x44,
↪0x28,0x10,0x00};
unsigned char right[] = {0x00,0x10,0x28,0x44,0x10,0x28,0x44,0x10,0x28,0x44,0x00,0x00,
↪0x00,0x00,0x00,0x00};
unsigned char STOP01[] = {0x2E,0x2A,0x3A,0x00,0x02,0x3E,0x02,0x00,0x3E,0x22,0x3E,0x00,
↪0x3E,0x0A,0x0E,0x00};
unsigned char clear[] = {0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
↪0x00,0x00,0x00,0x00};
unsigned char speed_a[] =
{0x00,0x40,0x20,0x10,0x08,0x04,0x02,0xff,0x02,0x04,0x08,0x10,0x20,0x40,0x00,0x00};
unsigned char speed_d[] =
{0x00,0x02,0x04,0x08,0x10,0x20,0x40,0xff,0x40,0x20,0x10,0x08,0x04,0x02,0x00,0x00};

int left_ctrl = 2;//define the direction control pins of group B motor
int left_pwm = 5;//define the PWM control pins of group B motor
int right_ctrl = 4;//define the direction control pins of group A motor
int right_pwm = 6;//define the PWM control pins of group A motor
int speeds = 150; //Set the initial speed to 150

const int servopin = A3;//set the pin of servo to A3

int L_pin = 11; //define the left tracking sensor pin as D11
int M_pin = 7; //define the middle tracking sensor pin as D7
int R_pin = 8; //define the right tracking sensor pin as D8
int L_val, M_val, R_val;

int trigPin = 12; //TRIG Pin be connected to D12
int echoPin = 13; //ECHO Pin be connected to D13
int distance, distance_l, distance_r;

char BLE_val;

void setup() {
  Serial.begin(9600);//Set baud rate to 9600
  pinMode(left_ctrl,OUTPUT);//set direction control pins of group B motor to OUTPUT
  pinMode(left_pwm,OUTPUT);//set PWM control pins of group B motor to OUTPUT
  pinMode(right_ctrl,OUTPUT);//set direction control pins of group A motor to OUTPUT
  pinMode(right_pwm,OUTPUT);//set PWM control pins of group A motor to OUTPUT
  servopulse(servopin,90);//the angle of servo is 90 degree
  delay(300);
  pinMode(L_pin, INPUT); //Tracking sensor pins are configured for input mode
```

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```
  pinMode(M_pin, INPUT);
  pinMode(R_pin, INPUT);
  pinMode(trigPin, OUTPUT); //define TRIG as the output mode
  pinMode(echoPin, INPUT); //define ECHO as the input mode
  pinMode(SCL_Pin,OUTPUT);// Set the clock pin to output
  pinMode(SDA_Pin,OUTPUT);//Set the data pin to output
  matrix_display(clear);
  matrix_display(start01); //display start01 expression pattern
}

void loop() {
   if(Serial.available()>0) {
    BLE_val = Serial.read();
    Serial.println(BLE_val);
  }
    switch(BLE_val)
    {
      case 'F' : car_front();
      matrix_display(clear);
      matrix_display(front);
      break;

      case 'B' : car_back();
      matrix_display(clear);
      matrix_display(back);
      break;

      case 'L' : car_left();
      matrix_display(clear);
      matrix_display(left);
      break;

      case 'R' : car_right();
      matrix_display(clear);
      matrix_display(right);
      break;

      case 'S' : car_Stop();
      matrix_display(clear);
      matrix_display(STOP01);
      break;

      case 'a' : speeds_a();
      matrix_display(clear);
      matrix_display(speed_a);
      break;

      case 'd' : speeds_d();
      matrix_display(clear);
      matrix_display(speed_d);
      break;
```

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```
      case  'U':  follow();  //Receiving 'U',enter follow mode
      break;
      case  'Y':  avoid(); //Receiving 'Y',enter obstacle avoidance mode
      break;
      case  'G':  confinement(); //Receiving 'G',enter confinement mode
      break;
      case  'X':  tracking(); //Receiving 'X',enter tracking mode
      break;
    }
}

void car_front()//define the state of going front
{
  digitalWrite(left_ctrl,HIGH);
  analogWrite(left_pwm,(255-speeds));
  digitalWrite(right_ctrl,HIGH);
  analogWrite(right_pwm,(255-speeds));
}
void car_back()//define the status of going back
{
  digitalWrite(left_ctrl,LOW);
  analogWrite(left_pwm,speeds);
  digitalWrite(right_ctrl,LOW);
  analogWrite(right_pwm,speeds);
}
void car_left()//set the status of left turning
{
  digitalWrite(left_ctrl, LOW);
  analogWrite(left_pwm, speeds);
  digitalWrite(right_ctrl, HIGH);
  analogWrite(right_pwm, (255-speeds));
}
void car_right()//set the status of right turning
{
  digitalWrite(left_ctrl, HIGH);
  analogWrite(left_pwm, (255-speeds));
  digitalWrite(right_ctrl, LOW);
  analogWrite(right_pwm, speeds);
}
void car_Stop()//define the state of stop
{
  digitalWrite(left_ctrl,LOW);
  analogWrite(left_pwm,0);
  digitalWrite(right_ctrl,LOW);
  analogWrite(right_pwm,0);
}

void speeds_a() { //rapidly growing function
  while (1) {
    Serial.println(speeds);  //display speed information
    if (speeds < 255) { //Up to 255
      matrix_display(clear);
```

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```
      matrix_display(speed_a);
      speeds++;
      delay(10);  //adjust the speed of growth
    }
    BLE_val = Serial.read();
    if (BLE_val == 'S') //Receive 'S',the car stops accelerating
    break;
  }
}
void speeds_d() { //velocity reduction function
  while (1) {
    Serial.println(speeds);  //display speed information
    if (speeds > 0) { //down to 0
      matrix_display(clear);
      matrix_display(speed_d);
      speeds--;
      delay(10);    //adjust the speed of deceleration
    }
    BLE_val = Serial.read();
    if (BLE_val == 'S') //Receive 'S',the car stops deceleration
    break;
}
}

int get_distance() {
  int distance = 0;
  digitalWrite(trigPin, LOW);     // send pulse through Trig/Pin, trigger HC-SR04␣
↪ranging, so that send out ultrasonic signal interface low level 2s
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);    // make ultrasonic signal interface high level 10s,␣
↪here is at least 10s
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);     // keep the ultrasonic signal interface low level
  distance = pulseIn(echoPin, HIGH) / 58; // read the pulse time and convert the pulse␣
↪time to the distance (unit: cm)
  Serial.println(distance);        //output distance value
  return distance;
}

void follow() {
  servopulse(servopin,90);
  delay(200);
  int follow_flag = 1;
  while (follow_flag) {
    distance = get_distance(); //call the ranging function
    if (distance < 8 ) {//If the distance is less than 8
      car_back();//the car goes back
      matrix_display(clear);
      matrix_display(back);
    }
    else if (distance >= 8 && distance < 13) { //If the distance is greater than or␣
↪equal to 8, it's less than 13
```

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```
      car_Stop();//stop
      matrix_display(clear);
      matrix_display(STOP01);
    }
    else if (distance >= 13 && distance <= 35 ) { //If the distance is greater than or equal to 13, it's less than 35
      car_front();//the car goes forward
      matrix_display(clear);
      matrix_display(front);
    }
    else {//If none of the above
      car_Stop();//stop
      matrix_display(clear);
      matrix_display(STOP01);
    }
    BLE_val = Serial.read();
    if (BLE_val == 'S') { //When S is received, the car stops
      follow_flag = 0;
      car_Stop();
    }
  }
}

void avoid() {
  int avoid_flag = 1;
  while (avoid_flag) {
    distance = get_distance(); //Call the ranging function
    if (distance > 0 && distance < 20) { //If the distance is less than 20 and greater than 0
      car_Stop();//stops
      matrix_display(clear);
      matrix_display(STOP01);   //the dot matrix displays a stop pattern
      delay(1000);
      servopulse(servopin,160); //bring the steering gear over 180 degrees
      delay(500);
      distance_l = get_distance(); //gets the left distance
      delay(100);
      servopulse(servopin,20); //turn the steering gear to 0 degrees
      delay(500);
      distance_r = get_distance(); //get the right distance
      delay(100);
      if (distance_l > distance_r) { //compare the distance, if the left is bigger than the right
        car_left();  //the car turns left
        matrix_display(clear);
        matrix_display(left);   //the dot matrix shows a left pattern
        servopulse(servopin,90);//the steering gear returns to 90 degrees
        delay(700);
        matrix_display(clear);
        matrix_display(front);   //the dot matrix displays a forward pattern
      }
      else { //Otherwise if the right is bigger than the left
```

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```
        car_right();//the car turns right
        matrix_display(clear);
        matrix_display(right);   //the dot matrix shows a left pattern
        servopulse(servopin,90);//the steering gear returns to 90 degrees
        delay(700);
        matrix_display(clear);
        matrix_display(front);   //the dot matrix displays a forward pattern
      }
    }
    else { //When the front distance is less than or equal to 10cm
      car_front();//the car goes forward
      matrix_display(clear);
      matrix_display(front);   //the dot matrix displays a forward pattern

    }
    BLE_val = Serial.read();
    if (BLE_val == 'S') {//When S is received, the car stops
      avoid_flag = 0;
      car_Stop();
    }
  }
}

void confinement() {
  int confinement_flag = 1;
  while (confinement_flag) {
    L_val = digitalRead(L_pin); //read the value of the left sensor
    M_val = digitalRead(M_pin); //read the value of the middle sensor
    R_val = digitalRead(R_pin); //read the value of the right sensor
    if ( L_val == 0 && M_val == 0 && R_val == 0 ) { //the car goes forward when no black␣
↪line is detected
      car_front();
    }
    else { //Otherwise, if any of the tracking sensors detect a black line, the goes␣
↪back and then turns left
      car_back();
      delay(500);
      car_left();
      delay(800);
    }
    BLE_val = Serial.read();
    if (BLE_val == 'S') { //When S is received, the car stops
      confinement_flag = 0;
      car_Stop();
    }
  }
}

void tracking() {
  int track_flag = 1;
  while (track_flag) {
    L_val = digitalRead(L_pin); //read the value of the left sensor
```

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```
    M_val = digitalRead(M_pin); //read the value of the middle sensor
    R_val = digitalRead(R_pin); //read the value of the right sensor
    if (M_val == 1) { //Black line detected in the middle
      if (L_val == 1 && R_val == 0) { //If a black line is detected on the left, but not␣
↪on the right, turn left
        car_left();
      }
      else if (L_val == 0 && R_val == 1) { //Otherwise, if a black line is detected on␣
↪the right and not on the left, turn right
        car_right();
      }
      else { //Otherwise, the car goes forward
        car_front();
      }
    }
    else { //no black lines detected in the middle
      if (L_val == 1 && R_val == 0) { //If a black line is detected on the left, but not␣
↪on the right, turn left
        car_right();
      }
      else if (L_val == 0 && R_val == 1) { //Otherwise, if a black line is detected on␣
↪the right and not on the left, turn right
        car_right();;
      }
      else { //Otherwise, stop
        car_Stop();
      }
    }
    BLE_val = Serial.read();
    if (BLE_val == 'S') { //When S is received, the car stops
      track_flag = 0;
      car_Stop();
    }
  }
}

void servopulse(int servopin,int myangle)//Steering gear running angle
{
  for(int i=0; i<30; i++)
  {
    int pulsewidth = (myangle*11)+500;
    digitalWrite(servopin,HIGH);
    delayMicroseconds(pulsewidth);
    digitalWrite(servopin,LOW);
    delay(20-pulsewidth/1000);
  }
}

//this function is used for dot matrix display
void matrix_display(unsigned char matrix_value[])
{
  IIC_start();  //the function that calls the data transfer start condition
```

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```
  IIC_send(0xc0);  //select address

  for (int i = 0; i < 16; i++) //the pattern data is 16 bytes
  {
    IIC_send(matrix_value[i]); //Transmit the data of the pattern
  }
  IIC_end();   //End pattern data transmission
  IIC_start();
  IIC_send(0x8A);  //Display control, select 4/16 pulse width
  IIC_end();
}
//Conditions under which data transmission begins
void IIC_start()
{
  digitalWrite(SDA_Pin, HIGH);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, LOW);
}
//Indicates the end of data transmission
void IIC_end()
{
  digitalWrite(SCL_Pin, LOW);
  digitalWrite(SDA_Pin, LOW);
  delayMicroseconds(3);
  digitalWrite(SCL_Pin, HIGH);
  delayMicroseconds(3);
  digitalWrite(SDA_Pin, HIGH);
  delayMicroseconds(3);
}
//transmit data
void IIC_send(unsigned char send_data)
{
  for (byte mask = 0x01; mask != 0; mask <<= 1) //Each byte has 8 bits and is checked␣
↪bit by bit starting at the lowest level
  {
    if (send_data & mask) { //Sets the high and low levels of SDA_Pin depending on␣
↪whether each bit of the byte is a 1 or a 0
      digitalWrite(SDA_Pin, HIGH);
    } else {
      digitalWrite(SDA_Pin, LOW);
    }
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, HIGH); //Pull the clock pin SCL_Pin high to stop data␣
↪transmission
    delayMicroseconds(3);
    digitalWrite(SCL_Pin, LOW); //pull the clock pin SCL_Pin low to change the SIGNAL of␣
↪SDA
  }
}
```

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```
//*****************************************************************************
```

5.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, power on the external power then turn the DIP switch to ON.

After the Bluetooth module is plugged into the APP and the mobile APP is successfully connected to the Bluetooth, the smart car can be controlled by the mobile APP. We can achieve the corresponding functions by pressing the corresponding buttons on the mobile APP.

CHAPTER

TEN

5. KIDSBLOCK TUTORIAL

10.1 Kidsblock - Read Me First

10.1.1 1 Download code file

Click to download the code file required for the courseClick to download

10.1.2 2 Software installation of Windows System

1Download kidsblock:https://wiki.kidsbits.cc/projects/KidsBlock/en/latest/

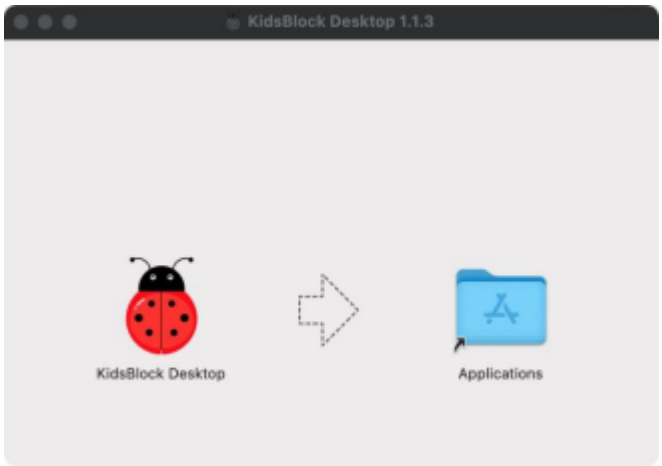
2software installation

3.operating software

First connect the development board to the computer

10.1.3 3 Software installation of Mac System

1.Download kidsblock:https://xiazai.keyesrobot.cn/KidsBlock.dmg



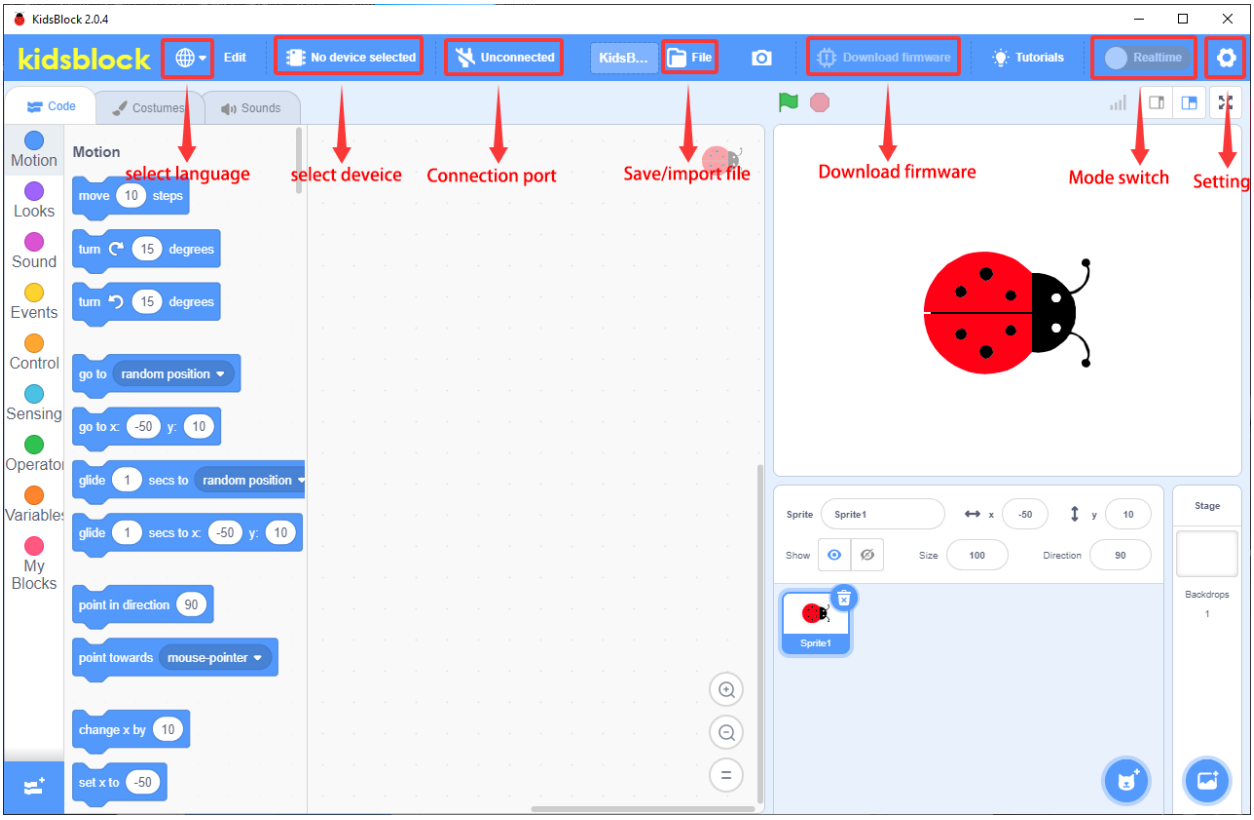
2.Click KidsBlock, and drag KidsBlock Desktop into Applications.

3.Wait for installing. KidsBlock icon will show up in Launchpad if it is installed successfully.

10.1.4 4 How to use KidsBlock

We demonstrate how to use KidsBlock on Windows system

1.Interface



2.Click to switch to different languages.

3.Click

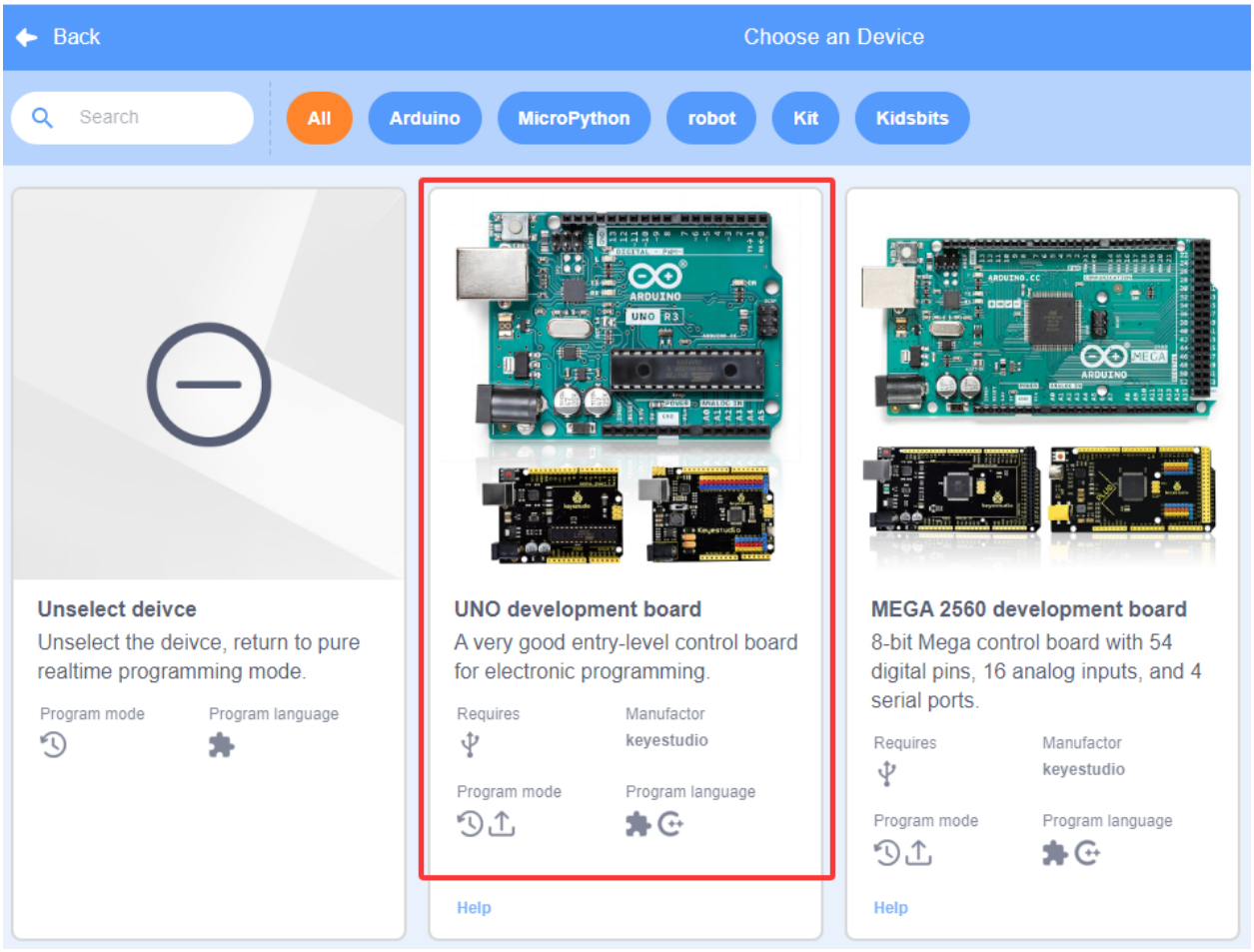
to enter the main page, select the control board needed. In this project, we select the "**UNO development board**" mainboard and click **Connect,** then it is connected. Click **Go to Editor** to return the code editor. Icon

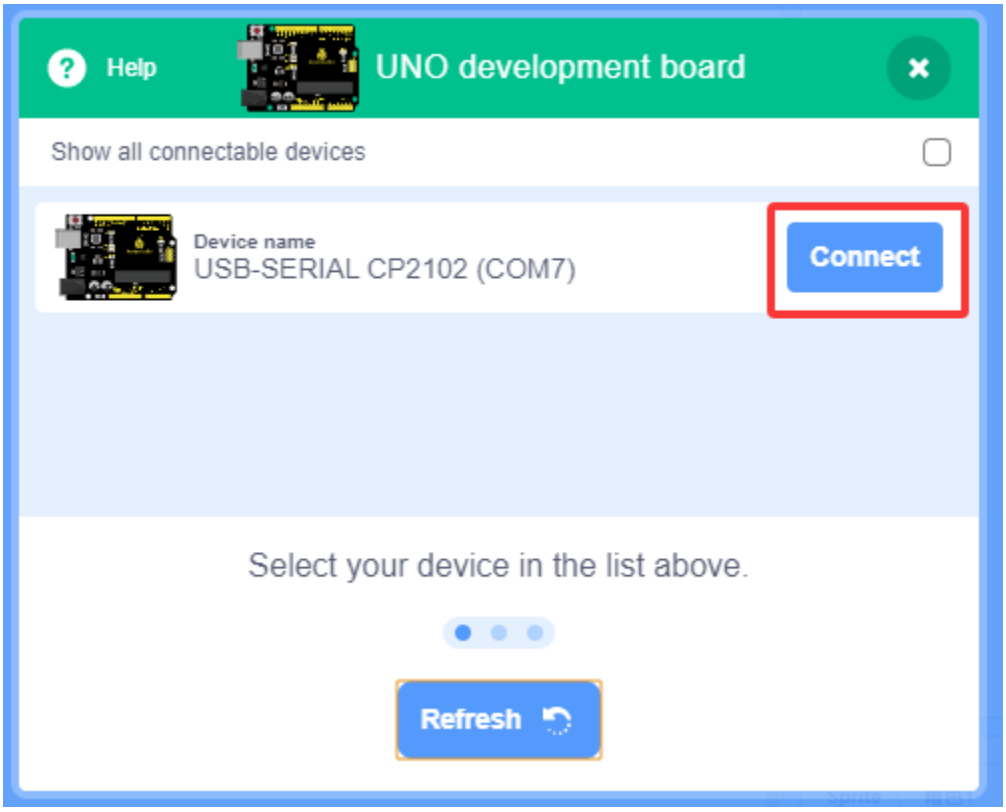
will change into

and

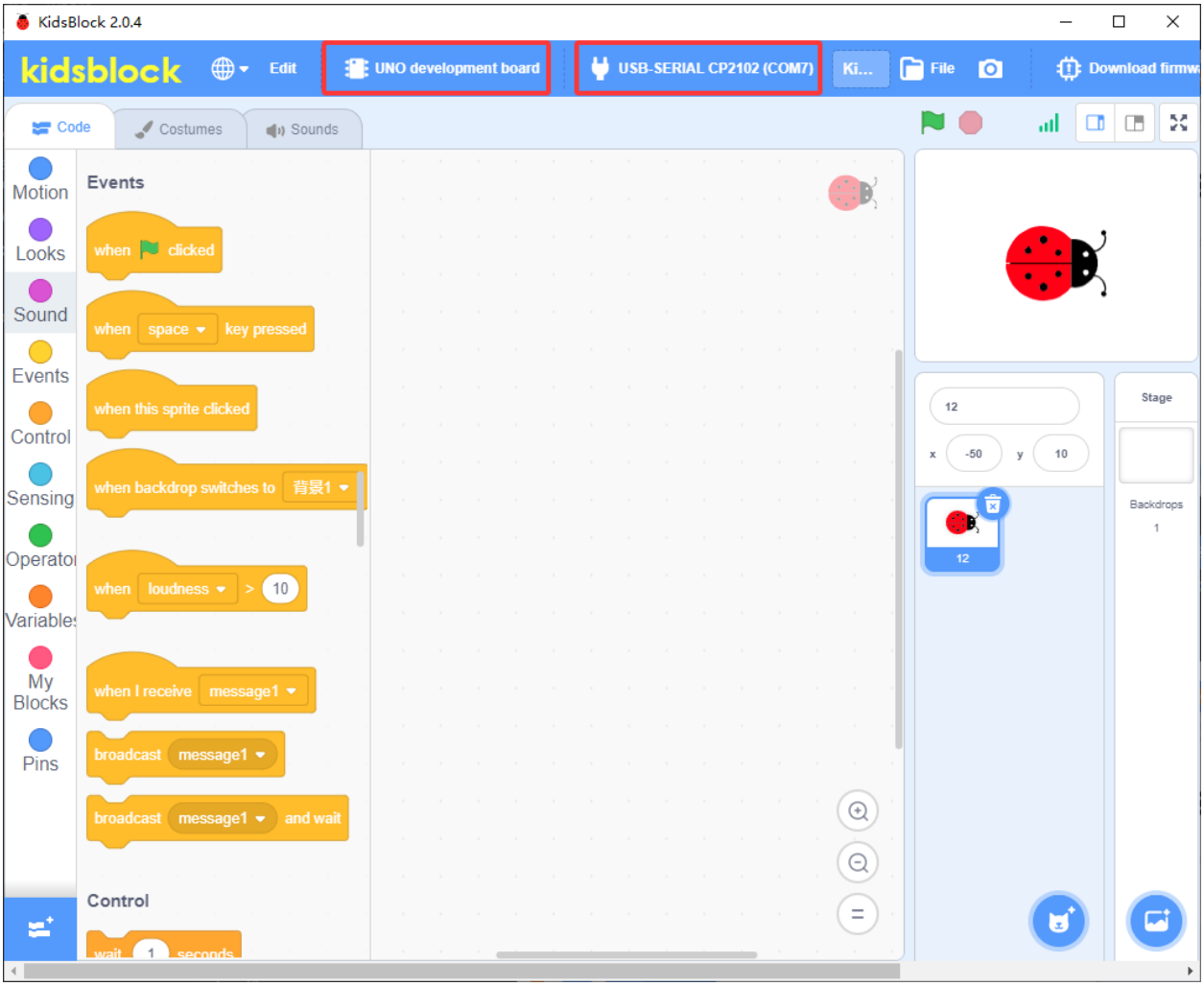
will change into

. This means the Uno Plus mainboard and portsCOM)are connected.




Help
UNO development board
Show all connectable devices
Device name
USB-SERIAL CP2102 (COM7)
Connect
Select your device in the list above.
Refresh

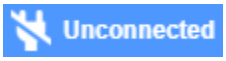
Help
UNO development board
keyestudio
Connected
Disconnect
Go to Editor



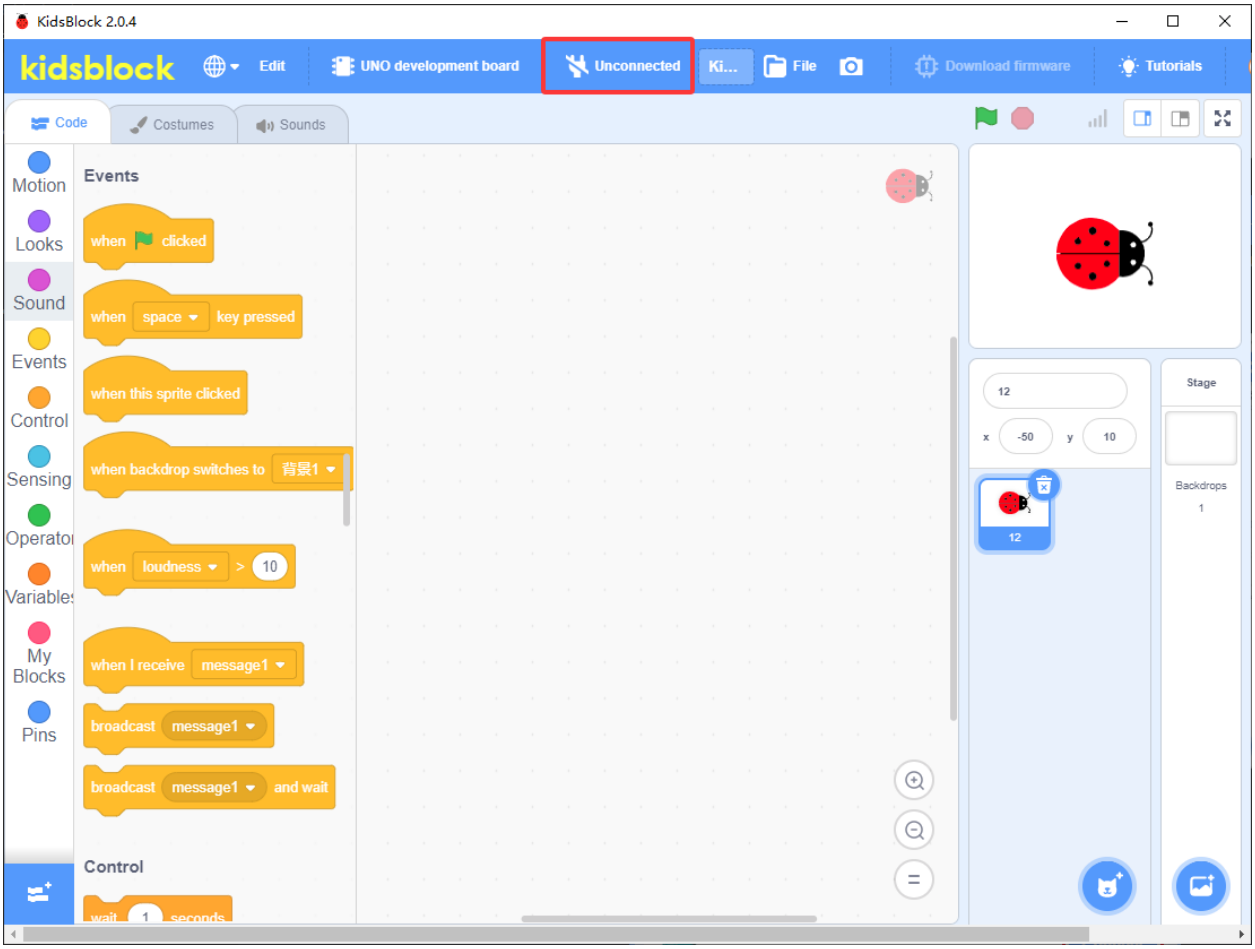
4.If the Uno Plus mainboard is connected , but

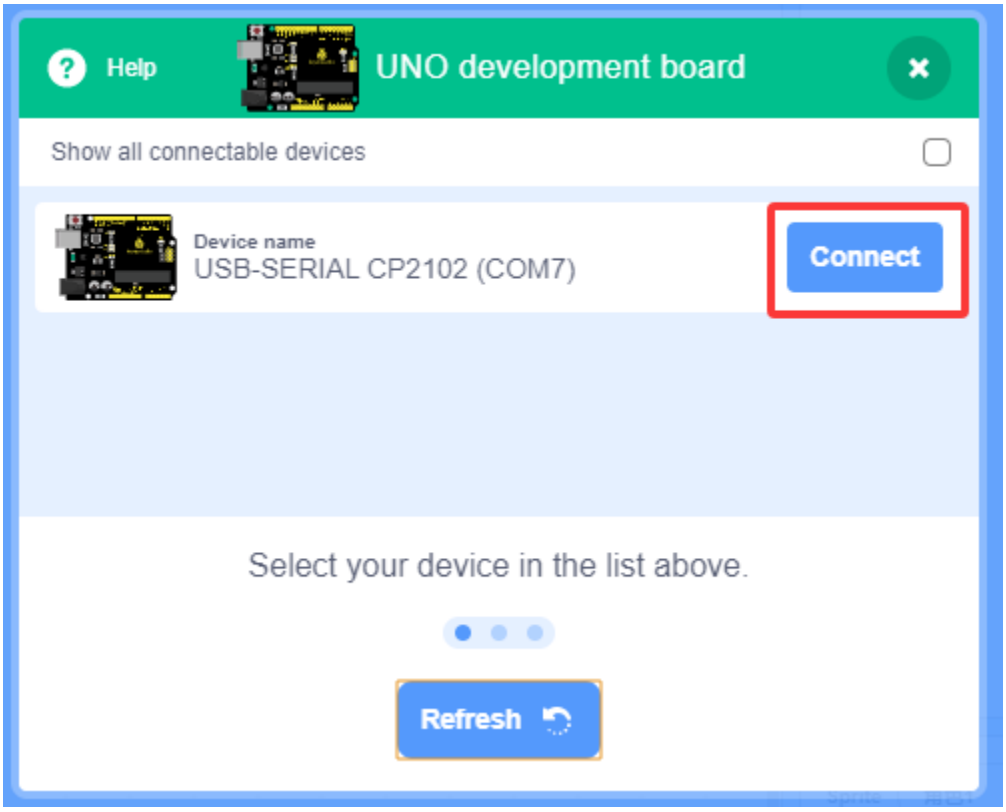
icondoesn't change into

. You need to click

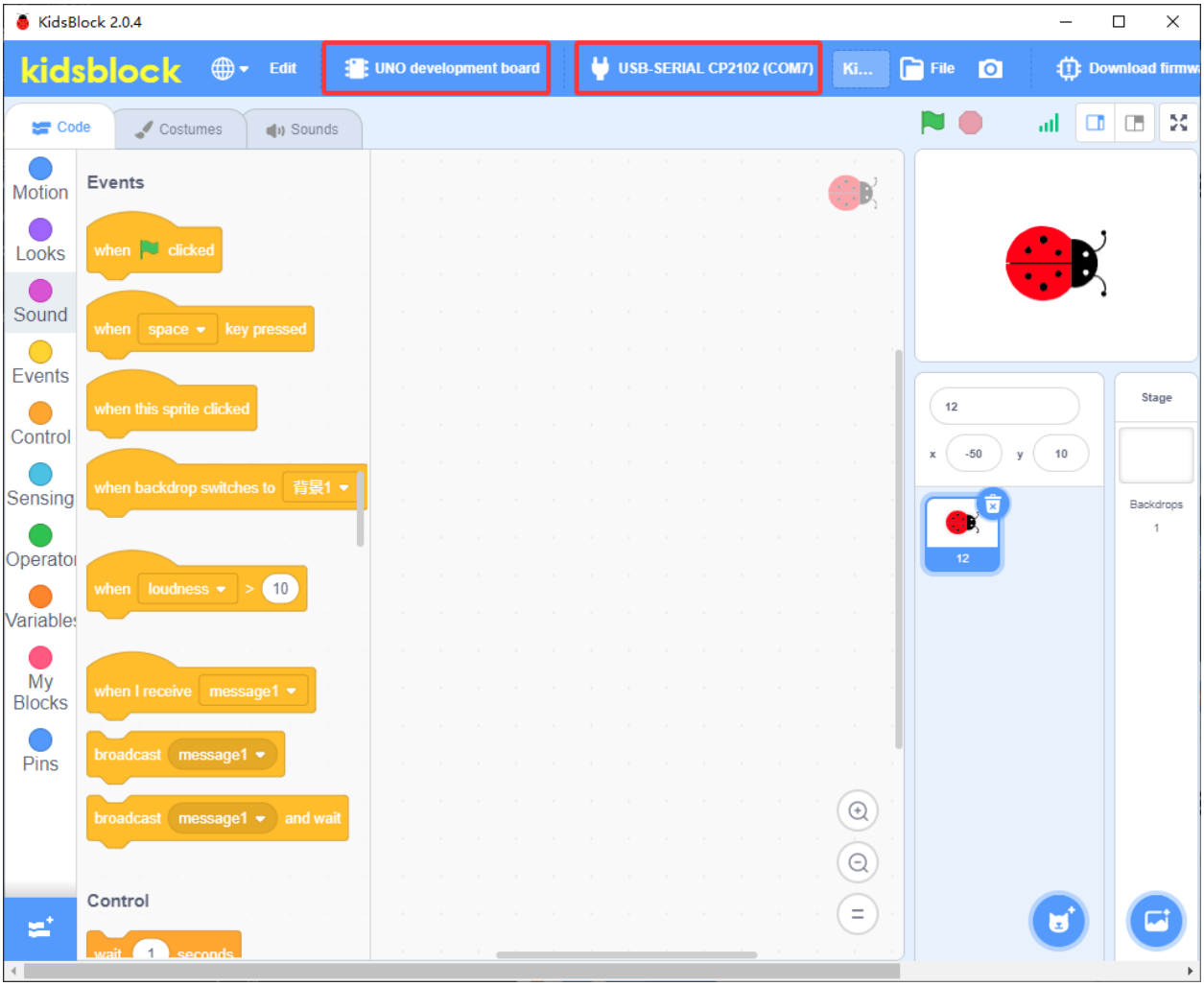
to connect the COM port. Click 

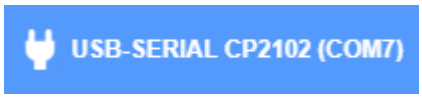
and then click **Connect**. Then you will find a page pop up, showing **Connected**.


KidsBlock 2.0.4
kidsblock
Edit
UNO development board
Unconnected
Ki...
File
Download firmware
Tutorials
Code
Costumes
Sounds
Motion
Looks
Sound
Events
Control
Sensing
Operato
Variable
My Blocks
Pins
Events
when clicked
when space key pressed
when this sprite clicked
when backdrop switches to 背景1
when loudness > 10
when I receive message1
broadcast message1
broadcast message1 and wait
Control
wait 1 seconds
12
x -50 y 10
Stage
Backdrops 1

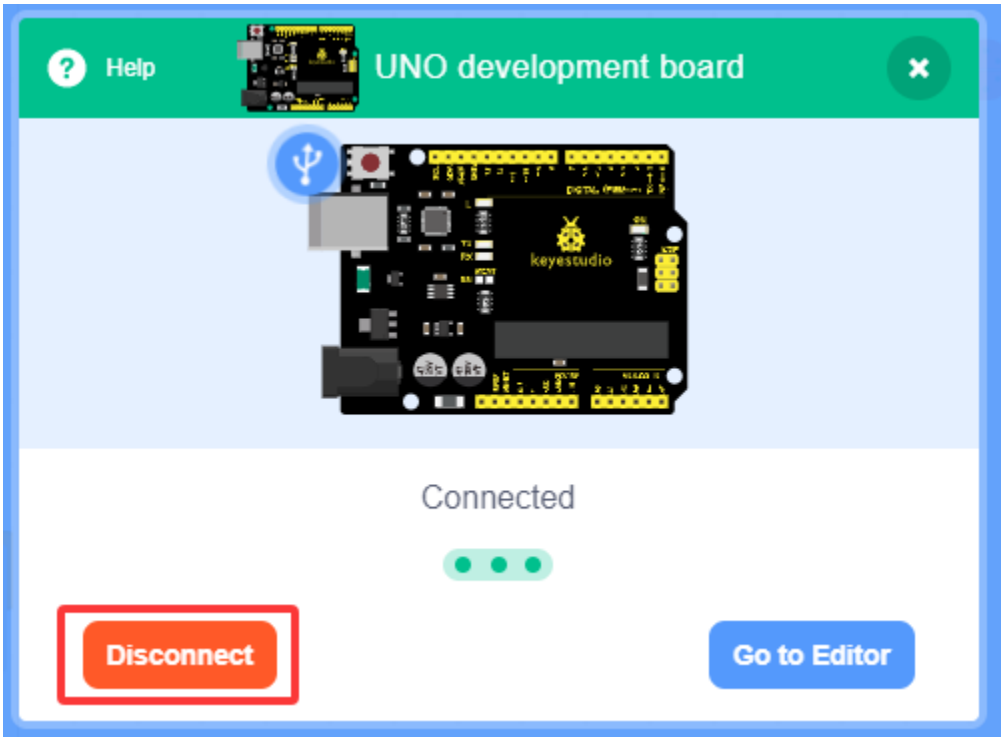

Help
UNO development board
Show all connectable devices
Device name
USB-SERIAL CP2102 (COM7)
Connect
Select your device in the list above.
Refresh

Help
UNO development board
keyestudio
Connected
Disconnect
Go to Editor



To disconnect the port, just click 

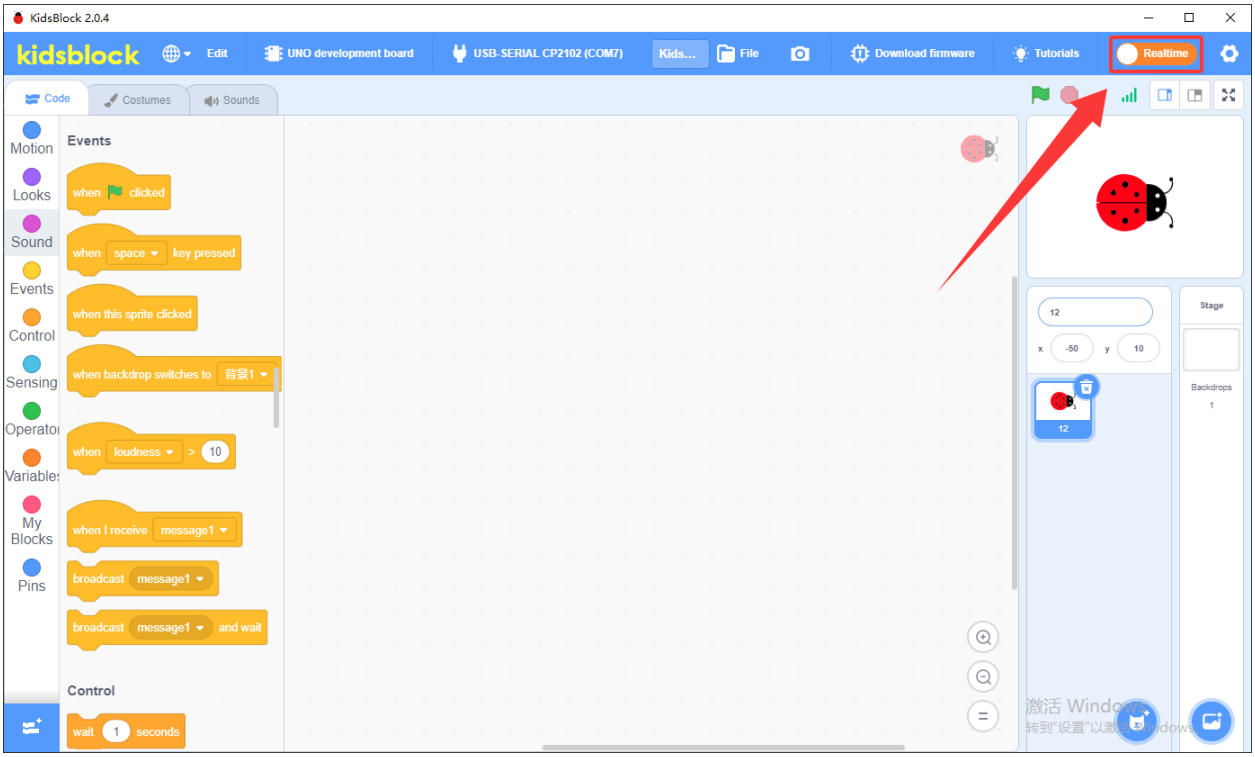
and**Disconnect**.

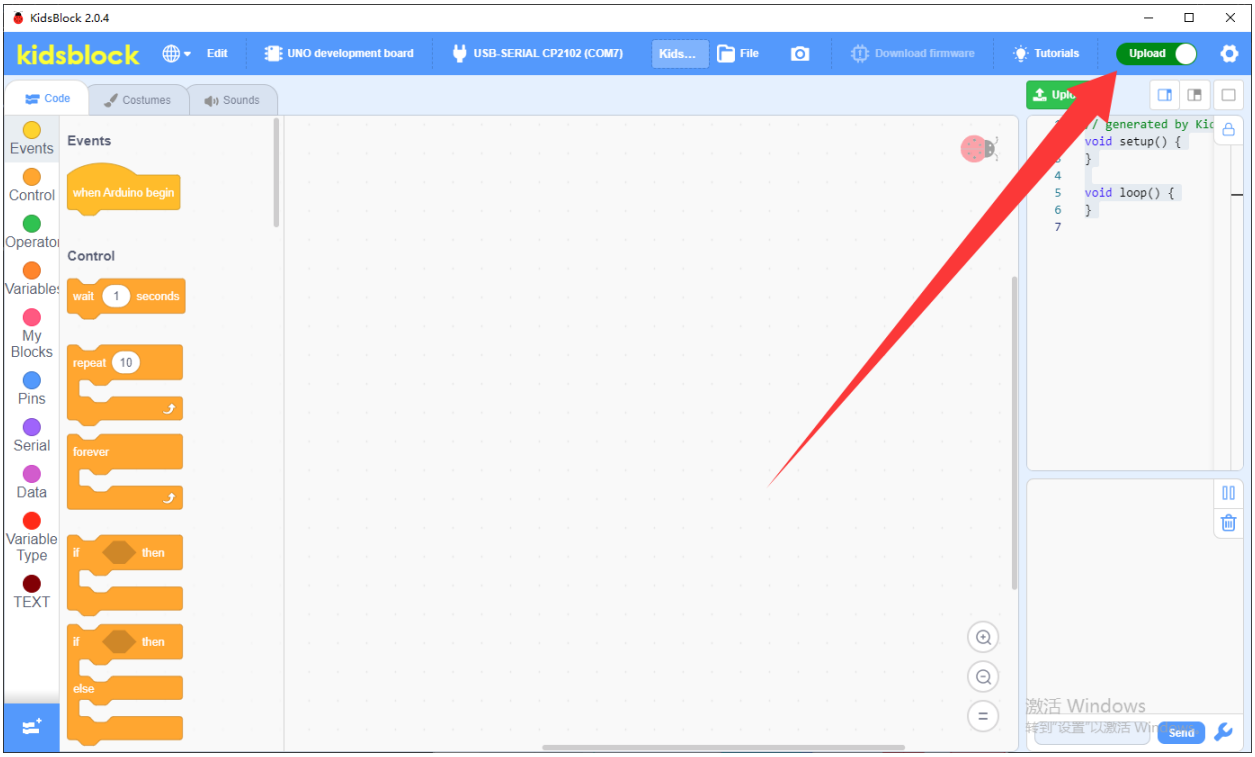


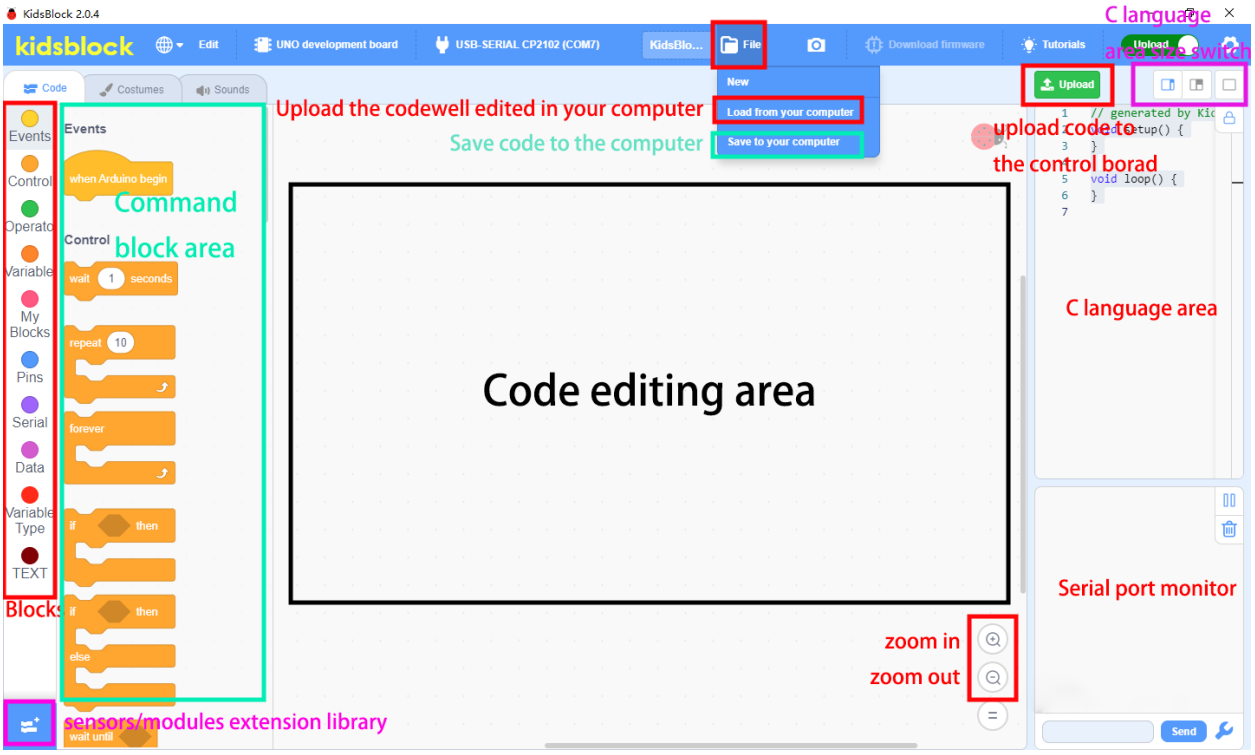
5.The Uno Plus mainboard and the COM port are connected, then click

will change into

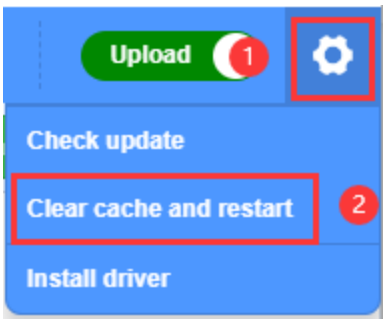
.







Note: if you want to update libraries of KidsBlock, click then Clear cache and restart.

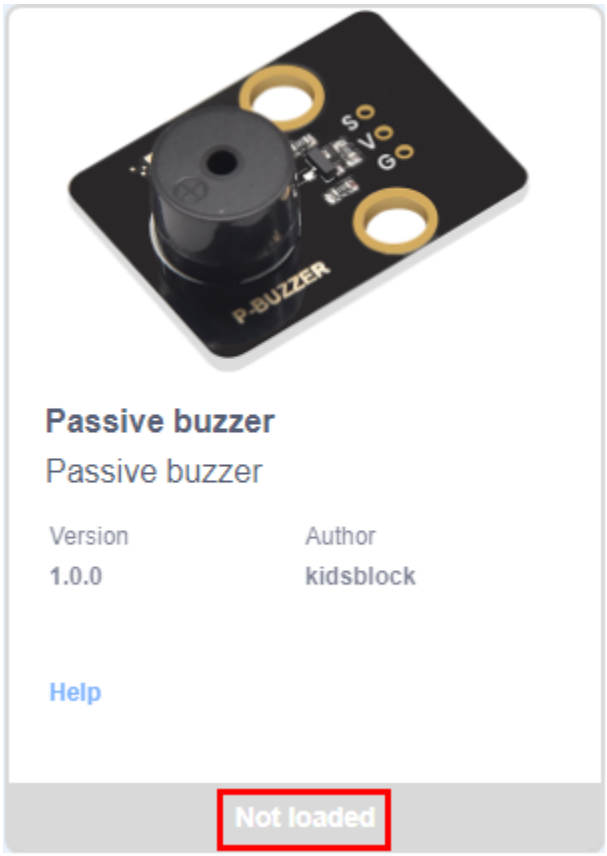


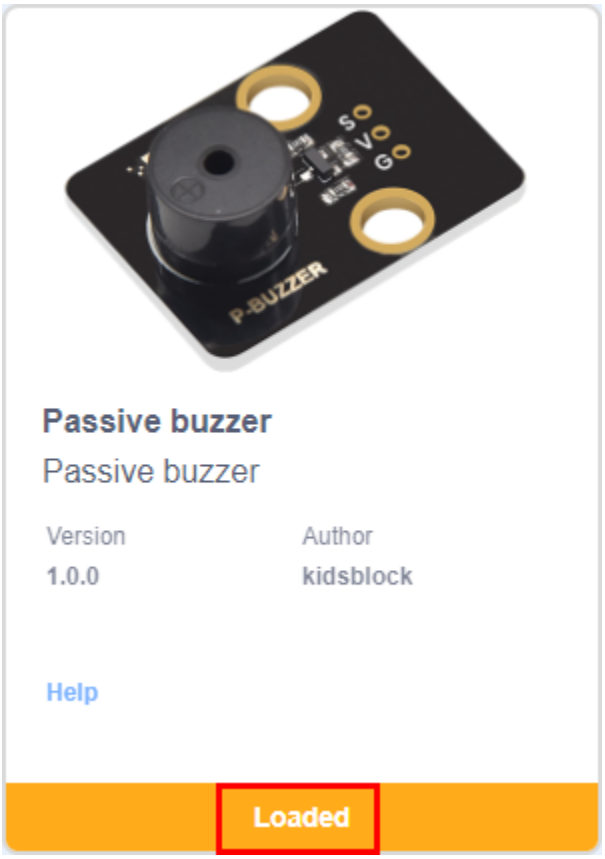
6. stands for extension libraries of sensors and modules. Click

to enter the page of extension libraries, click a sensor or module to add.

For example, if click the“passive buzzer”module

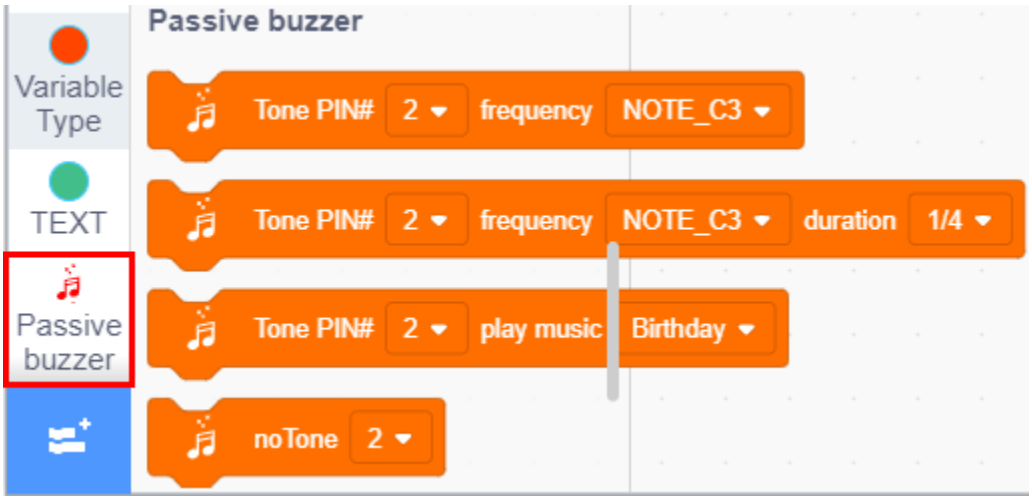
,“**Not loaded**”will change into “**Loaded**”. Then the passive buzzer is added.



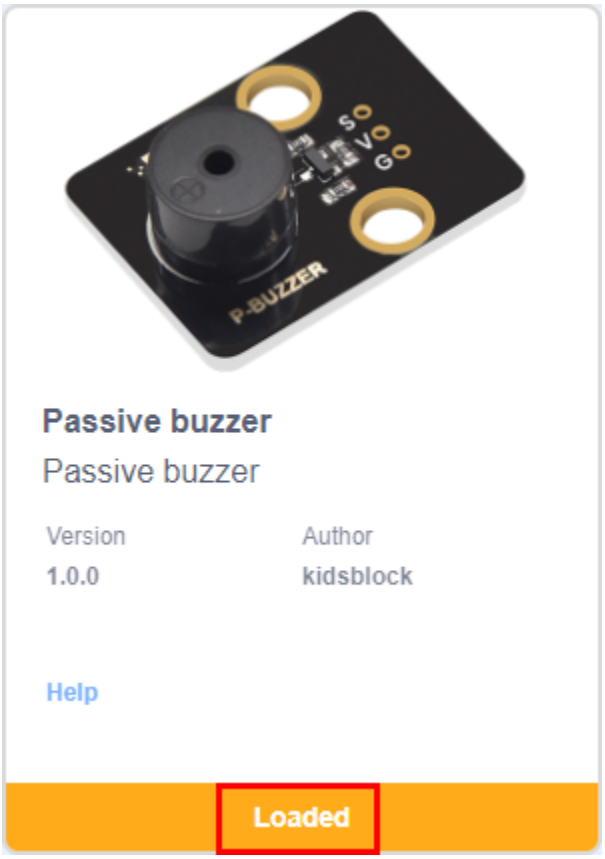


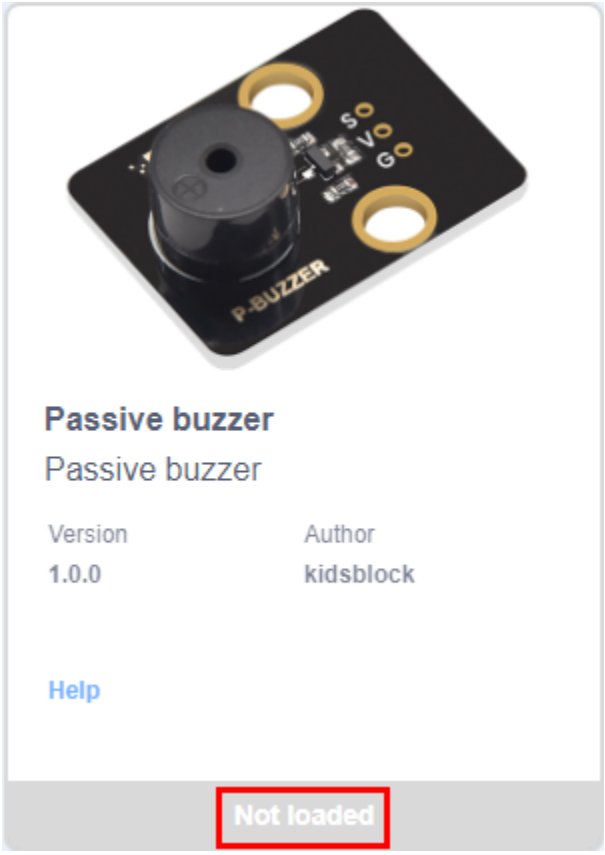
Click

to return the code editor. Then you can view the passive buzzer in the blocks area.



If you want to delete the passive buzzer, click to select the passive buzzer . Then "**Loaded**" will change into "**Not loaded**". Then the passive buzzer is deleted.

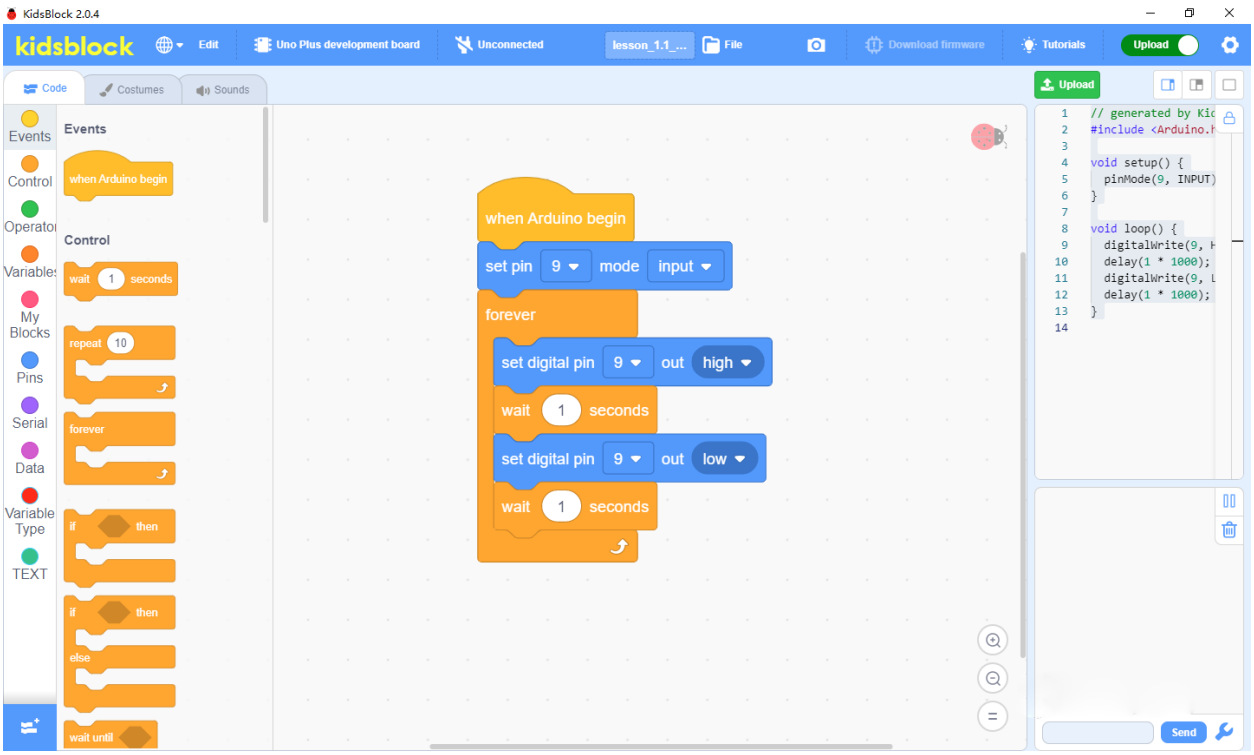

S
V
G
P-BUZZER
Passive buzzer
Passive buzzer
Version
1.0.0
Author
kidsblock
Help
Loaded


S
V
G
P-BUZZER
Passive buzzer
Passive buzzer
Version
1.0.0
Author
kidsblock
Help
Not loaded

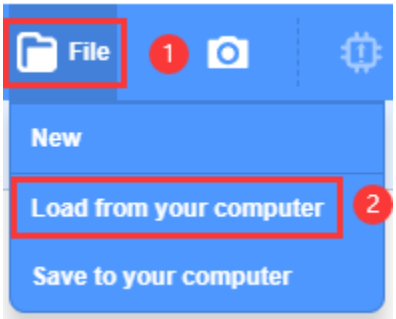
The way of deleting other sensors or modules is as same as the passive buzzer.

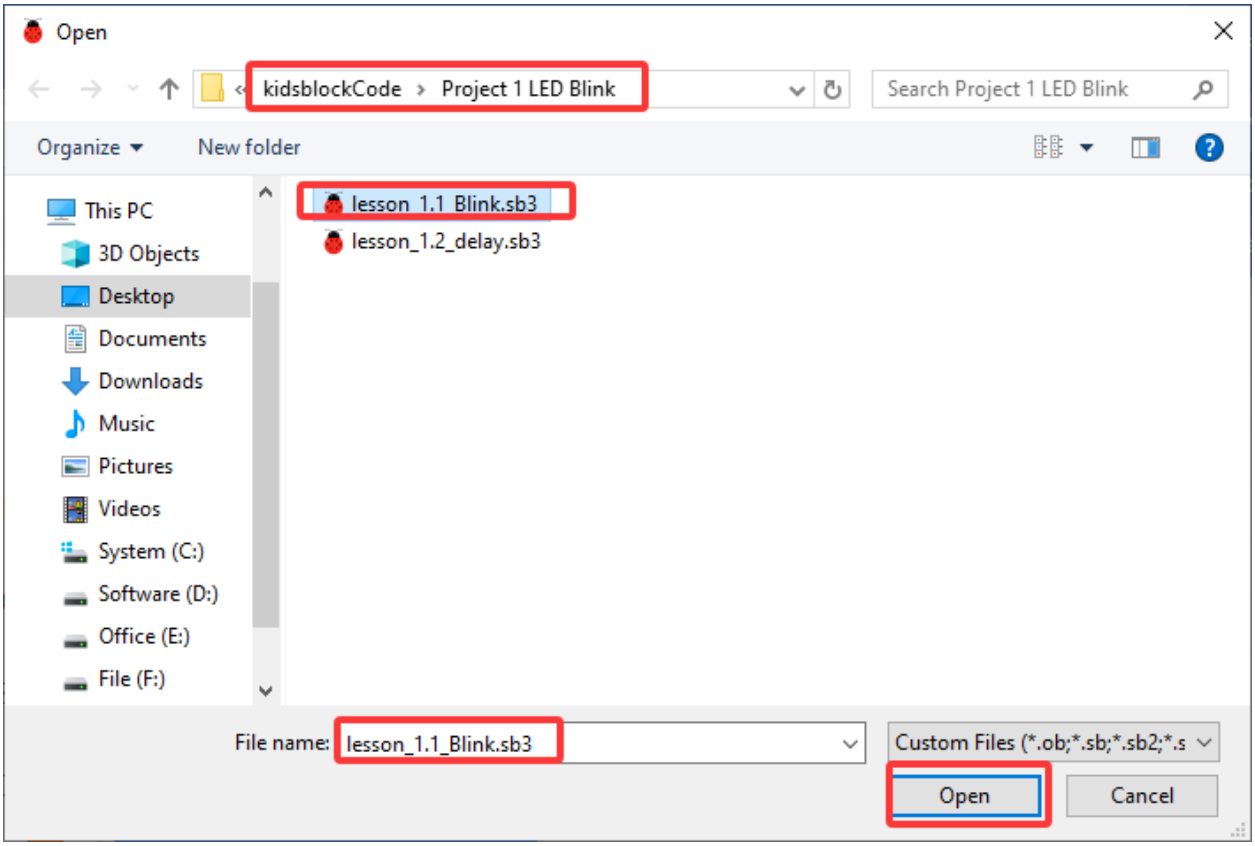
7.How to open SB3 type files

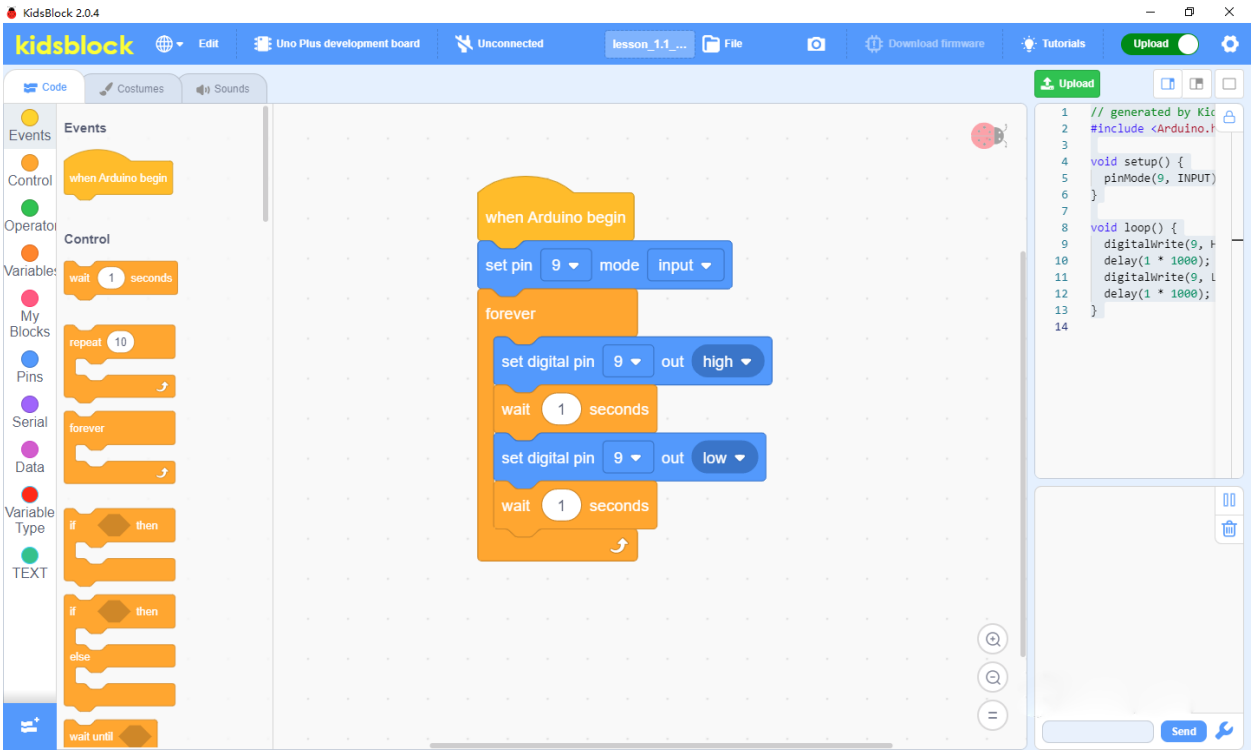
1). Double-click SB3 type files to open them. For instance, open lesson_1.1_Blink.sb3 , then we need to double-click lesson_1.1_Blink.sb3 .



2). Open Kidsblockclick **file and Load from your computer**then select the SB3 type file on the computer.for example: lesson_1.1_Blink.sb3).







NotePlease refer to the following link for MacOShttps://kidsblocksite.readthedocs.io/en/latest/

10.2 Kidsblock - Projects

10.2.1 Project 1 LED Blink

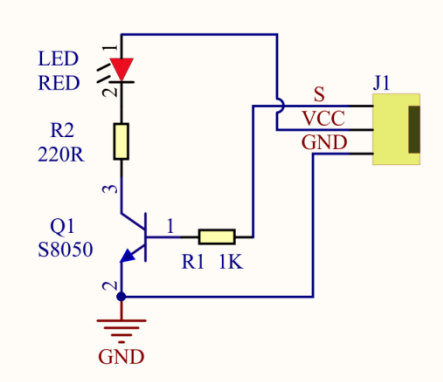
1.Description

For starters and enthusiasts, LED Blink is a fundamental program. LED, the abbreviation of light emitting diodes, consists of Ga, As, P, N chemical compounds and so on.

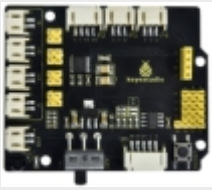
The LED can flash in diverse color by altering the delay time in the test code. When in control, power on GND and VCC, the LED will be on if S end is in high level, otherwise it will go off.

2.Specification

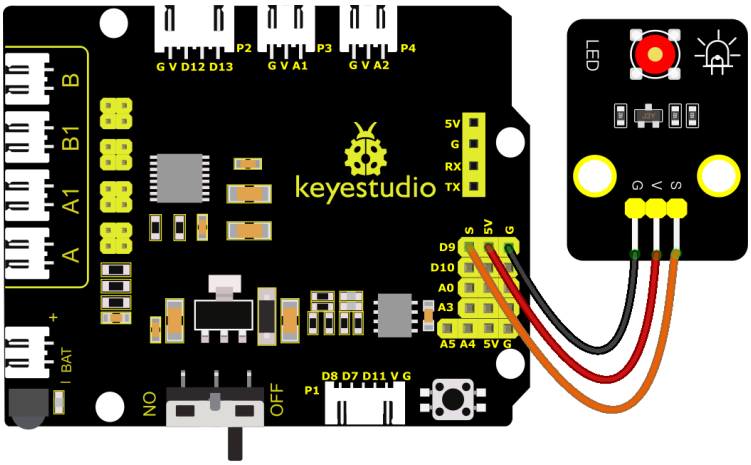
- Control interface: digital port
- Working voltage: DC 3.3-5V
- Pin spacing: 2.54mm
- LED display color: red



3.Components

Development Board *1	8833 Motor Driver *1	Red LED Module*1
		
3P F-F Dupont Wire*1	USB Cable*1	

4.Wiring Diagram



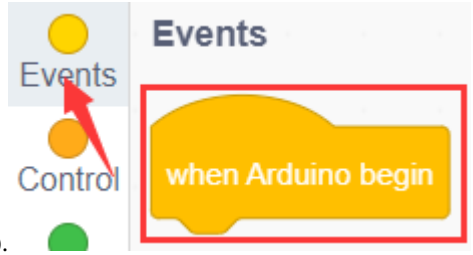
As can be seen from the above figure, the Keyestudio 8833 motor driver expansion board is stacked on the Keyestudio 4.0 development board.

The pin G, V and S of the LED module are connected to G, 5V and D9 of the expansion board respectively.

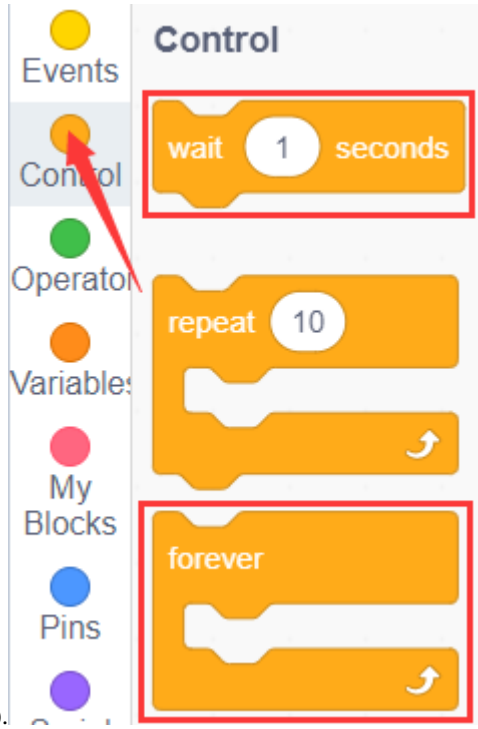
5.Test Code

You can drag blocks to edit. Blocks listed below are for your reference.

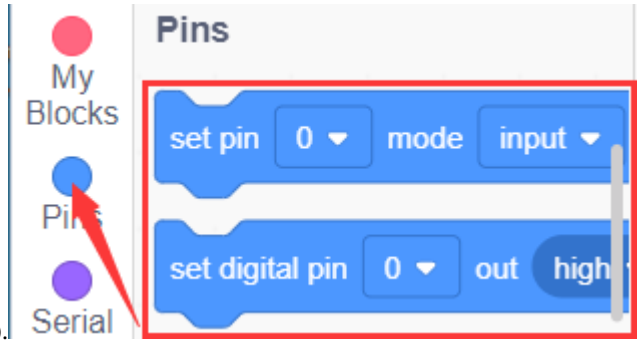
(1).



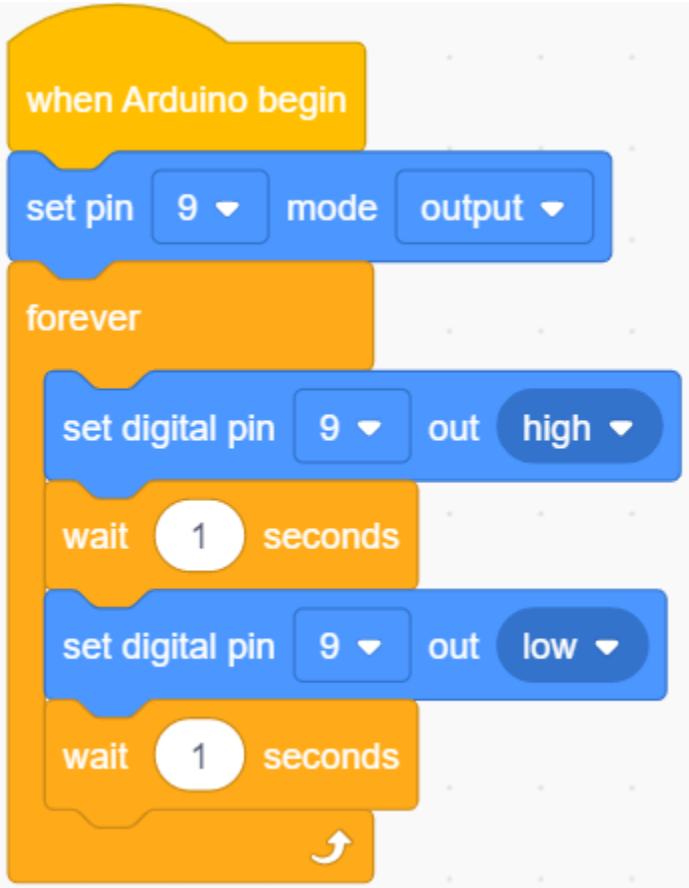
(2).



(3).



Complete Test Code



6.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, and use a USB cable to connect the computer to power the board. After powering on, you will see the LED connected to the D9 will be on and off.

7.Extension Practice

Next, we look to change the frequency of LED flicker by changing the wait time.

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, and use a USB cable to connect the computer to power the board. The test result shows that the LED flashes faster.

10.2.2 Project 2: Adjust LED Brightness

1.Description

In previous lesson, we control LED on and off and make it blink.

In this project, we will control LED's brightness through PWM simulating breathing effect.

PWM is a means of controlling the analog output via digital means. Digital control is used to generate square waves with different duty cycles (a signal that constantly switches between high and low levels) to control the analog output. In general, the input voltages of ports are 0V and 5V.

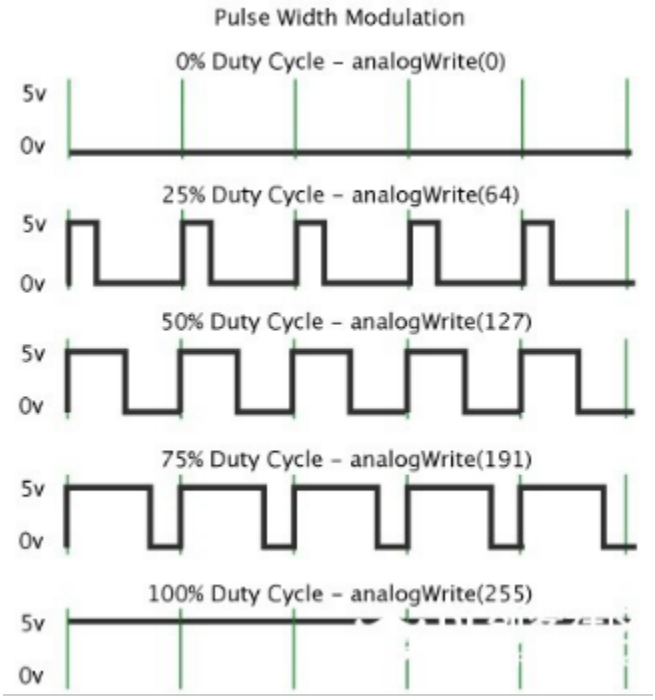
What if the 3V is required? Or a switch among 1V, 3V and 3.5V? We cannot change resistors constantly. For this reason, we resort to PWM.

For the Arduino digital port voltage output, there are only LOW and HIGH, which correspond to the voltage output of 0V and 5V. You can define LOW as 0 and HIGH as 1, and let the Arduino output five hundred 0 or 1 signals within 1s.

If all of the output five hundred are 1, that is 5V; if all of which are 0, that is 0V. If output 010101010101 in this way then the output port is 2.5V, which is like showing movie. The movie we watch are not completely continuous. It actually outputs 25 pictures per second. In this case, the human can't see it, neither does PWM. If we want different voltage, we need to control the ratio of 0 and 1. The more 0,1 signals output per unit time, the more accurate the control.

PWM is a technology that uses digital methods to obtain analog quantities. Digital control enables to form a square wave, the square wave signal only has on and off two states (high and low). A voltage ranging from 0 to 5V can be

simulated by controlling the ratio of on to off duration. The time spent on (technically called high level) is called pulse width, so PWM is also called pulse width modulation.

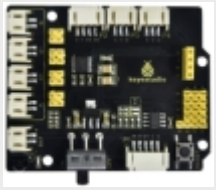


The green vertical bars represent one period of the square wave. The value written in each analogWrite(value) corresponds to a percentage, which is also called Duty Cycle. This percentage refers to the ratio of time occupied by the high level in a cycle, that is, duty cycle = high level time/cycle time.

In the figure, from top to bottom, the duty cycle of the first square wave is 0%, and the corresponding value is 0, and the LED brightness is the lowest, that is, off state. The longer the high level lasts, the brighter it will be. Therefore, the value of the last duty cycle of 100% is 255, and the LED is the brightest. 50% is the brightest half, and 25% is darker.

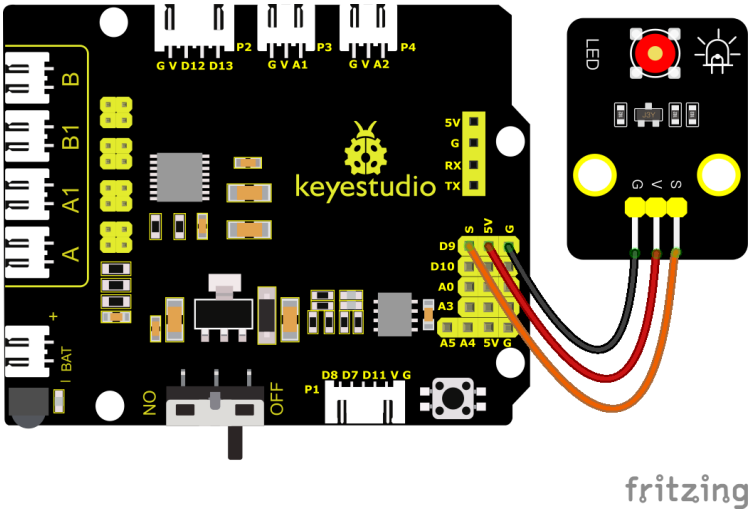
PWM is more used to adjust the brightness of LED lights or the rotation speed of the motors, and the wheels speed driven by the motors can be easily controlled. When playing with some Arduino robots, the benefits of PWM can be better reflected.

2.Components

Development Board *1	8833 Motor Driver *1	Red LED Module*1
		
3P F-F Dupont Wire*1	USB Cable*1	

3.Wiring Diagram

Keep the wiring-up unchanged.

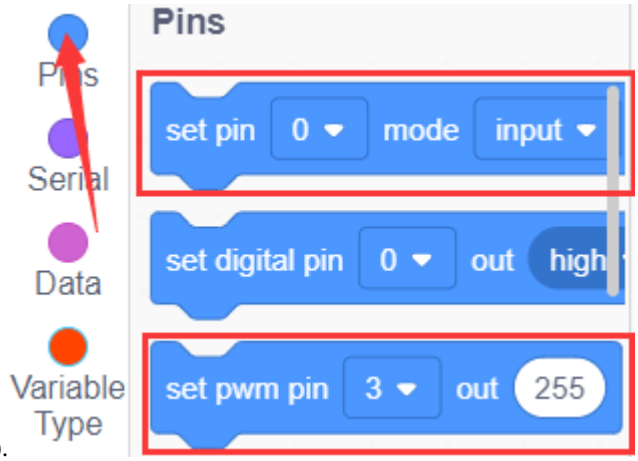


4.Test Code

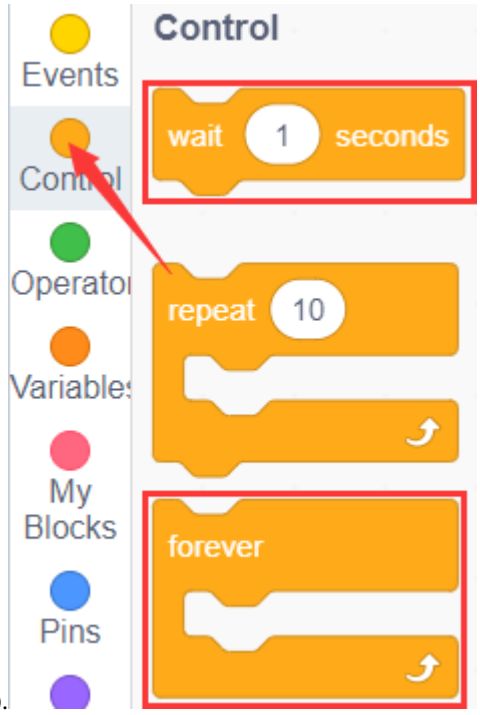
You can drag blocks to edit. Blocks listed below are for your reference.

(1).

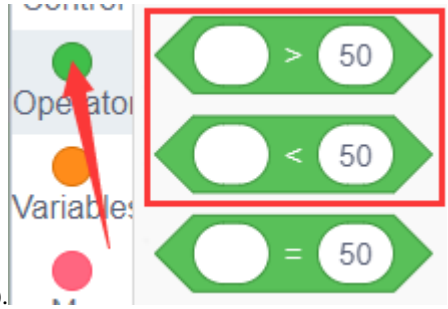
(2).



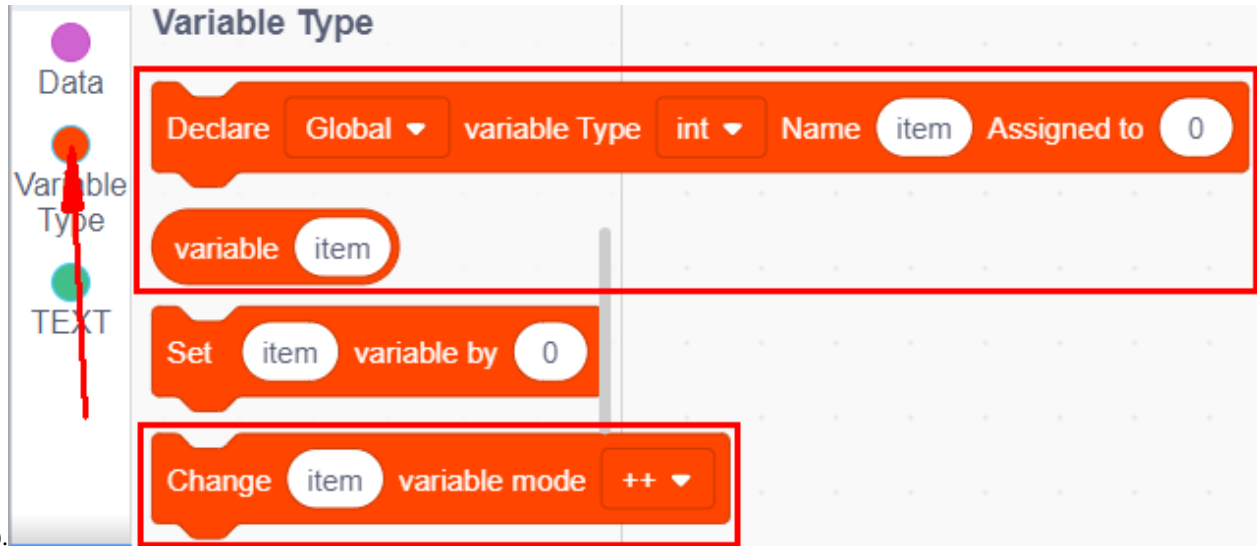
(3).

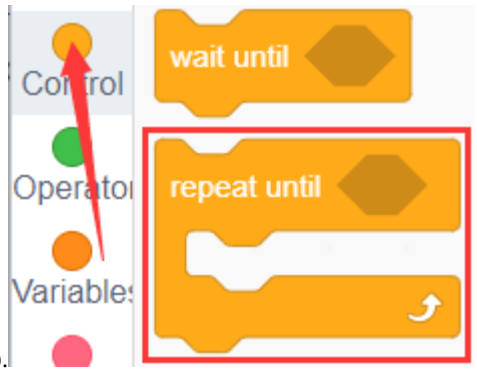

Control
Events
Control
Operator
Variables
My Blocks
Pins
wait 1 seconds
repeat 10
forever

(4).

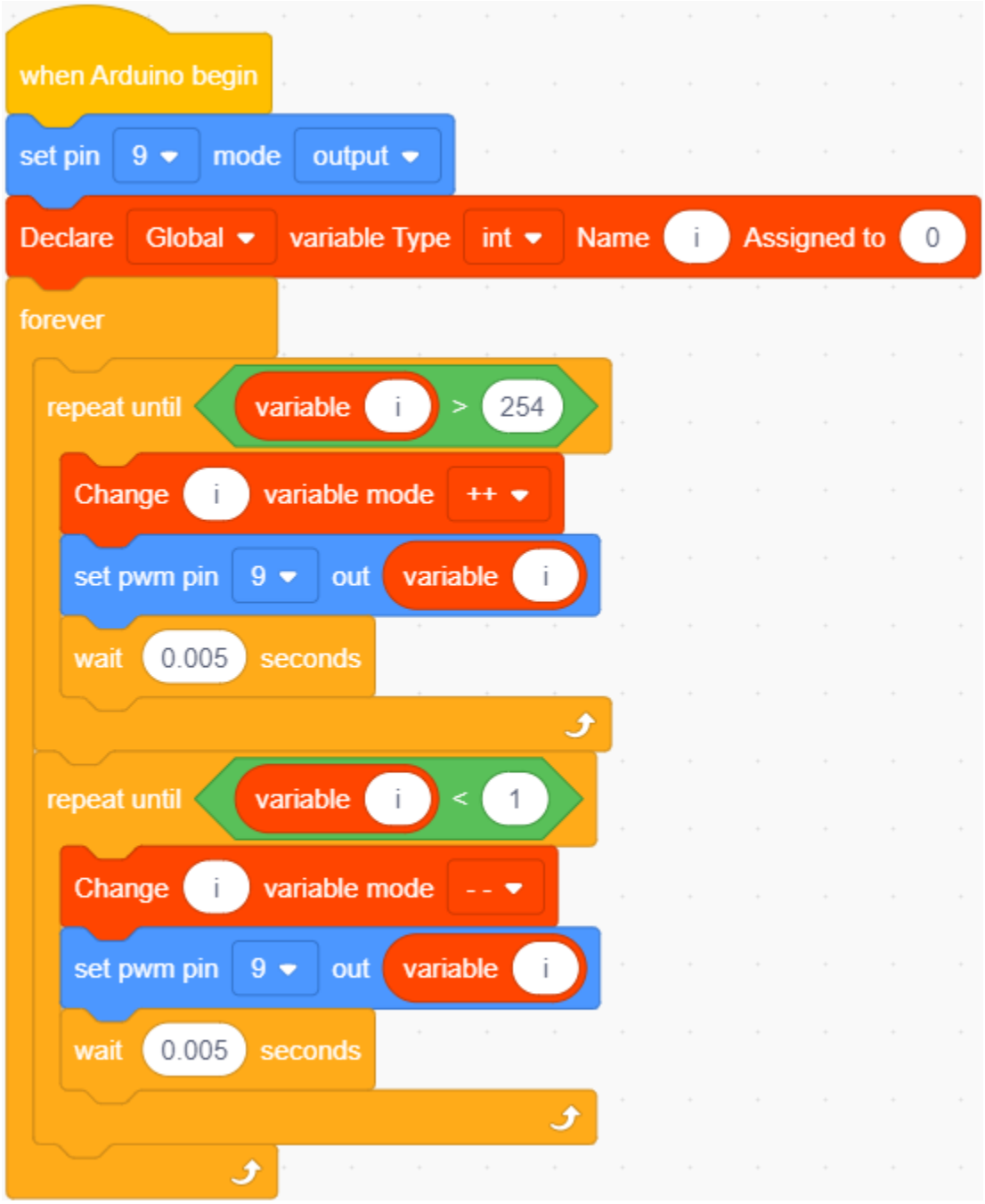

Operator
Variables
> 50
< 50
= 50

(5).


Variable Type
Data
Variable Type
TEXT
Declare Global variable Type int Name item Assigned to 0
variable item
Set item variable by 0
Change item variable mode ++

(6). 

Complete Test Code

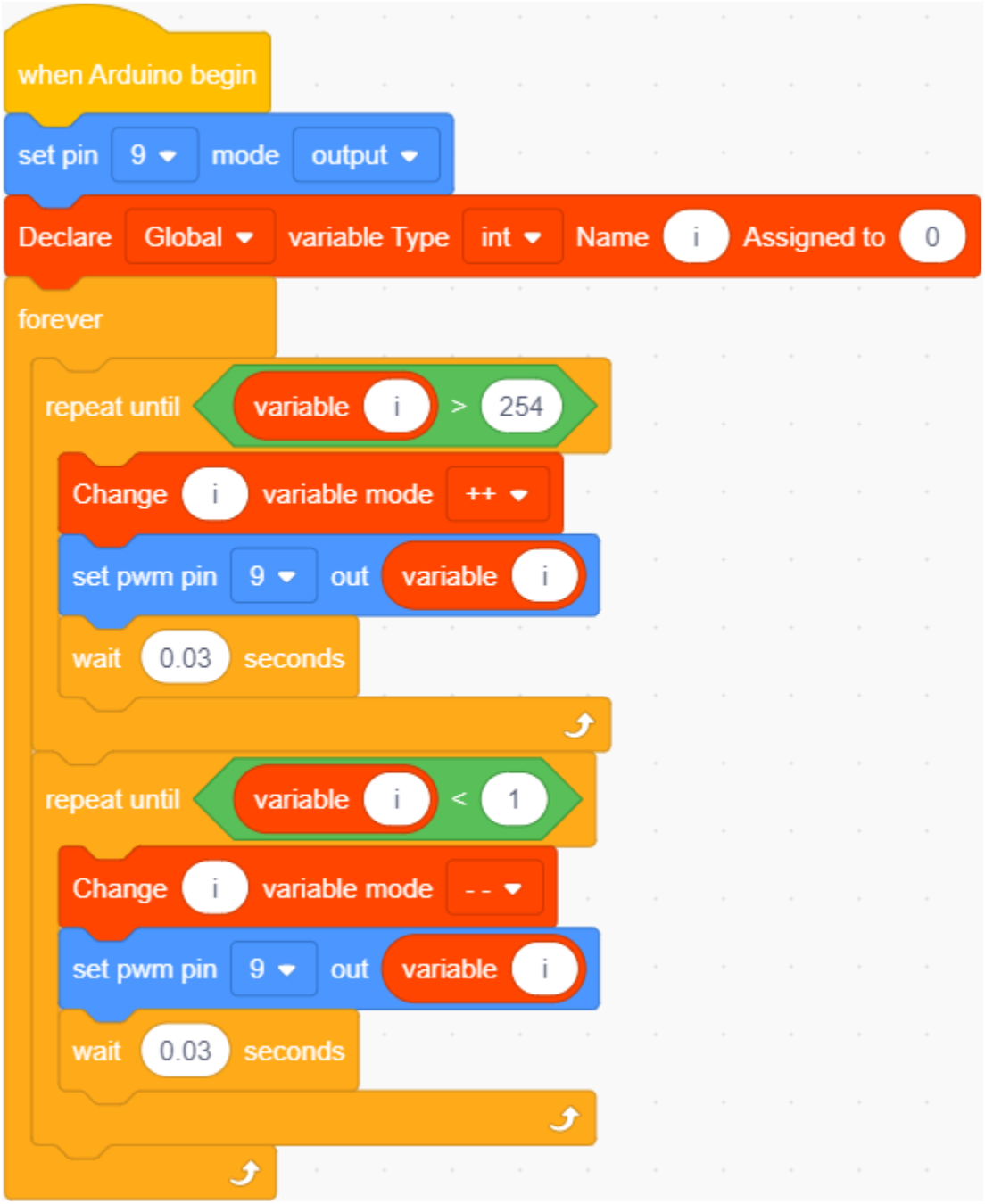


5.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, and use a USB cable to connect the computer to power the board. After powering on, you will see that the LED gradually changes from bright to dark, like human's breath, rather than turning on and off immediately.

6.Extension Practice

Keep pins of the LED unchanged, then change code(values behind wait)



Upload the code to the development board, then the LED will blink more slowly.

10.2.3 Project 3: Line Tracking Sensor

1.Description

The tracking sensor is actually an infrared sensor. The component used here is the TCRT5000 infrared tube. Its working principle is to use different reflectivity of infrared light to colors, then convert the strength of the reflected signal into a current signal.

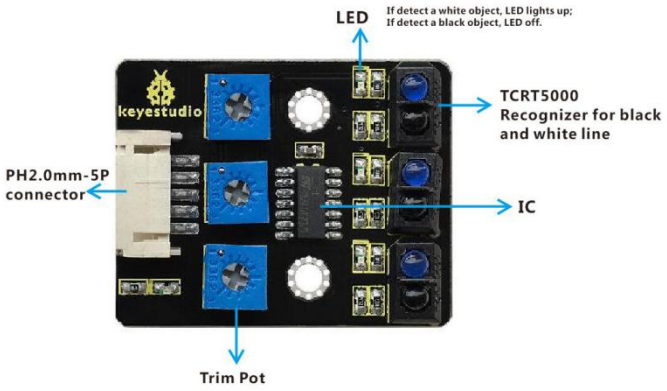
During the process of detection, black is active at HIGH level while white is active at LOW level. The detection height is 0-3 cm.

Keyestudio 3-channel line tracking module has integrated 3 sets of TCRT5000 infrared tubes on a board, which is more convenient for wiring and control.

By rotating the adjustable potentiometer on the sensor, it can adjust the detection sensitivity of the sensor.

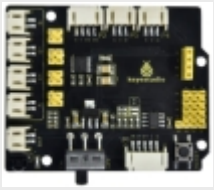
2.Specification

- Operating Voltage: 3.3-5V (DC)
- Interface: 5PIN
- Output Signal: Digital signal
- Detection Height: 0-3 cm

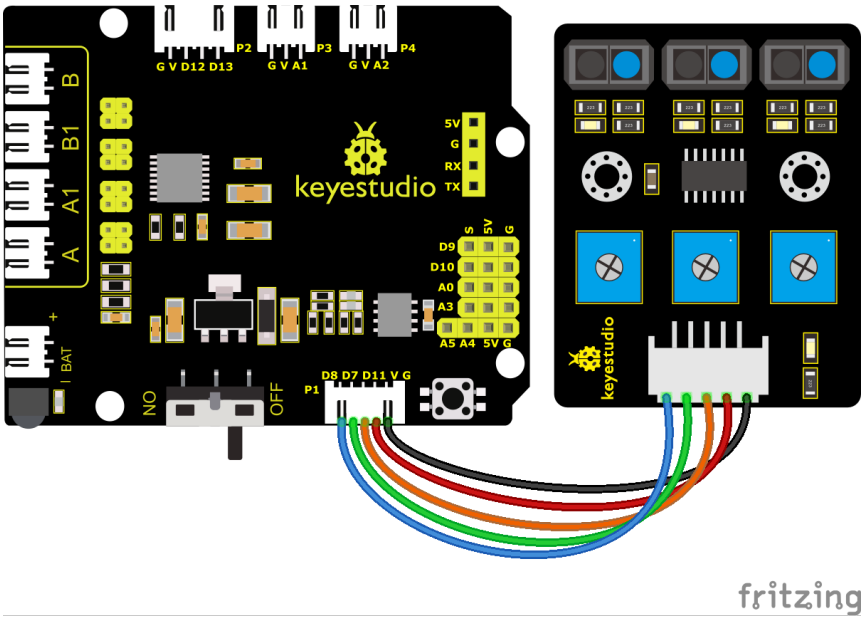


Note: Before testing, rotate the potentiometer on the sensor to adjust the detection sensitivity. The sensitivity is best when adjusting the LED to a threshold between ON and OFF.

3.Components

Development Board *1	8833 Motor Driver *1	Red LED Module*1	Line Tracking Sensor*1
			
5P Dupont Wire*1	USB Cable*1	3P Dupont Wire*1	

4.Wiring Diagram

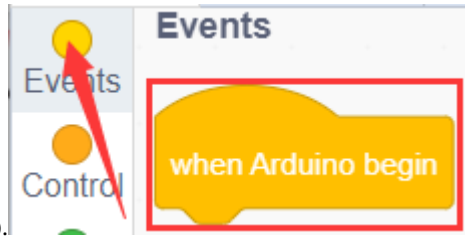


G, V, S1, S2 and S3 of the line tracking sensor are connected to GGND), VVCC), D11, D7 and D8 of the sensor expansion board.

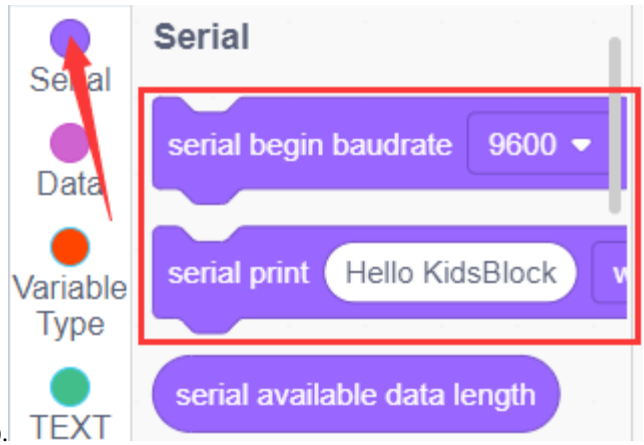
5.Test Code

You can drag blocks to edit. Blocks listed below are for your reference.

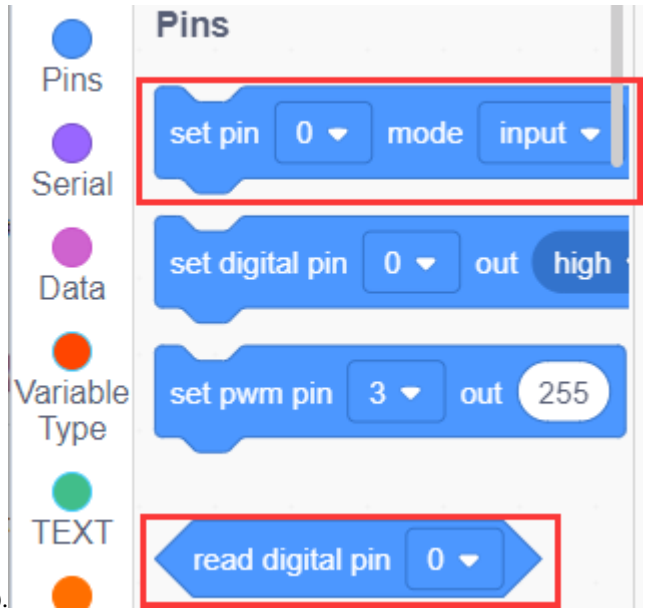
(1).



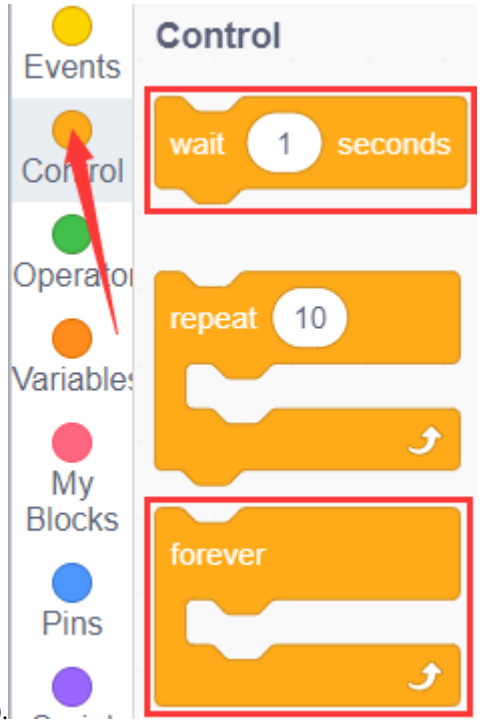
(2).


Serial
Serial
Data
Variable
Type
TEXT
serial begin baudrate 9600
serial print Hello KidsBlock
serial available data length

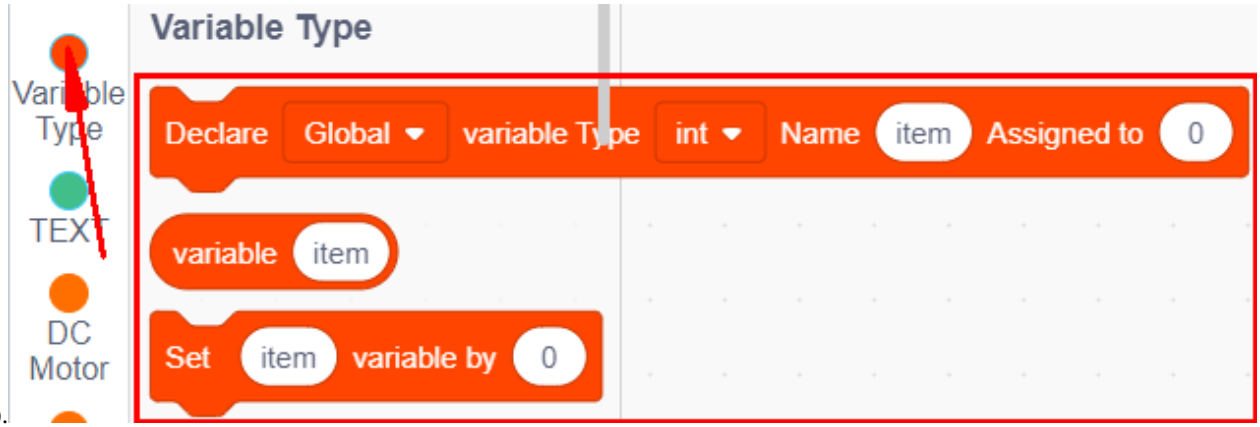
(3).


Pins
Pins
Serial
Data
Variable
Type
TEXT
set pin 0 mode input
set digital pin 0 out high
set pwm pin 3 out 255
read digital pin 0

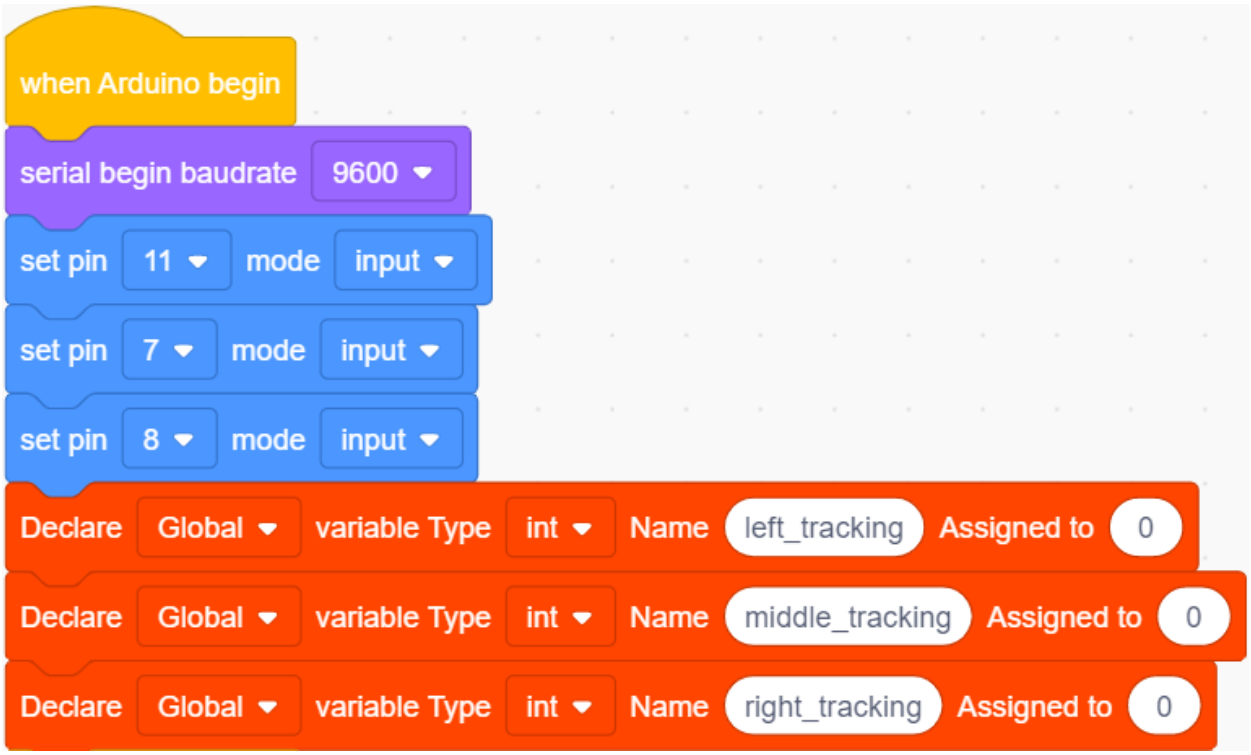
(4).



(5).



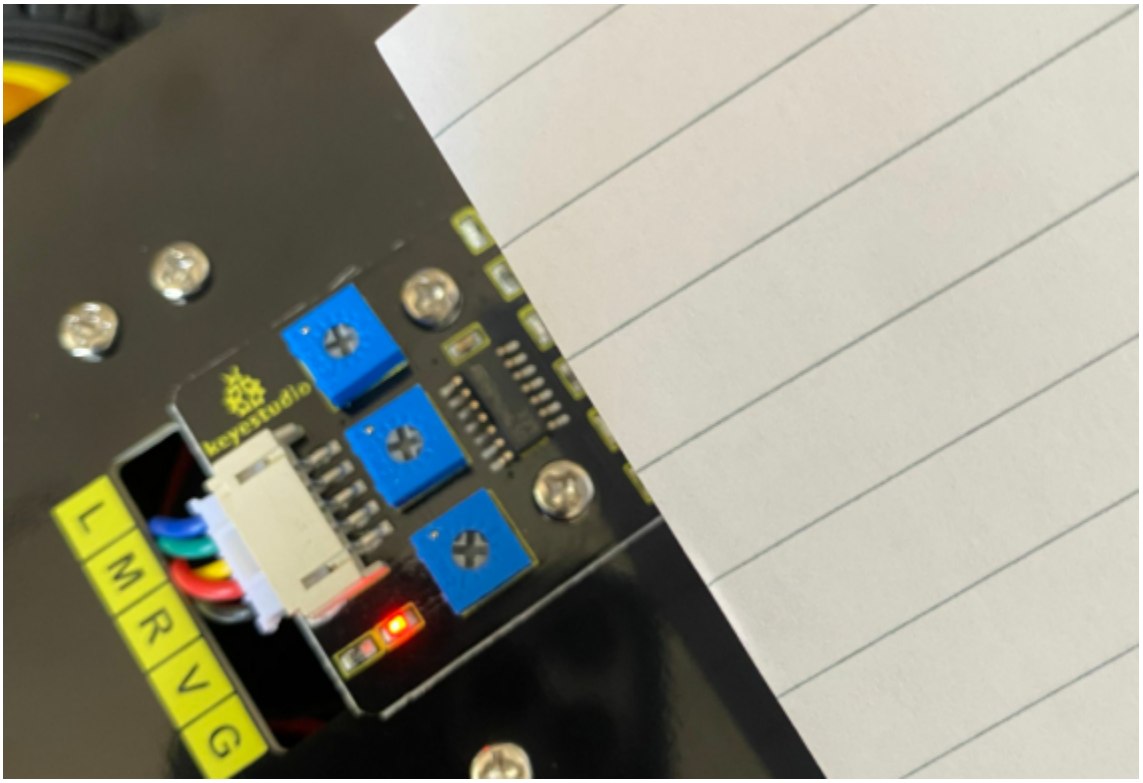
Complete Test Code

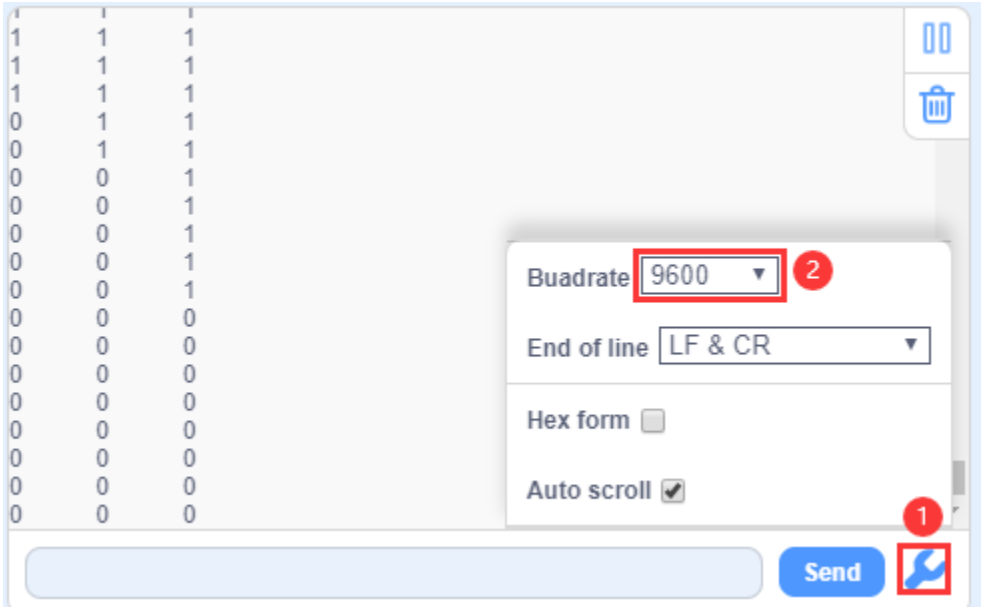

when Arduino begin
serial begin baudrate 9600
set pin 11 mode input
set pin 7 mode input
set pin 8 mode input
Declare Global variable Type int Name left_tracking Assigned to 0
Declare Global variable Type int Name middle_tracking Assigned to 0
Declare Global variable Type int Name right_tracking Assigned to 0

6.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, and use a USB cable to connect the computer to power the board.

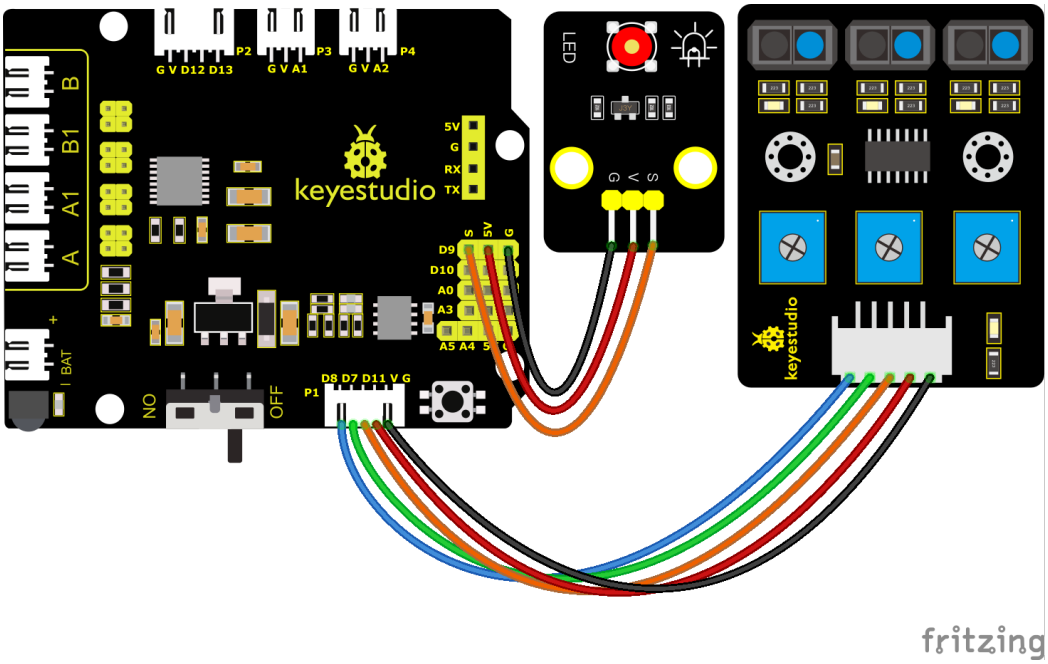
After powering on, click to set the baud rate to 9600 and you will view status of three line tracking sensors. When no signals are received, the value is 1. If we cover the sensor with a white paper, the value will be 0.





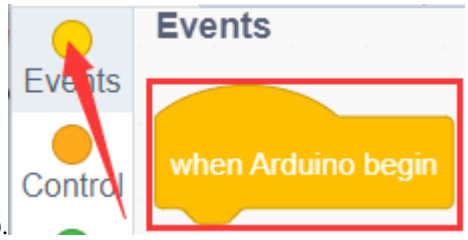
7.Extension Practice

After knowing its working principle, you can connect an LED to D9 so as to control LED by it.

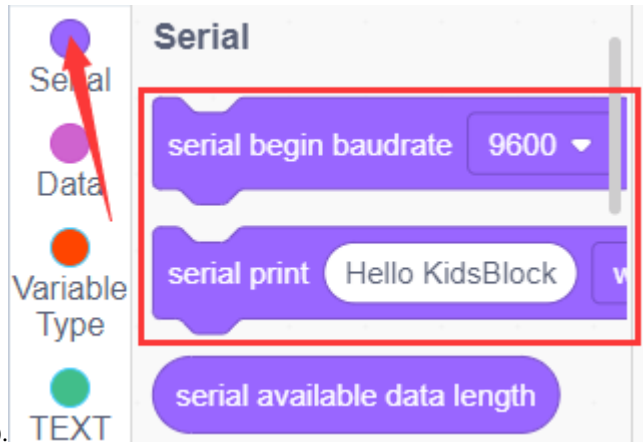


You can drag blocks to edit. Blocks listed below are for your reference.

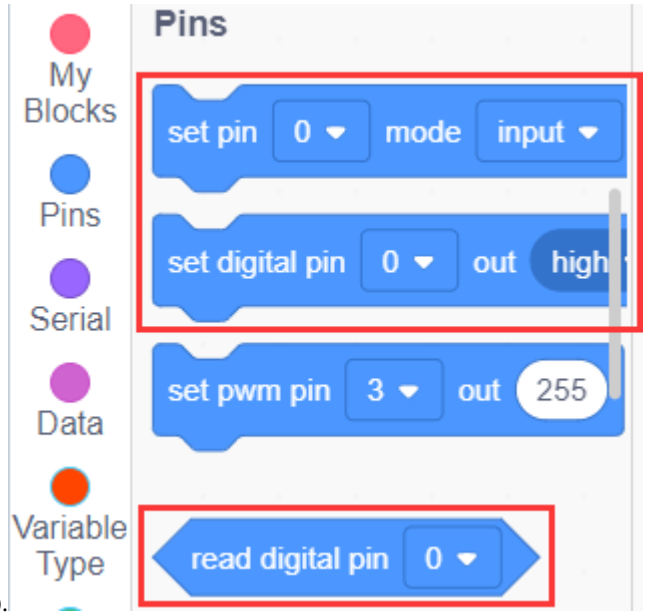
(1).



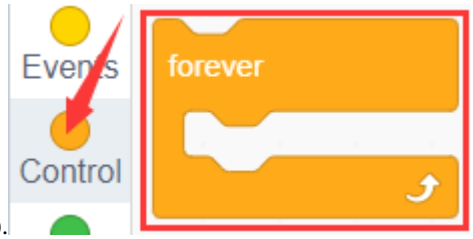
(2).



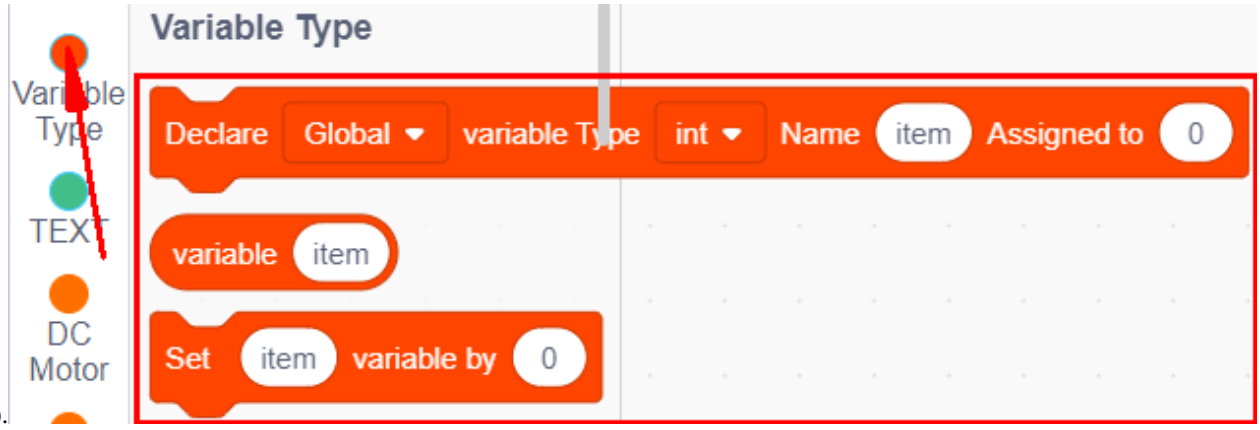
(3).


My Blocks
Pins
Serial
Data
Variable Type
Pins
set pin 0 mode input
set digital pin 0 out high
set pwm pin 3 out 255
read digital pin 0

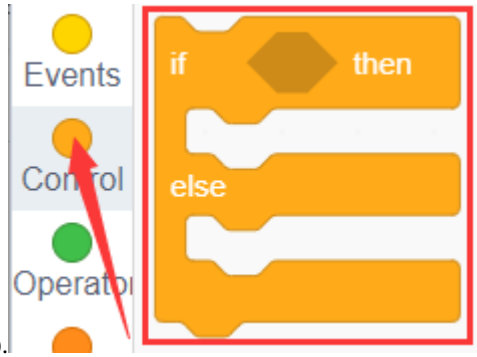
(4).


Events
Control
forever

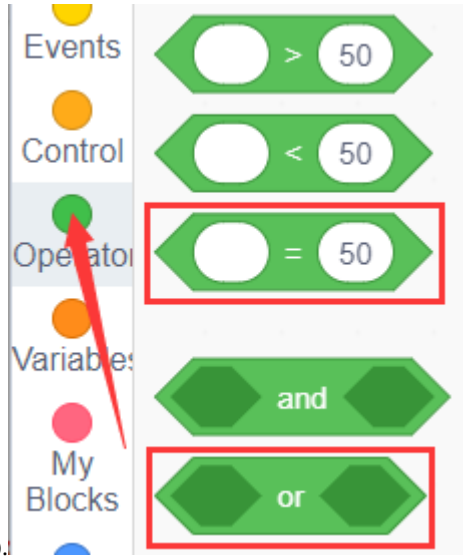
(5).


Variable Type
TEXT
DC Motor
Variable Type
Declare Global variable Type int Name item Assigned to 0
variable item
Set item variable by 0

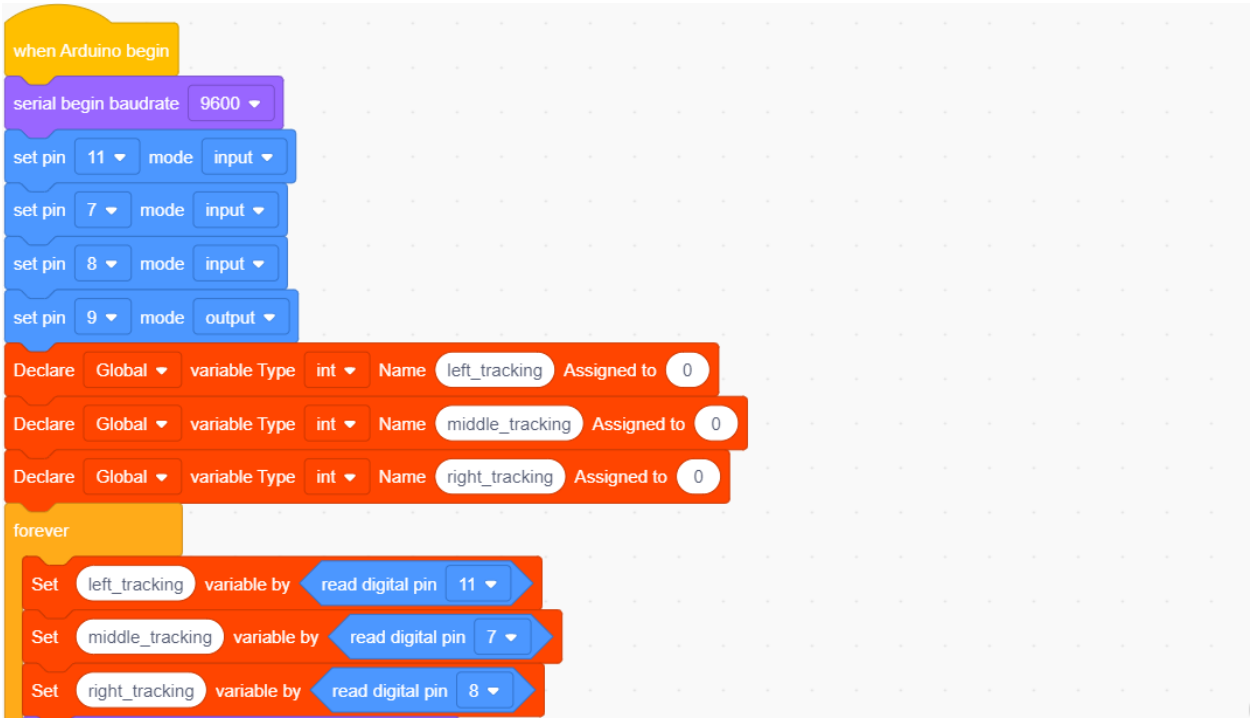
(6).

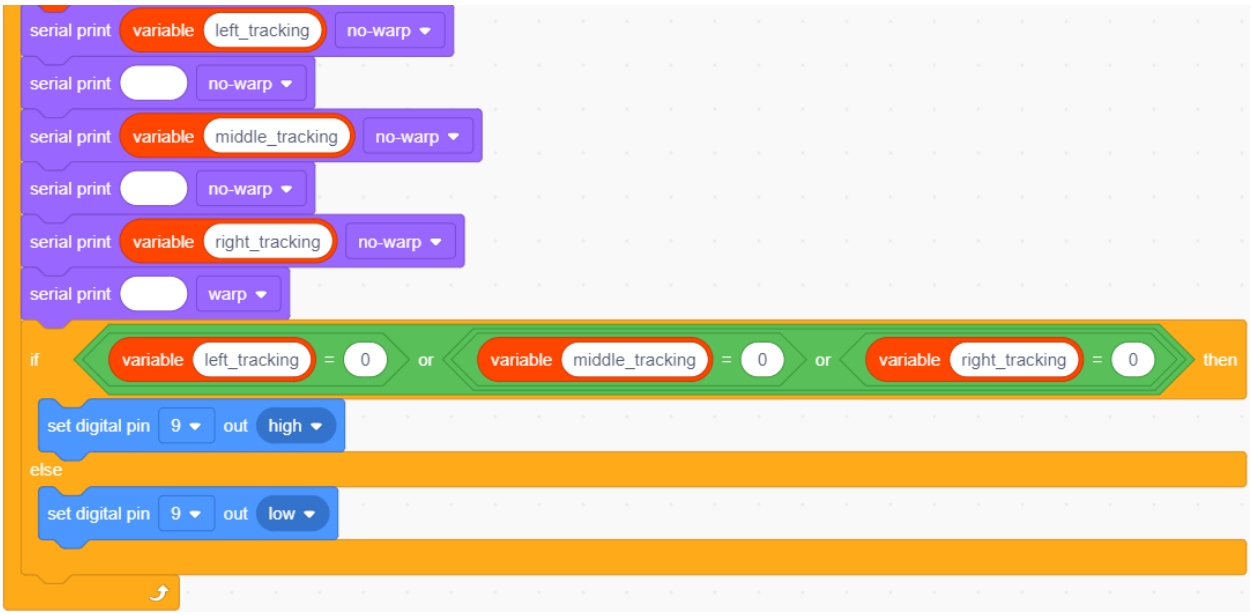

Events
Control
Operator
if then
else

(7).



Complete Test Code





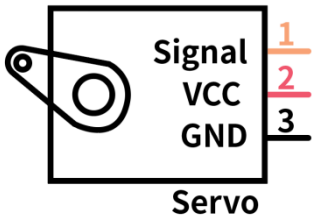
After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, and use a USB cable to connect the computer to power the board.

After powering on, make a paper close to the sensor, then we can find the LED light up when covering the line tracking sensor.

10.2.4 Project 4 Servo Control

1.Description

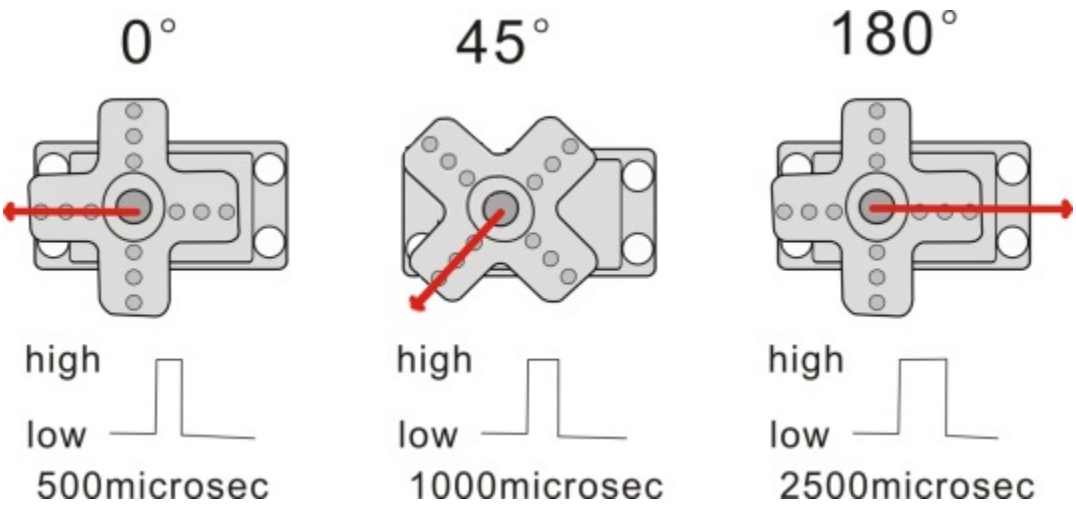
Servo motor is a position control rotary actuator. It mainly consists of a housing, a circuit board, a core-less motor, a gear and a position sensor. Its working principle is that the servo receives the signal sent by MCUs or receivers and produces a reference signal with a period of 20ms and width of 1.5ms, then compares the acquired DC bias voltage to the voltage of the potentiometer and obtain the voltage difference output.



In general, servo has three lines in brown, red and orange. The brown wire is grounded, the red one is a positive pole line and the orange one is a signal line.

The rotation angle of servo motor is controlled by regulating the duty cycle of PWM (Pulse-Width Modulation) signal. The standard cycle of PWM signal is 20ms (50Hz). Theoretically, the width is distributed between 1ms-2ms, but in

fact, it's between 0.5ms-2.5ms. The width corresponds the rotation angle from 0° to 180°. But note that for different brand motors, the same signal may have different rotation angles.



The corresponding servo angles are shown below:

High level time	Servo angle
0.5ms	0 degree
1ms	45 degree
1.5ms	90 degree
2ms	135 degree
2.5ms	180 degree

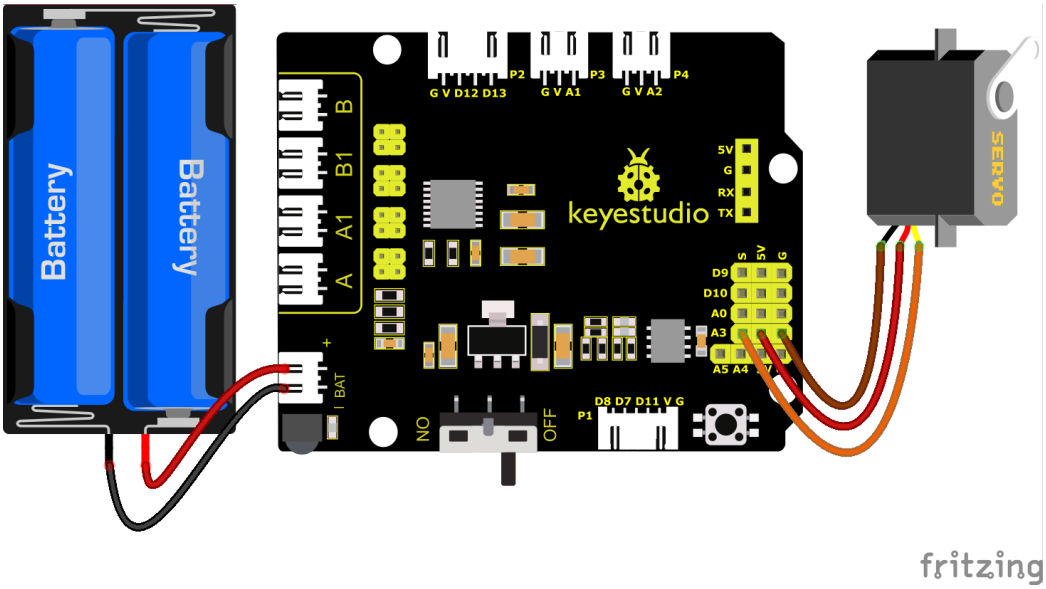
2.Specification

- Working voltage: DC 4.8V ~ 6V
- Operating angle range: about 180 ° (at 500 → 2500 sec)
- Pulse width range: 500 → 2500 sec
- No-load speed: 0.12 ± 0.01 sec / 60 (DC 4.8V) 0.1 ± 0.01 sec / 60(DC 6V)
- No-load current: 200 ± 20mA (DC 4.8V) 220 ± 20mA (DC 6V)
- Stopping torque: 1.3 ± 0.01kg · cm (DC 4.8V) 1.5 ± 0.1kg · cm (DC6V)
- Stop current: 850mA (DC 4.8V) 1000mA (DC 6V)
- Standby current: 3 ± 1mA (DC 4.8V) 4 ± 1mA (DC 6V)

3.Components

Development Board *1	8833 Motor Driver *1	Servo*1
18650 Battery Holder*1	USB Cable*1	18650 Battery*2self-provided

4.Wiring Diagram



Wiring note: The servo is connected to GGNDVVCCand A3, the brown line of servo is linked with Gnd(G), the red one is connected to 5v(V) and the orange one is attached to A3.

The servo is obliged to connect to the external power due to its high demand for driving servo current. Generally, the current of development board is not big enough. If without connecting the external power, the development board could be burnt.

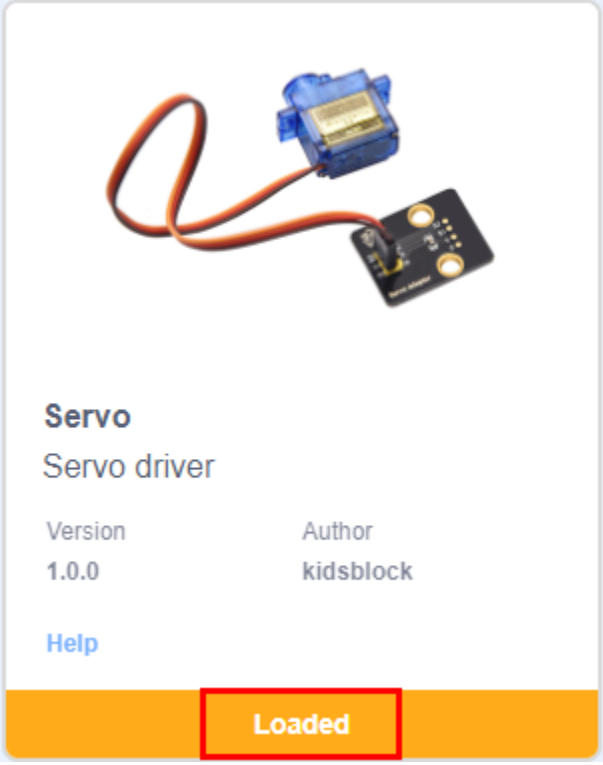
5.Test Code

Before writing the code, it is necessary to import the servo library file. The specific steps are as follows:

Click

to enter the extension library interface of sensors/modules/components, then look for "**Servo**".

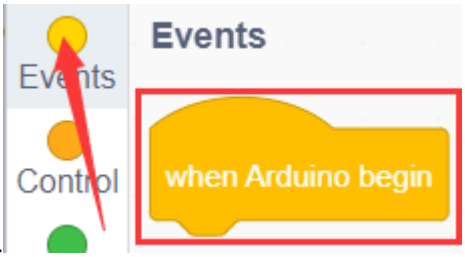
component and click it. In this way, "**Not Loaded**" changes to "**loaded**", indicating that "**Servo**" component was added successfully.



Click Back to go back to the code editor, and in the modules area you can see the "**Servo**" component directive block added.

You can drag blocks to edit. Blocks listed below are for your reference.

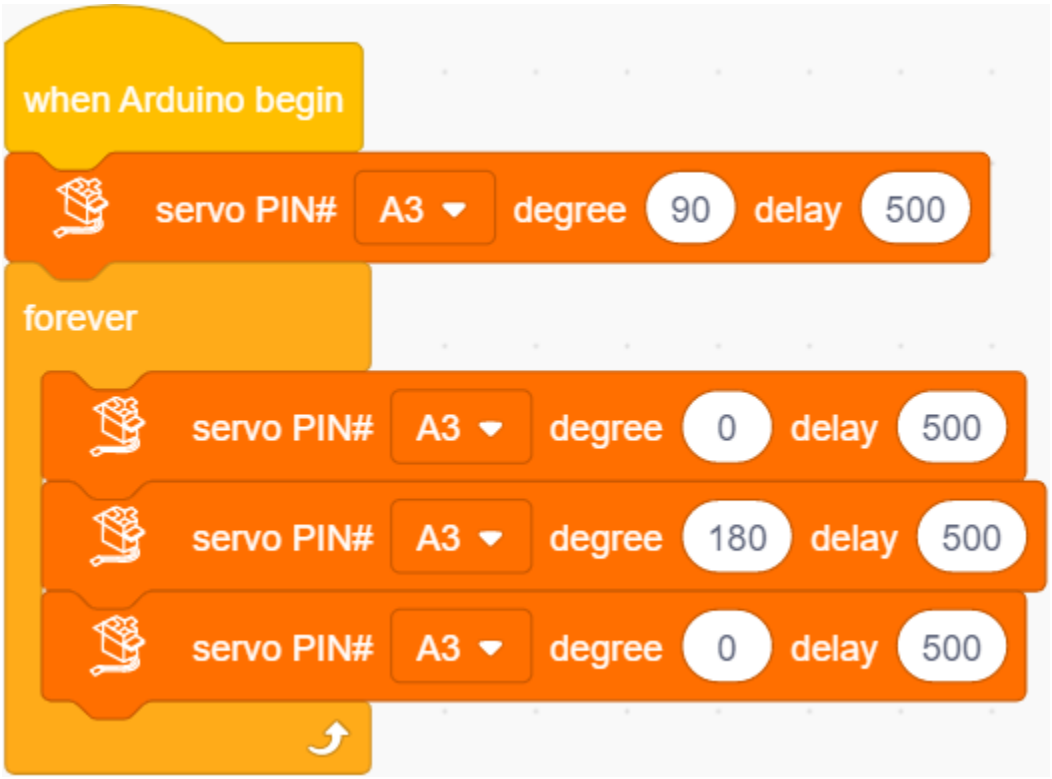
(1).



(2).

(3).

Complete Test Code



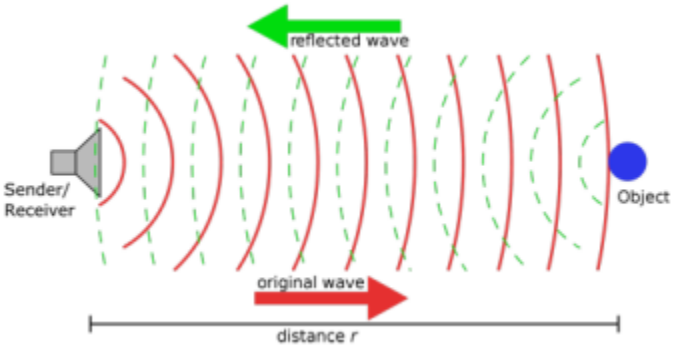
6.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, and power on the external power. After powering on, turn the dip switch to the "ON" end, then servo will swing in the range of 0° to 180°.

10.2.5 Project 5 Ultrasonic Sensor

1.Description

The HC-SR04 ultrasonic sensor uses sonar to determine distance to an object like what bats do. It offers excellent non-contact range detection with high accuracy and stable readings in an easy-to-use package. It comes complete with an ultrasonic transmitter and receiver modules.

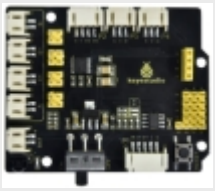


The HC-SR04 or the ultrasonic sensor is being used in a wide range of electronics projects for creating obstacle detection and distance measuring application as well as various other applications. Here we have brought the simple method to measure the distance with arduino and an ultrasonic sensor and how to use the ultrasonic sensor with Arduino.

2.Specification

- Working Voltage :+5V DC
- Quiescent Current : <2mA
- Working Current: 15mA
- Effectual Angle: <15°
- Distance Range : 2cm – 300 cm
- Precision : 0.3 cm
- Measuring Angle: 30 degree
- Trigger Input Pulse width: 10uS

3.Components

Development Board *1	8833 Motor Driver *1	Red LED Module*1	Ultrasonic Sensor*1
			
4P Dupont Wire*1	USB Cable*1	3P Dupont Wire*1	

4.Working Principle

As the above picture shown, it is like two eyes. One is transmitting end, the other is receiving end.

The ultrasonic module will emit the ultrasonic waves after triggering a signal. When the ultrasonic waves encounter the object and are reflected back, the module outputs an echo signal, so it can determine the distance of the object from the time difference between the trigger signal and the echo signal.

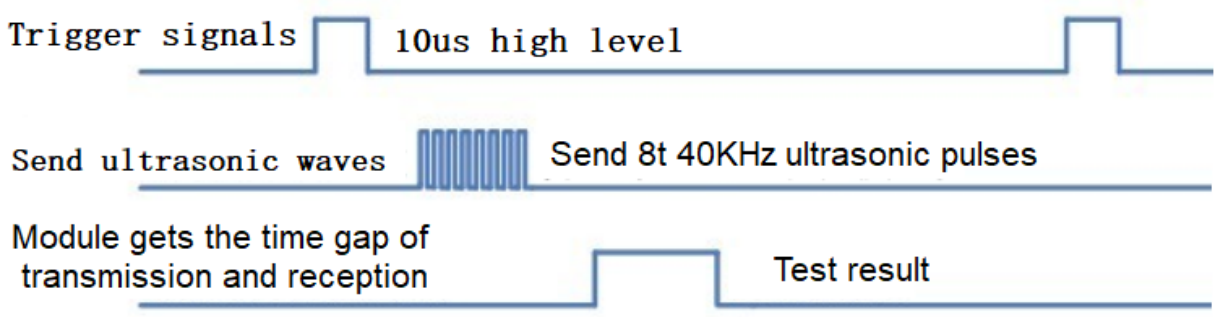
The t is the time that emitting signal meets obstacle and returns. And the propagation speed of sound in the air is about 343m/s, and distance = speed * time. However, the ultrasonic wave emits and comes back, which is 2 times of distance. Therefore, it needs to be divided by 2, the distance measured by ultrasonic wave = (speed * time)/2.

Use method and chart of ultrasonic module:

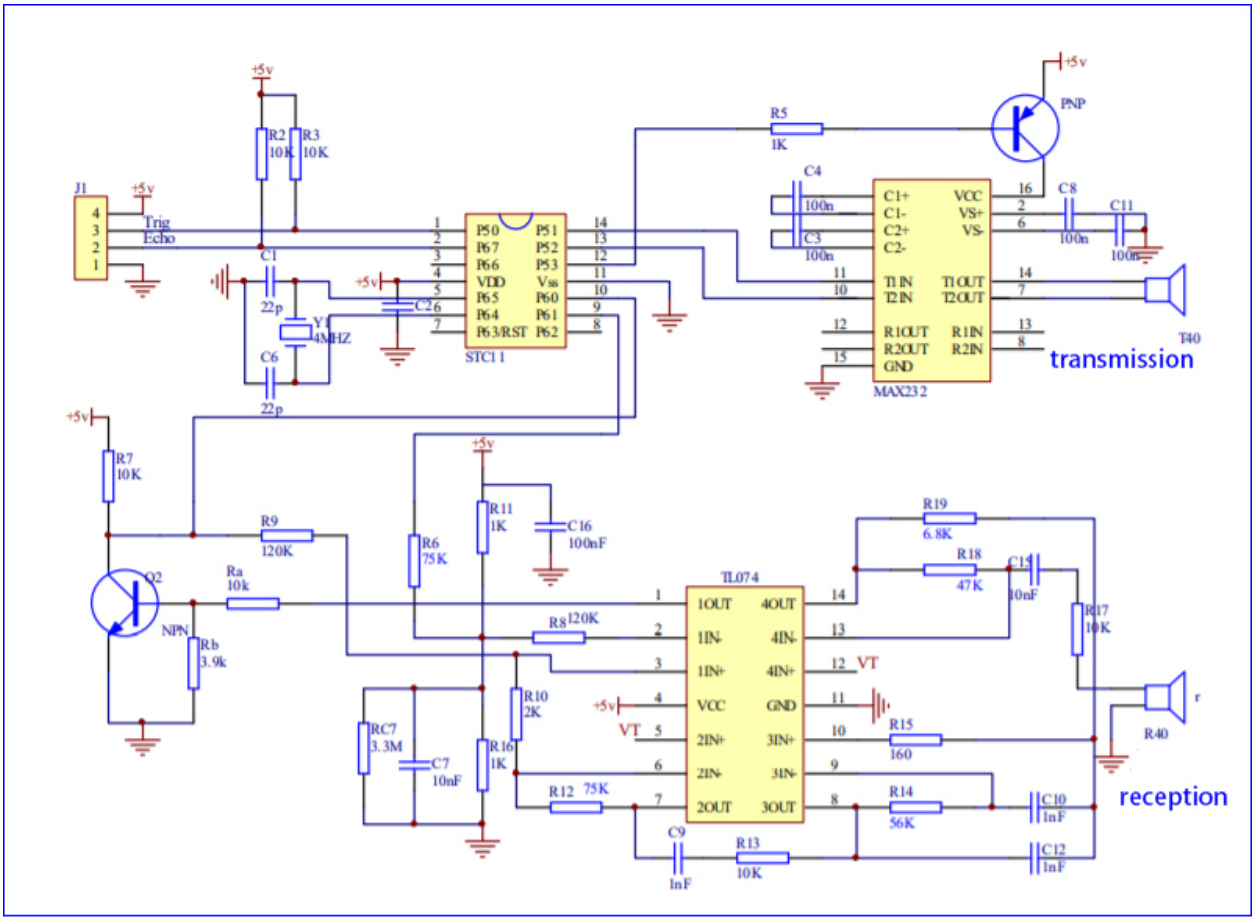
1).Use the GPIO pin to give a high level signal of at least 10s to the Trig pin of SR04, which can trigger it to detect distance.

2).After triggering, the module will automatically send eight 40KHz ultrasonic pulses and detect whether there is a signal return. This step will be completed automatically by the module.

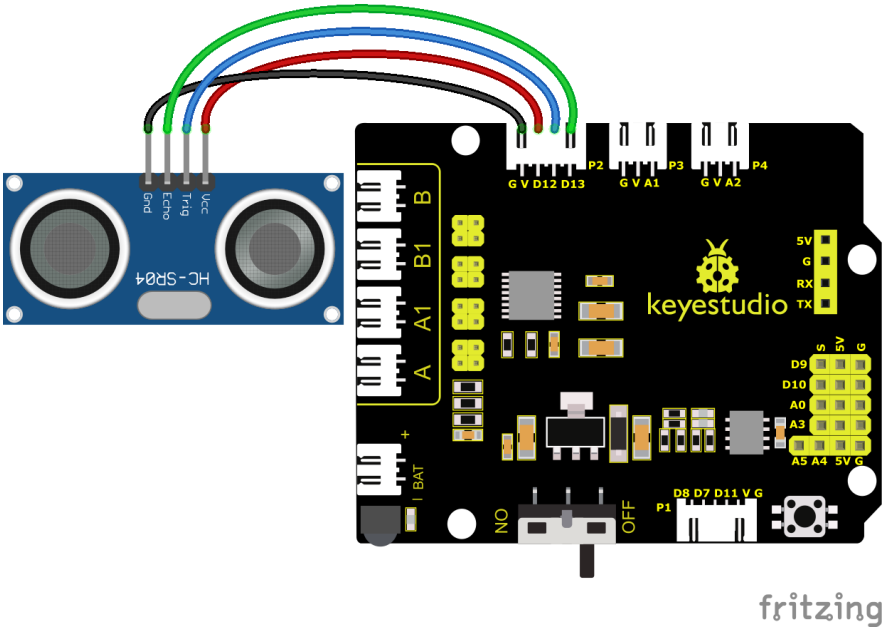
3).If the signal returns, the Echo pin will output a high level, and the duration of the high level is the time from the transmission of the ultrasonic wave to the return.



Circuit diagram of ultrasonic sensor:



5.Wiring Diagram



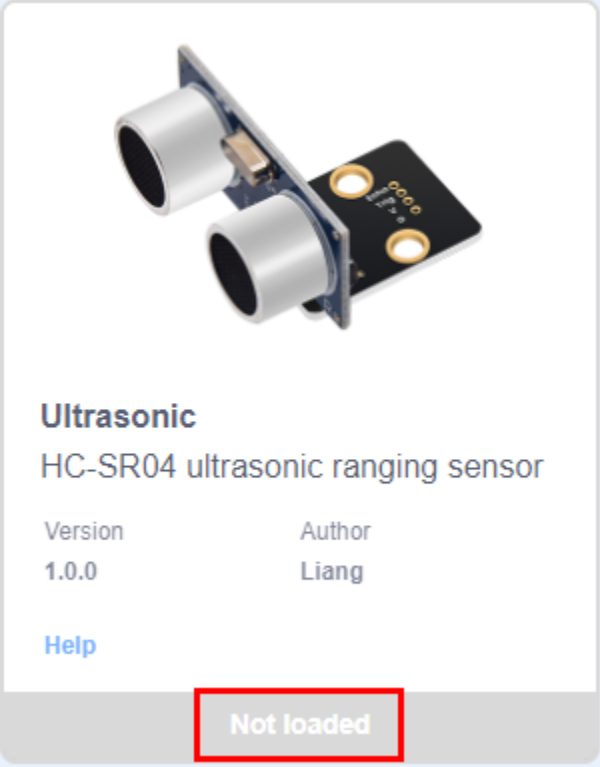
VCC, Trig, Echo and Gnd of the ultrasonic sensor are connected to 5V(V), D12, D13 and Gnd(G)

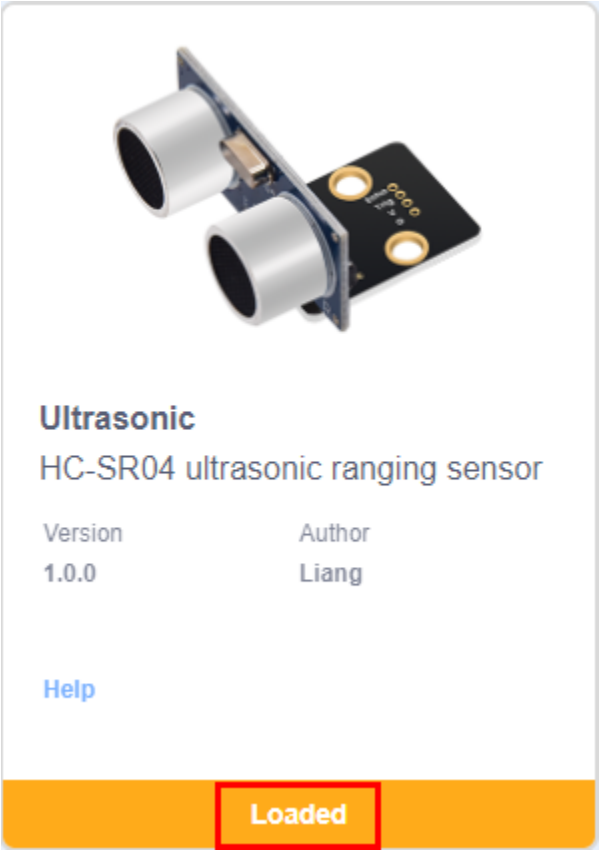
6.Test Code

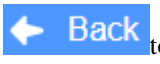
Before writing the code, it is necessary to import the library file of the ultrasonic sensor. The specific steps are as follows:

Click

to enter the extension library interface of sensors/modules/components, then search for "**Ultrasonic**" sensor and click it. In this way, "**Not loaded**" changes to "**loaded**", indicating that "**Ultrasonic**" sensor was added successfully.





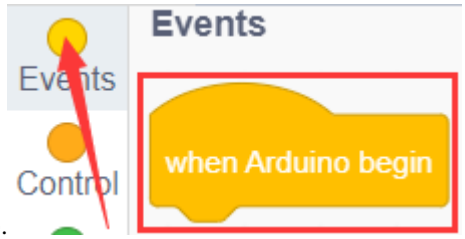
Click 

to return to the code editor interface, the instruction block of the added "**Ultrasonic**" sensor can be

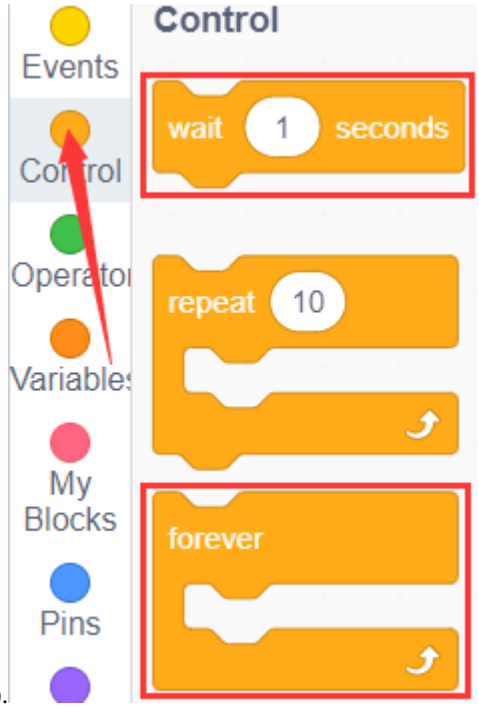
seen in the module area.

You can drag blocks to edit. Blocks listed below are for your reference.

1).

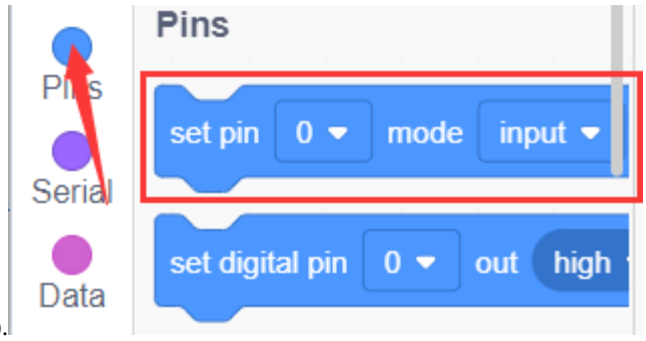


(2).

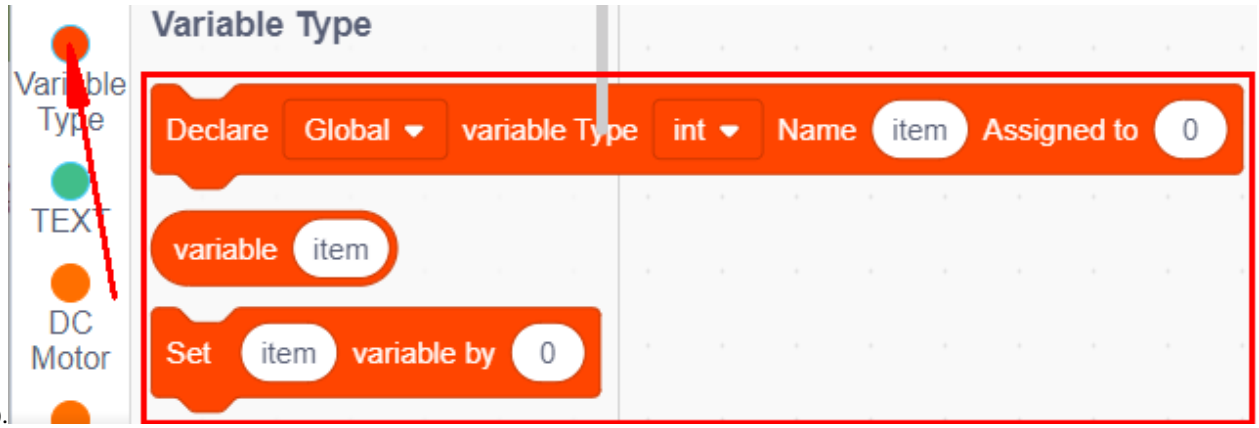


(3).

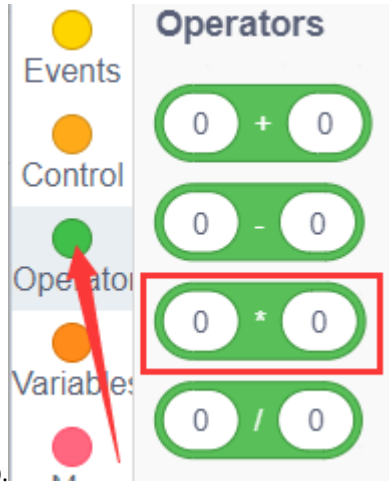
(4).



(5).

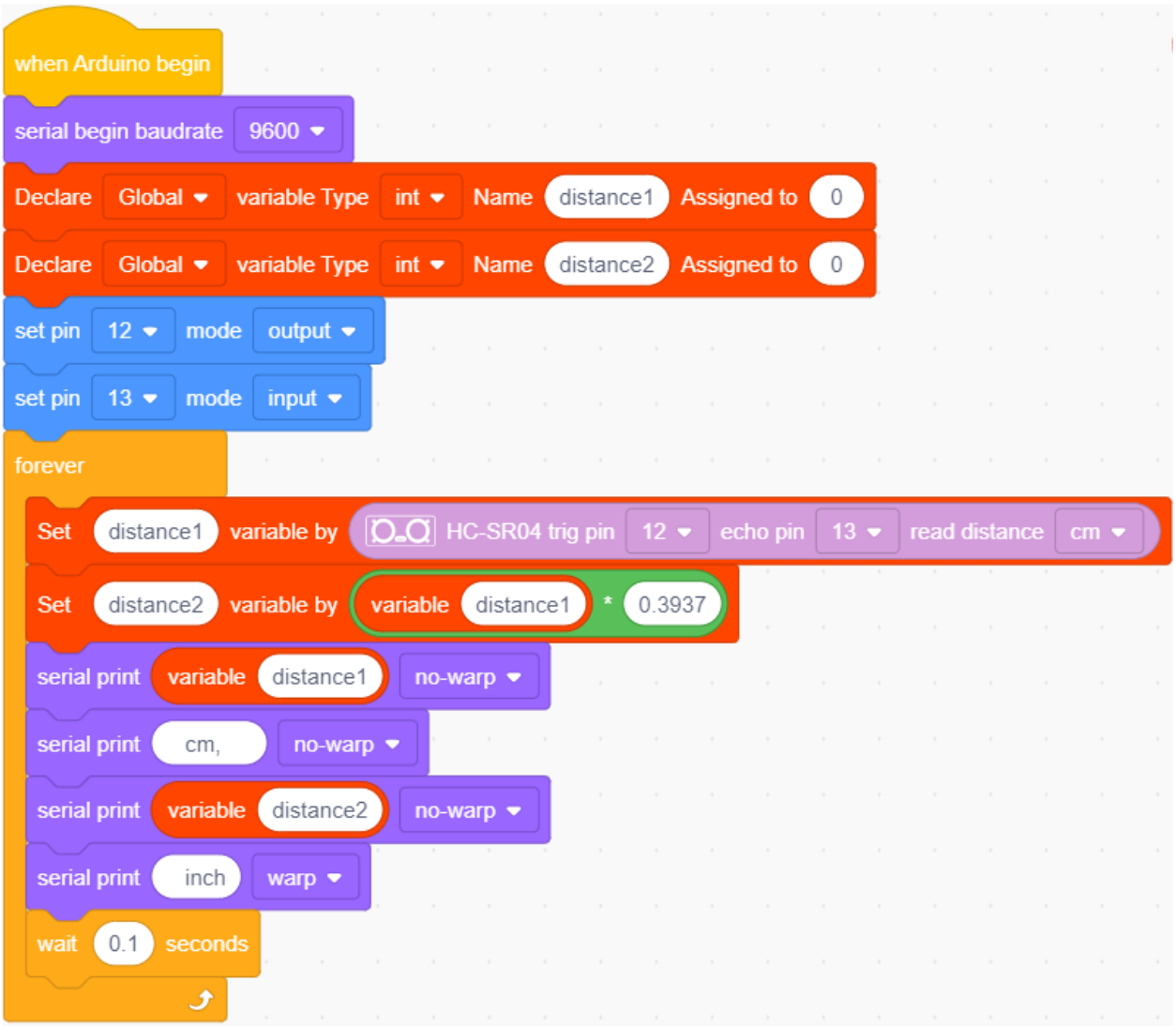


(6).



(7).

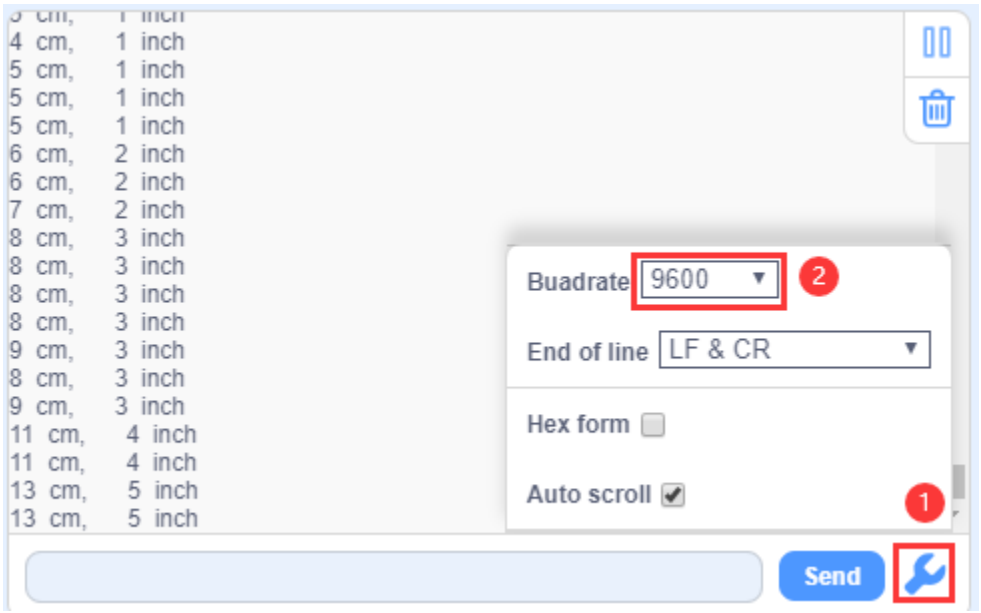
Complete Test Code



7.Test Result

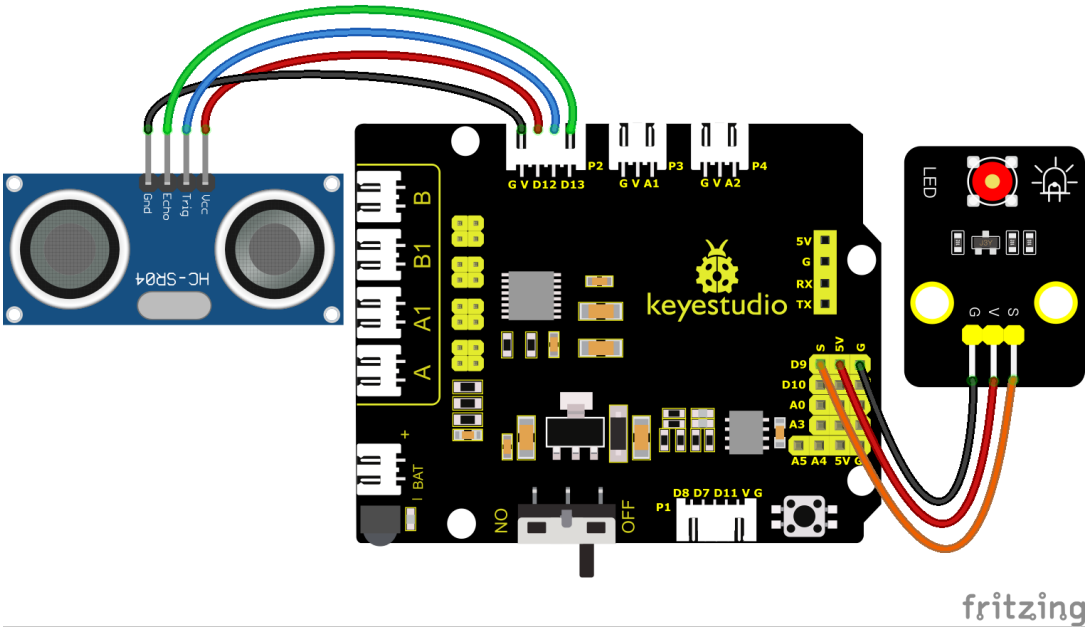
After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, then connect the computer via a USB cable to power the board. After powering on, click to set baud rate to 9600.

The detected distance will be displayed, and the unit is cm and inch. Hinder the ultrasonic sensor by hand, the displayed distance value gets smaller.



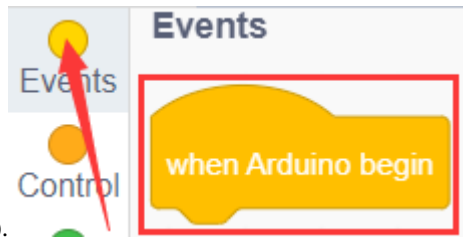
8.Extension Practice

We have just measured the distance displayed by the ultrasonic. How about controlling the LED with the measured distance? Let's try it and connect an LED light module to the D9 pin.

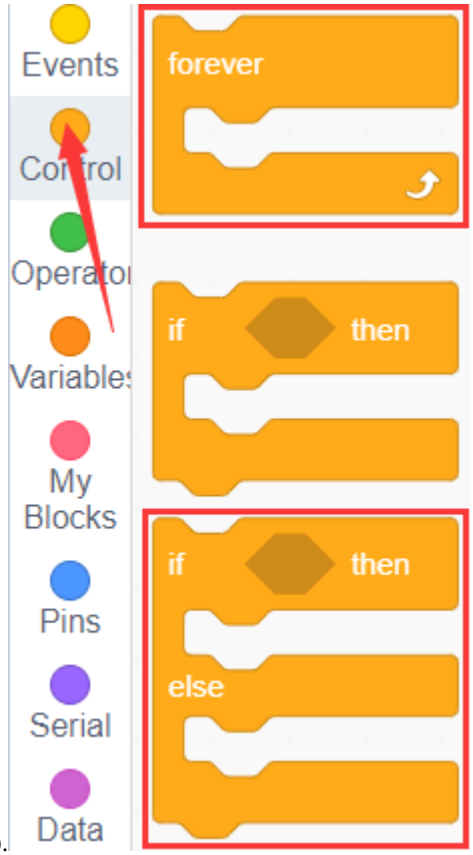


You can drag blocks to edit. Blocks listed below are for your reference.

(1).



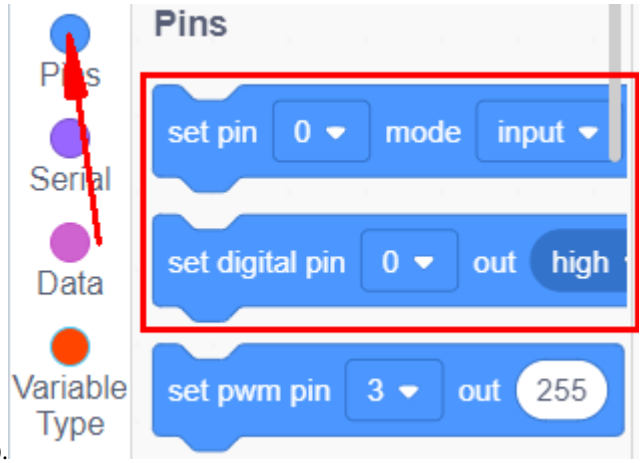
(2).


Events
Control
Operator
Variables
My Blocks
Pins
Serial
Data
forever
if then
if then
else

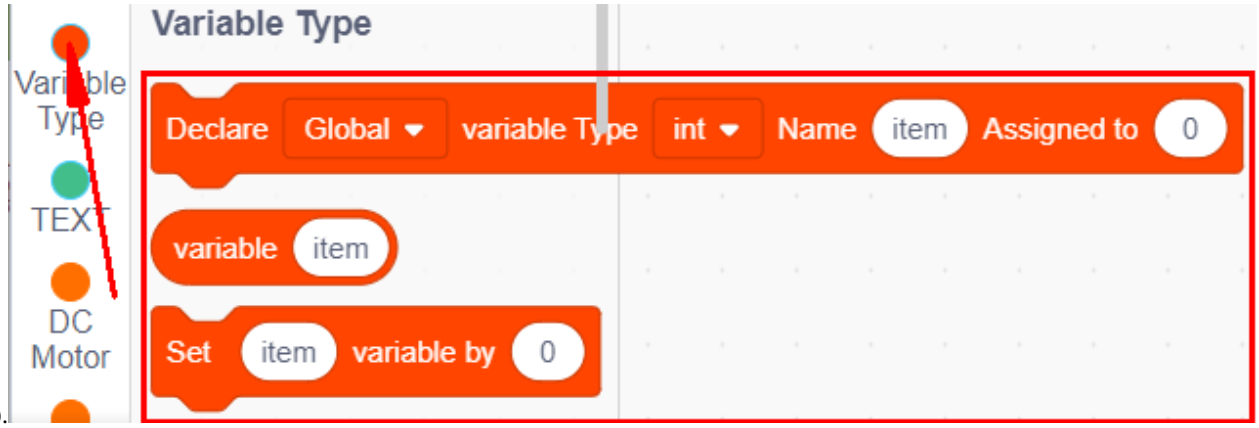
(3).

Serial
Data
Variable Type
Serial
serial begin baudrate 9600
serial print Hello KidsBlock

(4).


Pins
Serial
Data
Variable Type
Pins
set pin 0 mode input
set digital pin 0 out high
set pwm pin 3 out 255

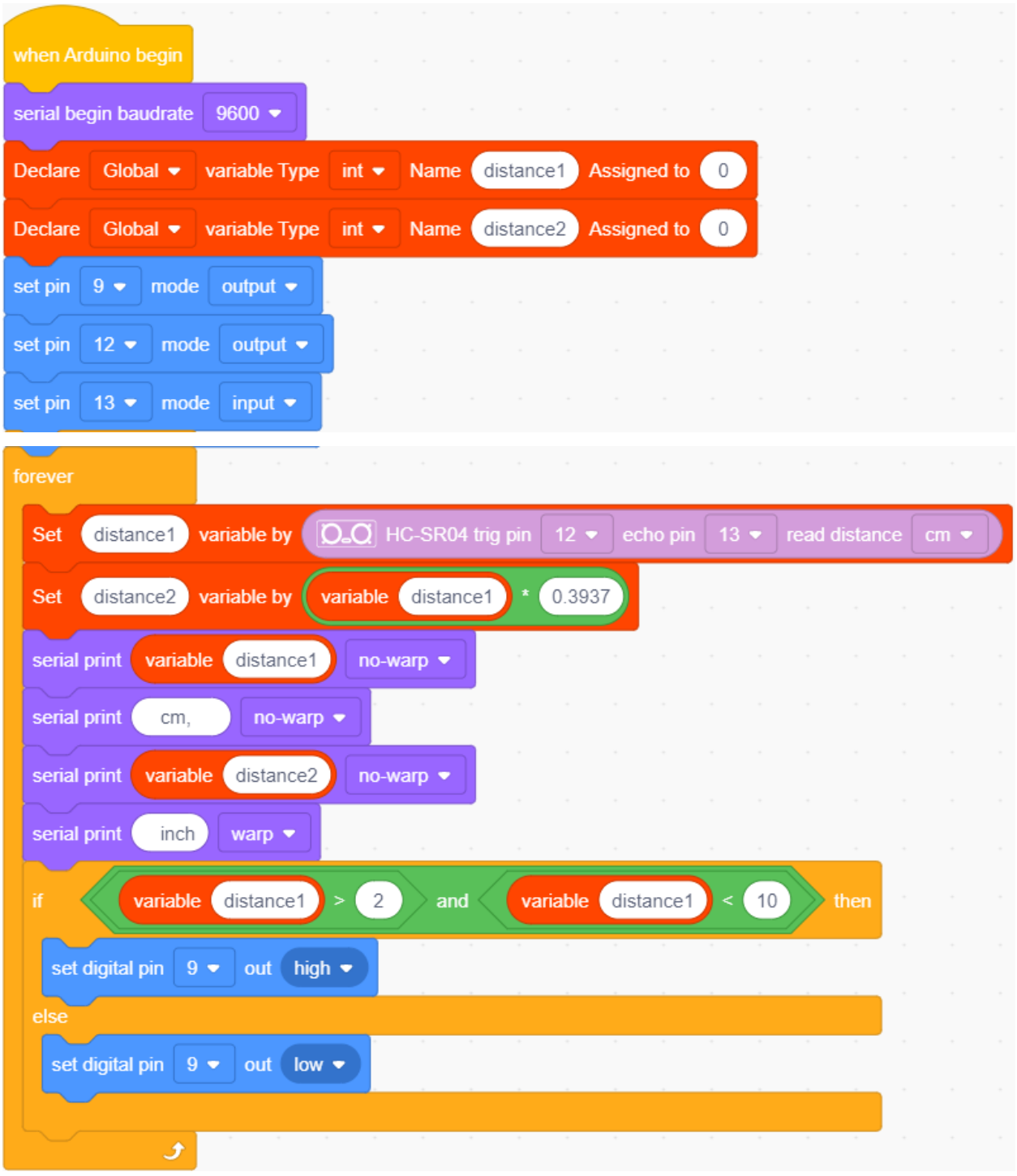
(5).



(6).

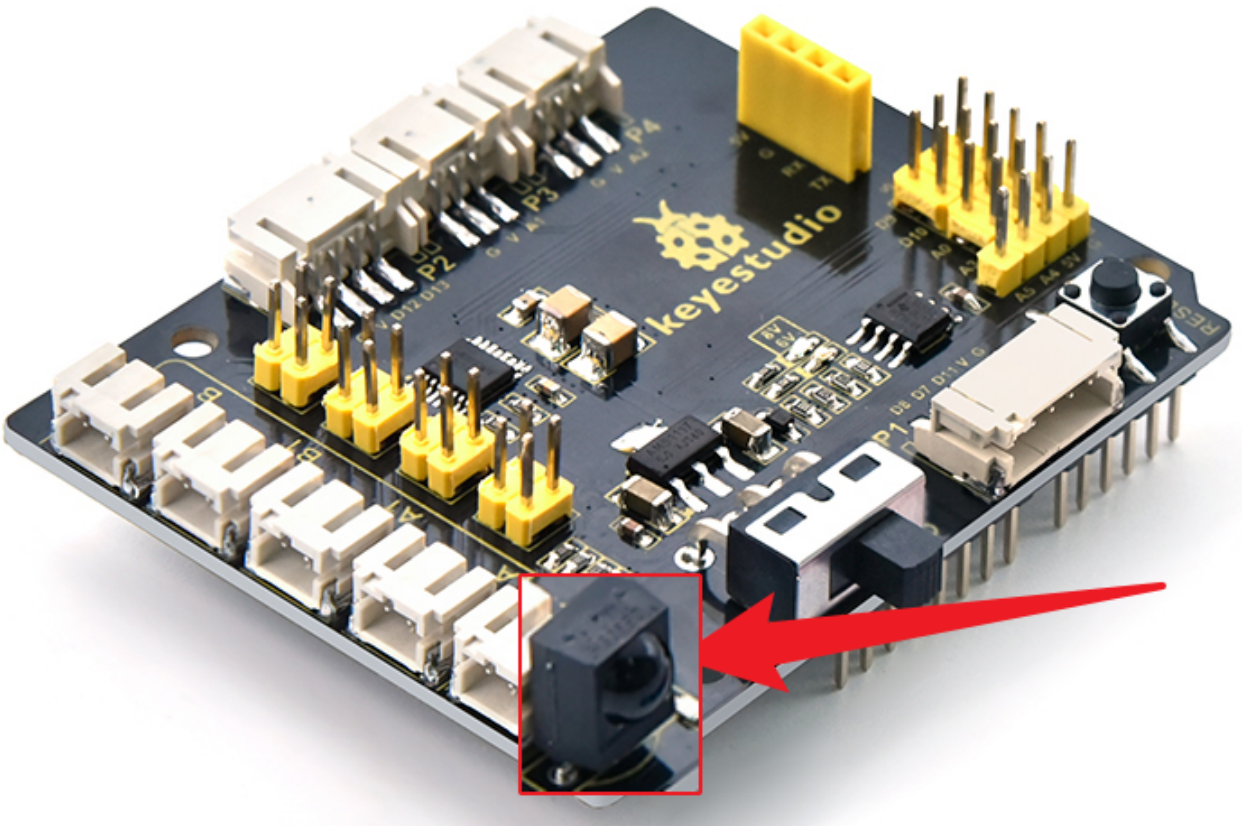
(7).

Complete Test Code



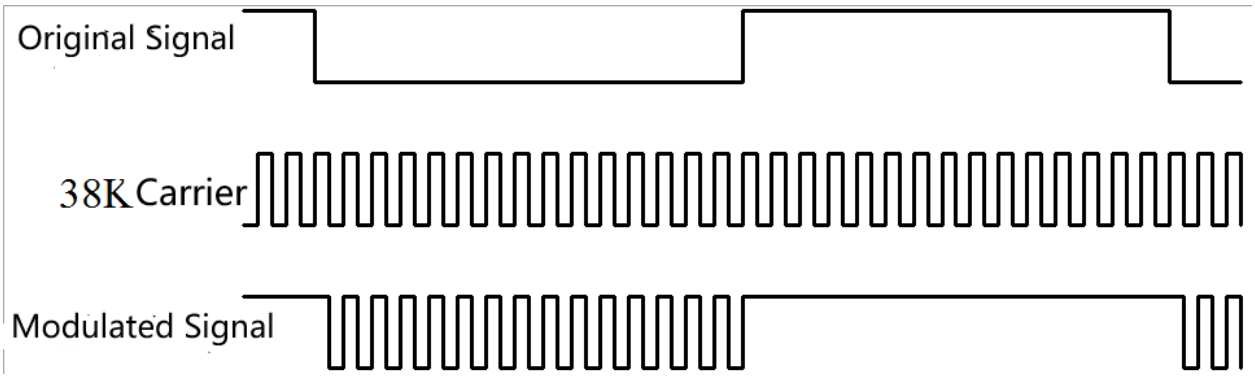
After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, then connect the computer via a USB cable to power the board. After powering on, block the ultrasonic sensor by hand(the distance is between 2-10cm), then check if the LED is on.

10.2.6 Project 6 IR Reception



1.Description

There is no doubt that infrared remote control is ubiquitous in daily life. It is used to control various household appliances, such as TVs, stereos, video recorders and satellite signal receivers. Infrared remote control is composed of infrared transmitting and infrared receiving systems, that is, an infrared remote control and infrared receiving module and a single-chip microcomputer capable of decoding.



The 38K infrared carrier signal emitted by remote controller is encoded by the encoding chip in the remote controller. It is composed of a section of pilot code, user code, user inverse code, data code, and data inverse code. The time interval of the pulse is used to distinguish whether it is 0 or 1 signal and the encoding is made up of these 0, 1 signals.

The user code of the same remote control is constant while the data code can distinguish the key.

When the remote control button is pressed, the remote control sends out an infrared carrier signal. When the IR receiver

receives the signal, the program will decode the carrier signal and determines which key is pressed. The MCU decodes the received 01 signal, thereby judging what key is pressed by the remote control.

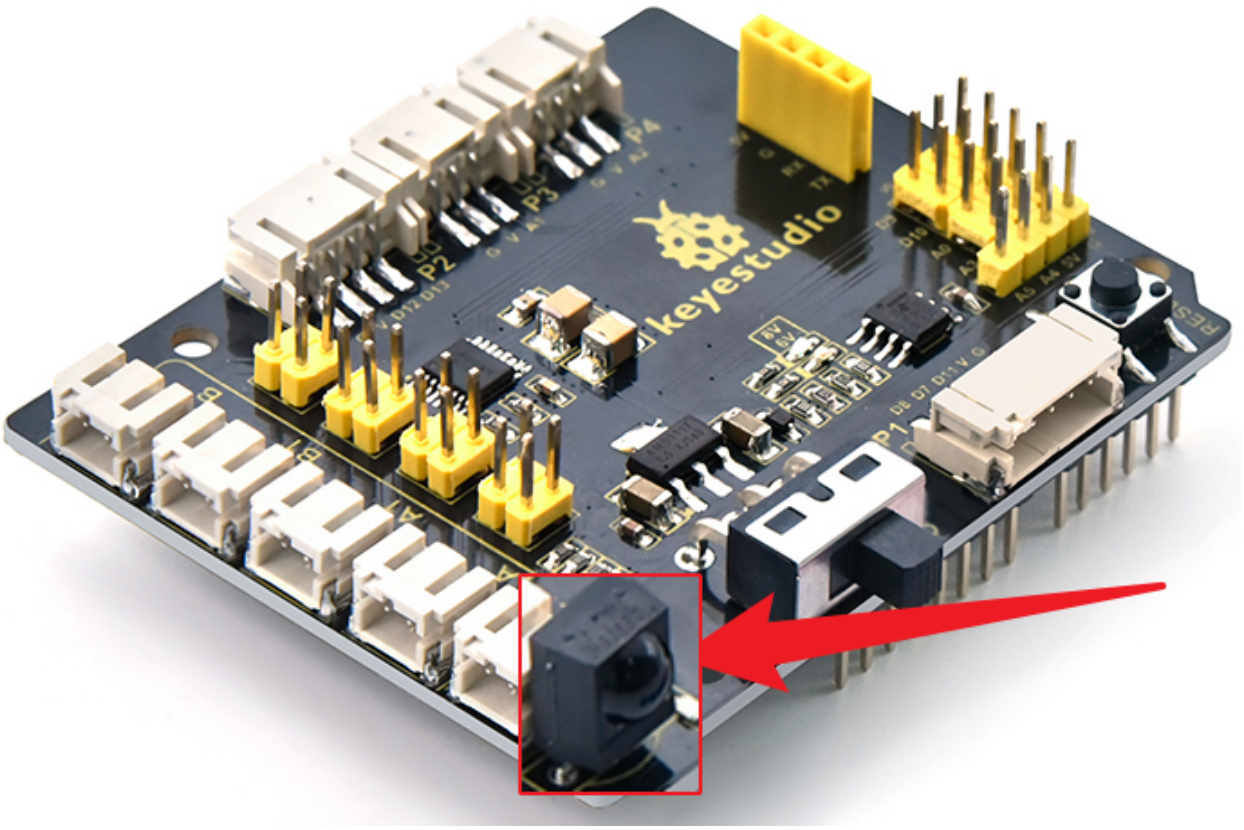
Infrared receiver we use is an infrared receiver module. Mainly composed of an infrared receiver head, which is a device that integrates reception, amplification, and demodulation. Its internal IC has completed demodulation, and can achieve from infrared reception to output and be compatible with TTL signals.

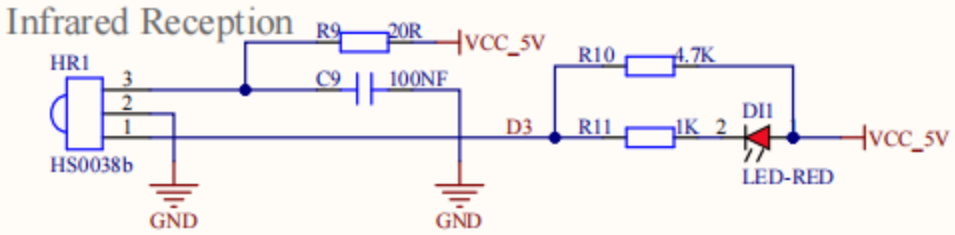
Additionally, it is suitable for infrared remote control and infrared data transmission. The infrared receiving module made by the receiver has only three pins, signal line, VCC and GND. It is very convenient to communicate with Arduino and other microcontrollers.

2.Specification

- Operating Voltage: 3.3-5VDC)
- Output Signal: Digital signal
- Receiving Angle: 90 degrees
- Frequency: 38khz
- Receiving Distance: 10m

The picture shows the real product and circuit diagram of the infrared receiver.





3.Components

Development Board *1	8833 Motor Driver *1	Red LED Module*1
3P F-F Dupont Wire*1	USB Cable*1	

Since the 8833 board integrates with the IR receiver, it doesn't need wiring up. Pins of IR receiver module are G(GND, VVCCand D3.

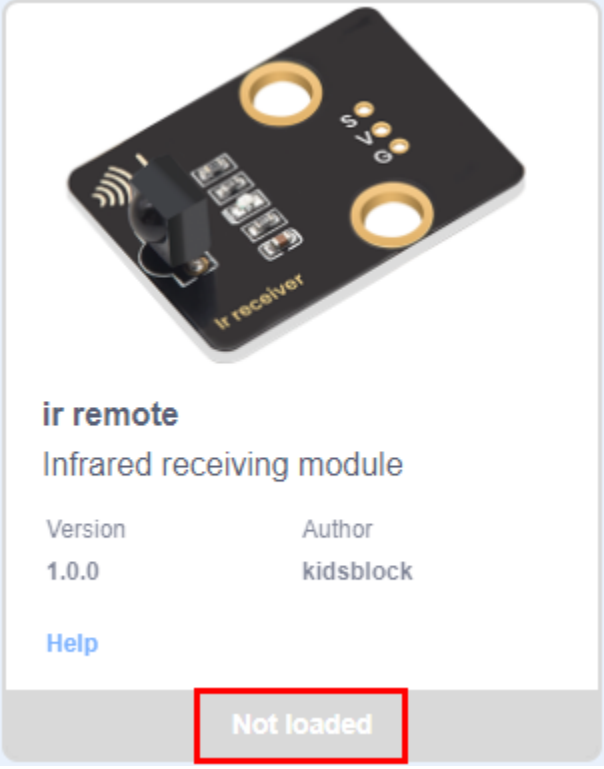
4.Test Code

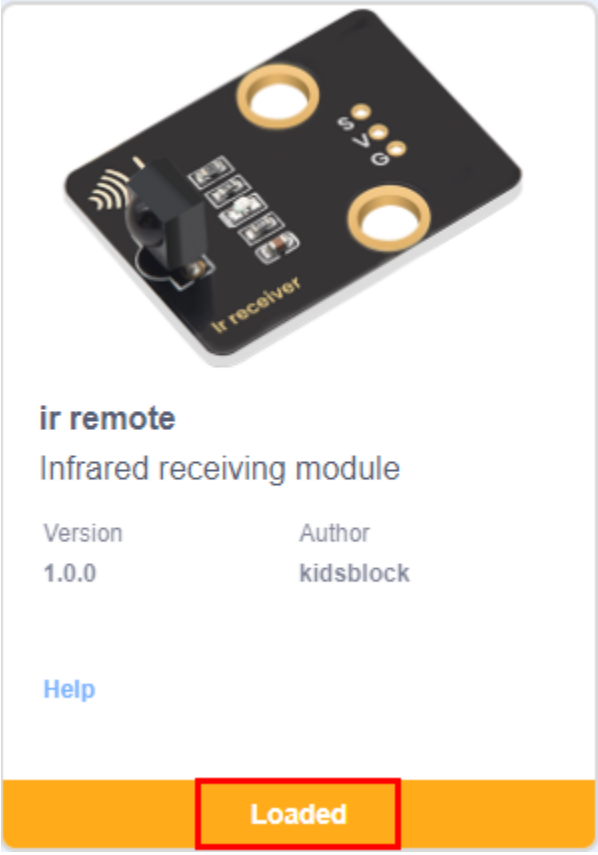
Please note: The infrared module shown in the software demonstration is already integrated into the expansion board and is not supplied separately. Consequently, you will not find the module depicted in the image below within the product.

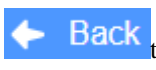
Before writing the code, it is necessary to import the library file of the IR receiver sensor. The specific steps are as follows:

Click

to enter the extension library interface of sensors/modules/components, then search for"**ir remote**" sensor and click it. In this way, "**Not loaded**" changes to "**loaded**", indicating that ir remote"sensor was added successfully.





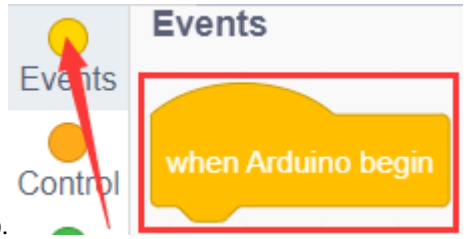
Click 

to return to the code editor interface, the instruction block of the added "**ir remote**"sensor can be

seen in the module area.

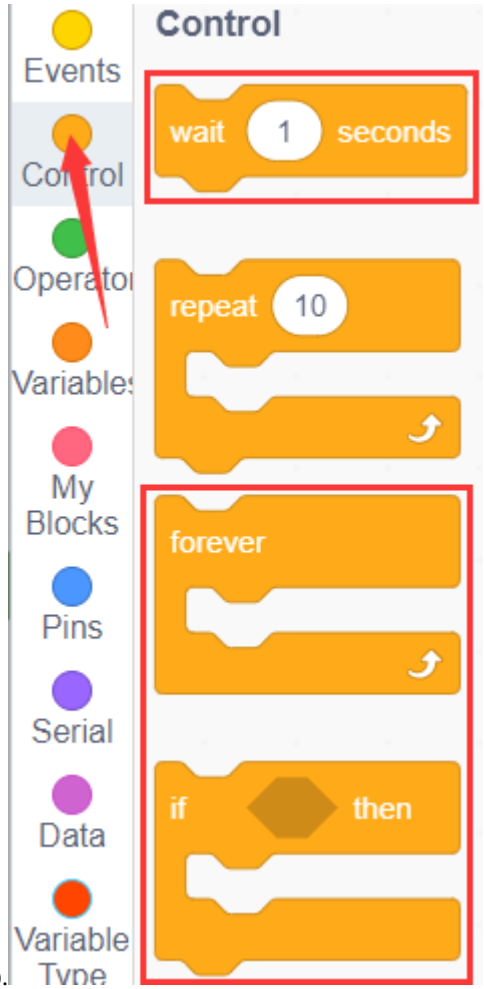
You can drag blocks to edit. Blocks listed below are for your reference.

(1).

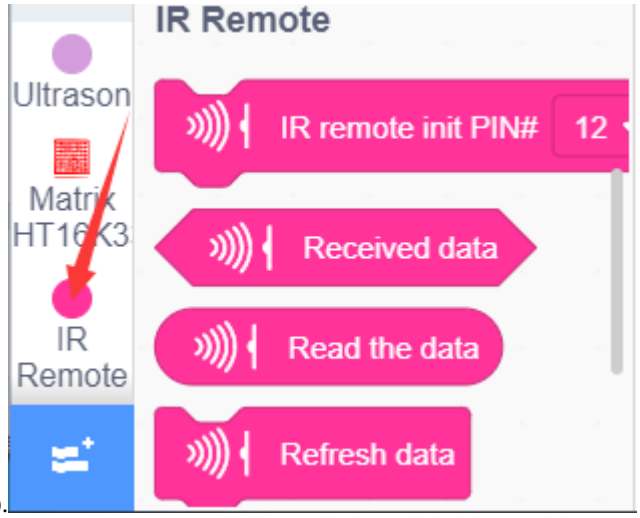


(2).

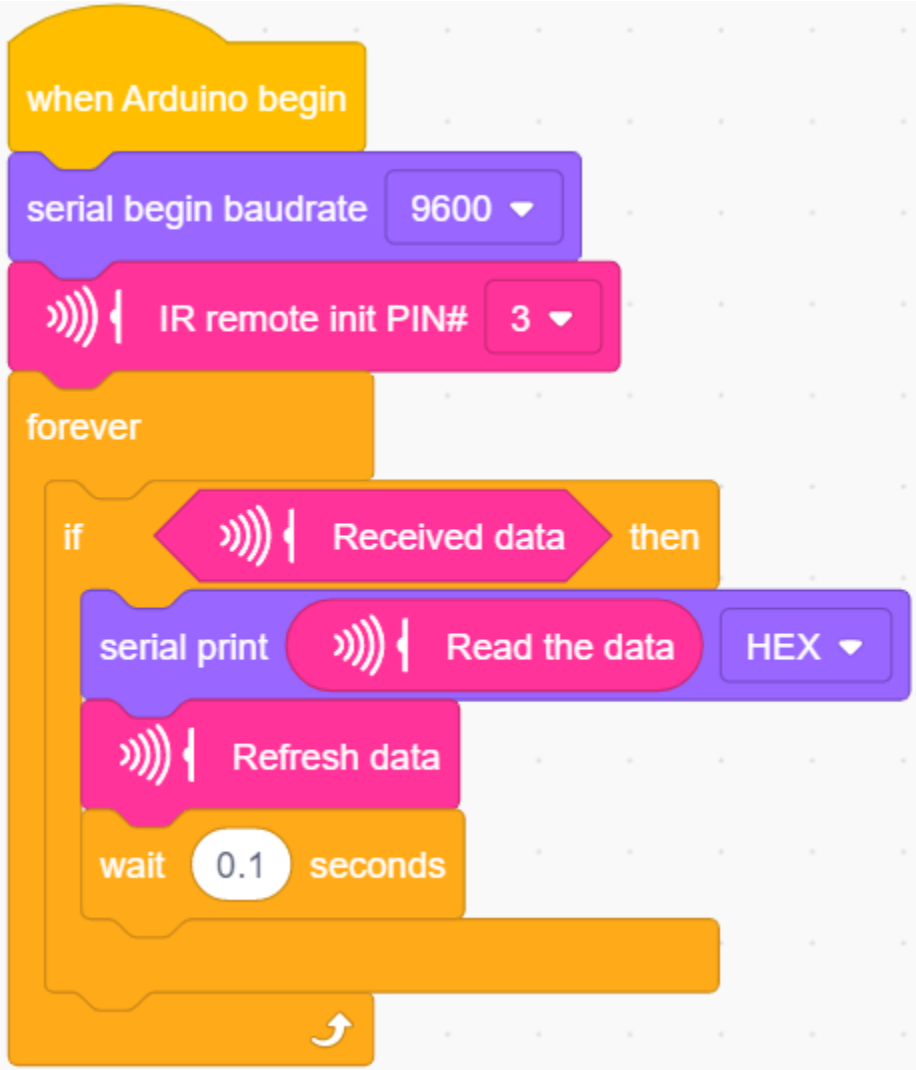
(3).



(4).



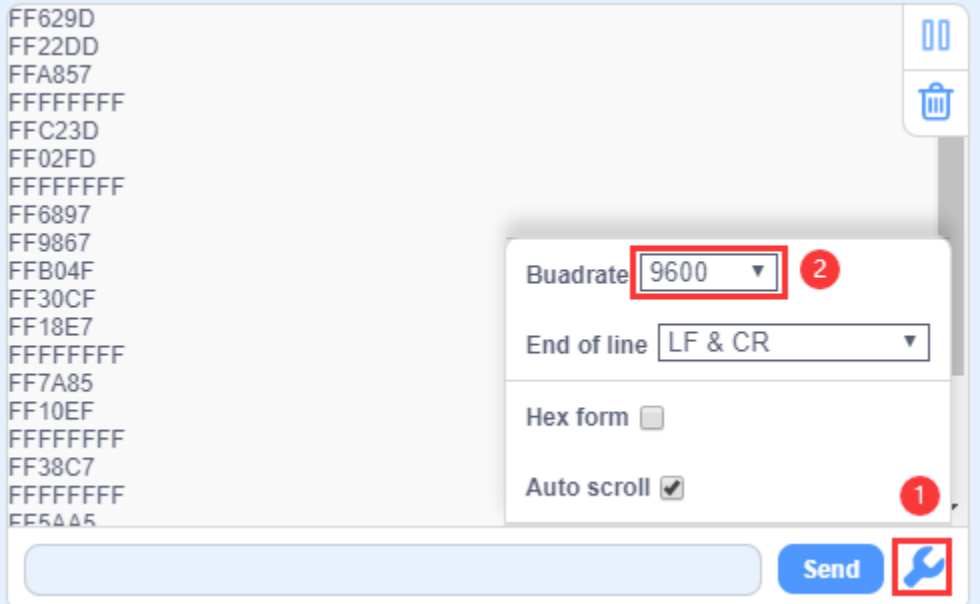
Complete Test Code



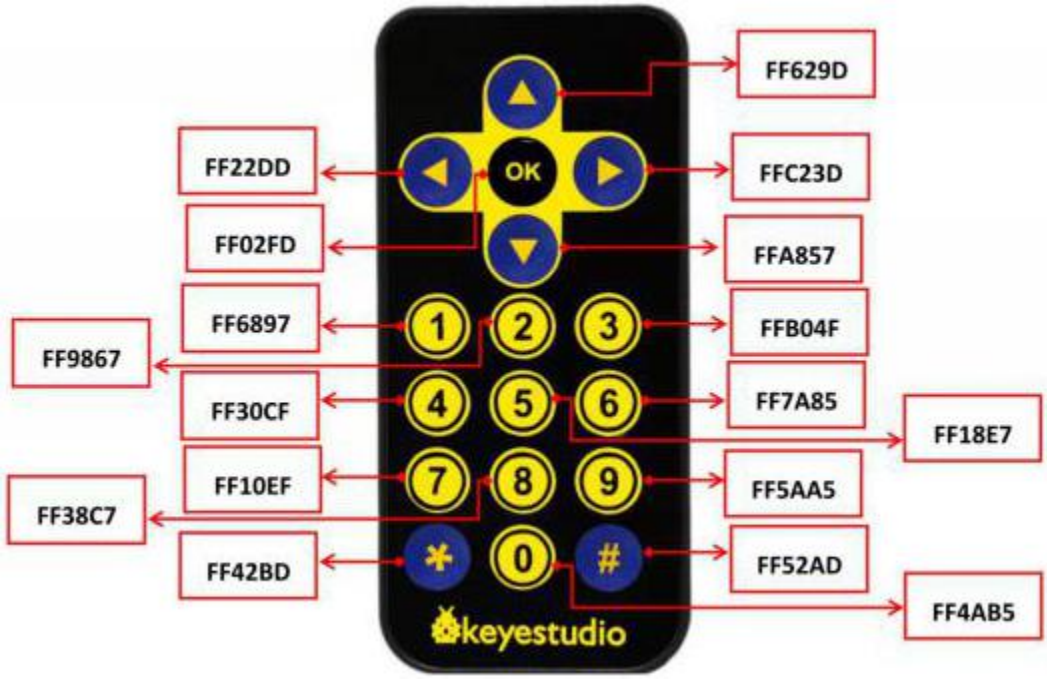
5.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, then connect the computer via a USB cable to power the board. After powering on, click to set baud rate to 9600.

Take out the remote control, and send signal to the infrared receiver sensor. You can see the key value of the corresponding key, if the key time is too long, FFFFFFFF is prone to garbled characters.



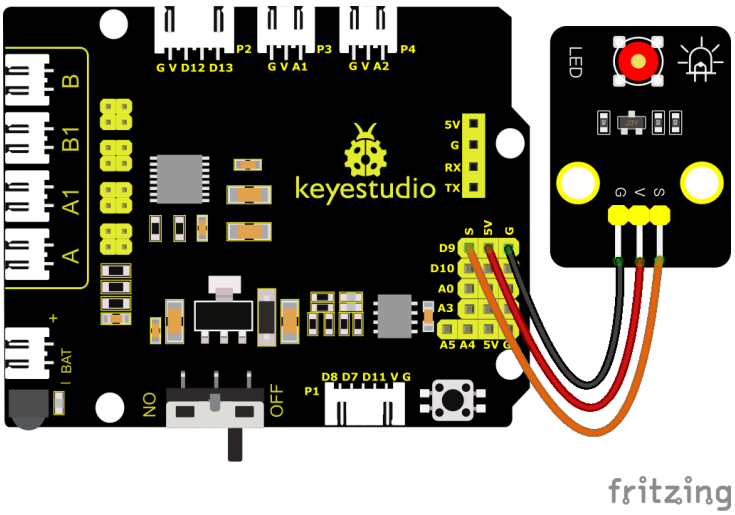
The keys value of remote control are shown below.



6.Extension Practice

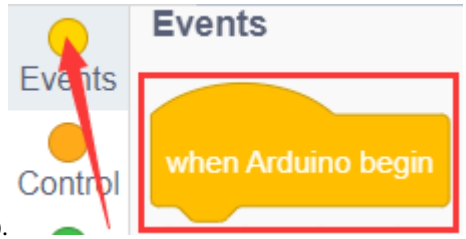
We have decoded the key value of the IR remote control. How about controlling LED by the measured value? We could design an experiment.

Attach an LED to D9, then press the keys of remote control to make LED light on and off.

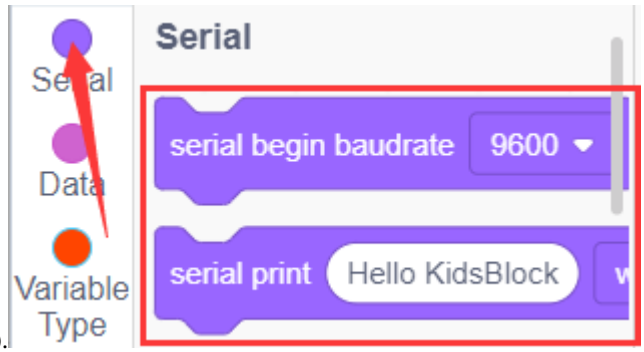


You can drag blocks to edit. Blocks listed below are for your reference.

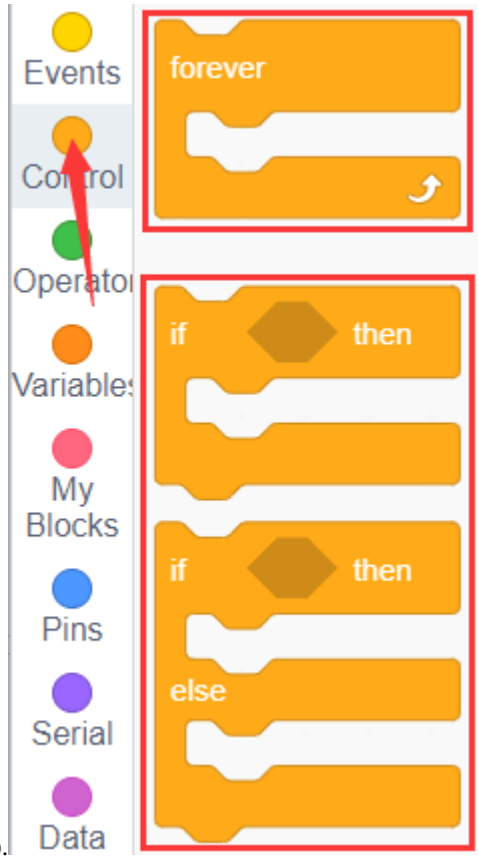
(1).



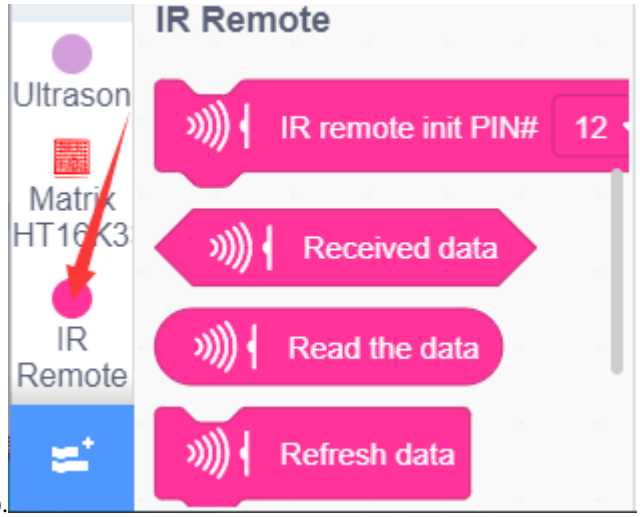
(2).



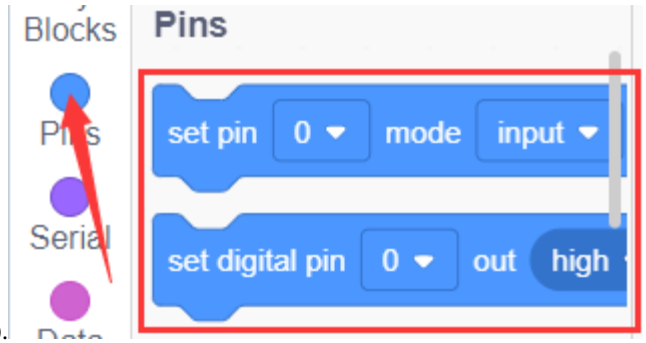
(3).


Events
Control
Operator
Variables
My Blocks
Pins
Serial
Data
forever
if then
if then
else

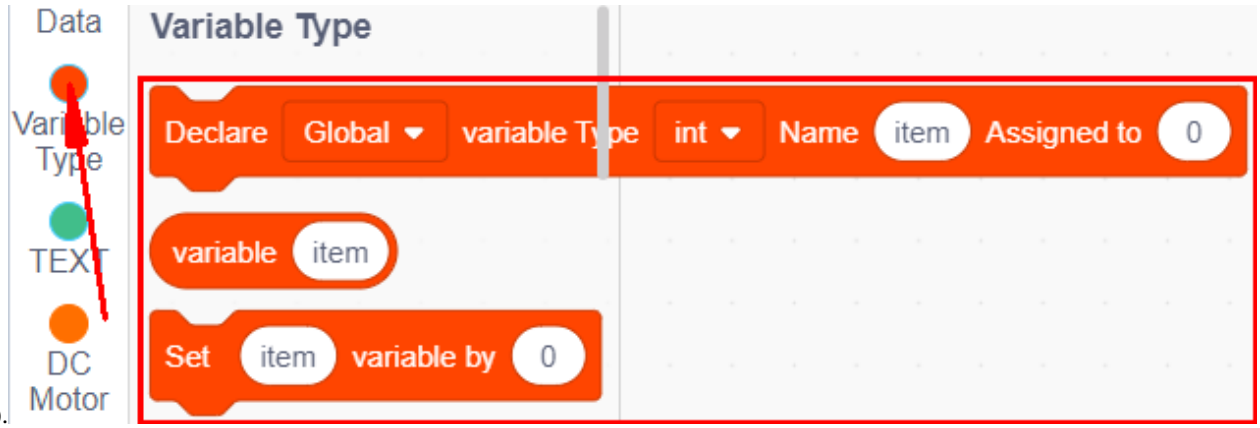
(4).


Ultrason
Matrix HT16K3
IR Remote
IR Remote
IR remote init PIN# 12
Received data
Read the data
Refresh data

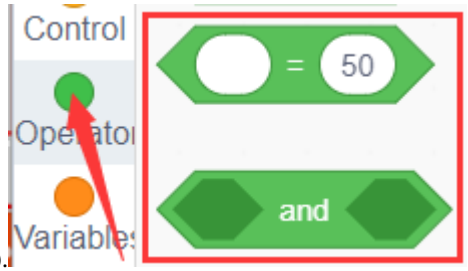
(5).



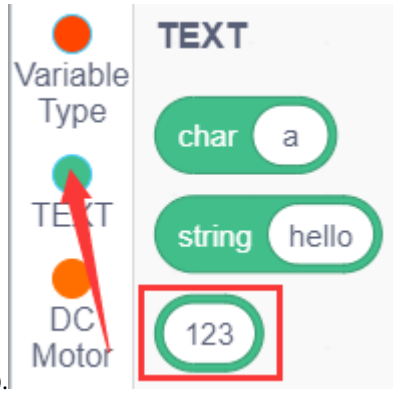
(6).



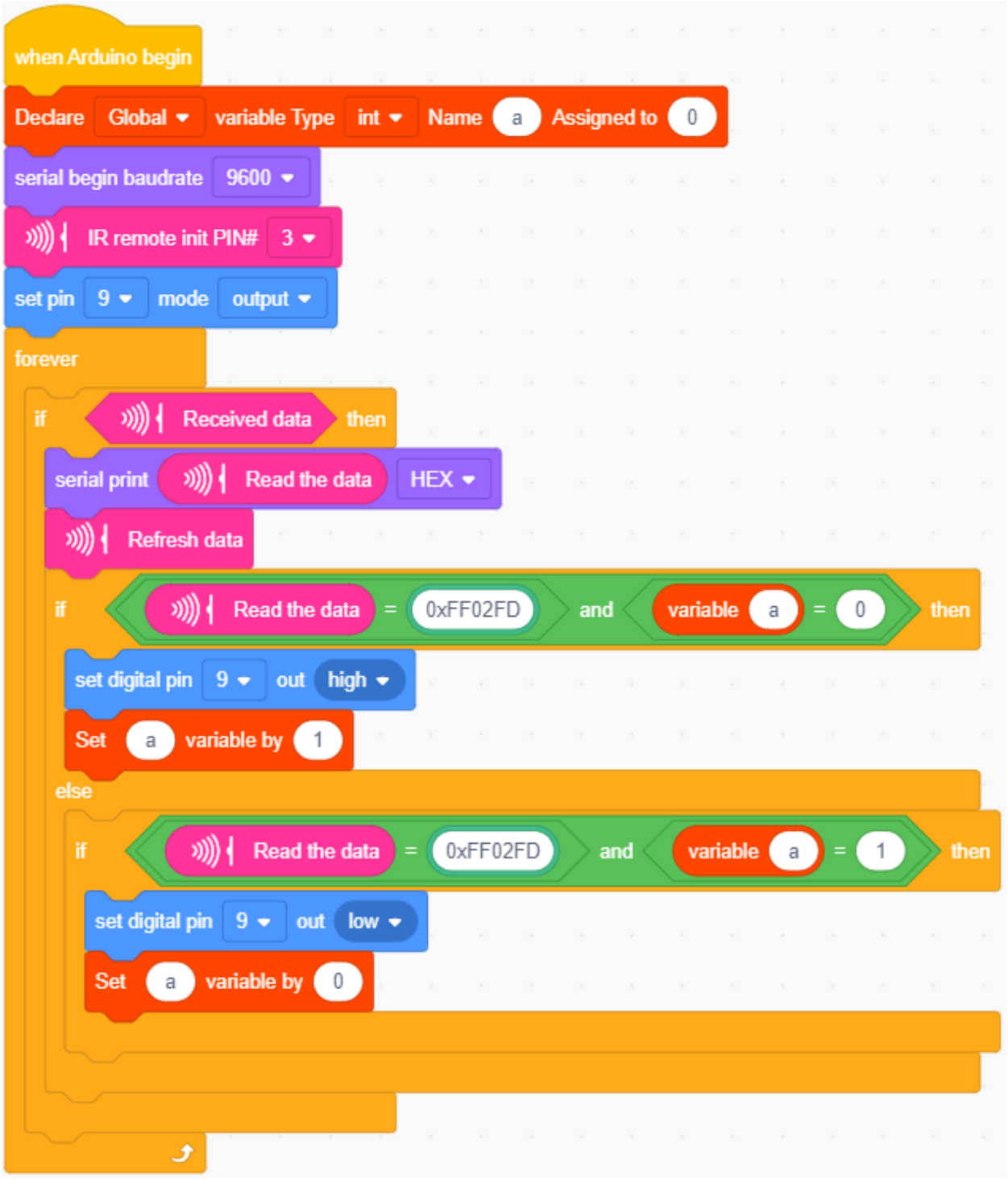
(7).



(8).



Complete Test Code



After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, then connect the computer via a USB cable to power the board. After powering on, press the "**OK**" key on remote control can make the LED on and off.

10.2.7 Project 7 Bluetooth Remote Control

1.Description

There is a DX-BT24 5.1 Bluetooth module in this kit. This bluetooth module comes with 256Kb space and complies with V5.1BLE bluetooth specification, which supports AT commands. Users can change parameters such as the baud rate and device name of the serial port as required.

Furthermore, it supports UART interface and bluetooth serial port transparent transmission, which also contains the advantages of low cost, small size, low power consumption and high sensitivity for sending and receiving. Notably, it solely needs a few peripheral components to realize its powerful functions.

2.Specification

- Bluetooth protocol: Bluetooth Specification V5.1 BLE
- Working distance: In an open environment, it can achieve 40m ultra-long distance communication
- Operating frequency: 2.4GHz ISM band
- Communication interface: UART
- Bluetooth certification: Accord with FCC CE ROHS REACH certification standard
- Serial port parameters: 9600, 8 data bits, 1 stop bit, invalid bit, no flow control
- Power: 5V DC
- Operating temperature: –10℃ to +65℃

3.Application

The DX-BT24 module also supports the BT5.1 BLE protocol, which can be directly connected to iOS devices with BLE Bluetooth function, and supports resident running of background programs. It is mainly used in the field of short-distance data wireless transmission. It enables to avoid cumbersome cable connections and can directly replace serial cables.

Successful application areas of BT24 modules:

※ Bluetooth wireless data transmission;

※ Mobile phone, computer peripheral equipment;

※ Handheld POS equipment;

※ Wireless data transmission of medical equipment;

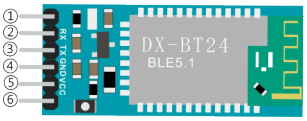
※ Smart home control;

※ Bluetooth printer;

※ Bluetooth remote control toys;

※ Shared bicycles;

Ports



STATEStatus pin

RXReceiving pin

TXsending pin

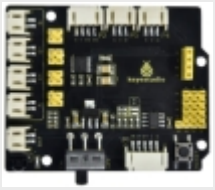
GNDGND

VCCPower

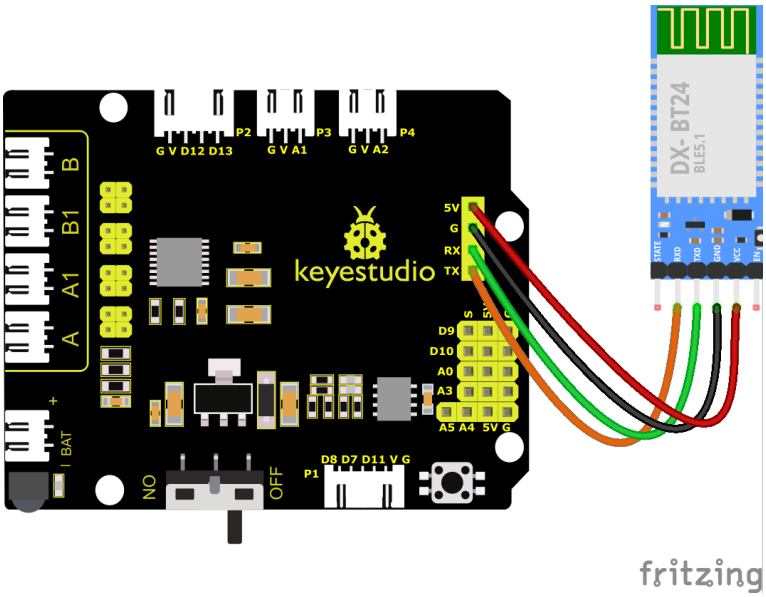
EN Enable pin

Connect the BT module to the development board.

4.Components

Development Board *1	8833 Motor Driver *1	Red LED Module*1
		
3P F-F Dupont Wire*1	USB Cable*1	DX-BT24 Bluetooth Module*1

5.Wiring Diagram



RXD, TXD, GND and VCC of the BT module are connected to TX, RX, G and 5V.

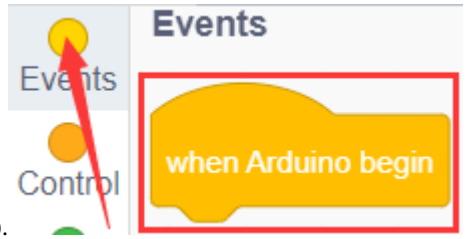
STATE and BRK of the BT module don't need connection.

Note: the direction of the BT module when inserting it onto the 8833 board. And don't insert it before uploading the code.

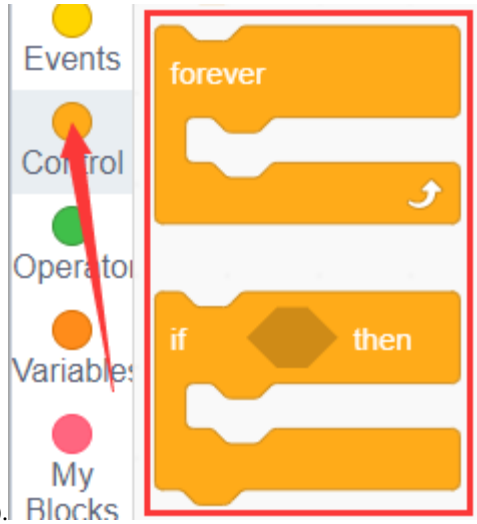
6.Test Code

You can drag blocks to edit. Blocks listed below are for your reference.

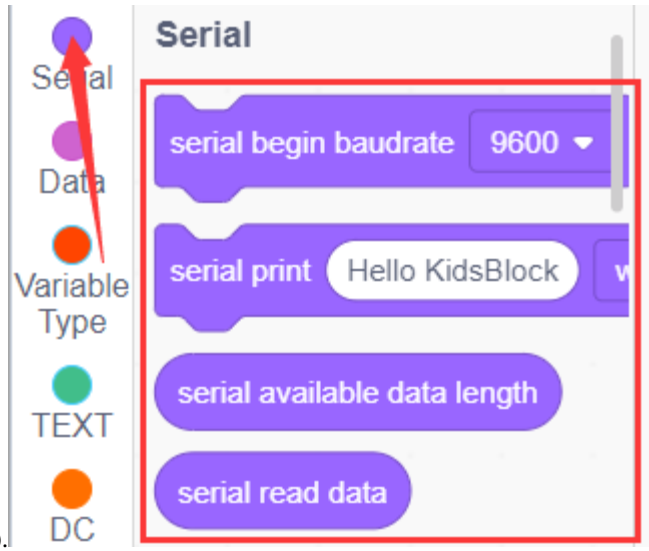
(1).



(2).

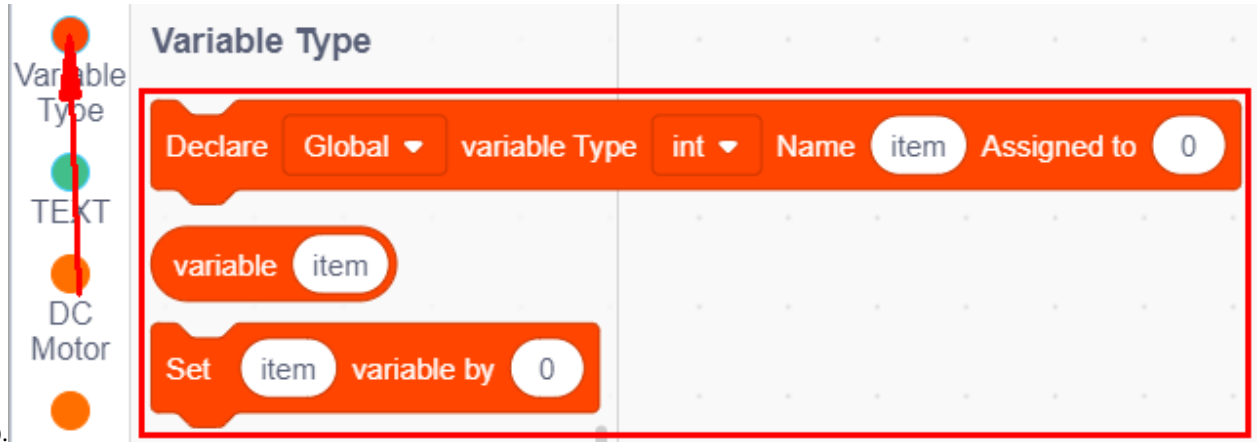


(3).



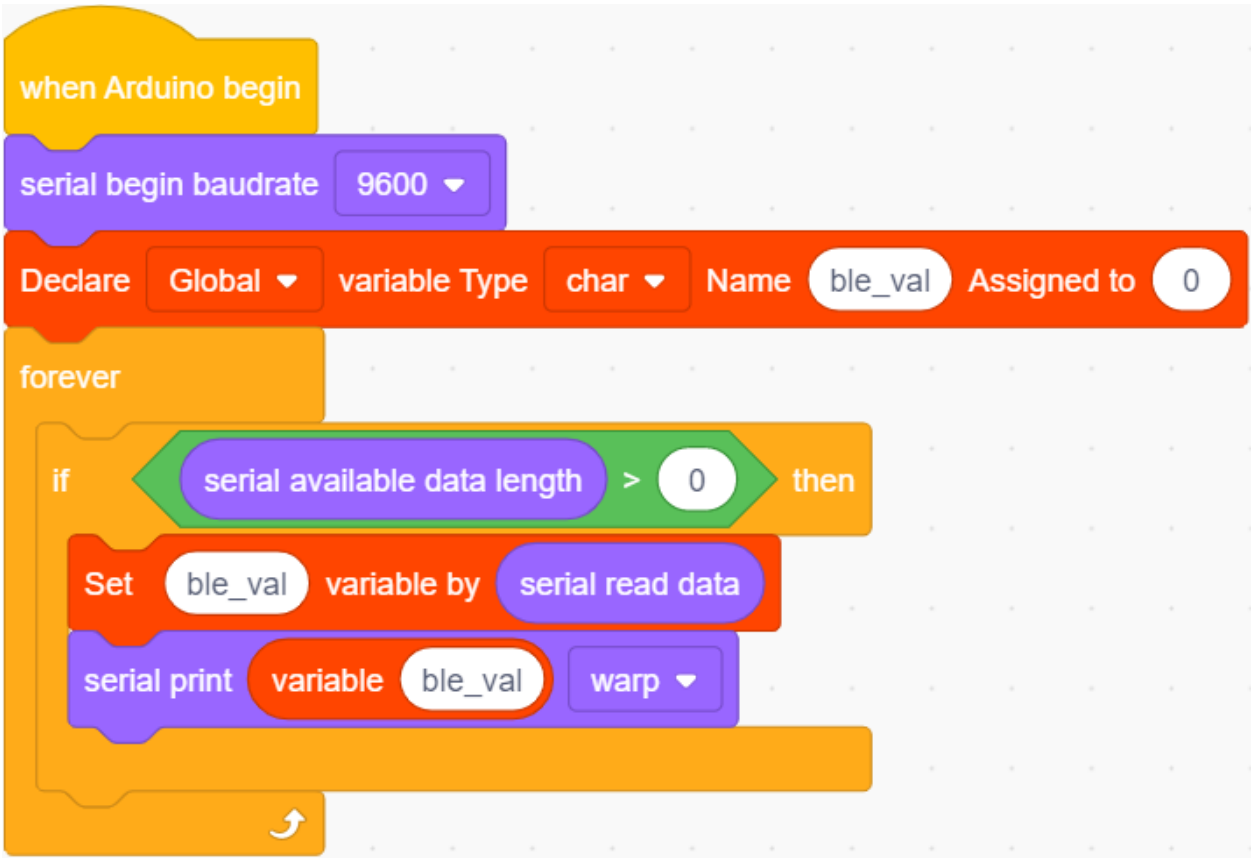
(4).

(5).



Complete Test Code

Note: Before uploading the test code, you need to remove the Bluetooth module, otherwise the code will fail to be uploaded.Connect the Bluetooth module after uploading the code successfully.



7.Test Result

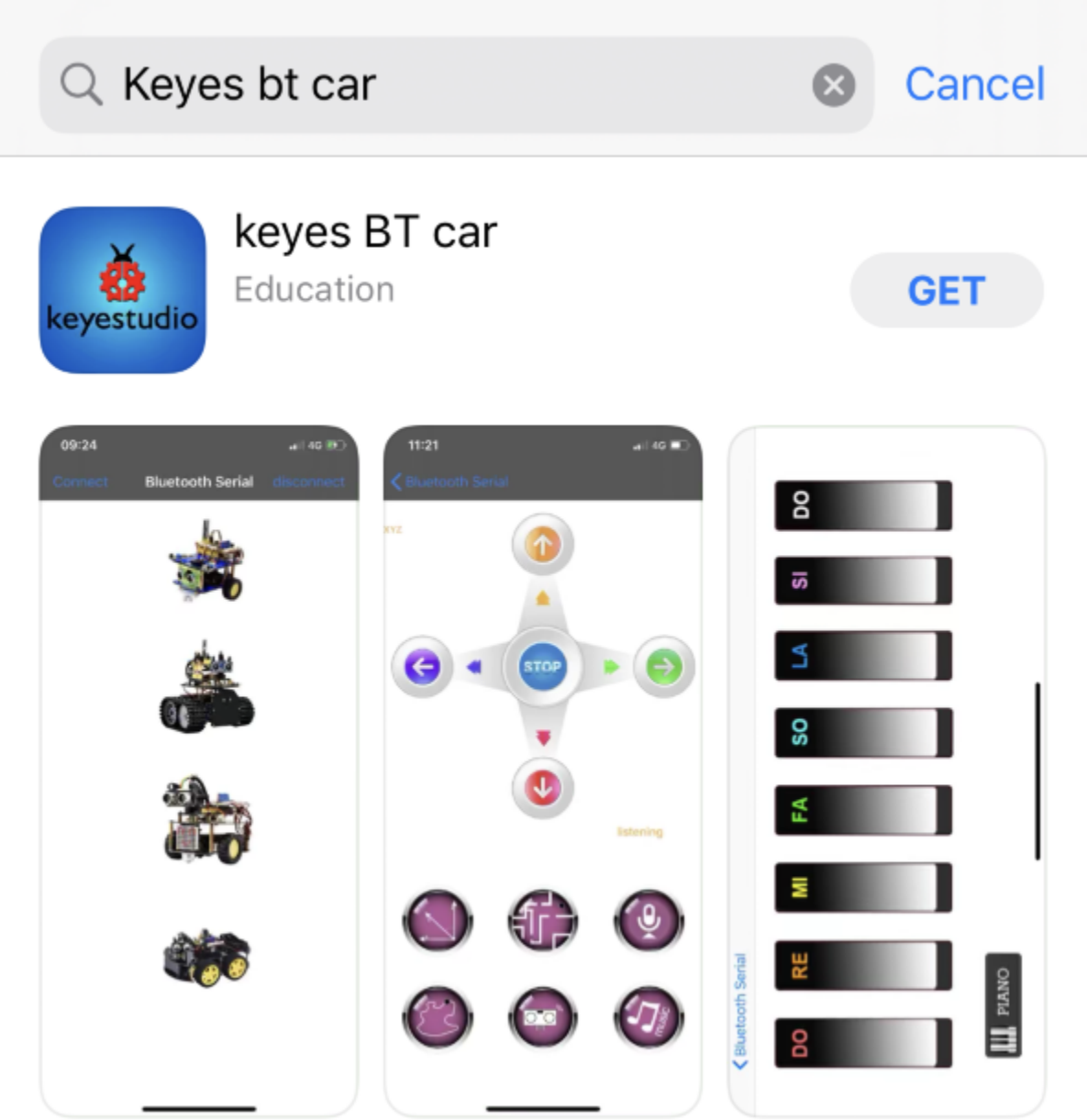
After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, then connect the computer via a USB cable to power the board. After powering on, insert the BT module and the LED will flash, then we need to download the BT app.

8.Download Bluetooth APP

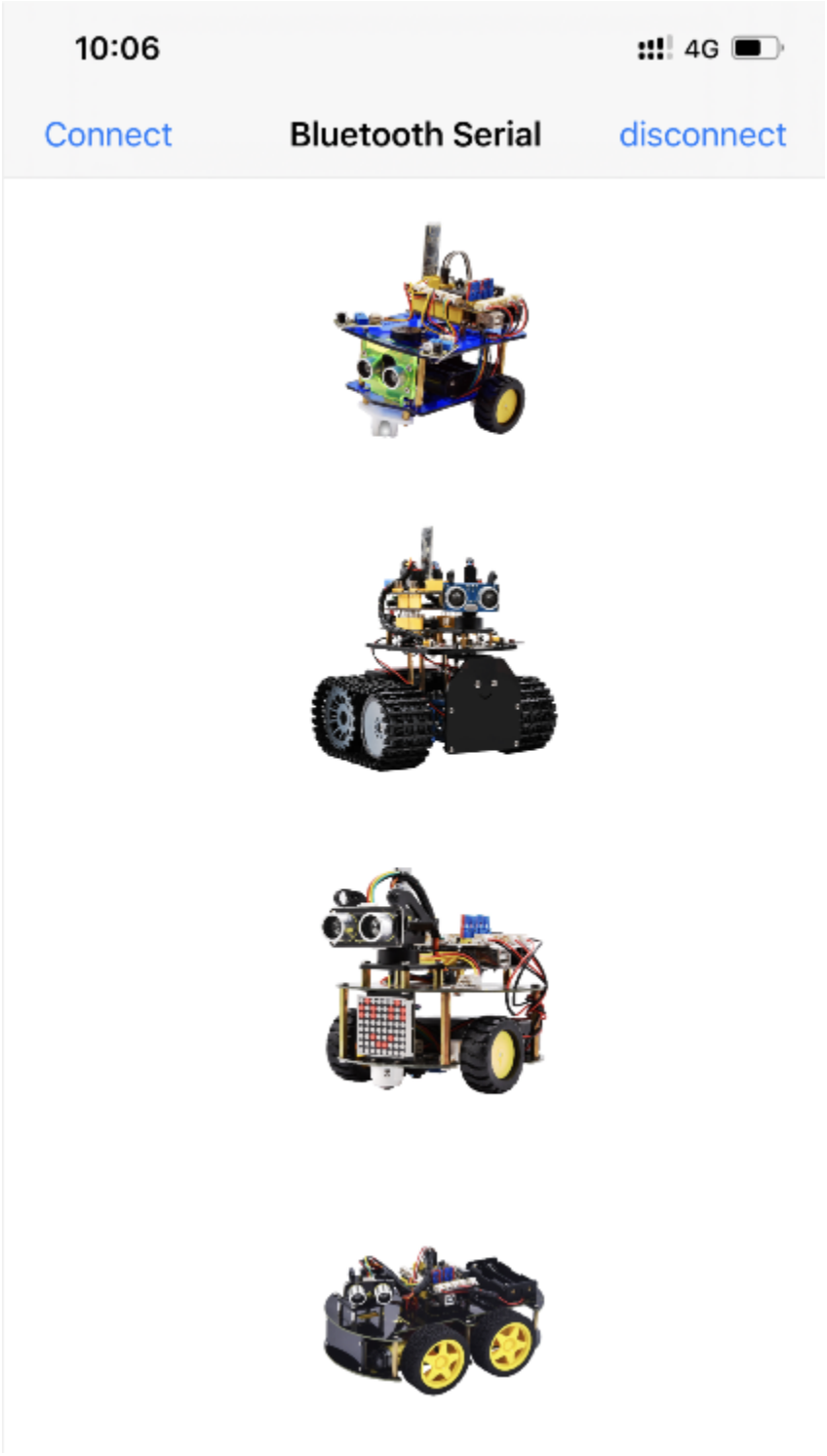
Apple system

(1).Open the App Store on the iPhone.

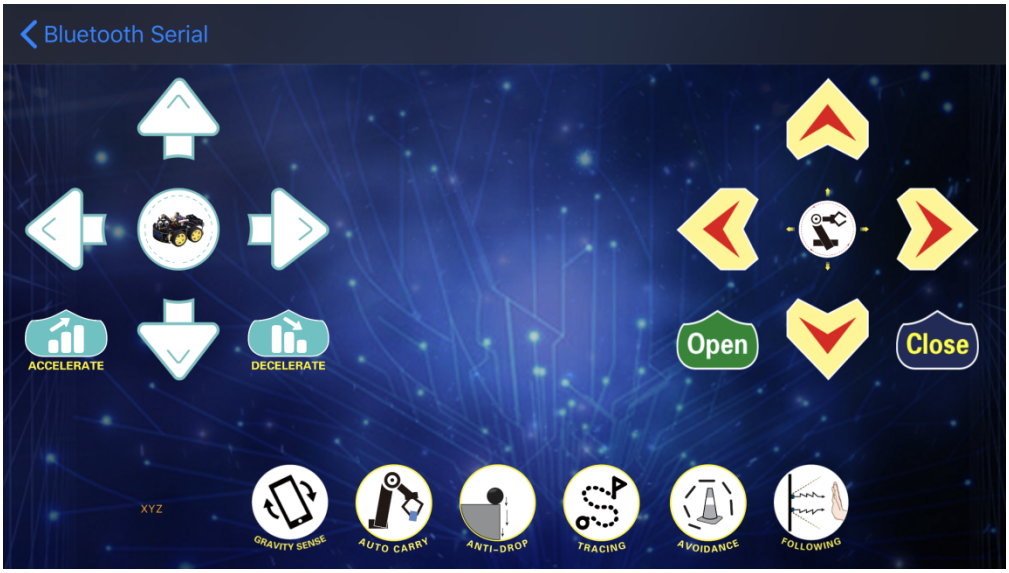
(2).Search keyes BT car and download the APP to your phone.



(3).After installation, enter its interface.

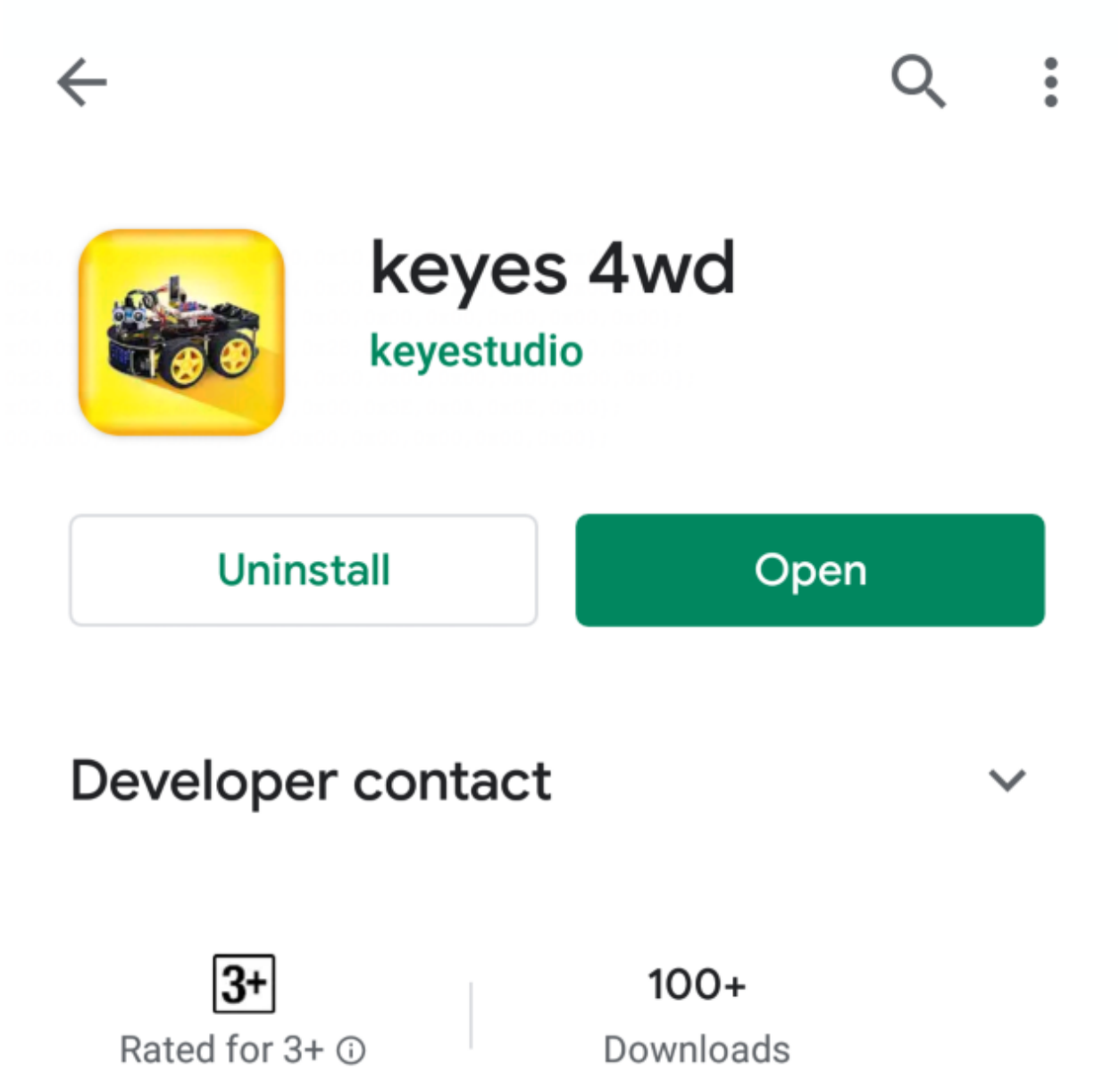


(4).Click "**Connect**" button in the upper left corner to automatically search for Bluetooth. When **BT24** is found, click "**Connect**" to connect Bluetooth, and then click to enter the control interface of 4WD smart car.



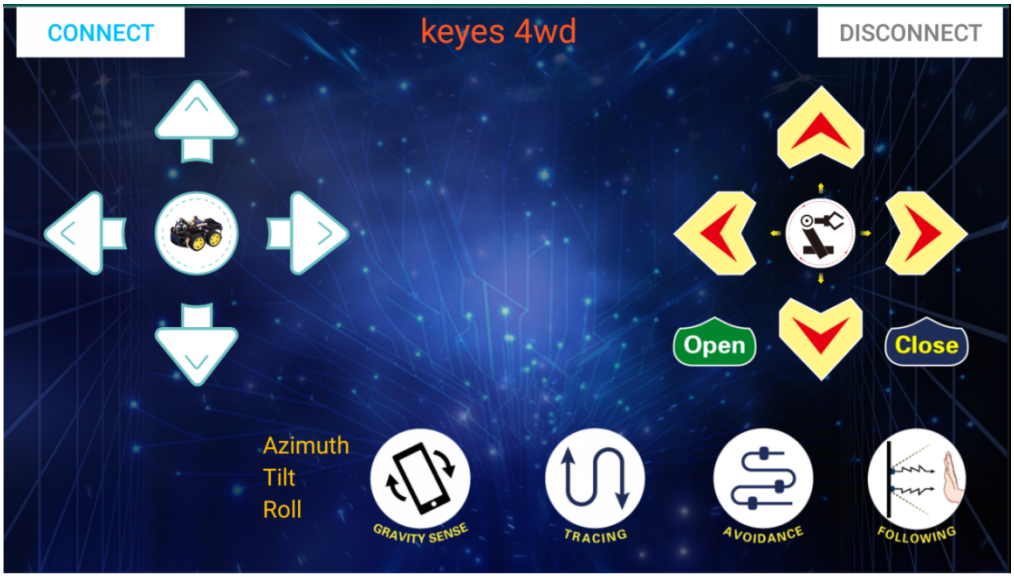
Android System

(1).Enter google play store to search for"**keyes 4wd**".

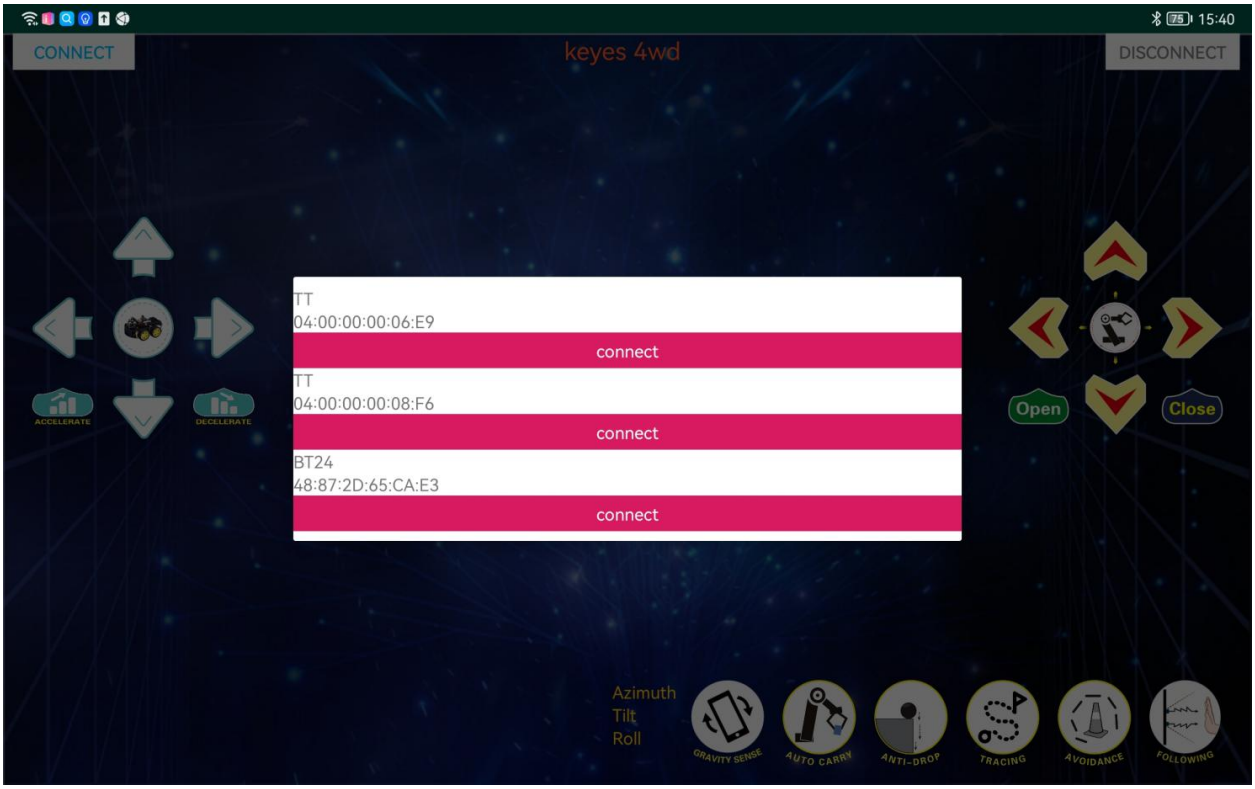


(2).The app icon is shown below after installation.

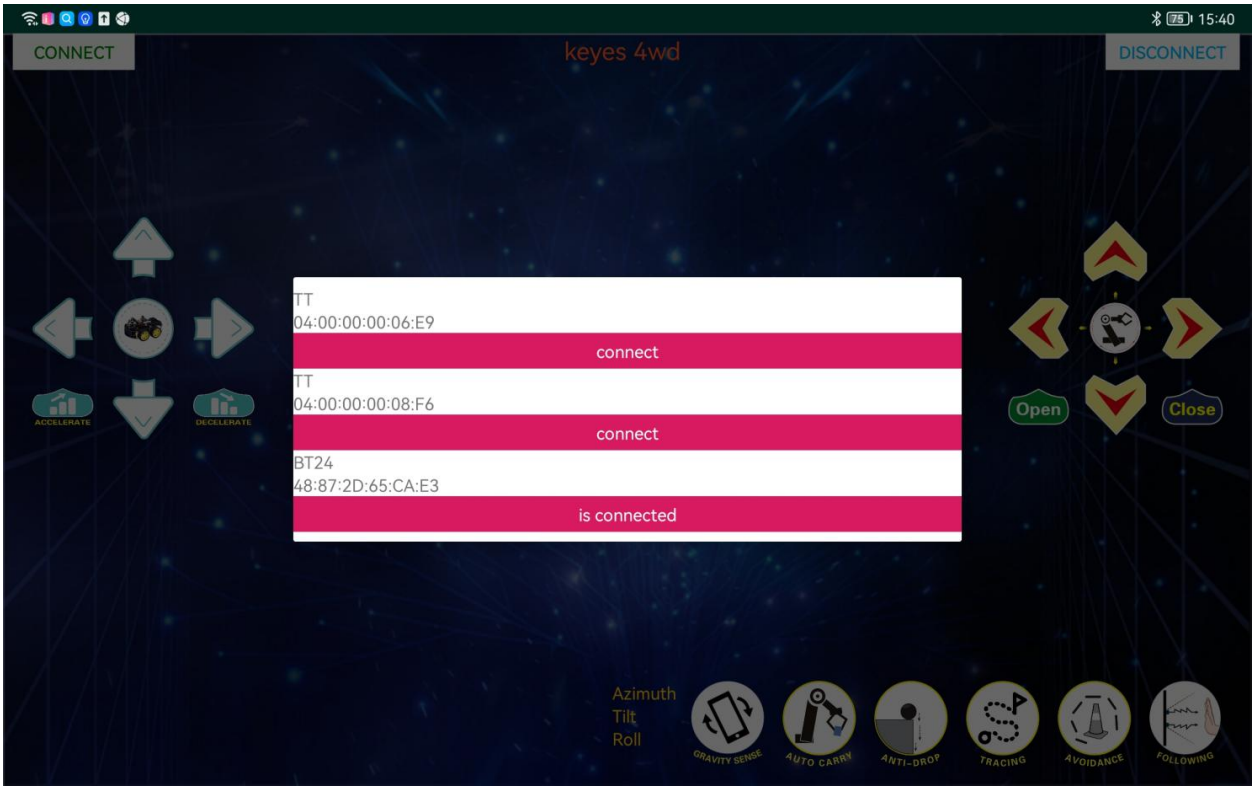
(3).Click app to enter the following page.



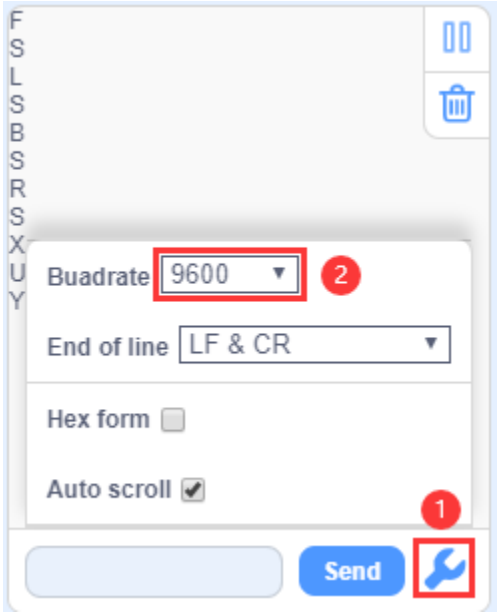
(4).After connecting Bluetooth, plug in power and LED indicator of Bluetooth module will flicker. Tap"Connect"to search the Bluetooth.



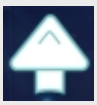
(5).When **BT24** is found, click "**connect**" to connect Bluetooth. When "**connect**" turns into "**is connected**", it indicates that the Bluetooth connection is successful. As shown in the picture below, the Bluetooth LED becomes will stay on.



(6).After connecting Bluetooth module, click to set baud rate to 9600. Pressing the button of the Bluetooth APP, and the corresponding characters will be displayed, as shown below:

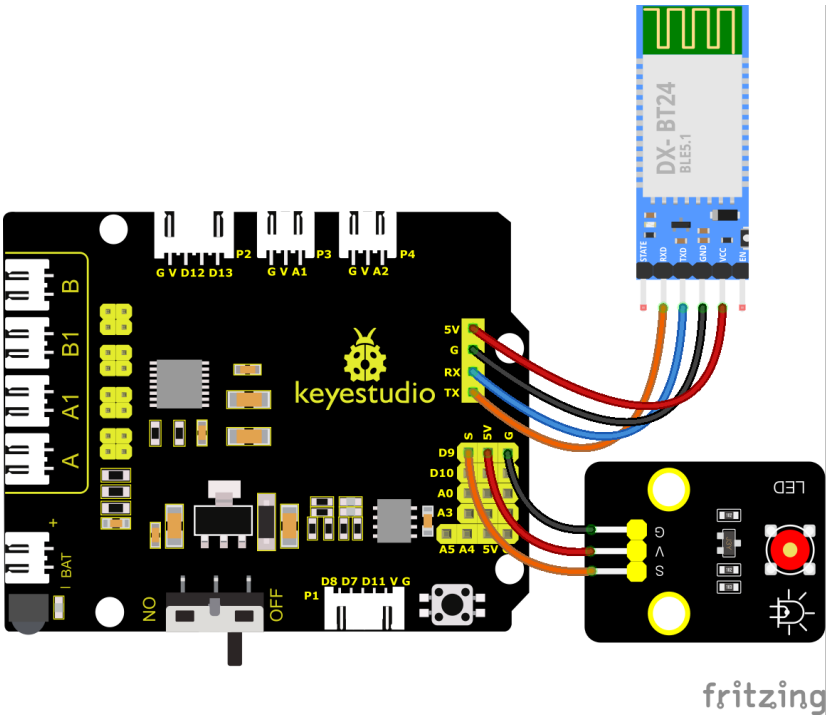


Key	Function
	Pair DX-BT24 5.1 Bluetooth module
	Disconnect Bluetooth

	Control character	Function
	Press: F Release: S	Press the button, the car goes front; release to stop
	Press: L Release: S	Press the button, the car turns left; release to stop
	Press: R Release: S	Press the button, the car turns right; release to stop
	Press: B Release: S	Press the button, the car goes back; release to stop
	Press: “a” Release: “S”	Click to speed up(maximum:255)
	Press: “d” Release: “S”	Click to slow down(minimum:0)
	Click to start the gravity sensing function of the mobile phone: click again to exit the gravity sensing control	
	Click to send“X”, click again to send“S”	Start line tracking function; click again to exit
	Click to send“Y”, click again to send“S”	Start ultrasonic avoiding function; click again to exit
	Click to send“U”, click again to send“S”	Start ultrasonic follow function; click again to exit
	Click to send“G”,click again to send“S”	Start restricting function; click again to exit

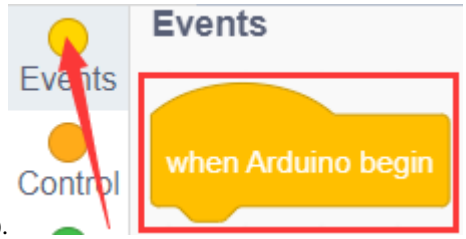
9.Extension Practice

Here we look to use the command sent by the mobile phone to turn on or off an LED light. Looking at the wiring diagram, an LED is connected to the D9 pin.

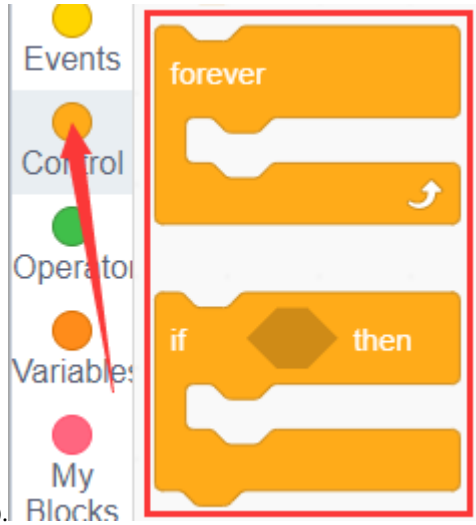


You can drag blocks to edit. Blocks listed below are for your reference.

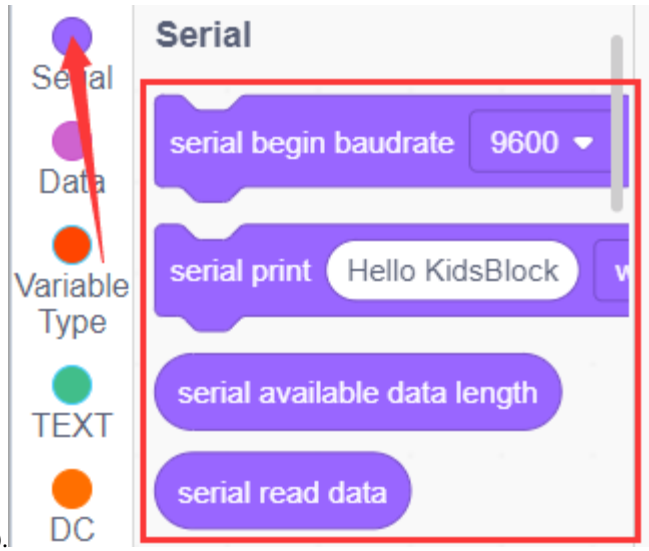
(1).



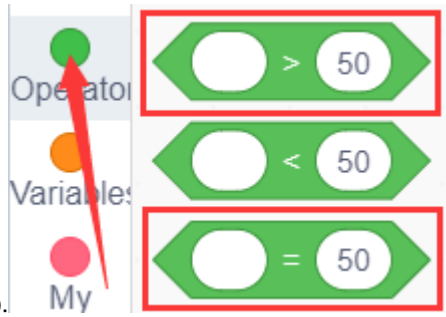
(2).



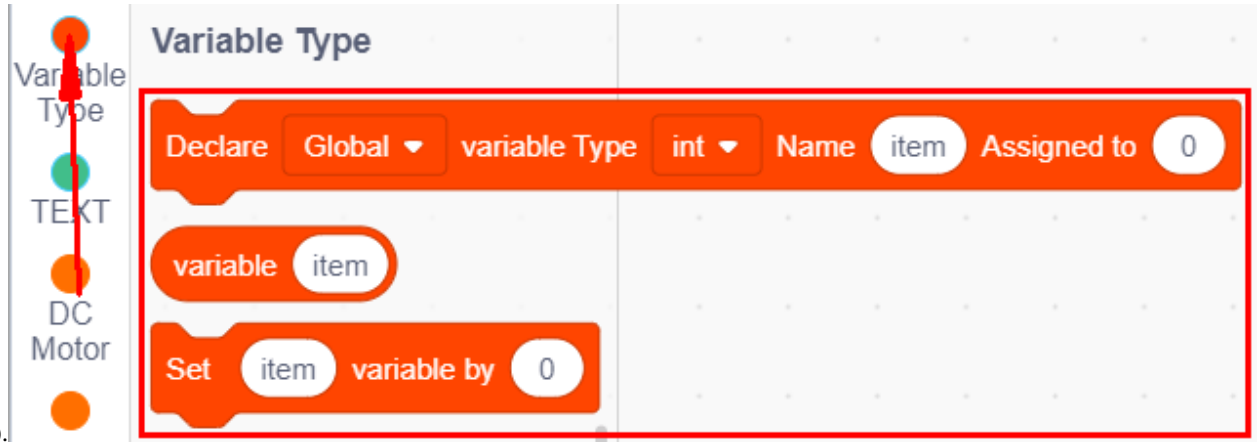
(3).


Serial
Serial
Data
Variable Type
TEXT
DC
serial begin baudrate 9600
serial print Hello KidsBlock
serial available data length
serial read data

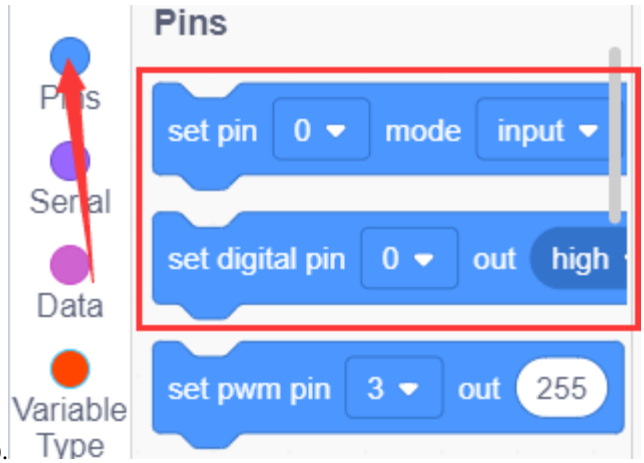
(4).


Operator
Variables
My
> 50
< 50
= 50

(5).

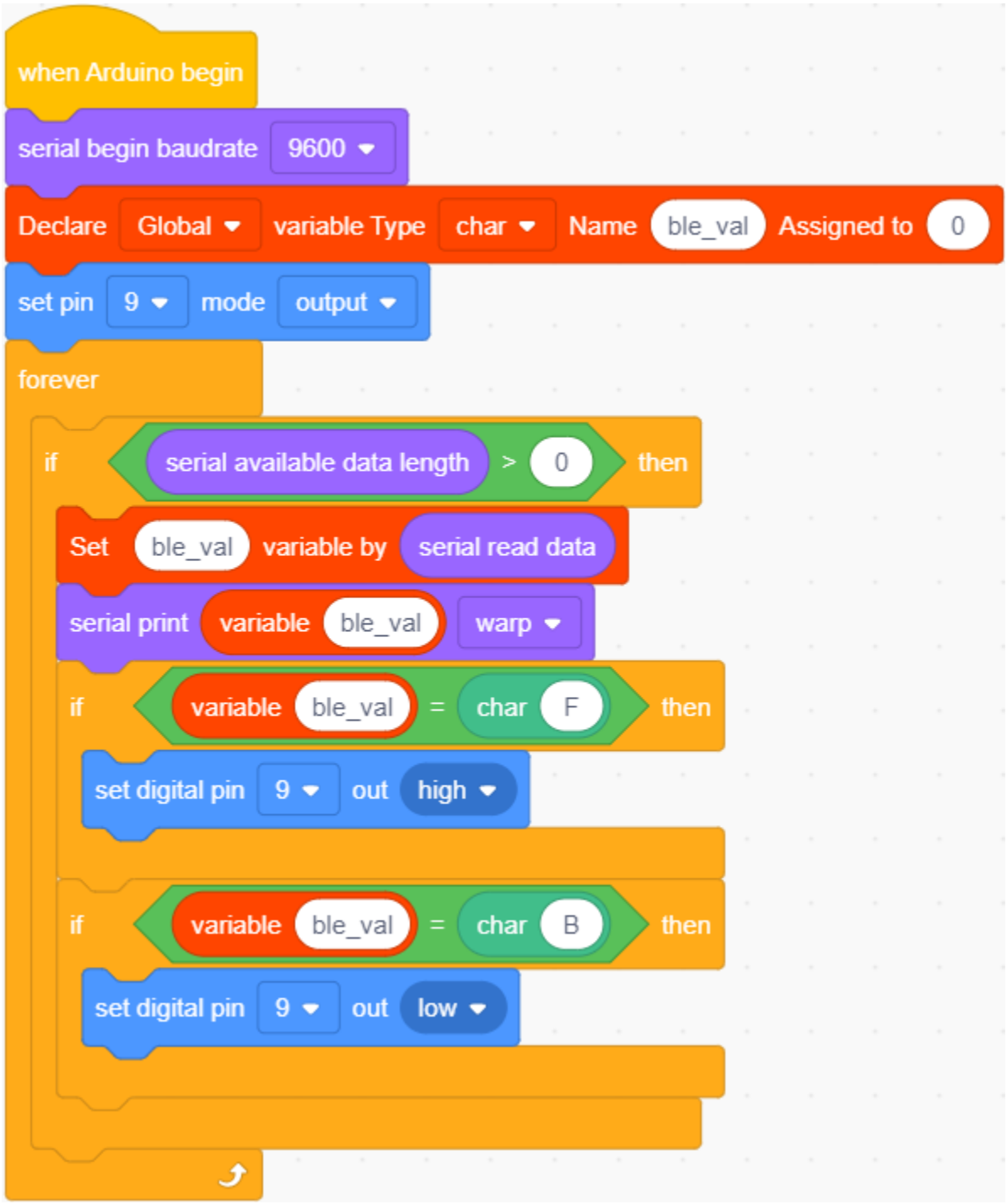

Variable Type
Variable Type
TEXT
DC Motor
Declare Global variable Type int Name item Assigned to 0
variable item
Set item variable by 0

(6).



(7).

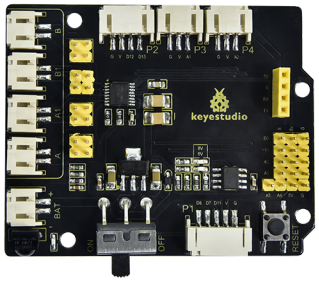
Complete Test Code



After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, then connect the computer via a USB cable to power the board. After powering on, click and

to control the LED turn on and turn off.

10.2.8 Project 8 Motor Driving and Speed Control



1.Description

There are many ways to drive motors. Our car uses the most commonly used DRV8833 motor driver chip, which provides a two-channel bridge electric drive solution for toys, printers and other integrated motor applications.

When we stack the driver expansion board on the 4.0 development board and power on the BAT, then set the DIP switch to the ON end, the external power supply will power on the two boards at the same time. To facilitate wiring connections, the driver expansion board comes with an anti-reverse port (PH2.0-2P-3P-4P-5P). You can connect the motors, power supply, and sensor modules directly to the driver expansion board.

The Bluetooth interface of the driver expansion board is fully compatible with the DX-BT24 5.1 Bluetooth module. When connecting the Bluetooth module, you solely need to plug it into the corresponding interface. At the same time, 2.54 row pins are used to draw out some unused digital and analog ports on the driver expansion board, making it accessible for you to add other sensors and carry out extension experiments.

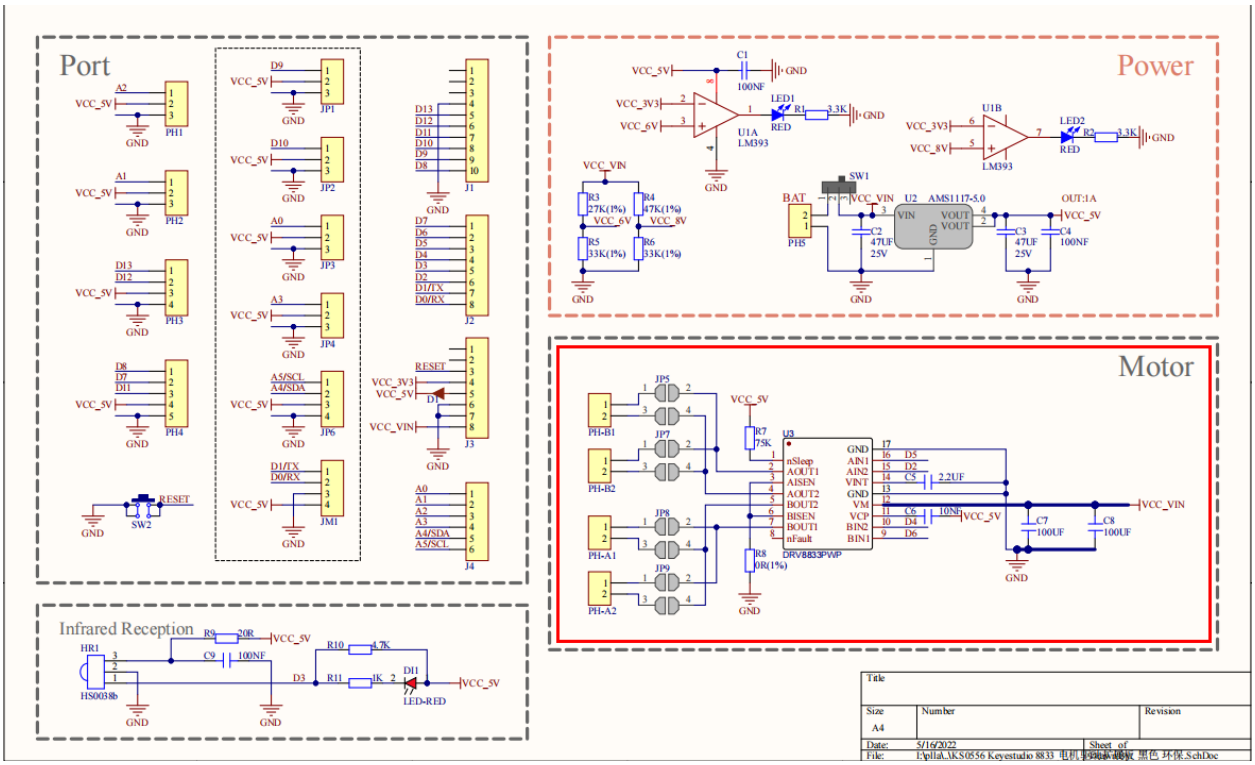
The expansion board can be connected to four DC motors. When the jumper cap is connected by default, the motors of ports A and A1 and B and B1 are connected in parallel and have the same motion law. 8 jumper caps can be used to control the rotation direction of the 4 motor interfaces.

For example, when the 2 jumper caps in front of B1 of the M1 motor change from transverse connection to longitudinal connection, the rotation direction of M1 motor will be opposite to the original rotation direction.

2.Specification

- Input voltage for logicDC 5V
- Input voltage for drivingDC 6-9 V
- Working current for logic<36mA
- Working current for driving<2A
- Maximum power dissipation25WT=75℃
- Input level for control signal high level is 2.3V<Vin<5V low level is -0.3V<Vin<1.5V
- Working temperature-25130℃

3.Keyestudio 8833 motor driver expansion board

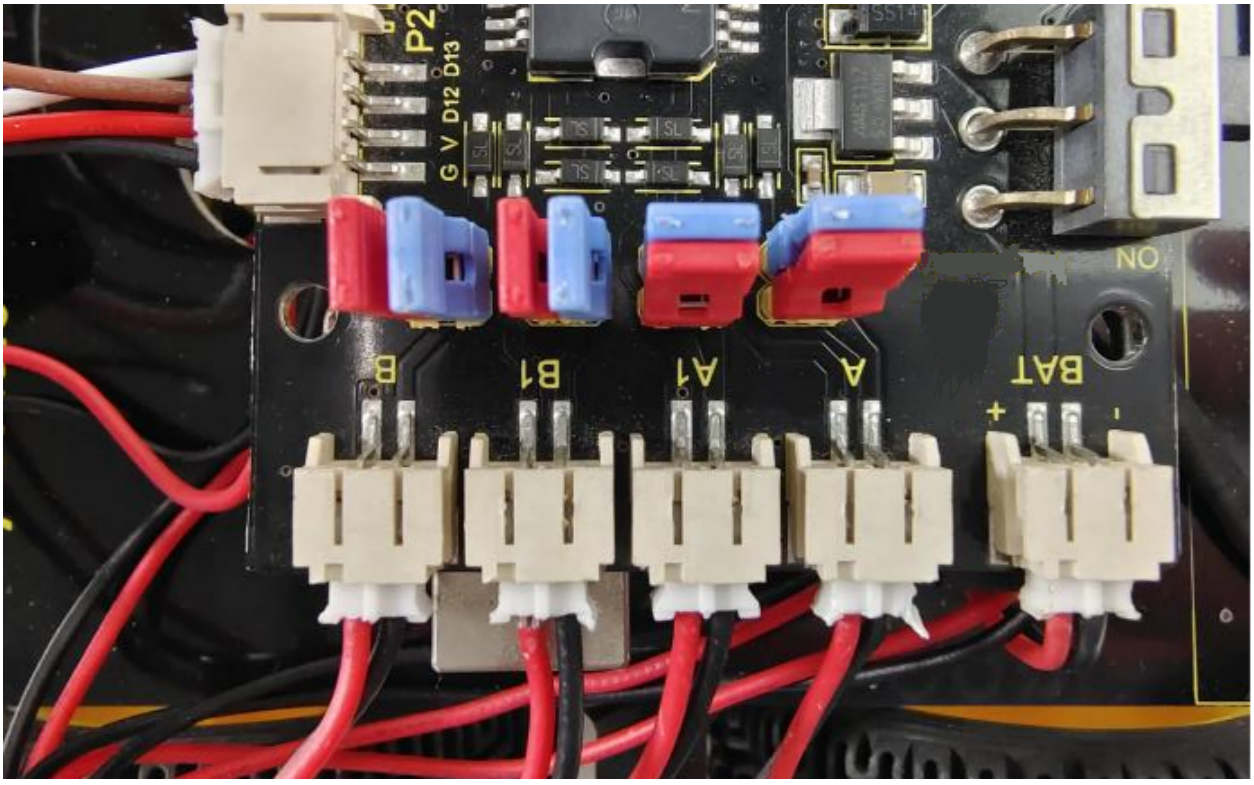


Working Principle

We use the same side parallel connection mode for the four motors, which can be regarded as two groups of motors. As shown in the wiring diagram, B and B1 are a group, and A and A1 are a group.

The motors in the same group should rotate in the same direction. If they are different, please adjust the corresponding jumper caps next to the terminal to change the direction.

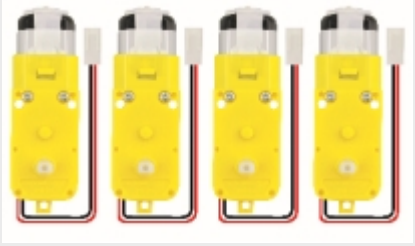
As shown below, if the directions of A and A1 are different, adjust the direction of jumper caps until the motor movement direction of the same group is consistent.



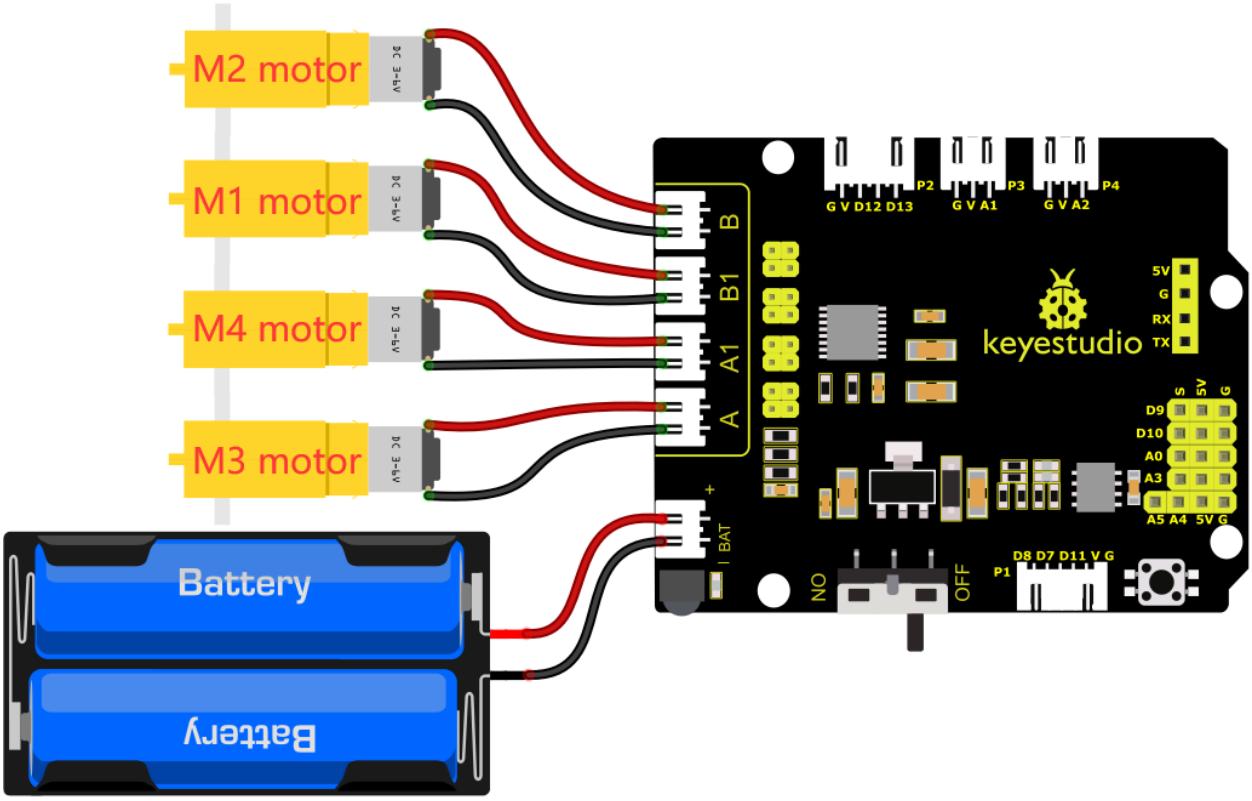
From the above diagram, it is known that the direction pin of A motor is D4, the speed pin is D6; D2 is the direction pin of B motor; and D6 is speed pin.

PWM drives the robot car. The PWM value is in the range of 0-255. When we set the direction to HIGH, the smaller the PWM number, the faster the rotation of the motor.

4.Components

Development Board *1	8833 Motor Driver *1	USB Cable*1
18650 Battery Holder*1	Motor*4	18650 Battery *2self-provided
		

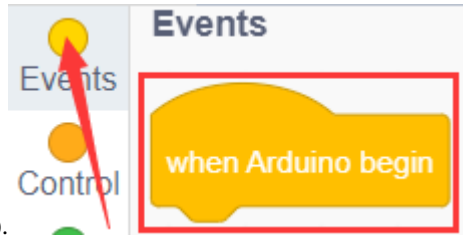
5.Wiring Diagram



Connect the power supply to the BAT port.

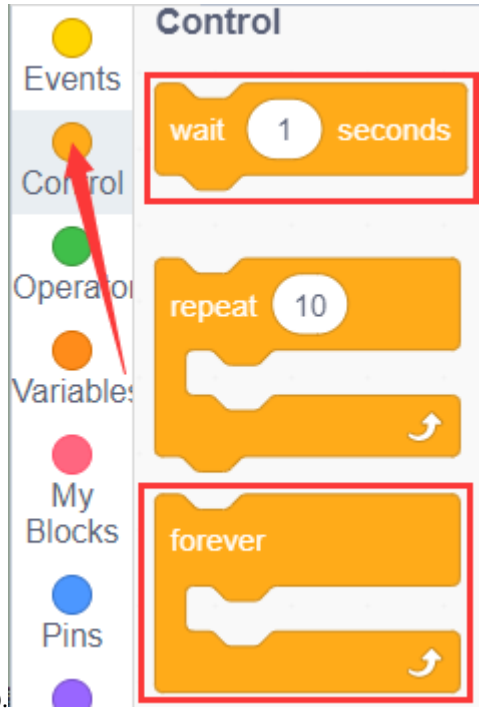
6.Test Code

You can drag blocks to edit. Blocks listed below are for your reference

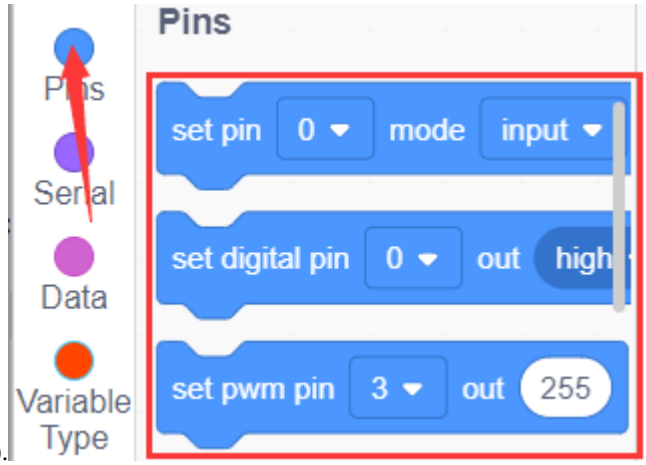


(1).

(2).



(3).



Complete Test Code

when Arduino begin
set pin 2 mode output
set pin 5 mode output
set pin 4 mode output
set pin 6 mode output
forever
set digital pin 2 out high
set pwm pin 5 out 55
set digital pin 4 out high
set pwm pin 6 out 55
wait 2 seconds
set digital pin 2 out low
set pwm pin 5 out 200
set digital pin 4 out low
set pwm pin 6 out 200
wait 2 seconds

7.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, then power on the external power and turn the DIP switch to ON, the car will go forward for 2s, back for 2s, turn left for 2s and right for 2s and stop for 2s.

8.Code Explanation

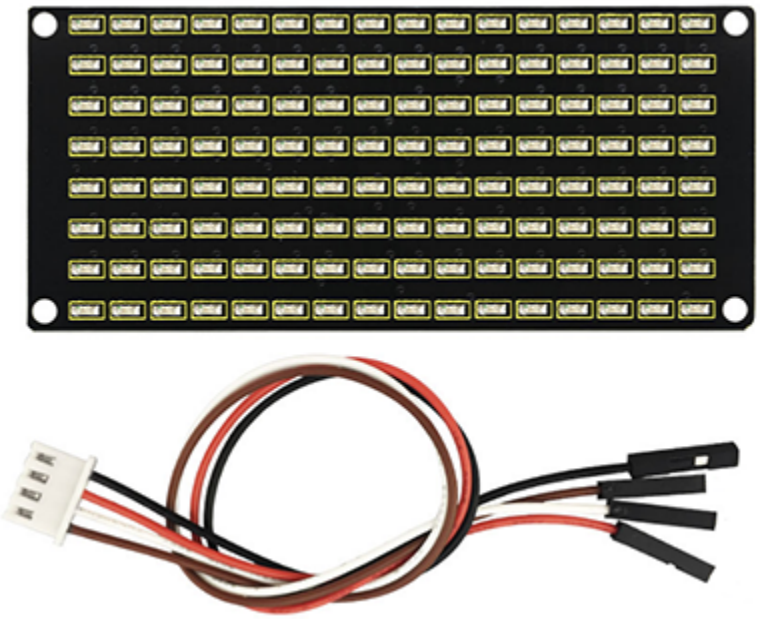
Adjust the speed that PWM controls the motor, hook up in the same way.

Complete Test Code

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, then power on the external power and turn the DIP switch to ON, then you find the speed of the motor is much slower.

Note: Low battery will lead to slow motor speed.

10.2.9 Project 9 Facial Expression LED Board



1.Description

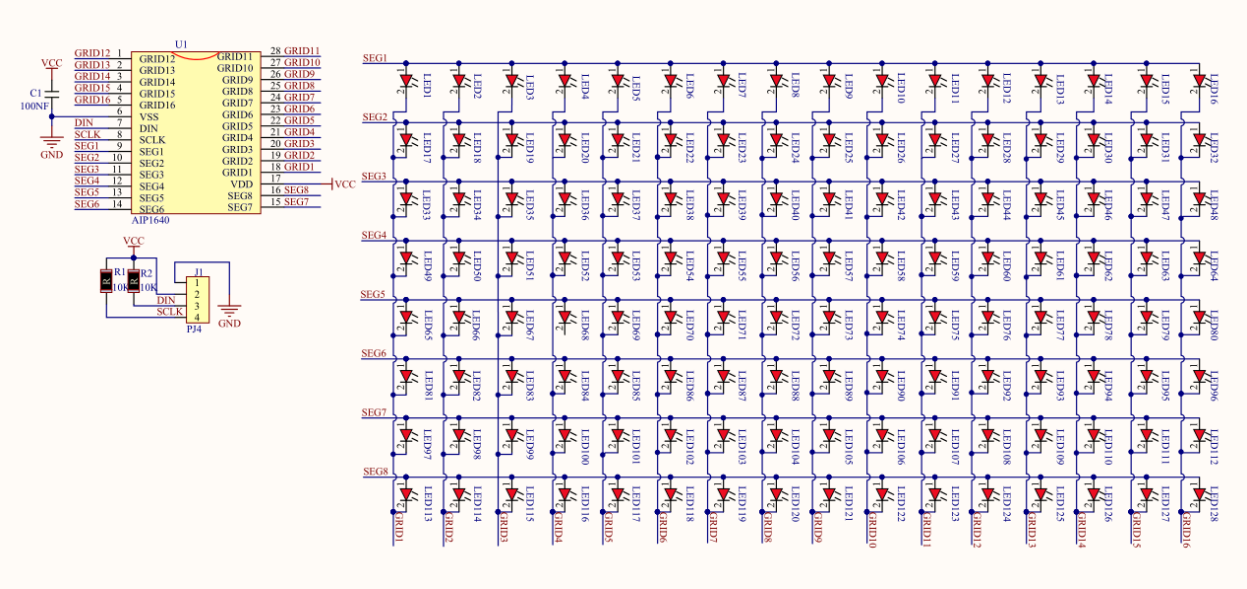
How fun it is if a expression board is added to the robot. And the Keyestudio 8*16 LED board can do the trick. With the help of it, you could design facial expressions, images, patterns and other displays by yourselves.

The 8*16 LED board comes with 128 LEDs. The data of the microprocessor(Arduino) communicates with the AiP1640 through a two-wire bus interface. Therefore, it can control the on and off of 128 LEDs on the module, so as to make the dot matrix on the module to display the pattern you need. A HX-2.54 4Pin cable is provided for your convenience of wiring.

2.Specification

- Working voltage: DC 3.3-5V
- Power loss: 400mW
- Oscillation frequency: 450KHz
- Drive current: 200mA
- Working temperature: -40~80℃
- Communication mode: I2C

3.Circuit Diagram



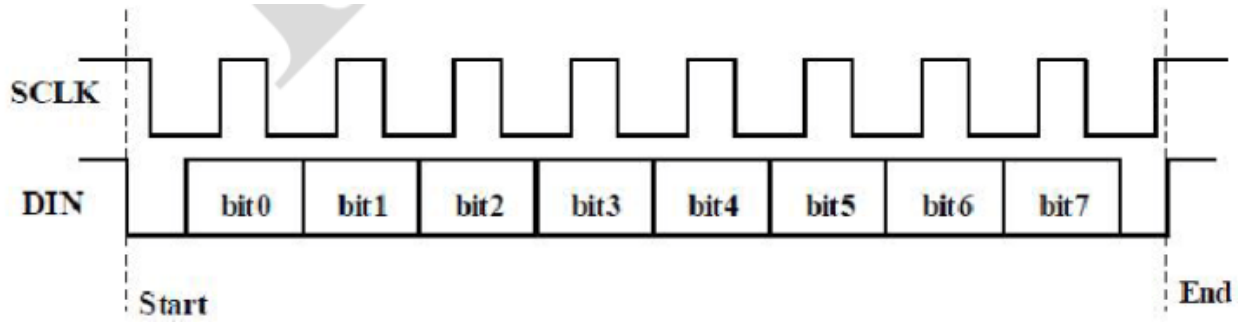
4.Working Principle

How to control each LED of the 8*16 dot matrix? It is known that each byte has 8 bits and each bit is 0 or 1. when it is 0, LED is off while when it is 1 LED is on. One byte can control one column of the LED,and naturally 16 bytes can control 16 columns of LEDs, that's the 8*16 dot matrix.

5.Pins description and communication protocol

The data of the microprocessor (Arduino) communicates with the AiP1640 through a two-wire bus cable.

The communication protocol diagram is as follows (SCLK) is SCL, (DIN) is SDA.



The starting condition for data input: SCL is high level and SDA changes from high to low.

For data command setting, there are methods as shown in the figure below.

In our sample program, select the way to **add 1 to the address automatically**, the binary value is 0100 0000 and the corresponding hexadecimal value is 0x40.

B7	B6	B5	B4	B3	B2	B1	B0	Description
0	1	Irrelevant choice, fill in 0			0	Irrelevant choice, fill in 0		add 1 to the address automatically
0	1				1			
0	1			0				Fixed address
0	1			1				Universal mode

For address command setting, the address can be selected as shown below.

The first 00H is selected in our sample program, and the binary number 1100 0000 corresponds to the hexadecimal 0xc0.

B7	B6	B5	B4	B3	B2	B1	B0	Display address
1	1	Irrelevant choice, fill in 0		0	0	0	0	00H
1	1			0	0	0	1	01H
1	1			0	0	1	0	02H
1	1			0	0	1	1	03H
1	1			0	1	0	0	04H
1	1			0	1	0	1	05H
1	1			0	1	1	0	06H
1	1			0	1	1	1	07H
1	1			1	0	0	0	08H
1	1			1	0	0	1	09H
1	1			1	0	1	0	0AH
1	1			1	0	1	1	0BH
1	1			1	1	0	0	0CH
1	1			1	1	0	1	0DH
1	1			1	1	1	0	0EH
1	1			1	1	1	1	0FH

The requirement for data input is that when SCL is at high level when inputting data, the signal on SDA must remain unchanged. Only when the clock signal on SCL is at low level, can the signal on SDA be changed. The input of data is the low bit first, and the high bit later.

The condition for the end of data transmission is that when SCL is at low level, SDA at low level and SCL at high level, the level of SDA becomes high.

Display control, set different pulse width, pulse width can be selected as shown in the figure below.

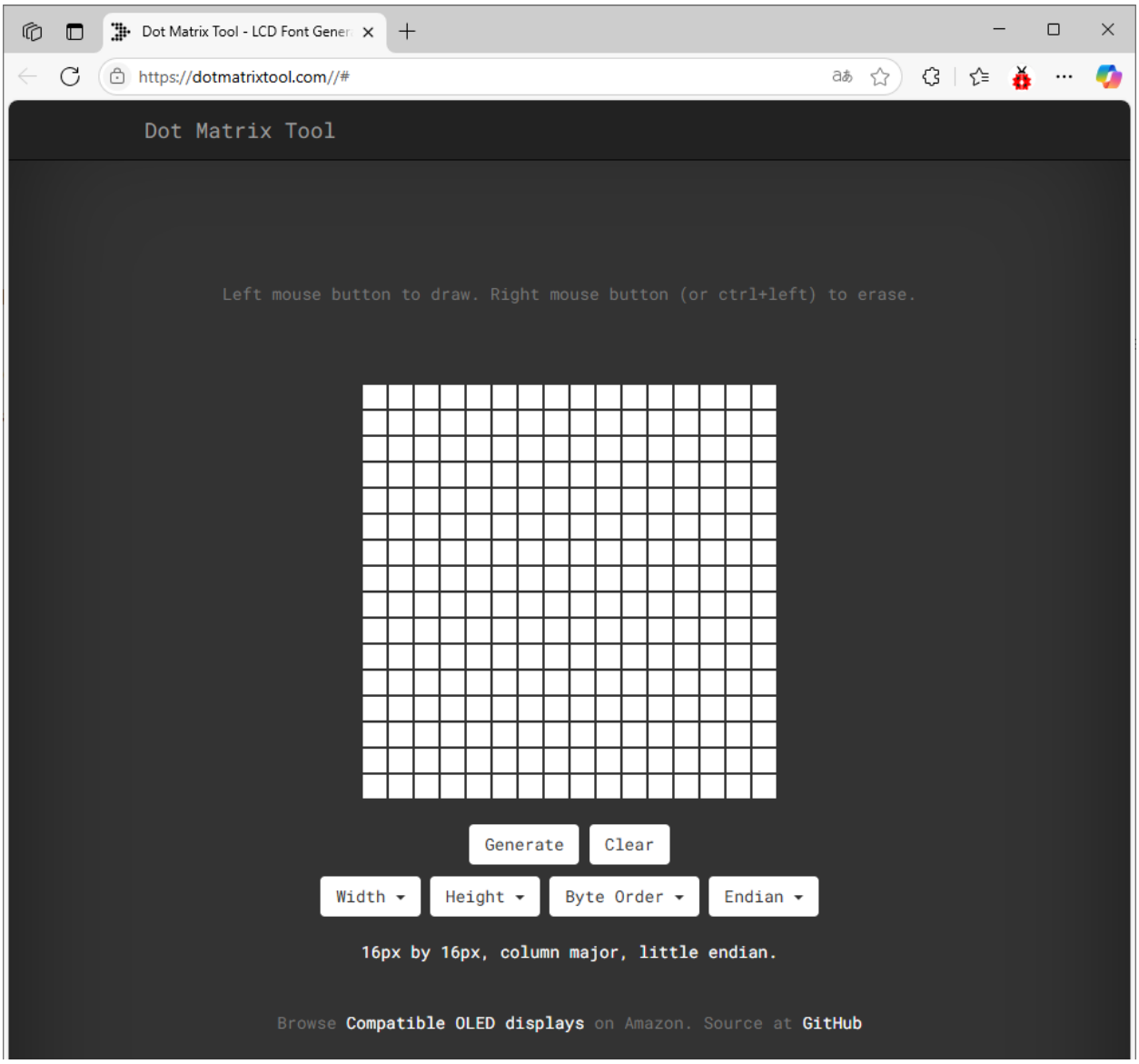
In the example, the pulse width is 4/16, and the hexadecimal corresponding to 1000 1010 is 0x8A.

B7	B6	B5	B4	B3	B2	B1	B0	Function	Description
1	0	Irrelevant choice, fill in 0		1	0	0	0	Clear quantity setting (Brightness setting)	Set pulse width to 1/16
1	0			1	0	0	1		Set pulse width to 2/16
1	0			1	0	1	0		Set pulse width to 4/16
1	0			1	0	1	1		Set pulse width to 10/16
1	0			1	1	0	0		Set pulse width to 11/16
1	0			1	1	0	1		Set pulse width to 12/16
1	0			1	1	1	0		Set pulse width to 13/16
1	0			1	1	1	1		Set pulse width to 14/16
1	0			0	X	X	X	Display switch setting	On
1	0			1	X	X	X		off

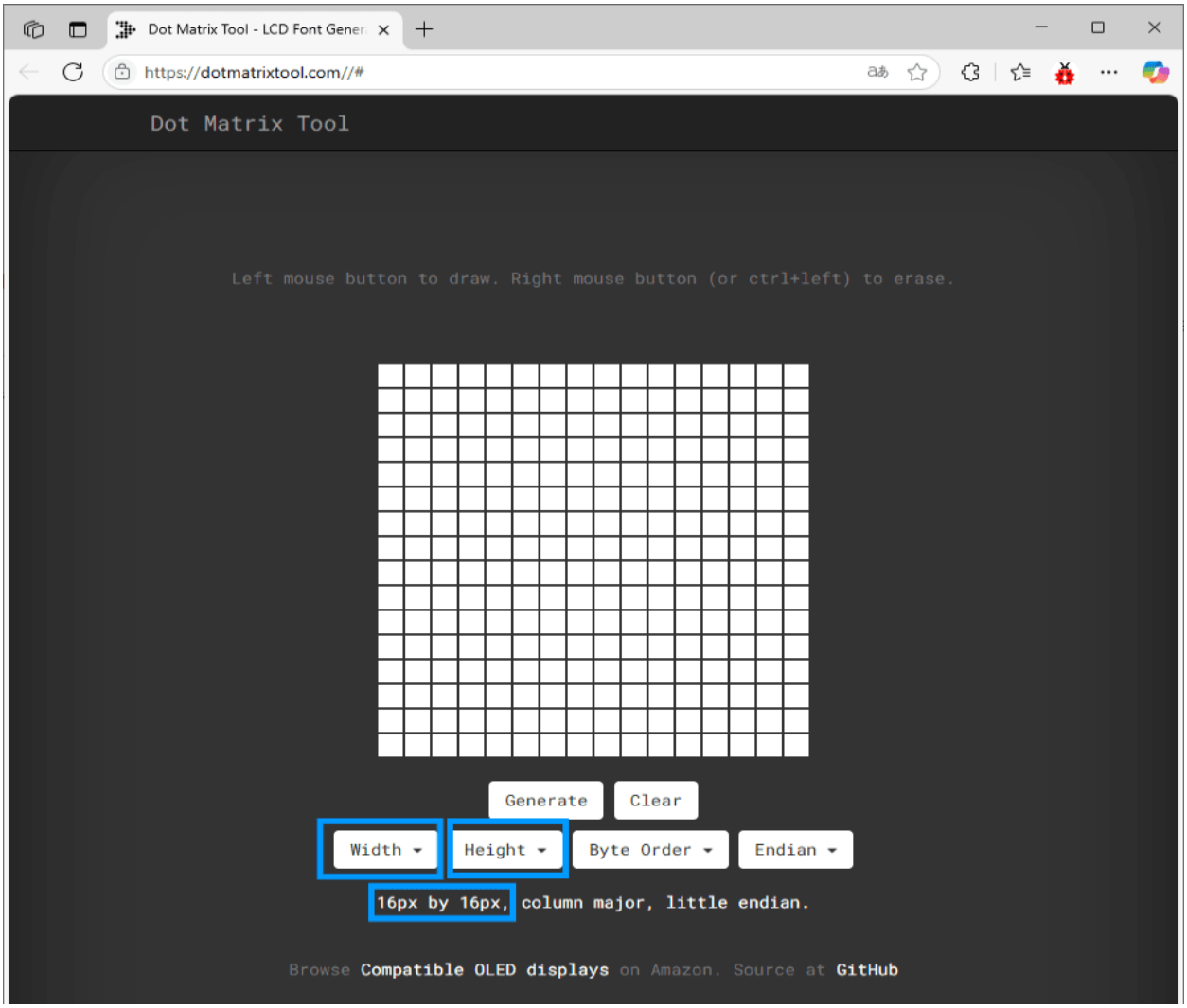
Instructions for the use of modulus tool

The dot matrix tool uses the online version, and the link is :http://dotmatrixtool.com/#

Enter the link and the page appears as shown below

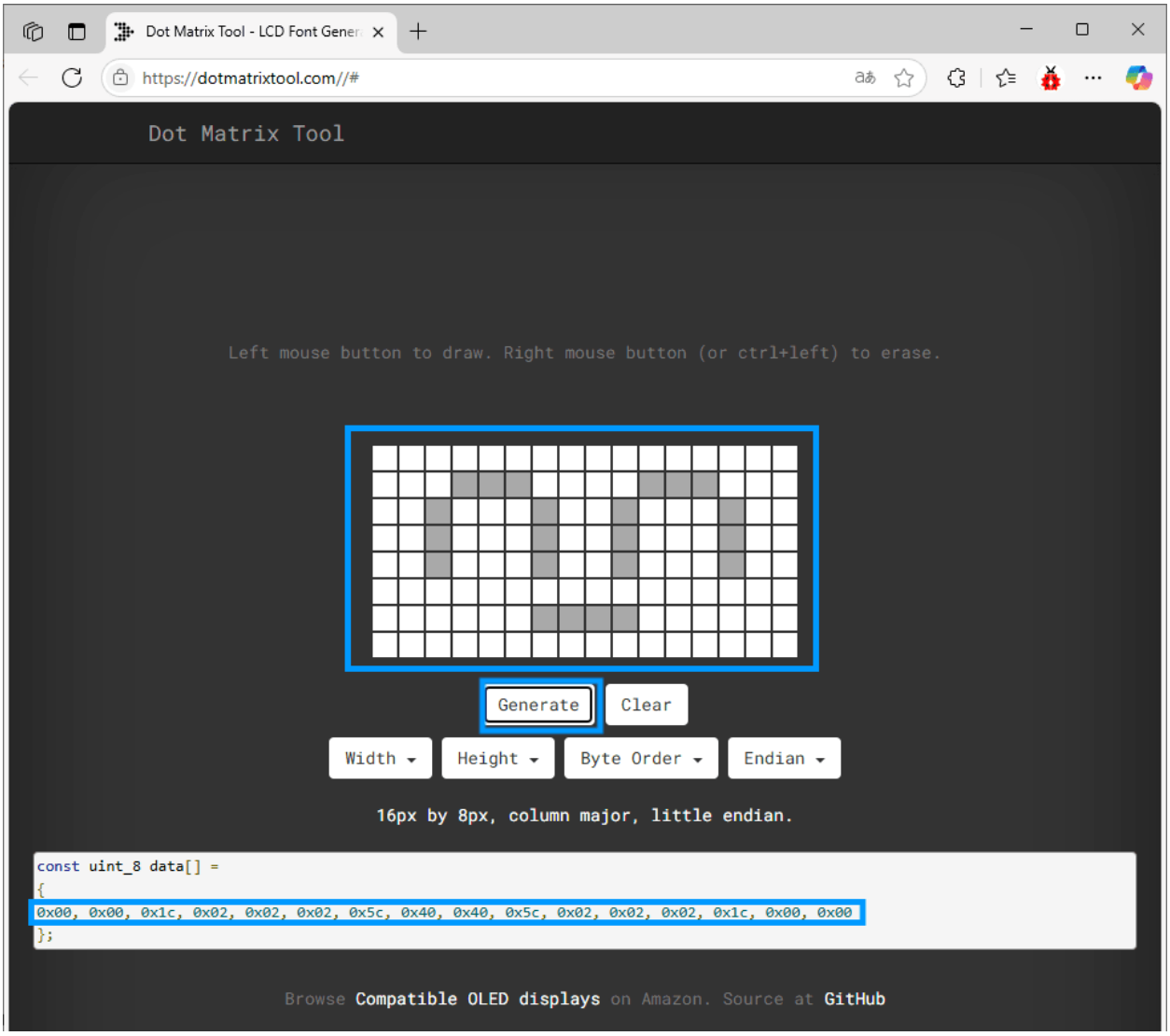


The dot matrix is 8*16, so adjust the height to 8 and width to 16, as shown in the figure below.

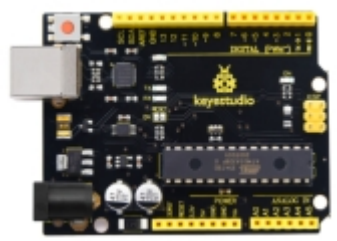
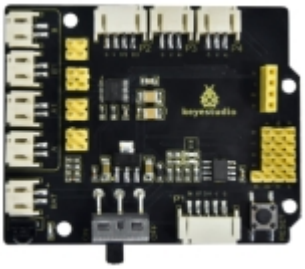
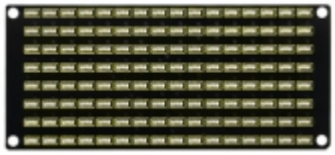


Generate hexadecimal data from the pattern

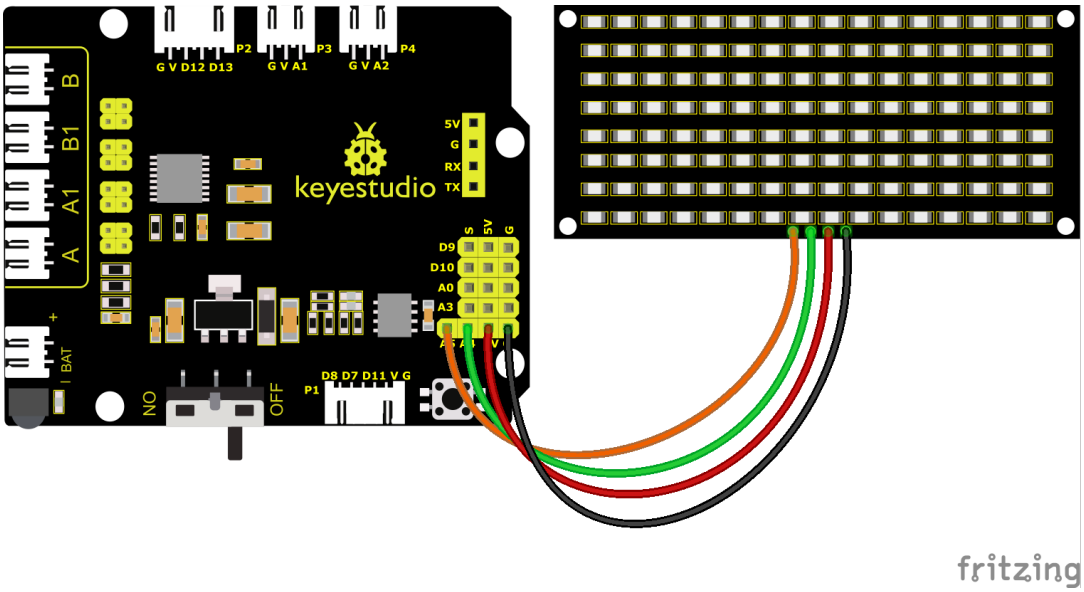
As shown in the figure below, press the left mouse button to select, right click to cancel; draw the pattern you want, click Generate, and the hexadecimal data we need will be generated.


Dot Matrix Tool - LCD Font Gener
https://dotmatrixtool.com//#
Dot Matrix Tool
Left mouse button to draw. Right mouse button (or ctrl+left) to erase.
Generate
Clear
Width
Height
Byte Order
Endian
16px by 8px, column major, little endian.
const uint_8 data[] =
{
0x00, 0x00, 0x1c, 0x02, 0x02, 0x02, 0x5c, 0x40, 0x40, 0x5c, 0x02, 0x02, 0x02, 0x1c, 0x00, 0x00
};
Browse Compatible OLED displays on Amazon. Source at GitHub

6.Components

Development Board *1	8833 Motor Driver *1	8x16 LED Panel*1
		
USB Cable*1	HX-2.54 4P Dupont Wire 200mm *1	

7.Wiring Diagram



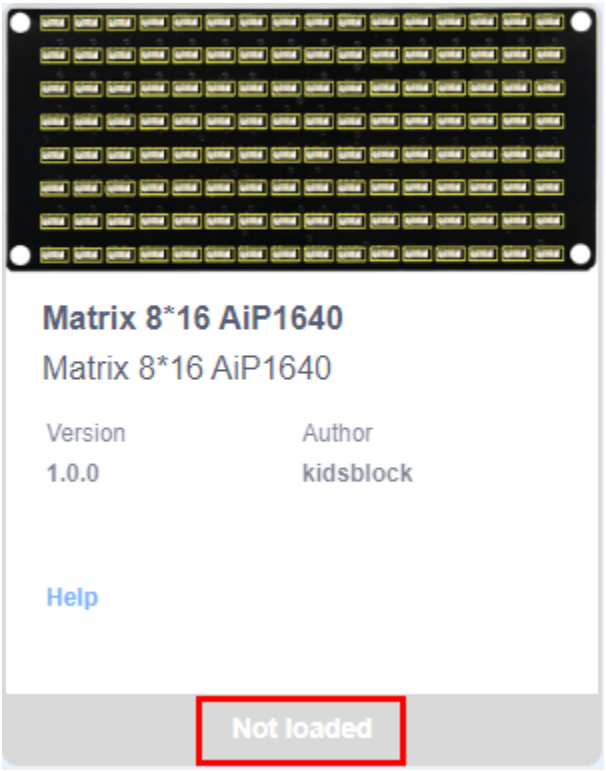
The GND, VCC, SDA, and SCL of the 8x16 LED light board are respectively connected to the keyestudio sensor expansion board-(GND), + (VCC), A4, A5 for two-wire serial communication.

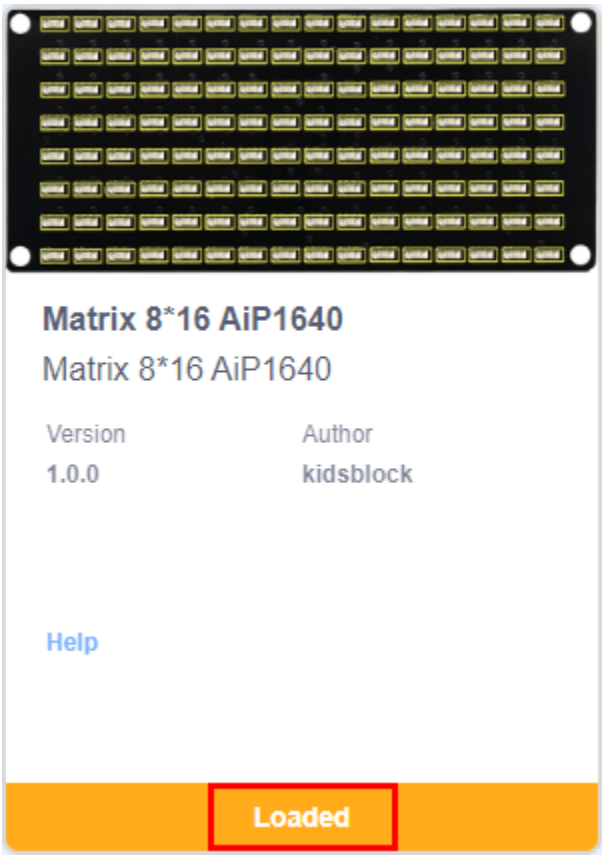
(Note: Though it is connected with the IIC pin of Arduino, this module is not for IIC communication. And the IO port here is to simulate I2C communication and can be connected with any two pins).

8.Test Code

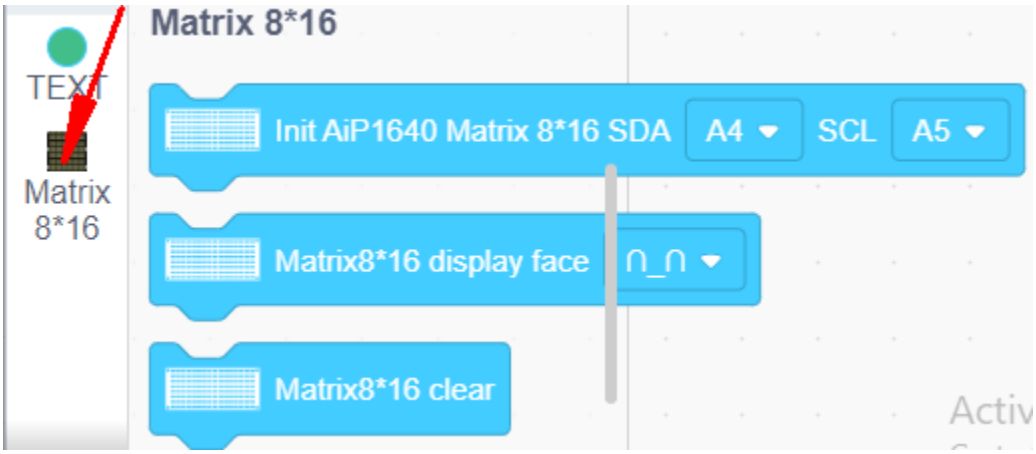
Before writing the code, it is necessary to import the library file of the 8x16 LED board. The specific steps are as follows:

Click to enter the extension library interface of sensors/modules/components, then search for"**Matrix 8*16 Aip1640**"module and click it. In this way, "**Not loaded**" changes to "**loaded**", indicating that"**Matrix 8*16 Aip1640**"module was added successfully.

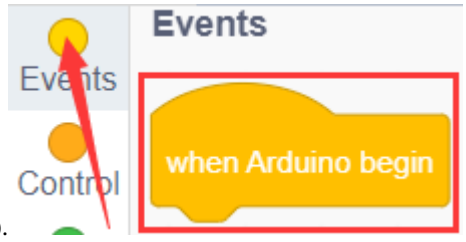




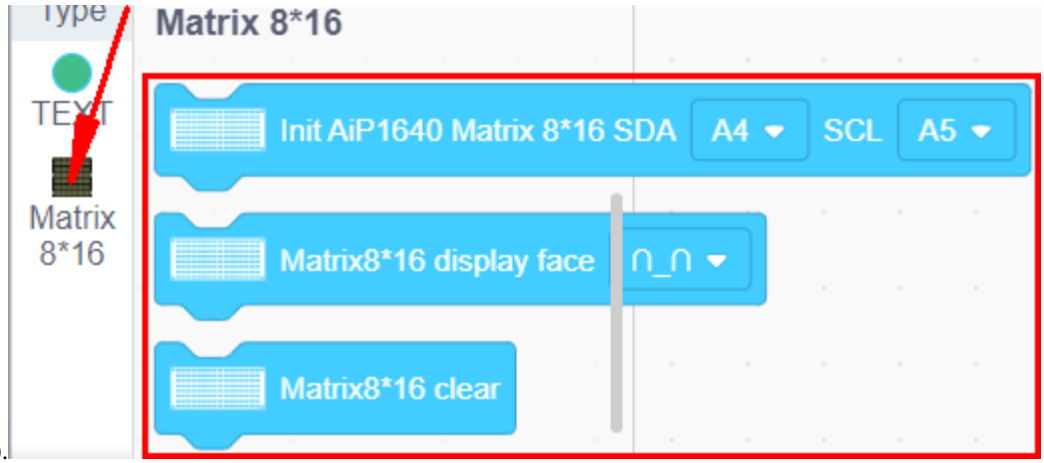
Click Back to return to the code editor interface, the instruction block of the added "**Matrix 8*16 Aip1640**"module can be seen in the module area.



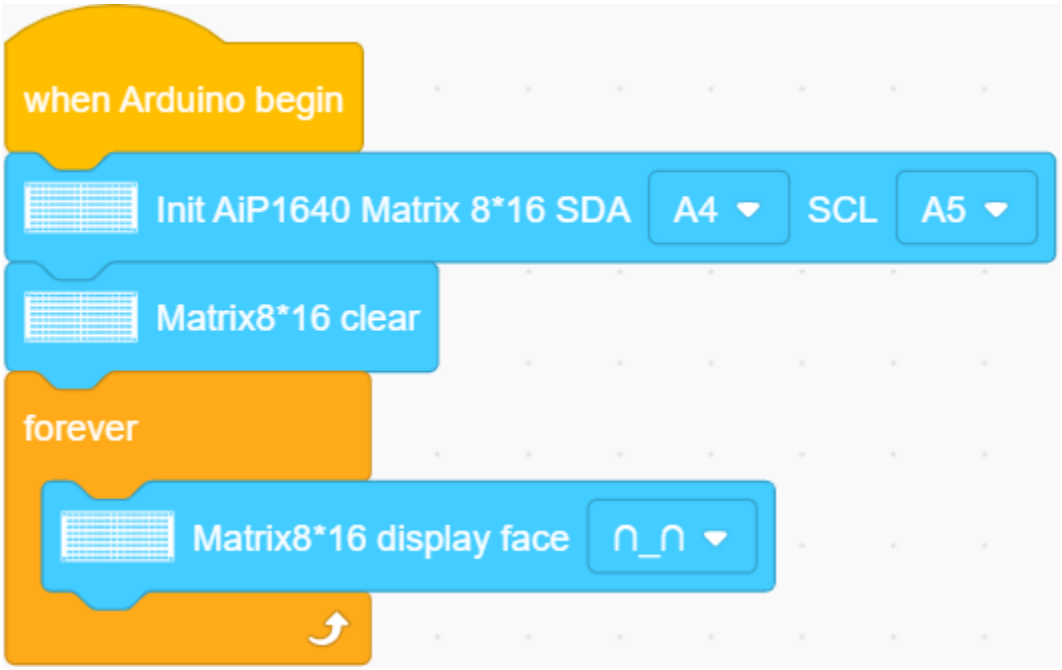
You can drag blocks to edit. Blocks listed below are for your reference.



(1).

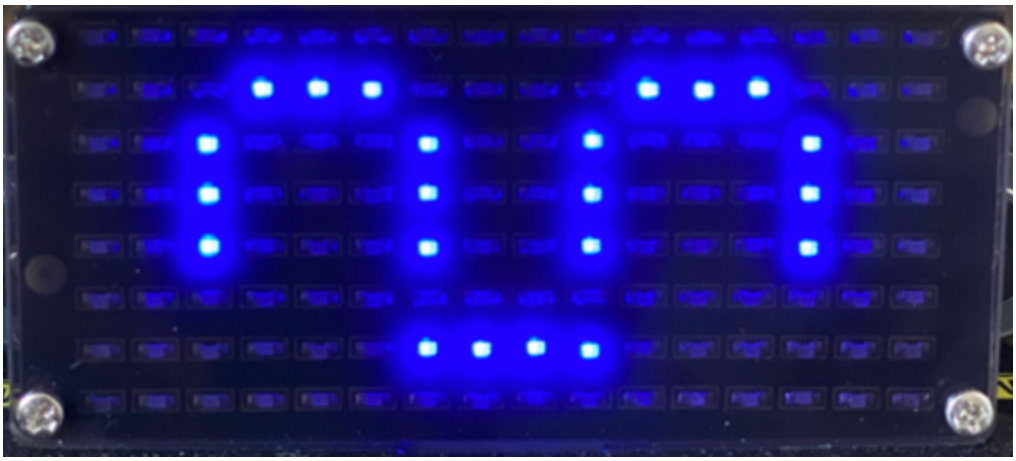
(2). 

Complete Test Code



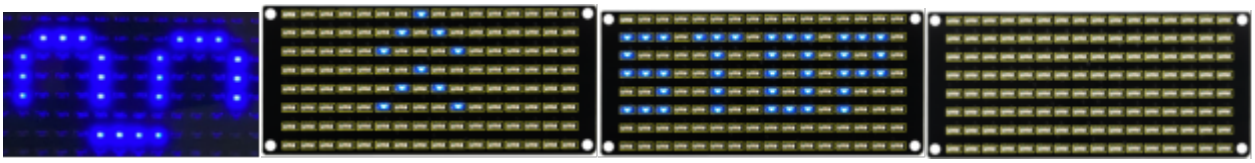
9.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, then turn the DIP switch to ON, a smile-shaped pattern will be displayed on the LED board.

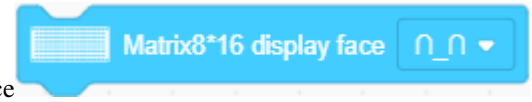


10.Code Explanation

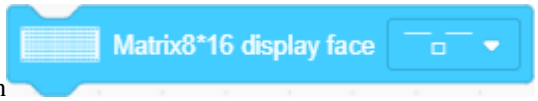
We use the modulus tool we just learned, http://dotmatrixtool.com/#, to make the dot matrix display the start pattern, going forward, and stop and then clear the pattern. The time interval is 2000 ms.



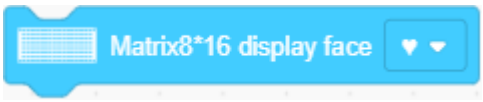
Instruction block for smiley face



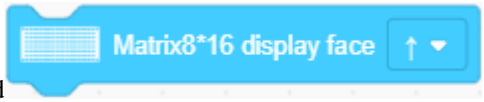
Instruction block for expression



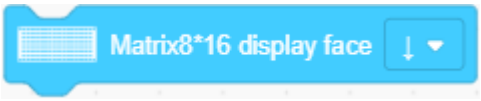
Instruction block for heart



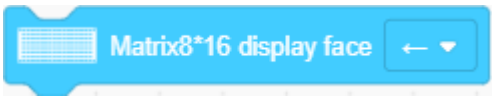
Instruction block for going forward



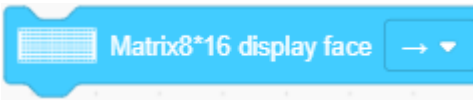
Instruction block for **stepping back**



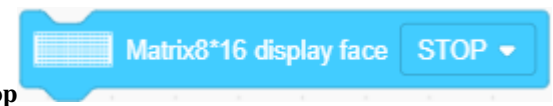
Instruction block for **turning left**



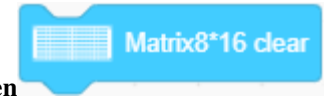
Instruction block for **turning right**

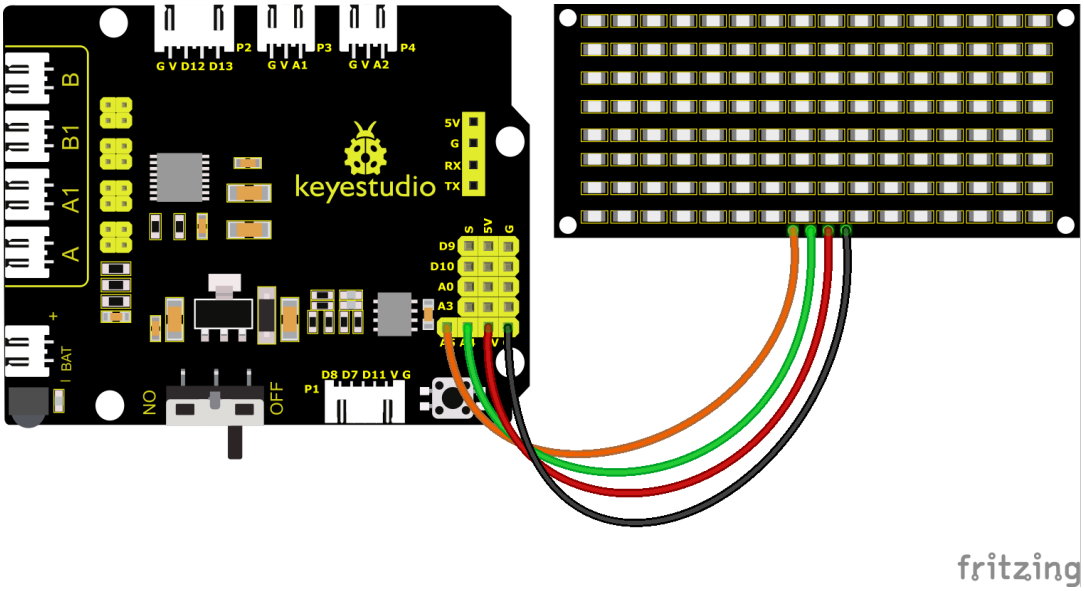


Instruction block for **stop**



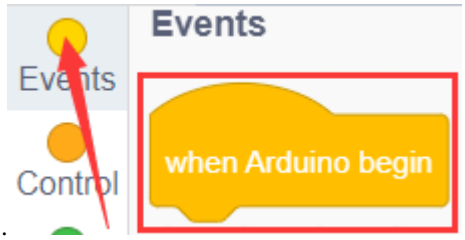
Instruction block for **clearing screen**



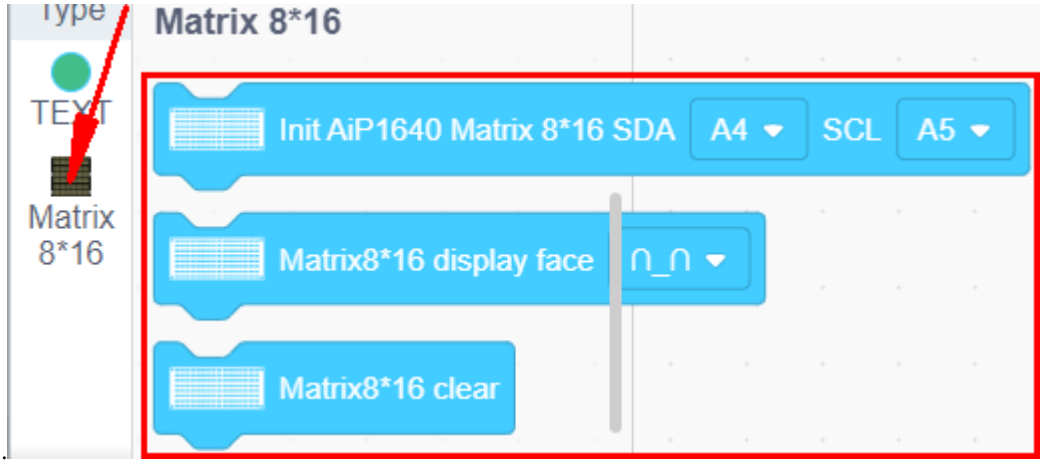


You can drag blocks to edit. Blocks listed below are for your reference.

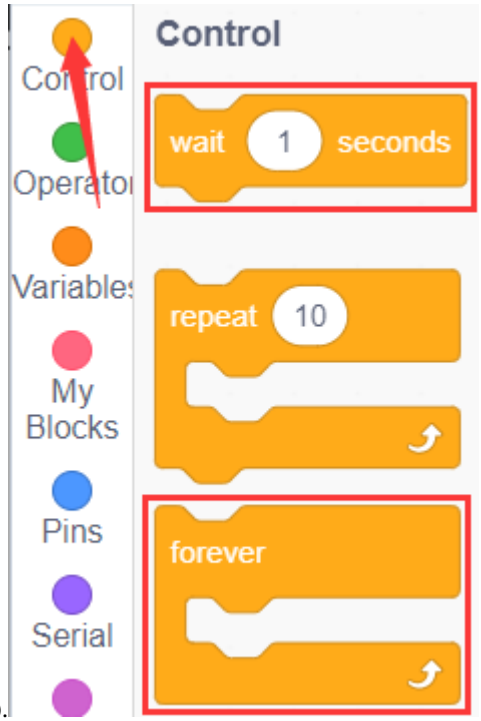
1).



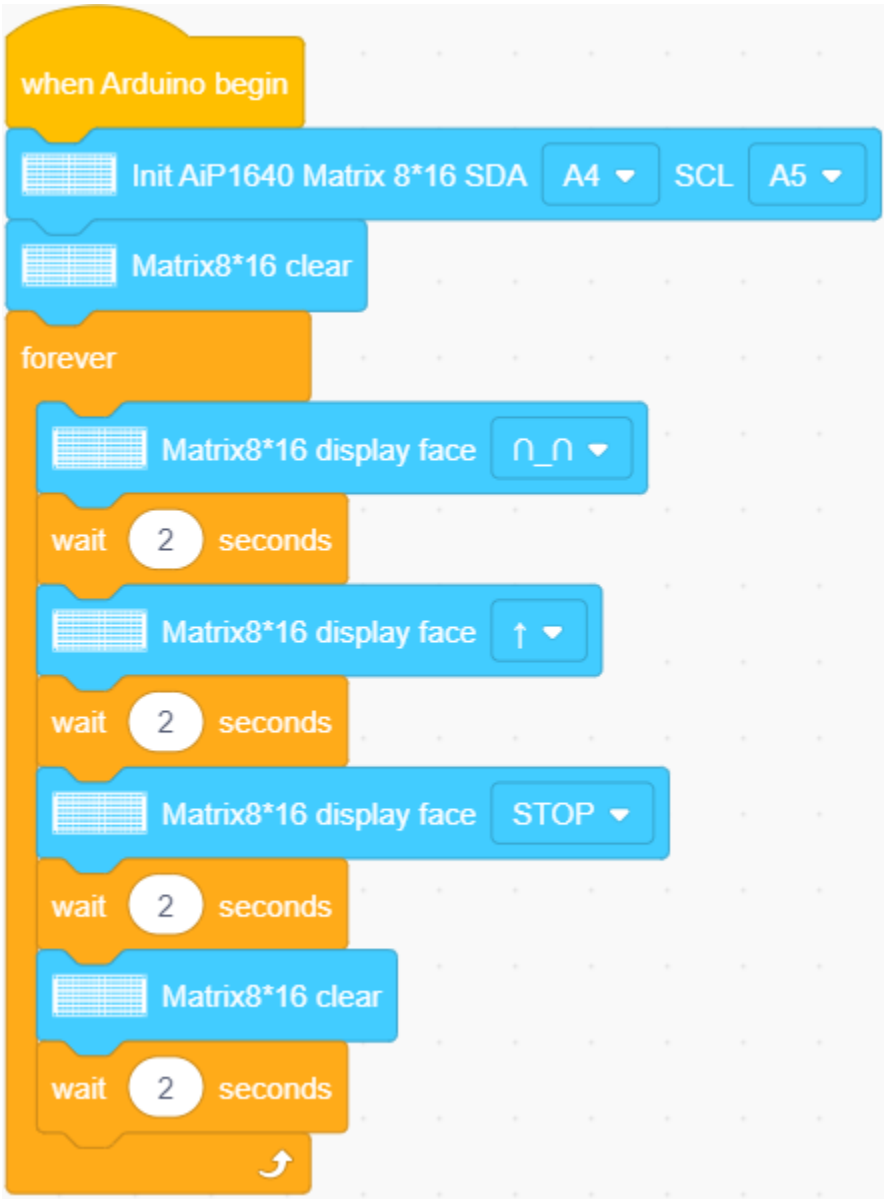
2).



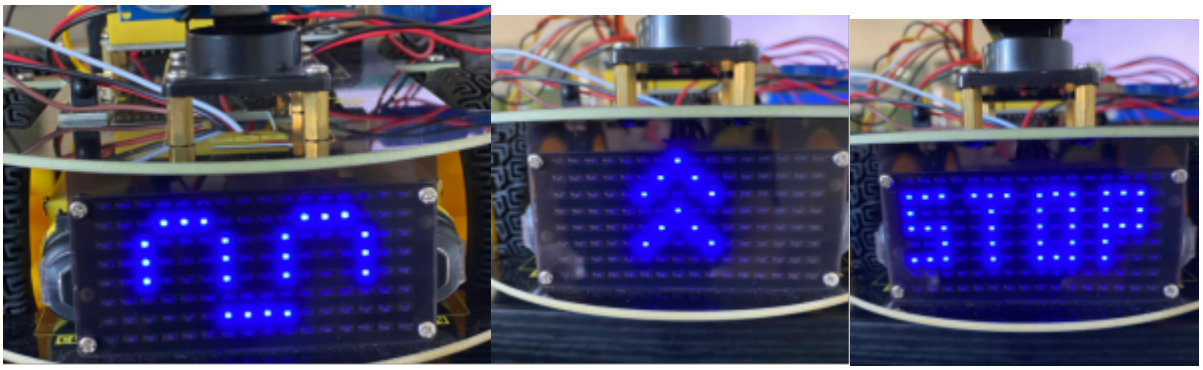
(3).



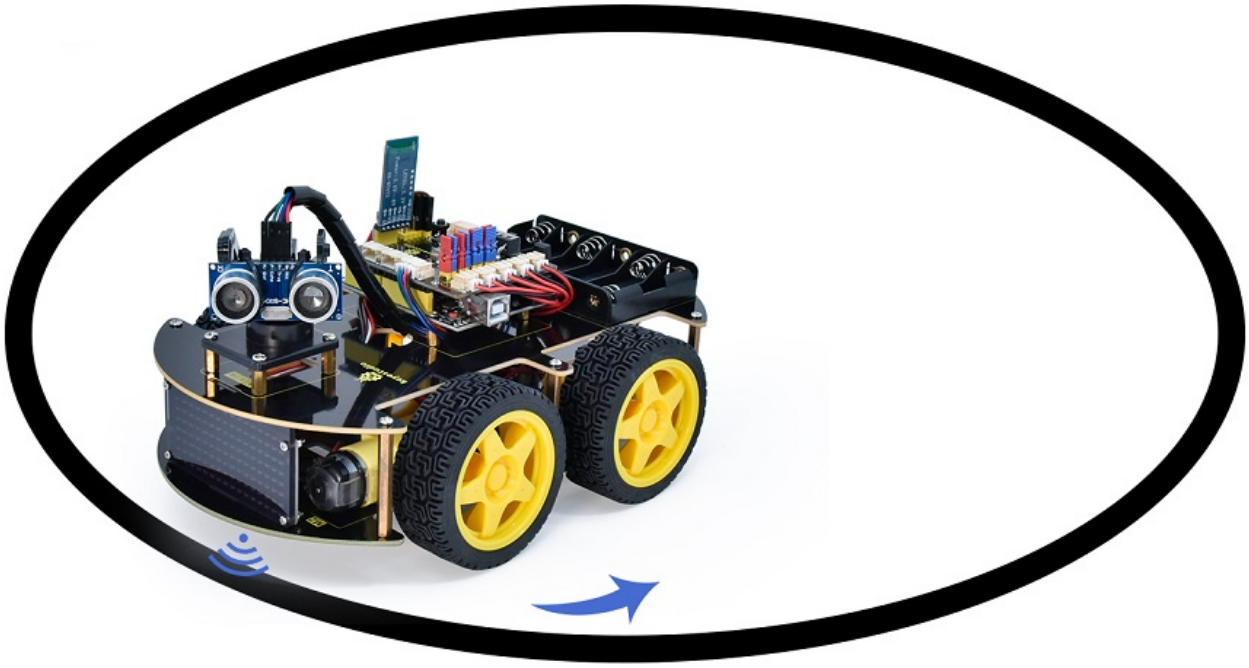
Complete Test Code



After uploading test code, the facial expression board shows these patterns orderly and repeats this sequence.



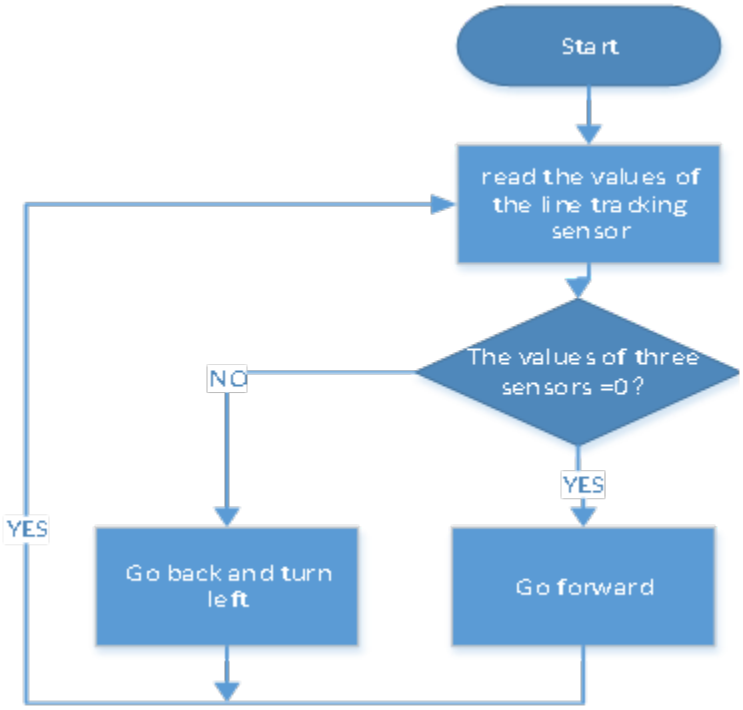
10.2.10 Project 10 Restricting Smart Car



1.Description

In this project, we look to combine the knowledge of a line tracking sensor and motor driver modules to make a restricting smart car. In the experiment, we aim to use the line tracking sensor to detect whether there is a black line around the smart car, and then control the rotation of the two motors according to the detection results in a way that lock the smart car in a circle drawn in black line.

2.Flow Chart

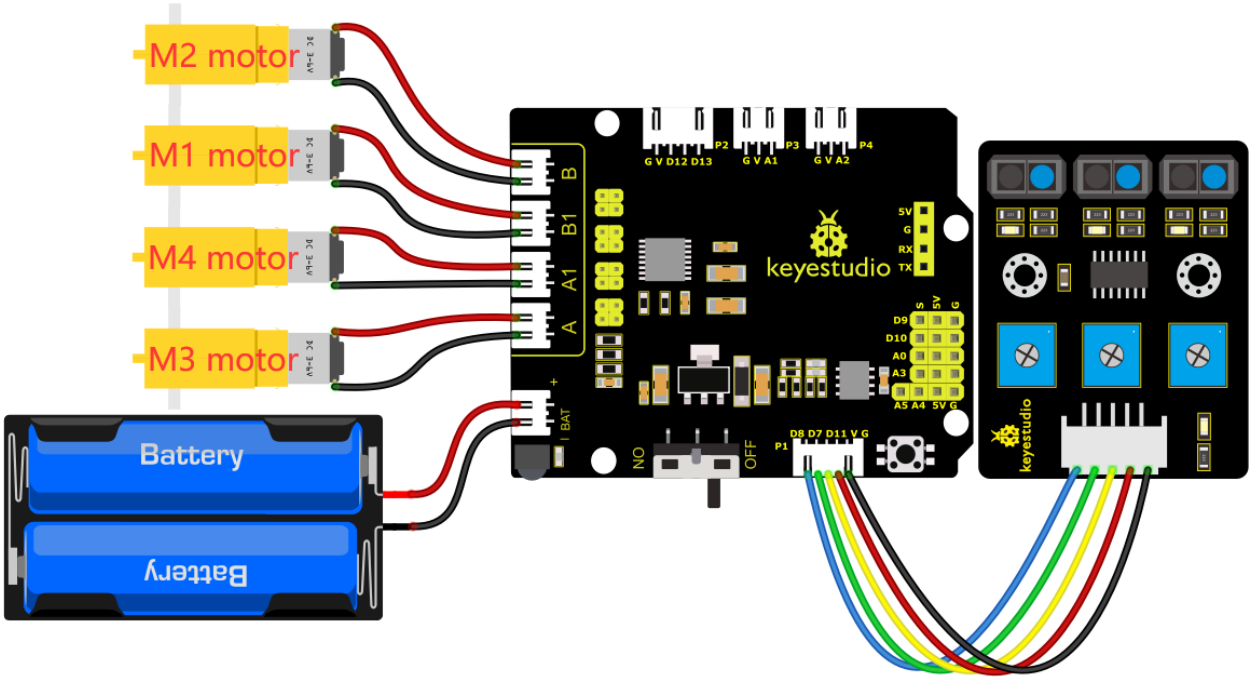


The specific logic of the restricting 4WD smart car is shown in the table.

Detection	Line-tracking sensor in the middle	Black line detected: in high level
		White line detected: in low level
	Line-tracking sensor on the left	Black line detected: in high level
		White line detected: in low level
	Line-tracking sensor on the right	Black line detected: in high level
		White line detected: in low level

Condition	Movement
All the three line-tracking sensors don' t detect black lines	Move forward
Any of the three line-tracking sensors detect black lines	Step back Then rotate left

3.Wiring Diagram

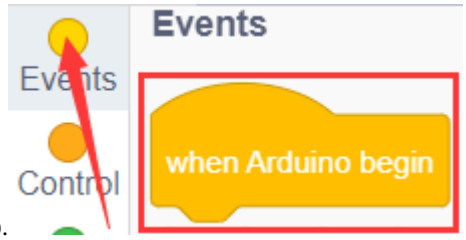


G, V, S1, S2 and S3 of the line tracking sensor are connected to GGND), VVCC), D11, D7 and D8 of the sensor expansion board.

The power is connected to the BAT port

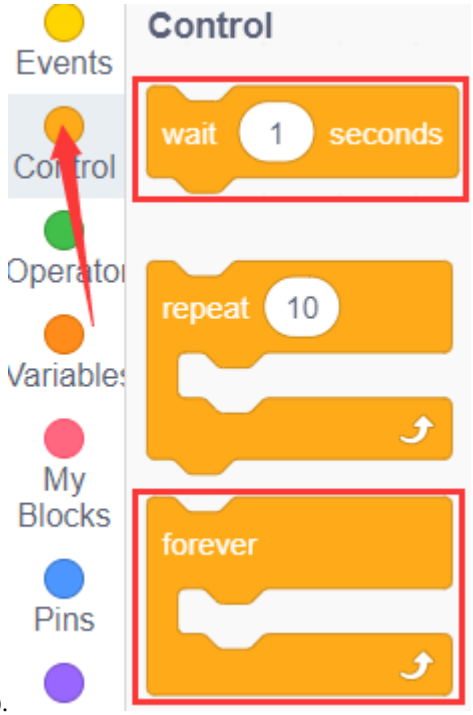
4.Test Code

You can drag blocks to edit. Blocks listed below are for your reference.

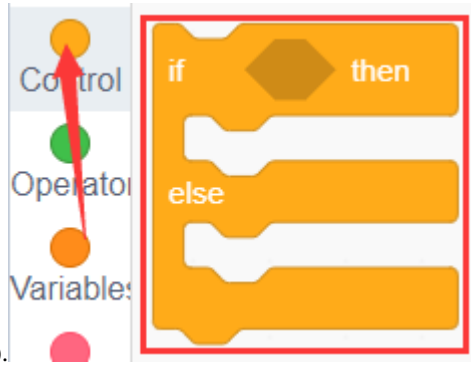


(1).

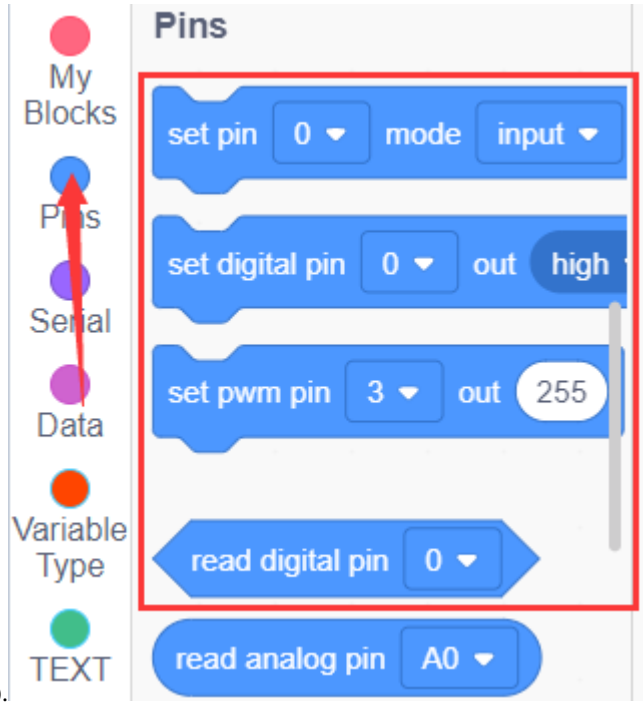
(2).


Control
Events
Control
Operator
Variables
My Blocks
Pins
wait 1 seconds
repeat 10
forever

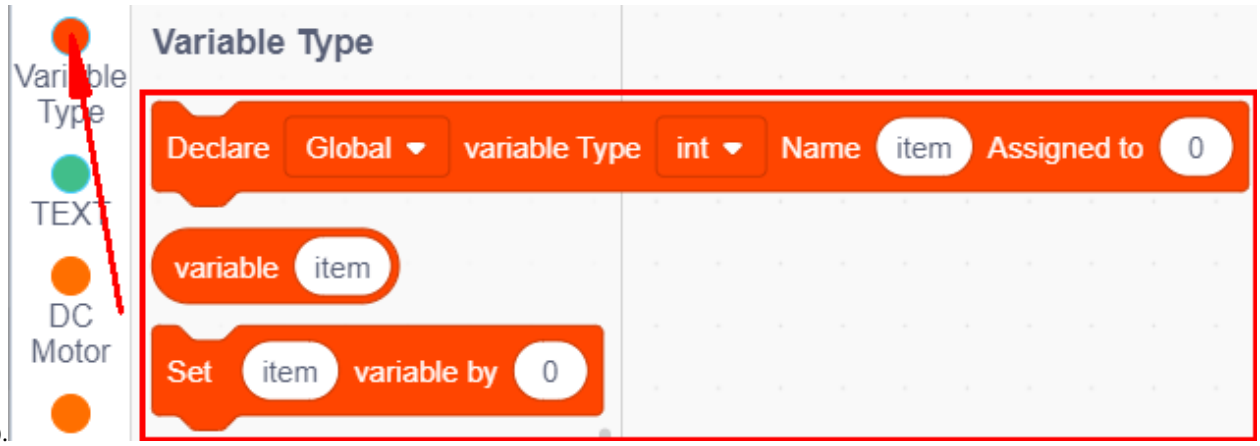
(3).


Control
Operator
Variables
if then
else

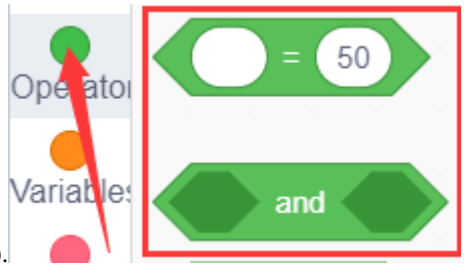
(4).



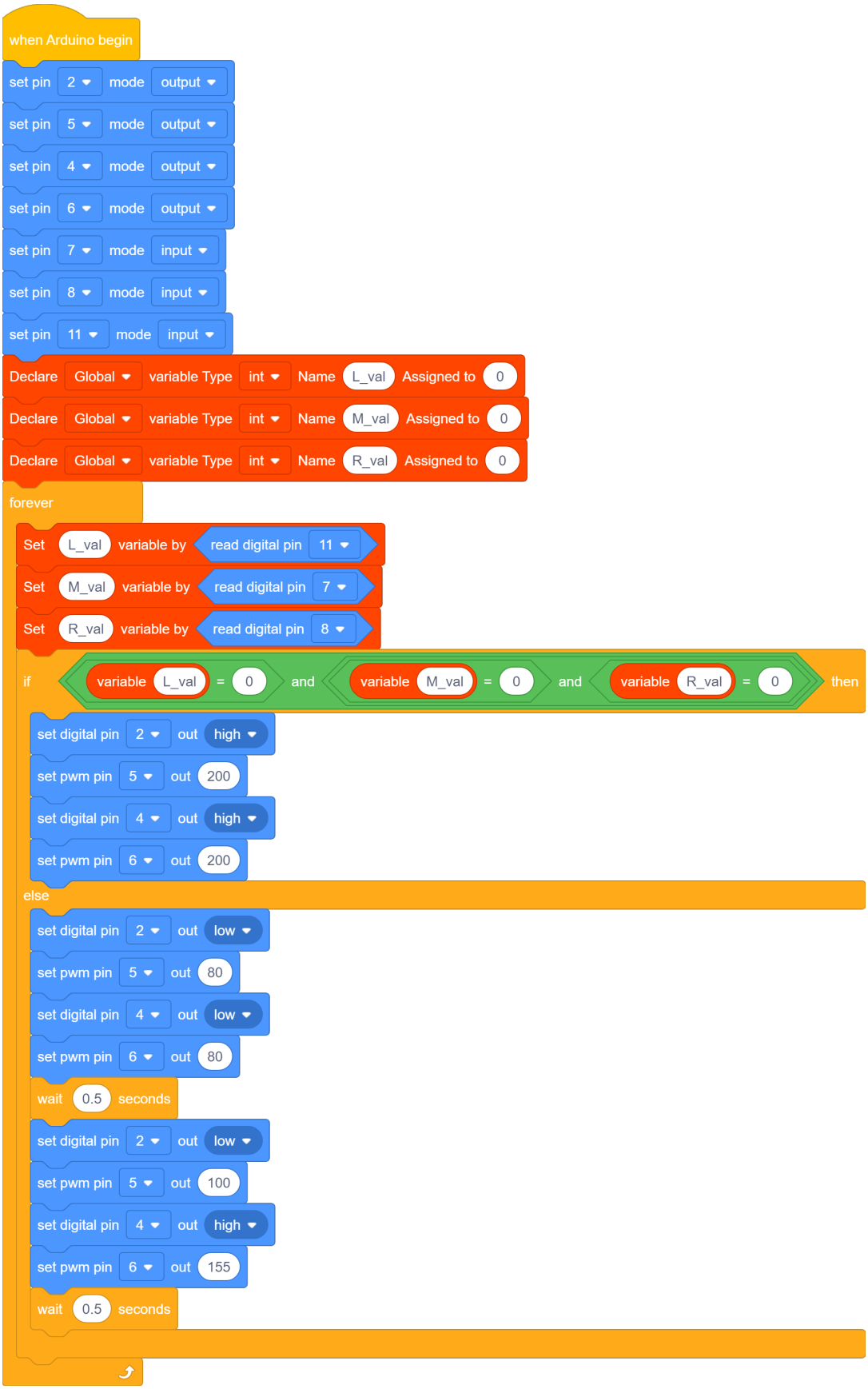
(5).



(6).



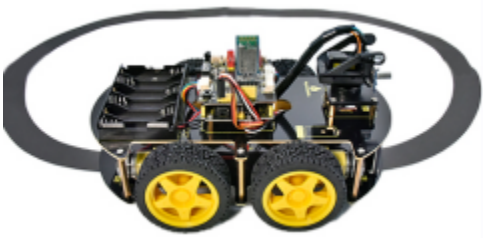
Complete Test Code


when Arduino begin
set pin 2 mode output
set pin 5 mode output
set pin 4 mode output
set pin 6 mode output
set pin 7 mode input
set pin 8 mode input
set pin 11 mode input
Declare Global variable Type int Name L_val Assigned to 0
Declare Global variable Type int Name M_val Assigned to 0
Declare Global variable Type int Name R_val Assigned to 0
forever
Set L_val variable by read digital pin 11
Set M_val variable by read digital pin 7
Set R_val variable by read digital pin 8
if variable L_val = 0 and variable M_val = 0 and variable R_val = 0 then
set digital pin 2 out high
set pwm pin 5 out 200
set digital pin 4 out high
set pwm pin 6 out 200
else
set digital pin 2 out low
set pwm pin 5 out 80
set digital pin 4 out low
set pwm pin 6 out 80
wait 0.5 seconds
set digital pin 2 out low
set pwm pin 5 out 100
set digital pin 4 out high
set pwm pin 6 out 155
wait 0.5 seconds

5.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, power on the external power then turn the DIP switch to ON. Put the smart car in the black circle, then it will move solely in the circle.

10.2.11 Project 11 Line Tracking Smart Car

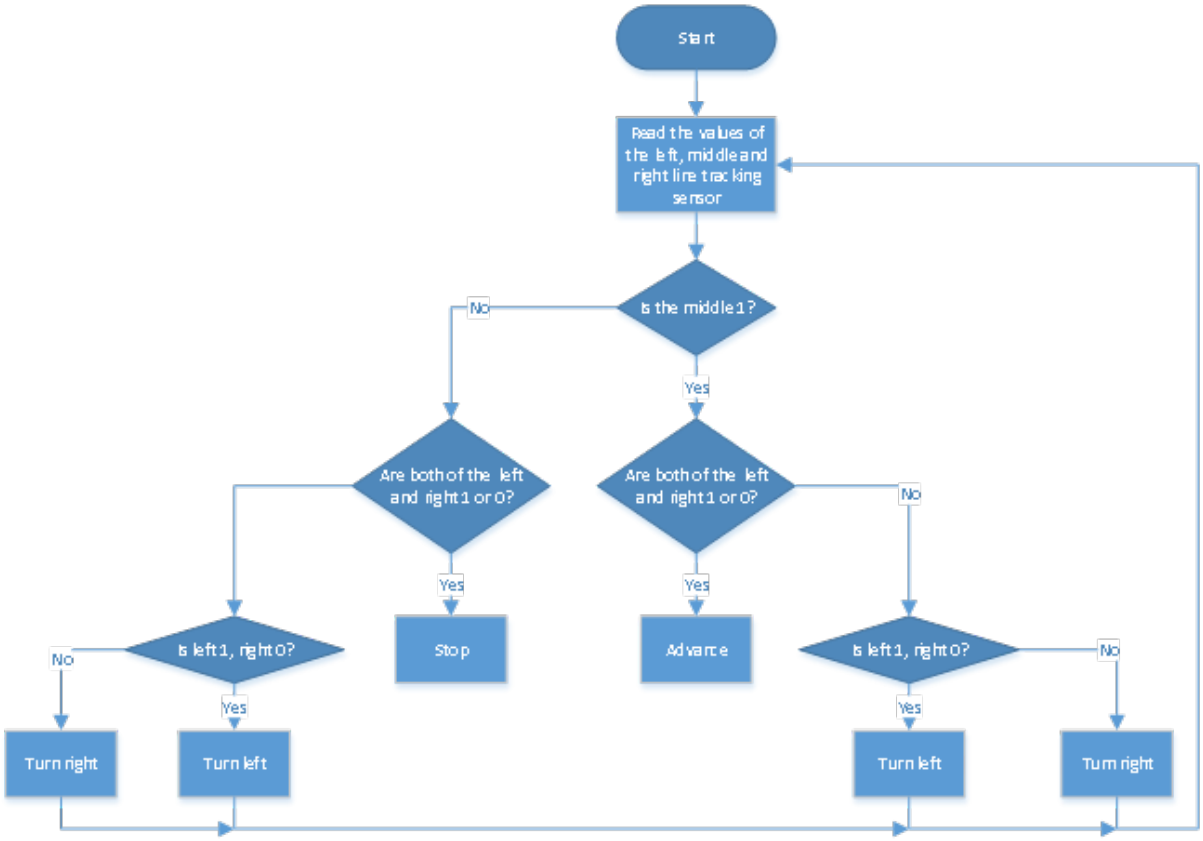


1.Description

Based on the working principle of the line tracking sensor, we empower to make a line tracking smart car.

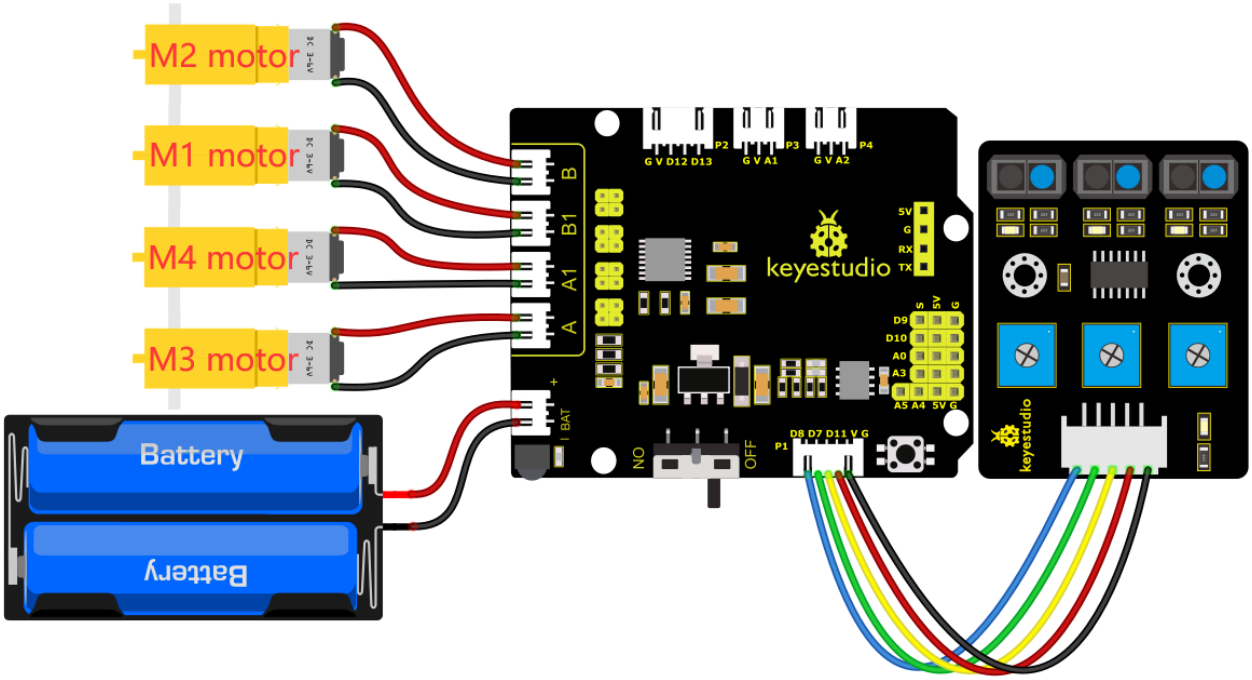
In this project, we detect whether there is a black line at the bottom of the smart car through a line tracking sensor, and then control the rotation of the two groups of motors according to the detection results in a way that controls the smart car to walk along the black line.

2.Flow Chart



Detection	Middle tracking sensor	detects black line: HIGH
		detects white line: LOW
	Left tracking sensor	detects black line: HIGH
		detects white line: LOW
	Right tracking sensor	detects black line: HIGH
		detects white line: LOW
Condition 1	Condition 2 Detect the left and the right tracking sensor	Status
Middle tracking sensor detects black line	left tracking sensor detects black line; right sensor detects white line	Turn left
	left tracking sensor detects white line; right sensor detects black line	Turn right
	left and right tracking sensor detect black line	Advance
	left and right tracking sensor detect white line	Advance
Middle tracking sensor detects white line	left tracking sensor detects black line; right sensor detects white line	Turn left
	left tracking sensor detects white line; right sensor detects black line	Turn right
	left and right tracking sensor detect black line	Stop
	left and right tracking sensor detect white line	Stop

3.Wiring Diagram

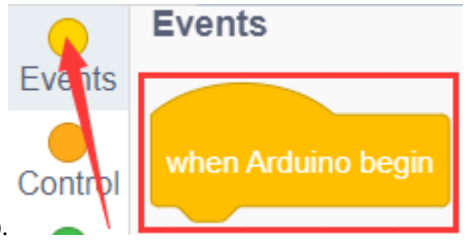


G, V, S1, S2 and S3 of the line tracking sensor are connected to GGND), VVCC), D11, D7 and D8 of the sensor expansion board.

The power is connected to the BAT port

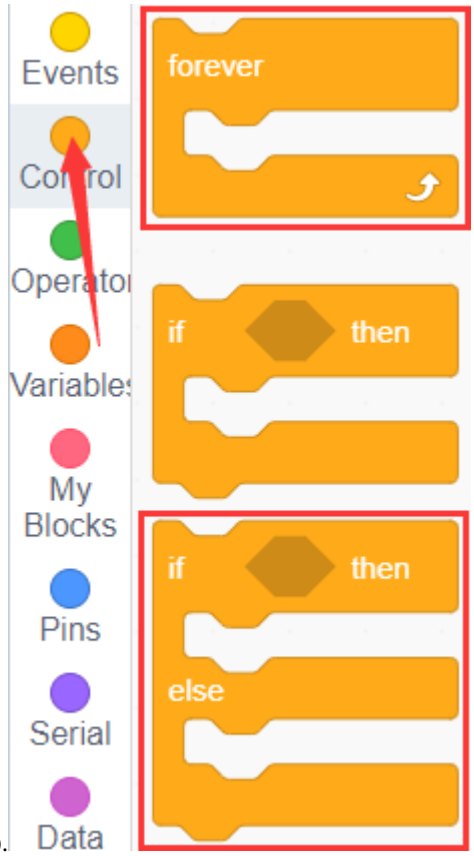
4.Test Code

You can drag blocks to edit. Blocks listed below are for your reference

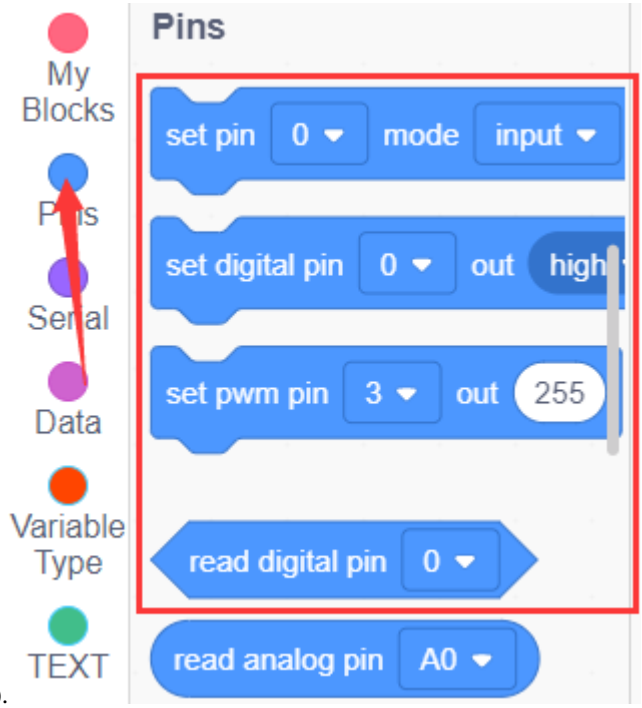


(1).

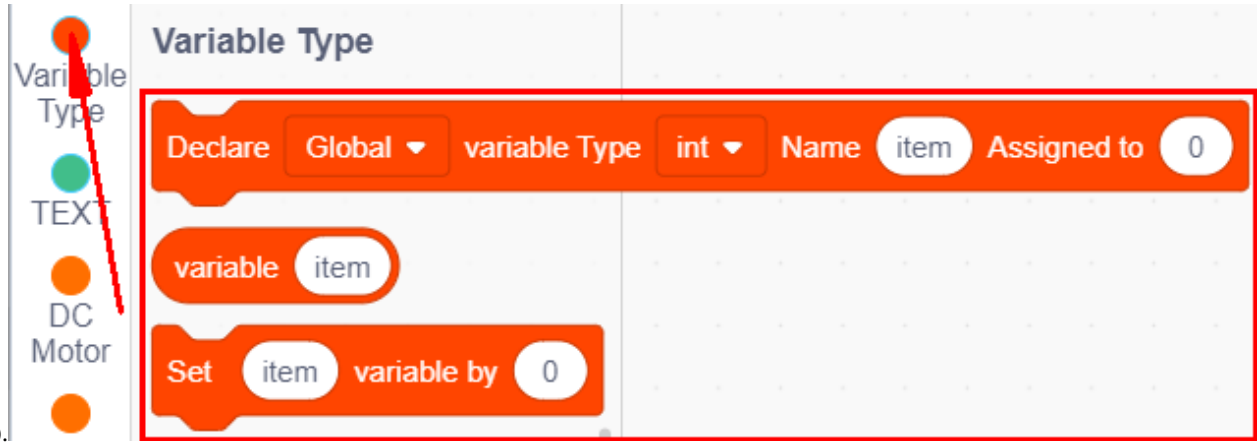
(2).


Events
Control
Operator
Variables
My Blocks
Pins
Serial
Data
forever
if then
if then
else

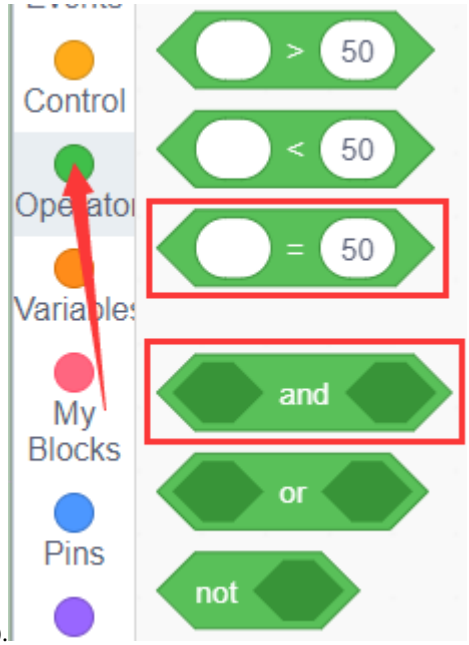
(3).


My Blocks
Pins
Serial
Data
Variable Type
TEXT
Pins
set pin 0 mode input
set digital pin 0 out high
set pwm pin 3 out 255
read digital pin 0
read analog pin A0

(4).

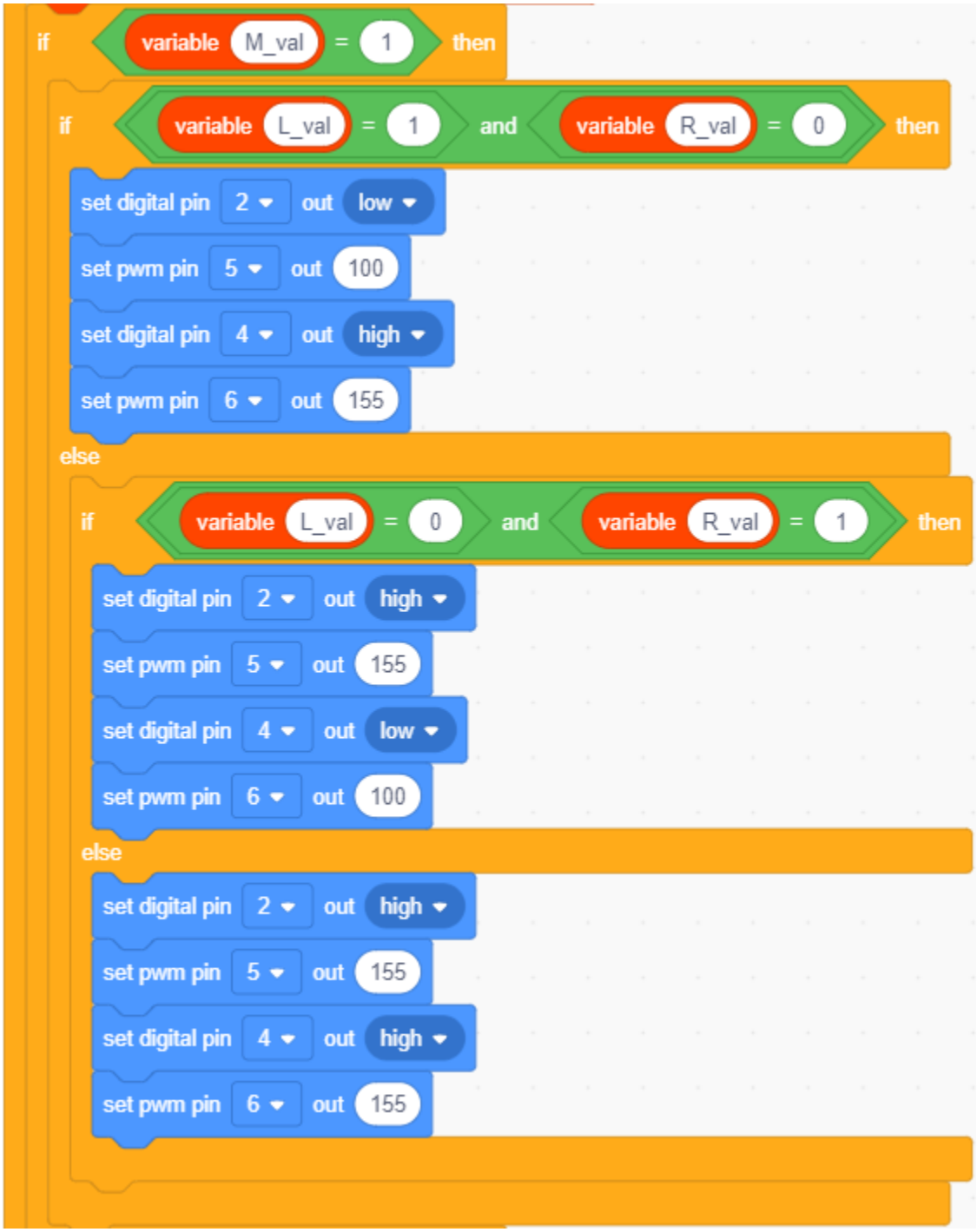


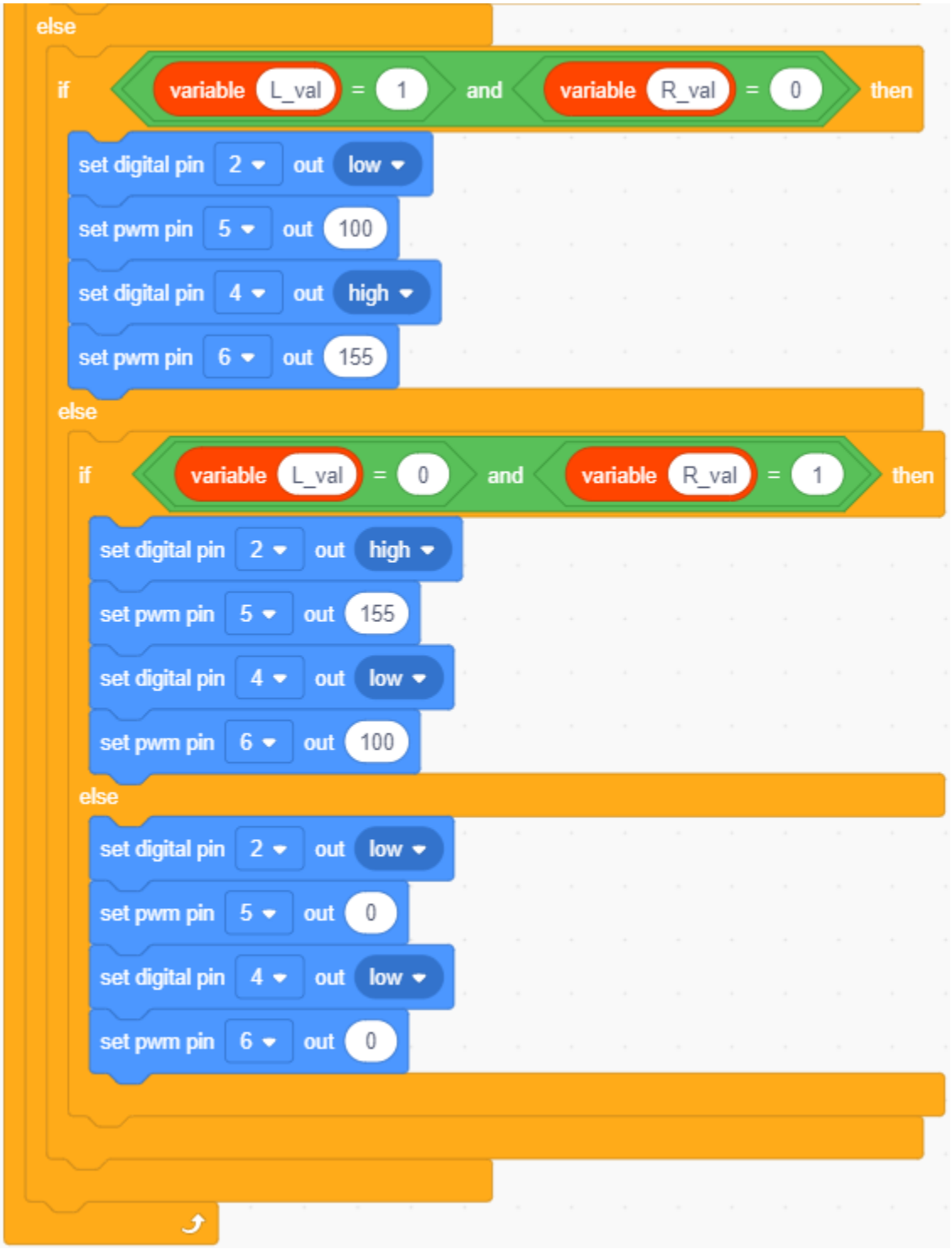
(5).



Complete Test Code

when Arduino begin
set pin 2 mode output
set pin 5 mode output
set pin 4 mode output
set pin 6 mode output
set pin 7 mode input
set pin 8 mode input
set pin 11 mode input
Declare Global variable Type int Name L_val Assigned to 0
Declare Global variable Type int Name M_val Assigned to 0
Declare Global variable Type int Name R_val Assigned to 0
forever
Set L_val variable by read digital pin 11
Set M_val variable by read digital pin 7
Set R_val variable by read digital pin 8

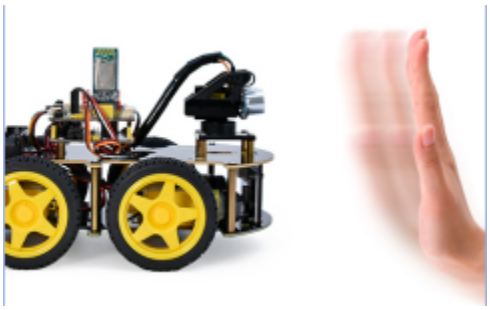

if variable M_val = 1 then
if variable L_val = 1 and variable R_val = 0 then
set digital pin 2 out low
set pwm pin 5 out 100
set digital pin 4 out high
set pwm pin 6 out 155
else
if variable L_val = 0 and variable R_val = 1 then
set digital pin 2 out high
set pwm pin 5 out 155
set digital pin 4 out low
set pwm pin 6 out 100
else
set digital pin 2 out high
set pwm pin 5 out 155
set digital pin 4 out high
set pwm pin 6 out 155



5.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, power on the external power then turn the DIP switch to ON. Then the smart car will walk along the lines.

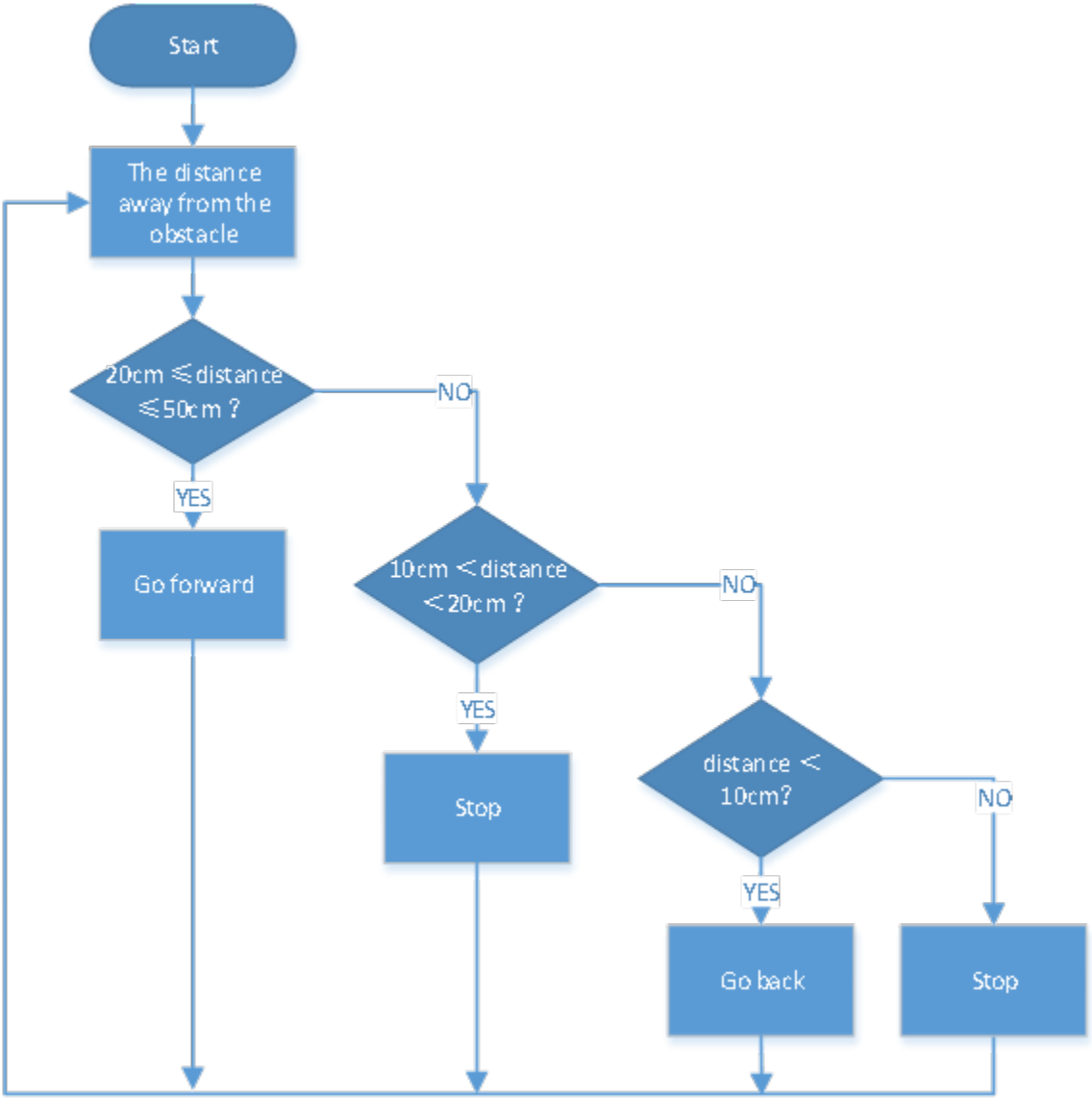
10.2.12 Project 12 Ultrasonic Following Smart Car



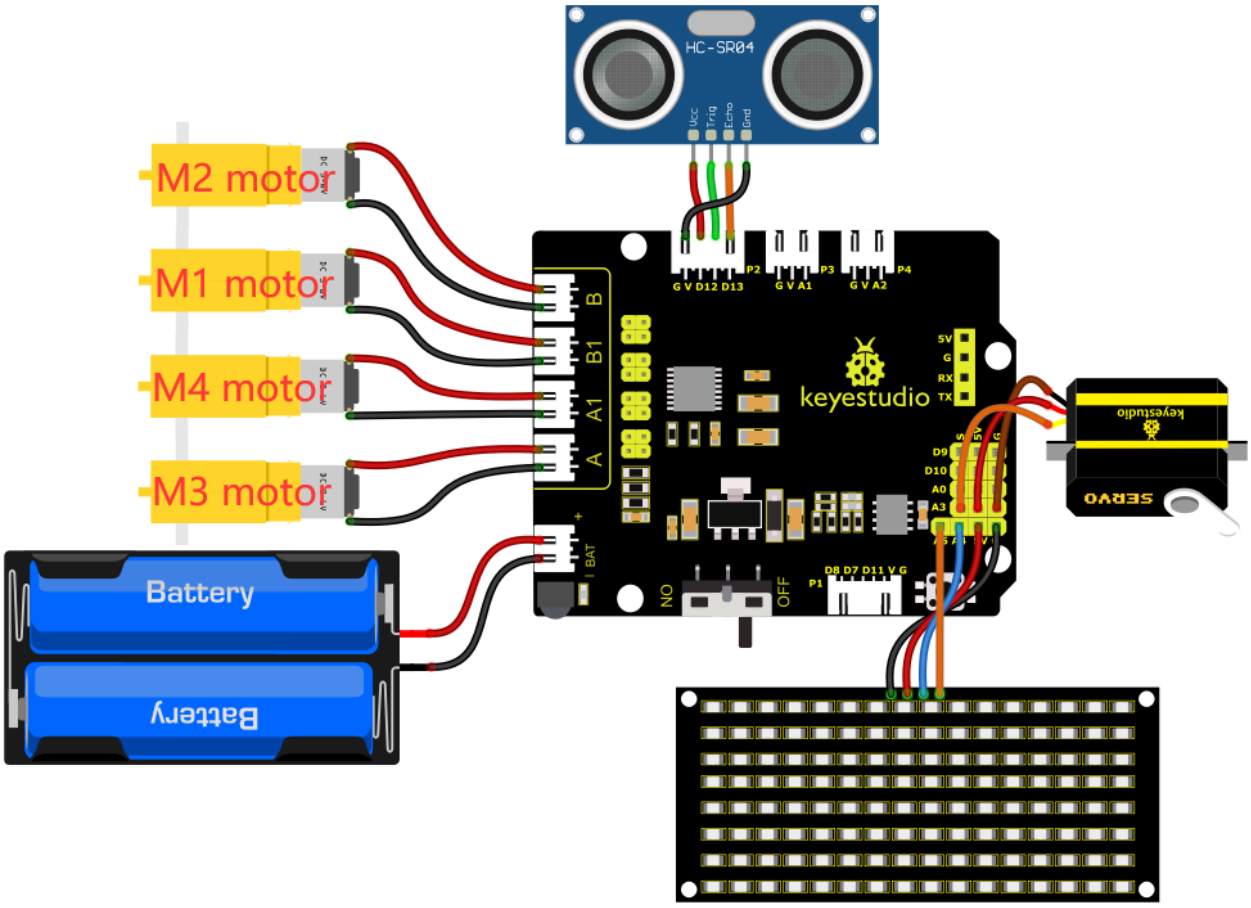
1.Description

In this project, we will look to detect the distance between the 4WD smart car and the obstacles ahead through an ultrasonic sensor to drive two motors in a way that make the car move and make the 8*8 LED board show a smile facial pattern.

2.Flow Chart


Start
The distance away from the obstacle
20cm ≤distance ≤50cm ?
NO
YES
Go forward
10cm <distance <20cm ?
NO
YES
Stop
distance < 10cm?
NO
YES
Go back
Stop

3.Wiring Diagram



Wiring up

1). GND, VCC, SDA and SCL of the 8*8 LED board are connected to GGND), VVCC), A4 and A5 of the expansion board.

2). VCC, Trig, Echo and Gnd of the ultrasonic sensor are connected to 5V(V), D12(S), D13(S) and Gnd(G)

3). The servo is connected to G, V and A3. The brown wire is interfaced with Gnd(G), the red wire is interfaced with 5V(V) and the orange wire is interfaced with A3.

4). The power is connected to the BAT port

4.Test Code

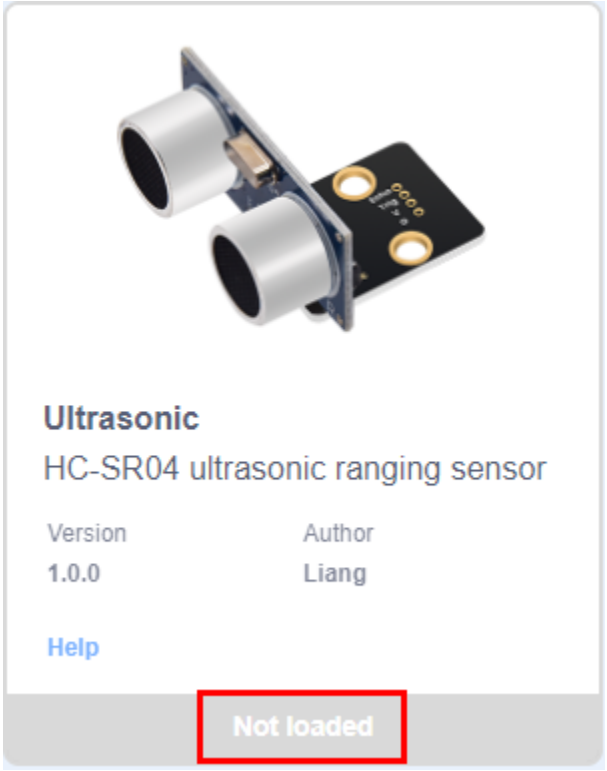
Before writing the code, it is necessary to import the library files of the ultrasonic sensor, 8x16 LED board and the servo. The specific steps are as follows:

Click

to enter the extension library interface of sensors/modules/components, then search

for“Ultrasonic”sensor and click it.

In this way, “**Not loaded**” changes to “**loaded**”, indicating that the“**Ultrasonic**”sensor was added successfully.

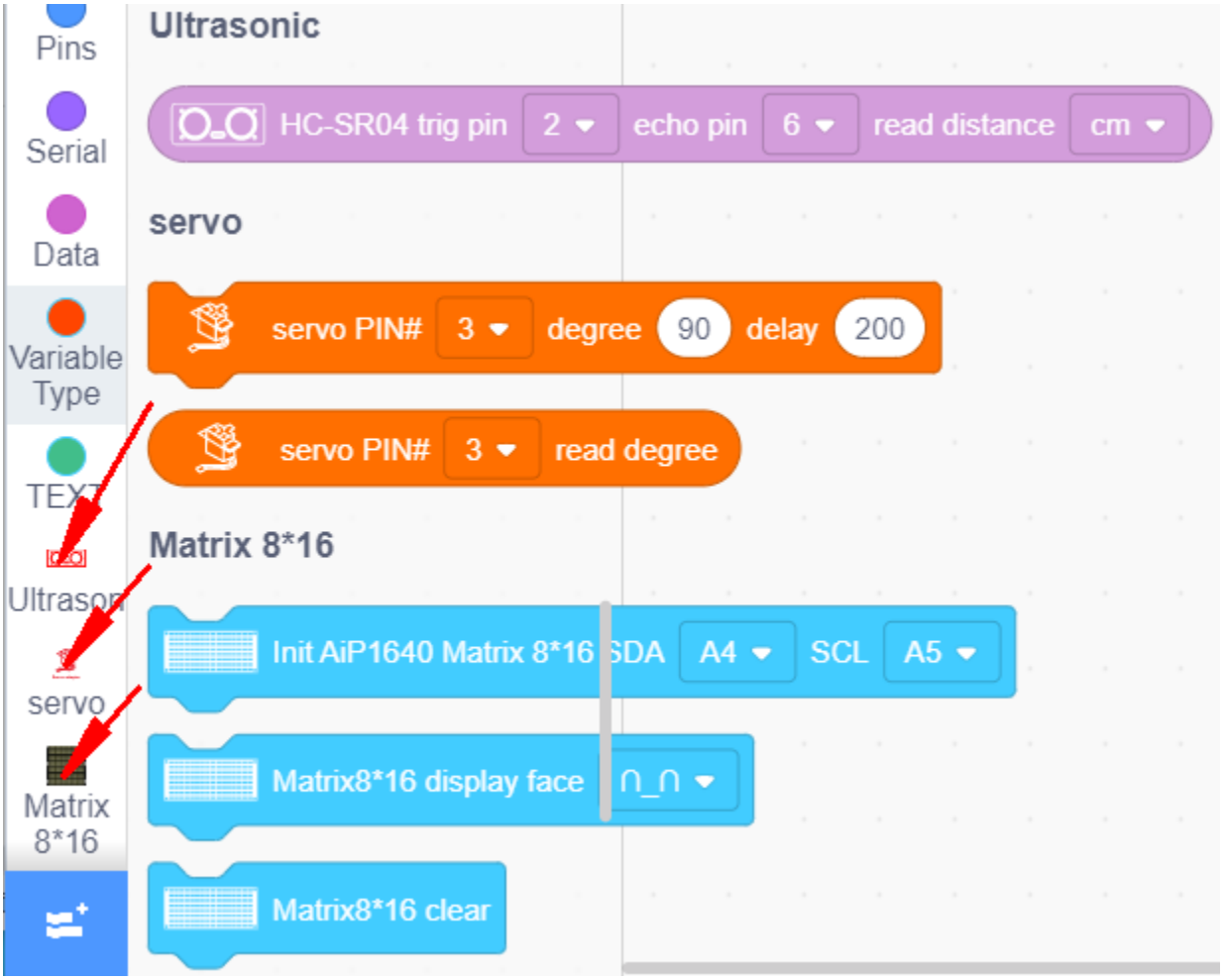


media/A284.png

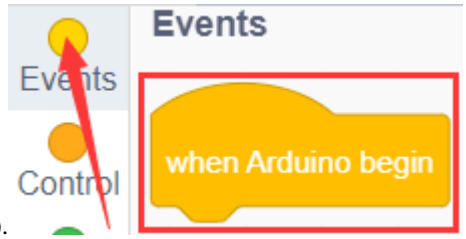
The 8x16 LED board and servo library files are added in the same way as the ultrasonic sensor.

Click

to return to the code editor interface, the instruction block of the added “**Ultrasonic**”sensor,“**Matrix 8*16 Aip1640**”module and “**Servo**”component can be seen in the module area.

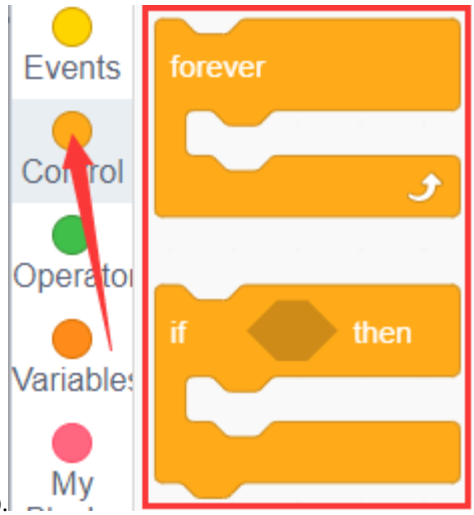


You can drag blocks to edit. Blocks listed below are for your reference

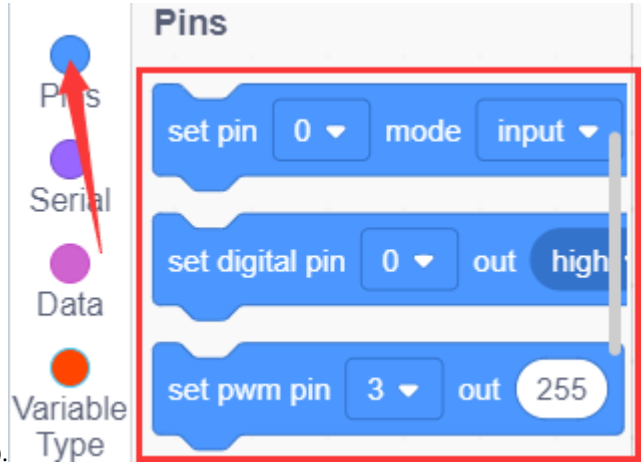


(1).

(2).


Events
Control
Operator
Variables
My
forever
if
then

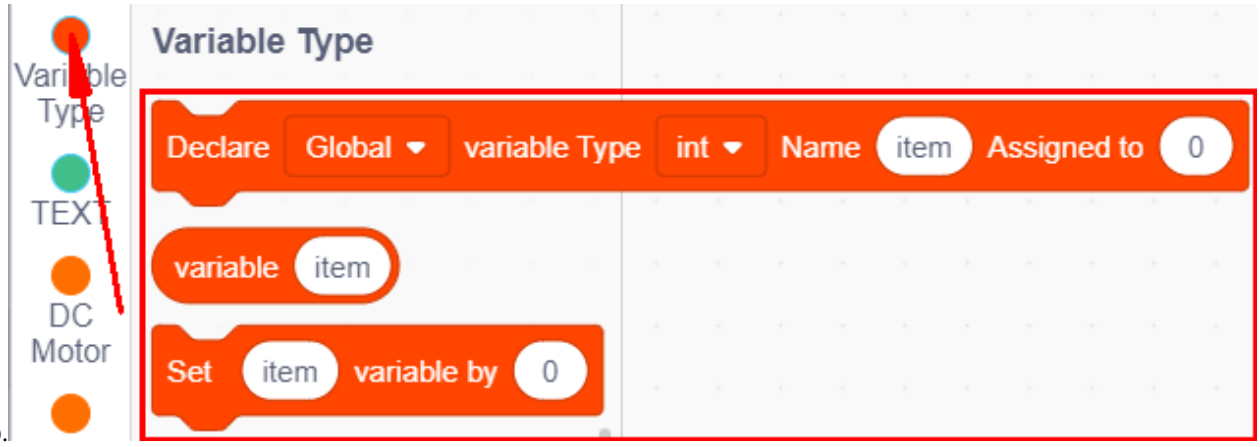
(3).


Pins
Serial
Data
Variable
Type
Pins
set pin 0 mode input
set digital pin 0 out high
set pwm pin 3 out 255

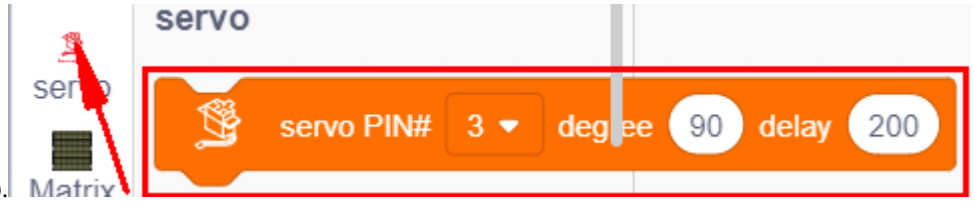
(4).

Serial
Data
Variable
Type
Serial
serial begin baudrate 9600
serial print Hello KidsBlock

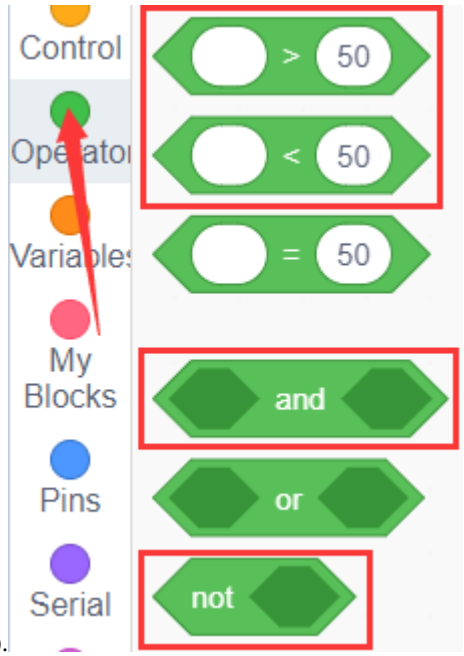
(5).


Variable Type
Declare Global variable Type int Name item Assigned to 0
variable item
Set item variable by 0
DC Motor

(6).


servo
servo PIN# 3 90 delay 200

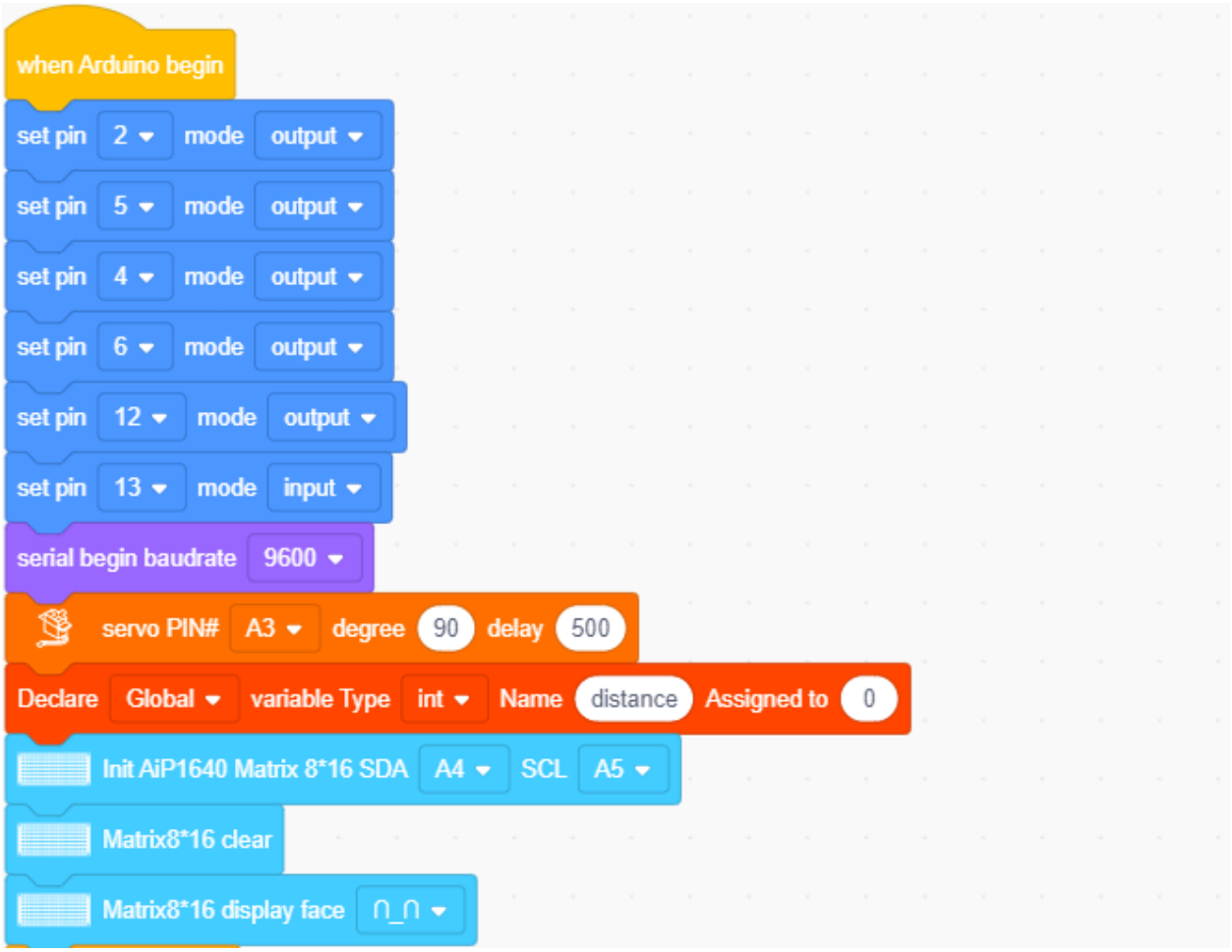
(7).

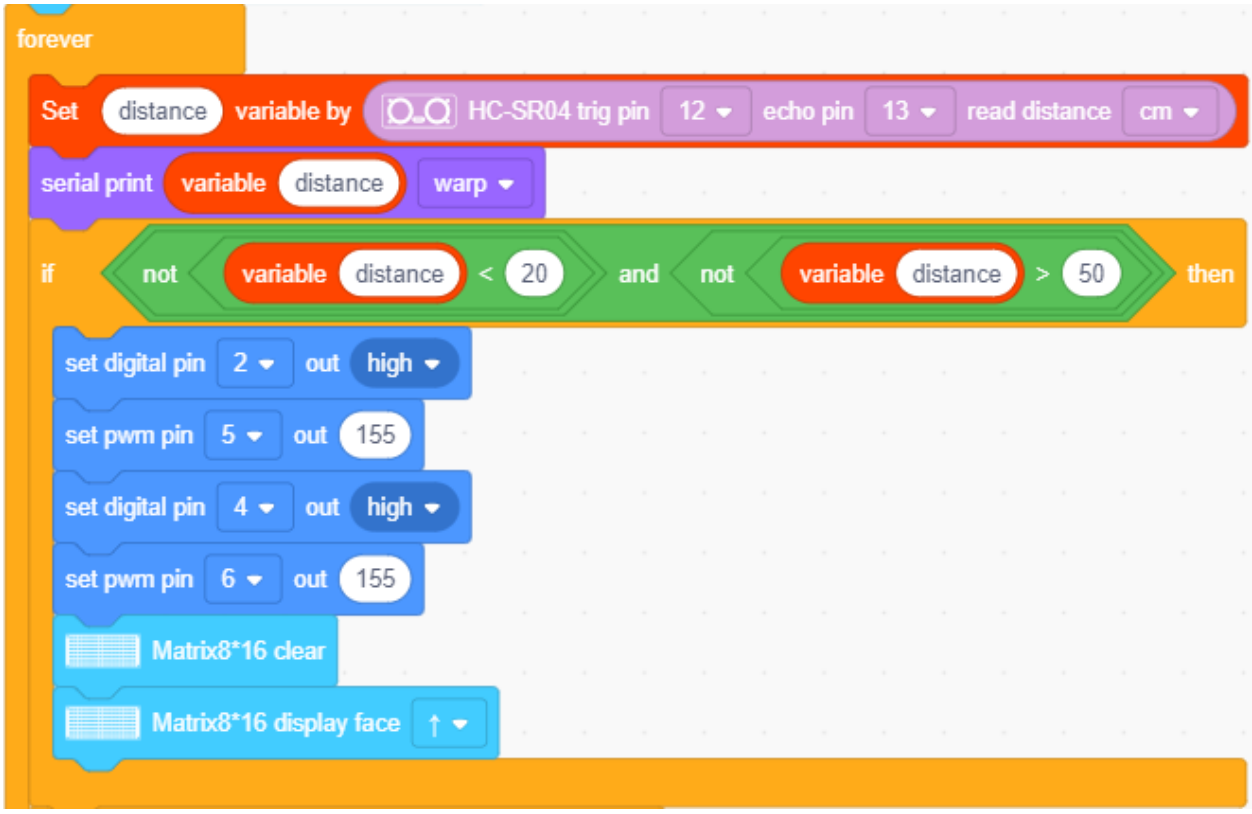

Control
> 50
< 50
= 50
My Blocks
and
Pins
or
Serial
not

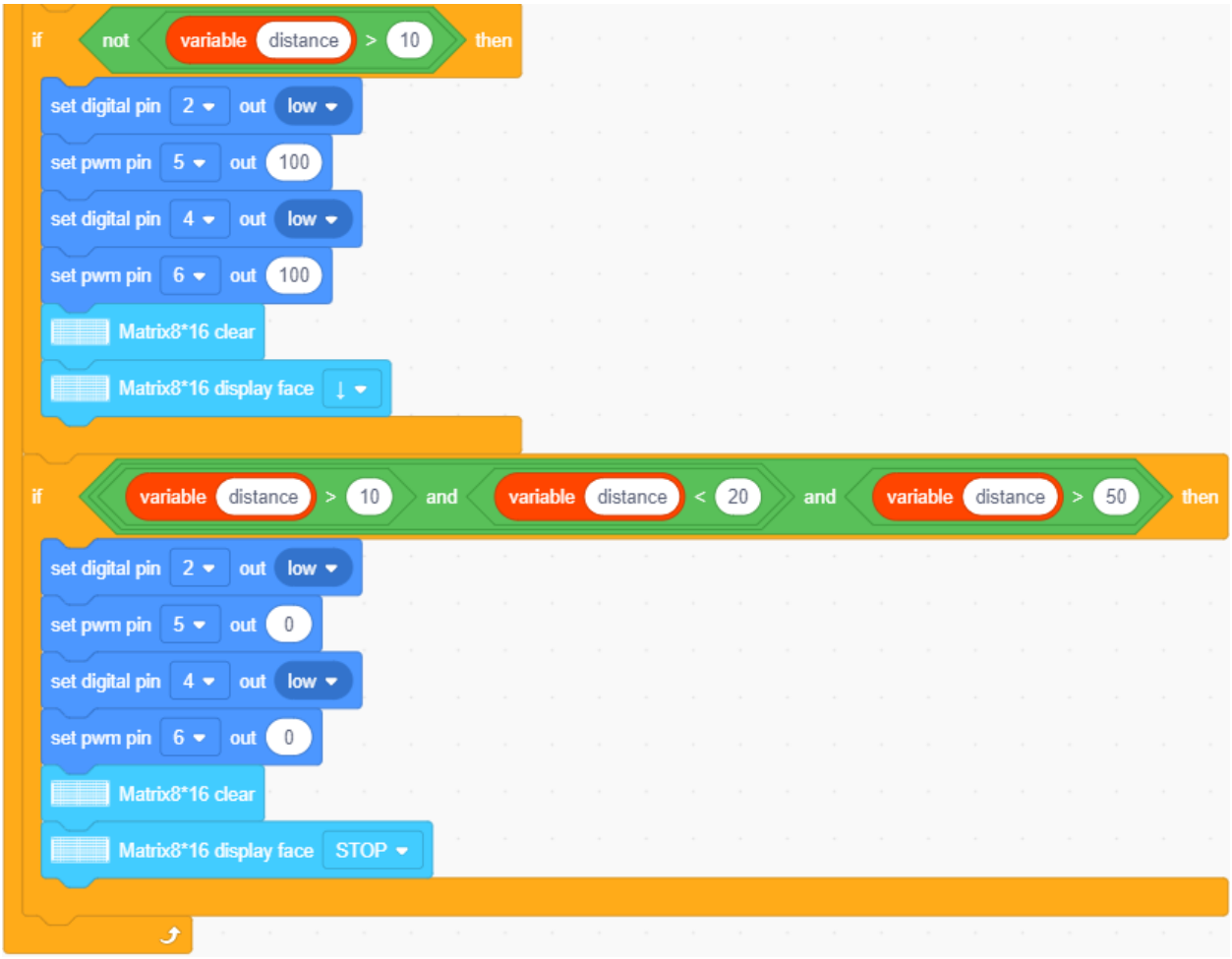
(8).

(9).

Complete Test Code




forever
Set distance variable by HC-SR04 trig pin 12 echo pin 13 read distance cm
serial print variable distance warp
if not variable distance < 20 and not variable distance > 50 then
set digital pin 2 out high
set pwm pin 5 out 155
set digital pin 4 out high
set pwm pin 6 out 155
Matrix8*16 clear
Matrix8*16 display face



5.Test Result

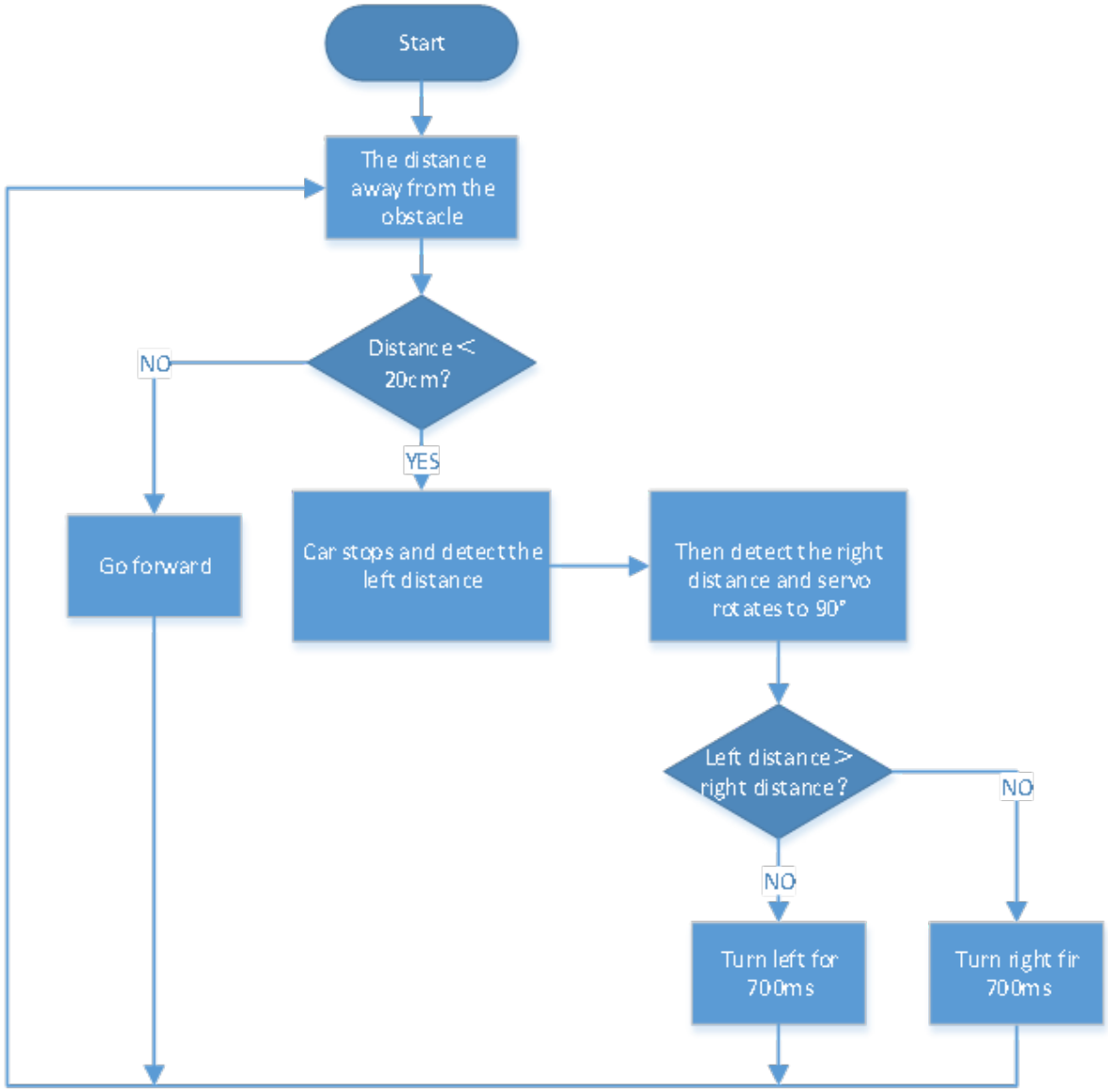
After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, power on the external power then turn the DIP switch to ON. Set the servo to 90°the smart car will move with the obstacles and the 8X16 LED board will show“smile”.

10.2.13 Project 13 Ultrasonic Obstacle Avoidance Smart Car

1.Description

In this project, we aim to make an ultrasonic obstacle avoidance smart car. We will use the ultrasonic to detect the distance from the obstacle, which can be used to control the servo to rotate so as to make the car move. Meanwhile, the 8X16 LED board will display the corresponding status pattern.

2.Flow Chart

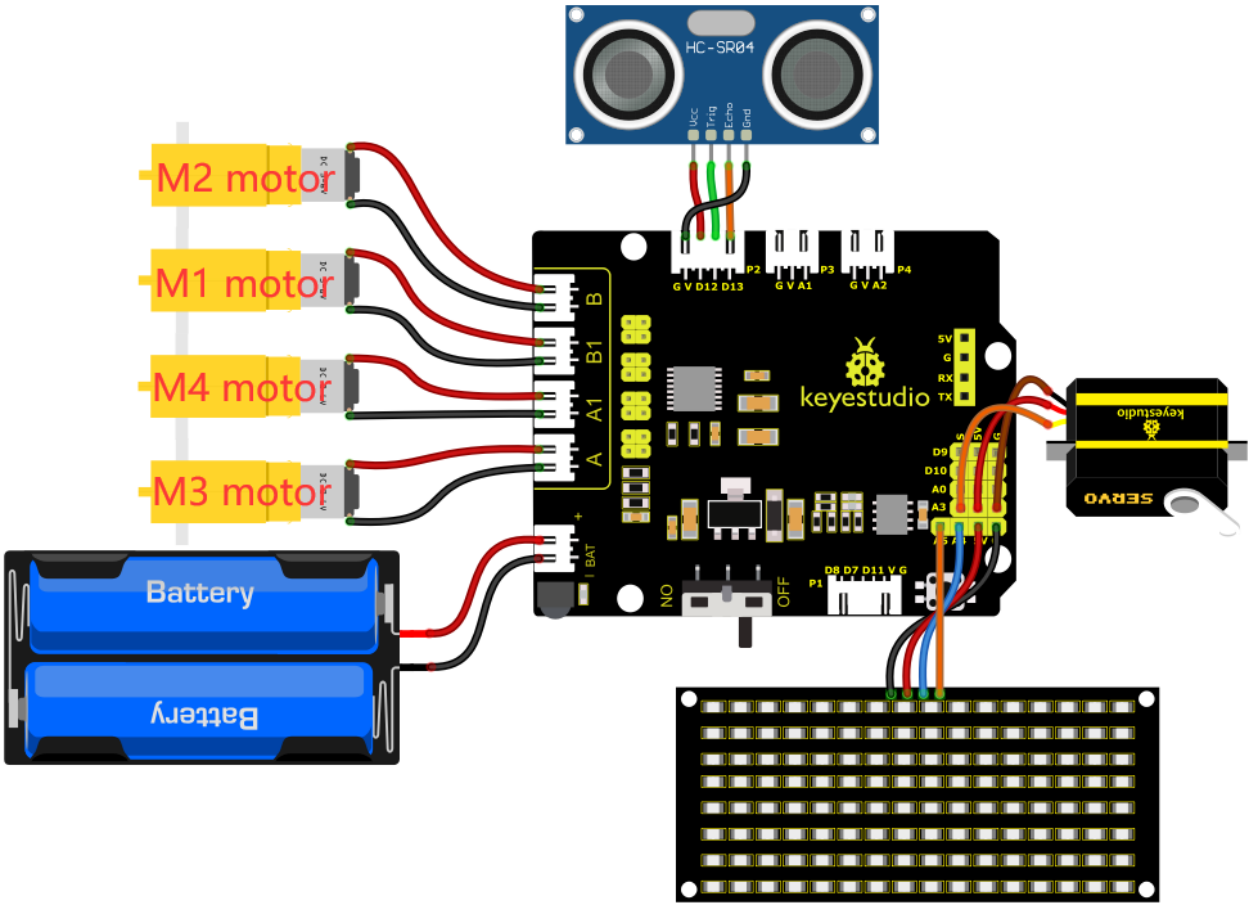


The specific logic of ultrasonic obstacle avoidance smart car is shown below:

<table>
<tr><td rowspan="3">Detection</td><td colspan="3">measured distance of front obstacle
set servo to 90°</td><td>distance (unit: cm)</td></tr>
<tr><td colspan="3">measured distance of left obstacle (set servo to 160°)</td><td>a1 (unit: cm)</td></tr>
<tr><td colspan="3">measured distance of right obstacle
(set servo to 20°)</td><td>a2 (unit: cm)</td></tr>
<tr><td>Setting</td><td colspan="4">set the initial angle of servo to 90°</td></tr>
<tr><td>Condition1</td><td colspan="4">Status</td></tr>
<tr><td rowspan="8">a < 20</td><td colspan="4">Stop for 1000ms; set the angle of servo to 160°, read a1, delay in 500ms;
set the angle of servo to 20°, read a2, delay in 500ms</td></tr>
<tr><td colspan="2">Condition 2</td><td colspan="2">Status</td></tr>
<tr><td colspan="2" rowspan="3">a1 < 50
or
a2 < 50</td><td colspan="2">Compare a1 and a2</td></tr>
<tr><td>Condition 3</td><td>Status</td></tr>
<tr><td>a1 > a2</td><td>Set the angle of servo to 90°, rotate to left for 700ms and go front</td></tr>
<tr><td></td><td>a1 < a2</td><td colspan="2">Set the angle of servo to 90°, rotate to right for 700ms, go front</td></tr>
<tr><td>Condition 2</td><td colspan="3">Status</td></tr>
<tr><td rowspan="2">a1 ≥ 50
and
a2 ≥ 50</td><td rowspan="2">Random</td><td colspan="2">Set the angle of servo to 90°, rotate to left for 700ms and go front</td></tr>
<tr><td colspan="2">Set the angle of servo to 90°, rotate to right for 700ms, go front</td></tr>
</table>

Condition1	Status
a ≥ 20	Go forward

3.Wiring Diagram



1). GND, VCC, SDA and SCL of the 8*8 LED board module are connected to GGND), VVCC), A4 and A5 of the expansion board.

2). VCC, Trig, Echo and Gnd of the ultrasonic sensor are connected to 5V(V), D12(S), D13(S) and Gnd(G).

3). The servo is connected to G, V and A3. The brown wire is interfaced with Gnd(G), the red wire is interfaced with 5V(V) and the orange wire is interfaced with A3.

4). The power is connected to the BAT port

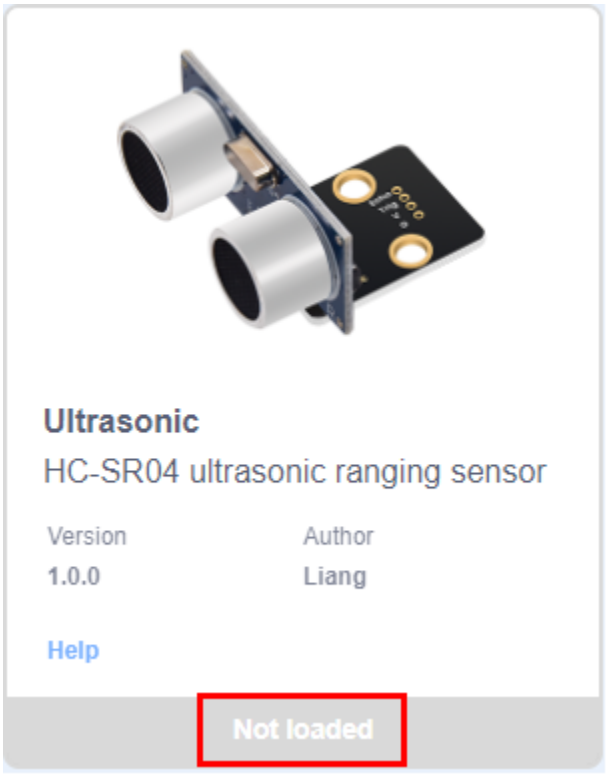
4.Test Code

Before writing the code, it is necessary to import the library files of the ultrasonic sensor , 8x16 LED board and the servo. The specific steps are as follows:

Click

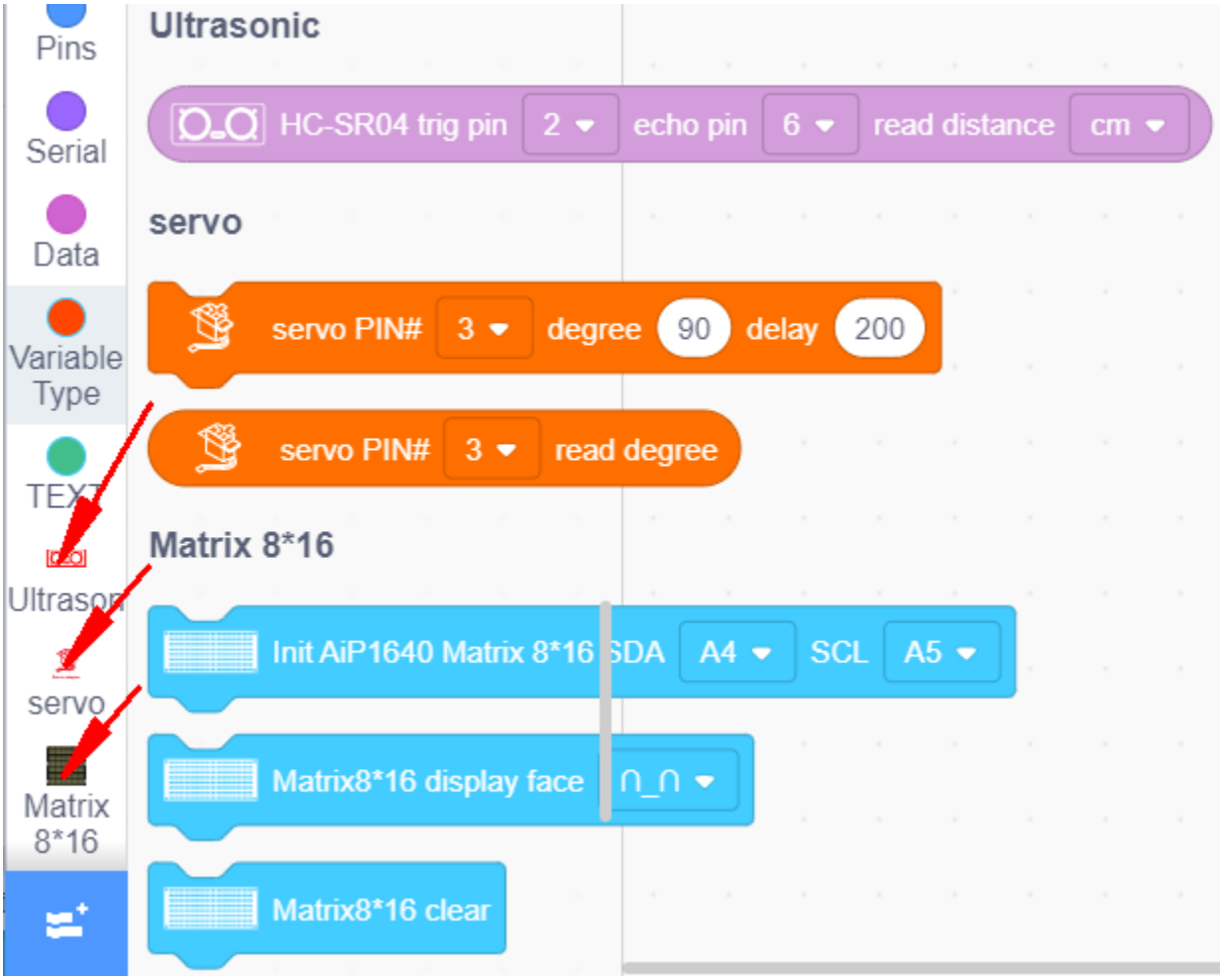
to enter the extension library interface of sensors/modules/components, then search for“Ultrasonic”sensor and click it. In this way, “**Not loaded**” changes to “**loaded**”, indicating that

the“**Ultrasonic**”sensor was added successfully.



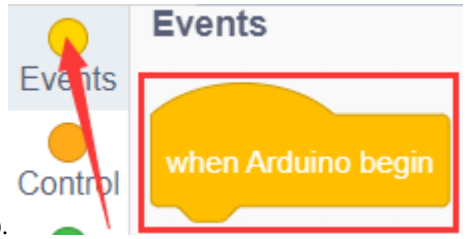
media/A284.png

Click Back to return to the code editor interface, the instruction block of the added “**Ultrasonic”sensor**,“**Matrix 8*16 Aip1640**”module and “**Servo**”component can be seen in the module area.

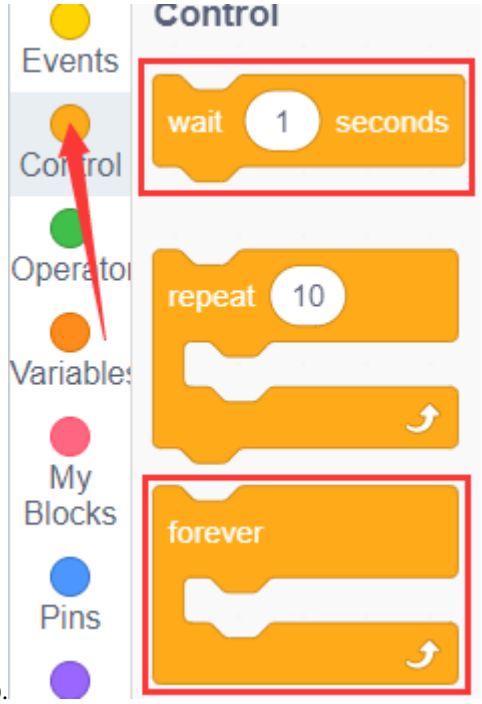


You can drag blocks to edit. Blocks listed below are for your reference.

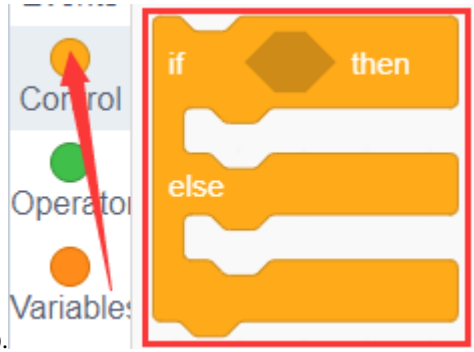
(1).



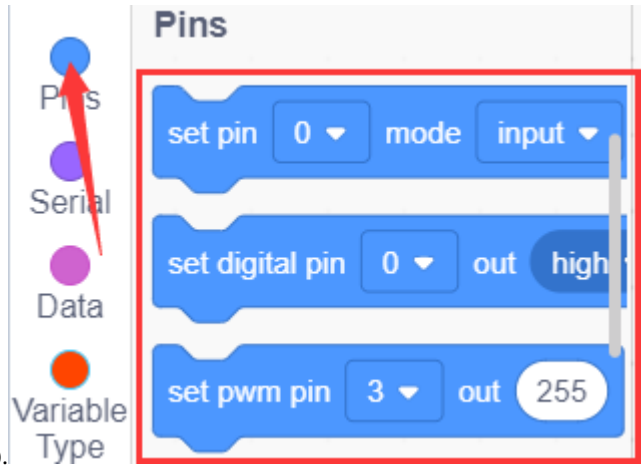
(2).


Control
Events
Control
Operator
Variables
My Blocks
Pins
wait 1 seconds
repeat 10
forever

(3).


Control
Operator
Variables
if then
else

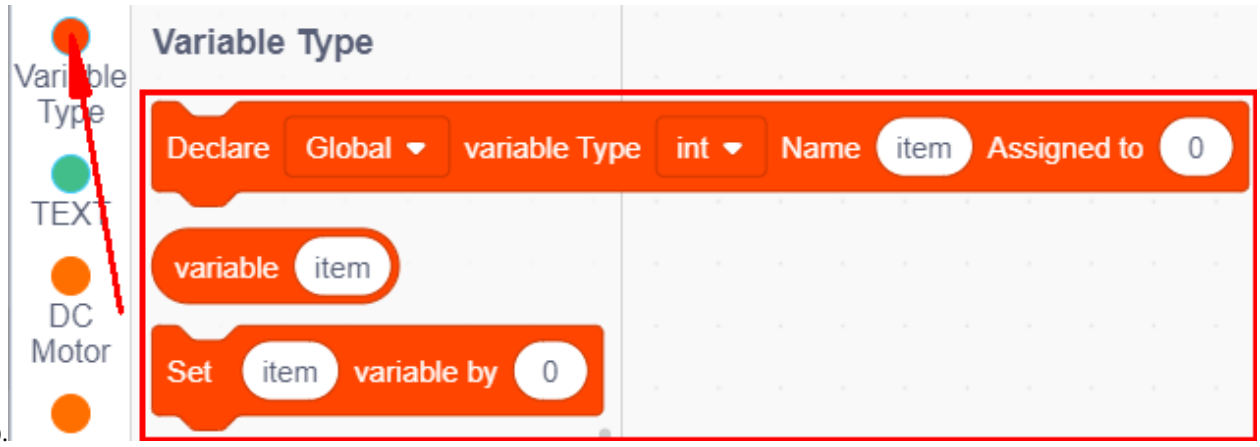
(4).


Pins
Pins
Serial
Data
Variable Type
set pin 0 mode input
set digital pin 0 out high
set pwm pin 3 out 255

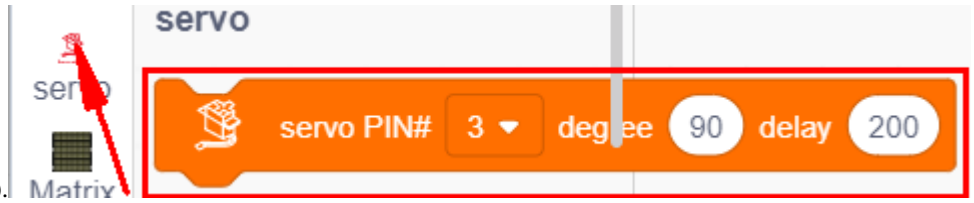
(5).

Serial
Serial
Data
Variable Type
serial begin baudrate 9600
serial print Hello KidsBlock

(6).


Variable Type
Variable Type
TEXT
DC Motor
Declare Global variable Type int Name item Assigned to 0
variable item
Set item variable by 0

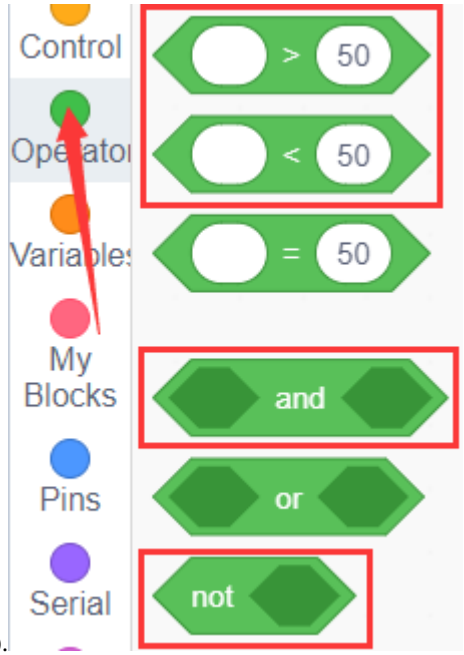
(7).


servo
servo
Matrix
servo PIN# 3 degree 90 delay 200

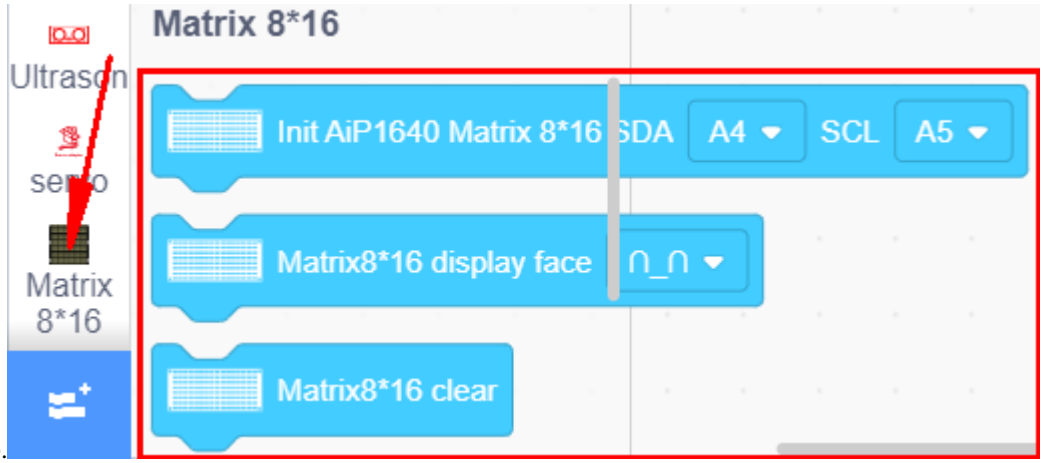
(8).

Ultrasonic
Ultrasonic
HC-SR04 trig pin 2 echo pin 6 read distance cm

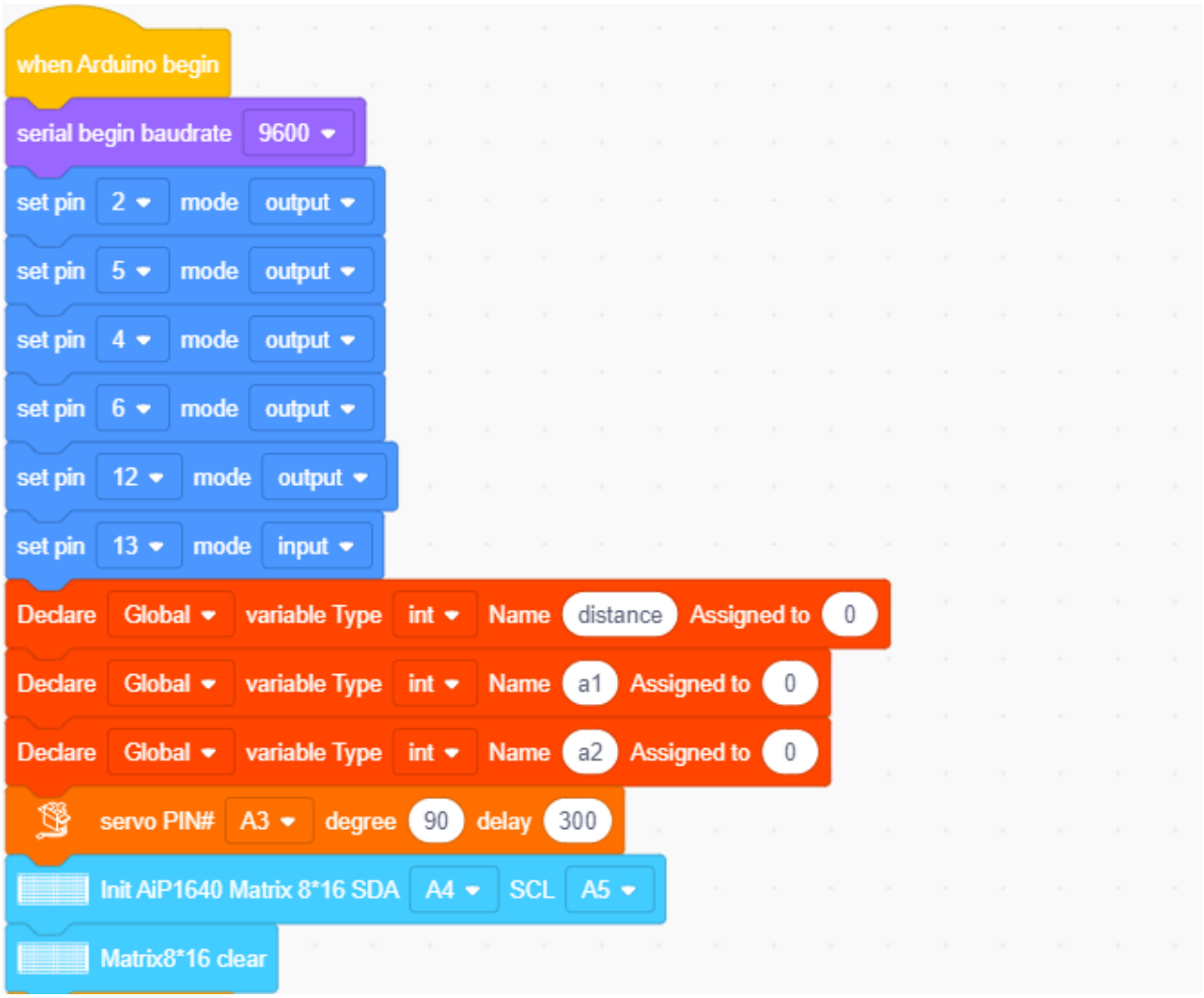
(9).

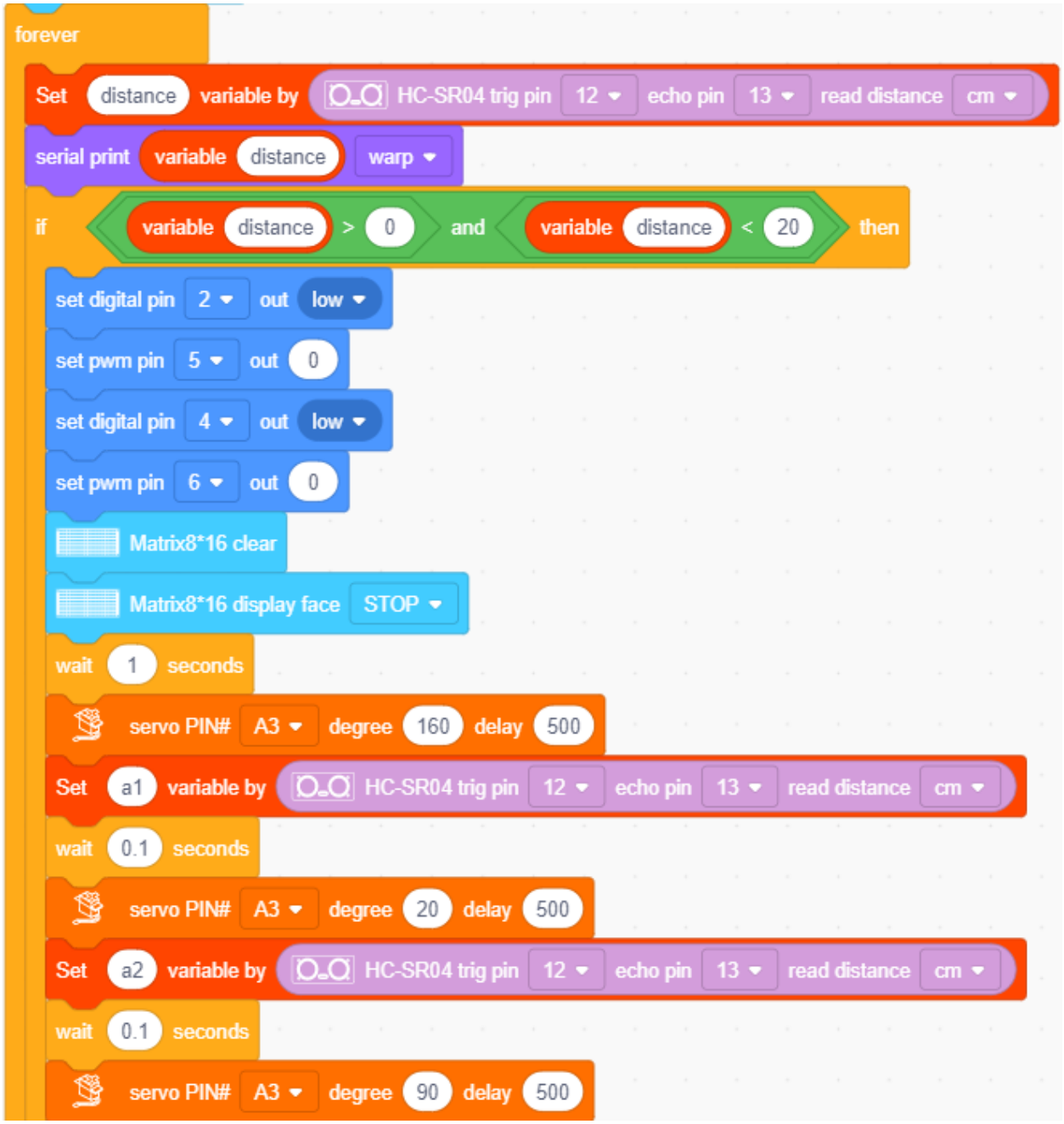


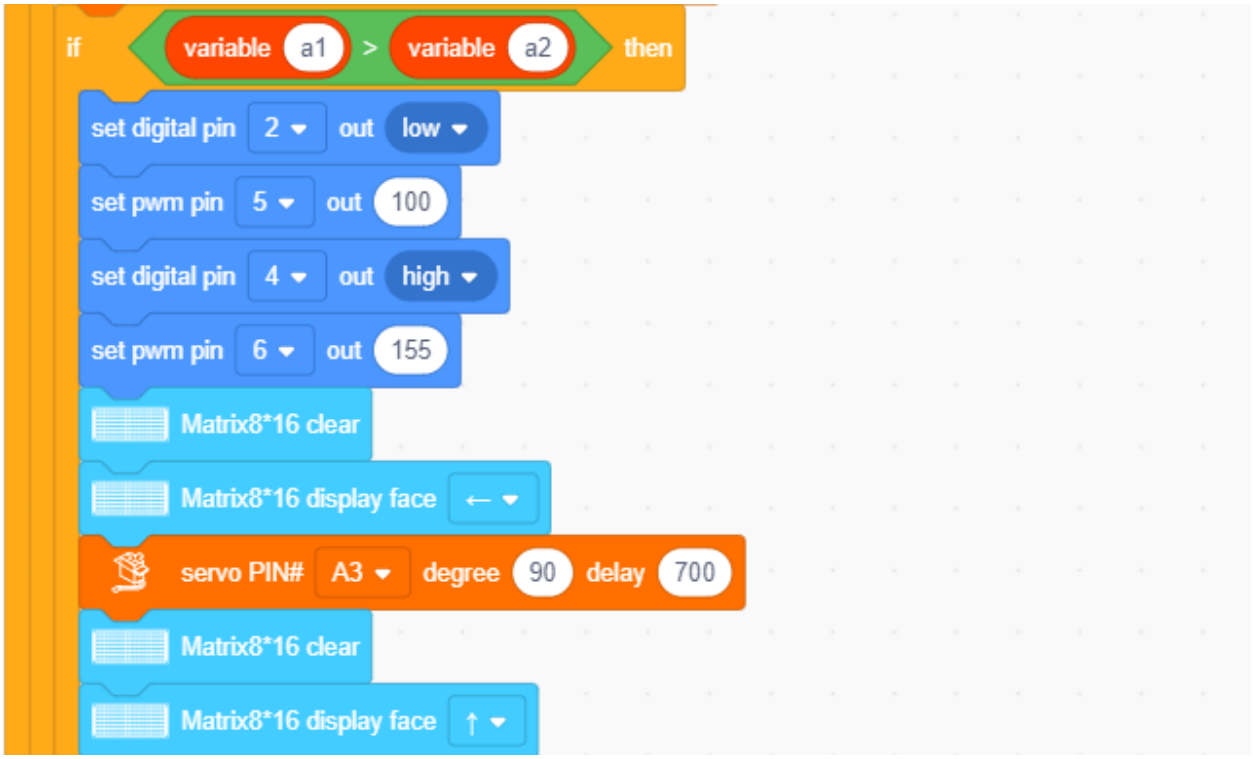
(10).

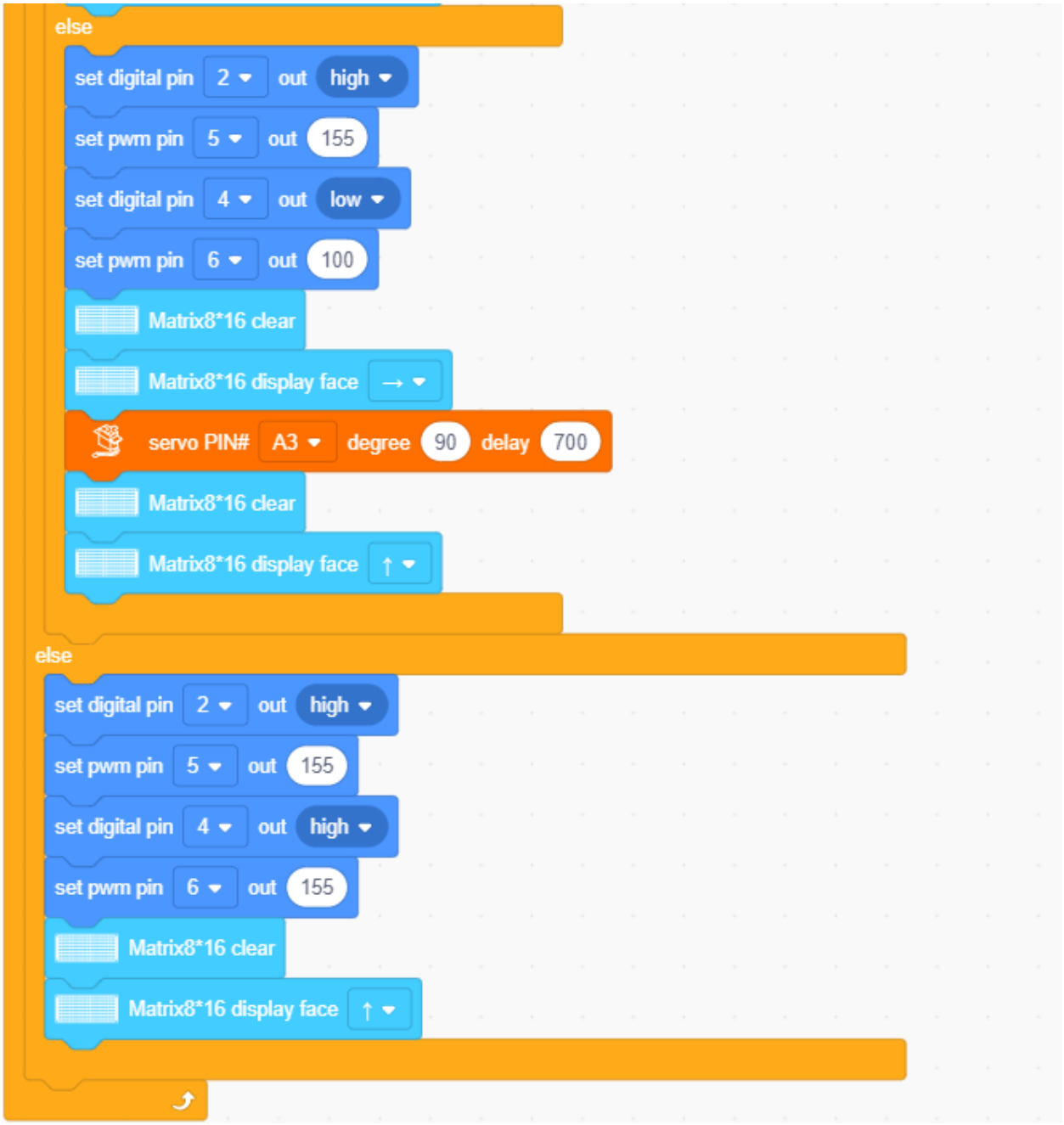


Complete Test Code


when Arduino begin
serial begin baudrate 9600
set pin 2 mode output
set pin 5 mode output
set pin 4 mode output
set pin 6 mode output
set pin 12 mode output
set pin 13 mode input
Declare Global variable Type int Name distance Assigned to 0
Declare Global variable Type int Name a1 Assigned to 0
Declare Global variable Type int Name a2 Assigned to 0
servo PIN# A3 degree 90 delay 300
Init AiP1640 Matrix 8*16 SDA A4 SCL A5
Matrix8*16 clear


forever
Set distance variable by HC-SR04 trig pin 12 echo pin 13 read distance cm
serial print variable distance warp
if variable distance > 0 and variable distance < 20 then
set digital pin 2 out low
set pwm pin 5 out 0
set digital pin 4 out low
set pwm pin 6 out 0
Matrix8*16 clear
Matrix8*16 display face STOP
wait 1 seconds
servo PIN# A3 degree 160 delay 500
Set a1 variable by HC-SR04 trig pin 12 echo pin 13 read distance cm
wait 0.1 seconds
servo PIN# A3 degree 20 delay 500
Set a2 variable by HC-SR04 trig pin 12 echo pin 13 read distance cm
wait 0.1 seconds
servo PIN# A3 degree 90 delay 500


if
variable a1 > variable a2
then
set digital pin 2 out low
set pwm pin 5 out 100
set digital pin 4 out high
set pwm pin 6 out 155
Matrix8*16 clear
Matrix8*16 display face ←
servo PIN# A3 degree 90 delay 700
Matrix8*16 clear
Matrix8*16 display face ↑

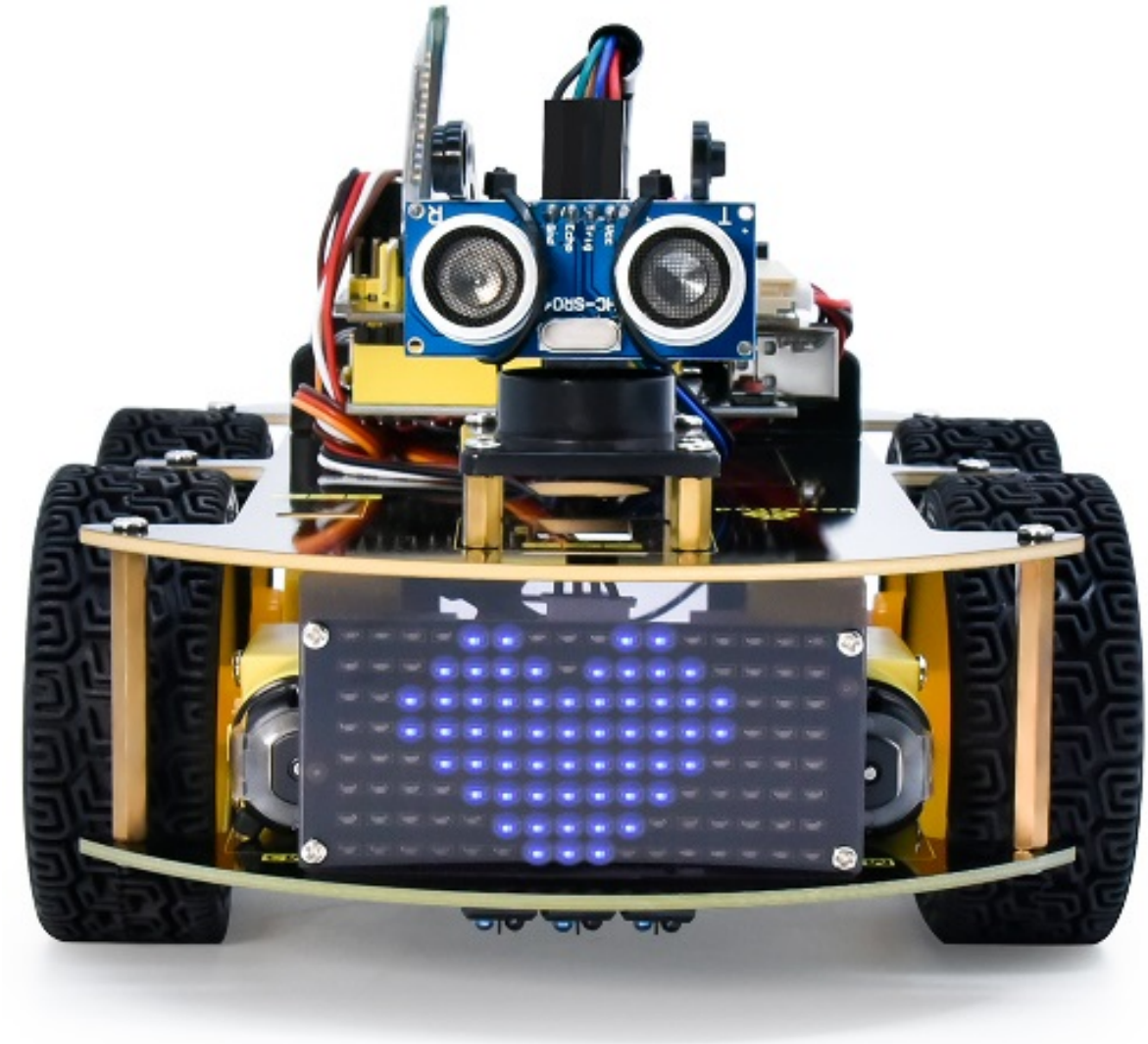


5.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, power on the external power then turn the DIP switch to ON.

The smart car moves forward and automatically avoids obstacles. When there is no road ahead, the servo will drive the ultrasonic sensor to scan the left, middle and right distances, and the car will turn to the open side. Meanwhile, the 8X16 LED board will display the corresponding status pattern.

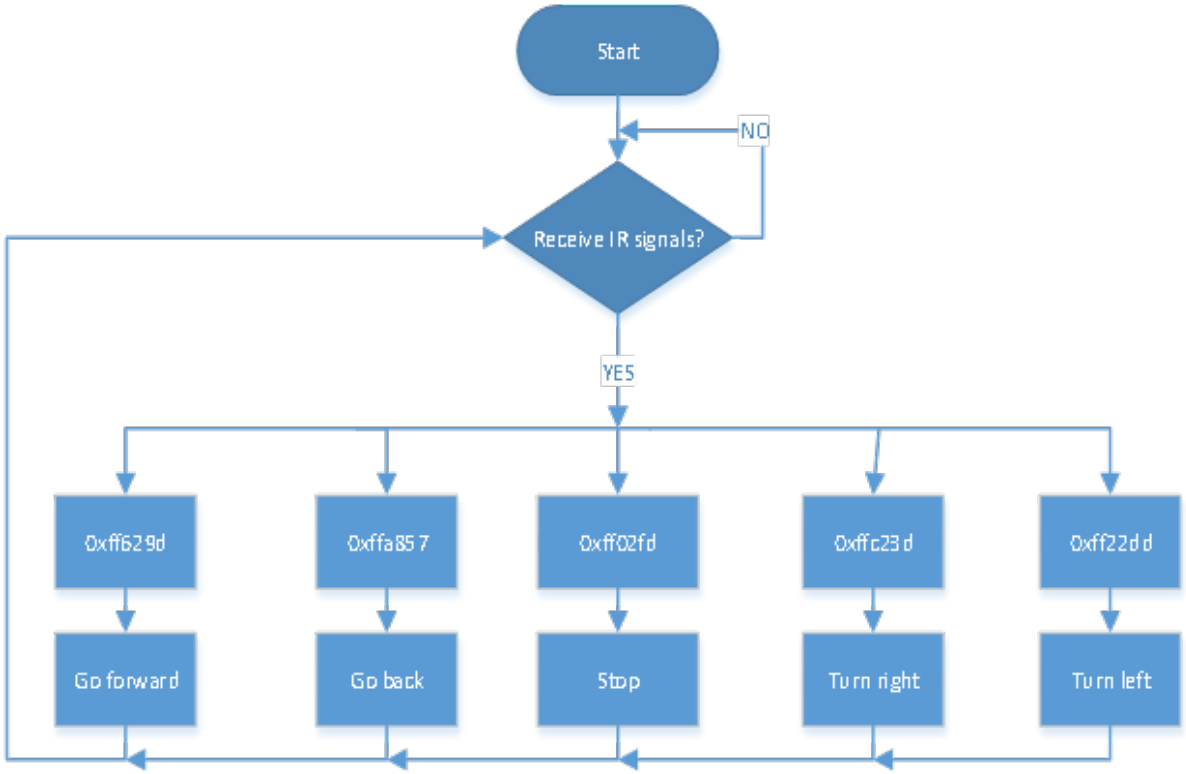
10.2.14 Project 14 IR Remote Control Smart Car



1.Description

In this project, we will make an IR remote control smart car and press the button on the IR remote control to drive the car to move.

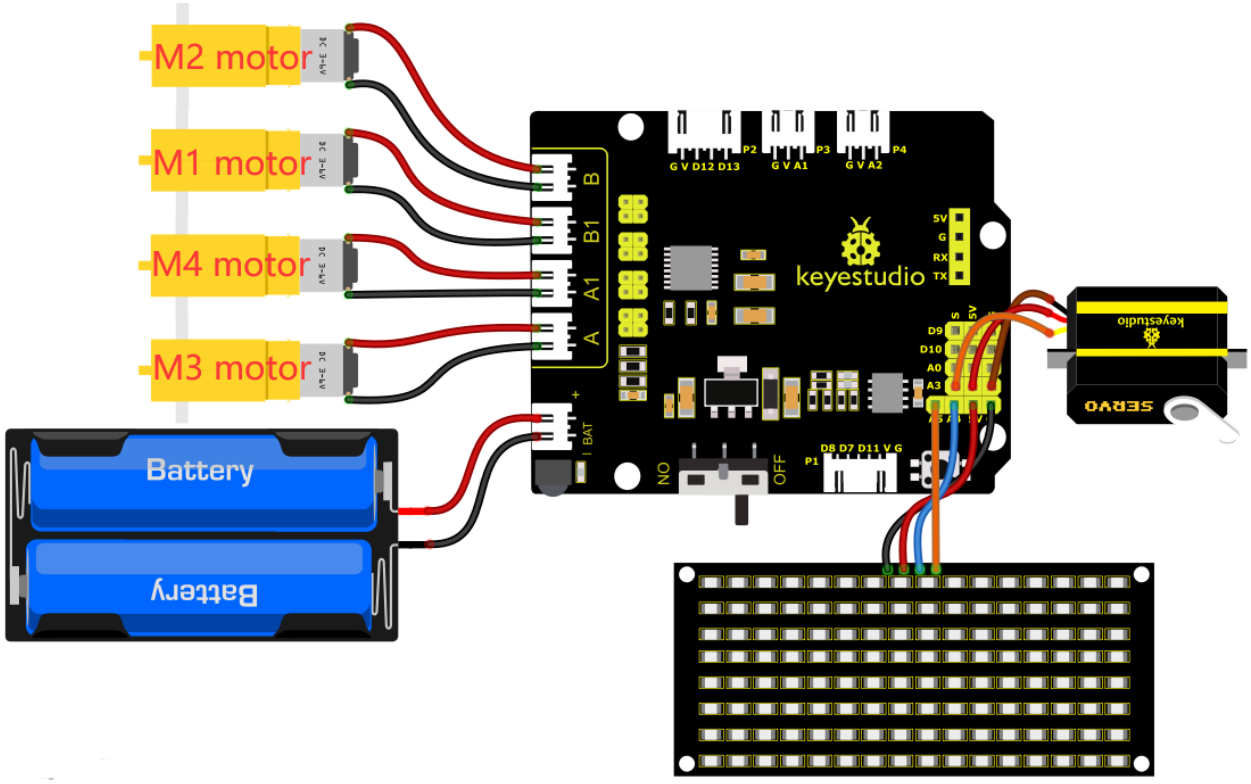
2.Flow Chart



The specific logic of IR remote control smart car is shown below:

Initial setup		LED board displays smile face
Remote control	Key value	Key state
	FF629D	Go front8*8 LED board shows front icon
	FFA857	Back8*8 LED board shows back icon
	FF22DD	Rotate to left8*8 LED board shows leftward icon
	FFC23D	Rotate to right8*8 LED board shows rightward icon
	FF02FD	Stop8*8 LED board shows“STOP”

3.Wiring Diagram



1). GND, VCC, SDA and SCL of the 8*8 LED board module are connected to GGND), VVCC), A4 and A5 of the expansion board.

2). As the IR receiver is integrated on the 8833 motor driver expansion board, there is no need for additional wiring. The pins of the IR receiver on the 8833 board are G (GND), V (VCC) and D3 respectively.

3). The servo is connected to G, V and A3. The brown wire is interfaced with Gnd(G), the red wire is interfaced with 5V(V) and the orange wire is interfaced with A3.

4). The power is connected to the BAT port

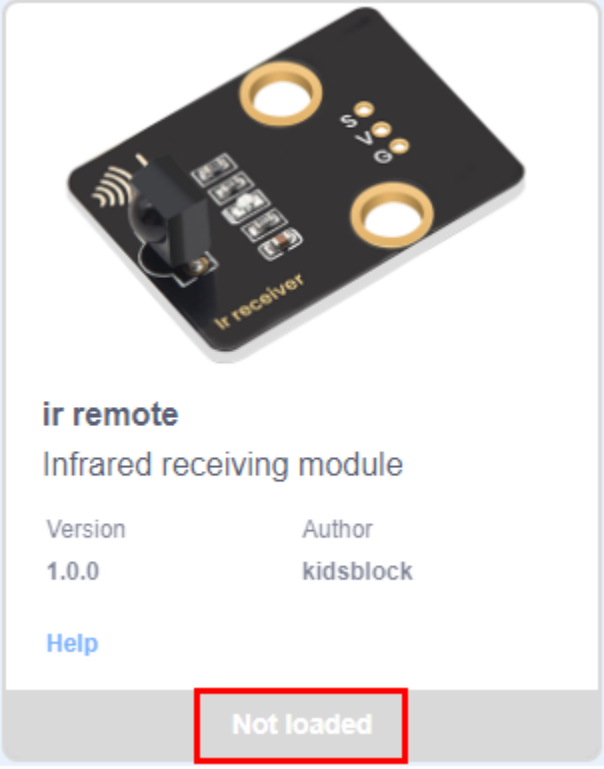
4.Test Code

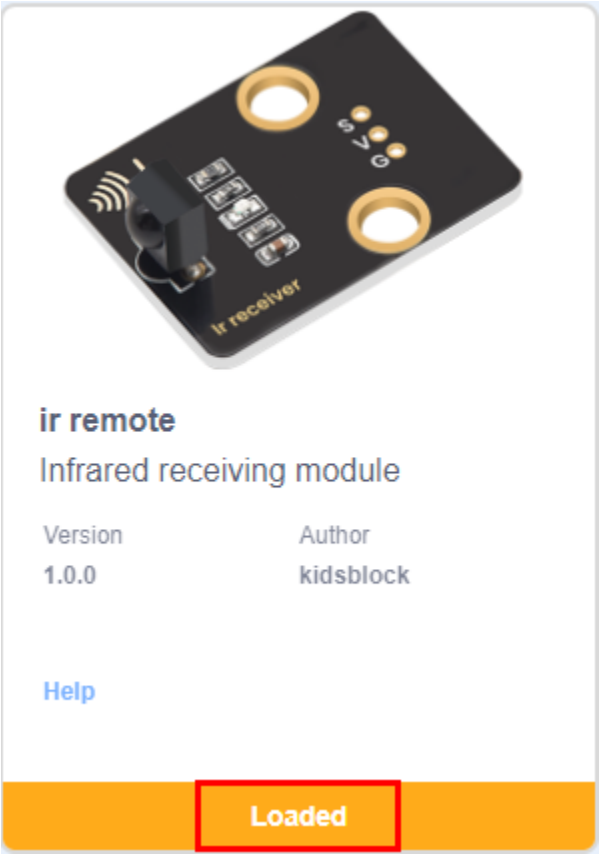
Please note: The infrared module shown in the software demonstration is already integrated into the expansion board and is not supplied separately. Consequently, you will not find the module depicted in the image below within the product.

Before writing the code, it is necessary to import the library files of the ultrasonic sensor, 8x16 LED board and the servo. The specific steps are as follows:

Click

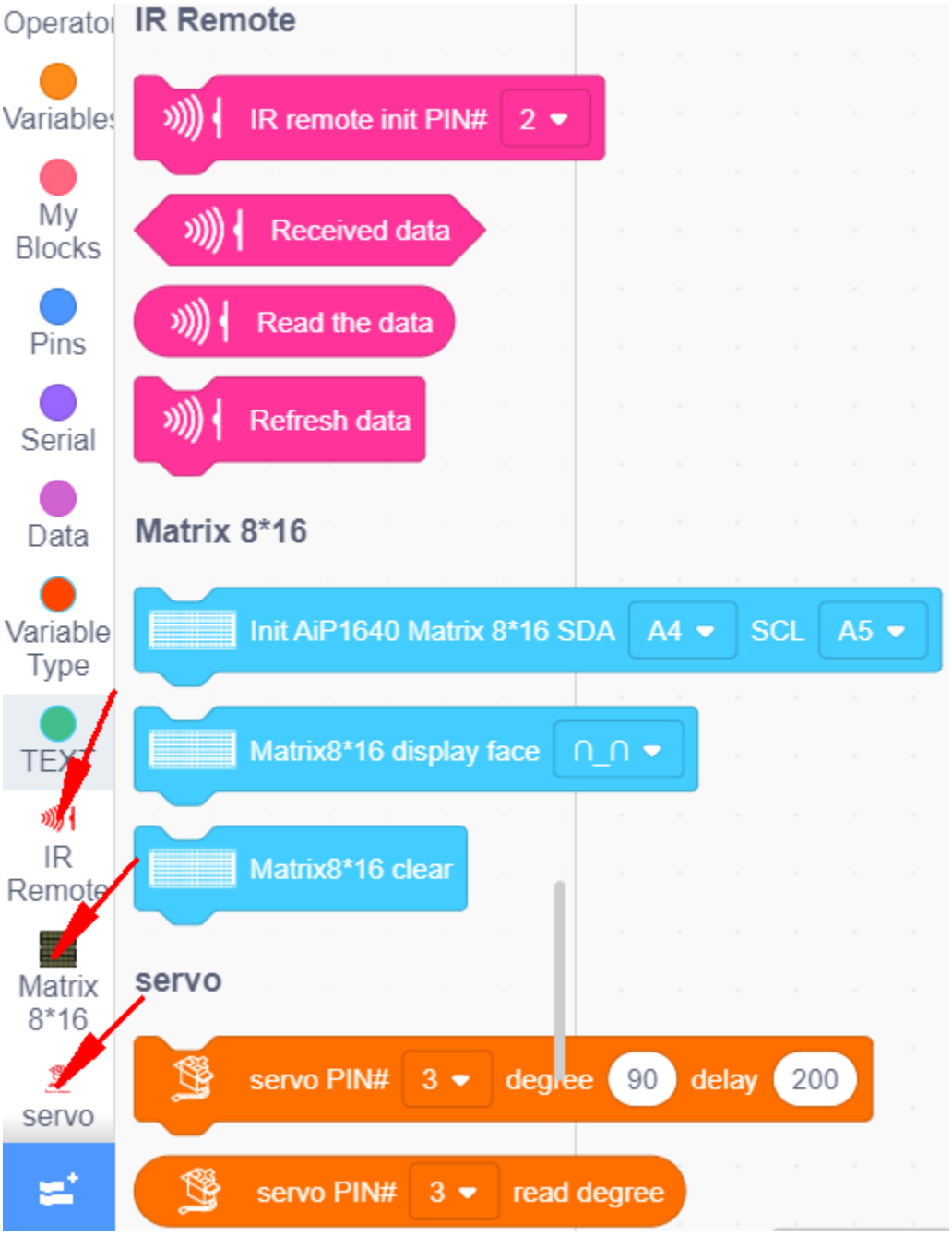
to enter the extension library interface of sensors/modules/components, then search for“ir remote”sensor and click it. In this way, “**Not loaded**” changes to “**loaded**”, indicating that the“**ir remote**”sensor was added successfully.



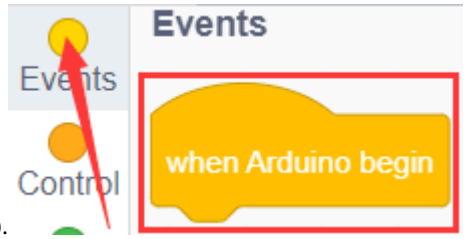


Click Back to return to the code editor interface, the instruction block of the added "**ir remote**"sensor,"**Matrix**

8*16 Aip1640"module and "**Servo**"component can be seen in the module area.

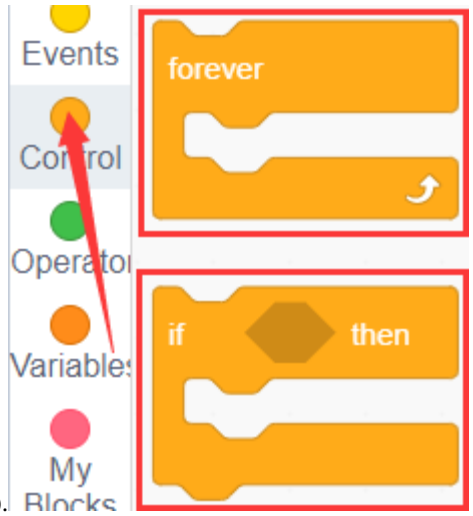


You can drag blocks to edit. Blocks listed below are for your reference

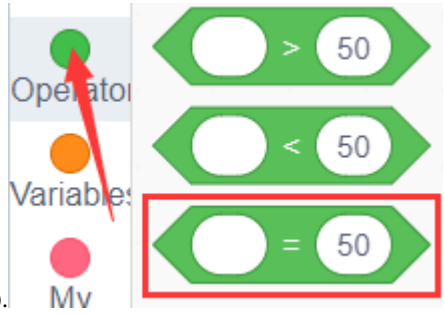


(1).

(2).


Events
Control
Operator
Variables
My Blocks
forever
if
then

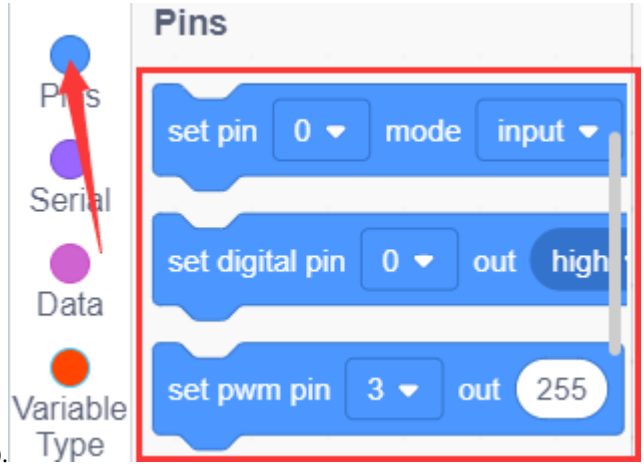
(3).


Operator
Variables
My
> 50
< 50
= 50

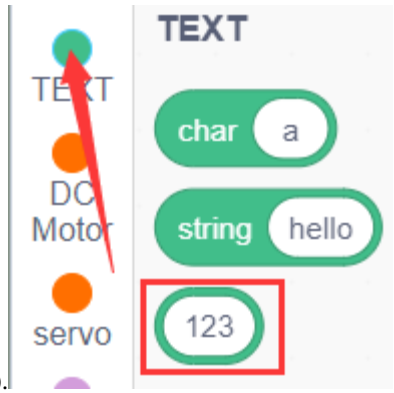
(4).

Serial
Data
Variable Type
Serial
serial begin baudrate 9600
serial print Hello KidsBlock

(5).

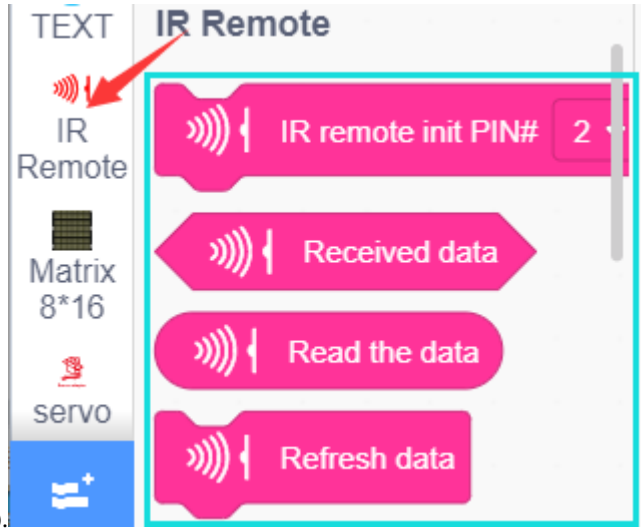

Pins
Serial
Data
Variable Type
Pins
set pin 0 mode input
set digital pin 0 out high
set pwm pin 3 out 255

(6).



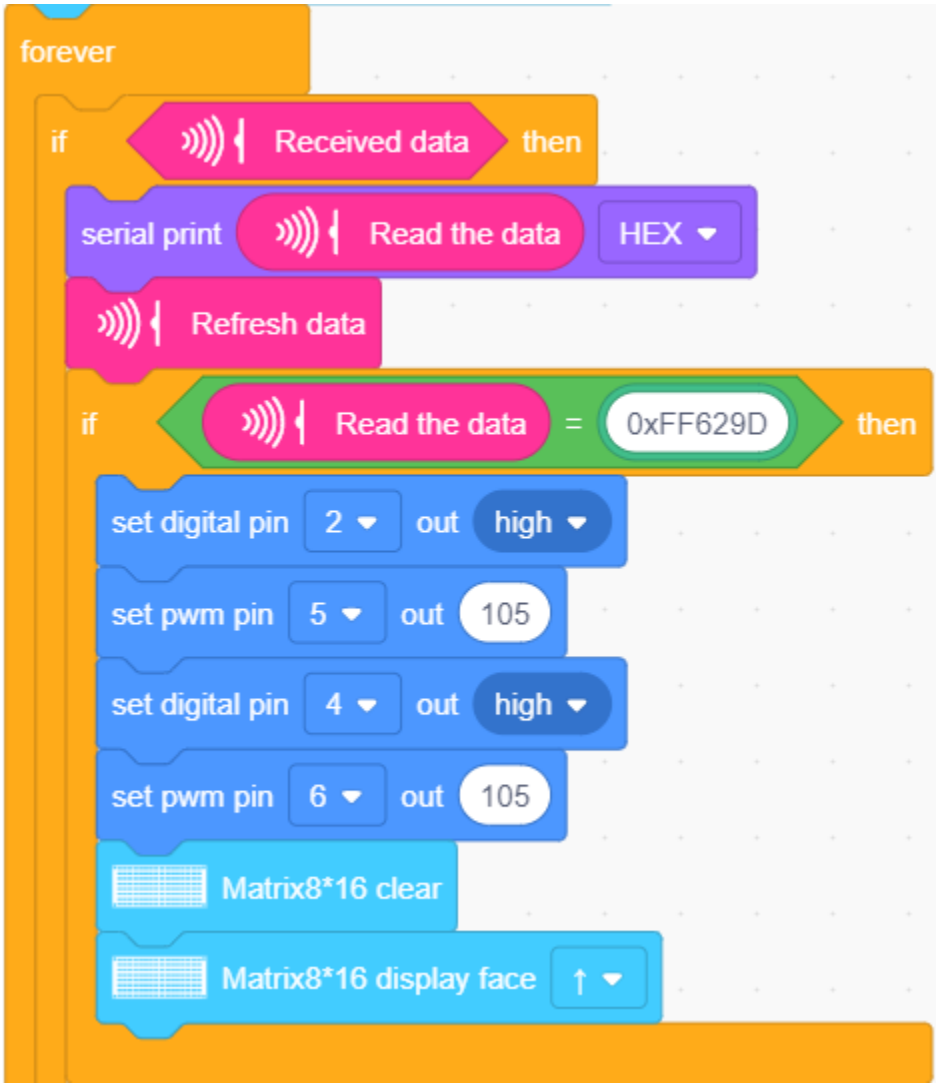
(7).

(8).

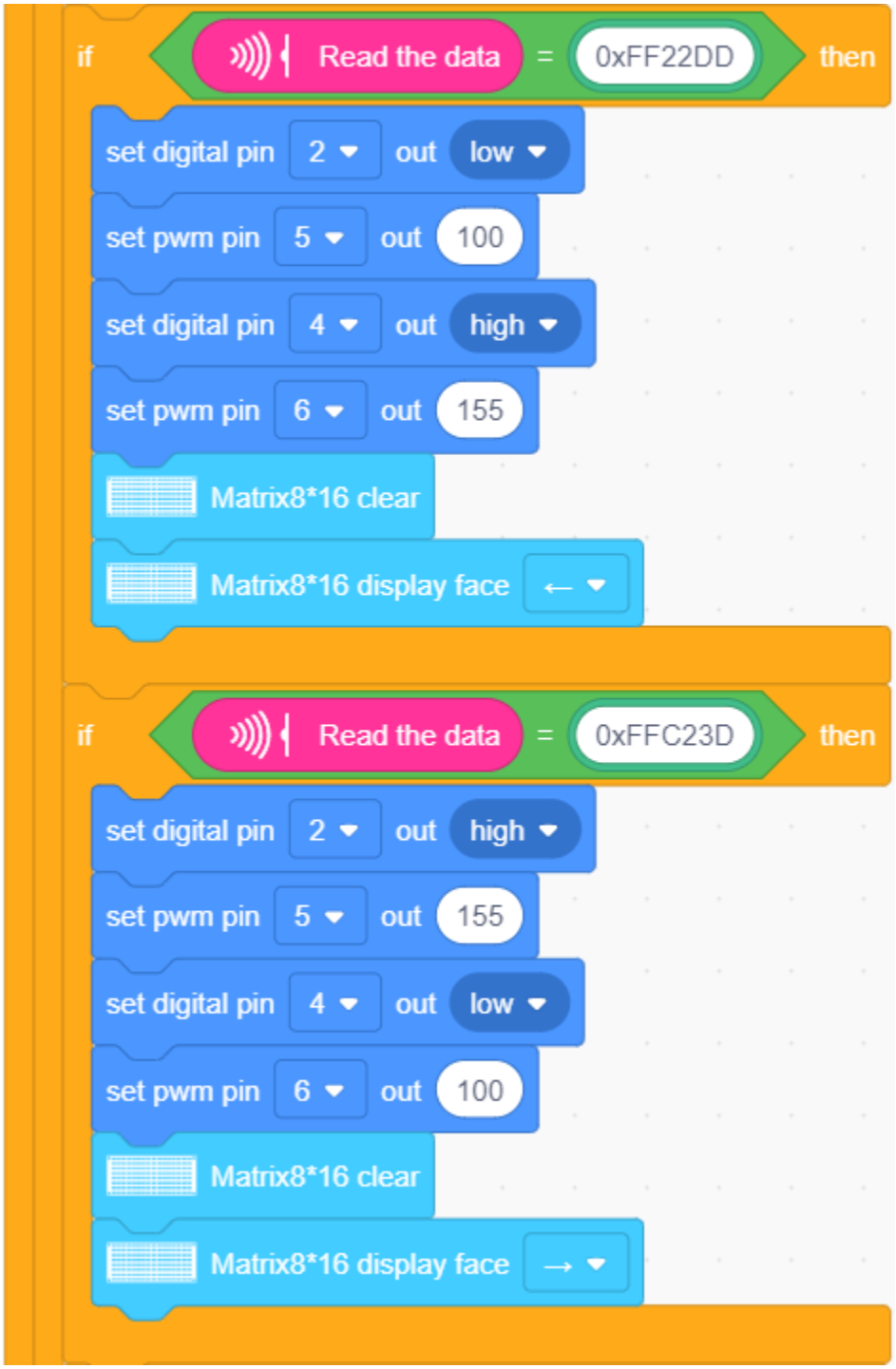


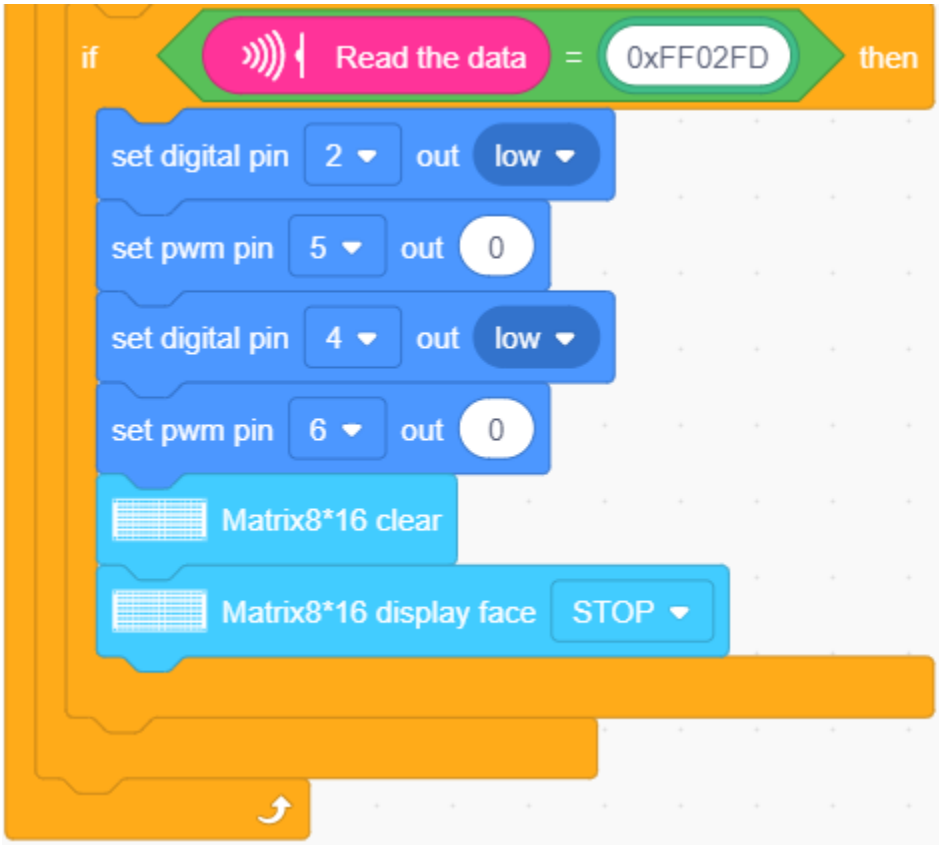
Complete Test Code

when Arduino begin
serial begin baudrate 9600
IR remote init PIN# 3
set pin 2 mode output
set pin 5 mode output
set pin 4 mode output
set pin 6 mode output
servo PIN# A3 degree 90 delay 300
Init AiP1640 Matrix 8*16 SDA A4 SCL A5
Matrix8*16 clear
Matrix8*16 display face ∩_∩


forever
if
Received data
then
serial print
Read the data
HEX
Refresh data
if
Read the data
=
0xFF629D
then
set digital pin
2
out
high
set pwm pin
5
out
105
set digital pin
4
out
high
set pwm pin
6
out
105
Matrix8*16 clear
Matrix8*16 display face

if
Read the data
=
0xFFA857
then
set digital pin 2 out low
set pwm pin 5 out 150
set digital pin 4 out low
set pwm pin 6 out 150
Matrix8*16 clear
Matrix8*16 display face
if
Read the data
=
0xFF22DD
then
set digital pin 2 out low
set pwm pin 5 out 100
set digital pin 4 out high
set pwm pin 6 out 155
Matrix8*16 clear
Matrix8*16 display face


if
Read the data
=
0xFF22DD
then
set digital pin
2
out
low
set pwm pin
5
out
100
set digital pin
4
out
high
set pwm pin
6
out
155
Matrix8*16 clear
Matrix8*16 display face
←
if
Read the data
=
0xFFC23D
then
set digital pin
2
out
high
set pwm pin
5
out
155
set digital pin
4
out
low
set pwm pin
6
out
100
Matrix8*16 clear
Matrix8*16 display face
→



5.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, power on the external power then turn the DIP switch to ON. Then we enable to use the IR remote control drive the car to move to and the 8X16 LED board will display the corresponding status pattern.

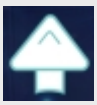
10.2.15 Project 15 Bluetooth Control Smart Car

1.Description

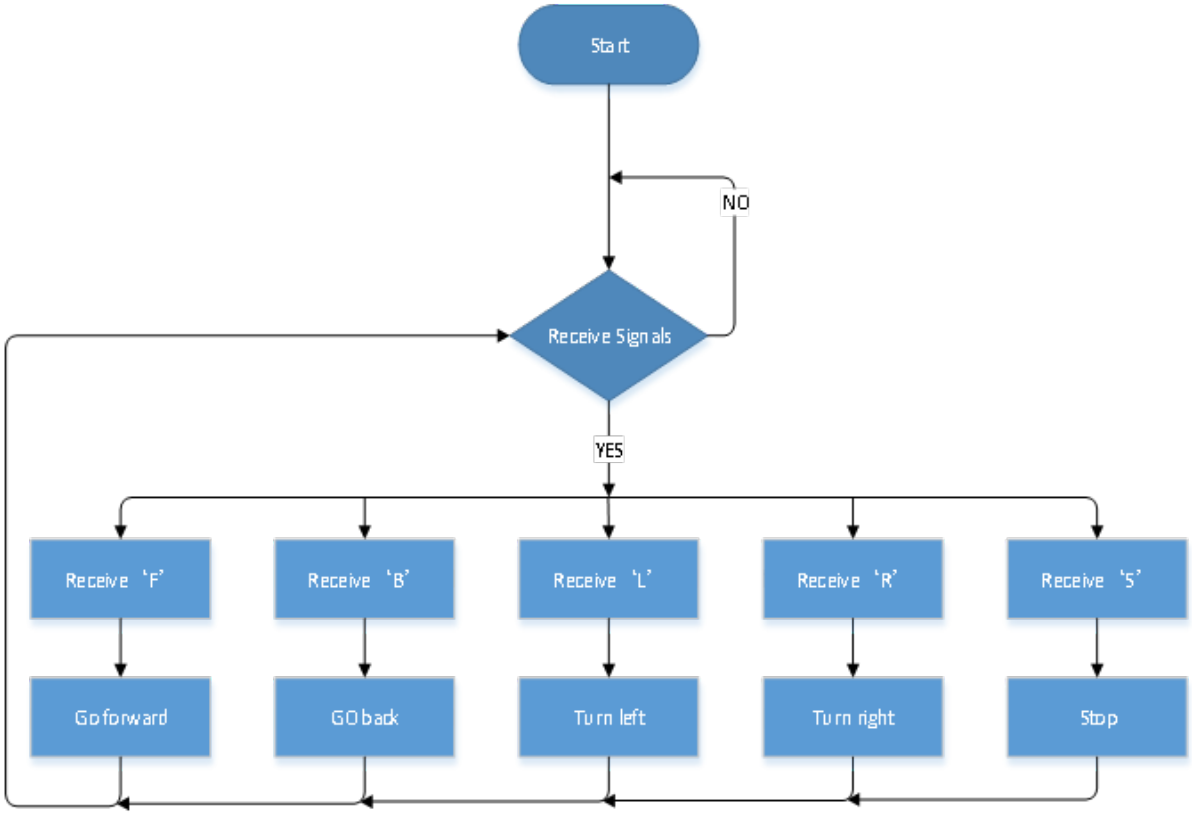
We've learned the basic knowledge of Bluetooth. And in this lesson, we will make a Bluetooth control smart car. In this project, we aim to regard the mobile phone as the transmitter (host), and the smart car connected to the BT24 Bluetooth module (slave) as the receiver and use the mobile APP to control the smart car via the Bluetooth.

2.APP Control Button

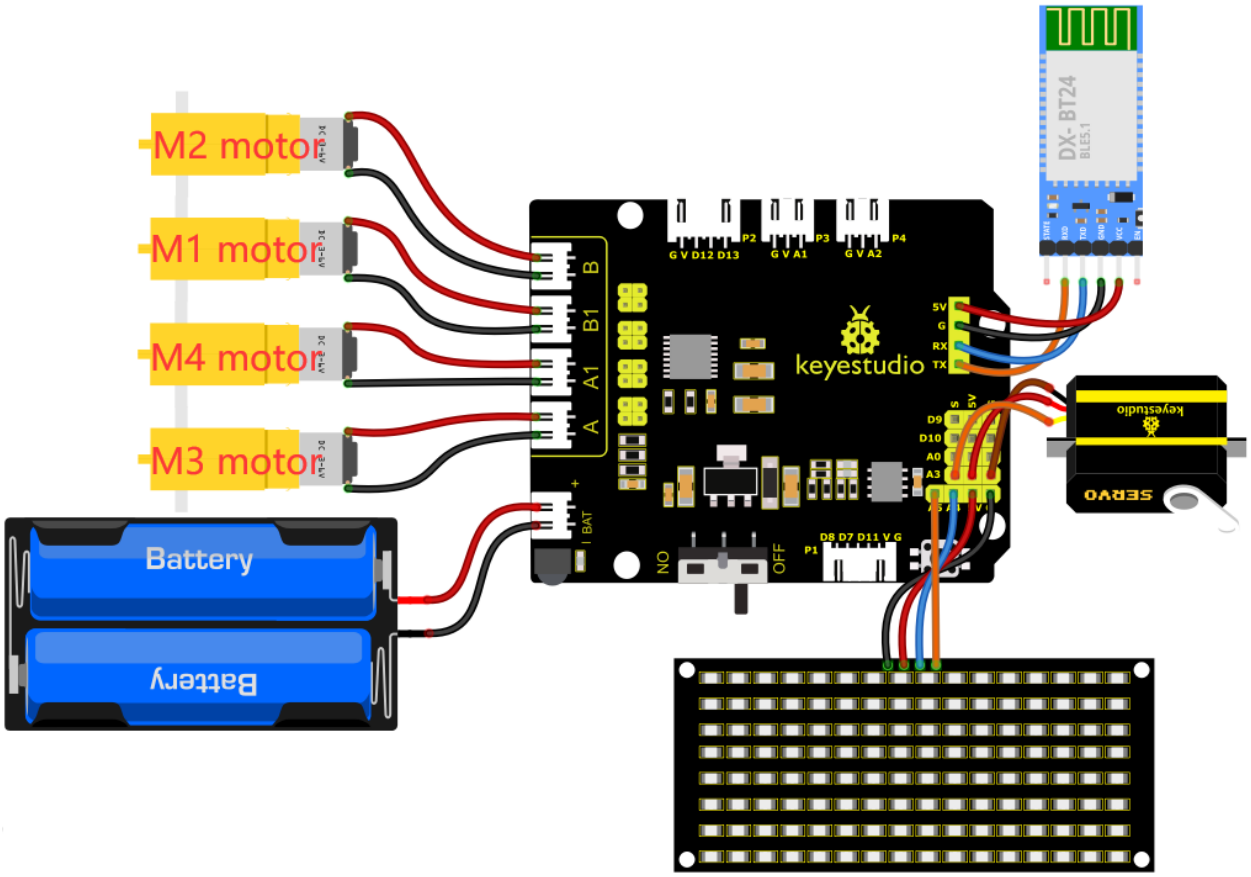
Key	Function
	Pair DX-BT24 5.1 Bluetooth module
	Disconnect Bluetooth

	Control character	Function
	Press: F Release: S	Press the button, the car goes front; release to stop
	Press: L Release: S	Press the button, the car turns left; release to stop
	Press: R Release: S	Press the button, the car turns right; release to stop
	Press: B Release: S	Press the button, the car goes back; release to stop
	Press: “a” Release: “S”	Click to speed up(maximum:255)
	Press: “d” Release: “S”	Click to slow down(minimum:0)
	Click to start the gravity sensing function of the mobile phone: click again to exit the gravity sensing control	
	Click to send“X”, click again to send“S”	Start line tracking function; click again to exit
	Click to send“Y”, click again to send“S”	Start ultrasonic avoiding function; click again to exit
	Click to send“U”, click again to send“S”	Start ultrasonic follow function; click again to exit
	Click to send“G”,click again to send“S”	Start restricting function; click again to exit

3.Flow Chart


Start
NO
Receive Signals
YES
Receive 'F'
Receive 'B'
Receive 'L'
Receive 'R'
Receive 'S'
Go forward
GO back
Turn left
Turn right
Stop

4.Wiring Diagram



1). GND, VCC, SDA and SCL of the 8*8 LED board are connected to GGND), VVCC), A4 and A5 of the expansion board.

2). The RXD, TXD, GND and VCC of the Bluetooth module are respectively connected to TX, RX, G and 5V on the 8833 motor driver expansion board, while the STATE and BRK pins of the Bluetooth module do not need to be connected.

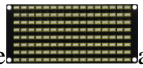
3). The servo is connected to G, V and A3. The brown wire is interfaced with Gnd(G), the red wire is interfaced with 5V(V) and the orange wire is interfaced with A3.

4). The power is connected to the BAT port

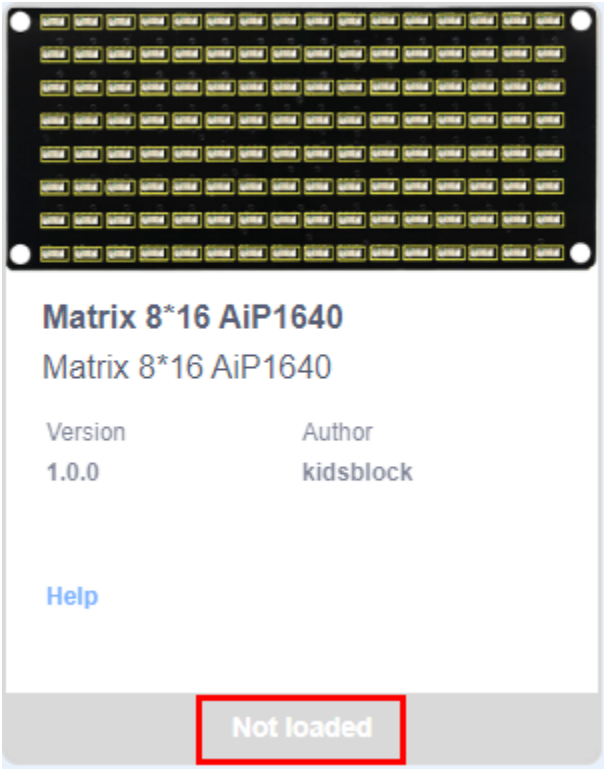
5.Test Code

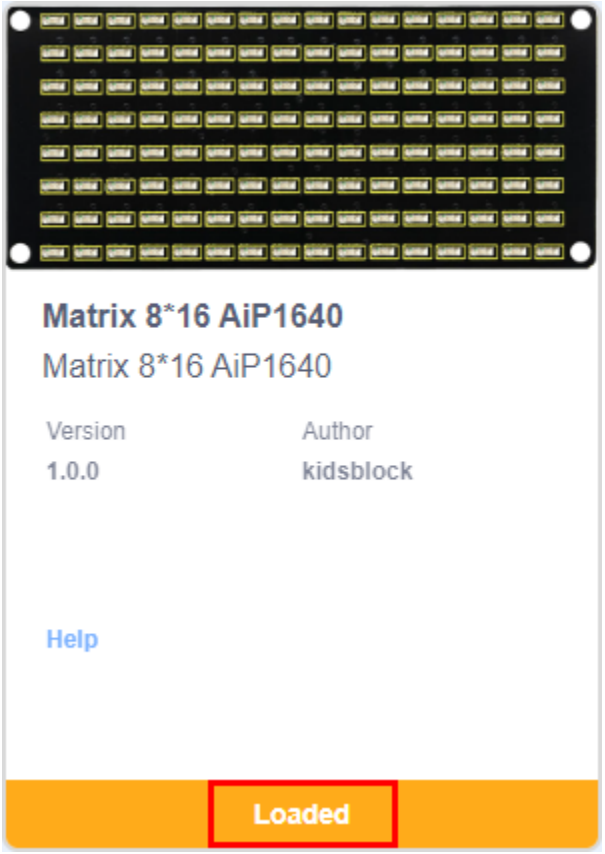
Before writing the code, it is necessary to import the library files of the 8x16 LED board and the servo. The specific steps are as follows:

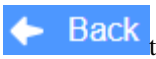
Click

to enter the extension library interface of sensors/modules/components, then search for"**Matrix 8*16 Aip1640**"module  and click it. In this way, "**Not loaded**" changes to "**loaded**", indicating that the"**Matrix**

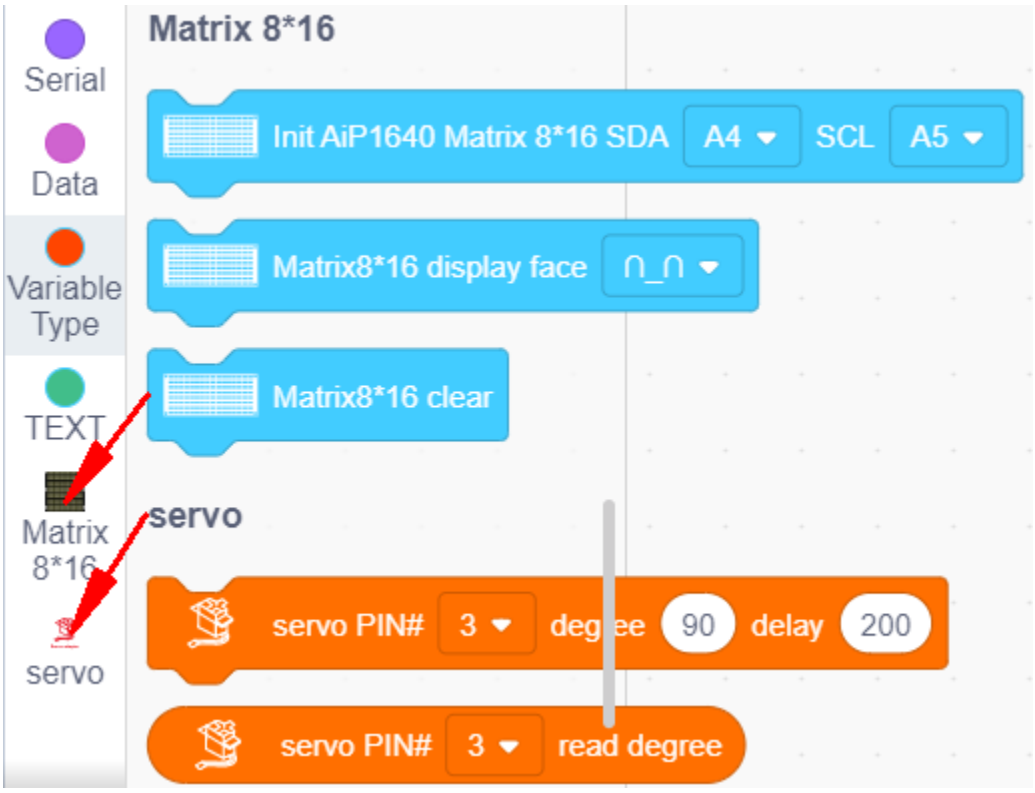
8*16 Aip1640"module was added successfully.





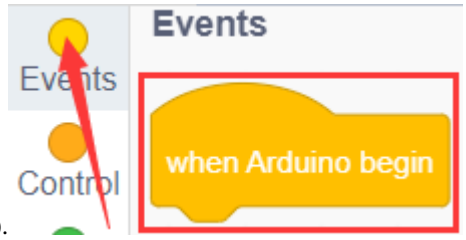
Click 

to return to the code editor interface, the instruction block of the added"**Matrix 8*16 Aip1640**"module and "**Servo**"component can be seen in the module area.

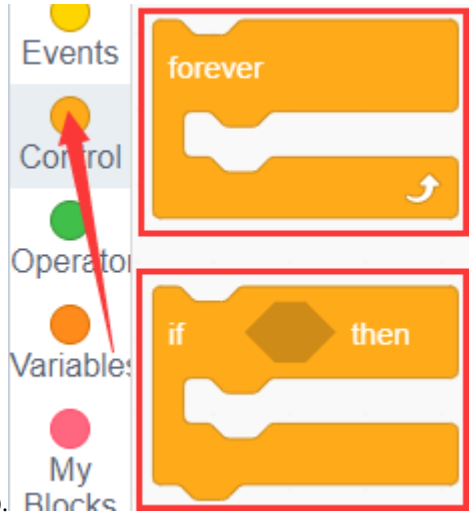


You can drag blocks to edit. Blocks listed below are for your reference.

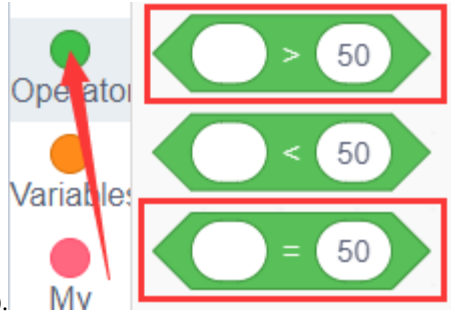
(1).



(2).



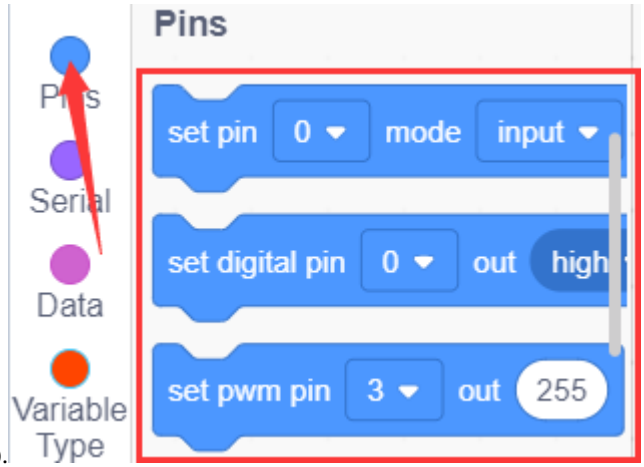
(3).


Operator
Variables
My
> 50
< 50
= 50

(4).

Serial
Data
Variable Type
Serial
serial begin baudrate 9600
serial print Hello KidsBlock

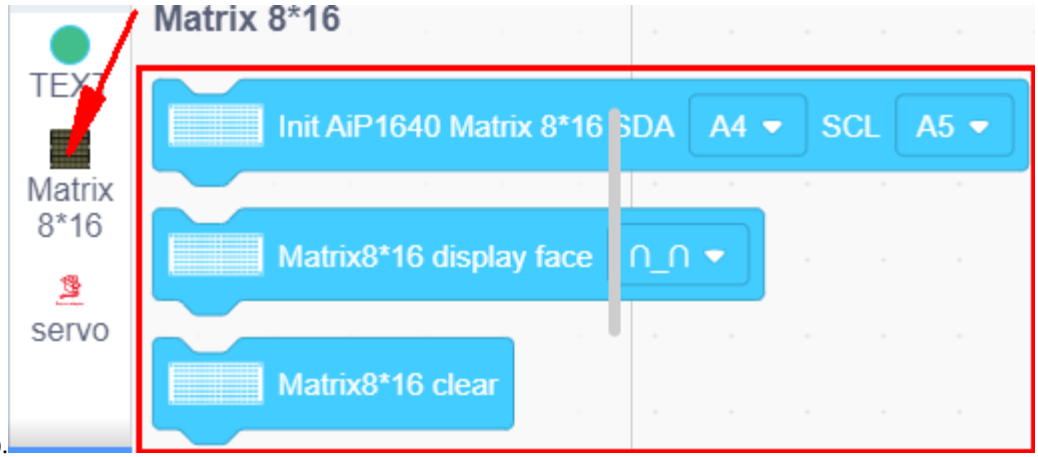
(5).


Pins
Serial
Data
Variable Type
Pins
set pin 0 mode input
set digital pin 0 out high
set pwm pin 3 out 255

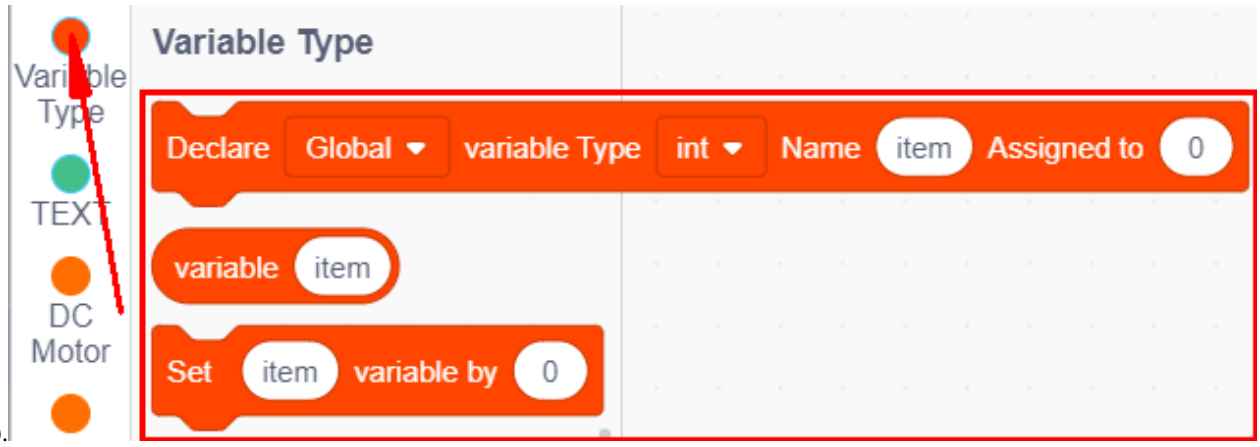
(6).

TEXT
Matrix 8*16
TEXT
char a
string hello

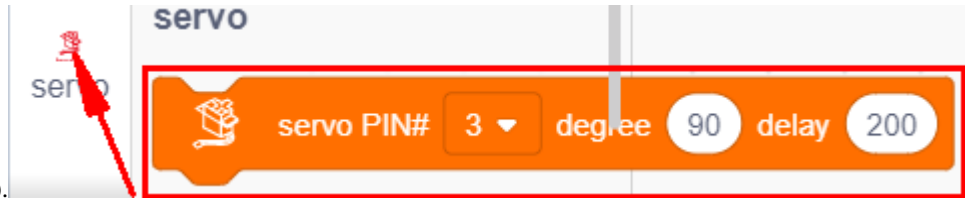
(7).



(8).

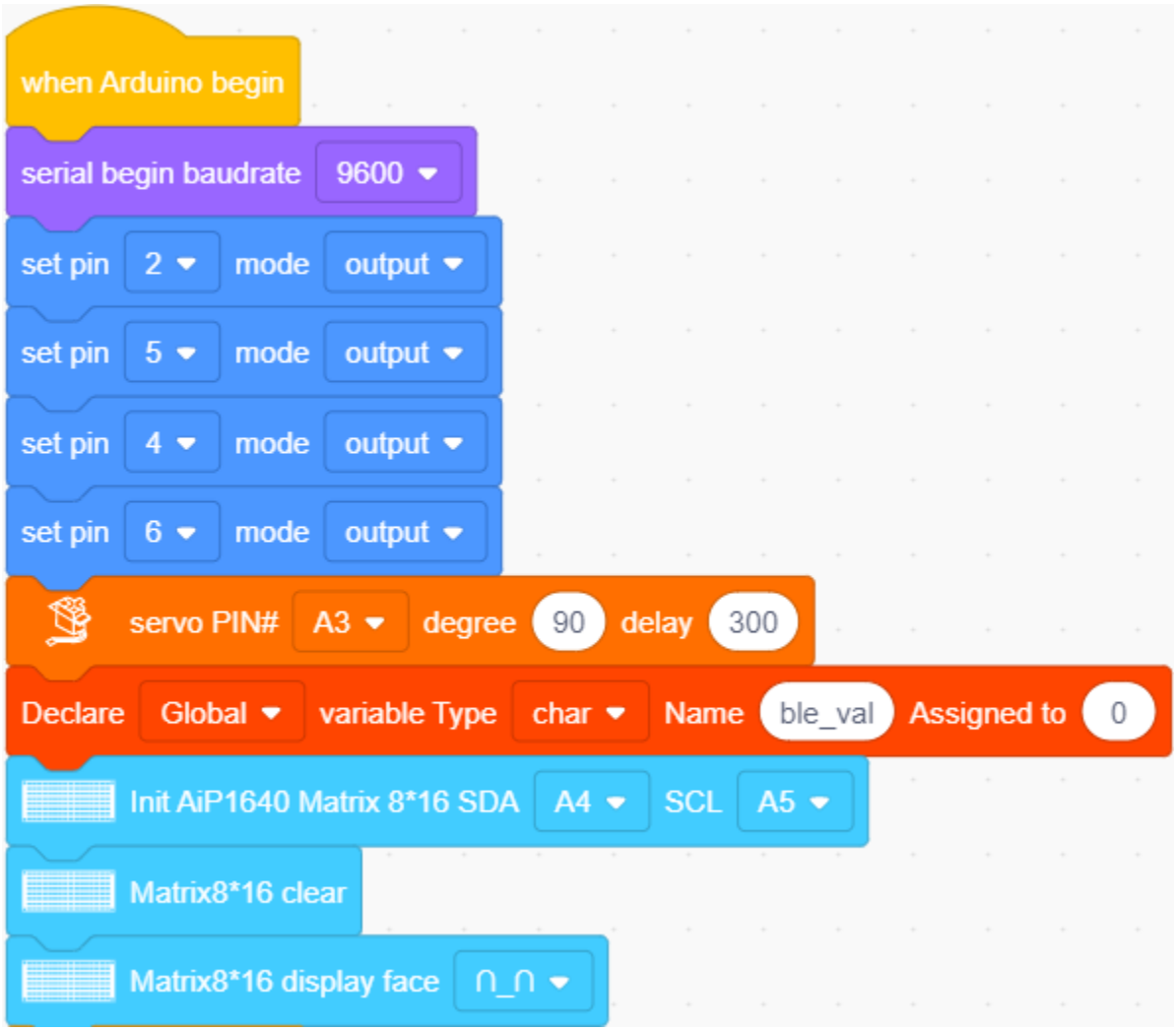


(9).

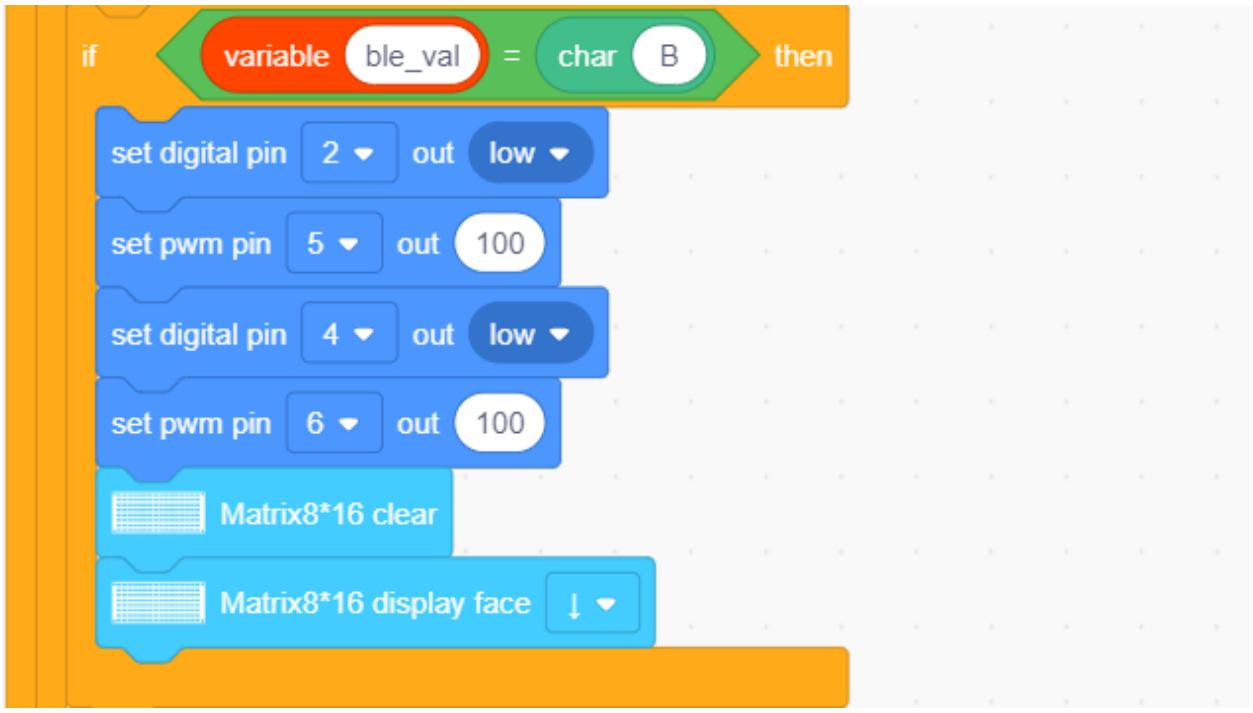


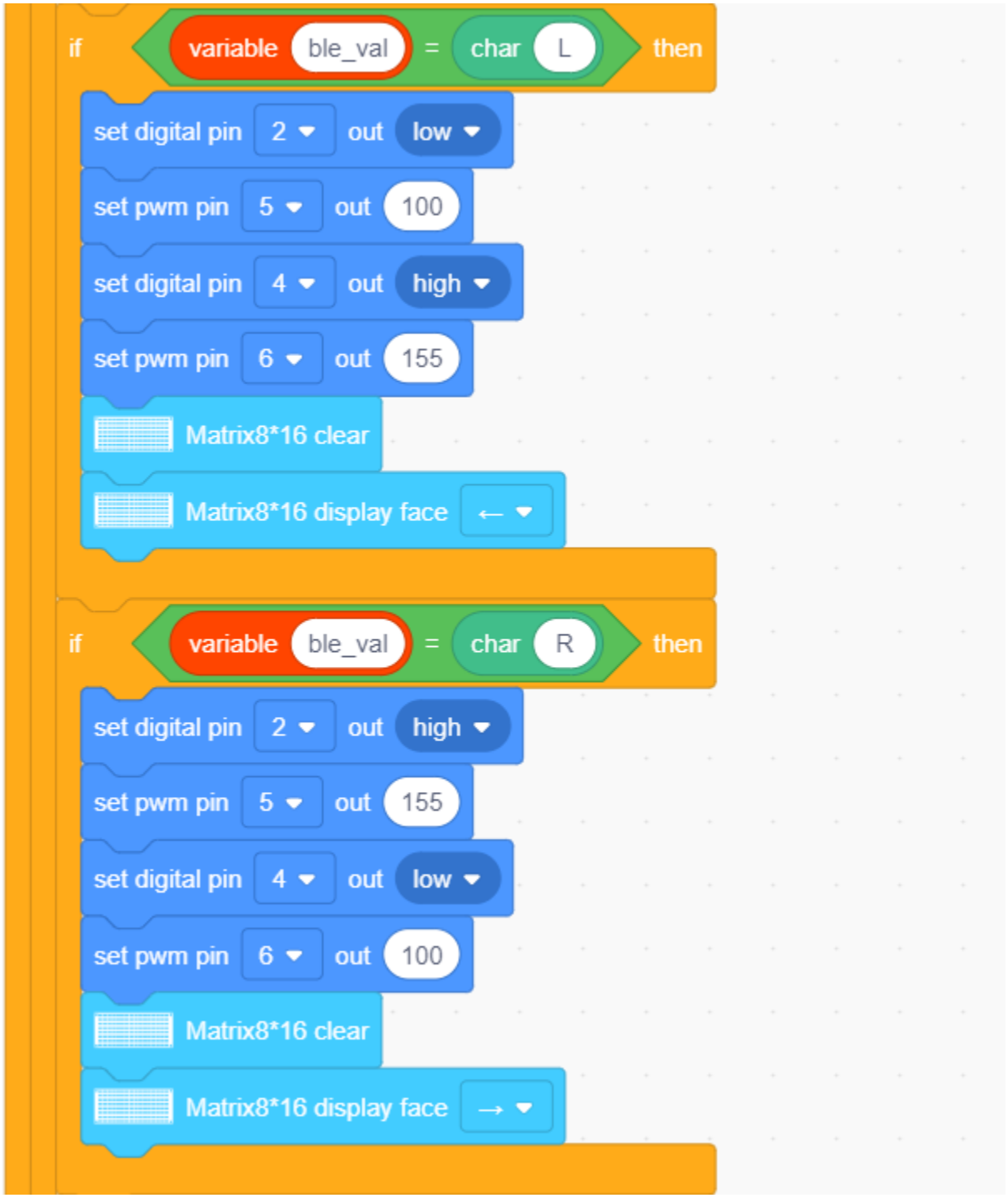
Complete Test Code

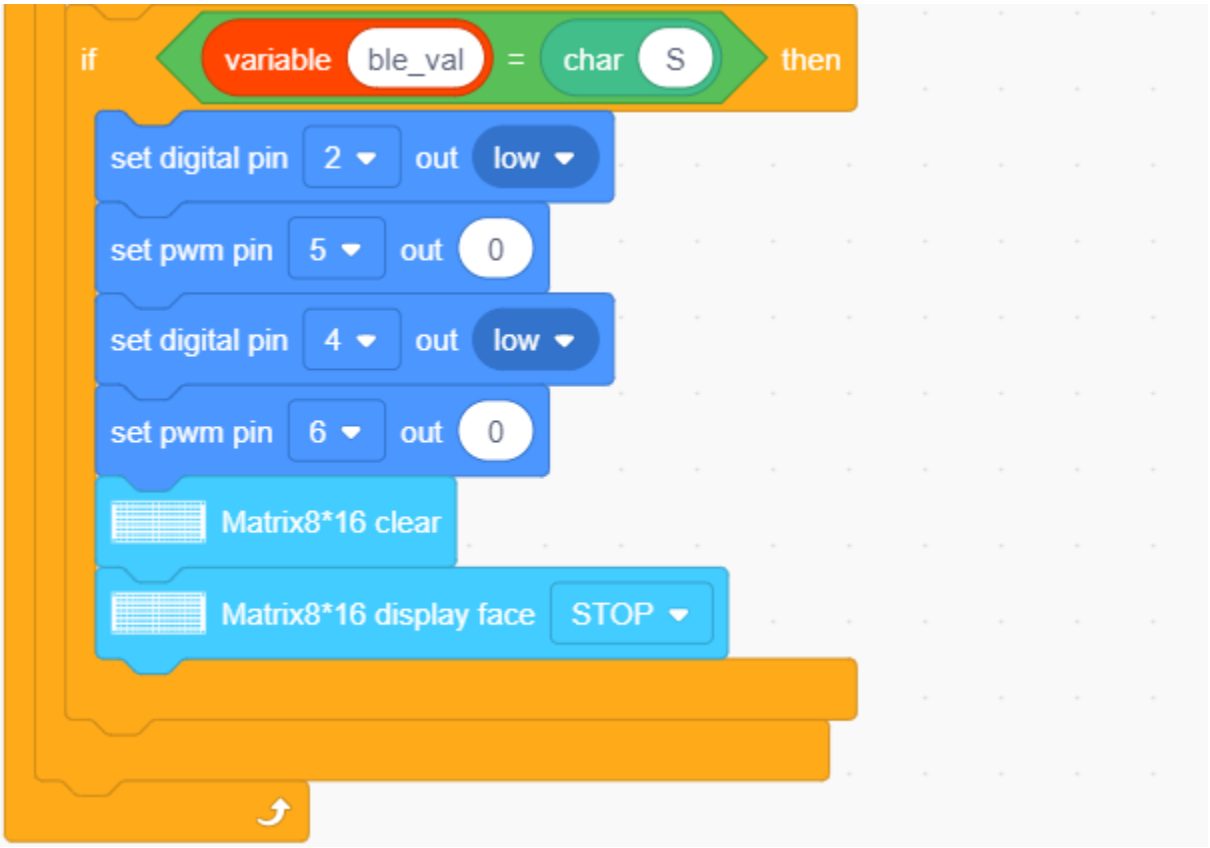
Note: Before uploading the test code, you need to remove the Bluetooth module, otherwise the code will fail to be uploaded.Connect the Bluetooth module after uploading the code successfully.


when Arduino begin
serial begin baudrate 9600
set pin 2 mode output
set pin 5 mode output
set pin 4 mode output
set pin 6 mode output
servo PIN# A3 degree 90 delay 300
Declare Global variable Type char Name ble_val Assigned to 0
Init AiP1640 Matrix 8*16 SDA A4 SCL A5
Matrix8*16 clear
Matrix8*16 display face ∩_∩

forever
if
serial available data length
>
0
then
Set
ble_val
variable by
serial read data
serial print
variable
ble_val
warp
if
variable
ble_val
=
char
F
then
set digital pin
2
out
high
set pwm pin
5
out
155
set digital pin
4
out
high
set pwm pin
6
out
155
Matrix8*16 clear
Matrix8*16 display face


if
variable
ble_val
=
char
B
then
set digital pin
2
out
low
set pwm pin
5
out
100
set digital pin
4
out
low
set pwm pin
6
out
100
Matrix8*16 clear
Matrix8*16 display face


if
variable ble_val = char L then
set digital pin 2 out low
set pwm pin 5 out 100
set digital pin 4 out high
set pwm pin 6 out 155
Matrix8*16 clear
Matrix8*16 display face ←
if
variable ble_val = char R then
set digital pin 2 out high
set pwm pin 5 out 155
set digital pin 4 out low
set pwm pin 6 out 100
Matrix8*16 clear
Matrix8*16 display face →



6.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, power on the external power then turn the DIP switch to ON.

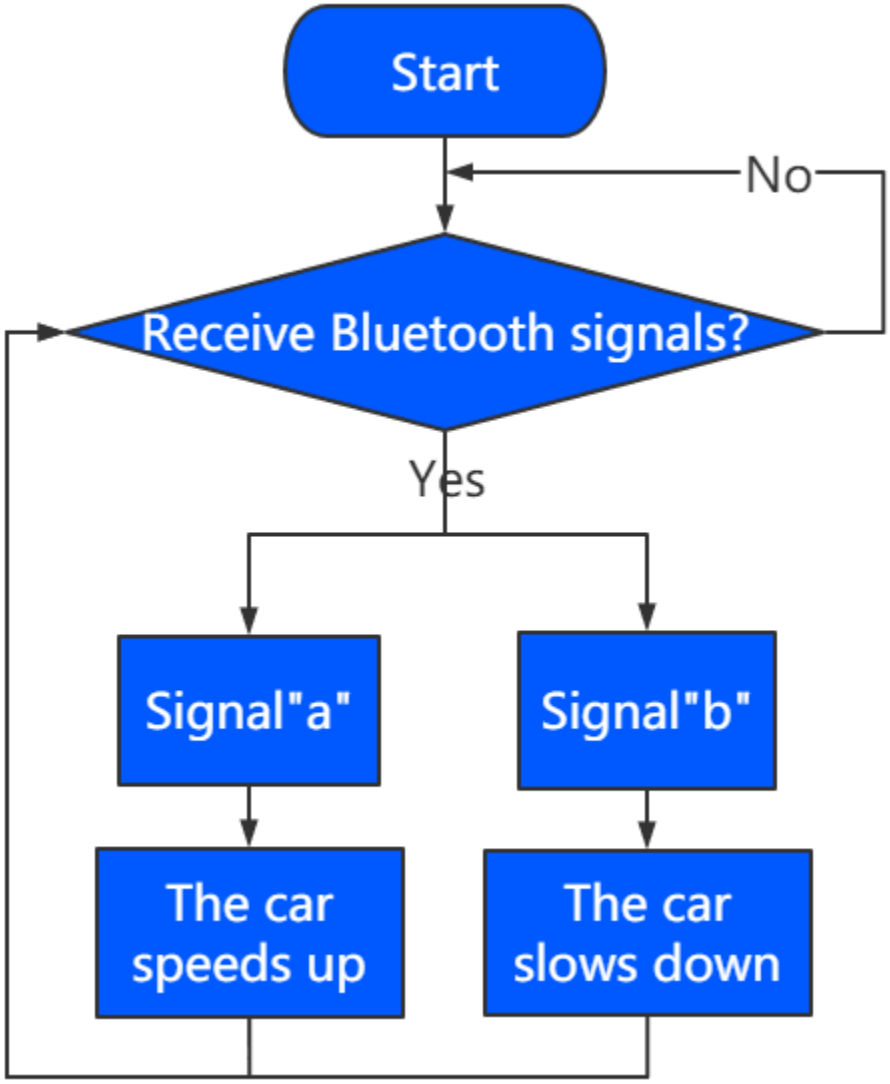
Inset the BT module and open your cellphone to connect the Bluetooth to control the smart car. The can will move forward, backward, turn left and right and stop. Also the 8*8 LED board will show the corresponding patterns.

10.2.16 Project 16 Bluetooth Speed Control Smart Car

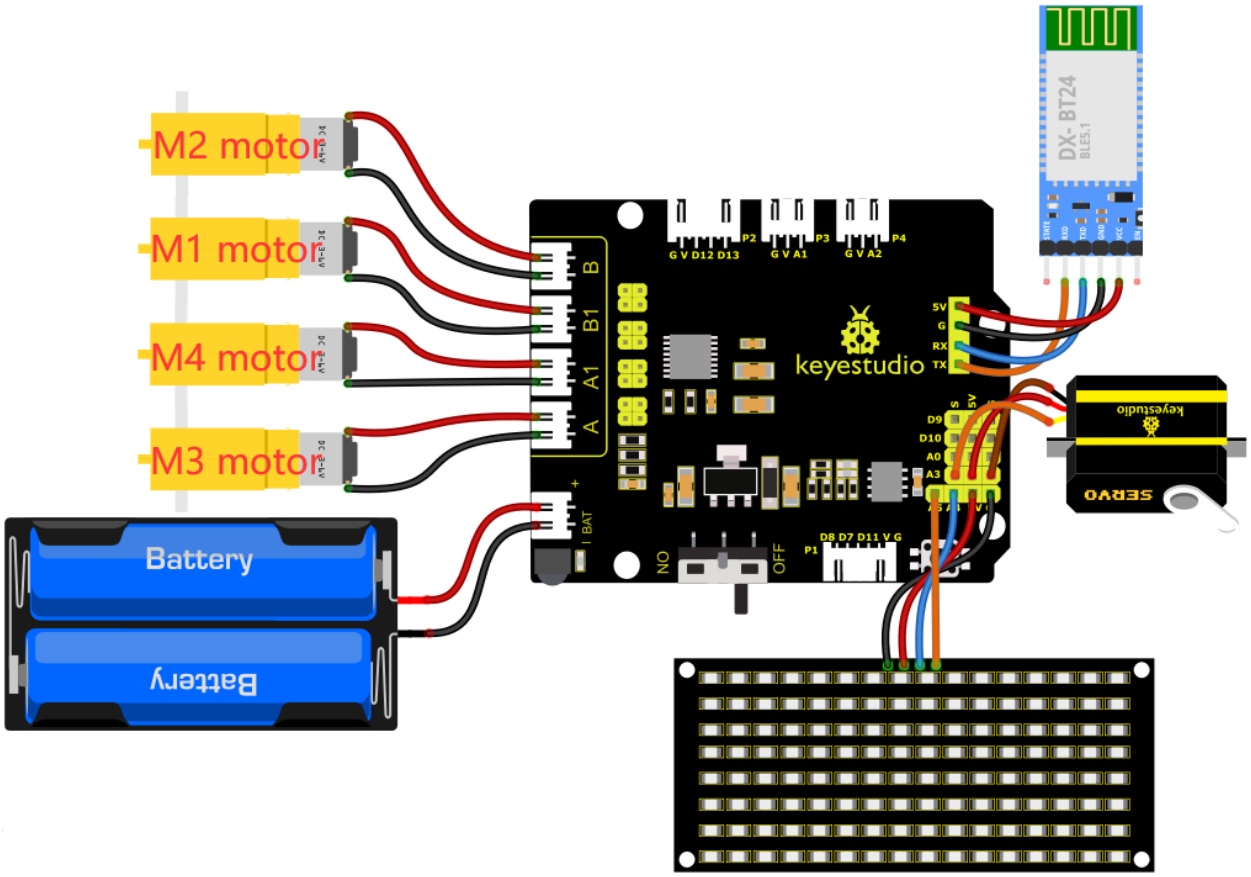
1.Description

In this project, we will use a Bluetooth to adjust the speed of the smart car. We empower to define a variable speeds and change it to change the speed of the smart car.

2.Flow Chart


Start
No
Receive Bluetooth signals?
Yes
Signal"a"
Signal"b"
The car speeds up
The car slows down

3.Wiring Diagram



1). GND, VCC, SDA and SCL of the 8*8 LED board are connected to GGND), VVCC), A4 and A5 of the expansion board.

2). The RXD, TXD, GND and VCC of the Bluetooth module are respectively connected to TX, RX, G and 5V on the 8833 motor driver expansion board, while the STATE and BRK pins of the Bluetooth module do not need to be connected.

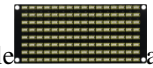
3). The servo is connected to G, V and A3. The brown wire is interfaced with Gnd(G), the red wire is interfaced with 5V(V) and the orange wire is interfaced with A3.

4). The power is connected to the BAT port

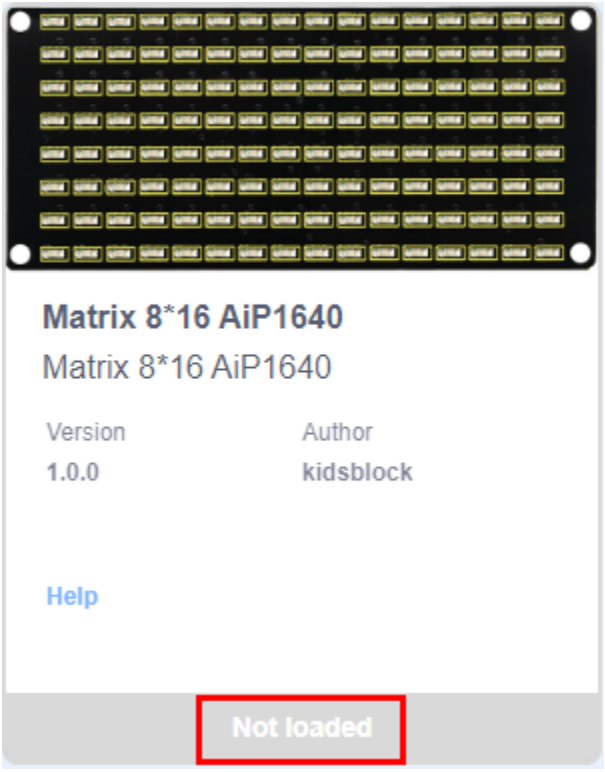
4.Test Code

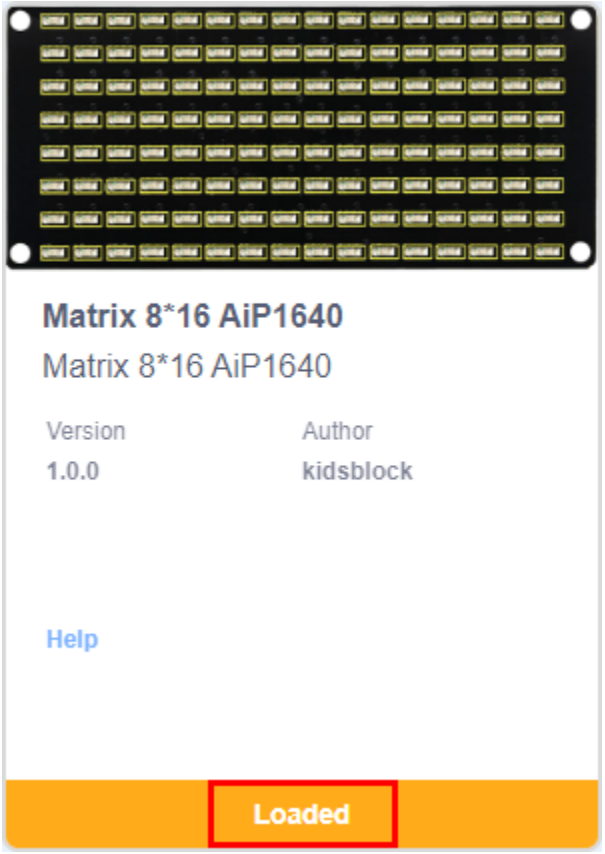
Before writing the code, it is necessary to import the library files of the 8x16 LED board and the servo. The specific steps are as follows:

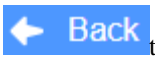
Click

to enter the extension library interface of sensors/modules/components, then search for“Matrix 8*16 Aip1640”module  and click it. In this way, “**Not loaded**” changes to “**loaded**”, indicating that the“**Matrix**

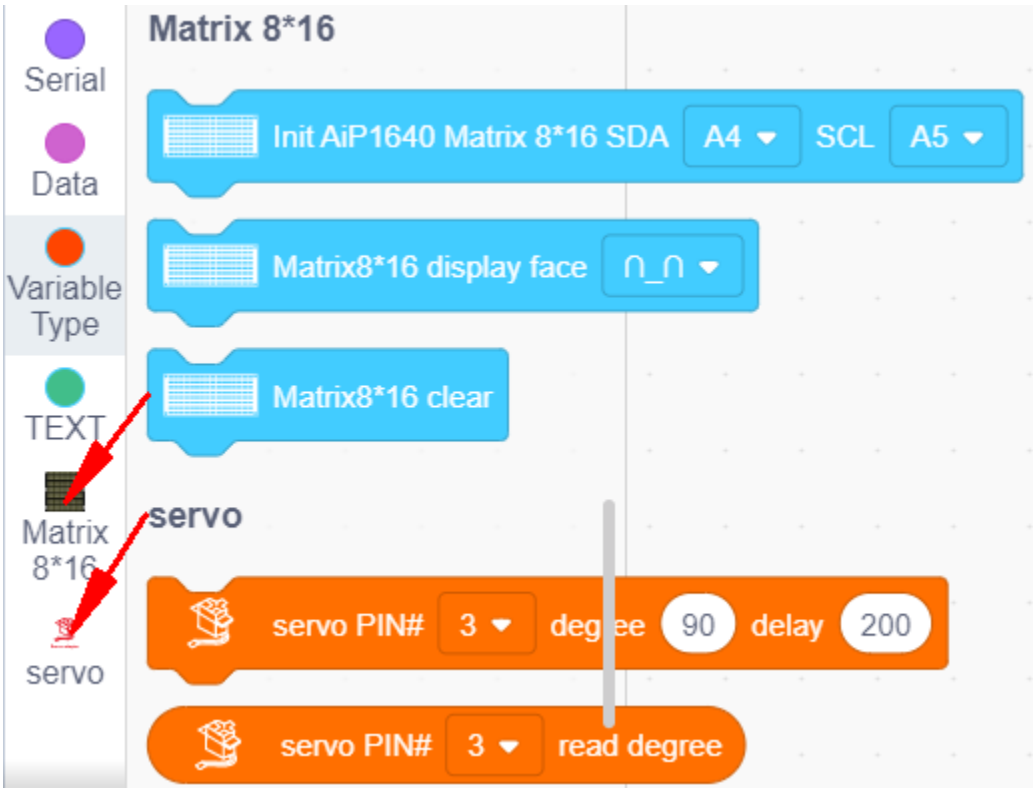
8*16 Aip1640"module was added successfully.





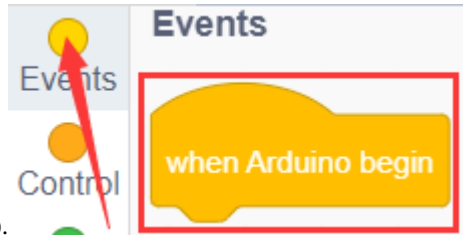
Click 

to return to the code editor interface, the instruction block of the added "**Matrix 8*16 Aip1640**"module and "**Servo**"component can be seen in the module area.

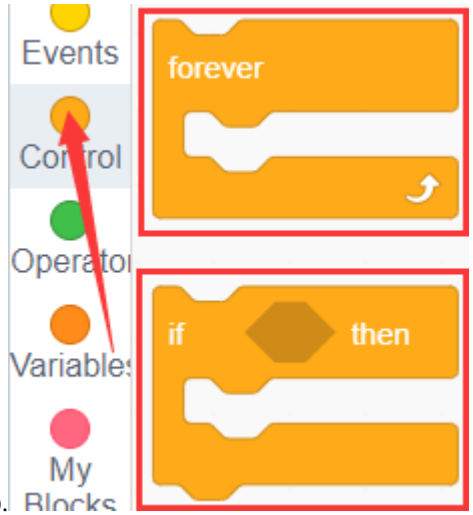


You can drag blocks to edit. Blocks listed below are for your reference

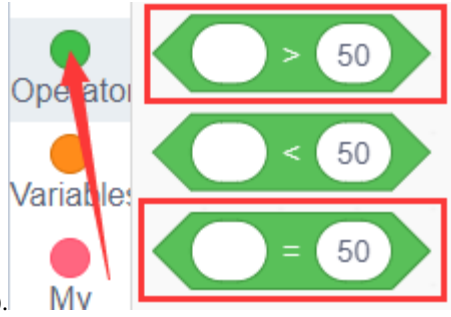
(1).



(2).



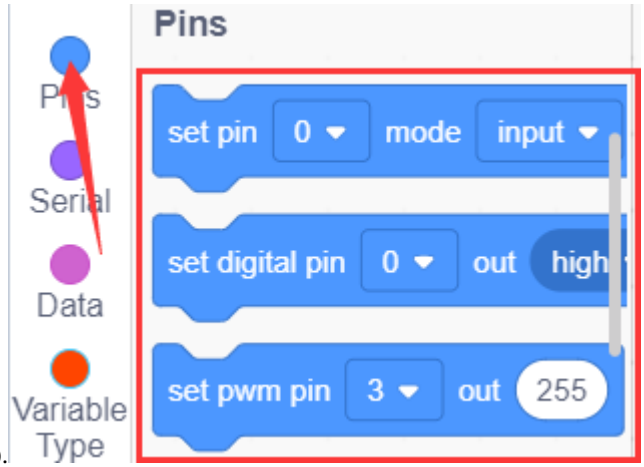
(3).


Operator
Variables
Mv
> 50
< 50
= 50

(4).

Serial
Data
Variable
Type
Serial
serial begin baudrate 9600
serial print Hello KidsBlock

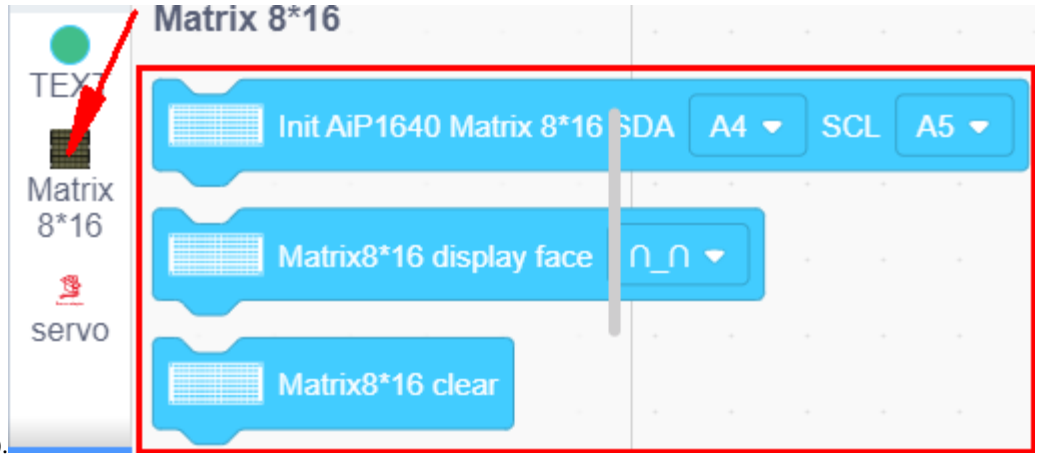
(5).


Pins
Serial
Data
Variable
Type
Pins
set pin 0 mode input
set digital pin 0 out high
set pwm pin 3 out 255

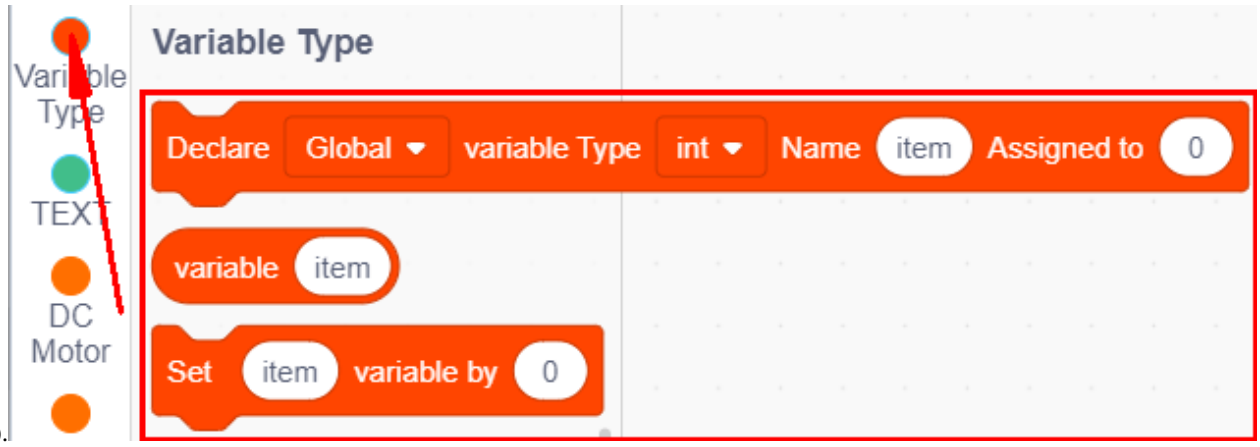
(6).

TEXT
Matrix
8*16
TEXT
char a
string hello

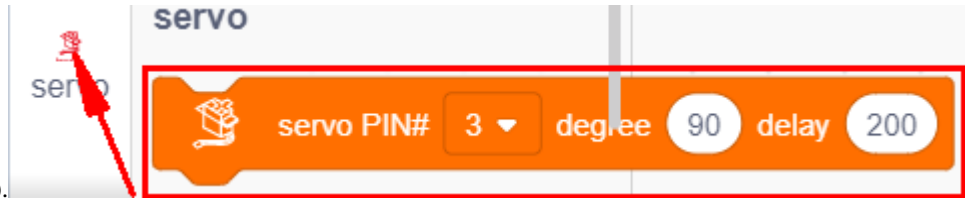
(7).



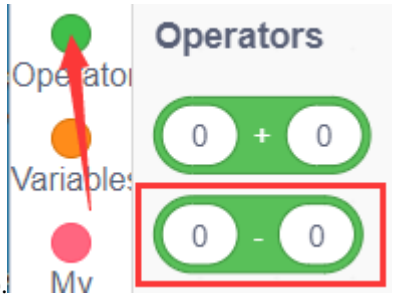
(8).



(9).

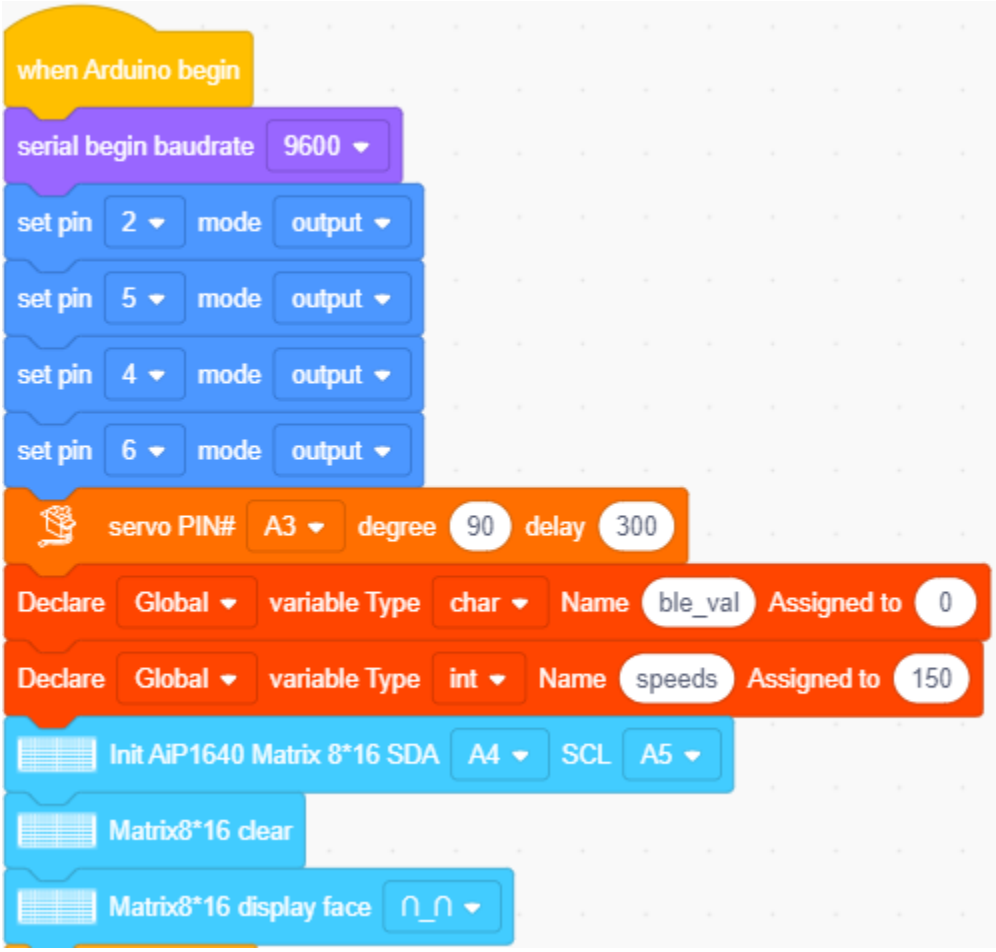


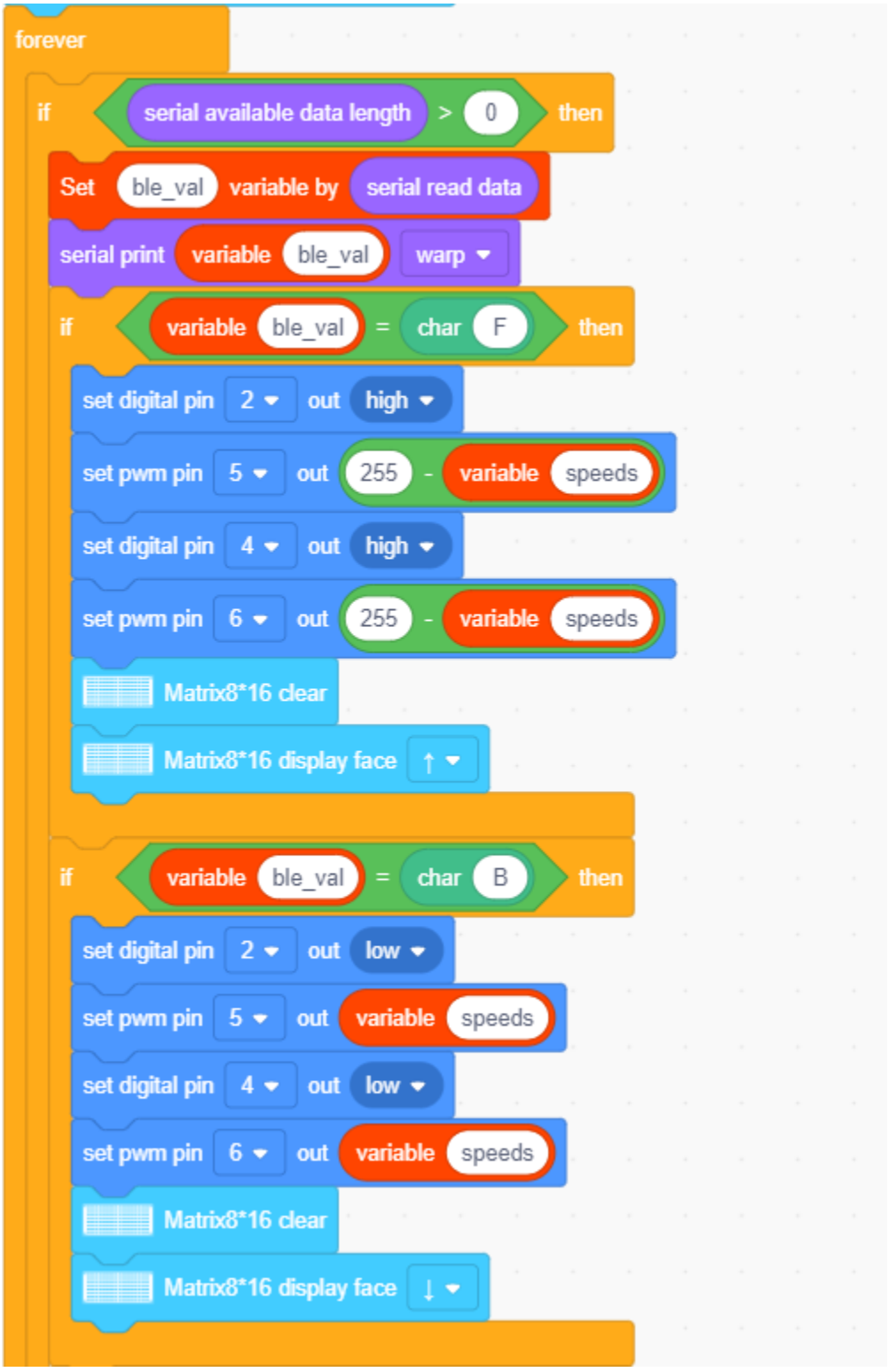
(10).

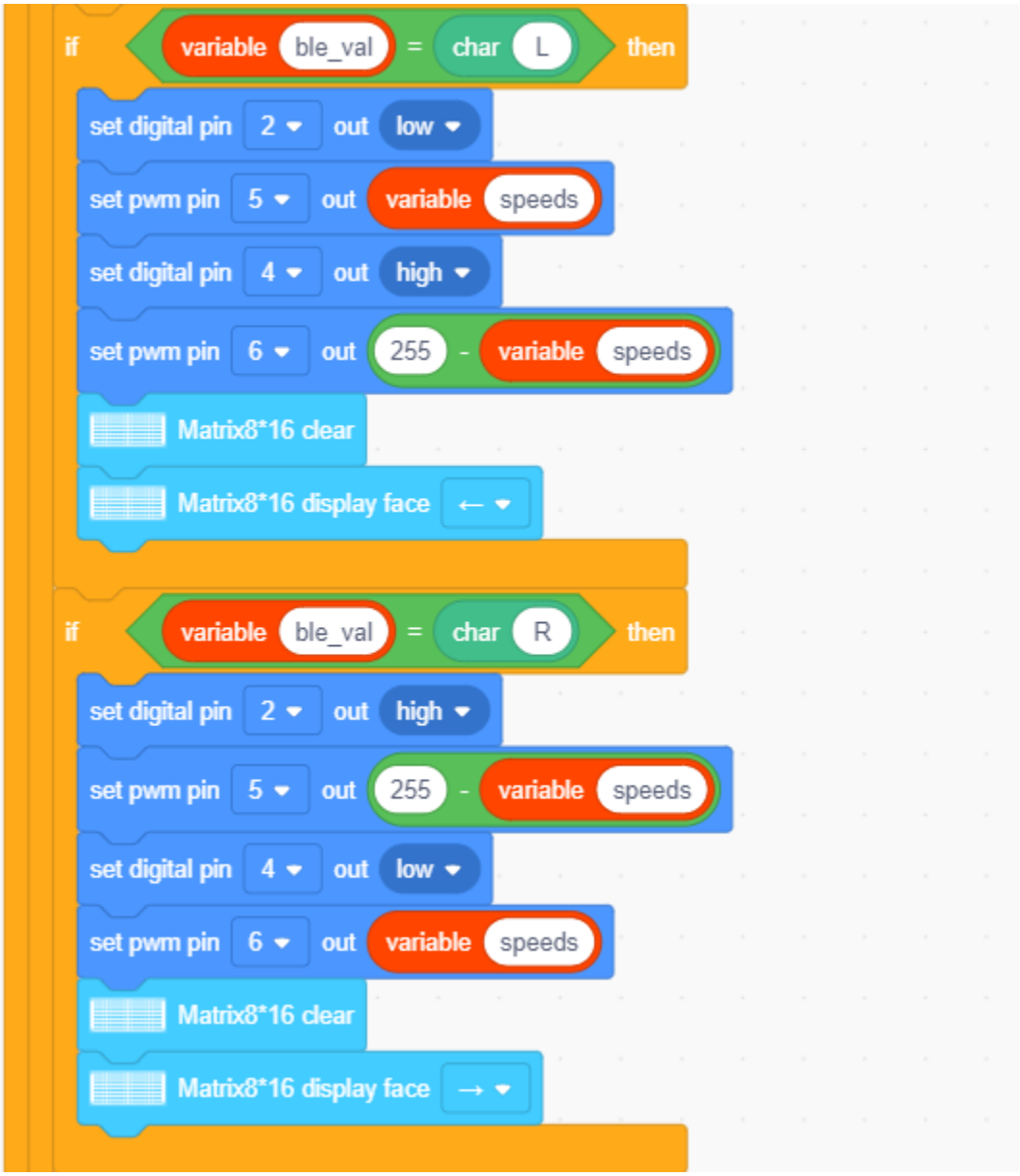


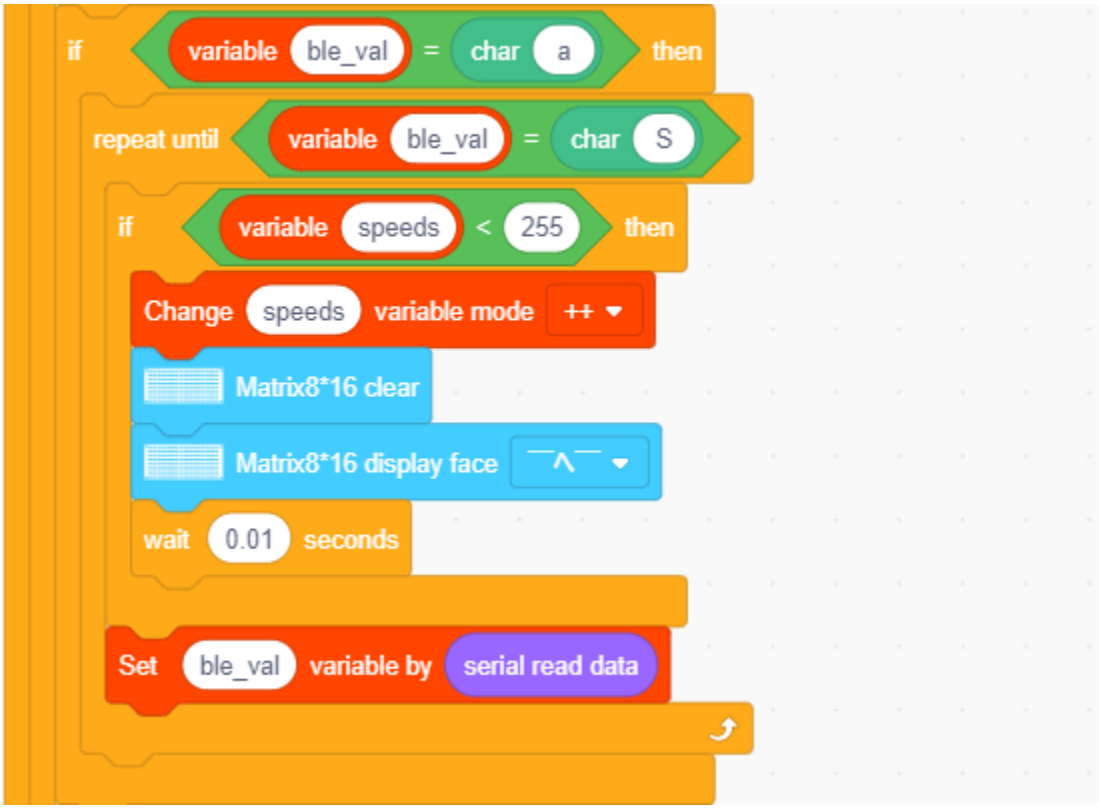
Complete Test Code

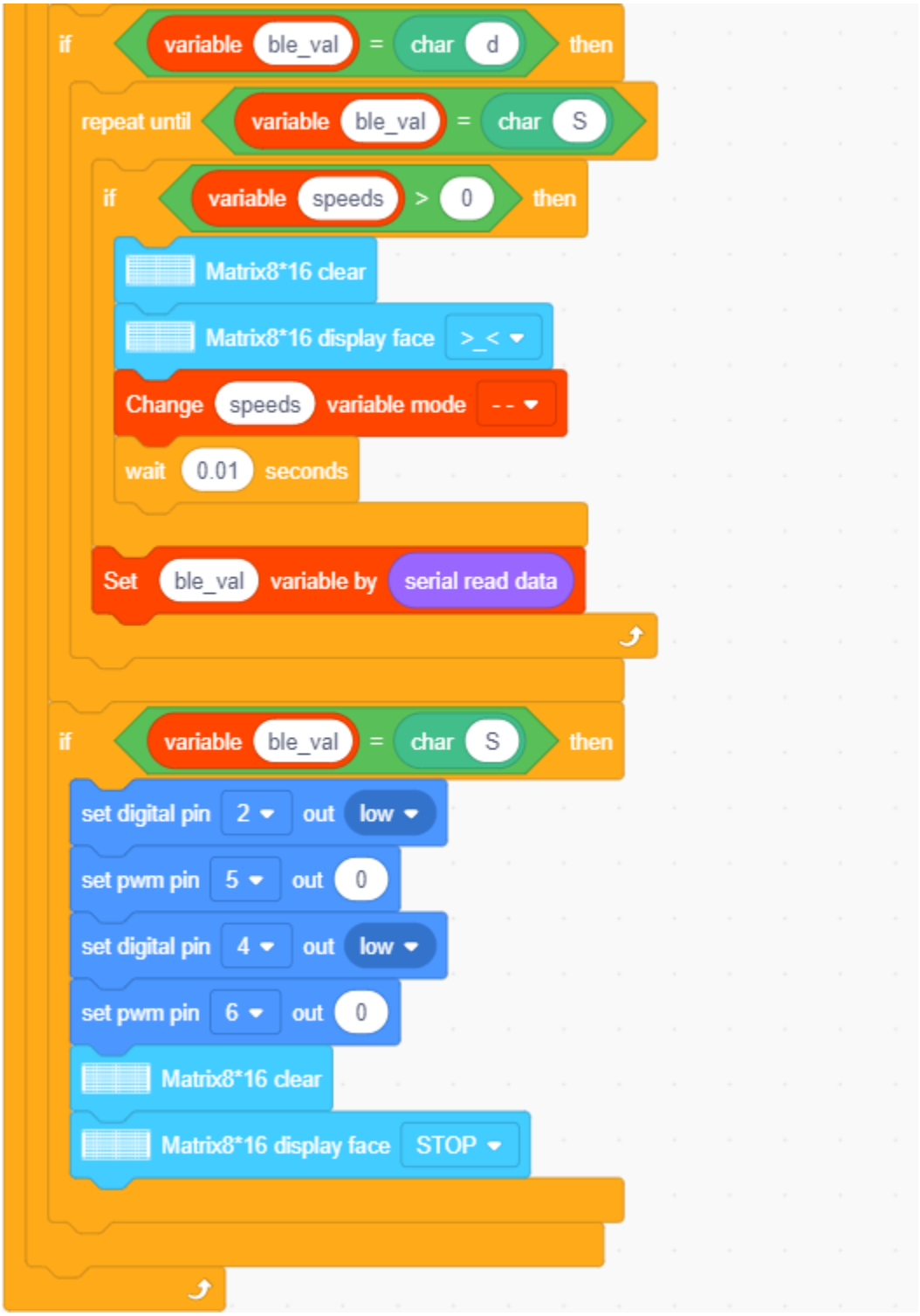
Note: Before uploading the test code, you need to remove the Bluetooth module, otherwise the code will fail to be uploaded.Connect the Bluetooth module after uploading the code successfully.

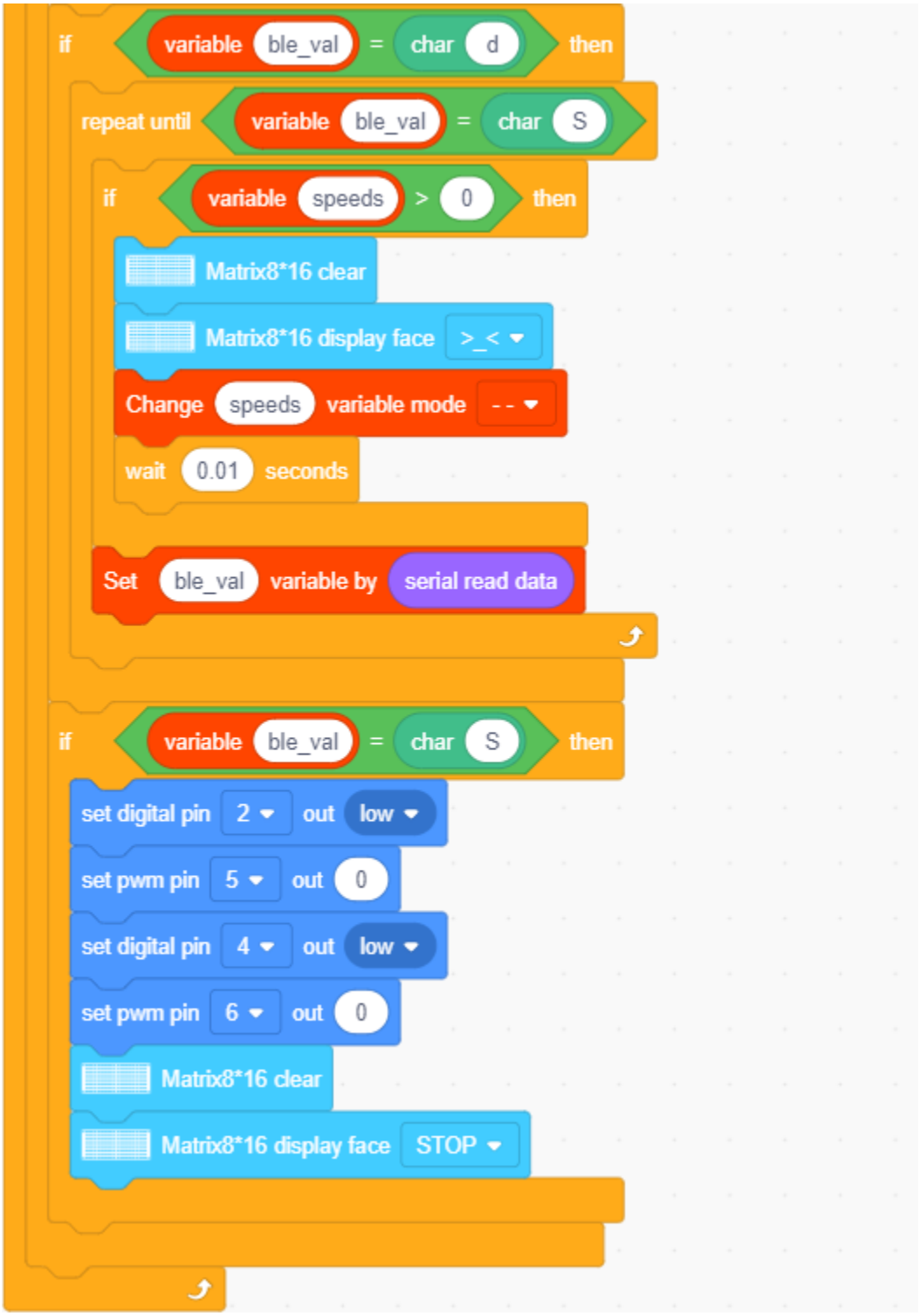

when Arduino begin
serial begin baudrate 9600
set pin 2 mode output
set pin 5 mode output
set pin 4 mode output
set pin 6 mode output
servo PIN# A3 degree 90 delay 300
Declare Global variable Type char Name ble_val Assigned to 0
Declare Global variable Type int Name speeds Assigned to 150
Init AiP1640 Matrix 8*16 SDA A4 SCL A5
Matrix8*16 clear
Matrix8*16 display face ∩_∩


forever
if
serial available data length
>
0
then
Set
ble_val
variable by
serial read data
serial print
variable
ble_val
warp
if
variable
ble_val
=
char
F
then
set digital pin
2
out
high
set pwm pin
5
out
255
-
variable
speeds
set digital pin
4
out
high
set pwm pin
6
out
255
-
variable
speeds
Matrix8*16 clear
Matrix8*16 display face
if
variable
ble_val
=
char
B
then
set digital pin
2
out
low
set pwm pin
5
out
variable
speeds
set digital pin
4
out
low
set pwm pin
6
out
variable
speeds
Matrix8*16 clear
Matrix8*16 display face


if
variable ble_val = char L then
set digital pin 2 out low
set pwm pin 5 out variable speeds
set digital pin 4 out high
set pwm pin 6 out 255 - variable speeds
Matrix8*16 clear
Matrix8*16 display face ←
if
variable ble_val = char R then
set digital pin 2 out high
set pwm pin 5 out 255 - variable speeds
set digital pin 4 out low
set pwm pin 6 out variable speeds
Matrix8*16 clear
Matrix8*16 display face →


if
variable ble_val = char a then
repeat until variable ble_val = char S
if variable speeds < 255 then
Change speeds variable mode ++
Matrix8*16 clear
Matrix8*16 display face ‾^‾
wait 0.01 seconds
Set ble_val variable by serial read data


if variable ble_val = char d then
repeat until variable ble_val = char S
if variable speeds > 0 then
Matrix8*16 clear
Matrix8*16 display face >_<
Change speeds variable mode --
wait 0.01 seconds
Set ble_val variable by serial read data
if variable ble_val = char S then
set digital pin 2 out low
set pwm pin 5 out 0
set digital pin 4 out low
set pwm pin 6 out 0
Matrix8*16 clear
Matrix8*16 display face STOP



5.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, power on the external power then turn the DIP switch to ON. Pairing the APP with Bluetooth, the smart car can be controlled to move by the APP.

Press

, the car will speed up, press

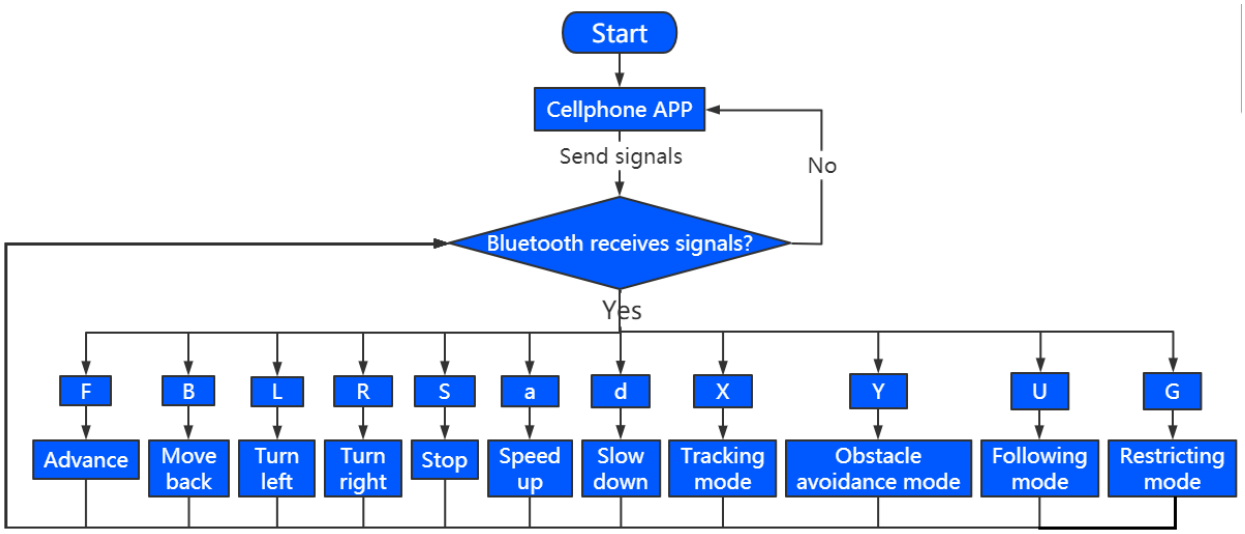
, the car will slow down, and the 8*16 LED board will display the corresponding status pattern of the smart car.

10.2.17 Project 17 Multi-purpose Bluetooth Smart Car

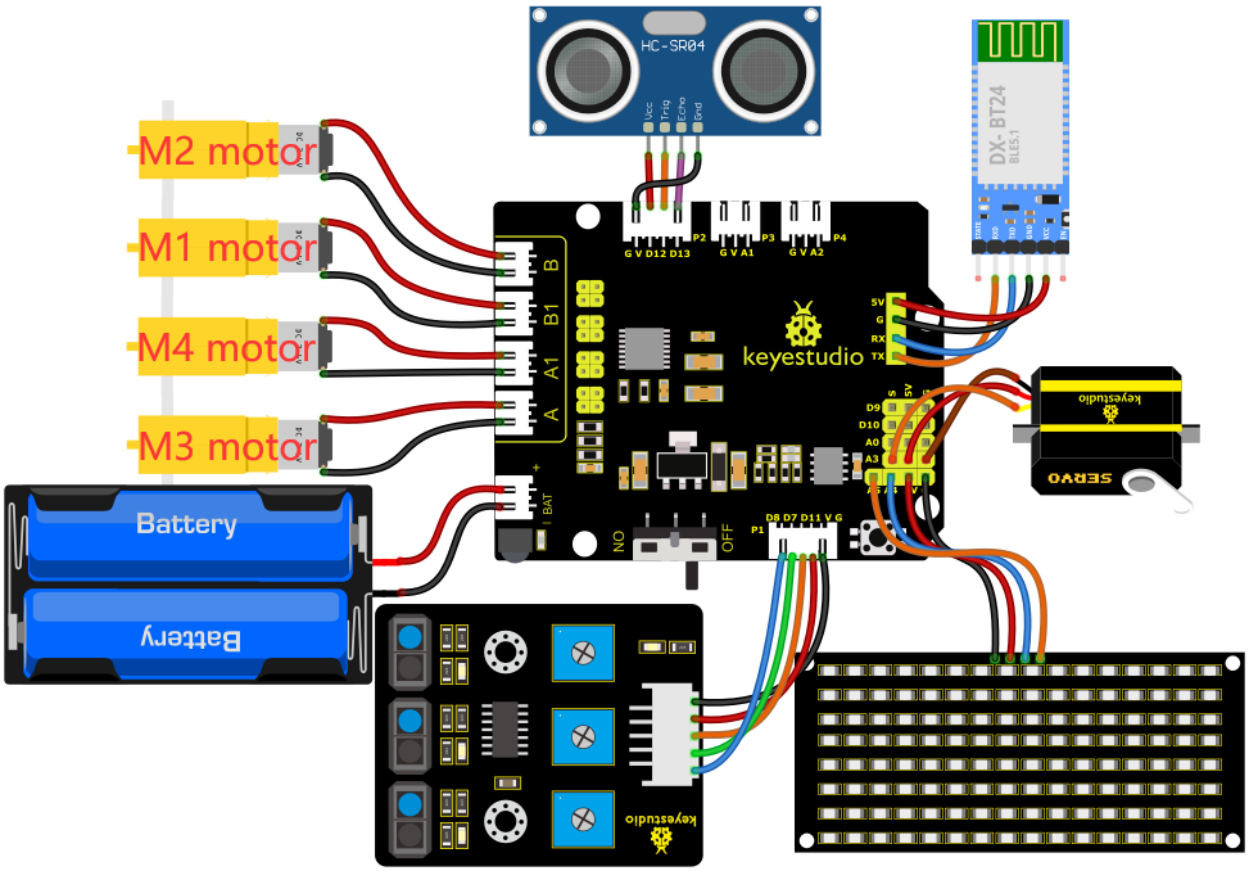
1.Description

In previous projects, the car only performs a single function. However, in this lesson, we will integrate all of its functions via a Bluetooth.

2.Flow Chart



3.Wiring Diagram



1). GND, VCC, SDA and SCL of the 8*8 LED board are connected to GGND), VVCC), A4 and A5 of the expansion board.

2). The RXD, TXD, GND and VCC of the Bluetooth module are respectively connected to TX, RX, G and 5V on the 8833 motor driver expansion board, while the STATE and BRK pins of the Bluetooth module do not need to be

connected.

3). The servo is connected to G, V and A3. The brown wire is interfaced with Gnd(G), the red wire is interfaced with 5V(V) and the orange wire is interfaced with A3.

4). G, V, S1, S2 and S3 of the line tracking sensor are connected to GGND), VVCC), D11, D7 and D8 of the sensor expansion board.

5). VCC, Trig, Echo and Gnd of the ultrasonic sensor are connected to 5V(V), D12(S), D13(S) and Gnd(G).

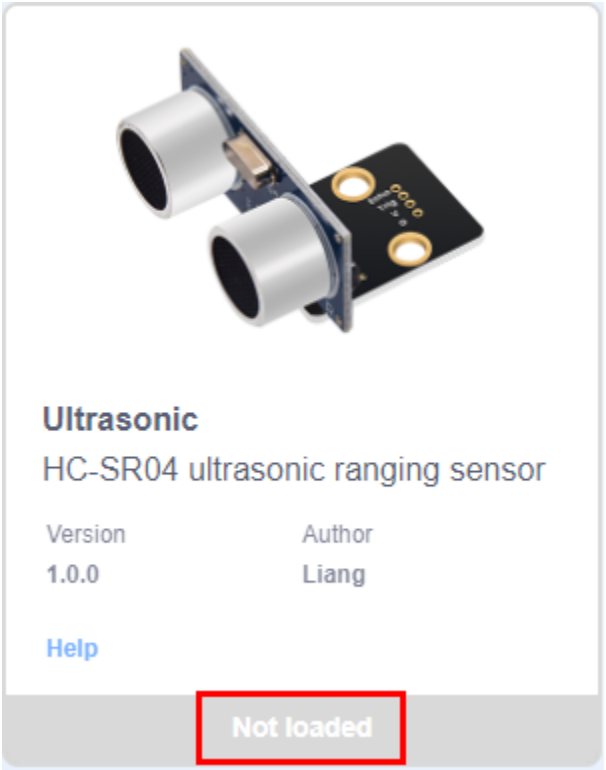
6). The power is connected to the BAT port

4.Test Code

Before writing the code, it is necessary to import the library files of the ultrasonic sensor, 8x16 LED board and the servo. The specific steps are as follows:

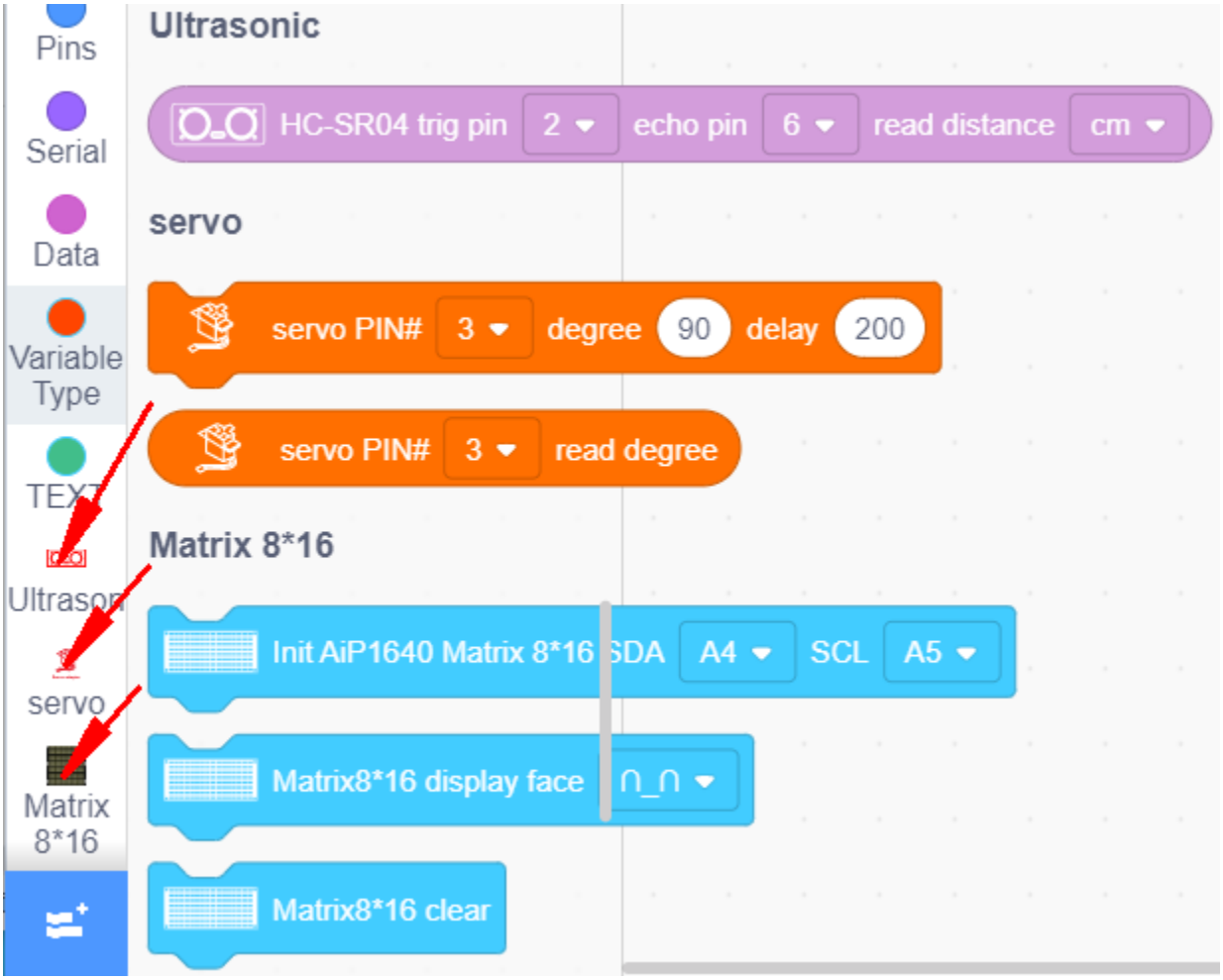
Click

to enter the extension library interface of sensors/modules/components, then search for"**Ultrasonic**"sensor and click it. In this way, "**Not loaded**" changes to "**loaded**", indicating that the"**Ultrasonic**"sensor was added successfully.



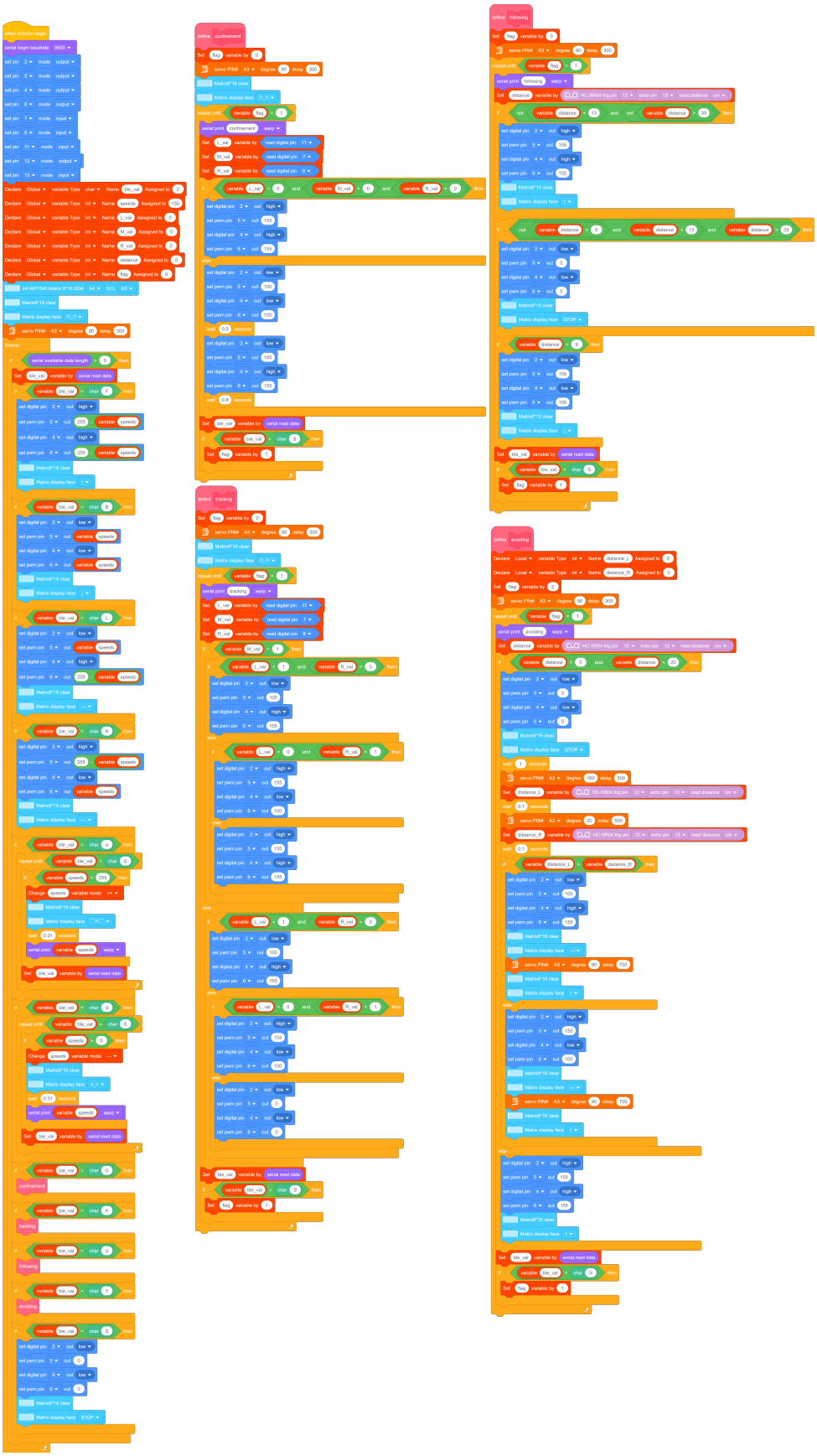
Click

to return to the code editor interface, the instruction block of the added "**Ultrasonic**"sensor,"**Matrix 8*16 Aip1640**"module and "**Servo**"component can be seen in the module area.



Complete Test Code

Note: Before uploading the test code, you need to remove the Bluetooth module, otherwise the code will fail to be uploaded.Connect the Bluetooth module after uploading the code successfully.



5.Test Result

After successfully uploading the code to the V4.0 board, connect the wirings according to the wiring diagram, power on the external power then turn the DIP switch to ON.

After the Bluetooth module is plugged into the APP and the mobile APP is successfully connected to the Bluetooth, the smart car can be controlled by the mobile APP. We can achieve the corresponding functions by pressing the corresponding buttons on the mobile APP.